

0902.0431: equation evidence

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_ EQ0515 [conf 0.002]	85	0.002	$\begin{aligned} & \& \lambda_0 - \lambda_1 = 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \\ & 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad \& \lambda_0 - \lambda_2 = 1 \\ & \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad \& \lambda_0 - \lambda_3 = 1 \\ & \quad \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad \& \lambda_1 - \lambda_2 = 0 \\ & \quad \quad \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad \& \lambda_1 - \lambda_3 = 0 \\ & \quad \quad \quad \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad \& \lambda_2 - \lambda_3 = 0 \end{aligned}$	$\begin{aligned} & \lambda_0 - \lambda_1 = 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \\ & \lambda_0 - \lambda_2 = 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad \lambda_0 + \lambda_1 = 1 \quad 2 \quad \frac{1}{1} = \frac{1}{1} \quad \frac{1}{\lambda_1 - \lambda_2} = 0 \quad \frac{1}{1} \quad 0 \quad 0 \quad \lambda_0 + \lambda_3 = 1 \quad \frac{1}{0} \quad \frac{2}{1} \\ & \lambda_0 - \lambda_3 = 1 \quad 1 \quad 1 \quad 0 \quad 2 \end{aligned}$	$\begin{aligned} \lambda_0 - \lambda_1 &= 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 & \lambda_0 + \lambda_1 &= 1 \quad 2 \quad 2 \quad 1 \quad 1 \quad 1 \\ \lambda_0 - \lambda_2 &= 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 & \lambda_0 + \lambda_2 &= 1 \quad 1 \quad 2 \quad 1 \quad 1 \quad 1 \\ \lambda_0 - \lambda_3 &= 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 & \lambda_0 + \lambda_3 &= 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \\ \lambda_1 - \lambda_2 &= 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 & \lambda_1 + \lambda_2 &= 0 \quad 1 \quad 2 \quad 1 \quad 1 \quad 1 \\ \lambda_1 - \lambda_3 &= 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 & \lambda_1 + \lambda_3 &= 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \\ \lambda_2 - \lambda_3 &= 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 & \lambda_2 + \lambda_3 &= 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_ EQ1026 [conf 0.018]	171	■ 0.018	$\left.\begin{array}{l} \lambda_0-\lambda_1=2 \quad 3 \quad 4 \quad 3 \quad 2 \quad 1 \quad 0 \quad 2 \\ \lambda_0-\lambda_2=2 \quad 3 \quad 4 \quad 6 \quad 5 \quad 4 \quad 3 \quad 2 \\ \lambda_0-\lambda_3=2 \quad 3 \quad 4 \quad 3 \quad 2 \quad 2 \quad 1 \quad 2 \\ \lambda_1-\lambda_2=0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \\ \lambda_1-\lambda_3=0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \\ \lambda_2-\lambda_3=0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \end{array}\right\} \lambda_0+\lambda_1=2 \quad 4 \quad 6 \quad 5 \quad 4 \quad 3 \quad 2 \quad 3 \\ \left.\begin{array}{l} \lambda_0+\lambda_2=2 \quad 4 \quad 6 \\ \lambda_0+\lambda_3=2 \quad 4 \quad 6 \quad 5 \quad 4 \quad 2 \quad 1 \quad 3 \\ \lambda_1+\lambda_2=0 \quad 1 \quad 2 \quad 2 \quad 2 \quad 2 \quad 1 \quad 1 \\ \lambda_1+\lambda_3=0 \quad 1 \quad 2 \quad 2 \quad 2 \quad 1 \quad 1 \quad 1 \\ \lambda_2+\lambda_3=0 \quad 1 \quad 2 \quad 2 \quad 2 \quad 1 \quad 0 \quad 1 \end{array}\right\}$	$\left.\begin{array}{l} \lambda_0-\lambda_1=2 \quad 3 \quad 4 \quad 3 \quad 2 \quad 1 \quad 0 \quad 2 \\ \lambda_0-\lambda_2=2 \quad 3 \quad 4 \quad 6 \quad 5 \quad 4 \quad 3 \quad 2 \\ \lambda_0-\lambda_3=2 \quad 3 \quad 4 \quad 3 \quad 2 \quad 2 \quad 1 \quad 2 \\ \lambda_1-\lambda_2=0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \\ \lambda_1-\lambda_3=0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \\ \lambda_2-\lambda_3=0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \end{array}\right\} \lambda_0+\lambda_1=2 \quad 4 \quad 6 \quad 5 \quad 4 \quad 3 \quad 2 \quad 3 \\ \left.\begin{array}{l} \lambda_0+\lambda_2=2 \quad 4 \quad 6 \\ \lambda_0+\lambda_3=2 \quad 4 \quad 6 \quad 5 \quad 4 \quad 2 \quad 1 \quad 3 \\ \lambda_1+\lambda_2=0 \quad 1 \quad 2 \quad 2 \quad 2 \quad 2 \quad 1 \quad 1 \\ \lambda_1+\lambda_3=0 \quad 1 \quad 2 \quad 2 \quad 2 \quad 1 \quad 1 \quad 1 \\ \lambda_2+\lambda_3=0 \quad 1 \quad 2 \quad 2 \quad 2 \quad 1 \quad 0 \quad 1 \end{array}\right\}$	
0902.0431_ EQ0518 [conf 0.021]	86	■ 0.021	$\begin{array}{l} \begin{array}{l} \frac{1}{2}(\lambda_0+\lambda_1-\lambda_2-\lambda_3)-\frac{1}{2}(\mu_1-\mu_2)=1 \quad 2 \quad 2 \quad 2 \quad 1 \quad 1 \quad c \\ \frac{1}{2}(-\lambda_0+\lambda_1-\lambda_2+\lambda_3)-\frac{1}{2}(\mu_1-\mu_2)=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \\ \frac{1}{2}(-\lambda_0+\lambda_1+\lambda_2-\lambda_3)-\frac{1}{2}(\mu_1-\mu_2)=0 \quad 1 \quad 2 \quad 1 \quad 1 \quad 1 \\ \frac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)-\frac{1}{2}(\mu_1-\mu_2)=1 \quad 2 \quad 3 \quad 2 \\ \frac{1}{2}(-\lambda_0-\lambda_1+\lambda_2+\lambda_3)-\frac{1}{2}(\mu_1-\mu_2)=0 \quad 0 \end{array} \\ \begin{array}{l} \frac{1}{2}(\lambda_0+\lambda_1-\lambda_2-\lambda_3)-\frac{1}{2}(\mu_1-\mu_2)=1 \quad 2 \quad 2 \quad 2 \quad 1 \quad 1 \quad c \\ \frac{1}{2}(-\lambda_0+\lambda_1-\lambda_2+\lambda_3)-\frac{1}{2}(\mu_1-\mu_2)=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \\ \frac{1}{2}(-\lambda_0+\lambda_1+\lambda_2-\lambda_3)-\frac{1}{2}(\mu_1-\mu_2)=0 \quad 1 \quad 2 \quad 1 \quad 1 \quad 1 \\ \frac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)-\frac{1}{2}(\mu_1-\mu_2)=1 \quad 2 \quad 3 \quad 2 \\ \frac{1}{2}(-\lambda_0-\lambda_1+\lambda_2+\lambda_3)-\frac{1}{2}(\mu_1-\mu_2)=0 \quad 0 \end{array} \end{array}$	$\begin{array}{l} \frac{1}{2}(\lambda_0+\lambda_1-\lambda_2-\lambda_3)-\frac{1}{2}(\mu_1-\mu_2)=1 \quad 2 \quad 2 \quad 2 \quad 1 \quad 1 \quad c \\ \frac{1}{2}(-\lambda_0+\lambda_1-\lambda_2+\lambda_3)-\frac{1}{2}(\mu_1-\mu_2)=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \\ \frac{1}{2}(-\lambda_0+\lambda_1+\lambda_2-\lambda_3)-\frac{1}{2}(\mu_1-\mu_2)=0 \quad 1 \quad 2 \quad 1 \quad 1 \quad 1 \\ \frac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)-\frac{1}{2}(\mu_1-\mu_2)=1 \quad 2 \quad 3 \quad 2 \\ \frac{1}{2}(-\lambda_0-\lambda_1+\lambda_2+\lambda_3)-\frac{1}{2}(\mu_1-\mu_2)=0 \quad 0 \end{array}$	$\begin{array}{l} \frac{1}{2}(\lambda_0+\lambda_1-\lambda_2-\lambda_3)-\frac{1}{2}(\mu_1-\mu_2)=1 \quad 2 \quad 2 \quad 1 \quad 1 \quad 0 \\ \frac{1}{2}(-\lambda_0+\lambda_1-\lambda_2+\lambda_3)-\frac{1}{2}(\mu_1-\mu_2)=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \frac{1}{2}(-\lambda_0+\lambda_1+\lambda_2-\lambda_3)-\frac{1}{2}(\mu_1-\mu_2)=0 \quad 1 \quad 2 \quad 1 \quad 1 \quad 0 \\ \frac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)-\frac{1}{2}(\mu_1-\mu_2)=1 \quad 2 \quad 3 \quad 2 \quad 2 \quad 1 \\ \frac{1}{2}(-\lambda_0-\lambda_1+\lambda_2+\lambda_3)-\frac{1}{2}(\mu_1-\mu_2)=0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0. \end{array}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0982.0431_ EQ1033 [conf 0.026]	174	■ 0.026	$\begin{array}{l} \lambda_0+\frac{1}{2}\mu_1+\frac{1}{3}\nu+r=2 \quad 4 \quad 6 \quad 4 \quad 3 \quad 2 \quad 1 \quad 3 \\ \lambda_0+\frac{1}{2}\mu_1+\frac{1}{3}\nu-r=2 \quad 4 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 3 \\ \lambda_0-\frac{1}{2}\mu_1-\frac{1}{3}\nu+r=2 \quad 3 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 2 \\ \lambda_0-\frac{1}{2}\mu_1-\frac{1}{3}\nu-r=2 \quad 3 \quad 4 \quad 4 \quad 3 \quad 2 \quad 1 \quad 2 \\ \lambda_1+\frac{1}{2}\mu_1+\frac{1}{3}\nu+r=0 \quad 1 \quad 2 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \\ \lambda_1+\frac{1}{2}\mu_1+\frac{1}{3}\nu-r=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \\ \lambda_1-\frac{1}{2}\mu_1-\frac{1}{3}\nu+r=0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_1-\frac{1}{2}\mu_1-\frac{1}{3}\nu-r=0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_2+\frac{1}{2}\mu_1+\frac{1}{3}\nu+r=0 \quad 1 \quad 2 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_2+\frac{1}{2}\mu_1+\frac{1}{3}\nu-r=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_2-\frac{1}{2}\mu_1-\frac{1}{3}\nu+r=0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ \lambda_2-\frac{1}{2}\mu_1-\frac{1}{3}\nu-r=0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ \lambda_3+\frac{1}{2}\mu_1+\frac{1}{3}\nu+r=0 \quad 1 \quad 2 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1 \\ \lambda_3+\frac{1}{2}\mu_1+\frac{1}{3}\nu-r=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1 \\ \lambda_3-\frac{1}{2}\mu_1-\frac{1}{3}\nu+r=0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \\ \lambda_3-\frac{1}{2}\mu_1-\frac{1}{3}\nu-r=0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \\ \lambda_1+\lambda_2-\lambda_3+\frac{1}{2}\mu_2+\frac{1}{3}\nu+r=1 \quad 2 \quad 4 \quad 3 \quad 2 \quad 2 \quad 1 \quad 2 \\ \lambda_1+\lambda_2-\lambda_3+\frac{1}{2}\mu_2+\frac{1}{3}\nu-r=1 \quad 2 \quad 3 \quad 3 \quad 2 \quad 2 \quad 1 \quad 2 \end{array}$		

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_ EQ1031 [conf 0.036]	173	■ 0.036	$\begin{aligned} & \lambda_2 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ & \lambda_2 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ & \lambda_3 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \\ & \lambda_3 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ & 0 \end{aligned}$	$\begin{aligned} & \lambda_2 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ & \lambda_2 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ & \lambda_3 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \\ & \lambda_3 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1 \end{aligned}$	
0902.0431_ EQ0761 [conf 0.036]	127	■ 0.036	$\begin{aligned} & \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1 \\ & \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu = 1 \quad 2 \quad 3 \quad 3 \quad 2 \quad 1 \quad 2 \\ & \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu = 0 \quad 1 \quad 2 \quad 3 \quad 2 \quad 1 \quad 2 \\ & \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu = 1 \quad 1 \quad 2 \quad 3 \quad 2 \quad 1 \quad 2 \\ & \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu = 1 \quad 2 \quad 2 \quad 3 \quad 2 \quad 1 \quad 2 \\ & \frac{1}{2}(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu = 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 1 \\ & \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu = 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1 \\ & \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1 \\ & \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu = 1 \quad 1 \quad 2 \quad 2 \quad 1 \quad 0 \quad 1 \\ & \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 + \frac{2}{3}\nu = 1 \quad 1 \quad 1 \quad 2 \quad 1 \quad 0 \quad 1 \\ & \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}\mu_3 - \frac{2}{3}\nu = 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \\ & \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu = 1 \quad 2 \quad 2 \quad 2 \quad 1 \quad 0 \quad 1 \\ & \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 2 \quad 1 \quad 0 \quad 1 \\ & \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu = 0 \quad 1 \quad 2 \quad 2 \quad 1 \quad 0 \quad 1 \\ & \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu = 1 \quad 2 \quad 3 \quad 4 \quad 3 \quad 1 \quad 2 \\ & \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 2 \quad 1 \quad 0 \quad 1. \end{aligned}$	$\begin{aligned} & \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1 \quad \textbf{[U+0930][U+094D]} \quad 0 \quad 0 \quad 1 \quad 2. \\ & \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu = 1 \quad 2 \quad 3 \quad 3 \quad 2 \end{aligned}$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_ EQ0516 [conf 0.041]	85	■0.041	$\begin{aligned} & \& \lambda_0 + \frac{1}{2}(\mu_2 - \mu_3) = 1 \quad 1 \quad 1 \quad 1 \quad a \quad 0 \\ & \& \lambda_1 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 1 \quad 1 \quad 1 \quad a \\ & \& \lambda_2 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 1 \quad 1 \\ & \& \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 0 \\ & \& \lambda_0 - \frac{1}{2}(\mu_2 - \mu_3) = 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ & \& \lambda_1 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 1 \quad 1 \quad 1 \quad 0 \\ & \& \lambda_2 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 1 \quad 1 \\ & \& \lambda_3 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 0 \quad 0 \end{aligned}$	$\begin{aligned} & \lambda_0 + \frac{1}{2}(\mu_2 - \mu_3) = 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ & \lambda_1 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ & \lambda_2 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \\ & \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \\ & \lambda_0 - \frac{1}{2}(\mu_2 - \mu_3) = 1 \quad 1 \quad 1 \quad 0 \quad 1 \quad 1 \\ & \lambda_1 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 1 \quad 1 \quad 0 \quad 1 \quad 1 \\ & \lambda_2 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 1 \\ & \lambda_3 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 1 \end{aligned}$	
0902.0431_ EQ0425 [conf 0.082]	70	■0.082	$\begin{aligned} & \& A \times A^* \times A \times A^* = A(X \times Y) \times A^* \\ & \& \left((PMA^*)^* (PNA^*) \right) = AM^* P^* PNA = A(M^*N)A, \\ & \& (PMA^*) (AYA^*) = P(MY)A^*, \\ & \& \overline{PMA^* \times PNA^*} = \overline{{}^t \tilde{P}(M \times N) {}^t \tilde{A}^*} = \overline{PM} \times \overline{NA^*}, \text{ etc.} \end{aligned}$	$\begin{aligned} & AXA^* \times AYA^* = A(X \times Y)A^* \\ & (PMA^*)^* (PNA^*) = AM^* P^* PNA = A(M^*N)A, \\ & (PMA^*) (AYA^*) = P(MY)A^*, \\ & \overline{PMA^* \times PNA^*} = \overline{{}^t \tilde{P}(M \times N) {}^t \tilde{A}^*} = \overline{PM} \times \overline{NA^*}, \text{ etc.} \end{aligned}$	$\begin{aligned} & AXA^* \times AYA^* = A(X \times Y)A^*, \\ & (PMA^*)^* (PNA^*) = AM^* P^* PNA = A(M^*N)A, \\ & (PMA^*) (AYA^*) = P(MY)A^*, \\ & \overline{PMA^* \times PNA^*} = \overline{{}^t \tilde{P}(M \times N) {}^t \tilde{A}^*} = \overline{PM} \times \overline{NA^*}, \text{ etc.} \end{aligned}$
0902.0431_ EQ1059 [conf 0.093]	180	■0.093	$\begin{aligned} & \& \alpha \in (E_8^C)_{1,1-,1-} \mid \tau \tilde{\lambda} \alpha = \alpha \tau \tilde{\lambda} \\ & \& \alpha \in E_7^C \mid \tau \tilde{\lambda} \alpha = \alpha \tau \tilde{\lambda} \} \text{ (Proposition 5.7.1)} \\ & \& \alpha \in E_7^C \mid \tau \lambda \alpha = \alpha \tau \lambda \} \text{ (by the correspondence to Proposition 5.7.1)} \\ & \& = E_7 \text{ (Lemma 4.3.3.(4)).} \end{aligned}$	$\begin{aligned} & = \left\{ \alpha \in (E_8^C)_{1,1-,1-} \mid \tau \tilde{\lambda} \alpha = \alpha \tau \tilde{\lambda} \right\} \\ & = \left\{ \alpha \in E_7^C \mid \tau \tilde{\lambda} \alpha = \alpha \tau \tilde{\lambda} \right\} \text{ (Proposition 5.7.1)} \\ & = \left\{ \alpha \in E_7^C \mid \tau \lambda \alpha = \alpha \tau \lambda \right\} \text{ (by the correspondence to Proposition 5.7.1)} \\ & = E_7 \text{ (Lemma 4.3.3.(4)).} \end{aligned}$	$\begin{aligned} & = \{ \alpha \in (E_8^C)_{1,1-,1-} \mid \tau \tilde{\lambda} \alpha = \alpha \tau \tilde{\lambda} \} \\ & = \{ \alpha \in E_7^C \mid \tau \tilde{\lambda} \alpha = \alpha \tau \tilde{\lambda} \} \text{ (Proposition 5.7.1)} \\ & = \{ \alpha \in E_7^C \mid \tau \lambda \alpha = \alpha \tau \lambda \} \text{ (by the correspondence to Proposition 5.7.1)} \\ & = E_7 \text{ (Lemma 4.3.3.(4)).} \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_ E08757 [conf 0.095]	126	■ 0.095	$\begin{aligned} & \backslash \text{begin}\{\text{aligned}\} \ \& \ \backslash \text{lambda}_{\{0\}}-\backslash \text{frac}\{1\}{2}\backslash \text{left}(\backslash \text{mu}_{\{2\}}-\backslash \text{mu}_{\{3\}}\backslash \text{right})=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ & \backslash \text{quad} \ 1 \quad \backslash \text{quad} \ 1 \quad \backslash \text{quad} \ 1 \quad \backslash \text{quad} \ 1 \quad \backslash \text{quad} \ 1 \quad \backslash \text{quad} \ 0 \quad \backslash \text{quad} \ 0 \quad \backslash \backslash \ \& \ \backslash \text{lambda}_{\{1\}}-\backslash \text{frac}\{1\}{2}\backslash \text{left}(\backslash \text{mu}_{\{2\}}-\backslash \text{mu}_{\{3\}}\backslash \text{right})=0 \quad \backslash \text{quad} \ 1 \quad \backslash \text{quad} \ 1 \quad \backslash \text{quad} \ 1 \quad \backslash \text{quad} \ 1 \quad \backslash \text{quad} \ 1 \quad \backslash \text{quad} \ a \quad \backslash \backslash \ \& \ \backslash \text{lambda}_{\{2\}}-\backslash \text{frac}\{1\}{2}\backslash \text{left}(\backslash \text{mu}_{\{2\}}-\backslash \text{mu}_{\{3\}}\backslash \text{right})=0 \quad \backslash \text{quad} \ 0 \quad \backslash \text{quad} \ 1 \quad \backslash \text{quad} \ 1 \quad \backslash \text{quad} \ 1 \quad \backslash \backslash \ \& \ \backslash \text{lambda}_{\{3\}}-\backslash \text{frac}\{1\}{2}\backslash \text{left}(\backslash \text{mu}_{\{2\}}-\backslash \text{mu}_{\{3\}}\backslash \text{right})=0 \quad \backslash \text{quad} \ 0 \quad \backslash \text{quad} \ 0 \quad \backslash \backslash \ \& \ 0 \end{aligned}$	$\begin{aligned} \lambda_0 - \frac{1}{2}(\mu_2 - \mu_3) &= 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ \lambda_1 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad a \\ \lambda_2 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 \quad 0 \quad 1 \quad 1 \quad 1 \\ \lambda_3 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 \quad 0 \quad 0 \\ 0 \end{aligned}$	$\begin{aligned} \lambda_0 - \frac{1}{2}(\mu_2 - \mu_3) &= 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ \lambda_1 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ \lambda_2 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ \lambda_3 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \end{aligned}$
0902.0431_E08922	153	■ 0.102	$f^{-1}\left(\sum_{i<j<k}x_{ijk}e_i\wedge e_j\wedge e_k\right)=\left(\begin{array}{ccc}h(x_{156}) & h(x_{164},\bar{x}_{256}) & h(x_{145},\bar{x}_{356}) \\ h(x_{256},\bar{x}_{164}) & h(x_{264}) & h(x_{245},\bar{x}_{364}) \\ h(x_{356},\bar{x}_{145}) & h(x_{364},\bar{x}_{245}) & h(x_{345}) \\ h(x_{423}) & h(x_{431},\bar{x}_{523}) & h(x_{412},\bar{x}_{623}) \\ h(x_{523},\bar{x}_{431}) & h(x_{531}) & h(x_{512},\bar{x}_{631}) \\ h(x_{623},\bar{x}_{412}) & h(x_{631},\bar{x}_{512}) & h(x_{612})\end{array}\right),$	$f^{-1}\left(\sum_{i<j<k}x_{ijk}e_i\wedge e_j\wedge e_k\right)=\left(\begin{array}{ccc}h(x_{156}) & h(x_{164},\bar{x}_{256}) & h(x_{145},\bar{x}_{356}) \\ h(x_{256},\bar{x}_{164}) & h(x_{264}) & h(x_{245},\bar{x}_{364}) \\ h(x_{356},\bar{x}_{145}) & h(x_{364},\bar{x}_{245}) & h(x_{345}) \\ h(x_{423}) & h(x_{431},\bar{x}_{523}) & h(x_{412},\bar{x}_{623}) \\ h(x_{523},\bar{x}_{431}) & h(x_{531}) & h(x_{512},\bar{x}_{631}) \\ h(x_{623},\bar{x}_{412}) & h(x_{631},\bar{x}_{512}) & h(x_{612}) \\ h(x_{123}) \\ h(x_{456})\end{array}\right),$	$f^{-1}\left(\sum_{i<j<k}x_{ijk}e_i\wedge e_j\wedge e_k\right)=\left(\begin{array}{ccc}h(x_{156}) & h(x_{164},\bar{x}_{256}) & h(x_{145},\bar{x}_{356}) \\ h(x_{256},\bar{x}_{164}) & h(x_{264}) & h(x_{245},\bar{x}_{364}) \\ h(x_{356},\bar{x}_{145}) & h(x_{364},\bar{x}_{245}) & h(x_{345}) \\ h(x_{423}) & h(x_{431},\bar{x}_{523}) & h(x_{412},\bar{x}_{623}) \\ h(x_{523},\bar{x}_{431}) & h(x_{531}) & h(x_{512},\bar{x}_{631}) \\ h(x_{623},\bar{x}_{412}) & h(x_{631},\bar{x}_{512}) & h(x_{612}) \\ h(x_{123}) \\ h(x_{456})\end{array}\right),$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00756	125	■ 0.119	$\begin{array}{l} \backslash begin{array}{llllllllll} \lambda_0 - \lambda_1 = 1 \\ \& 0 \& 0 \& 0 \& 0 \& 0 \& 0 \& 0 \& \lambda_0 + \lambda_1 = 1 \\ 2 \& 2 \& 2 \& 2 \& 2 \& 1 \backslash \backslash \lambda_0 - \lambda_2 = 1 \& 1 \& \\ 0 \& 0 \& 0 \& 0 \& 0 \& 0 \& \lambda_0 + \lambda_2 = 1 \& 1 \& 2 \\ \& 2 \& 2 \& 1 \backslash \backslash \lambda_0 - \lambda_3 = 1 \& 1 \& 1 \& 0 \\ \& 0 \& 0 \& 0 \& 0 \& \lambda_0 + \lambda_3 = 1 \& 1 \& 1 \& 2 \& \\ 2 \& 1 \backslash \backslash \lambda_1 - \lambda_2 = 0 \& 1 \& 0 \& 0 \& 0 \& 0 \& \\ 0 \& 0 \& \lambda_1 + \lambda_2 = 0 \& 1 \& 2 \& 2 \& 2 \& 1 \\ \backslash \backslash \lambda_1 - \lambda_3 = 0 \& 1 \& 1 \& 0 \& 0 \& 0 \& 0 \& \\ \& \lambda_1 + \lambda_3 = 0 \& 1 \& 1 \& 2 \& 2 \& 1 \backslash \backslash \\ \lambda_2 - \lambda_3 = 0 \& 0 \& 1 \& 0 \& 0 \& 0 \& 0 \& \\ \lambda_2 + \lambda_3 = 0 \& 0 \& 1 \& 2 \& 2 \& 1 \backslash \backslash \& \& \\ \lambda_0 + \frac{1}{2}(\mu_2 - \mu_3) = 1 \& 1 \& 1 \& 1 \& 1 \& 1 \& \\ \lambda_1 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \& 1 \& 1 \& 1 \& 1 \& 1 \& \\ \lambda_2 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \& 0 \& 1 \& 1 \& 1 \& 1 \& \\ +\frac{1}{2}(\mu_2 - \mu_3) = 0 \& 0 \& 0 \& 1 \& 1 \& 1 \& 1 \end{array}$	$\begin{array}{l} \lambda_0 - \lambda_1 = 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad \lambda_0 + \lambda_1 = 1 \quad 2 \quad 2 \quad 2 \quad 2 \quad 1 \\ \lambda_0 - \lambda_2 = 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad \lambda_0 + \lambda_2 = 1 \quad 1 \quad 2 \quad 2 \quad 2 \quad 1 \\ \lambda_0 - \lambda_3 = 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad \lambda_0 + \lambda_3 = 1 \quad 1 \quad 1 \quad 2 \quad 2 \quad 1 \\ \lambda_1 - \lambda_2 = 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad \lambda_1 + \lambda_2 = 0 \quad 1 \quad 2 \quad 2 \quad 2 \quad 1 \\ \lambda_1 - \lambda_3 = 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad \lambda_1 + \lambda_3 = 0 \quad 1 \quad 1 \quad 2 \quad 2 \quad 1 \\ \lambda_2 - \lambda_3 = 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad \lambda_2 + \lambda_3 = 0 \quad 0 \quad 1 \quad 2 \quad 2 \quad 1 \\ \\ \lambda_0 + \frac{1}{2}(\mu_2 - \mu_3) = 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \\ \lambda_1 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \\ \lambda_2 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \\ +\frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \end{array}$	$\begin{array}{l} \lambda_0 - \lambda_1 = 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad \lambda_0 + \lambda_1 = 1 \quad 2 \quad 2 \quad 2 \quad 2 \quad 1 \quad 1 \\ \lambda_0 - \lambda_2 = 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad \lambda_0 + \lambda_2 = 1 \quad 1 \quad 2 \quad 2 \quad 2 \quad 1 \quad 1 \\ \lambda_0 - \lambda_3 = 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad \lambda_0 + \lambda_3 = 1 \quad 1 \quad 1 \quad 2 \quad 2 \quad 1 \quad 1 \\ \lambda_1 - \lambda_2 = 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad \lambda_1 + \lambda_2 = 0 \quad 1 \quad 2 \quad 2 \quad 2 \quad 1 \quad 1 \\ \lambda_1 - \lambda_3 = 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad \lambda_1 + \lambda_3 = 0 \quad 1 \quad 1 \quad 2 \quad 2 \quad 1 \quad 1 \\ \lambda_2 - \lambda_3 = 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad \lambda_2 + \lambda_3 = 0 \quad 0 \quad 1 \quad 2 \quad 2 \quad 1 \quad 1 \\ \\ \lambda_0 + \frac{1}{2}(\mu_2 - \mu_3) = 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \\ \lambda_1 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \\ \lambda_2 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \\ \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \end{array}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_EQ1187	200	■ 0.183	<pre> \begin{aligned} & \&= (\boldsymbol{v}, \boldsymbol{w})(\boldsymbol{a}, *(\boldsymbol{b} \wedge \boldsymbol{x})) - (\boldsymbol{a}_1, \boldsymbol{w})(\boldsymbol{w} \wedge \boldsymbol{a}_2, *(\boldsymbol{b} \wedge \boldsymbol{x})) \\ & + (\boldsymbol{a}_2, \boldsymbol{w})(\boldsymbol{w} \wedge \boldsymbol{a}_1, *(\boldsymbol{b} \wedge \boldsymbol{x})) - \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) \\ & = \frac{2}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) - (\boldsymbol{a}_1, \boldsymbol{w})(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_2, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5) \\ & \quad + (\boldsymbol{a}_2, \boldsymbol{w})(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_1, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5), \\ & ((\boldsymbol{b} \times *(\boldsymbol{a} \wedge \boldsymbol{x}))\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{x} \wedge \boldsymbol{a}, \boldsymbol{w} \wedge *(\boldsymbol{b} \wedge \boldsymbol{v})) - \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{v}, \boldsymbol{w})(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{v}) + (\boldsymbol{a}_1, \boldsymbol{w})(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_2, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5) \\ & - (\boldsymbol{a}_2, \boldsymbol{w})(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_1, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5) - \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}). \end{aligned} </pre>	(not rendered)	$ \begin{aligned} &= (\boldsymbol{v}, \boldsymbol{w})(\boldsymbol{a}, *(\boldsymbol{b} \wedge \boldsymbol{x})) - (\boldsymbol{a}_1, \boldsymbol{w})(\boldsymbol{w} \wedge \boldsymbol{a}_2, *(\boldsymbol{b} \wedge \boldsymbol{x})) \\ &+ (\boldsymbol{a}_2, \boldsymbol{w})(\boldsymbol{w} \wedge \boldsymbol{a}_1, *(\boldsymbol{b} \wedge \boldsymbol{x})) - \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) \\ &= \frac{2}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) - (\boldsymbol{a}_1, \boldsymbol{w})(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_2, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5) \\ &\quad + (\boldsymbol{a}_2, \boldsymbol{w})(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_1, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5), \\ &((\boldsymbol{b} \times *(\boldsymbol{a} \wedge \boldsymbol{x}))\boldsymbol{v}, \boldsymbol{w}) \\ &= (\boldsymbol{x} \wedge \boldsymbol{a}, \boldsymbol{w} \wedge *(\boldsymbol{b} \wedge \boldsymbol{v})) - \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) \\ &= (\boldsymbol{v}, \boldsymbol{w})(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{v}) + (\boldsymbol{a}_1, \boldsymbol{w})(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_2, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5) \\ &- (\boldsymbol{a}_2, \boldsymbol{w})(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_1, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5) - \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}). \end{aligned} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00830	138	■ 0.186	<pre>\begin{aligned} \kappa(X, Y, \xi, \eta) = & \kappa \\ \left(\left(\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right), \left(\begin{array}{ccc} \eta_1 & y_3 & \bar{y}_2 \\ \bar{y}_3 & \eta_2 & y_1 \\ y_2 & \bar{y}_1 & \eta_3 \end{array}\right), \xi, \eta\right) \\ & \left(\left(\begin{array}{ccc} -\xi_1 & 0 & 0 \\ 0 & \xi_2 & x_1 \\ 0 & \bar{x}_1 & \xi_3 \end{array}\right), \left(\begin{array}{ccc} \eta_1 & 0 & 0 \\ 0 & -\eta_2 & -y_1 \\ 0 & -\bar{y}_1 & -\eta_3 \end{array}\right), -\xi, \eta\right), \\ \mu(X, Y, \xi, \eta) = & \left(\left(\begin{array}{ccc} \eta & 0 & 0 \\ 0 & \eta_3 & -y_1 \\ 0 & -\bar{y}_1 & \eta_2 \end{array}\right), \left(\begin{array}{ccc} \xi & 0 & 0 \\ 0 & \xi_3 & -x_1 \\ 0 & -\bar{x}_1 & \xi_2 \end{array}\right), \eta_1, \xi_1\right). \end{aligned}</pre>	$\kappa(X, Y, \xi, \eta) = \kappa\left(\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right), \left(\begin{array}{ccc} \eta_1 & y_3 & \bar{y}_2 \\ \bar{y}_3 & \eta_2 & y_1 \\ y_2 & \bar{y}_1 & \eta_3 \end{array}\right), \xi, \eta\right)$ $= \left(\left(\begin{array}{ccc} -\xi_1 & 0 & 0 \\ 0 & \xi_2 & x_1 \\ 0 & \bar{x}_1 & \xi_3 \end{array}\right), \left(\begin{array}{ccc} \eta_1 & 0 & 0 \\ 0 & -\eta_2 & -y_1 \\ 0 & -\bar{y}_1 & -\eta_3 \end{array}\right), -\xi, \eta\right),$ $\mu(X, Y, \xi, \eta) = \left(\left(\begin{array}{ccc} \eta & 0 & 0 \\ 0 & \eta_3 & -y_1 \\ 0 & -\bar{y}_1 & \eta_2 \end{array}\right), \left(\begin{array}{ccc} \xi & 0 & 0 \\ 0 & \xi_3 & -x_1 \\ 0 & -\bar{x}_1 & \xi_2 \end{array}\right), \eta_1, \xi_1\right).$	$\kappa(X, Y, \xi, \eta) = \kappa\left(\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right), \left(\begin{array}{ccc} \eta_1 & y_3 & \bar{y}_2 \\ \bar{y}_3 & \eta_2 & y_1 \\ y_2 & \bar{y}_1 & \eta_3 \end{array}\right), \xi, \eta\right)$ $= \left(\left(\begin{array}{ccc} -\xi_1 & 0 & 0 \\ 0 & \xi_2 & x_1 \\ 0 & \bar{x}_1 & \xi_3 \end{array}\right), \left(\begin{array}{ccc} \eta_1 & 0 & 0 \\ 0 & -\eta_2 & -y_1 \\ 0 & -\bar{y}_1 & -\eta_3 \end{array}\right), -\xi, \eta\right),$ $\mu(X, Y, \xi, \eta) = \left(\left(\begin{array}{ccc} \eta & 0 & 0 \\ 0 & \eta_3 & -y_1 \\ 0 & -\bar{y}_1 & \eta_2 \end{array}\right), \left(\begin{array}{ccc} \xi & 0 & 0 \\ 0 & \xi_3 & -x_1 \\ 0 & -\bar{x}_1 & \xi_2 \end{array}\right), \eta_1, \xi_1\right).$
0902.0431_E01041	177	■ 0.204	<pre>\begin{aligned} & \left(\alpha_i, \alpha_i\right)=\frac{1}{30}, \quad i=1,2,3,4,5,6,7,8, \\ & \left(\alpha_i, \alpha_{i+1}\right)=-\frac{1}{60}, \quad i=1,2, \cdots, 6, \quad\left(\alpha_3, \alpha_8\right)=-\frac{1}{60}, \\ & \left(\alpha_i, \alpha_j\right)=0, \quad \text { otherwise } \\ & \left(-\mu,-\mu\right)=\frac{1}{30}, \quad\left(-\mu, \alpha_7\right)=-\frac{1}{60}, \quad\left(-\mu, \alpha_i\right)=0, \quad i=1,2,3,4,5,6,8, \end{aligned}</pre>	$\left(\alpha_i, \alpha_i\right)=\frac{1}{30}, \quad i=1,2,3,4,5,6,7,8,$ $\left(\alpha_i, \alpha_{i+1}\right)=-\frac{1}{60}, \quad i=1,2, \cdots, 6, \quad\left(\alpha_3, \alpha_8\right)=-\frac{1}{60}$ $\left(\alpha_i, \alpha_j\right)=0, \quad \text { otherwise }$ $\left(-\mu,-\mu\right)=\frac{1}{30}, \quad\left(-\mu, \alpha_7\right)=-\frac{1}{60}, \quad\left(-\mu, \alpha_i\right)=0, \quad i=1,2,3,4,5,6,8,$	$\left(\alpha_i, \alpha_i\right)=\frac{1}{30}, \quad i=1,2,3,4,5,6,7,8,$ $\left(\alpha_i, \alpha_{i+1}\right)=-\frac{1}{60}, \quad i=1,2, \cdots, 6, \quad\left(\alpha_3, \alpha_8\right)=-\frac{1}{60},$ $\left(\alpha_i, \alpha_j\right)=0, \quad \text { otherwise },$ $\left(-\mu,-\mu\right)=\frac{1}{30}, \quad\left(-\mu, \alpha_7\right)=-\frac{1}{60}, \quad\left(-\mu, \alpha_i\right)=0, \quad i=1,2,3,4,5,6,8,$
0902.0431_E00340	54	■ 0.226	<pre>\begin{array}{l} \lambda_0=\alpha_1+\alpha_2+ \\ \lambda_1=\alpha_2+\alpha_3 \\ \lambda_3=\alpha_3 \\ \lambda_3=\alpha_1+2 \alpha_2+3 \alpha_3+2 \alpha_4, \end{array}</pre>	$\lambda_0 = \alpha_1 + \alpha_2 + \alpha_3$ $\lambda_1 = \alpha_2 + \alpha_3$ $\lambda_3 = \alpha_3$ $\lambda_3 = \alpha_1 + 2\alpha_2 + 3\alpha_3 + 2\alpha_4,$	$\lambda_0 = \alpha_1 + \alpha_2 + \alpha_3$ $\lambda_1 = \alpha_2 + \alpha_3$ $\lambda_3 = \alpha_3$ $\lambda_3 = \alpha_1 + 2\alpha_2 + 3\alpha_3 + 2\alpha_4,$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00937	155	■ 0.238	$\begin{aligned} & \& P=(0,0,1,0) \\ & \& \xrightarrow{f} \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 \\ & \& \xrightarrow{D} D\mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 + \mathbf{e}_1 \wedge D\mathbf{e}_2 \wedge \mathbf{e}_3 + \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge D\mathbf{e}_3 \\ & \& + \left(b_{11} + \frac{\nu}{3}\right) \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 - \bar{l}_{11} \mathbf{e}_4 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 - \bar{l}_{12} \mathbf{e}_5 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 - \bar{l}_{13} \mathbf{e}_6 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 \\ & \& + \left(b_{22} + \frac{\nu}{3}\right) \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 - \bar{l}_{21} \mathbf{e}_4 \wedge \mathbf{e}_3 \wedge \mathbf{e}_1 - \bar{l}_{22} \mathbf{e}_1 \wedge \mathbf{e}_5 \wedge \mathbf{e}_3 - \bar{l}_{23} \mathbf{e}_1 \wedge \mathbf{e}_6 \wedge \mathbf{e}_3 \\ & \& + \left(b_{33} + \frac{\nu}{3}\right) \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 - \bar{l}_{31} \mathbf{e}_4 \wedge \mathbf{e}_1 \wedge \mathbf{e}_2 - \bar{l}_{32} \mathbf{e}_5 \wedge \mathbf{e}_1 \wedge \mathbf{e}_2 - \bar{l}_{33} \mathbf{e}_6 \wedge \mathbf{e}_1 \wedge \mathbf{e}_2 \end{aligned}$	$\begin{aligned} & P=(0,0,1,0) \\ & \xrightarrow{f} \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 \\ & \xrightarrow{D} D\mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 + \mathbf{e}_1 \wedge D\mathbf{e}_2 \wedge \mathbf{e}_3 + \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge D\mathbf{e}_3 \\ & + \left(b_{11} + \frac{\nu}{3}\right) \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 - \bar{l}_{11} \mathbf{e}_4 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 - \bar{l}_{12} \mathbf{e}_5 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 - \bar{l}_{13} \mathbf{e}_6 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 \\ & + \left(b_{22} + \frac{\nu}{3}\right) \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 - \bar{l}_{21} \mathbf{e}_4 \wedge \mathbf{e}_3 \wedge \mathbf{e}_1 - \bar{l}_{22} \mathbf{e}_1 \wedge \mathbf{e}_5 \wedge \mathbf{e}_3 - \bar{l}_{23} \mathbf{e}_1 \wedge \mathbf{e}_6 \wedge \mathbf{e}_3 \\ & + \left(b_{33} + \frac{\nu}{3}\right) \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 - \bar{l}_{31} \mathbf{e}_4 \wedge \mathbf{e}_1 \wedge \mathbf{e}_2 - \bar{l}_{32} \mathbf{e}_5 \wedge \mathbf{e}_1 \wedge \mathbf{e}_2 - \bar{l}_{33} \mathbf{e}_6 \wedge \mathbf{e}_1 \wedge \mathbf{e}_2 \\ & \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ & \xrightarrow{f^{-1}} \begin{pmatrix} -h(\bar{l}_{11}) & -h(\bar{l}_{21}, l_{12}) & -h(\bar{l}_{31}, l_{13}) \\ -h(\bar{l}_{12}, l_{21}) & -h(\bar{l}_{22}) & -h(\bar{l}_{32}, l_{23}) \\ -h(\bar{l}_{13}, l_{31}) & -h(\bar{l}_{23}, l_{32}) & -h(\bar{l}_{33}) \end{pmatrix} = \begin{pmatrix} h(\nu) \\ 0 \\ -i\nu e_1 \\ 0 \end{pmatrix} h(L) \\ & = \begin{pmatrix} \phi_C(B, C) + \frac{i\nu e_1}{3} & -2\tau h(L) & 0 & h(L) \\ 2h(L) & \tau \phi_C(B, C)\tau - \frac{i\nu e_1}{3} & -\tau h(L) & 0 \\ 0 & h(L) & -i\nu e_1 & 0 \\ -\tau h(L) & 0 & 0 & i\nu e_1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \\ & = \Phi(\phi_C(B, C), h(L), -\tau h(L), -i\nu e_1) P. \end{aligned}$	$\begin{aligned} & P=(0,0,1,0) \\ & \xrightarrow{f} \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 \\ & \xrightarrow{D} D\mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 + \mathbf{e}_1 \wedge D\mathbf{e}_2 \wedge \mathbf{e}_3 + \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge D\mathbf{e}_3 \\ & + \left(b_{11} + \frac{\nu}{3}\right) \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 - \bar{l}_{11} \mathbf{e}_4 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 - \bar{l}_{12} \mathbf{e}_5 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 - \bar{l}_{13} \mathbf{e}_6 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 \\ & + \left(b_{22} + \frac{\nu}{3}\right) \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 - \bar{l}_{21} \mathbf{e}_4 \wedge \mathbf{e}_3 \wedge \mathbf{e}_1 - \bar{l}_{22} \mathbf{e}_1 \wedge \mathbf{e}_5 \wedge \mathbf{e}_3 - \bar{l}_{23} \mathbf{e}_1 \wedge \mathbf{e}_6 \wedge \mathbf{e}_3 \\ & + \left(b_{33} + \frac{\nu}{3}\right) \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3 - \bar{l}_{31} \mathbf{e}_4 \wedge \mathbf{e}_1 \wedge \mathbf{e}_2 - \bar{l}_{32} \mathbf{e}_5 \wedge \mathbf{e}_1 \wedge \mathbf{e}_2 - \bar{l}_{33} \mathbf{e}_6 \wedge \mathbf{e}_1 \wedge \mathbf{e}_2 \\ & \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ & \xrightarrow{f^{-1}} \begin{pmatrix} -h(\bar{l}_{11}) & -h(\bar{l}_{21}, l_{12}) & -h(\bar{l}_{31}, l_{13}) \\ -h(\bar{l}_{12}, l_{21}) & -h(\bar{l}_{22}) & -h(\bar{l}_{32}, l_{23}) \\ -h(\bar{l}_{13}, l_{31}) & -h(\bar{l}_{23}, l_{32}) & -h(\bar{l}_{33}) \end{pmatrix} = \begin{pmatrix} h(\nu) \\ 0 \\ -\tau h(L) \\ -i\nu e_1 \\ 0 \end{pmatrix} h(L) \\ & = \begin{pmatrix} \phi_C(B, C) + \frac{i\nu e_1}{3} & -2\tau h(L) & 0 \\ 2h(L) & \tau \phi_C(B, C)\tau - \frac{i\nu e_1}{3} & -\tau h(L) \\ 0 & h(L) & 0 \\ -\tau h(L) & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \\ & = \Phi(\phi_C(B, C), h(L), -\tau h(L), -i\nu e_1) P. \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00206	35	■ 0.294	$\begin{aligned} & \& \kappa\left(\alpha_1, \alpha_2, \alpha_3\right)=\left(\kappa \alpha_1, \kappa \alpha_3, \kappa \alpha_2\right), \\ & \& \pi\left(\alpha_1, \alpha_2, \alpha_3\right)=\left(\kappa \alpha_3, \kappa \alpha_2, \kappa \alpha_1\right), \\ & \& \nu\left(\alpha_1, \alpha_2, \alpha_3\right)=\left(\alpha_2, \alpha_3, \alpha_1\right), \end{aligned}$	$\begin{aligned} & \kappa\left(\alpha_1, \alpha_2, \alpha_3\right)=\left(\kappa \alpha_1, \kappa \alpha_3, \kappa \alpha_2\right), \\ & \pi\left(\alpha_1, \alpha_2, \alpha_3\right)=\left(\kappa \alpha_3, \kappa \alpha_2, \kappa \alpha_1\right), \\ & \nu\left(\alpha_1, \alpha_2, \alpha_3\right)=\left(\alpha_2, \alpha_3, \alpha_1\right), \end{aligned}$	$\begin{aligned} \kappa\left(\alpha_1, \alpha_2, \alpha_3\right) & =\left(\kappa \alpha_1, \kappa \alpha_3, \kappa \alpha_2\right), \\ \pi\left(\alpha_1, \alpha_2, \alpha_3\right) & =\left(\kappa \alpha_3, \kappa \alpha_2, \kappa \alpha_1\right), \\ \nu\left(\alpha_1, \alpha_2, \alpha_3\right) & =\left(\alpha_2, \alpha_3, \alpha_1\right), \end{aligned}$
0902.0431_E00655	108	■ 0.299	$\begin{aligned} & \& \left(P M \tau h(A, B)^*\right)^*\left(P N \tau h(A, B)^*\right)=\tau h(A, B) M^* P^* P N \tau h(A, B)^* \\ & \& =\tau\left(h(A, B) \tau\left(M^* N\right) h(A, B)^*\right), \\ & \& \frac{\left(P M \tau h(A, B)^*\right)\left(h(A, B) Y h(A, B)^*\right)}{\left(P M \tau h(A, B)^*\right) \times\left(P N \tau h(A, B)^*\right)}=\frac{\tau\left(P \tau(M Y) \tau h(A, B)^*\right)}{\bar{t} \bar{P}(M \times N) \tau^t\left(h(A, B)^*\right)^{\sim}} \\ & \& =P(\overline{M \times N}) h(A, B)^*=\tau\left(P(\overline{M \times N}) \tau h(A, B)^*\right) . \end{aligned}$	$\begin{aligned} & \left(P M \tau h(A, B)^*\right)^*\left(P N \tau h(A, B)^*\right)=\tau h(A, B) M^* P^* P N \tau h(A, B)^* \\ & =\tau\left(h(A, B) \tau\left(M^* N\right) h(A, B)^*\right), \\ & \frac{\left(P M \tau h(A, B)^*\right)\left(h(A, B) Y h(A, B)^*\right)}{\left(P M \tau h(A, B)^*\right) \times\left(P N \tau h(A, B)^*\right)}=\frac{\tau\left(P \tau(M Y) \tau h(A, B)^*\right)}{\bar{t} \bar{P}(M \times N) \tau^t\left(h(A, B)^*\right)^{\sim}}, \\ & =P(\overline{M \times N}) h(A, B)^*=\tau\left(P(\overline{M \times N}) \tau h(A, B)^*\right) . \end{aligned}$	$\begin{aligned} (P M \tau h(A, B)^*)^*(P N \tau h(A, B)^*) & =\tau h(A, B) M^* P^* P N \tau h(A, B)^* \\ & =\tau\left(h(A, B) \tau\left(M^* N\right) h(A, B)^*\right), \\ \frac{(P M \tau h(A, B)^*)(h(A, B) Y h(A, B)^*)}{(P M \tau h(A, B)^*) \times(P N \tau h(A, B)^*)} & =\frac{\tau(P \tau(M Y) \tau h(A, B)^*)}{\bar{t} \bar{P}(M \times N) \tau^t(h(A, B)^*)^{\sim}}, \\ & =P(\overline{M \times N}) h(A, B)^*=\tau(P(\overline{M \times N}) \tau h(A, B)^*). \end{aligned}$
0902.0431_E00101	20	■ 0.306	$\begin{aligned} & \& \lambda_1-\lambda_2=\alpha_1, \quad \lambda_1-\lambda_3=2 \alpha_1+3 \alpha_2, \quad \lambda_2-\lambda_3=\alpha_1+3 \alpha_2, \\ & \& \lambda_1=\alpha_1+\alpha_2, \quad \lambda_2=\alpha_2, \quad \lambda_3=\alpha_1+2 \alpha_2. \end{aligned}$	$\begin{aligned} \lambda_1-\lambda_2 & =\alpha_1, & \lambda_1-\lambda_3 & =2 \alpha_1+3 \alpha_2, & \lambda_2-\lambda_3 & =\alpha_1+3 \alpha_2, \\ \lambda_1 & =\alpha_1+\alpha_2, & \lambda_2 & =\alpha_2, & & -\lambda_3 & =\alpha_1+2 \alpha_2. \end{aligned}$	$\begin{aligned} \lambda_1-\lambda_2 & =\alpha_1, & \lambda_1-\lambda_3 & =2 \alpha_1+3 \alpha_2, & \lambda_2-\lambda_3 & =\alpha_1+3 \alpha_2, \\ \lambda_1 & =\alpha_1+\alpha_2, & \lambda_2 & =\alpha_2, & & -\lambda_3 & =\alpha_1+2 \alpha_2. \end{aligned}$
0902.0431_E01154	195	■ 0.318	$\begin{aligned} & \& +(\boldsymbol{v}, \boldsymbol{w})(\boldsymbol{u}, \boldsymbol{a})-\sum_{i=1}^3 \sum_{j=1}^3\left(\boldsymbol{u}_i \wedge \boldsymbol{v}_j \wedge \boldsymbol{v}_{j+1}, \boldsymbol{w}\right)\left(\boldsymbol{u}_{i+1} \wedge \boldsymbol{u}_{i+2} \wedge \boldsymbol{v}_{j+2}, \boldsymbol{a}\right) \\ & \& =-(\boldsymbol{u} \wedge \boldsymbol{v}, \boldsymbol{w} \wedge \boldsymbol{a})=-\left(*\left(*(\boldsymbol{u} \wedge \boldsymbol{v}) \wedge \boldsymbol{w}\right), \boldsymbol{a}\right) \end{aligned}$	$\begin{aligned} & +(\boldsymbol{v}, \boldsymbol{w})(\boldsymbol{u}, \boldsymbol{a})-\sum_{i=1}^3 \sum_{j=1}^3\left(\boldsymbol{u}_i \wedge \boldsymbol{v}_j \wedge \boldsymbol{v}_{j+1}, \boldsymbol{w}\right)\left(\boldsymbol{u}_{i+1} \wedge \boldsymbol{u}_{i+2} \wedge \boldsymbol{v}_{j+2}, \boldsymbol{a}\right) \\ & =-(\boldsymbol{u} \wedge \boldsymbol{v}, \boldsymbol{w} \wedge \boldsymbol{a})=-\left(*\left(*(\boldsymbol{u} \wedge \boldsymbol{v}) \wedge \boldsymbol{w}\right), \boldsymbol{a}\right) \end{aligned}$	$\begin{aligned} & +(\boldsymbol{v}, \boldsymbol{w})(\boldsymbol{u}, \boldsymbol{a})-\sum_{i=1}^3 \sum_{j=1}^3\left(\boldsymbol{u}_i \wedge \boldsymbol{v}_j \wedge \boldsymbol{v}_{j+1}, \boldsymbol{w}\right)\left(\boldsymbol{u}_{i+1} \wedge \boldsymbol{u}_{i+2} \wedge \boldsymbol{v}_{j+2}, \boldsymbol{a}\right) \\ & =-(\boldsymbol{u} \wedge \boldsymbol{v}, \boldsymbol{w} \wedge \boldsymbol{a})=-\left(*\left(*(\boldsymbol{u} \wedge \boldsymbol{v}) \wedge \boldsymbol{w}\right), \boldsymbol{a}\right) . \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_EQ1189	200	■ 0.322	$\begin{aligned} & ((\boldsymbol{a} \times (\boldsymbol{x} \wedge \boldsymbol{y}))\boldsymbol{v}, \boldsymbol{w}) = (\boldsymbol{x} \wedge \boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{v} \wedge \boldsymbol{a}) - \frac{3}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y})(\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{x}, \boldsymbol{v})(\boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{a}) - (\boldsymbol{y}, \boldsymbol{v})(\boldsymbol{x} \wedge \boldsymbol{w}, \boldsymbol{a}) + \frac{2}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y})(\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{x}, \boldsymbol{v})(*(\boldsymbol{a} \wedge \boldsymbol{y}), \boldsymbol{w}) - (\boldsymbol{y}, \boldsymbol{v})(*(\boldsymbol{a} \wedge \boldsymbol{x}), \boldsymbol{w}) + \frac{2}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y})(\boldsymbol{v}, \boldsymbol{w}). \end{aligned}$	$\begin{aligned} & ((\boldsymbol{a} \times (\boldsymbol{x} \wedge \boldsymbol{y}))\boldsymbol{v}, \boldsymbol{w}) = (\boldsymbol{x} \wedge \boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{v} \wedge \boldsymbol{a}) - \frac{3}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y})(\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{x}, \boldsymbol{v})(\boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{a}) - (\boldsymbol{y}, \boldsymbol{v})(\boldsymbol{x} \wedge \boldsymbol{w}, \boldsymbol{a}) + \frac{2}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y})(\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{x}, \boldsymbol{v})(*(\boldsymbol{a} \wedge \boldsymbol{y}), \boldsymbol{w}) - (\boldsymbol{y}, \boldsymbol{v})(*(\boldsymbol{a} \wedge \boldsymbol{x}), \boldsymbol{w}) + \frac{2}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y})(\boldsymbol{v}, \boldsymbol{w}). \end{aligned}$	$\begin{aligned} & ((\boldsymbol{a} \times (\boldsymbol{x} \wedge \boldsymbol{y}))\boldsymbol{v}, \boldsymbol{w}) = (\boldsymbol{x} \wedge \boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{v} \wedge \boldsymbol{a}) - \frac{3}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y})(\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{x}, \boldsymbol{v})(\boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{a}) - (\boldsymbol{y}, \boldsymbol{v})(\boldsymbol{x} \wedge \boldsymbol{w}, \boldsymbol{a}) + \frac{2}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y})(\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{x}, \boldsymbol{v})(*(\boldsymbol{a} \wedge \boldsymbol{y}), \boldsymbol{w}) - (\boldsymbol{y}, \boldsymbol{v})(*(\boldsymbol{a} \wedge \boldsymbol{x}), \boldsymbol{w}) + \frac{2}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y})(\boldsymbol{v}, \boldsymbol{w}). \end{aligned}$
0902.0431_EQ1157	195	■ 0.323	$\begin{aligned} & \& \theta = D \sum_{I \not\supseteq 9} u_I \boldsymbol{e}_I = 3 \sum_{I \not\supseteq 9} u_I \boldsymbol{e}_I - 6 \sum_{I \supseteq 9} u_I \boldsymbol{e}_I, \\ & \& \theta = -^t D \boldsymbol{v} = -3 \sum_{J \not\supseteq 9} v_J \boldsymbol{e}_J + 6 \sum_{J \supseteq 9} v_J \boldsymbol{e}_J, \end{aligned}$	$\begin{aligned} 0 &= D\boldsymbol{u} = \sum_I u_I D\boldsymbol{e}_I = 3 \sum_{I \not\supseteq 9} u_I \boldsymbol{e}_I - 6 \sum_{I \supseteq 9} u_I \boldsymbol{e}_I, \\ 0 &= -^t D\boldsymbol{v} = -3 \sum_{J \not\supseteq 9} v_J \boldsymbol{e}_J + 6 \sum_{J \supseteq 9} v_J \boldsymbol{e}_J, \end{aligned}$	$\begin{aligned} 0 &= D\boldsymbol{u} = \sum_I u_I D\boldsymbol{e}_I = 3 \sum_{I \not\supseteq 9} u_I \boldsymbol{e}_I - 6 \sum_{I \supseteq 9} u_I \boldsymbol{e}_I, \\ 0 &= -^t D\boldsymbol{v} = -3 \sum_{J \not\supseteq 9} v_J \boldsymbol{e}_J + 6 \sum_{J \supseteq 9} v_J \boldsymbol{e}_J, \end{aligned}$
0902.0431_EQ0460	75	■ 0.325	$\begin{aligned} & \mathfrak{a} \supset [\mathfrak{a}, \mathfrak{e}_6^C] \supset [\mathfrak{f}_4^C, \widetilde{\mathfrak{J}}_0^C] = \mathfrak{f}_4^C \widetilde{\mathfrak{J}}_0^C \text{ (Lemma 3.2.2) } = \widetilde{\mathfrak{J}}_0^C \text{ (Proposition 2.4.6.(2)). } \\ & \mathfrak{a} \supset [\mathfrak{a}, \mathfrak{e}_6^C] \supset [\mathfrak{f}_4^C, \widetilde{\mathfrak{J}}_0^C] = \mathfrak{f}_4^C \widetilde{\mathfrak{J}}_0^C \text{ (Lemma 3.2.2) } = \widetilde{\mathfrak{J}}_0^C \text{ (Proposition 2.4.6.(2)). } \end{aligned}$	$\mathfrak{a} \supset [\mathfrak{a}, \mathfrak{e}_6^C] \supset [\mathfrak{f}_4^C, \widetilde{\mathfrak{J}}_0^C] = \mathfrak{f}_4^C \widetilde{\mathfrak{J}}_0^C \text{ (Lemma 3.2.2) } = \widetilde{\mathfrak{J}}_0^C \text{ (Proposition 2.4.6.(2)). }$	$\mathfrak{a} \supset [\mathfrak{a}, \mathfrak{e}_6^C] \supset [\mathfrak{f}_4^C, \widetilde{\mathfrak{J}}_0^C] = \mathfrak{f}_4^C \widetilde{\mathfrak{J}}_0^C \text{ (Lemma 3.2.2) } = \widetilde{\mathfrak{J}}_0^C \text{ (Proposition 2.4.6.(2)). }$

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0902.0431_E08802	134	■ 0.338	$\left(\begin{array}{c} \left(\begin{array}{ccc} \cos a_1 a_2 \frac{\sin a_2 }{ a_2 } a_3 \frac{\sin a_3 }{ a_3 } & 0 & 0 \\ 0 & a_1 \frac{\sin a_1 }{ a_1 } \cos a_2 a_3 \frac{\sin a_3 }{ a_3 } & 0 \\ 0 & 0 & a_1 \frac{\sin a_1 }{ a_1 } a_2 \frac{\sin a_2 }{ a_2 } \cos a_3 \end{array}\right) \\ \left(\begin{array}{ccc} a_1 \frac{\sin a_1 }{ a_1 } \cos a_2 \cos a_3 & 0 & 0 \\ 0 & \cos a_1 a_2 \frac{\sin a_2 }{ a_2 } \cos a_3 & 0 \\ 0 & 0 & \cos a_1 \cos a_2 a_3 \frac{\sin a_3 }{ a_3 } \end{array}\right) \\ \cos a_1 \cos a_2 \cos a_3 \\ a_1 \frac{\sin a_1 }{ a_1 } a_2 \frac{\sin a_2 }{ a_2 } a_3 \frac{\sin a_3 }{ a_3 } \end{array}\right)$	$\left(\begin{array}{ccc} \cos a_1 a_2 \frac{\sin a_2 }{ a_2 } a_3 \frac{\sin a_3 }{ a_3 } & 0 & 0 \\ 0 & a_1 \frac{\sin a_1 }{ a_1 } \cos a_2 a_3 \frac{\sin a_3 }{ a_3 } & 0 \\ 0 & 0 & a_1 \frac{\sin a_1 }{ a_1 } a_2 \frac{\sin a_2 }{ a_2 } \cos a_3 \end{array}\right)$ $\left(\begin{array}{ccc} a_1 \frac{\sin a_1 }{ a_1 } \cos a_2 \cos a_3 & 0 & 0 \\ 0 & \cos a_1 a_2 \frac{\sin a_2 }{ a_2 } \cos a_3 & 0 \\ 0 & 0 & \cos a_1 \cos a_2 a_3 \frac{\sin a_3 }{ a_3 } \end{array}\right)$ $\cos a_1 \cos a_2 \cos a_3 $ $a_1 \frac{\sin a_1 }{ a_1 } a_2 \frac{\sin a_2 }{ a_2 } a_3 \frac{\sin a_3 }{ a_3 }$	
0902.0431_E08875	145	■ 0.339	$\begin{aligned} & \varphi_2(A) \left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array} \right) & \left(\begin{array}{ccc} \eta_1 & y_3 & \bar{y}_2 \\ \bar{y}_3 & \eta_2 & y_1 \\ y_2 & \bar{y}_1 & \eta_3 \end{array} \right), \xi, \eta \\ & = \left(\begin{array}{ccc} \xi'_1 & x'_3 & \bar{x}'_2 \\ \bar{x}'_3 & \xi'_2 & x'_1 \\ x'_2 & \bar{x}'_1 & \xi'_3 \end{array} \right), \left(\begin{array}{ccc} \eta'_1 & y'_3 & \bar{y}'_2 \\ \bar{y}'_3 & \eta'_2 & y'_1 \\ y'_2 & \bar{y}'_1 & \eta'_3 \end{array} \right), \xi', \eta' \end{aligned}$	$\varphi_2(A) \left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array} \right), \left(\begin{array}{ccc} \eta_1 & y_3 & \bar{y}_2 \\ \bar{y}_3 & \eta_2 & y_1 \\ y_2 & \bar{y}_1 & \eta_3 \end{array} \right), \xi, \eta$ $= \left(\begin{array}{ccc} \xi'_1 & x'_3 & \bar{x}'_2 \\ \bar{x}'_3 & \xi'_2 & x'_1 \\ x'_2 & \bar{x}'_1 & \xi'_3 \end{array} \right), \left(\begin{array}{ccc} \eta'_1 & y'_3 & \bar{y}'_2 \\ \bar{y}'_3 & \eta'_2 & y'_1 \\ y'_2 & \bar{y}'_1 & \eta'_3 \end{array} \right), \xi', \eta' \right),$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00941	156	0.372	<pre>\begin{aligned} & P=\left(F_1(1), 0,0,0\right) \\ & \& \xrightarrow{f} \boldsymbol{e}_2 \wedge \boldsymbol{e}_4 \wedge \boldsymbol{e}_5 + \boldsymbol{e}_3 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_4 \\ & \xrightarrow{D} (De_2 \wedge e_4 \wedge e_5 + e_2 \wedge De_4 \wedge e_5 + e_2 \wedge e_4 \wedge De_5) \\ & + (De_3 \wedge e_6 \wedge e_4 + e_3 \wedge De_6 \wedge e_4 + e_3 \wedge e_6 \wedge De_4) \\ & = \left(b_{12}e_1 \wedge e_4 \wedge e_5 + \left(b_{22} + \frac{\nu}{3}\right)e_2 \wedge e_4 \wedge e_5 - \bar{b}_{23}e_3 \wedge e_4 \wedge e_5 - \bar{l}_{23}e_6 \wedge e_4 \wedge e_5 \right. \\ & + l_{11}e_2 \wedge e_1 \wedge e_5 + l_{31}e_2 \wedge e_3 \wedge e_5 + \left(c_{11} - \frac{\nu}{3}\right)e_2 \wedge e_4 \wedge e_5 - \bar{c}_{13}e_2 \wedge e_6 \wedge e_5 \\ & + l_{12}e_2 \wedge e_4 \wedge e_1 + l_{32}e_2 \wedge e_4 \wedge e_3 + \left(c_{22} - \frac{\nu}{3}\right)e_3 \wedge e_6 \wedge e_4 - \bar{c}_{23}e_2 \wedge e_4 \wedge e_6 \Big) \\ & + \left(b_{13}e_1 \wedge e_6 \wedge e_4 + b_{23}e_2 \wedge e_6 \wedge e_4 + \left(b_{33} + \frac{\nu}{3}\right)e_3 \wedge e_6 \wedge e_4 - \bar{l}_{32}e_5 \wedge e_6 \wedge e_4 \right. \\ & + l_{13}e_3 \wedge e_1 \wedge e_4 + l_{23}e_3 \wedge e_2 \wedge e_4 + c_{23}e_3 \wedge e_5 \wedge e_4 + \left(c_{33} - \frac{\nu}{3}\right)e_3 \wedge e_6 \wedge e_4 \\ & + l_{11}e_3 \wedge e_6 \wedge e_1 + l_{21}e_3 \wedge e_6 \wedge e_2 + \left(c_{11} - \frac{\nu}{3}\right)e_3 \wedge e_6 \wedge e_4 - \bar{c}_{12}e_3 \wedge e_6 \wedge e_5 \Big) \\ & \xrightarrow{f^{-1}} \left(\begin{pmatrix} 0 & h(b_{13}, c_{13}) & 0 \\ * & h(b_{23} + \bar{c}_{23}) & h(b_{22} - c_{33} - \frac{\nu}{3}, -b_{33} + c_{33} + \frac{\nu}{3}) \\ * & * & -h(\bar{b}_{23} + c_{23}) \end{pmatrix} \right) \end{aligned}</pre>	$P = (F_1(1), 0, 0, 0)$ $\xrightarrow{f} \boldsymbol{e}_2 \wedge \boldsymbol{e}_4 \wedge \boldsymbol{e}_5 + \boldsymbol{e}_3 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_4$ $\xrightarrow{D} (De_2 \wedge e_4 \wedge e_5 + e_2 \wedge De_4 \wedge e_5 + e_2 \wedge e_4 \wedge De_5)$ $+ (De_3 \wedge e_6 \wedge e_4 + e_3 \wedge De_6 \wedge e_4 + e_3 \wedge e_6 \wedge De_4)$ $= \left(b_{12}e_1 \wedge e_4 \wedge e_5 + \left(b_{22} + \frac{\nu}{3}\right)e_2 \wedge e_4 \wedge e_5 - \bar{b}_{23}e_3 \wedge e_4 \wedge e_5 - \bar{l}_{23}e_6 \wedge e_4 \wedge e_5\right.$ $+ l_{11}e_2 \wedge e_1 \wedge e_5 + l_{31}e_2 \wedge e_3 \wedge e_5 + \left(c_{11} - \frac{\nu}{3}\right)e_2 \wedge e_4 \wedge e_5 - \bar{c}_{13}e_2 \wedge e_6 \wedge e_5$ $+ l_{12}e_2 \wedge e_4 \wedge e_1 + l_{32}e_2 \wedge e_4 \wedge e_3 + \left(c_{22} - \frac{\nu}{3}\right)e_3 \wedge e_6 \wedge e_4 - \bar{c}_{23}e_2 \wedge e_4 \wedge e_6 \Big)$ $+ \left(b_{13}e_1 \wedge e_6 \wedge e_4 + b_{23}e_2 \wedge e_6 \wedge e_4 + \left(b_{33} + \frac{\nu}{3}\right)e_3 \wedge e_6 \wedge e_4 - \bar{l}_{32}e_5 \wedge e_6 \wedge e_4\right.$ $+ l_{13}e_3 \wedge e_1 \wedge e_4 + l_{23}e_3 \wedge e_2 \wedge e_4 + c_{23}e_3 \wedge e_5 \wedge e_4 + \left(c_{33} - \frac{\nu}{3}\right)e_3 \wedge e_6 \wedge e_4$ $+ l_{11}e_3 \wedge e_6 \wedge e_1 + l_{21}e_3 \wedge e_6 \wedge e_2 + \left(c_{11} - \frac{\nu}{3}\right)e_3 \wedge e_6 \wedge e_4 - \bar{c}_{12}e_3 \wedge e_6 \wedge e_5 \Big)$ $\xrightarrow{f^{-1}} \left(\begin{pmatrix} 0 & h(b_{13}, c_{13}) & 0 \\ * & h(b_{23} + \bar{c}_{23}) & h(b_{22} - c_{33} - \frac{\nu}{3}, -b_{33} + c_{33} + \frac{\nu}{3}) \\ * & * & -h(\bar{b}_{23} + c_{23}) \end{pmatrix} \right)$	

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0902.0431_E00093	19	■ 0.382	$\begin{aligned} H_{\{1\}} H_{\{1\}} e_{\{2\}} &= H_{\{1\}} e_{\{3\}} = -e_{\{2\}}, \\ &\quad \end{aligned}$	$H_1 H_1 e_2 = H_1 e_3 = -e_2,$	$\begin{aligned} H_1 H_1 e_2 &= H_1 e_3 = -e_2, & H_1 H_1 e_3 &= -H_1 e_2 = -e_3, \\ H_1 H_1 e_4 &= -H_1 e_5 = -e_4, & H_1 H_1 e_5 &= H_1 e_4 = -e_4, \\ H_1 H_1 e_i &= 0 & \text{otherwise.} \end{aligned}$
0902.0431_E00371	61	■ 0.388	$\begin{aligned} &\operatorname{tr}(X) = \operatorname{tr}(\alpha X) \quad (\text{Lemma 2.2.1.(2)}) = \sum_{i=1}^3 \xi_i, \\ (X, X) &= (\alpha X, \alpha X) \quad (\text{Lemma 2.2.4}) = \sum_{i=1}^3 \xi_i^2, \\ \operatorname{tr}(X, X, X) &= \operatorname{tr}(\alpha X, \alpha X, \alpha X) \quad (\text{Lemma 2.2.4}) = \sum_{i=1}^3 \xi_i^3. \end{aligned}$	$\begin{aligned} \operatorname{tr}(X) &= \operatorname{tr}(\alpha X) \quad (\text{Lemma 2.2.1.(2)}) = \sum_{i=1}^3 \xi_i, \\ (X, X) &= (\alpha X, \alpha X) \quad (\text{Lemma 2.2.4}) = \sum_{i=1}^3 \xi_i^2, \\ \operatorname{tr}(X, X, X) &= \operatorname{tr}(\alpha X, \alpha X, \alpha X) \quad (\text{Lemma 2.2.4}) = \sum_{i=1}^3 \xi_i^3. \end{aligned}$	$\begin{aligned} \operatorname{tr}(X) &= \operatorname{tr}(\alpha X) \quad (\text{Lemma 2.2.1.(2)}) = \sum_{i=1}^3 \xi_i, \\ (X, X) &= (\alpha X, \alpha X) \quad (\text{Lemma 2.2.4}) = \sum_{i=1}^3 \xi_i^2, \\ \operatorname{tr}(X, X, X) &= \operatorname{tr}(\alpha X, \alpha X, \alpha X) \quad (\text{Lemma 2.2.4}) = \sum_{i=1}^3 \xi_i^3. \end{aligned}$
0902.0431_E00602	100	■ 0.396	$\begin{aligned} &\left. \begin{aligned} &\left(\begin{array}{rrl} \mu_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \mu_2 & m_1 \\ m_2 & \bar{m}_1 & \mu_3 \end{array} \right) \\ &\quad \end{aligned} \right \begin{aligned} &\mu_i \in \mathbf{R} \\ &m_i, a_i \in \mathbf{H} \end{aligned} \end{aligned}$	$\begin{aligned} (\mathfrak{J}^C)_{\tau\gamma} &= \{X \in \mathfrak{J}^C \mid \tau\gamma X = X\} \\ &= \left\{ \left(\begin{array}{rrl} \mu_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \mu_2 & m_1 \\ m_2 & \bar{m}_1 & \mu_3 \end{array} \right) + i \left(\begin{array}{rrl} 0 & a_3 e_4 & -a_2 e_4 \\ -a_3 e_4 & 0 & a_1 e_4 \\ a_2 e_4 & -a_1 e_4 & 0 \end{array} \right) \right\} \end{aligned}$	$\begin{aligned} (\mathfrak{J}^C)_{\tau\gamma} &= \{X \in \mathfrak{J}^C \mid \tau\gamma X = X\} \\ &= \left\{ \left(\begin{array}{rrl} \mu_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \mu_2 & m_1 \\ m_2 & \bar{m}_1 & \mu_3 \end{array} \right) + i \left(\begin{array}{rrl} 0 & a_3 e_4 & -a_2 e_4 \\ -a_3 e_4 & 0 & a_1 e_4 \\ a_2 e_4 & -a_1 e_4 & 0 \end{array} \right) \right\} \end{aligned}$

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0902.0431_E00990	149	■ 0.397	$\begin{aligned} P = & (X, Y, \xi, \eta) \\\stackrel{\chi}{\longrightarrow} & k\left(gX - \frac{\xi}{2}E\right)J + e_1k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J \\ \longrightarrow & e_1k(T)k\left(gX - \frac{\xi}{2}E\right)J - k(T)k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J \\ & + e_1k\left(gX - \frac{\xi}{2}E\right)J^tk(T) - k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J^tk(T) \\ = & k(-Tg(\gamma Y) - g(\gamma Y)T + \eta T)J + e_1k(T(gX) + (gX)T - \xi T)J \\ = & k(-2gA \circ g(\gamma Y) + \eta gA)J + e_1k(2gA \circ gX - \xi gA)J \\ = & k\left(-2g(\gamma A \times Y) - \frac{1}{2}(A, Y)E + \eta gA\right)J \\ & + e_1k\left(2g(\gamma A \times \gamma X) + \frac{1}{2}(\gamma A, X)E - \xi gA\right)J \text{ (Lemma 3.12.1)} \\ = & \chi\left(\begin{array}{c} -2\gamma A \times Y + \eta A \\ 2A \times X - \xi \gamma A \\ (A, Y) \\ (-\gamma A, X) \end{array}\right) = \chi\left(\left(\begin{array}{cccc} 0 & -2\gamma A & 0 & A \\ 2A & 0 & -\gamma A & 0 \\ 0 & A & 0 & 0 \\ -\gamma A & 0 & 0 & 0 \end{array}\right)\left(\begin{array}{c} X \\ Y \\ \xi \\ \eta \end{array}\right)\right) \\ = & \chi(\Phi(0, A, -\gamma A, 0)P). \end{aligned}$	$\begin{aligned} P = & (X, Y, \xi, \eta) \\\xrightarrow{\chi} & k\left(gX - \frac{\xi}{2}E\right)J + e_1k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J \\ \longrightarrow & e_1k(T)k\left(gX - \frac{\xi}{2}E\right)J - k(T)k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J \\ & + e_1k\left(gX - \frac{\xi}{2}E\right)J^tk(T) - k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J^tk(T) \\ = & k(-Tg(\gamma Y) - g(\gamma Y)T + \eta T)J + e_1k(T(gX) + (gX)T - \xi T)J \\ = & k(-2gA \circ g(\gamma Y) + \eta gA)J + e_1k(2gA \circ gX - \xi gA)J \\ = & k\left(-2g(\gamma A \times Y) - \frac{1}{2}(A, Y)E + \eta gA\right)J \\ & + e_1k\left(2g(\gamma A \times \gamma X) + \frac{1}{2}(\gamma A, X)E - \xi gA\right)J \text{ (Lemma 3.12.1)} \\ = & \chi\left(\begin{array}{c} -2\gamma A \times Y + \eta A \\ 2A \times X - \xi \gamma A \\ (A, Y) \\ (-\gamma A, X) \end{array}\right) = \chi\left(\left(\begin{array}{cccc} 0 & -2\gamma A & 0 & A \\ 2A & 0 & -\gamma A & 0 \\ 0 & A & 0 & 0 \\ -\gamma A & 0 & 0 & 0 \end{array}\right)\left(\begin{array}{c} X \\ Y \\ \xi \\ \eta \end{array}\right)\right) \\ = & \chi(\Phi(0, A, -\gamma A, 0)P) \end{aligned}$	$P = (X, Y, \xi, \eta)$ $\xrightarrow{\chi} k\left(gX - \frac{\xi}{2}E\right)J + e_1k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J$ $\longrightarrow e_1k(T)k\left(gX - \frac{\xi}{2}E\right)J - k(T)k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J$ $+ e_1k\left(gX - \frac{\xi}{2}E\right)J^tk(T) - k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J^tk(T)$ $= k(-Tg(\gamma Y) - g(\gamma Y)T + \eta T)J + e_1k(T(gX) + (gX)T - \xi T)J$ $= k(-2gA \circ g(\gamma Y) + \eta gA)J + e_1k(2gA \circ gX - \xi gA)J$ $= k\left(-2g(\gamma A \times Y) - \frac{1}{2}(A, Y)E + \eta gA\right)J$ $+ e_1k\left(2g(\gamma A \times \gamma X) + \frac{1}{2}(\gamma A, X)E - \xi gA\right)J \text{ (Lemma 3.12.1)}$ $= \chi\left(\begin{array}{c} -2\gamma A \times Y + \eta A \\ 2A \times X - \xi \gamma A \\ (A, Y) \\ (-\gamma A, X) \end{array}\right) = \chi\left(\left(\begin{array}{cccc} 0 & -2\gamma A & 0 & A \\ 2A & 0 & -\gamma A & 0 \\ 0 & A & 0 & 0 \\ -\gamma A & 0 & 0 & 0 \end{array}\right)\left(\begin{array}{c} X \\ Y \\ \xi \\ \eta \end{array}\right)\right)$ $= \chi(\Phi(0, A, -\gamma A, 0)P).$
0902.0431_E01080	183	■ 0.403	$S_{\{1\}} \tau S_{\{2\}} - S_{\{2\}} \tau S_{\{1\}} - \frac{1}{8}(S_{\{1\}} \tau S_{\{2\}} - S_{\{2\}} \tau S_{\{1\}}) \in \mathfrak{su}(8),$	$S_1 \tau S_2 - S_2 \tau S_1 - \frac{1}{8}(S_1 \tau S_2 - S_2 \tau S_1) \in \mathfrak{su}(8),$	$S_1 \tau S_2 - S_2 \tau S_1 - \frac{1}{8}(S_1 \tau S_2 - S_2 \tau S_1) \in \mathfrak{su}(8),$
0902.0431_E01030	173	■ 0.410	$\begin{aligned} & \lambda_0 + \frac{1}{2} \mu_1 - \frac{2}{3} \nu = 2 \quad 4 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 2 \\ & \lambda_0 - \frac{1}{2} \mu_1 + \frac{2}{3} \nu = 2 \quad 3 \quad 5 \quad 4 \\ & \lambda_1 + \frac{1}{2} \mu_1 - \frac{2}{3} \nu = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ & \lambda_1 - \frac{1}{2} \mu_1 + \frac{2}{3} \nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \end{aligned}$	$\lambda_0 + \frac{1}{2} \mu_1 - \frac{2}{3} \nu = 2 \quad 4 \quad 5 \quad 4 \quad 3 \quad 2$ $\lambda_0 - \frac{1}{2} \mu_1 + \frac{2}{3} \nu = 2 \quad 3 \quad 5 \quad 4$ $\lambda_1 + \frac{1}{2} \mu_1 - \frac{2}{3} \nu = 0 \quad 1 \quad 1 \quad 1 \quad 2$ $\lambda_1 - \frac{1}{2} \mu_1 + \frac{2}{3} \nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1$	$\lambda_0 + \frac{1}{2} \mu_1 - \frac{2}{3} \nu = 2 \quad 4 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 2$ $\lambda_0 - \frac{1}{2} \mu_1 + \frac{2}{3} \nu = 2 \quad 3 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 3$ $\lambda_1 + \frac{1}{2} \mu_1 - \frac{2}{3} \nu = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0$ $\lambda_1 - \frac{1}{2} \mu_1 + \frac{2}{3} \nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1$

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0902.0431_E00492	81	■ 0.411	$\delta \left(\begin{array}{ccc} \xi_1 & x_{12} & x_{13} \\ x_{21} & \xi_2 & x_{23} \\ x_{31} & x_{32} & \xi_3 \end{array} \right) = \left(\begin{array}{ccc} 0 & \delta_3 x_{12} & \bar{\delta}_2 \bar{x}_{13} \\ \bar{\delta}_3 \bar{x}_{21} & 0 & \delta_1 x_{23} \\ \delta_2 x_{31} & \bar{\delta}_1 \bar{x}_{32} & 0 \end{array} \right).$		$\delta \left(\begin{array}{ccc} \xi_1 & x_{12} & x_{13} \\ x_{21} & \xi_2 & x_{23} \\ x_{31} & x_{32} & \xi_3 \end{array} \right) = \left(\begin{array}{ccc} 0 & \delta_3 x_{12} & \bar{\delta}_2 \bar{x}_{13} \\ \bar{\delta}_3 \bar{x}_{21} & 0 & \delta_1 x_{23} \\ \delta_2 x_{31} & \bar{\delta}_1 \bar{x}_{32} & 0 \end{array} \right).$
0902.0431_E01097	186	■ 0.420	$\begin{aligned} \zeta[(icE), l(S)I] &= \zeta(2l(icS)I) \\ &= \Theta(0, 4\lambda\gamma\chi^{-1}(icS), 4\chi^{-1}(icS), 0, 0, 0) \\ &= \Theta(0, 4\chi^{-1}(cS), -4\lambda\gamma\chi^{-1}(cS), 0, 0, 0) \quad (\text{Lemma 5.8.3}) \\ &= [\Theta(0, 0, 0, 0, 2c, -2c), \Theta(0, 2\lambda\gamma\chi^{-1}S, 2\chi^{-1}S, 0, 0, 0)] \\ &= [\zeta l(icE), \zeta(l(S)I)]. \end{aligned}$	$\begin{aligned} \zeta[(icE), l(S)I] &= \zeta(2l(icS)I) \\ &= \Theta(0, 4\lambda\gamma\chi^{-1}(icS), 4\chi^{-1}(icS), 0, 0, 0) \\ &= \Theta(0, 4\chi^{-1}(cS), -4\lambda\gamma\chi^{-1}(cS), 0, 0, 0) \quad (\text{Lemma 5.8.3}) \\ &= [\Theta(0, 0, 0, 0, 2c, -2c), \Theta(0, 2\lambda\gamma\chi^{-1}S, 2\chi^{-1}S, 0, 0, 0)] \\ &= [\zeta l(icE), \zeta(l(S)I)]. \end{aligned}$	$\begin{aligned} \zeta[(icE), l(S)I] &= \zeta(2l(icS)I) \\ &= \Theta(0, 4\lambda\gamma\chi^{-1}(icS), 4\chi^{-1}(icS), 0, 0, 0) \\ &= \Theta(0, 4\chi^{-1}(cS), -4\lambda\gamma\chi^{-1}(cS), 0, 0, 0) \quad (\text{Lemma 5.8.3}) \\ &= [\Theta(0, 0, 0, 0, 2c, -2c), \Theta(0, 2\lambda\gamma\chi^{-1}S, 2\chi^{-1}S, 0, 0, 0)] \\ &= [\zeta l(icE), \zeta(l(S)I)]. \end{aligned}$
0902.0431_E00694	115	■ 0.434	$\mathfrak{e}_7^C \ni f_r \Phi f_r^{-1} = \begin{pmatrix} g & r^{-1}l & r^{-2}C & rA \\ rk & h & r^{-1}B & r^2D \\ r^2c & ra & \nu & r^3\lambda \\ r^{-1}b & r^{-2}d & r^{-3}k & \mu \end{pmatrix}$		$\mathfrak{e}_7^C \ni f_r \Phi f_r^{-1} = \begin{pmatrix} g & r^{-1}l & r^{-2}C & rA \\ rk & h & r^{-1}B & r^2D \\ r^2c & ra & \nu & r^3\lambda \\ r^{-1}b & r^{-2}d & r^{-3}k & \mu \end{pmatrix}$
0902.0431_E00642	106	■ 0.434	$\begin{aligned} &h(a, b) + h(a', b') = h(a + a', b + b'), \quad h(ca, cb) = ch(a, b), c \in \mathbf{C}, \\ &h(A, B) + h(A', B') = h(A + A', B + B'), \quad h(cA, cB) = h(A, B), c \in \mathbf{C}. \end{aligned}$	$\begin{aligned} h(a, b) + h(a', b') &= h(a + a', b + b'), \quad h(ca, cb) = ch(a, b), c \in \mathbf{C} \\ h(A, B) + h(A', B') &= h(A + A', B + B'), \quad h(cA, cB) = h(A, B), c \in \mathbf{C}. \end{aligned}$	$\begin{aligned} h(a, b) + h(a', b') &= h(a + a', b + b'), \quad h(ca, cb) = ch(a, b), c \in \mathbf{C}, \\ h(A, B) + h(A', B') &= h(A + A', B + B'), \quad h(cA, cB) = h(A, B), c \in \mathbf{C}. \end{aligned}$
0902.0431_E00558	94	■ 0.435	$\begin{array}{l} 1 \longrightarrow Spin(9) \longrightarrow (E_6)_{E_1} \longrightarrow S^9 \longrightarrow * \\ \quad \quad \quad \downarrow p' \quad \quad \quad \downarrow p \quad \quad \quad \downarrow = \\ 1 \longrightarrow SO(9) \longrightarrow SO(10) \longrightarrow S^9 \longrightarrow * \end{array}$		$\begin{array}{l} 1 \longrightarrow Spin(9) \longrightarrow (E_6)_{E_1} \longrightarrow S^9 \longrightarrow * \\ \quad \quad \quad \downarrow p' \quad \quad \quad \downarrow p \quad \quad \quad \downarrow = \\ 1 \longrightarrow SO(9) \longrightarrow SO(10) \longrightarrow S^9 \longrightarrow * \end{array}$

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0902.0431_E00332	53	■0.438	$\begin{aligned} & \pm\left(\lambda_k-\lambda_l\right), \pm\left(\lambda_k+\lambda_l\right), \quad 0 \leq k<l \leq 3, \\ & \pm \lambda_0, \quad \pm \lambda_1, \quad \pm \lambda_2, \quad \pm \lambda_3, \\ & \pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2-\lambda_3\right), \pm \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2-\lambda_3\right), \\ & \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2+\lambda_3\right), \pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right), \\ & \pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2-\lambda_3\right), \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2-\lambda_3\right), \\ & \pm \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right), \pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) . \end{aligned}$	$\begin{aligned} & \pm\left(\lambda_k-\lambda_l\right), \pm\left(\lambda_k+\lambda_l\right), \quad 0 \leq k<l \leq 3, \\ & \pm \lambda_0, \quad \pm \lambda_1, \quad \pm \lambda_2, \quad \pm \lambda_3, \\ & \pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2-\lambda_3\right), \pm \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2-\lambda_3\right), \\ & \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2+\lambda_3\right), \pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right), \\ & \pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2-\lambda_3\right), \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2-\lambda_3\right), \\ & \pm \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right), \pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) . \end{aligned}$	
0902.0431_EQ1211	203	■0.448	$\begin{aligned} & \exp (\operatorname{ad}(Z, 0))(C, D, \boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}, \boldsymbol{w} \otimes \boldsymbol{d}) \\ & =(\exp (\operatorname{ad}(Z)) C, D,((\exp Z) \boldsymbol{x}) \otimes \boldsymbol{a}, \\ & \quad((\exp Z) \boldsymbol{b}) \otimes \boldsymbol{y},((\exp (-^t Z)) \boldsymbol{c} \otimes \boldsymbol{z},((\exp (-^t Z)) \boldsymbol{w}) \otimes \boldsymbol{d}) \\ & =(\operatorname{Ad}(\exp Z) C, D,((\exp Z) \boldsymbol{x}) \otimes \boldsymbol{a}, \\ & \quad((\exp Z) \boldsymbol{b}) \otimes \boldsymbol{y},\left({}^t(\exp Z)^{-1} \boldsymbol{c}\right) \otimes \boldsymbol{z},\left({}^t(\exp Z)^{-1} \boldsymbol{w}\right) \otimes \boldsymbol{d}) \\ & =\varphi_1(\exp Z)(C, D, \boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}, \boldsymbol{w} \otimes \boldsymbol{d}) . \end{aligned}$	$\begin{aligned} & \exp (\operatorname{ad}(Z, 0))(C, D, \boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}, \boldsymbol{w} \otimes \boldsymbol{d}) \\ & =(\exp (\operatorname{ad}(Z)) C, D,((\exp Z) \boldsymbol{x}) \otimes \boldsymbol{a}, \\ & \quad((\exp Z) \boldsymbol{b}) \otimes \boldsymbol{y},\left({}^t(\exp Z)^{-1} \boldsymbol{c}\right) \otimes \boldsymbol{z},\left({}^t(\exp Z)^{-1} \boldsymbol{w}\right) \otimes \boldsymbol{d}) \\ & =\varphi_1(\exp Z)(C, D, \boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}, \boldsymbol{w} \otimes \boldsymbol{d}) . \end{aligned}$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E01039	177	■0.452	$\begin{aligned} H_{\alpha_3} &= \left(0, 0, 0, \frac{1}{60}, 0, 0\right), \\ H_{\alpha_4} &= \left(\Phi\left(\frac{1}{90}(-E_1 + 2E_2 - E_3)^\sim, 0, 0, -\frac{1}{120}\right), 0, 0, -\frac{1}{120}, 0, 0\right), \\ H_{\alpha_5} &= \left(\Phi\left(\frac{1}{60}(H_3 - (E_2 - E_3)^\sim), 0, 0, 0\right), 0, 0, 0, 0, 0\right), \\ H_{\alpha_6} &= \left(\Phi\left(\frac{1}{60}(H_2 - H_3), 0, 0, 0\right), 0, 0, 0, 0, 0\right), \\ H_{\alpha_7} &= \left(\Phi\left(\frac{1}{60}(H_1 - H_2), 0, 0, 0\right), 0, 0, 0, 0, 0\right), \\ H_{\alpha_8} &= \left(\Phi\left(0, 0, 0, \frac{1}{40}\right), 0, 0, -\frac{1}{120}, 0, 0\right). \end{aligned}$	$\begin{aligned} H_{\alpha_3} &= \left(0, 0, 0, \frac{1}{60}, 0, 0\right), \\ H_{\alpha_4} &= \left(\Phi\left(\frac{1}{90}(-E_1 + 2E_2 - E_3)^\sim, 0, 0, -\frac{1}{120}\right), 0, 0, -\frac{1}{120}, 0, 0\right), \\ H_{\alpha_5} &= \left(\Phi\left(\frac{1}{60}(H_3 - (E_2 - E_3)^\sim), 0, 0, 0\right), 0, 0, 0, 0, 0\right), \\ H_{\alpha_6} &= \left(\Phi\left(\frac{1}{60}(H_2 - H_3), 0, 0, 0\right), 0, 0, 0, 0, 0\right), \\ H_{\alpha_7} &= \left(\Phi\left(\frac{1}{60}(H_1 - H_2), 0, 0, 0\right), 0, 0, 0, 0, 0\right), \\ H_{\alpha_8} &= \left(\Phi\left(0, 0, 0, \frac{1}{40}\right), 0, 0, -\frac{1}{120}, 0, 0\right). \end{aligned}$	
0902.0431_E00939	156	■0.471	$\left.\begin{array}{l} \xrightarrow{\quad} \left(\begin{array}{ccc} h(b_{11} - c_{11} - \frac{\nu}{3}) & -h(c_{12}, b_{12}) & -h(c_{13}, b_{13}) \\ -h(\bar{b}_{12}, \bar{c}_{12}) & 0 & 0 \\ -h(\bar{b}_{13}, \bar{c}_{13}) & 0 & 0 \end{array} \right) \\ \begin{array}{l} = \left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & h(l_{33}) & -h(l_{23}, \bar{l}_{32}) \\ 0 & -h(l_{32}, \bar{l}_{23}) & h(l_{22}) \end{array} \right) \\ 0 \\ -h(\bar{l}_{11}) \end{array} \right) \end{array} \right\}.$	$\xrightarrow{f^{-1}} \left(\begin{array}{ccc} \left(\begin{array}{ccc} h(b_{11} - c_{11} - \frac{\nu}{3}) & -h(c_{12}, b_{12}) & -h(c_{13}, b_{13}) \\ -h(\bar{b}_{12}, \bar{c}_{12}) & 0 & 0 \\ -h(\bar{b}_{13}, \bar{c}_{13}) & 0 & 0 \end{array} \right) \\ \left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & h(l_{33}) & -h(l_{23}, \bar{l}_{32}) \\ 0 & -h(l_{32}, \bar{l}_{23}) & h(l_{22}) \end{array} \right) \\ 0 \\ -h(\bar{l}_{11}) \end{array} \right)$	$= \left(\begin{array}{c} \phi_C(B, C)E_1 + \frac{i\nu e_1}{3}E_1 \\ 2h(L) \times E_1 \\ 0 \\ -(\tau h(L), E_1) \end{array} \right) = \Phi(\phi_C(B, C), h(L), -\tau h(L), -i\nu e_1) \begin{pmatrix} E_1 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E01150	194	0.504	$\begin{aligned} & (\boldsymbol{u} \times *(\boldsymbol{v} \wedge \boldsymbol{w}) + \boldsymbol{v} \times *(\boldsymbol{w} \wedge \boldsymbol{u}) + \boldsymbol{w} \times *(\boldsymbol{u} \wedge \boldsymbol{v})) \boldsymbol{x} \\ & = \sum_{j=1}^9 (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_{j-1} \wedge \boldsymbol{x} \wedge \boldsymbol{u}_{j+1} \wedge \cdots \wedge \boldsymbol{u}_9, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_9) \boldsymbol{u}_j \\ & \quad - (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_9, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_9) \boldsymbol{x} = \text{(ii)} \end{aligned}$	$(\boldsymbol{u} \times *(\boldsymbol{v} \wedge \boldsymbol{w}) + \boldsymbol{v} \times *(\boldsymbol{w} \wedge \boldsymbol{u}) + \boldsymbol{w} \times *(\boldsymbol{u} \wedge \boldsymbol{v})) \boldsymbol{x}$ $= \sum_{j=1}^9 (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_{j-1} \wedge \boldsymbol{x} \wedge \boldsymbol{u}_{j+1} \wedge \cdots \wedge \boldsymbol{u}_9, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_9) \boldsymbol{u}_j$ $- (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_9, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_9) \boldsymbol{x} = \text{(ii)}$	$(\boldsymbol{u} \times *(\boldsymbol{v} \wedge \boldsymbol{w}) + \boldsymbol{v} \times *(\boldsymbol{w} \wedge \boldsymbol{u}) + \boldsymbol{w} \times *(\boldsymbol{u} \wedge \boldsymbol{v})) \boldsymbol{x}$ $= \sum_{j=1}^9 (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_{j-1} \wedge \boldsymbol{x} \wedge \boldsymbol{u}_{j+1} \wedge \cdots \wedge \boldsymbol{u}_9, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_9) \boldsymbol{u}_j$ $- (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_9, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_9) \boldsymbol{x} = \text{(ii)}.$
0902.0431_E00317	51	0.506	$\left(\left(\left[\delta_1, \delta_2 \right] \right)_4 + \left(\delta_1, [\delta, \delta_2] \right)_4 = 0, \quad \delta, \delta_i \in \mathfrak{f}_4^C.$	$([\delta, \delta_1], \delta_2)_4 + (\delta_1, [\delta, \delta_2])_4 = 0, \quad \delta, \delta_i \in \mathfrak{f}_4^C.$	$([\delta, \delta_1], \delta_2)_4 + (\delta_1, [\delta, \delta_2])_4 = 0, \quad \delta, \delta_i \in \mathfrak{f}_4^C.$
0902.0431_E00593	99	0.507	$\begin{aligned} & \langle \varphi(A)M, \varphi(A)N \rangle = \langle k_J(\varphi(A)M), k_J(\varphi(A)N) \rangle \quad (\text{Lemma 3.11.1}) \\ & = \langle A(k_J(M))^t A, A(k_J(N))^t A \rangle = \langle k_J(M), k_J(N) \rangle \quad (\text{because } A \in SU(6)) \\ & = 2\langle M, N \rangle \end{aligned}$	$2\langle \varphi(A)M, \varphi(A)N \rangle = \langle k_J(\varphi(A)M), k_J(\varphi(A)N) \rangle \quad (\text{Lemma 3.11.1})$ $= \langle A(k_J(M))^t A, A(k_J(N))^t A \rangle = \langle k_J(M), k_J(N) \rangle \quad (\text{because } A \in SU(6))$ $= 2\langle M, N \rangle$	$2\langle \varphi(A)M, \varphi(A)N \rangle = \langle k_J(\varphi(A)M), k_J(\varphi(A)N) \rangle \quad (\text{Lemma 3.11.1})$ $= \langle A(k_J(M))^t A, A(k_J(N))^t A \rangle = \langle k_J(M), k_J(N) \rangle \quad (\text{because } A \in SU(6))$ $= 2\langle M, N \rangle,$
0902.0431_E00764	128	0.507	$\begin{aligned} & B_7(h, h') = \frac{3}{2} B_6 \left(\sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim}, \sum_{k=0}^3 \lambda_k' H_k + \left(\sum_{j=1}^3 \mu_j' E_j \right)^{\sim} \right) + 24\nu\nu' \\ & = \frac{3}{2} 12 \left(2 \sum_{k=0}^3 \lambda_k \lambda_k' + \sum_{j=1}^3 \mu_j \mu_j' \right) + 24\nu\nu' \quad (\text{Theorem 3.6.5}) \\ & = 6 \left(6 \sum_{k=0}^3 \lambda_k \lambda_k' + 3 \sum_{j=1}^3 \mu_j \mu_j' + 4\nu\nu' \right). \end{aligned}$	$B_7(h, h') = \frac{3}{2} B_6 \left(\sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim}, \sum_{k=0}^3 \lambda_k' H_k + \left(\sum_{j=1}^3 \mu_j' E_j \right)^{\sim} \right) + 24\nu\nu'$ $= \frac{3}{2} 12 \left(2 \sum_{k=0}^3 \lambda_k \lambda_k' + \sum_{j=1}^3 \mu_j \mu_j' \right) + 24\nu\nu' \quad (\text{Theorem 3.6.5})$ $= 6 \left(6 \sum_{k=0}^3 \lambda_k \lambda_k' + 3 \sum_{j=1}^3 \mu_j \mu_j' + 4\nu\nu' \right).$	$B_7(h, h') = \frac{3}{2} B_6 \left(\sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim}, \sum_{k=0}^3 \lambda_k' H_k + \left(\sum_{j=1}^3 \mu_j' E_j \right)^{\sim} \right) + 24\nu\nu'$ $= \frac{3}{2} 12 \left(2 \sum_{k=0}^3 \lambda_k \lambda_k' + \sum_{j=1}^3 \mu_j \mu_j' \right) + 24\nu\nu' \quad (\text{Theorem 3.6.5})$ $= 6 \left(6 \sum_{k=0}^3 \lambda_k \lambda_k' + 3 \sum_{j=1}^3 \mu_j \mu_j' + 4\nu\nu' \right).$
0902.0431_E00051	13	0.507	$\left(D_2 x \right) y + x \left(D_3^{\prime} y \right) = (\kappa D_1^{\prime})(xy), \quad x, y \in \mathfrak{C},$	$(D_2 x) y + x (D_3' y) = (\kappa D_1')(xy), \quad x, y \in \mathfrak{C},$	$(D_2 x) y + x (D_3' y) = (\kappa D_1')(xy), \quad x, y \in \mathfrak{C},$

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0902.0431_E00702 (iii)	116	■ 0.508	$\begin{aligned} & 3(\phi X, X, X) = 3(g(X), X, X) - (\nu + 2\mu)(X, X, X) \\ & \quad = (\mu X \times X + h(X \times X), X) + (g(X), X \times X) - 3(\nu + 2\mu) \det X \\ & \quad = 3\mu \det X + 3(\nu + \mu) \det X - 3(\nu + 2\mu) \det X = 0 \end{aligned}$	$3(\phi X, X, X) = 3(g(X), X, X) - (\nu + 2\mu)(X, X, X) \\ = (\mu X \times X + h(X \times X), X) + (g(X), X \times X) - 3(\nu + 2\mu) \det X \\ = 3\mu \det X + 3(\nu + \mu) \det X - 3(\nu + 2\mu) \det X = 0$	$3(\phi X, X, X) = 3(g(X), X, X) - (\nu + 2\mu)(X, X, X) \\ = (\mu X \times X + h(X \times X), X) + (g(X), X \times X) - 3(\nu + 2\mu) \det X \\ = 3\mu \det X + 3(\nu + \mu) \det X - 3(\nu + 2\mu) \det X = 0.$
0902.0431_E00903	150	■ 0.510	$\alpha P = (X, Y, \xi, \eta), \quad X, Y \quad \text{are real diagonal, } \xi > 0.$	$\alpha P = (X, Y, \xi, \eta), \quad X, Y \quad \text{are real diagonal, } \xi > 0.$	$\alpha P = (X, Y, \xi, \eta), \quad X, Y \quad \text{are real diagonal, } \xi > 0.$
0902.0431_E01101	187	■ 0.512	$\begin{gathered} \mathfrak{e}_{\{8\}^C} = \left(\mathfrak{e}_{\{8\}^C} \right)_{\tilde{\lambda}\gamma} \oplus \left(\mathfrak{e}_{\{8\}^C} \right)_{-\tilde{\lambda}\gamma}, \\ \left(\mathfrak{e}_{\{8\}^C} \right)_{\tilde{\lambda}\gamma} = \left\{ \Theta \in \text{Der}(\mathfrak{e}_{\{8\}^C}) \mid \tilde{\lambda}\gamma\Theta = \Theta\tilde{\lambda}\gamma \right\} \cong \left\{ R \in \mathfrak{e}_{\{8\}^C} \mid \tilde{\lambda}\gamma R = R \right\} = (\mathfrak{e}_{\{8\}^C})^{\tilde{\lambda}\gamma}, \\ \left(\mathfrak{e}_{\{8\}^C} \right)_{-\tilde{\lambda}\gamma} = \left\{ \Theta \in \text{Der}(\mathfrak{e}_{\{8\}^C}) \mid \tilde{\lambda}\gamma\Theta = -\Theta\tilde{\lambda}\gamma \right\} \cong \left\{ R \in \mathfrak{e}_{\{8\}^C} \mid \tilde{\lambda}\gamma R = -R \right\}, \end{gathered}$	$\mathfrak{e}_8^C = (\mathfrak{e}_8^C)_{\tilde{\lambda}\gamma} \oplus (\mathfrak{e}_8^C)_{-\tilde{\lambda}\gamma}, \\ (\mathfrak{e}_8^C)_{\tilde{\lambda}\gamma} = \{ \Theta \in \text{Der}(\mathfrak{e}_8^C) \mid \tilde{\lambda}\gamma\Theta = \Theta\tilde{\lambda}\gamma \} \cong \{ R \in \mathfrak{e}_8^C \mid \tilde{\lambda}\gamma R = R \} = (\mathfrak{e}_8^C)^{\tilde{\lambda}\gamma}, \\ (\mathfrak{e}_8^C)_{-\tilde{\lambda}\gamma} = \{ \Theta \in \text{Der}(\mathfrak{e}_8^C) \mid \tilde{\lambda}\gamma\Theta = -\Theta\tilde{\lambda}\gamma \} \cong \{ R \in \mathfrak{e}_8^C \mid \tilde{\lambda}\gamma R = -R \},$	$\mathfrak{e}_8^C = (\mathfrak{e}_8^C)_{\tilde{\lambda}\gamma} \oplus (\mathfrak{e}_8^C)_{-\tilde{\lambda}\gamma}, \\ (\mathfrak{e}_8^C)_{\tilde{\lambda}\gamma} = \{ \Theta \in \text{Der}(\mathfrak{e}_8^C) \mid \tilde{\lambda}\gamma\Theta = \Theta\tilde{\lambda}\gamma \} \cong \{ R \in \mathfrak{e}_8^C \mid \tilde{\lambda}\gamma R = R \} = (\mathfrak{e}_8^C)^{\tilde{\lambda}\gamma}, \\ (\mathfrak{e}_8^C)_{-\tilde{\lambda}\gamma} = \{ \Theta \in \text{Der}(\mathfrak{e}_8^C) \mid \tilde{\lambda}\gamma\Theta = -\Theta\tilde{\lambda}\gamma \} \cong \{ R \in \mathfrak{e}_8^C \mid \tilde{\lambda}\gamma R = -R \},$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00878	145	■ 0.528	$\begin{aligned} & \binom{\xi_1'}{\eta'} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_1 \\ \eta \end{pmatrix}, \quad \binom{\xi_2'}{\eta_2'} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_2 \\ \eta_2 \end{pmatrix}, \\ & \binom{\xi_3'}{\eta_3'} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_3 \\ \eta_3 \end{pmatrix}, \quad \binom{\xi_4'}{\eta_4'} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_4 \\ \eta_4 \end{pmatrix}, \\ & \binom{x_1'}{y_1'} = \begin{pmatrix} -\nu & \tau a \\ -a & \nu \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \quad \binom{x_2'}{y_2'} = \begin{pmatrix} x_3' \\ y_3' \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \end{aligned}$	$\begin{pmatrix} \xi_1' \\ \eta' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_1 \\ \eta \end{pmatrix}, \quad \begin{pmatrix} \xi_2' \\ \eta_2' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_2 \\ \eta_2 \end{pmatrix},$ $\begin{pmatrix} \xi_3' \\ \eta_3' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_3 \\ \eta_3 \end{pmatrix}, \quad \begin{pmatrix} \xi_4' \\ \eta_4' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_4 \\ \eta_4 \end{pmatrix},$ $\begin{pmatrix} x_1' \\ y_1' \end{pmatrix} = \begin{pmatrix} -\nu & \tau a \\ -a & \nu \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \quad \begin{pmatrix} x_2' \\ y_2' \end{pmatrix} = \begin{pmatrix} x_3' \\ y_3' \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$	$\begin{pmatrix} \xi_1' \\ \eta' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_1 \\ \eta \end{pmatrix}, \quad \begin{pmatrix} \xi_2' \\ \eta_2' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_2 \\ \eta_2 \end{pmatrix},$ $\begin{pmatrix} \xi_3' \\ \eta_3' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_3 \\ \eta_3 \end{pmatrix}, \quad \begin{pmatrix} \xi_4' \\ \eta_4' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_4 \\ \eta_4 \end{pmatrix},$ $\begin{pmatrix} x_1' \\ y_1' \end{pmatrix} = \begin{pmatrix} -\nu & \tau a \\ -a & \nu \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \quad \begin{pmatrix} x_2' \\ y_2' \end{pmatrix} = \begin{pmatrix} x_3' \\ y_3' \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$
0902.0431_E00945	157	■ 0.528	$\begin{aligned} & \overline{f^{-1}(D(fP))} = \overline{\Phi(\phi_{\mathbf{C}}(B, C), h(L), -\tau h(L), -i\nu e_1)P} \\ & = \Phi(\overline{\phi_{\mathbf{C}}(B, C)}, \overline{h(L)}, -\tau \overline{h(L)}, -i\nu e_1) \overline{P} \\ & = \Phi(\phi_{\mathbf{C}}(\bar{C}, \bar{B}), h(\bar{L}^*), -\tau h(\bar{L}^*), -i\nu e_1) \bar{P} \\ & = f^{-1}(\overline{((\text{Ad } J_3) D)}(f\bar{P})). \end{aligned}$	$\begin{aligned} \overline{f^{-1}(D(fP))} &= \overline{\Phi(\phi_{\mathbf{C}}(B, C), h(L), -\tau h(L), -i\nu e_1)P} \\ &= \Phi(\overline{\phi_{\mathbf{C}}(B, C)}, \overline{h(L)}, -\tau \overline{h(L)}, -i\nu e_1) \overline{P} \\ &= \Phi(\phi_{\mathbf{C}}(\bar{C}, \bar{B}), h(\bar{L}^*), -\tau h(\bar{L}^*), -i\nu e_1) \bar{P} \\ &= f^{-1}(\overline{((\text{Ad } J_3) D)}(f\bar{P})). \end{aligned}$	$\begin{aligned} \overline{f^{-1}(D(fP))} &= \overline{\Phi(\phi_{\mathbf{C}}(B, C), h(L), -\tau h(L), -i\nu e_1)P} \\ &= \Phi(\overline{\phi_{\mathbf{C}}(B, C)}, \overline{h(L)}, -\tau \overline{h(L)}, -i\nu e_1) \overline{P} \\ &= \Phi(\phi_{\mathbf{C}}(\bar{C}, \bar{B}), h(\bar{L}^*), -\tau h(\bar{L}^*), -i\nu e_1) \bar{P} \\ &= f^{-1}(\overline{((\text{Ad } J_3) D)}(f\bar{P})). \end{aligned}$
0902.0431_E01082	184	■ 0.533	$\begin{aligned} & \begin{aligned} & \left(k \left(g X_1 - \frac{\xi_1}{2} E \right) + i k \left(g(\gamma Y_1) - \frac{\eta_1}{2} E \right) \right) \left(k \left(g X_2 - \frac{\xi_2}{2} E \right) - i k \left(g(\gamma Y_2) - \frac{\eta_2}{2} E \right) \right) J^2 \\ & = -k \left(\left(g X_1 - \frac{\xi_1}{2} E \right) \left(g X_2 - \frac{\xi_2}{2} E \right) + \left(g(\gamma Y_1) - \frac{\eta_1}{2} E \right) \left(g(\gamma Y_2) - \frac{\eta_2}{2} E \right) \right) \\ & \quad - i k \left(\left(g(\gamma Y_1) - \frac{\eta_1}{2} E \right) \left(g X_2 - \frac{\xi_2}{2} E \right) - \left(g X_1 - \frac{\xi_1}{2} E \right) \left(g(\gamma Y_2) - \frac{\eta_2}{2} E \right) \right). \end{aligned} \end{aligned}$	$\begin{aligned} & \left(k \left(g X_1 - \frac{\xi_1}{2} E \right) + i k \left(g(\gamma Y_1) - \frac{\eta_1}{2} E \right) \right) \left(k \left(g X_2 - \frac{\xi_2}{2} E \right) - i k \left(g(\gamma Y_2) - \frac{\eta_2}{2} E \right) \right) J^2 \\ & = -k \left(\left(g X_1 - \frac{\xi_1}{2} E \right) \left(g X_2 - \frac{\xi_2}{2} E \right) + \left(g(\gamma Y_1) - \frac{\eta_1}{2} E \right) \left(g(\gamma Y_2) - \frac{\eta_2}{2} E \right) \right) \\ & \quad - i k \left(\left(g(\gamma Y_1) - \frac{\eta_1}{2} E \right) \left(g X_2 - \frac{\xi_2}{2} E \right) - \left(g X_1 - \frac{\xi_1}{2} E \right) \left(g(\gamma Y_2) - \frac{\eta_2}{2} E \right) \right). \end{aligned}$	$\begin{aligned} & \left(k \left(g X_1 - \frac{\xi_1}{2} E \right) + i k \left(g(\gamma Y_1) - \frac{\eta_1}{2} E \right) \right) \left(k \left(g X_2 - \frac{\xi_2}{2} E \right) - i k \left(g(\gamma Y_2) - \frac{\eta_2}{2} E \right) \right) J^2 \\ & = -k \left(\left(g X_1 - \frac{\xi_1}{2} E \right) \left(g X_2 - \frac{\xi_2}{2} E \right) + \left(g(\gamma Y_1) - \frac{\eta_1}{2} E \right) \left(g(\gamma Y_2) - \frac{\eta_2}{2} E \right) \right) \\ & \quad - i k \left(\left(g(\gamma Y_1) - \frac{\eta_1}{2} E \right) \left(g X_2 - \frac{\xi_2}{2} E \right) - \left(g X_1 - \frac{\xi_1}{2} E \right) \left(g(\gamma Y_2) - \frac{\eta_2}{2} E \right) \right). \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00989	165	■ 0.535	$\begin{aligned} & \left[\Phi, \Phi_1 \right] + P \times Q_1 - P_1 \times Q \\ & \Phi P_1 - \Phi_1 P + r P_1 - r_1 P + s Q_1 - s_1 Q \\ & \Phi Q_1 - \Phi_1 Q - r Q_1 + r_1 Q + t P_1 - t_1 P \\ & -\frac{1}{8} \{P, Q_1\} + \frac{1}{8} \{P_1, Q\} + s t_1 - s_1 t \\ & \frac{1}{4} \{P, P_1\} + 2 r s_1 - 2 r_1 s \\ & -\frac{1}{4} \{Q, Q_1\} - 2 r t_1 + 2 r_1 t \end{aligned}$	$= \left(\begin{array}{c} [\Phi, \Phi_1] + P \times Q_1 - P_1 \times Q \\ \Phi P_1 - \Phi_1 P + r P_1 - r_1 P + s Q_1 - s_1 Q \\ \Phi Q_1 - \Phi_1 Q - r Q_1 + r_1 Q + t P_1 - t_1 P \\ -\frac{1}{8} \{P, Q_1\} + \frac{1}{8} \{P_1, Q\} + s t_1 - s_1 t \\ \frac{1}{4} \{P, P_1\} + 2 r s_1 - 2 r_1 s \\ -\frac{1}{4} \{Q, Q_1\} - 2 r t_1 + 2 r_1 t \end{array} \right), \left(\begin{array}{c} \Phi_2 \\ P_2 \\ Q_2 \\ r_2 \\ s_2 \\ t_2 \end{array} \right)_8$ $= \dots \left(\text{using } ([\Phi, \Phi_1], \Phi_2)_7 + (\Phi_1, [\Phi, \Phi_2])_7 = 0 \text{ (Lemma 4.5.1.(1))}, \right.$ $\left. (\Phi, P \times Q)_7 = \{\Phi P, Q\} \text{ (Lemma 4.5.1.(2)) etc.} \right) \dots$ $= -(R_1, [R, R_2])_8.$	$= \left(\begin{array}{c} [\Phi, \Phi_1] + P \times Q_1 - P_1 \times Q \\ \Phi P_1 - \Phi_1 P + r P_1 - r_1 P + s Q_1 - s_1 Q \\ \Phi Q_1 - \Phi_1 Q - r Q_1 + r_1 Q + t P_1 - t_1 P \\ -\frac{1}{8} \{P, Q_1\} + \frac{1}{8} \{P_1, Q\} + s t_1 - s_1 t \\ \frac{1}{4} \{P, P_1\} + 2 r s_1 - 2 r_1 s \\ -\frac{1}{4} \{Q, Q_1\} - 2 r t_1 + 2 r_1 t \end{array} \right), \left(\begin{array}{c} \Phi_2 \\ P_2 \\ Q_2 \\ r_2 \\ s_2 \\ t_2 \end{array} \right)_8$ $= \dots \left(\text{using } ([\Phi, \Phi_1], \Phi_2)_7 + (\Phi_1, [\Phi, \Phi_2])_7 = 0 \text{ (Lemma 4.5.1.(1))}, \right.$ $\left. (\Phi, P \times Q)_7 = \{\Phi P, Q\} \text{ (Lemma 4.5.1.(2)) etc.} \right) \dots$ $= -(R_1, [R, R_2])_8.$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00696	115	0.535	$\begin{array}{cc} \Phi_{-3}=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \kappa & 0 \\ 0 & 0 & C & 0 \end{array}\right), & \Phi_3=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \lambda \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \\ \Phi_{-2}=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & d & 0 & 0 \\ 0 & l & 0 & 0 \end{array}\right), & \Phi_2=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ c & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \\ \Phi_{-1}=\left(\begin{array}{cccc} 0 & 0 & B & 0 \\ 0 & 0 & 0 & 0 \\ b & 0 & 0 & 0 \end{array}\right), & \Phi_1=\left(\begin{array}{cccc} 0 & 0 & 0 & A \\ k & 0 & 0 & 0 \\ 0 & a & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \\ \Phi_0=\left(\begin{array}{cccc} g & 0 & 0 & 0 \\ 0 & h & 0 & 0 \\ 0 & 0 & \nu & 0 \\ 0 & 0 & 0 & \mu \end{array}\right). \end{array}$	$\Phi_{-3}=\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \kappa & 0 \\ 0 & 0 & C & 0 \end{pmatrix}, \quad \Phi_3=\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \lambda \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$ $\Phi_{-2}=\begin{pmatrix} 0 & 0 & C & 0 \\ 0 & 0 & 0 & 0 \\ 0 & d & 0 & 0 \\ 0 & l & 0 & 0 \end{pmatrix}, \quad \Phi_2=\begin{pmatrix} 0 & 0 & 0 & 0 \\ c & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$ $\Phi_{-1}=\begin{pmatrix} 0 & 0 & B & 0 \\ 0 & 0 & 0 & 0 \\ b & 0 & 0 & 0 \end{pmatrix}, \quad \Phi_1=\begin{pmatrix} 0 & 0 & 0 & A \\ k & 0 & 0 & 0 \\ 0 & a & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$ $\Phi_0=\begin{pmatrix} g & 0 & 0 & 0 \\ 0 & h & 0 & 0 \\ 0 & 0 & \nu & 0 \\ 0 & 0 & 0 & \mu \end{pmatrix}.$	
0902.0431_E01193	201	0.542	$(*(*(\mathfrak{a} \wedge \mathfrak{c}) \wedge \mathfrak{b}), \mathfrak{d}) &= (\mathfrak{a} \wedge \mathfrak{c}, \mathfrak{b} \wedge \mathfrak{d}) \\ &= (\mathfrak{a}, \mathfrak{b})(\mathfrak{c}, \mathfrak{d}) - (\mathfrak{a}_1 \wedge \mathfrak{c}_1, \mathfrak{b})(\mathfrak{a}_2 \wedge \mathfrak{c}_2, \mathfrak{d}) + (\mathfrak{a}_1 \wedge \mathfrak{c}_2, \mathfrak{b})(\mathfrak{a}_2 \wedge \mathfrak{c}_1, \mathfrak{d}) \\ &\quad + (\mathfrak{a}_2 \wedge \mathfrak{c}_1, \mathfrak{b})(\mathfrak{a}_1 \wedge \mathfrak{c}_2, \mathfrak{d}) - (\mathfrak{a}_2 \wedge \mathfrak{c}_2, \mathfrak{b})(\mathfrak{a}_1 \wedge \mathfrak{c}_1, \mathfrak{d}) + (\mathfrak{c}, \mathfrak{b})(\mathfrak{a}, \mathfrak{d}). \end{aligned}$	$(*(*(\mathfrak{a} \wedge \mathfrak{c}) \wedge \mathfrak{b}), \mathfrak{d}) &= (\mathfrak{a} \wedge \mathfrak{c}, \mathfrak{b} \wedge \mathfrak{d}) \\ &= (\mathfrak{a}, \mathfrak{b})(\mathfrak{c}, \mathfrak{d}) - (\mathfrak{a}_1 \wedge \mathfrak{c}_1, \mathfrak{b})(\mathfrak{a}_2 \wedge \mathfrak{c}_2, \mathfrak{d}) + (\mathfrak{a}_1 \wedge \mathfrak{c}_2, \mathfrak{b})(\mathfrak{a}_2 \wedge \mathfrak{c}_1, \mathfrak{d}) \\ &\quad + (\mathfrak{a}_2 \wedge \mathfrak{c}_1, \mathfrak{b})(\mathfrak{a}_1 \wedge \mathfrak{c}_2, \mathfrak{d}) - (\mathfrak{a}_2 \wedge \mathfrak{c}_2, \mathfrak{b})(\mathfrak{a}_1 \wedge \mathfrak{c}_1, \mathfrak{d}) + (\mathfrak{c}, \mathfrak{b})(\mathfrak{a}, \mathfrak{d}). \end{aligned}$	$(*(*(\mathfrak{a} \wedge \mathfrak{c}) \wedge \mathfrak{b}), \mathfrak{d}) &= (\mathfrak{a} \wedge \mathfrak{c}, \mathfrak{b} \wedge \mathfrak{d}) \\ &= (\mathfrak{a}, \mathfrak{b})(\mathfrak{c}, \mathfrak{d}) - (\mathfrak{a}_1 \wedge \mathfrak{c}_1, \mathfrak{b})(\mathfrak{a}_2 \wedge \mathfrak{c}_2, \mathfrak{d}) + (\mathfrak{a}_1 \wedge \mathfrak{c}_2, \mathfrak{b})(\mathfrak{a}_2 \wedge \mathfrak{c}_1, \mathfrak{d}) \\ &\quad + (\mathfrak{a}_2 \wedge \mathfrak{c}_1, \mathfrak{b})(\mathfrak{a}_1 \wedge \mathfrak{c}_2, \mathfrak{d}) - (\mathfrak{a}_2 \wedge \mathfrak{c}_2, \mathfrak{b})(\mathfrak{a}_1 \wedge \mathfrak{c}_1, \mathfrak{d}) + (\mathfrak{c}, \mathfrak{b})(\mathfrak{a}, \mathfrak{d}). \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00221	36	0.542	$\begin{aligned} (X, Y) &= \sum_{i=1}^3 \left(\xi_i \eta_i + 2(x_i, y_i) \right), \\ \operatorname{tr}(X, Y, Z) &= \sum_{i=1}^3 \left(\xi_i \eta_i \zeta_i + R(x_i y_{i+1} z_{i+2} + x_i z_{i+1} y_{i+2}) \right. \\ &\quad \left. + \xi_i ((y_{i+1}, z_{i+1}) + (y_{i+2}, z_{i+2})) + \eta_i ((z_{i+1}, x_{i+1}) + (z_{i+2}, x_{i+2})) \right. \\ &\quad \left. + \zeta_i ((x_{i+1}, y_{i+1}) + (x_{i+2}, y_{i+2})) \right), \end{aligned}$	$(X, Y) = \sum_{i=1}^3 (\xi_i \eta_i + 2(x_i, y_i)),$ $\operatorname{tr}(X, Y, Z) = \sum_{i=1}^3 (\xi_i \eta_i \zeta_i + R(x_i y_{i+1} z_{i+2} + x_i z_{i+1} y_{i+2})$ $+ \xi_i ((y_{i+1}, z_{i+1}) + (y_{i+2}, z_{i+2})) + \eta_i ((z_{i+1}, x_{i+1}) + (z_{i+2}, x_{i+2}))$ $+ \zeta_i ((x_{i+1}, y_{i+1}) + (x_{i+2}, y_{i+2}))),$	$(X, Y) = \sum_{i=1}^3 (\xi_i \eta_i + 2(x_i, y_i)),$ $\operatorname{tr}(X, Y, Z) = \sum_{i=1}^3 (\xi_i \eta_i \zeta_i + R(x_i y_{i+1} z_{i+2} + x_i z_{i+1} y_{i+2})$ $+ \xi_i ((y_{i+1}, z_{i+1}) + (y_{i+2}, z_{i+2})) + \eta_i ((z_{i+1}, x_{i+1}) + (z_{i+2}, x_{i+2}))$ $+ \zeta_i ((x_{i+1}, y_{i+1}) + (x_{i+2}, y_{i+2}))),$
0902.0431_E00960	159	0.542	$\begin{aligned} &\beta(P_C + (M + N)) = P_C + Q(M + N) = P_C + (QM + QN) \\ &= \psi(Q, E)(P_C + (M + N)), \quad P_C + (M + N) \in \mathfrak{P}^C \end{aligned}$	$\beta(P_C + (M + N)) = P_C + Q(M + N) = P_C + (QM + QN)$ $= \psi(Q, E)(P_C + (M + N)), \quad P_C + (M + N) \in \mathfrak{P}^C$	$\beta(P_C + (M + N)) = P_C + Q(M + N) = P_C + (QM + QN)$ $= \psi(Q, E)(P_C + (M + N)), \quad P_C + (M + N) \in \mathfrak{P}^C$
0902.0431_E00005	6	0.542	$\begin{array}{l} \overline{x_0 + \sum_{i=1}^7 x_i e_i} = x_0 - \sum_{i=1}^7 x_i e_i, \quad \left(\sum_{i=0}^7 x_i e_i, \sum_{i=0}^7 y_i e_i \right) = \sum_{i=0}^7 x_i y_i, \\ x = \sqrt{(x, x)}, \quad R\left(x_0 + \sum_{i=1}^7 x_i e_i\right) = x_0. \end{array}$	$\overline{x_0 + \sum_{i=1}^7 x_i e_i} = x_0 - \sum_{i=1}^7 x_i e_i, \quad \left(\sum_{i=0}^7 x_i e_i, \sum_{i=0}^7 y_i e_i \right) = \sum_{i=0}^7 x_i y_i$ $ x = \sqrt{(x, x)}, \quad R\left(x_0 + \sum_{i=1}^7 x_i e_i\right) = x_0.$	$\overline{x_0 + \sum_{i=1}^7 x_i e_i} = x_0 - \sum_{i=1}^7 x_i e_i, \quad \left(\sum_{i=0}^7 x_i e_i, \sum_{i=0}^7 y_i e_i \right) = \sum_{i=0}^7 x_i y_i,$ $ x = \sqrt{(x, x)}, \quad R\left(x_0 + \sum_{i=1}^7 x_i e_i\right) = x_0.$
0902.0431_E00097	19	0.544	$\begin{array}{l} \pm(\lambda_1 - \lambda_2) : \pm(G_{24} + G_{35}) + i(-G_{25} + G_{34}), \\ \pm(\lambda_1 - \lambda_3) : \pm(G_{26} + G_{37}) + i(-G_{27} + G_{36}), \\ \pm(\lambda_2 - \lambda_3) : \pm(G_{46} + G_{57}) + i(-G_{47} + G_{56}). \end{array}$	$\pm(\lambda_1 - \lambda_2) : \pm(G_{24} + G_{35}) + i(-G_{25} + G_{34}),$ $\pm(\lambda_1 - \lambda_3) : \pm(G_{26} + G_{37}) + i(-G_{27} + G_{36}),$ $\pm(\lambda_2 - \lambda_3) : \pm(G_{46} + G_{57}) + i(-G_{47} + G_{56}).$	$\pm(\lambda_1 - \lambda_2) : \pm(G_{24} + G_{35}) + i(-G_{25} + G_{34}),$ $\pm(\lambda_1 - \lambda_3) : \pm(G_{26} + G_{37}) + i(-G_{27} + G_{36}),$ $\pm(\lambda_2 - \lambda_3) : \pm(G_{46} + G_{57}) + i(-G_{47} + G_{56}).$
0902.0431_E00949	158	0.546	$\begin{aligned} &\mu \left(\begin{array}{cc} M_{11} & M_{12} \\ M_{21} & M_{22} \end{array} \right) \\ &= \left(\frac{(M_{21} - M_{12})e_1}{2} + i \frac{M_{21} + M_{12}}{2}, \frac{(M_{22} + M_{11})e_1}{2} + i \frac{M_{22} - M_{11}}{2} \right), \end{aligned}$	$\mu \left(\begin{array}{cc} M_{11} & M_{12} \\ M_{21} & M_{22} \end{array} \right)$ $= \left(\frac{(M_{21} - M_{12})e_1}{2} + i \frac{M_{21} + M_{12}}{2}, \frac{(M_{22} + M_{11})e_1}{2} + i \frac{M_{22} - M_{11}}{2} \right)$	$\mu \left(\begin{array}{cc} M_{11} & M_{12} \\ M_{21} & M_{22} \end{array} \right)$ $= \left(\frac{(M_{21} - M_{12})e_1}{2} + i \frac{M_{21} + M_{12}}{2}, \frac{(M_{22} + M_{11})e_1}{2} + i \frac{M_{22} - M_{11}}{2} \right),$
0902.0431_E00297	47	0.546	$x \xrightarrow{G_{i0}} x_i e_0 - x_0 e_i \xrightarrow{G_{j0}(j \neq i)} x_i e_j \xrightarrow{x_j^{-1} G_{0j}} e_0 = 1.$	$x \xrightarrow{G_{i0}} x_i e_0 - x_0 e_i \xrightarrow{G_{j0}(j \neq i)} x_i e_j \xrightarrow{x_j^{-1} G_{0j}} e_0 = 1.$	$x \xrightarrow{G_{i0}} x_i e_0 - x_0 e_i \xrightarrow{G_{j0}(j \neq i)} x_i e_j \xrightarrow{x_j^{-1} G_{0j}} e_0 = 1.$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00524	87	0.546	$\begin{aligned} & \left(\alpha_i, \alpha_i\right)=\frac{1}{12}, \quad i=1,2,3,4,5,6 \\ & \left(\alpha_1, \alpha_2\right)=\left(\alpha_2, \alpha_3\right)=\left(\alpha_3, \alpha_4\right)=\left(\alpha_3, \alpha_5\right)=\left(\alpha_5, \alpha_6\right)=-\frac{1}{24}, \\ & \left(\alpha_i, \alpha_j\right)=0, \quad \text { otherwise, } \\ & \left(-\mu,-\mu\right)=\frac{1}{12}, \quad\left(-\mu, \alpha_4\right)=-\frac{1}{24}, \quad\left(-\mu, \alpha_i\right)=0, \quad i=1,2,3,5,6, \end{aligned}$	$\begin{aligned} & \left(\alpha_i, \alpha_i\right)=\frac{1}{12}, \quad i=1,2,3,4,5,6 \\ & \left(\alpha_1, \alpha_2\right)=\left(\alpha_2, \alpha_3\right)=\left(\alpha_3, \alpha_4\right)=\left(\alpha_3, \alpha_5\right)=\left(\alpha_5, \alpha_6\right)=-\frac{1}{24} \\ & \left(\alpha_i, \alpha_j\right)=0, \quad \text { otherwise, } \\ & \left(-\mu,-\mu\right)=\frac{1}{12}, \quad\left(-\mu, \alpha_4\right)=-\frac{1}{24}, \quad\left(-\mu, \alpha_i\right)=0, \quad i=1,2,3,5,6, \end{aligned}$	$\begin{aligned} & \left(\alpha_i, \alpha_i\right)=\frac{1}{12}, \quad i=1,2,3,4,5,6, \\ & \left(\alpha_1, \alpha_2\right)=\left(\alpha_2, \alpha_3\right)=\left(\alpha_3, \alpha_4\right)=\left(\alpha_3, \alpha_5\right)=\left(\alpha_5, \alpha_6\right)=-\frac{1}{24}, \\ & \left(\alpha_i, \alpha_j\right)=0, \quad \text { otherwise, } \\ & \left(-\mu,-\mu\right)=\frac{1}{12}, \quad\left(-\mu, \alpha_4\right)=-\frac{1}{24}, \quad\left(-\mu, \alpha_i\right)=0, \quad i=1,2,3,5,6, \end{aligned}$
0902.0431_E00920	152	0.547	$\begin{aligned} & \left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right),\left(\begin{array}{ccc} \eta_1 & y_3 & \bar{y}_2 \\ \bar{y}_3 & \eta_2 & y_1 \\ y_2 & \bar{y}_1 & \eta_3 \end{array}\right), \xi, \eta\right)=\sum_{i<j<k} x_{i j k} \boldsymbol{e}_i \wedge \boldsymbol{e}_j \wedge \boldsymbol{e}_k \end{aligned}$	$f\left(\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right),\left(\begin{array}{ccc} \eta_1 & y_3 & \bar{y}_2 \\ \bar{y}_3 & \eta_2 & y_1 \\ y_2 & \bar{y}_1 & \eta_3 \end{array}\right), \xi, \eta\right)=\sum_{i<j<k} x_{i j k} \boldsymbol{e}_i \wedge \boldsymbol{e}_j \wedge \boldsymbol{e}_k$	$f\left(\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right),\left(\begin{array}{ccc} \eta_1 & y_3 & \bar{y}_2 \\ \bar{y}_3 & \eta_2 & y_1 \\ y_2 & \bar{y}_1 & \eta_3 \end{array}\right), \xi, \eta\right)=\sum_{i<j<k} x_{i j k} \boldsymbol{e}_i \wedge \boldsymbol{e}_j \wedge \boldsymbol{e}_k$
0902.0431_E00833	139	0.551	$\begin{aligned} & \left(\frac{\boldsymbol{e}_7}{\kappa}\right)^{\kappa, \mu}=\left\{\Phi \in \mathfrak{e}_7 \mid \kappa \Phi=\Phi \kappa, \mu \Phi=\Phi \mu\right\} \\ & =\left\{\Phi\left(\phi \cdot A,-\tau A, \nu\right) \in \mathfrak{e}_7 \mid \begin{array}{l} \phi \in\left(\mathfrak{e}_6\right)^{\sigma}, A \in\left(\mathfrak{J}^C\right)_{\sigma},\left(E_1, A\right)=0, \\ \nu=-\frac{3}{2}\left(\phi E_1, E_1\right) \end{array}\right\} . \end{aligned}$	$\begin{aligned} & \left(\mathfrak{e}_7\right)^{\kappa, \mu}=\left\{\Phi \in \mathfrak{e}_7 \mid \kappa \Phi=\Phi \kappa, \mu \Phi=\Phi \mu\right\} \\ & =\left\{\Phi\left(\phi \cdot A,-\tau A, \nu\right) \in \mathfrak{e}_7 \mid \begin{array}{l} \phi \in\left(\mathfrak{e}_6\right)^{\sigma}, A \in\left(\mathfrak{J}^C\right)_{\sigma},\left(E_1, A\right)=0, \\ \nu=-\frac{3}{2}\left(\phi E_1, E_1\right) \end{array}\right\} . \end{aligned}$	$\begin{aligned} & \left(\mathfrak{e}_7\right)^{\kappa, \mu}=\left\{\Phi \in \mathfrak{e}_7 \mid \kappa \Phi=\Phi \kappa, \mu \Phi=\Phi \mu\right\} \\ & =\left\{\Phi\left(\phi \cdot A,-\tau A, \nu\right) \in \mathfrak{e}_7 \mid \begin{array}{l} \phi \in\left(\mathfrak{e}_6\right)^{\sigma}, A \in\left(\mathfrak{J}^C\right)_{\sigma},\left(E_1, A\right)=0, \\ \nu=-\frac{3}{2}\left(\phi E_1, E_1\right) \end{array}\right\} . \end{aligned}$
0902.0431_E01036	176	0.553	$\begin{aligned} & B_8\left(\tilde{h}, \tilde{h}^{\prime}\right)=60 \sum_{k=0}^3 \lambda_k \lambda_k^{\prime}+30 \sum_{j=1}^3 \mu_j \mu_j^{\prime}+40 \nu \nu^{\prime}+120 r r^{\prime}, \end{aligned}$	$B_8\left(\tilde{h}, \tilde{h}^{\prime}\right)=60 \sum_{k=0}^3 \lambda_k \lambda_k^{\prime}+30 \sum_{j=1}^3 \mu_j \mu_j^{\prime}+40 \nu \nu^{\prime}+120 r r^{\prime},$	$B_8\left(\tilde{h}, \tilde{h}^{\prime}\right)=60 \sum_{k=0}^3 \lambda_k \lambda_k^{\prime}+30 \sum_{j=1}^3 \mu_j \mu_j^{\prime}+40 \nu \nu^{\prime}+120 r r^{\prime},$

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0902.0431_E00953	158	0.553	$\begin{aligned} & \psi_*(D)(\mu\widetilde{M}) \quad \& = \\ & \Phi(\phi_C(B, C), h(L), -\tau h(L), -ive_1)(M, N) \\ & \left(\begin{array}{cccc} \phi_C(B, C) + \frac{1}{3}ive_1 & -2\tau h(L) & 0 & h(L) \\ 2h(L) & \tau\phi_C(B, C)\tau - \frac{1}{3}ive_1 & -\tau h(L) & 0 \\ 0 & h(L) & -ive_1 & 0 \\ -\tau h(L) & 0 & 0 & ive_1 \end{array} \right) \begin{pmatrix} M \\ N \\ 0 \\ 0 \end{pmatrix} \\ & = \left(\begin{array}{c} \phi_C(B, C)M + \frac{1}{3}ive_1M - 2\tau h(L) \times N \\ 2h(l) \times M + \tau\phi_C(B, C)\tau N - \frac{1}{3}ive_1N \\ (h(L), N) \\ -(\tau h(L), M) \end{array} \right) \\ & = \left(\begin{array}{c} -Mh(B, C) + N\tau h(L) + \frac{1}{3}ive_1M \\ -Mh(L) - N\tau h(B, C) - \frac{1}{3}ive_1N \\ 0 \\ 0 \end{array} \right) \end{aligned}$	$\begin{aligned} & \psi_*(D)(\mu\widetilde{M}) \\ & = \Phi(\phi_C(B, C), h(L), -\tau h(L), -ive_1)(M, N) \\ & = \left(\begin{array}{cccc} \phi_C(B, C) + \frac{1}{3}ive_1 & -2\tau h(L) & 0 & h(L) \\ 2h(L) & \tau\phi_C(B, C)\tau - \frac{1}{3}ive_1 & -\tau h(L) & 0 \\ 0 & h(L) & -ive_1 & 0 \\ -\tau h(L) & 0 & 0 & ive_1 \end{array} \right) \begin{pmatrix} M \\ N \\ 0 \\ 0 \end{pmatrix} \\ & = \left(\begin{array}{c} \phi_C(B, C)M + \frac{1}{3}ive_1M - 2\tau h(L) \times N \\ 2h(l) \times M + \tau\phi_C(B, C)\tau N - \frac{1}{3}ive_1N \\ (h(L), N) \\ -(\tau h(L), M) \end{array} \right) \\ & = \left(\begin{array}{c} -Mh(B, C) + N\tau h(L) + \frac{1}{3}ive_1M \\ -Mh(L) - N\tau h(B, C) - \frac{1}{3}ive_1N \\ 0 \\ 0 \end{array} \right) \end{aligned}$	$\begin{aligned} & \psi_*(D)(\mu\widetilde{M}) \\ & = \Phi(\phi_C(B, C), h(L), -\tau h(L), -ive_1)(M, N) \\ & = \left(\begin{array}{cccc} \phi_C(B, C) + \frac{1}{3}ive_1 & -2\tau h(L) & 0 & h(L) \\ 2h(L) & \tau\phi_C(B, C)\tau - \frac{1}{3}ive_1 & -\tau h(L) & 0 \\ 0 & h(L) & -ive_1 & 0 \\ -\tau h(L) & 0 & 0 & ive_1 \end{array} \right) \begin{pmatrix} M \\ N \\ 0 \\ 0 \end{pmatrix} \\ & = \left(\begin{array}{c} \phi_C(B, C)M + \frac{1}{3}ive_1M - 2\tau h(L) \times N \\ 2h(l) \times M + \tau\phi_C(B, C)\tau N - \frac{1}{3}ive_1N \\ (h(L), N) \\ -(\tau h(L), M) \end{array} \right) \\ & = \left(\begin{array}{c} -Mh(B, C) + N\tau h(L) + \frac{1}{3}ive_1M \\ -Mh(L) - N\tau h(B, C) - \frac{1}{3}ive_1N \\ 0 \\ 0 \end{array} \right) \end{aligned}$
0902.0431_E00440	72	0.554	$F_{4(4)} = \left(F_4^C \right)^{\tau\gamma}, \quad F_{4(-20)} = \left(F_4^C \right)^{\tau\sigma}.$	$F_{4(4)} = \left(F_4^C \right)^{\tau\gamma}, \quad F_{4(-20)} = \left(F_4^C \right)^{\tau\sigma}.$	$F_{4(4)} = \left(F_4^C \right)^{\tau\gamma}, \quad F_{4(-20)} = \left(F_4^C \right)^{\tau\sigma}.$
0902.0431_E01174	197	0.554	$\begin{aligned} & \mathfrak{g}_0 = \mathfrak{sl}(5, C) \oplus \mathfrak{sl}(5, C), \\ & \mathfrak{g}_1 = C^5 \otimes \Lambda^2(C^5) = \mathfrak{g}_{-1}, \quad \mathfrak{g}_2 = \Lambda^2(C^5) \otimes C^5 = \mathfrak{g}_{-2}, \end{aligned}$	$\begin{aligned} & \mathfrak{g}_0 = \mathfrak{sl}(5, C) \oplus \mathfrak{sl}(5, C), \\ & \mathfrak{g}_1 = C^5 \otimes \Lambda^2(C^5) = \mathfrak{g}_{-1}, \quad \mathfrak{g}_2 = \Lambda^2(C^5) \otimes C^5 = \mathfrak{g}_{-2}, \end{aligned}$	$\begin{aligned} & \mathfrak{g}_0 = \mathfrak{sl}(5, C) \oplus \mathfrak{sl}(5, C), \\ & \mathfrak{g}_1 = C^5 \otimes \Lambda^2(C^5) = \mathfrak{g}_{-1}, \quad \mathfrak{g}_2 = \Lambda^2(C^5) \otimes C^5 = \mathfrak{g}_{-2}, \end{aligned}$
0902.0431_E00037	11	0.555	$\begin{aligned} & 1 \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 1 & 2 & 3 \end{array} \right), \quad \kappa \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 2 & 1 & 3 \end{array} \right), \quad \pi \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 3 & 2 & 1 \end{array} \right), \\ & \nu \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 2 & 3 & 1 \end{array} \right), \quad \nu^2 \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 3 & 1 & 2 \end{array} \right), \quad \nu\pi \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 1 & 3 & 2 \end{array} \right) \end{aligned}$	$\begin{aligned} & 1 \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 1 & 2 & 3 \end{array} \right), \quad \kappa \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 2 & 1 & 3 \end{array} \right), \quad \pi \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 3 & 2 & 1 \end{array} \right), \\ & \nu \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 2 & 3 & 1 \end{array} \right), \quad \nu^2 \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 3 & 1 & 2 \end{array} \right), \quad \nu\pi \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 1 & 3 & 2 \end{array} \right) \end{aligned}$	$\begin{aligned} & 1 \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 1 & 2 & 3 \end{array} \right), \quad \kappa \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 2 & 1 & 3 \end{array} \right), \quad \pi \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 3 & 2 & 1 \end{array} \right), \\ & \nu \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 2 & 3 & 1 \end{array} \right), \quad \nu^2 \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 3 & 1 & 2 \end{array} \right), \quad \nu\pi \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 1 & 3 & 2 \end{array} \right) \end{aligned}$

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0902.0431_E08370	61	■ 0.557	$\begin{aligned} & \backslash begin{aligned} \& \ \eta_{\{1\}}(t)^2+\eta_{\{2\}}(t)^2+ \\ & \ \eta_{\{3\}}(t)^2 \ \backslash \& \ =\xi_{\{1\}}\{ \}^2+\backslash left(\frac{ \\ \xi_{\{2\}}+\xi_{\{3\}}\}{2}+\frac{\xi_{\{2\}}-\xi_{\{3\}}\}{2} \ \cos 2 \ t+ \\ & \ \backslash left x_{\{1\}}\right \ \sin 2 \ t\right)^2 \ \backslash \& \ \quad +\backslash left(\\ & \ \frac{\xi_{\{2\}}+\xi_{\{3\}}\}{2}-\frac{\xi_{\{2\}}-\xi_{\{3\}}\}{2} \ \cos 2 t- x_{\{1\}}\sin 2 t) \\ & \ \backslash cos 2 \ t-\backslash left x_{\{1\}}\right \ \sin 2 \ t\right)^2 \ \backslash \\ & \ \& \ =\xi_{\{1\}}\{ \}^2+2\backslash left(\frac{\xi_{\{2\}}+\xi_{\{3\}}\}{2} \\ & \ \right)^2+2\backslash left(\frac{\xi_{\{2\}}-\xi_{\{3\}}\}{2} \ \cos 2 \ t+ \\ & \ \backslash left x_{\{1\}}\right \ \sin 2 \ t\right)^2 \ \backslash \& \ =\xi_{\{1\}}\{ \\ & \ \}^2+2\backslash left(\frac{\xi_{\{2\}}+\xi_{\{3\}}\}{2}\right)^2+2\backslash left(\\ & \ \backslash left(\frac{\xi_{\{2\}}-\xi_{\{3\}}\}{2}\right)^2\right)^2+\backslash left x_{\{1\}} \\ & \ \right ^2\right) \ \sin ^2\left(2 \ t+t_{\{0\}}\right) \backslash left(\\ & \ \text { { for some } } t_{\{0\}} \ \backslash in \ \boldsymbol{\mathbb{R}} \right) \ \backslash \& \ \\ & \ \& \ \leq \xi_{\{1\}}^2+2\backslash left(\frac{\xi_{\{2\}}+\xi_{\{3\}}\}{2} \\ & \ \right)^2+2\backslash left(\backslash left(\frac{\xi_{\{2\}}-\xi_{\{3\}}\}{2} \\ & \ \right)^2+\backslash left x_{\{1\}}\right ^2\right) \ \backslash \& \ =\{ \\ & \ \xi_{\{1\}}^2+\xi_{\{2\}}^2+\xi_{\{3\}}^2+2\backslash left x_{\{1\}} \\ & \ \right ^2 \ \end{aligned}$	$\begin{aligned} & \eta_1(t)^2+\eta_2(t)^2+\eta_3(t)^2 \\ & =\xi_1^2+\left(\frac{\xi_2+\xi_3}{2}+\frac{\xi_2-\xi_3}{2} \cos 2 t+\left x_1\right \sin 2 t\right)^2 \\ & \quad +\left(\frac{\xi_2+\xi_3}{2}-\frac{\xi_2-\xi_3}{2} \cos 2 t-\left x_1\right \sin 2 t\right)^2 \\ & =\xi_1^2+2\left(\frac{\xi_2+\xi_3}{2}\right)^2+2\left(\frac{\xi_2-\xi_3}{2} \cos 2 t+\left x_1\right \sin 2 t\right)^2 \\ & =\xi_1^2+2\left(\frac{\xi_2+\xi_3}{2}\right)^2+2\left(\left(\frac{\xi_2-\xi_3}{2}\right)^2+\left x_1\right ^2\right) \sin ^2\left(2 t+t_0\right) \text { (for some } t_0 \in \boldsymbol{R} \text {) } \\ & \leq \xi_1^2+2\left(\frac{\xi_2+\xi_3}{2}\right)^2+2\left(\left(\frac{\xi_2-\xi_3}{2}\right)^2+\left x_1\right ^2\right) \\ & =\xi_1^2+\xi_2^2+\xi_3^2+2\left x_1\right ^2 \end{aligned}$	$\begin{aligned} & \eta_1(t)^2+\eta_2(t)^2+\eta_3(t)^2 \\ & =\xi_1^2+\left(\frac{\xi_2+\xi_3}{2}+\frac{\xi_2-\xi_3}{2} \cos 2 t+\left x_1\right \sin 2 t\right)^2 \\ & \quad +\left(\frac{\xi_2+\xi_3}{2}-\frac{\xi_2-\xi_3}{2} \cos 2 t-\left x_1\right \sin 2 t\right)^2 \\ & =\xi_1^2+2\left(\frac{\xi_2+\xi_3}{2}\right)^2+2\left(\frac{\xi_2-\xi_3}{2} \cos 2 t+\left x_1\right \sin 2 t\right)^2 \\ & =\xi_1^2+2\left(\frac{\xi_2+\xi_3}{2}\right)^2+2\left(\left(\frac{\xi_2-\xi_3}{2}\right)^2+\left x_1\right ^2\right) \sin ^2\left(2 t+t_0\right) \text { (for some } t_0 \in \boldsymbol{R} \text {) } \\ & \leq \xi_1^2+2\left(\frac{\xi_2+\xi_3}{2}\right)^2+2\left(\left(\frac{\xi_2-\xi_3}{2}\right)^2+\left x_1\right ^2\right) \\ & =\xi_1^2+\xi_2^2+\xi_3^2+2\left x_1\right ^2 \end{aligned}$
0902.0431_E08674 (i)	112	■ 0.558	$2(P \times R) Q-(P \times Q) R-(R \times Q) P+\frac{3}{8}\{Q, R\} P-\frac{3}{8}\{P, Q\} R=0 .$	$2(P \times R) Q-(P \times Q) R-(R \times Q) P+\frac{3}{8}\{Q, R\} P-\frac{3}{8}\{P, Q\} R=0 .$	$2(P \times R) Q-(P \times Q) R-(R \times Q) P+\frac{3}{8}\{Q, R\} P-\frac{3}{8}\{P, Q\} R=0 .$
0902.0431_E01188	200	■ 0.562	$\begin{aligned} & \backslash begin{aligned} (a \times *(b \wedge x)) \ v \& \ + (b \times *(a \wedge x)) \ v = (x, w) \ *(a \wedge b) - \frac{1}{5} \ *(a \wedge b), x) v \\ & \ \backslash \& \ = \& \ (x \times *(a \wedge b)) \ v . \\ & \ \end{aligned}$	$\begin{aligned} & (a \times *(b \wedge x)) v+(b \times *(a \wedge x)) v=(x, w) *(a \wedge b)-\frac{1}{5} *(a \wedge b), x) v \\ & \quad =\left(x \times *(a \wedge b)\right) v . \end{aligned}$	$\begin{aligned} & (\boldsymbol{a} \times *(\boldsymbol{b} \wedge \boldsymbol{x})) \boldsymbol{v}+(\boldsymbol{b} \times *(\boldsymbol{a} \wedge \boldsymbol{x})) \boldsymbol{v}=(\boldsymbol{x}, \boldsymbol{w}) *(\boldsymbol{a} \wedge \boldsymbol{b})-\frac{1}{5} *(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x}) \boldsymbol{v} \\ & \quad =(\boldsymbol{x} \times *(\boldsymbol{a} \wedge \boldsymbol{b})) \boldsymbol{v} . \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E01095	186	■ 0.566	$\begin{aligned} & \backslash begin{aligned} \backslash zeta[l \& \backslash left.\backslash left(S_{\{1\}\backslash right}) \\ & I, \backslash left(S_{\{2\}\backslash right}) I\backslash right]=\backslash zeta \backslash left(S_{\{1\}} \\ & \backslash tau S_{\{2\}}-S_{\{2\}} \backslash tau S_{\{1\}\backslash right}) \backslash \backslash = \& \backslash zeta\backslash left(l \\ & \backslash left(S_{\{1\}} \backslash tau S_{\{2\}}-S_{\{2\}} \backslash tau S_{\{1\}}-\backslash frac{1}{8}) \\ & \backslash operatorname{tr}\backslash left(S_{\{1\}} \backslash tau S_{\{2\}}-S_{\{2\}} \backslash tau \\ & S_{\{1\}\backslash right}) E\backslash right)\backslash right. \backslash \backslash \& \backslash left.\backslash quad+\backslash left(\\ & \backslash frac{1}{8}) \backslash operatorname{tr}\backslash left(S_{\{1\}} \backslash tau S_{\{2\}}-S_{\{2\}} \\ & \backslash tau S_{\{1\}\backslash right}) E\backslash right)\backslash right) \backslash \backslash = \& \backslash Theta\backslash left(\\ & \backslash varphi_{\{*\}}\backslash left(S_{\{1\}} \backslash tau S_{\{2\}}-S_{\{2\}} \backslash tau S_{\{1\}}- \\ & \backslash frac{1}{8}) \backslash operatorname{tr}\backslash left(S_{\{1\}} \backslash tau S_{\{2\}}-S_{\{2\}} \\ & \backslash tau S_{\{1\}\backslash right}) E\backslash right), \& 0,0,0\backslash right. \backslash \backslash \& \backslash left. \\ & \backslash quad-\backslash frac{i}{4}) \backslash operatorname{tr}\backslash left(S_{\{1\}} \\ & \backslash tau S_{\{2\}}-S_{\{2\}} \backslash tau S_{\{1\}\backslash right}), \backslash frac{i}{4}) \\ & \backslash operatorname{tr}\backslash left(S_{\{1\}} \backslash tau S_{\{2\}}-S_{\{2\}} \backslash tau S_{\{1\}} \\ & \backslash right)\backslash right) \backslash \backslash = \& \backslash Theta\backslash left(4\backslash left(\backslash lambda \backslash gamma \\ & \backslash chi^{-1} S_{\{1\}} \backslash times \backslash chi^{-1} S_{\{2\}}-\backslash lambda \backslash gamma \\ & \backslash chi^{-1} S_{\{2\}} \backslash times \backslash chi^{-1} S_{\{1\}\backslash right}), \& 0,0,0 \\ & \backslash right. \backslash \backslash \& \backslash left.\backslash quad\backslash left\{\backslash chi^{-1} S_{\{1\}}, \backslash chi^{-1} S_{\{2\}} \\ & S_{\{2\}\backslash right}\}-\backslash left\{\backslash chi^{-1} S_{\{1\}}, \backslash chi^{-1} S_{\{2\}} \\ & \backslash right\}\backslash right)(\backslash text { Lemma 5.8.3}). \backslash end{aligned} \end{aligned}$	$\begin{aligned} & \zeta[l(S_1)I, l(S_2)I] = \zeta l(S_1\tau S_2 - S_2\tau S_1) \\ & = \zeta \left(l \left(S_1\tau S_2 - S_2\tau S_1 - \frac{1}{8} \operatorname{tr}(S_1\tau S_2 - S_2\tau S_1) E \right) \right. \\ & \quad \left. + l \left(\frac{1}{8} \operatorname{tr}(S_1\tau S_2 - S_2\tau S_1) E \right) \right) \\ & = \Theta \left(\varphi_* \left(S_1\tau S_2 - S_2\tau S_1 - \frac{1}{8} \operatorname{tr}(S_1\tau S_2 - S_2\tau S_1) E \right), 0, 0, 0 \right. \\ & \quad \left. - \frac{i}{4} \operatorname{tr}(S_1\tau S_2 - S_2\tau S_1), \frac{i}{4} \operatorname{tr}(S_1\tau S_2 - S_2\tau S_1) \right) \\ & = \Theta \left(4 \left(\lambda\gamma\chi^{-1}S_1 \times \chi^{-1}S_2 - \lambda\gamma\chi^{-1}S_2 \times \chi^{-1}S_1 \right), 0, 0, 0 \right. \\ & \quad \left. \{ \chi^{-1}S_1, \chi^{-1}S_2 \} - \{ \chi^{-1}S_1, \chi^{-1}S_2 \} \right) \text{ (Lemma 5.8.3)}. \end{aligned}$	$\begin{aligned} & \zeta[l(S_1)I, l(S_2)I] = \zeta l(S_1\tau S_2 - S_2\tau S_1) \\ & = \zeta \left(l \left(S_1\tau S_2 - S_2\tau S_1 - \frac{1}{8} \operatorname{tr}(S_1\tau S_2 - S_2\tau S_1) E \right) \right. \\ & \quad \left. + l \left(\frac{1}{8} \operatorname{tr}(S_1\tau S_2 - S_2\tau S_1) E \right) \right) \\ & = \Theta \left(\varphi_* \left(S_1\tau S_2 - S_2\tau S_1 - \frac{1}{8} \operatorname{tr}(S_1\tau S_2 - S_2\tau S_1) E \right), 0, 0, 0, \right. \\ & \quad \left. - \frac{i}{4} \operatorname{tr}(S_1\tau S_2 - S_2\tau S_1), \frac{i}{4} \operatorname{tr}(S_1\tau S_2 - S_2\tau S_1) \right) \\ & = \Theta(4(\lambda\gamma\chi^{-1}S_1 \times \chi^{-1}S_2 - \lambda\gamma\chi^{-1}S_2 \times \chi^{-1}S_1), 0, 0, 0, \\ & \quad \{ \chi^{-1}S_1, \chi^{-1}S_2 \} - \{ \chi^{-1}S_1, \chi^{-1}S_2 \}) \text{ (Lemma 5.8.3)}. \end{aligned}$
0902.0431_E01179	199	■ 0.568	$\begin{aligned} & \backslash begin{aligned} (*(\backslash boldsymbol{a} \& \backslash wedge (*(\\ & \backslash boldsymbol{b} \backslash wedge \backslash boldsymbol{x})))+*(\backslash boldsymbol{b} \\ & \backslash wedge (*(\backslash boldsymbol{a} \backslash wedge \backslash boldsymbol{x}))+ \\ & \backslash boldsymbol{x} \backslash wedge *(\backslash boldsymbol{a} \backslash wedge \\ & \backslash boldsymbol{b}), \backslash boldsymbol{c})) \backslash \backslash \backslash quad \& =(* \\ & \backslash boldsymbol{b} \backslash wedge \backslash boldsymbol{x}, \backslash boldsymbol{c} \\ & \backslash wedge \backslash boldsymbol{a}))+(* \backslash boldsymbol{a} \backslash wedge \\ & \backslash boldsymbol{x}, \backslash boldsymbol{b} \backslash wedge \backslash boldsymbol{c}))+(* \\ & \backslash boldsymbol{c} \backslash wedge \backslash boldsymbol{x}, \backslash boldsymbol{a} \\ & \backslash wedge \backslash boldsymbol{b})) \backslash \backslash \backslash quad \& =(* \backslash boldsymbol{x}, \\ & * \backslash boldsymbol{b} \backslash wedge *(\backslash boldsymbol{c} \backslash wedge \\ & \backslash boldsymbol{a}))+* \backslash boldsymbol{a} \backslash wedge *(\backslash boldsymbol{b} \\ & \backslash wedge \backslash boldsymbol{c}))+* \backslash boldsymbol{c} \backslash wedge *(\\ & \backslash boldsymbol{a} \backslash wedge \backslash boldsymbol{b}))=0 . \backslash end{aligned} \end{aligned}$	$\begin{aligned} & (*(a \wedge *(b \wedge x)) + *(b \wedge *(a \wedge x)) + x \wedge *(a \wedge b), c) \\ & = *(b \wedge x, c \wedge a) + *(a \wedge x, b \wedge c) + *(c \wedge x, a \wedge b) \\ & = *(x, *b \wedge *(c \wedge a) + *a \wedge *(b \wedge c) + *c \wedge *(a \wedge b)) = 0. \end{aligned}$	$\begin{aligned} & (*(a \wedge *(b \wedge x)) + *(b \wedge *(a \wedge x)) + x \wedge *(a \wedge b), c) \\ & = *(b \wedge x, c \wedge a) + *(a \wedge x, b \wedge c) + *(c \wedge x, a \wedge b) \\ & = *(x, *b \wedge *(c \wedge a) + *a \wedge *(b \wedge c) + *c \wedge *(a \wedge b)) = 0. \end{aligned}$
0902.0431_E00454	74	■ 0.570	$=(\delta X, Y)-(\widetilde{T} X, Y)=((\delta-\widetilde{T}) X, Y) \quad X, Y \in \mathfrak{J}^C$	$=(\delta X, Y)-(\widetilde{T} X, Y)=((\delta-\widetilde{T}) X, Y) \quad X, Y \in \mathfrak{J}^C.$	$=(\delta X, Y)-(\widetilde{T} X, Y)=((\delta-\widetilde{T}) X, Y) \quad X, Y \in \mathfrak{J}^C.$
0902.0431_E01000	166	■ 0.572	$\begin{aligned} & \backslash begin{aligned} \& \backslash lambda(\backslash Phi, P, Q, r, s, t)=\backslash left(\\ & \backslash lambda \backslash Phi \backslash lambda^{-1}, \backslash lambda P, \backslash lambda Q, r, s, t \\ & \backslash right) \backslash \backslash \& \backslash lambda^{\backslash prime}(\backslash Phi, P, Q, r, s, t)=(\backslash Phi, \\ & Q, -P, -r, -t, -s), \backslash end{aligned} \end{aligned}$	$\begin{aligned} & \lambda(\Phi, P, Q, r, s, t) = (\lambda\Phi\lambda^{-1}, \lambda P, \lambda Q, r, s, t) \\ & \lambda'(\Phi, P, Q, r, s, t) = (\Phi, Q, -P, -r, -t, -s), \end{aligned}$	$\begin{aligned} & \lambda(\Phi, P, Q, r, s, t) = (\lambda\Phi\lambda^{-1}, \lambda P, \lambda Q, r, s, t), \\ & \lambda'(\Phi, P, Q, r, s, t) = (\Phi, Q, -P, -r, -t, -s), \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
9902.0431_E08688	114	■ 0.589	$\begin{aligned} & \left[\Phi_1, \Phi_2 \right] P = \Phi_1 \Phi_2 P - \Phi_2 \Phi_1 P \\ & = \cdots \text{ (using the formula of } A \vee B \text{ (Lemma 3.4.1) etc.) } \cdots \\ & = \Phi P \end{aligned}$	$[\Phi_1, \Phi_2] P = \Phi_1 \Phi_2 P - \Phi_2 \Phi_1 P$ $= \cdots \text{ (using the formula of } A \vee B \text{ (Lemma 3.4.1) etc.) } \cdots$ $= \Phi P$	$[\Phi_1, \Phi_2] P = \Phi_1 \Phi_2 P - \Phi_2 \Phi_1 P$ $= \cdots \text{ (using the formula of } A \vee B \text{ (Lemma 3.4.1) etc.) } \cdots$ $= \Phi P.$
9902.0431_E08350	57	■ 0.589	$\begin{aligned} & R(\alpha_1 x_1, \alpha_2 x_2, \alpha_3 x_3) = R(\overline{\alpha_3 x_1 x_2}, \alpha_3 x_3) \\ & = R(\alpha_3 \overline{x_1 x_2}, \alpha_3 x_3) = R(\overline{x_1 x_2}, x_3) = R(x_1 x_2 x_3) \end{aligned}$	$R((\alpha_1 x_1)(\alpha_2 x_2)(\alpha_3 x_3)) = R\left(\overline{\alpha_3(x_1 x_2)}\right) \alpha_3 x_3$ $= (\alpha_3 \overline{x_1 x_2}, \alpha_3 x_3) = (\overline{x_1 x_2}, x_3) = R(x_1 x_2 x_3)$	$R((\alpha_1 x_1)(\alpha_2 x_2)(\alpha_3 x_3)) = R(\overline{(\alpha_3(x_1 x_2))} \alpha_3 x_3)$ $= (\alpha_3 \overline{x_1 x_2}, \alpha_3 x_3) = (\overline{x_1 x_2}, x_3) = R(x_1 x_2 x_3),$
9902.0431_E08870	144	■ 0.590	$\begin{array}{l} 1 \rightarrow \text{Spin}(11) \rightarrow (E_7)^{\kappa, \mu} \rightarrow S^{11} \rightarrow * \\ \quad \downarrow p' \quad \quad \downarrow p \quad \quad \downarrow = \\ 1 \rightarrow SO(11) \rightarrow SO(12) \rightarrow S^{11} \rightarrow * \end{array}$	$\begin{array}{l} 1 \rightarrow \text{Spin}(11) \rightarrow (E_7)^{\kappa, \mu} \rightarrow S^{11} \rightarrow * \\ \quad \downarrow p' \quad \quad \downarrow p \quad \quad \downarrow = \\ 1 \rightarrow SO(11) \rightarrow SO(12) \rightarrow S^{11} \rightarrow * \end{array}$	$\begin{array}{ccccccc} 1 & \longrightarrow & Spin(11) & \longrightarrow & (E_7)^{\kappa, \mu} & \longrightarrow & S^{11} \longrightarrow * \\ & & \downarrow p' & & \downarrow p & & \downarrow = \\ 1 & \longrightarrow & SO(11) & \longrightarrow & SO(12) & \longrightarrow & S^{11} \longrightarrow * \end{array}$
9902.0431_E08324	52	■ 0.591	$\begin{aligned} & \widetilde{A}_1(1) \widetilde{A}_1(1) E_2 = -\frac{1}{2} \widetilde{A}_1(1) F_1(1) = -\frac{1}{2} (E_2 - E_3), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) E_3 = \frac{1}{2} \widetilde{A}_1(1) F_1(1) = \frac{1}{2} (E_2 - E_3), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) F_1(1) = \widetilde{A}_1(1) (E_2 - E_3) = -F_1(1), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) F_2(e_i) = \frac{1}{2} \widetilde{A}_1(1) F_3(e_i) = -\frac{1}{4} F_2(e_i), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) F_3(e_i) = -\frac{1}{2} \widetilde{A}_1(1) F_3(\overline{e_i}) = -\frac{1}{4} F_3(e_i), \\ & \text{the others} = 0. \end{aligned}$	$\begin{aligned} & \widetilde{A}_1(1) \widetilde{A}_1(1) E_2 = -\frac{1}{2} \widetilde{A}_1(1) F_1(1) = -\frac{1}{2} (E_2 - E_3), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) E_3 = \frac{1}{2} \widetilde{A}_1(1) F_1(1) = \frac{1}{2} (E_2 - E_3), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) F_1(1) = \widetilde{A}_1(1) (E_2 - E_3) = -F_1(1), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) F_2(e_i) = \frac{1}{2} \widetilde{A}_1(1) F_3(e_i) = -\frac{1}{4} F_2(e_i), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) F_3(e_i) = -\frac{1}{2} \widetilde{A}_1(1) F_3(\overline{e_i}) = -\frac{1}{4} F_3(e_i), \\ & \text{the others} = 0. \end{aligned}$	$\begin{aligned} & \widetilde{A}_1(1) \widetilde{A}_1(1) E_2 = -\frac{1}{2} \widetilde{A}_1(1) F_1(1) = -\frac{1}{2} (E_2 - E_3), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) E_3 = \frac{1}{2} \widetilde{A}_1(1) F_1(1) = \frac{1}{2} (E_2 - E_3), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) F_1(1) = \widetilde{A}_1(1) (E_2 - E_3) = -F_1(1), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) F_2(e_i) = \frac{1}{2} \widetilde{A}_1(1) F_3(\overline{e_i}) = -\frac{1}{4} F_2(e_i), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) F_3(e_i) = -\frac{1}{2} \widetilde{A}_1(1) F_3(\overline{e_i}) = -\frac{1}{4} F_3(e_i), \\ & \text{the others} = 0. \end{aligned}$

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0902.0431_EQ1083	184	■ 0.593	$\begin{aligned} S_1 \tau S_2 - S_2 \tau S_1 &= \\ &= -k \left(g X_1 g X_2 - g X_2 g X_1 + g(\gamma Y_1) g(\gamma Y_2) - g(\gamma Y_2) g(\gamma Y_1) \right) \\ &\quad - i k \left(g(\gamma Y_1) g X_2 - g(\gamma Y_2) g X_1 - g X_1 g(\gamma Y_2) + g X_2 g(\gamma Y_1) \right) \\ &\quad - \eta_1 g X_2 + \eta_2 g X_1 + \xi_1 g(\gamma Y_2) - \xi_2 g(\gamma Y_1) + \frac{i}{2} (\xi_1 \eta_2 - \eta_2 \xi_1) E \\ &= k \left(-[g X_1, g X_2] - [g(\gamma Y_1), g(\gamma Y_2)] \right) \\ &\quad + i k \left(g(2\gamma X_1 \times Y_2 - 2\gamma X_2 \times Y_1 + \eta_1 X_2 - \eta_2 X_1 - \xi_1 \gamma Y_2 + \xi_2 \gamma Y_1) \right) \\ &\quad + \frac{i}{2} ((X_1, Y_2) - (X_2, Y_1) + \xi_1 \eta_2 - \xi_2 \eta_1) E \quad (\text{Lemma 3.12.1}) \end{aligned}$	$\begin{aligned} S_1 \tau S_2 - S_2 \tau S_1 &= \\ &= -k \left(g X_1 g X_2 - g X_2 g X_1 + g(\gamma Y_1) g(\gamma Y_2) - g(\gamma Y_2) g(\gamma Y_1) \right) \\ &\quad - i k \left(g(\gamma Y_1) g X_2 - g(\gamma Y_2) g X_1 - g X_1 g(\gamma Y_2) + g X_2 g(\gamma Y_1) \right) \\ &\quad - \eta_1 g X_2 + \eta_2 g X_1 + \xi_1 g(\gamma Y_2) - \xi_2 g(\gamma Y_1) + \frac{i}{2} (\xi_1 \eta_2 - \eta_2 \xi_1) E \\ &= k \left(-[g X_1, g X_2] - [g(\gamma Y_1), g(\gamma Y_2)] \right) \\ &\quad + i k \left(g(2\gamma X_1 \times Y_2 - 2\gamma X_2 \times Y_1 + \eta_1 X_2 - \eta_2 X_1 - \xi_1 \gamma Y_2 + \xi_2 \gamma Y_1) \right) \\ &\quad + \frac{i}{2} ((X_1, Y_2) - (X_2, Y_1) + \xi_1 \eta_2 - \xi_2 \eta_1) E \quad (\text{Lemma 3.12.1}) \end{aligned}$	$\begin{aligned} S_1 \tau S_2 - S_2 \tau S_1 &= \\ &= -k(g X_1 g X_2 - g X_2 g X_1 + g(\gamma Y_1) g(\gamma Y_2) - g(\gamma Y_2) g(\gamma Y_1)) \\ &\quad - i k(g(\gamma Y_1) g X_2 - g(\gamma Y_2) g X_1 - g X_1 g(\gamma Y_2) + g X_2 g(\gamma Y_1)) \\ &\quad - \eta_1 g X_2 + \eta_2 g X_1 + \xi_1 g(\gamma Y_2) - \xi_2 g(\gamma Y_1) + \frac{i}{2} (\xi_1 \eta_2 - \eta_2 \xi_1) E \\ &= k(-[g X_1, g X_2] - [g(\gamma Y_1), g(\gamma Y_2)]) \\ &\quad + i k(g(2\gamma X_1 \times Y_2 - 2\gamma X_2 \times Y_1 + \eta_1 X_2 - \eta_2 X_1 - \xi_1 \gamma Y_2 + \xi_2 \gamma Y_1)) \\ &\quad + \frac{i}{2} ((X_1, Y_2) - (X_2, Y_1) + \xi_1 \eta_2 - \xi_2 \eta_1) E \quad (\text{Lemma 3.12.1}) \end{aligned}$
0902.0431_EQ0678	112	■ 0.595	$\begin{aligned} &\left(\begin{array}{c} X \\ \frac{1}{\eta} (X \times X) \\ \frac{1}{\eta^2} \det X \\ \eta \end{array} \right), \quad \left(\begin{array}{c} \frac{1}{\xi} (Y \times Y) \\ Y \\ \xi \\ \frac{1}{\xi^2} \det Y \end{array} \right), \quad \mathbf{i} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{l} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \\ &\left(\begin{array}{c} X \\ \frac{1}{\eta} (X \times X) \\ \frac{1}{\eta^2} \det X \\ \eta \end{array} \right), \quad \left(\begin{array}{c} \frac{1}{\xi} (Y \times Y) \\ Y \\ \xi \\ \frac{1}{\xi^2} \det Y \end{array} \right), \quad \mathbf{i} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{l} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \end{aligned}$	$\left(\begin{array}{c} X \\ \frac{1}{\eta} (X \times X) \\ \frac{1}{\eta^2} \det X \\ \eta \end{array} \right), \quad \left(\begin{array}{c} \frac{1}{\xi} (Y \times Y) \\ Y \\ \xi \\ \frac{1}{\xi^2} \det Y \end{array} \right), \quad \mathbf{i} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{l} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$	$\left(\begin{array}{c} X \\ \frac{1}{\eta} (X \times X) \\ \frac{1}{\eta^2} \det X \\ \eta \end{array} \right), \quad \left(\begin{array}{c} \frac{1}{\xi} (Y \times Y) \\ Y \\ \xi \\ \frac{1}{\xi^2} \det Y \end{array} \right), \quad \mathbf{i} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{l} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$

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0902.0431_E01155	195	0.623	$\mathfrak{e}_{\{8\}}^C = \mathfrak{sl}(9, C) \oplus \mathfrak{q}, \quad \mathfrak{q} = \Lambda^3(C^9) \oplus \Lambda^3(C^9).$	$\mathfrak{e}_8^C = \mathfrak{sl}(9, C) \oplus \mathfrak{q}, \quad \mathfrak{q} = \Lambda^3(C^9) \oplus \Lambda^3(C^9).$	$\mathfrak{e}_8^C = \mathfrak{sl}(9, C) \oplus \mathfrak{q}, \quad \mathfrak{q} = \Lambda^3(C^9) \oplus \Lambda^3(C^9).$
0902.0431_E00319	51	0.625	$B_4(\delta_1, \delta_2) = 9(\delta_1, \delta_2)_4 = 3 \operatorname{tr}(\delta_1 \delta_2), \quad \delta_1, \delta_2 \in \mathfrak{f}_4^C.$	$B_4(\delta_1, \delta_2) = 9(\delta_1, \delta_2)_4 = 3 \operatorname{tr}(\delta_1 \delta_2), \quad \delta_1, \delta_2 \in \mathfrak{f}_4^C.$	$B_4(\delta_1, \delta_2) = 9(\delta_1, \delta_2)_4 = 3 \operatorname{tr}(\delta_1 \delta_2), \quad \delta_1, \delta_2 \in \mathfrak{f}_4^C.$
0902.0431_E00677	112	0.625	$\begin{aligned} \mathfrak{M}^C &= \left\{ P \in \mathfrak{P}^C \mid P \times P = 0, P \neq 0 \right\} \\ &= \left\{ \begin{array}{l} P = (X, Y, \xi, \eta) \in \mathfrak{P}^C \\ P \neq 0 \end{array} \mid \begin{array}{l} X \vee Y = 0, (X, Y) = 3\xi\eta, \\ X \times X = \eta Y, Y \times Y = \xi X \end{array} \right\}. \end{aligned}$	$\mathfrak{M}^C = \left\{ P \in \mathfrak{P}^C \mid P \times P = 0, P \neq 0 \right\} \\ = \left\{ \begin{array}{l} P = (X, Y, \xi, \eta) \in \mathfrak{P}^C \\ P \neq 0 \end{array} \mid \begin{array}{l} X \vee Y = 0, (X, Y) = 3\xi\eta, \\ X \times X = \eta Y, Y \times Y = \xi X \end{array} \right\}.$	$\mathfrak{M}^C = \left\{ P \in \mathfrak{P}^C \mid P \times P = 0, P \neq 0 \right\} \\ = \left\{ \begin{array}{l} P = (X, Y, \xi, \eta) \in \mathfrak{P}^C \\ P \neq 0 \end{array} \mid \begin{array}{l} X \vee Y = 0, (X, Y) = 3\xi\eta, \\ X \times X = \eta Y, Y \times Y = \xi X \end{array} \right\}.$
0902.0431_E00472	78	0.629	$\begin{aligned} &= \phi \left(\frac{1}{2} (B, X) A + \frac{1}{6} (A, B) X - 2B \times (A \times X) \right) - (A \vee B) \phi X \text{ (Lemma 3.4.1)} \\ &= \frac{1}{2} (B, X) \phi A + \frac{1}{6} (A, B) \phi X - 2\phi' B \times (A \times X) - 2B \times (\phi A \times X) \\ &\quad - 2B \times (A \times \phi X) - \frac{1}{2} (B, \phi X) A - \frac{1}{6} (A, B) \phi X + 2B \times (A \times \phi X) \text{ (Lemma 3.4.3)} \\ &= \frac{1}{2} (B, X) \phi A + \frac{1}{6} (\phi A, B) X - 2B \times (\phi A \times X) \\ &\quad + \frac{1}{2} (\phi' B, X) A + \frac{1}{6} (A, \phi' B) X - 2\phi' B \times (A \times X) \\ &= (\phi A \vee B) X + (A \vee \phi' B) X, \quad X \in \mathfrak{J}^C. \end{aligned}$	$\begin{aligned} &= \phi \left(\frac{1}{2} (B, X) A + \frac{1}{6} (A, B) X - 2B \times (A \times X) \right) - (A \vee B) \phi X \text{ (Lemma 3.4.1)} \\ &= \frac{1}{2} (B, X) \phi A + \frac{1}{6} (\phi A, B) X - 2B \times (\phi A \times X) \\ &\quad - 2B \times (A \times \phi X) - \frac{1}{2} (B, \phi X) A - \frac{1}{6} (A, B) \phi X + 2B \times (A \times \phi X) \text{ (Lemma 3.4.3)} \\ &= \frac{1}{2} (B, X) \phi A + \frac{1}{6} (\phi A, B) X - 2B \times (\phi A \times X) \\ &\quad + \frac{1}{2} (\phi' B, X) A + \frac{1}{6} (A, \phi' B) X - 2\phi' B \times (A \times X) \\ &= (\phi A \vee B) X + (A \vee \phi' B) X, \quad X \in \mathfrak{J}^C. \end{aligned}$	$\begin{aligned} &= \phi \left(\frac{1}{2} (B, X) A + \frac{1}{6} (A, B) X - 2B \times (A \times X) \right) - (A \vee B) \phi X \text{ (Lemma 3.4.1)} \\ &= \frac{1}{2} (B, X) \phi A + \frac{1}{6} (A, B) \phi X - 2\phi' B \times (A \times X) - 2B \times (\phi A \times X) \\ &\quad - 2B \times (A \times \phi X) - \frac{1}{2} (B, \phi X) A - \frac{1}{6} (A, B) \phi X + 2B \times (A \times \phi X) \text{ (Lemma 3.4.3)} \\ &= \frac{1}{2} (B, X) \phi A + \frac{1}{6} (\phi A, B) X - 2B \times (\phi A \times X) \\ &\quad + \frac{1}{2} (\phi' B, X) A + \frac{1}{6} (A, \phi' B) X - 2\phi' B \times (A \times X) \\ &= (\phi A \vee B) X + (A \vee \phi' B) X, \quad X \in \mathfrak{J}^C. \end{aligned}$
0902.0431_E00244 (5 Rightarrow4)	40	0.631	$\begin{aligned} &(\det(\alpha X)) \alpha X = (\alpha X \times \alpha X) \times (\alpha X \times \alpha X) \quad (\text{Lemma 2.1.1.}) \\ &= \alpha((X \times X) \times (X \times X)) = (\det X) \alpha X \quad (\text{Lemma 2.1.1.}) \end{aligned}$	$\begin{aligned} &(\det(\alpha X)) \alpha X = (\alpha X \times \alpha X) \times (\alpha X \times \alpha X) \quad (\text{Lemma 2.1.1.}) \\ &= \alpha((X \times X) \times (X \times X)) = (\det X) \alpha X \quad (\text{Lemma 2.1.1.}) \end{aligned}$	$\begin{aligned} &(\det(\alpha X)) \alpha X = (\alpha X \times \alpha X) \times (\alpha X \times \alpha X) \quad (\text{Lemma 2.1.1.}) \\ &= \alpha((X \times X) \times (X \times X)) = (\det X) \alpha X \quad (\text{Lemma 2.1.1.}) \end{aligned}$
0902.0431_E01093	185	0.633	$\begin{aligned} &[\zeta l(D), \zeta(l(S)I)] \\ &= [\Theta(\varphi_* D, 0, 0, 0, 0, 0), \Theta(0, 2\lambda\gamma\chi^{-1}S, 2\chi^{-1}S, 0, 0, 0)] \\ &= \Theta(0, 2(\varphi_* D)\lambda\gamma\chi^{-1}S, 2(\varphi_* D)\chi^{-1}S, 0, 0, 0) \end{aligned}$	$\begin{aligned} &[\zeta l(D), \zeta(l(S)I)] \\ &= [\Theta(\varphi_* D, 0, 0, 0, 0, 0), \Theta(0, 2\lambda\gamma\chi^{-1}S, 2\chi^{-1}S, 0, 0, 0)] \\ &= \Theta(0, 2(\varphi_* D)\lambda\gamma\chi^{-1}S, 2(\varphi_* D)\chi^{-1}S, 0, 0, 0) \end{aligned}$	$\begin{aligned} &[\zeta l(D), \zeta(l(S)I)] \\ &= [\Theta(\varphi_* D, 0, 0, 0, 0, 0), \Theta(0, 2\lambda\gamma\chi^{-1}S, 2\chi^{-1}S, 0, 0, 0)] \\ &= \Theta(0, 2(\varphi_* D)\lambda\gamma\chi^{-1}S, 2(\varphi_* D)\chi^{-1}S, 0, 0, 0). \end{aligned}$

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0902.0431_E00436	72	■ 0.637	$\begin{aligned} \mathfrak{J}\left(3, \mathfrak{C}'\right) &= \left\{X \in M\left(3, \mathfrak{C}'\right) \mid X^* = X\right\}, \\ \mathfrak{J}\left(1,2, \mathfrak{C}\right) &= \left\{X \in M\left(3, \mathfrak{C}\right) \mid I_1 X^* I_1 = X\right\}, \end{aligned}$	$\mathfrak{J}\left(3, \mathfrak{C}'\right)=\left\{X \in M\left(3, \mathfrak{C}'\right) \mid X^*=X\right\},$ $\mathfrak{J}\left(1,2, \mathfrak{C}\right)=\left\{X \in M\left(3, \mathfrak{C}\right) \mid I_1 X^* I_1=X\right\},$	$\mathfrak{J}\left(3, \mathfrak{C}'\right)=\left\{X \in M\left(3, \mathfrak{C}'\right) \mid X^*=X\right\},$ $\mathfrak{J}\left(1,2, \mathfrak{C}\right)=\left\{X \in M\left(3, \mathfrak{C}\right) \mid I_1 X^* I_1=X\right\},$
0902.0431_E01067	181	■ 0.642	$\begin{aligned} \left(E_8\right)^{\widetilde{\lambda} \gamma} &= \left\{\alpha \in E_8 \mid \widetilde{\lambda} \gamma \alpha=\alpha \widetilde{\lambda} \gamma\right\} \\ &= \left\{\alpha \in E_8 \mid \tau \gamma \alpha=\alpha \tau \gamma\right\}=\left(E_8\right)^{\tau \gamma} . \end{aligned}$	$\left(E_8\right)^{\widetilde{\lambda} \gamma}=\left\{\alpha \in E_8 \mid \widetilde{\lambda} \gamma \alpha=\alpha \widetilde{\lambda} \gamma\right\}$ $=\left\{\alpha \in E_8 \mid \tau \gamma \alpha=\alpha \tau \gamma\right\}=\left(E_8\right)^{\tau \gamma} .$	$\left(E_8\right)^{\widetilde{\lambda} \gamma}=\left\{\alpha \in E_8 \mid \widetilde{\lambda} \gamma \alpha=\alpha \widetilde{\lambda} \gamma\right\}$ $=\left\{\alpha \in E_8 \mid \tau \gamma \alpha=\alpha \tau \gamma\right\}=\left(E_8\right)^{\tau \gamma} .$
0902.0431_E00811	135	■ 0.647	$\begin{aligned} &\left(0, \omega' X, 0,0\right)=-\lambda\left(\omega' X, 0,0,0\right)=-\lambda \alpha(X, 0,0,0) \\ &=-\alpha \lambda(X, 0,0,0)=\alpha(0, X, 0,0)=\left(0, \omega'^{-1} X, 0,0\right) \end{aligned}$	$\left(0, \omega' X, 0,0\right)=-\lambda\left(\omega' X, 0,0,0\right)=-\lambda \alpha(X, 0,0,0)$ $=-\alpha \lambda(X, 0,0,0)=\alpha(0, X, 0,0)=\left(0, \omega'^{-1} X, 0,0\right)$	$\left(0, \omega' X, 0,0\right)=-\lambda\left(\omega' X, 0,0,0\right)=-\lambda \alpha(X, 0,0,0)$ $=-\alpha \lambda(X, 0,0,0)=\alpha(0, X, 0,0)=\left(0, \omega'^{-1} X, 0,0\right),$
0902.0431_E00477	79	■ 0.651	$\begin{aligned} &\left(\left[\delta, \delta_1\right]+\left[\widetilde{T}, \widetilde{T}_1\right]\right)+\left(\widetilde{\delta T_1}-\widetilde{\delta_1 T}\right), \delta_2+\widetilde{T}_2\right)_6 \quad(\text { Theorem 3.2.2 }) \\ &=\left([\delta, \delta_1], \delta_2\right)_4+\left([\widetilde{T}, \widetilde{T}_1], \delta_2\right)_4+\left(\delta T_1-\delta_1 T, T_2\right) \\ &=-\left(\delta_1,\left[\delta, \delta_2\right]\right)_4+\left(\delta_2 T, T_1\right)+\left(\delta T_1, T_2\right)-\left(\delta_1 T, T_2\right) \quad(\text { Lemma 2.5.2 }) \\ &=-\left(\delta_1,\left[\delta, \delta_2\right]\right)_4-\left(\delta_1 T, T_2\right)-\left(T_1, \delta T_2\right)+\left(T_1, \delta_2 T\right) \\ &=-\left(\delta_1+\widetilde{T}_1,\left[\delta, \delta_2\right]+\left[\widetilde{T}, \widetilde{T}_2\right]\right)+\left(\widetilde{\delta T_2}-\widetilde{\delta_2 T}\right)_6 \\ &=-\left(\delta_1+\widetilde{T}_1,\left[\delta+\widetilde{T}, \delta_2+\widetilde{T}_2\right]\right)_6 \\ &=-\left(\phi_1,\left[\phi, \phi_2\right]\right)_6 . \end{aligned}$	$\begin{aligned} &=\left(\left([\delta, \delta_1]+\left[\widetilde{T}, \widetilde{T}_1\right]\right)+\left(\widetilde{\delta T_1}-\widetilde{\delta_1 T}\right), \delta_2+\widetilde{T}_2\right)_6 \quad(\text { Theorem 3.2.2 }) \\ &=\left([\delta, \delta_1], \delta_2\right)_4+\left([\widetilde{T}, \widetilde{T}_1], \delta_2\right)_4+\left(\delta T_1-\delta_1 T, T_2\right) \\ &=-\left(\delta_1,\left[\delta, \delta_2\right]\right)_4+\left(\delta_2 T, T_1\right)+\left(\delta T_1, T_2\right)-\left(\delta_1 T, T_2\right) \quad(\text { Lemma 2.5.2 }) \\ &=-\left(\delta_1,\left[\delta, \delta_2\right]\right)_4-\left(\delta_1 T, T_2\right)-\left(T_1, \delta T_2\right)+\left(T_1, \delta_2 T\right) \\ &=-\left(\delta_1+\widetilde{T}_1,\left[\delta, \delta_2\right]+\left[\widetilde{T}, \widetilde{T}_2\right]\right)+\left(\widetilde{\delta T_2}-\widetilde{\delta_2 T}\right)_6 \\ &=-\left(\delta_1+\widetilde{T}_1,\left[\delta+\widetilde{T}, \delta_2+\widetilde{T}_2\right]\right)_6 \\ &=-\left(\phi_1,\left[\phi, \phi_2\right]\right)_6 . \end{aligned}$	$=\left(\left([\delta, \delta_1]+\left[\widetilde{T}, \widetilde{T}_1\right]\right)+\left(\widetilde{\delta T_1}-\widetilde{\delta_1 T}\right), \delta_2+\widetilde{T}_2\right)_6 \quad(\text { Theorem 3.2.2 })$ $=\left([\delta, \delta_1], \delta_2\right)_4+\left([\widetilde{T}, \widetilde{T}_1], \delta_2\right)_4+\left(\delta T_1-\delta_1 T, T_2\right)$ $=-\left(\delta_1,\left[\delta, \delta_2\right]\right)_4+\left(\delta_2 T, T_1\right)+\left(\delta T_1, T_2\right)-\left(\delta_1 T, T_2\right) \quad(\text { Lemma 2.5.2 })$ $=-\left(\delta_1,\left[\delta, \delta_2\right]\right)_4-\left(\delta_1 T, T_2\right)-\left(T_1, \delta T_2\right)+\left(T_1, \delta_2 T\right)$ $=-\left(\delta_1+\widetilde{T}_1,\left[\delta, \delta_2\right]+\left[\widetilde{T}, \widetilde{T}_2\right]\right)+\left(\widetilde{\delta T_2}-\widetilde{\delta_2 T}\right)_6$ $=-\left(\delta_1+\widetilde{T}_1,\left[\delta+\widetilde{T}, \delta_2+\widetilde{T}_2\right]\right)_6$ $=-\left(\phi_1,\left[\phi, \phi_2\right]\right)_6 .$

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0902.0431_E00522	86	0.651	$\begin{aligned} & H_{\alpha_1} = \frac{1}{24} (H_0 - H_1), \quad H_{\alpha_2} = \frac{1}{24} (H_1 - H_2), \quad H_{\alpha_3} = \frac{1}{24} (H_2 - H_3), \\ & H_{\alpha_4} = \frac{1}{24} (H_3 + (E_2 - E_3)^\sim), \\ & H_{\alpha_5} = \frac{1}{48} ((-H_0 - H_1 - H_2 + H_3) + 2(E_3 - E_1)^\sim), \\ & H_{\alpha_6} = \frac{1}{48} ((H_0 + H_1 + H_2 + H_3) + 2(E_1 - E_2)^\sim). \end{aligned}$	$\begin{aligned} H_{\alpha_1} &= \frac{1}{24} (H_0 - H_1), \quad H_{\alpha_2} = \frac{1}{24} (H_1 - H_2), \quad H_{\alpha_3} = \frac{1}{24} (H_2 - H_3), \\ H_{\alpha_4} &= \frac{1}{24} (H_3 + (E_2 - E_3)^\sim), \\ H_{\alpha_5} &= \frac{1}{48} ((-H_0 - H_1 - H_2 + H_3) + 2(E_3 - E_1)^\sim), \\ H_{\alpha_6} &= \frac{1}{48} ((H_0 + H_1 + H_2 + H_3) + 2(E_1 - E_2)^\sim). \end{aligned}$	$\begin{aligned} H_{\alpha_1} &= \frac{1}{24} (H_0 - H_1), \quad H_{\alpha_2} = \frac{1}{24} (H_1 - H_2), \quad H_{\alpha_3} = \frac{1}{24} (H_2 - H_3), \\ H_{\alpha_4} &= \frac{1}{24} (H_3 + (E_2 - E_3)^\sim), \\ H_{\alpha_5} &= \frac{1}{48} ((-H_0 - H_1 - H_2 + H_3) + 2(E_3 - E_1)^\sim), \\ H_{\alpha_6} &= \frac{1}{48} ((H_0 + H_1 + H_2 + H_3) + 2(E_1 - E_2)^\sim). \end{aligned}$
0902.0431_E00024	9	0.655	$\begin{aligned} & \left[F_{i0}, F_{0k} \right] x = (F_{i0} F_{0k} - F_{0k} F_{i0}) x = -\frac{1}{4} e_i (e_k x) + \frac{1}{4} e_k (e_i x) \\ & = \frac{1}{2} e_k (e_i x) = F_{ik} x, \end{aligned}$	$\begin{aligned} [F_{i0}, F_{0k}] x &= (F_{i0} F_{0k} - F_{0k} F_{i0}) x = -\frac{1}{4} e_i (e_k x) + \frac{1}{4} e_k (e_i x) \\ &= \frac{1}{2} e_k (e_i x) = F_{ik} x, \end{aligned}$	$\begin{aligned} [F_{i0}, F_{0k}] x &= (F_{i0} F_{0k} - F_{0k} F_{i0}) x = -\frac{1}{4} e_i (e_k x) + \frac{1}{4} e_k (e_i x) \\ &= \frac{1}{2} e_k (e_i x) = F_{ik} x, \end{aligned}$
0902.0431_E00021	9	0.656	$\begin{aligned} & \left[\kappa D_1, \kappa D_2 \right] x = (\kappa D_1) (\kappa D_2 x) - (\kappa D_2) (\kappa D_1 x) = \overline{D_1 (\kappa D_2 x)} - \overline{D_2 (\kappa D_1 x)} \\ & = \overline{D_1 D_2 x} - \overline{D_2 D_1 x} = \kappa (D_1 D_2 - D_2 D_1) x = \kappa [D_1, D_2] x, \quad x \in \mathfrak{C} \end{aligned}$	$\begin{aligned} [\kappa D_1, \kappa D_2] x &= (\kappa D_1) (\kappa D_2 x) - (\kappa D_2) (\kappa D_1 x) = \overline{D_1 (\kappa D_2 x)} - \overline{D_2 (\kappa D_1 x)} \\ &= \overline{D_1 D_2 x} - \overline{D_2 D_1 x} = \kappa (D_1 D_2 - D_2 D_1) x = \kappa [D_1, D_2] x, \quad x \in \mathfrak{C}. \end{aligned}$	$\begin{aligned} [\kappa D_1, \kappa D_2] x &= (\kappa D_1) (\kappa D_2 x) - (\kappa D_2) (\kappa D_1 x) = \overline{D_1 (\kappa D_2 x)} - \overline{D_2 (\kappa D_1 x)} \\ &= \overline{D_1 D_2 x} - \overline{D_2 D_1 x} = \kappa (D_1 D_2 - D_2 D_1) x = \kappa [D_1, D_2] x, \quad x \in \mathfrak{C}. \end{aligned}$
0902.0431_E01122	190	0.658	$\begin{aligned} & f(\Phi(\phi, A, B, \nu), (X, Y, \xi, \eta), (Z, W, \zeta, \omega), r, s, t)) \\ & = \left(\begin{array}{ccc} \frac{2}{3} \nu & -\frac{1}{2} \xi & \frac{1}{2} \zeta \\ \frac{1}{2} \omega & -\frac{1}{3} \nu - r & t \\ \frac{1}{2} \eta & s & -\frac{1}{3} \nu + r \end{array} \right), \phi, \left(\begin{array}{c} -2A \\ Z \\ X \end{array} \right), \left(\begin{array}{c} -2B \\ Y \\ -W \end{array} \right), \end{aligned}$	$\begin{aligned} & f(\Phi(\phi, A, B, \nu), (X, Y, \xi, \eta), (Z, W, \zeta, \omega), r, s, t)) \\ & = \left(\begin{array}{ccc} \frac{2}{3} \nu & -\frac{1}{2} \xi & \frac{1}{2} \zeta \\ \frac{1}{2} \omega & -\frac{1}{3} \nu - r & t \\ \frac{1}{2} \eta & s & -\frac{1}{3} \nu + r \end{array} \right), \phi, \left(\begin{array}{c} -2A \\ Z \\ X \end{array} \right), \left(\begin{array}{c} -2B \\ Y \\ -W \end{array} \right), \end{aligned}$	$\begin{aligned} & f(\Phi(\phi, A, B, \nu), (X, Y, \xi, \eta), (Z, W, \zeta, \omega), r, s, t)) \\ & = \left(\begin{array}{ccc} \frac{2}{3} \nu & -\frac{1}{2} \xi & \frac{1}{2} \zeta \\ \frac{1}{2} \omega & -\frac{1}{3} \nu - r & t \\ \frac{1}{2} \eta & s & -\frac{1}{3} \nu + r \end{array} \right), \phi, \left(\begin{array}{c} -2A \\ Z \\ X \end{array} \right), \left(\begin{array}{c} -2B \\ Y \\ -W \end{array} \right), \end{aligned}$

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0902.0431_E01015	169	■ 0.662	$\begin{gathered} \pm\left(\lambda_k-\lambda_l\right), \quad \pm\left(\lambda_k+\lambda_l\right), \quad 0 \leq k<l \leq 3, \\ \pm \lambda_k \pm \frac{1}{2}\left(\mu_2-\mu_3\right), \quad 0 \leq k \leq 3, \\ \pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) \pm \frac{1}{2}\left(\mu_3-\mu_1\right), \\ \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) \pm \frac{1}{2}\left(\mu_3-\mu_1\right), \\ \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) \pm \frac{1}{2}\left(\mu_3-\mu_1\right), \\ \pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) \pm \frac{1}{2}\left(\mu_3-\mu_1\right), \\ \pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) \pm \frac{1}{2}\left(\mu_1-\mu_2\right), \\ \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) \pm \frac{1}{2}\left(\mu_1-\mu_2\right), \\ \pm \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) \pm \frac{1}{2}\left(\mu_1-\mu_2\right), \\ \pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) \pm \frac{1}{2}\left(\mu_1-\mu_2\right), \\ \pm\left(\mu_k+\frac{2}{3} \nu\right), \quad 0 \leq k<l \leq 3, \\ \pm \lambda_k \pm\left(\frac{1}{2} \mu_1-\frac{2}{3} \nu\right), \quad 0 \leq k \leq 3, \\ \pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_2-\frac{2}{3} \nu\right), \\ \pm \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_2-\frac{2}{3} \nu\right), \\ \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) \pm\left(\frac{1}{2} \mu_2-\frac{2}{3} \nu\right), \\ \pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) \pm\left(\frac{1}{2} \mu_2-\frac{2}{3} \nu\right), \\ \pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_3-\frac{2}{3} \nu\right), \\ \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_3-\frac{2}{3} \nu\right), \\ \pm \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) \pm\left(\frac{1}{2} \mu_3-\frac{2}{3} \nu\right), \\ \pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) \pm\left(\frac{1}{2} \mu_3-\frac{2}{3} \nu\right), \\ \pm\left(\mu_j-\frac{1}{3} \nu+r\right), \quad 1 \leq j \leq 3, \\ \pm \lambda_k \pm\left(\frac{1}{2} \mu_1+\frac{1}{3} \nu\right) \pm r, \quad 0 \leq k \leq 3, \\ \pm \frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_2+\frac{1}{3} \nu\right) \pm r, \\ \pm \frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_2+\frac{1}{3} \nu\right) \pm r, \end{gathered}$	$\begin{aligned} &\pm\left(\lambda_k-\lambda_l\right), \quad \pm\left(\lambda_k+\lambda_l\right), \quad 0 \leq k<l \leq 3, \\ &\pm \lambda_k \pm \frac{1}{2}\left(\mu_2-\mu_3\right), \quad 0 \leq k \leq 3, \\ &\pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) \pm \frac{1}{2}\left(\mu_3-\mu_1\right), \\ &\pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) \pm \frac{1}{2}\left(\mu_3-\mu_1\right), \\ &\pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) \pm \frac{1}{2}\left(\mu_3-\mu_1\right), \\ &\pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) \pm \frac{1}{2}\left(\mu_3-\mu_1\right), \\ &\pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) \pm \frac{1}{2}\left(\mu_1-\mu_2\right), \\ &\pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) \pm \frac{1}{2}\left(\mu_1-\mu_2\right), \\ &\pm \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) \pm \frac{1}{2}\left(\mu_1-\mu_2\right), \\ &\pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) \pm \frac{1}{2}\left(\mu_1-\mu_2\right), \\ &\pm\left(\mu_k+\frac{2}{3} \nu\right), \quad 0 \leq k<l \leq 3, \\ &\pm \lambda_k \pm\left(\frac{1}{2} \mu_1-\frac{2}{3} \nu\right), \quad 0 \leq k \leq 3, \\ &\pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_2-\frac{2}{3} \nu\right), \\ &\pm \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_2-\frac{2}{3} \nu\right), \\ &\pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) \pm\left(\frac{1}{2} \mu_2-\frac{2}{3} \nu\right), \\ &\pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) \pm\left(\frac{1}{2} \mu_2-\frac{2}{3} \nu\right), \\ &\pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_3-\frac{2}{3} \nu\right), \\ &\pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_3-\frac{2}{3} \nu\right), \\ &\pm \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) \pm\left(\frac{1}{2} \mu_3-\frac{2}{3} \nu\right), \\ &\pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) \pm\left(\frac{1}{2} \mu_3-\frac{2}{3} \nu\right), \\ &\pm\left(\mu_j-\frac{1}{3} \nu+r\right), \quad 1 \leq j \leq 3, \\ &\pm \lambda_k \pm\left(\frac{1}{2} \mu_1+\frac{1}{3} \nu\right) \pm r, \quad 0 \leq k \leq 3, \\ &\pm \frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_2+\frac{1}{3} \nu\right) \pm r, \\ &\pm \frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_2+\frac{1}{3} \nu\right) \pm r, \end{aligned}$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00131	25	0.662	$\begin{aligned} (\alpha x)(\alpha y) &= \left(q m \bar{q} + (p a \bar{q}) e_4 \right) \left(q n \bar{q} + (p b \bar{q}) e_4 \right) \\ &= ((q m \bar{q})(q n \bar{q}) - (\overline{p b \bar{q}})(p a \bar{q})) + ((p a \bar{q})(\overline{q n \bar{q}}) + (p b \bar{q})(q m \bar{q})) e_4 \\ &= q(m n - \bar{b} a) \bar{q} + (p(a \bar{n} + b m) \bar{q}) e_4 \\ &= \varphi(p, q) ((m + a e_4)(n + b e_4)) = \alpha(x y). \end{aligned}$	$\begin{aligned} (\alpha x)(\alpha y) &= (q m \bar{q} + (p a \bar{q}) e_4) (q n \bar{q} + (p b \bar{q}) e_4) \\ &= ((q m \bar{q})(q n \bar{q}) - (\overline{p b \bar{q}})(p a \bar{q})) + ((p a \bar{q})(\overline{q n \bar{q}}) + (p b \bar{q})(q m \bar{q})) e_4 \\ &= q(m n - \bar{b} a) \bar{q} + (p(a \bar{n} + b m) \bar{q}) e_4 \\ &= \varphi(p, q) ((m + a e_4)(n + b e_4)) = \alpha(x y) \end{aligned}$	$\begin{aligned} (\alpha x)(\alpha y) &= (q m \bar{q} + (p a \bar{q}) e_4) (q n \bar{q} + (p b \bar{q}) e_4) \\ &= ((q m \bar{q})(q n \bar{q}) - (\overline{p b \bar{q}})(p a \bar{q})) + ((p a \bar{q})(\overline{q n \bar{q}}) + (p b \bar{q})(q m \bar{q})) e_4 \\ &= q(m n - \bar{b} a) \bar{q} + (p(a \bar{n} + b m) \bar{q}) e_4 \\ &= \varphi(p, q) ((m + a e_4)(n + b e_4)) = \alpha(x y). \end{aligned}$
0902.0431_E00734	121	0.666	$\left(\left(\Phi_1, \Phi_1 \right), \left(\Phi_1, \Phi_2 \right) \right)_7 + \left(\left(\Phi_1, \Phi_2 \right), \left(\Phi_1, \Phi_1 \right) \right)_7 = 0, \quad \Phi_i \in \mathfrak{e}_7^C,$	$([\Phi, \Phi_1], \Phi_2)_7 + (\Phi_1, [\Phi, \Phi_2])_7 = 0, \quad \Phi, \Phi_i \in \mathfrak{e}_7^C.$	$([\Phi, \Phi_1], \Phi_2)_7 + (\Phi_1, [\Phi, \Phi_2])_7 = 0, \quad \Phi, \Phi_i \in \mathfrak{e}_7^C.$
0902.0431_E00873	145	0.666	$\begin{aligned} &\operatorname{Spin}(12) / \{1, \sigma\} \cong SO(12) \\ &\operatorname{Spin}(12) / \{1, -1\} \cong \operatorname{Spin}(12) / \{1, -\sigma\} \cong \operatorname{Ss}(12) \end{aligned}$	$\operatorname{Spin}(12) / \{1, \sigma\} \cong SO(12)$ $\operatorname{Spin}(12) / \{1, -1\} \cong \operatorname{Spin}(12) / \{1, -\sigma\} \cong \operatorname{Ss}(12)$	$\operatorname{Spin}(12) / \{1, \sigma\} \cong SO(12),$ $\operatorname{Spin}(12) / \{1, -1\} \cong \operatorname{Spin}(12) / \{1, -\sigma\} \cong \operatorname{Ss}(12).$
0902.0431_E00649	107	0.672	$\varphi_*: \mathfrak{su}(3) \oplus \mathfrak{su}(3) \rightarrow \mathfrak{e}_{6,C}.$	$\varphi_*: \mathfrak{su}(3) \oplus \mathfrak{su}(3) \rightarrow \mathfrak{e}_{6,C}.$	$\varphi_*: \mathfrak{su}(3) \oplus \mathfrak{su}(3) \rightarrow \mathfrak{e}_{6,C}.$
0902.0431_E00774	130	0.675	$\begin{aligned} &\langle \alpha \dot{X}, \dot{1} \rangle = \langle \alpha \dot{X}, \alpha \dot{1} \rangle = \langle \dot{X}, \dot{1} \rangle = 0 \\ &\langle \alpha \dot{X}, 1 \rangle = \langle \alpha \dot{X}, \alpha 1 \rangle = \langle \dot{X}, 1 \rangle = 0. \end{aligned}$	$\begin{aligned} \langle \alpha \dot{X}, \dot{1} \rangle &= \langle \alpha \dot{X}, \alpha \dot{1} \rangle = \langle \dot{X}, \dot{1} \rangle = 0 \\ \langle \alpha \dot{X}, 1 \rangle &= \langle \alpha \dot{X}, \alpha 1 \rangle = \langle \dot{X}, 1 \rangle = 0. \end{aligned}$	$\begin{aligned} \langle \alpha \dot{X}, \dot{1} \rangle &= \langle \alpha \dot{X}, \alpha \dot{1} \rangle = \langle \dot{X}, \dot{1} \rangle = 0, \\ \langle \alpha \dot{X}, 1 \rangle &= \langle \alpha \dot{X}, \alpha 1 \rangle = \langle \dot{X}, 1 \rangle = 0. \end{aligned}$
0902.0431_E00142	26	0.683	$\overline{\alpha x} = \alpha \bar{x}, \quad x \in \mathfrak{C}^C.$	$\overline{\alpha x} = \alpha \bar{x}, \quad x \in \mathfrak{C}^C.$	$\overline{\alpha x} = \alpha \bar{x}, \quad x \in \mathfrak{C}^C.$
0902.0431_E01206	203	0.686	$\begin{aligned} &\left(\operatorname{tr} (C_1 \tau^t C_2) + \operatorname{tr} (D_1 \tau^t D_2) + (x_1, \tau x_2) (a_1, \tau a_2) + (y_1, \tau y_2) (b_1, \tau b_2) \right. \\ &\quad \left. + (z_1, \tau z_2) (c_1, \tau c_2) + (w_1, \tau w_2) (d_1, \tau d_2) \right). \end{aligned}$	$\langle R_1, R_2 \rangle = 60 \left(\operatorname{tr} (C_1 \tau^t C_2) + \operatorname{tr} (D_1 \tau^t D_2) + (x_1, \tau x_2) (a_1, \tau a_2) + (y_1, \tau y_2) (b_1, \tau b_2) \right. \\ \left. + (z_1, \tau z_2) (c_1, \tau c_2) + (w_1, \tau w_2) (d_1, \tau d_2) \right).$	$\langle R_1, R_2 \rangle = 60 \left(\operatorname{tr} (C_1 \tau^t C_2) + \operatorname{tr} (D_1 \tau^t D_2) + (x_1, \tau x_2) (a_1, \tau a_2) + (y_1, \tau y_2) (b_1, \tau b_2) \right. \\ \left. + (z_1, \tau z_2) (c_1, \tau c_2) + (w_1, \tau w_2) (d_1, \tau d_2) \right).$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E09765	128	0.686	<pre>\begin{aligned} & H_{\alpha_1}=\frac{1}{36} \Phi\left(H_0-H_1, 0,0,0\right), \quad \& H_{\alpha_2}=\frac{1}{36} \Phi\left(H_1-H_2, 0,0,0\right), \quad \& H_{\alpha_3}=\frac{1}{36} \Phi\left(H_2-H_3, 0,0,0\right), \\ & H_{\alpha_4}=\frac{1}{72} \Phi\left(\left(-H_0-H_1-H_2+H_3\right)+2\left(E_3-E_1\right)^{\sim}, 0,0,0\right), \quad \& H_{\alpha_5}=\frac{1}{72} \Phi\left(\left(H_0+H_1+H_2+H_3\right)+2\left(E_1-E_2\right)^{\sim}, 0,0,0\right), \\ & H_{\alpha_6}=\frac{1}{54} \Phi\left(\left(-E_1+2 E_2-E_3\right)^{\sim}, 0,0,\frac{3}{2}\right), \quad \& H_{\alpha_7}=\frac{1}{54} \Phi\left(\left(E_1+E_2-2 E_3\right)^{\sim}, 0,0,-\frac{3}{2}\right). \\ & H_{\alpha_7}=\frac{1}{54} \Phi\left(\left(E_1+E_2-2 E_3\right)^{\sim}, 0,0,-\frac{3}{2}\right). \end{aligned}</pre>	$\begin{aligned} H_{\alpha_1} &= \frac{1}{36} \Phi\left(H_0-H_1, 0,0,0\right), \\ H_{\alpha_2} &= \frac{1}{36} \Phi\left(H_1-H_2, 0,0,0\right), \\ H_{\alpha_3} &= \frac{1}{36} \Phi\left(H_2-H_3, 0,0,0\right), \\ H_{\alpha_4} &= \frac{1}{72} \Phi\left(\left(-H_0-H_1-H_2+H_3\right)+2\left(E_3-E_1\right)^{\sim}, 0,0,0\right), \\ H_{\alpha_5} &= \frac{1}{72} \Phi\left(\left(H_0+H_1+H_2+H_3\right)+2\left(E_1-E_2\right)^{\sim}, 0,0,0\right), \\ H_{\alpha_6} &= \frac{1}{54} \Phi\left(\left(-E_1+2 E_2-E_3\right)^{\sim}, 0,0,\frac{3}{2}\right), \\ H_{\alpha_7} &= \frac{1}{54} \Phi\left(\left(E_1+E_2-2 E_3\right)^{\sim}, 0,0,-\frac{3}{2}\right). \end{aligned}$	$\begin{aligned} H_{\alpha_1} &= \frac{1}{36} \Phi\left(H_0-H_1, 0,0,0\right), \\ H_{\alpha_2} &= \frac{1}{36} \Phi\left(H_1-H_2, 0,0,0\right), \\ H_{\alpha_3} &= \frac{1}{36} \Phi\left(H_2-H_3, 0,0,0\right), \\ H_{\alpha_4} &= \frac{1}{72} \Phi\left(\left(-H_0-H_1-H_2+H_3\right)+2\left(E_3-E_1\right)^{\sim}, 0,0,0\right), \\ H_{\alpha_5} &= \frac{1}{72} \Phi\left(\left(H_0+H_1+H_2+H_3\right)+2\left(E_1-E_2\right)^{\sim}, 0,0,0\right), \\ H_{\alpha_6} &= \frac{1}{54} \Phi\left(\left(-E_1+2 E_2-E_3\right)^{\sim}, 0,0,\frac{3}{2}\right), \\ H_{\alpha_7} &= \frac{1}{54} \Phi\left(\left(E_1+E_2-2 E_3\right)^{\sim}, 0,0,-\frac{3}{2}\right). \end{aligned}$
0902.0431_E09616	102	0.688	<pre>\begin{gathered} g(\gamma(X \times Y))=g(\gamma((M+\boldsymbol{a}) \times(N+\boldsymbol{b}))) \\ =g((M-\boldsymbol{a}) \times(N-\boldsymbol{b})) \\ =g\left(\left(M \times N-\frac{1}{2}\left(\boldsymbol{a}^* \boldsymbol{b}+\boldsymbol{b}^* \boldsymbol{a}\right)\right)+\frac{1}{2}(\boldsymbol{a} N+\boldsymbol{b} M)\right) \\ =\left(\frac{i}{2}(\boldsymbol{a} N+\boldsymbol{b} M)^* \quad M \times N-\frac{1}{2}\left(\boldsymbol{a}^* \boldsymbol{b}+\boldsymbol{b}^* \boldsymbol{a}\right)-\frac{1}{2}(\operatorname{tr}(M \times N)-(\boldsymbol{a}, \boldsymbol{b})) E\right) \\ =\dots \\ =g(M+\boldsymbol{a}) \circ g(N+\boldsymbol{b})-\left(\frac{1}{4}(M, N)-\frac{1}{2}(\boldsymbol{a}, \boldsymbol{b})\right) E \\ =g(M+\boldsymbol{a}) \circ g(N+\boldsymbol{b})-\frac{1}{4}(\gamma(M+\boldsymbol{a}), (N+\boldsymbol{b})) E \\ =g X \circ g Y-\frac{1}{4}(\gamma X, Y) E . \end{gathered}</pre>	$\begin{aligned} g(\gamma(X \times Y)) &= g(\gamma((M+\boldsymbol{a}) \times(N+\boldsymbol{b}))) \\ &= g((M-\boldsymbol{a}) \times(N-\boldsymbol{b})) \\ &= g\left(\left(M \times N-\frac{1}{2}\left(\boldsymbol{a}^* \boldsymbol{b}+\boldsymbol{b}^* \boldsymbol{a}\right)\right)+\frac{1}{2}(\boldsymbol{a} N+\boldsymbol{b} M)\right) \\ &= \left(\frac{i}{2}(\boldsymbol{a} N+\boldsymbol{b} M)^* \quad M \times N-\frac{1}{2}\left(\boldsymbol{a}^* \boldsymbol{b}+\boldsymbol{b}^* \boldsymbol{a}\right)-\frac{1}{2}(\operatorname{tr}(M \times N)-(\boldsymbol{a}, \boldsymbol{b})) E\right) \\ &= \dots \\ &= g(M+\boldsymbol{a}) \circ g(N+\boldsymbol{b})-\left(\frac{1}{4}(M, N)-\frac{1}{2}(\boldsymbol{a}, \boldsymbol{b})\right) E \\ &= g(M+\boldsymbol{a}) \circ g(N+\boldsymbol{b})-\frac{1}{4}(\gamma(M+\boldsymbol{a}), (N+\boldsymbol{b})) E \\ &= g X \circ g Y-\frac{1}{4}(\gamma X, Y) E . \end{aligned}$	$\begin{aligned} g(\gamma(X \times Y)) &= g(\gamma((M+\boldsymbol{a}) \times(N+\boldsymbol{b}))) \\ &= g((M-\boldsymbol{a}) \times(N-\boldsymbol{b})) \\ &= g\left(\left(M \times N-\frac{1}{2}\left(\boldsymbol{a}^* \boldsymbol{b}+\boldsymbol{b}^* \boldsymbol{a}\right)\right)+\frac{1}{2}(\boldsymbol{a} N+\boldsymbol{b} M)\right) \\ &= \left(\frac{1}{2} \operatorname{tr}(M \times N)-\frac{1}{2}(\boldsymbol{a}, \boldsymbol{b}) \quad \frac{i}{2}(\boldsymbol{a} N+\boldsymbol{b} M) \right. \\ &\quad \left. \frac{i}{2}(\boldsymbol{a} N+\boldsymbol{b} M)^* \quad M \times N-\frac{1}{2}\left(\boldsymbol{a}^* \boldsymbol{b}+\boldsymbol{b}^* \boldsymbol{a}\right)-\frac{1}{2}(\operatorname{tr}(M \times N)-(\boldsymbol{a}, \boldsymbol{b})) E\right) \\ &= \dots \\ &= g(M+\boldsymbol{a}) \circ g(N+\boldsymbol{b})-\left(\frac{1}{4}(M, N)-\frac{1}{2}(\boldsymbol{a}, \boldsymbol{b})\right) E \\ &= g(M+\boldsymbol{a}) \circ g(N+\boldsymbol{b})-\frac{1}{4}(\gamma(M+\boldsymbol{a}), (N+\boldsymbol{b})) E \\ &= g X \circ g Y-\frac{1}{4}(\gamma X, Y) E . \end{aligned}$

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0902.0431_E00383	63	0.688	$l=\left(E_{\{1\}},E_{\{1\}}\right)=\left(\alpha E_{\{1\}},\alpha E_{\{1\}}\right)=1+\xi_2^2+\xi_3^2+2\left x_1\right ^2,$	$1=(E_1,E_1)=(\alpha E_1,\alpha E_1)=1+\xi_2^2+\xi_3^2+2 x_1 ^2,$	$1=(E_1,E_1)=(\alpha E_1,\alpha E_1)=1+\xi_2^2+\xi_3^2+2 x_1 ^2,$
0902.0431_E01181	199	0.697	$\begin{aligned} & (x\wedge a,y\wedge b)=(x,y)(a,b)-\left(a_1,y\right)\left(x\wedge a_2,b\right)+\left(a_2,y\right)\left(x\wedge a_1,b\right)\\ & (*b\wedge x,*a\wedge y)=(a,y\wedge (*b\wedge x))=(a_1,y)\left(x\wedge a_2,b\right)-\left(a_2,y\right)\left(x\wedge a_1,b\right), \end{aligned}$	$\begin{aligned} (x\wedge a,y\wedge b) &= (x,y)(a,b)-(a_1,y)(x\wedge a_2,b)+(a_2,y)(x\wedge a_1,b) \\ (*b\wedge x,*a\wedge y) &= (a,y\wedge (*b\wedge x))=(a_1,y)(x\wedge a_2,b)-(a_2,y)(x\wedge a_1,b), \end{aligned}$	$\begin{aligned} (x\wedge a,y\wedge b) &= (x,y)(a,b)-(a_1,y)(x\wedge a_2,b)+(a_2,y)(x\wedge a_1,b), \\ (*b\wedge x,*a\wedge y) &= (a,y\wedge (*b\wedge x))=(a_1,y)(x\wedge a_2,b)-(a_2,y)(x\wedge a_1,b), \end{aligned}$
0902.0431_E00086	18	0.697	$\mathfrak{a}\supset [\mathfrak{a},\mathfrak{S}]\supset [\mathfrak{su}(3),\mathfrak{S}]=\mathfrak{S}\text{ (Lemma 1.6.1).}$	$\mathfrak{a}\supset [\mathfrak{a},\mathfrak{S}]\supset [\mathfrak{su}(3),\mathfrak{S}]=\mathfrak{S}\text{ (Lemma 1.6.1).}$	$\mathfrak{a}\supset [\mathfrak{a},\mathfrak{S}]\supset [\mathfrak{su}(3),\mathfrak{S}]=\mathfrak{S}\text{ (Lemma 1.6.1).}$
0902.0431_E00646	107	0.705	$\begin{aligned} & \varphi((A,B),1)X=h(A,B)Xh(A,B)^*, \\ & \varphi((A,B),\epsilon)X=h(A,B)\bar{X}h(A,B)^*, \end{aligned} \quad X\in(\mathfrak{J}_C)^C.$	$\begin{aligned} \varphi((A,B),1)X &= h(A,B)Xh(A,B)^*, \\ \varphi((A,B),\epsilon)X &= h(A,B)\bar{X}h(A,B)^*, \end{aligned} \quad X\in(\mathfrak{J}_C)^C.$	$\begin{aligned} \varphi((A,B),1)X &= h(A,B)Xh(A,B)^*, \\ \varphi((A,B),\epsilon)X &= h(A,B)\bar{X}h(A,B)^*, \end{aligned} \quad X\in(\mathfrak{J}_C)^C.$
0902.0431_E00976	162	0.710	$\begin{aligned} & \left[R_1,[R_2,R_3]\right]+\left[R_2,[R_3,R_1]\right]+\left[R_3,[R_1,R_2]\right] \\ & =\cdots(\text{using }[\Phi,P\times Q]=\Phi P\times Q+P\times\Phi Q\text{ (Proposition 4.3.2),} \\ & \quad (P\times R)Q-(Q\times R)P+\frac{1}{8}\{Q,R\}P-\frac{1}{8}\{P,R\}-\frac{1}{4}\{P,Q\}R=0 \\ & \quad \text{(Lemma 4.1.1.(3), }\{\Phi P,Q\}+\{P,\Phi Q\}=0\text{ (Proposition 4.2.2.(2)) etc.)}\cdots \\ & =0 \end{aligned}$	$\begin{aligned} & \left[R_1,[R_2,R_3]\right]+\left[R_2,[R_3,R_1]\right]+\left[R_3,[R_1,R_2]\right] \\ & =\cdots(\text{ using }[\Phi,P\times Q]=\Phi P\times Q+P\times\Phi Q\text{(Proposition 4.3.2)} \\ & \quad (P\times R)Q-(Q\times R)P+\frac{1}{8}\{Q,R\}P-\frac{1}{8}\{P,R\}-\frac{1}{4}\{P,Q\}R=0 \\ & \quad \text{(Lemma 4.1.1.(3), }\{\Phi P,Q\}+\{P,\Phi Q\}=0\text{ (Proposition 4.2.2.(2)) etc.)}\cdots \\ & =0 \end{aligned}$	$\begin{aligned} & \left[R_1,[R_2,R_3]\right]+\left[R_2,[R_3,R_1]\right]+\left[R_3,[R_1,R_2]\right] \\ & =\cdots(\text{using }[\Phi,P\times Q]=\Phi P\times Q+P\times\Phi Q\text{ (Proposition 4.3.2),} \\ & \quad (P\times R)Q-(Q\times R)P+\frac{1}{8}\{Q,R\}P-\frac{1}{8}\{P,R\}-\frac{1}{4}\{P,Q\}R=0 \\ & \quad \text{(Lemma 4.1.1.(3), }\{\Phi P,Q\}+\{P,\Phi Q\}=0\text{ (Proposition 4.2.2.(2)) etc.)}\cdots \\ & =0. \end{aligned}$
0902.0431_E00575	96	0.718	$\begin{aligned} & \{ \}^{\{t\}}\left(k_{\{J\}}(M)\right) \& =\{ \}^{\{t\}}((k(M))J)=-J^{\{t\}}(k(M))=-J^{\{t\}}\left(k\left(M_1\right)+i k\left(M_2\right)\right) \\ & \& =-\left(\tau^t\left(k\left(M_1\right)\right)+i \tau^t\left(k\left(M_2\right)\right)\right) J=-\left(k\left(M_1^*\right)+i k\left(M_2^*\right)\right) J \\ & \& =-\left(k\left(M_1\right)+i k\left(M_2\right)\right) J=-\left(k(M)\right) J=-k_J(M) \end{aligned}$	$\begin{aligned} {}^t(k_J(M)) &= {}^t((k(M))J)=-J^t(k(M))=-J^t(k(M_1)+ik(M_2)) \\ &= -(\tau^t(k(M_1))+i\tau^t(k(M_2)))J=-(k(M_1^*)+ik(M_2^*))J \\ &= -(k(M_1)+ik(M_2))J=-(k(M))J=-k_J(M) \end{aligned}$	$\begin{aligned} {}^t(k_J(M)) &= {}^t((k(M))J)=-J^t(k(M))=-J^t(k(M_1)+ik(M_2)) \\ &= -(\tau^t(k(M_1))+i\tau^t(k(M_2)))J=-(k(M_1^*)+ik(M_2^*))J \\ &= -(k(M_1)+ik(M_2))J=-(k(M))J=-k_J(M). \end{aligned}$
0902.0431_E00704	116	0.718	$2(\psi(X\times X)+\frac{1}{3}(2\nu+\mu)(X\times X))\times(X\times X)\\=(\det X)\left(\nu X+\phi X+\frac{1}{3}(\nu+2\mu)X\right)$	$2(\psi(X\times X)+\frac{1}{3}(2\nu+\mu)(X\times X))\times(X\times X)\\=(\det X)\left(\nu X+\phi X+\frac{1}{3}(\nu+2\mu)X\right)$	$2\Big(\psi(X\times X)+\frac{1}{3}(2\nu+\mu)(X\times X)\Big)\times(X\times X)\\=(\det X)\Big(\nu X+\phi X+\frac{1}{3}(\nu+2\mu)X\Big),$
0902.0431_E00082	17	0.719	$[\mathfrak{su}(3),\mathfrak{S}]=\mathfrak{S}.$	$[\mathfrak{su}(3),\mathfrak{S}]=\mathfrak{S}.$	$[\mathfrak{su}(3),\mathfrak{S}]=\mathfrak{S}.$

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0902.0431_E00095	19	■ 0.719	$\mathfrak{h}=\left\{\left(\begin{array}{ccc}\lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3\end{array}\right)\middle \lambda_i\in\boldsymbol{C},\lambda_1+\lambda_2+\lambda_3=0\right\}$	$\mathfrak{h}=\left\{\left(\begin{array}{ccc}\lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3\end{array}\right)\middle \lambda_i\in\boldsymbol{C},\lambda_1+\lambda_2+\lambda_3=0\right\}$	$\mathfrak{h}=\left\{\left(\begin{array}{ccc}\lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3\end{array}\right)\middle \lambda_i\in\boldsymbol{C},\lambda_1+\lambda_2+\lambda_3=0\right\}$
0902.0431_E00964	160	■ 0.719	$E_{7\{ \}}^{\{ \} C} \simeq E_7 \times \boldsymbol{R}^{133} \text{ .}$	$E_7^C \simeq E_7 \times \boldsymbol{R}^{133}.$	$E_7^C \simeq E_7 \times \boldsymbol{R}^{133}.$
0902.0431_E00197	33	■ 0.724	$\begin{aligned} & \widetilde{D}_4=\left(\alpha_1,\alpha_2,\alpha_3\right)\in SO(8)\times SO(8)\times SO(8)\mid \left(\alpha_1x\right)\left(\alpha_2y\right)=\overline{\alpha_3\left(\overline{xy}\right)},x,y\in \mathfrak{C}\} \\ & \widetilde{D}_4 \text{ is a compact group. } \end{aligned}$	$\begin{aligned} \widetilde{D}_4 &= \left\{ (\alpha_1, \alpha_2, \alpha_3) \in SO(8) \times SO(8) \times SO(8) \mid (\alpha_1 x) (\alpha_2 y) = \overline{\alpha_3(\overline{xy})}, x, y \in \mathfrak{C} \right\} \\ \widetilde{D}_4 &\text{ is a compact group.} \end{aligned}$	$\begin{aligned} \widetilde{D}_4 &= \{ (\alpha_1, \alpha_2, \alpha_3) \in SO(8) \times SO(8) \times SO(8) \mid (\alpha_1 x) (\alpha_2 y) = \overline{\alpha_3(\overline{xy})}, x, y \in \mathfrak{C} \}. \\ \widetilde{D}_4 &\text{ is a compact group.} \end{aligned}$
0902.0431_E00532	89	■ 0.724	$\begin{aligned} & \left \eta_1(t)\right ^2+\left \eta_2(t)\right ^2+\left \eta_3(t)\right ^2 \\ &= \left \xi_1\right ^2+\left \frac{\xi_2-\xi_3}{2}+\frac{\xi_2+\xi_3}{2} \cos t- q \sin t+i \nu \sin t\right ^2 \\ &+ \left -\frac{\xi_2-\xi_3}{2}+\frac{\xi_2+\xi_3}{2} \cos t- q \sin t+i \nu \sin t\right ^2 \\ &= \left \xi_1\right ^2+2\left(\frac{\xi_2-\xi_3}{2}\right)^2+2\left(\frac{\xi_2+\xi_3}{2} \cos t- q \sin t\right)^2+2 \nu^2 \sin ^2 t \\ &= \left \xi_1\right ^2+2\left(\frac{\xi_2-\xi_3}{2}\right)^2+2\left(\left(\frac{\xi_2+\xi_3}{2}\right)^2+ q ^2\right) \sin ^2\left(t+t_0\right)+2 \nu^2 \sin ^2 t \\ &\leq \left \xi_1\right ^2+\left \xi_2\right ^2+\left \xi_3\right ^2+2 q ^2+2 \nu^2 \cos ^2 t_0 \quad\left(\text { for some } t_0 \in \boldsymbol{R}\right) \end{aligned}$	$\begin{aligned} & \left \eta_1(t)\right ^2+\left \eta_2(t)\right ^2+\left \eta_3(t)\right ^2 \\ &= \left \xi_1\right ^2+\left \frac{\xi_2-\xi_3}{2}+\frac{\xi_2+\xi_3}{2} \cos t- q \sin t+i \nu \sin t\right ^2 \\ &+ \left -\frac{\xi_2-\xi_3}{2}+\frac{\xi_2+\xi_3}{2} \cos t- q \sin t+i \nu \sin t\right ^2 \\ &= \left \xi_1\right ^2+2\left(\frac{\xi_2-\xi_3}{2}\right)^2+2\left(\frac{\xi_2+\xi_3}{2} \cos t- q \sin t\right)^2+2 \nu^2 \sin ^2 t \\ &= \left \xi_1\right ^2+2\left(\frac{\xi_2-\xi_3}{2}\right)^2+2\left(\left(\frac{\xi_2+\xi_3}{2}\right)^2+ q ^2\right) \sin ^2\left(t+t_0\right)+2 \nu^2 \sin ^2 t \\ &\leq \left \xi_1\right ^2+\left \xi_2\right ^2+\left \xi_3\right ^2+2 q ^2+2 \nu^2 \cos ^2 t_0 \quad\left(\text { for some } t_0 \in \boldsymbol{R}\right) \end{aligned}$	$\begin{aligned} & \left \eta_1(t)\right ^2+\left \eta_2(t)\right ^2+\left \eta_3(t)\right ^2 \\ &= \left \xi_1\right ^2+\left \frac{\xi_2-\xi_3}{2}+\frac{\xi_2+\xi_3}{2} \cos t- q \sin t+i \nu \sin t\right ^2 \\ &+ \left -\frac{\xi_2-\xi_3}{2}+\frac{\xi_2+\xi_3}{2} \cos t- q \sin t+i \nu \sin t\right ^2 \\ &= \left \xi_1\right ^2+2\left(\frac{\xi_2-\xi_3}{2}\right)^2+2\left(\frac{\xi_2+\xi_3}{2} \cos t- q \sin t\right)^2+2 \nu^2 \sin ^2 t \\ &= \left \xi_1\right ^2+2\left(\frac{\xi_2-\xi_3}{2}\right)^2+2\left(\left(\frac{\xi_2+\xi_3}{2}\right)^2+ q ^2\right) \sin ^2\left(t+t_0\right)+2 \nu^2 \sin ^2 t \\ &\leq \left \xi_1\right ^2+\left \xi_2\right ^2+\left \xi_3\right ^2+2 q ^2+2 \nu^2 \cos ^2 t_0 \quad\left(\text { for some } t_0 \in \boldsymbol{R}\right) \end{aligned}$

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0902.0431_E01178	199	0.725	$\begin{aligned} &= \sum_{j=1}^5 \sum_{i=1}^6 (-1)^i (\mathbf{a}_1 \wedge \cdots \mathbf{a}_{i-1} \wedge \mathbf{a}_{i+1} \wedge \cdots \wedge \mathbf{a}_6, \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_5) \mathbf{a}_{ij} \mathbf{e}_j \\ &= - \sum_{j=1}^5 \det \begin{pmatrix} a_{1j} & a_{11} & \cdots & a_{15} \\ a_{2j} & a_{21} & \cdots & a_{25} \\ & & \cdots & \\ a_{6j} & a_{61} & \cdots & a_{65} \end{pmatrix} \mathbf{e}_j = 0. \end{aligned}$	$\begin{aligned} &= \sum_{j=1}^5 \sum_{i=1}^6 (-1)^i (\mathbf{a}_1 \wedge \cdots \mathbf{a}_{i-1} \wedge \mathbf{a}_{i+1} \wedge \cdots \wedge \mathbf{a}_6, \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_5) \mathbf{a}_{ij} \mathbf{e}_j \\ &= - \sum_{j=1}^5 \det \begin{pmatrix} a_{1j} & a_{11} & \cdots & a_{15} \\ a_{2j} & a_{21} & \cdots & a_{25} \\ & & \cdots & \\ a_{6j} & a_{61} & \cdots & a_{65} \end{pmatrix} \mathbf{e}_j = 0. \end{aligned}$	
0902.0431_E01148	194	0.732	$\begin{aligned} &((\mathbf{u} \times \mathbf{v})\mathbf{x}, \mathbf{y}) = (\mathbf{u} \wedge \mathbf{v}, \mathbf{x} \wedge \mathbf{y}) \\ &= -(\mathbf{x} \wedge \mathbf{u}, \mathbf{y} \wedge \mathbf{v}) + \frac{2}{3}(\mathbf{u}, \mathbf{v})(\mathbf{x}, \mathbf{y}) \\ &= (\mathbf{x} \wedge \mathbf{u}_2 \wedge \mathbf{u}_3, \mathbf{v})(\mathbf{u}_1, \mathbf{y}) - (\mathbf{x} \wedge \mathbf{u}_1 \wedge \mathbf{u}_3, \mathbf{v})(\mathbf{u}_2, \mathbf{y}) \\ &\quad + (\mathbf{x} \wedge \mathbf{u}_1 \wedge \mathbf{u}_2, \mathbf{v})(\mathbf{u}_3, \mathbf{y}) - \frac{1}{3}(\mathbf{u}, \mathbf{v})(\mathbf{x}, \mathbf{y}). \end{aligned}$	$\begin{aligned} &((\mathbf{u} \times \mathbf{v})\mathbf{x}, \mathbf{y}) = (\mathbf{u} \wedge \mathbf{v}, \mathbf{x} \wedge \mathbf{y}) + \frac{2}{3}(\mathbf{u}, \mathbf{v})(\mathbf{x}, \mathbf{y}) \\ &= -(\mathbf{x} \wedge \mathbf{u}, \mathbf{y} \wedge \mathbf{v}) + \frac{2}{3}(\mathbf{u}, \mathbf{v})(\mathbf{x}, \mathbf{y}) \\ &= (\mathbf{x} \wedge \mathbf{u}_2 \wedge \mathbf{u}_3, \mathbf{v})(\mathbf{u}_1, \mathbf{y}) - (\mathbf{x} \wedge \mathbf{u}_1 \wedge \mathbf{u}_3, \mathbf{v})(\mathbf{u}_2, \mathbf{y}) \\ &\quad + (\mathbf{x} \wedge \mathbf{u}_1 \wedge \mathbf{u}_2, \mathbf{v})(\mathbf{u}_3, \mathbf{y}) - \frac{1}{3}(\mathbf{u}, \mathbf{v})(\mathbf{x}, \mathbf{y}). \end{aligned}$	$\begin{aligned} ((\mathbf{u} \times \mathbf{v})\mathbf{x}, \mathbf{y}) &= (\mathbf{u} \wedge \mathbf{v}, \mathbf{x} \wedge \mathbf{y}) + \frac{2}{3}(\mathbf{u}, \mathbf{v})(\mathbf{x}, \mathbf{y}) \\ &= -(\mathbf{x} \wedge \mathbf{u}, \mathbf{y} \wedge \mathbf{v}) + \frac{2}{3}(\mathbf{u}, \mathbf{v})(\mathbf{x}, \mathbf{y}) \\ &= (\mathbf{x} \wedge \mathbf{u}_2 \wedge \mathbf{u}_3, \mathbf{v})(\mathbf{u}_1, \mathbf{y}) - (\mathbf{x} \wedge \mathbf{u}_1 \wedge \mathbf{u}_3, \mathbf{v})(\mathbf{u}_2, \mathbf{y}) \\ &\quad + (\mathbf{x} \wedge \mathbf{u}_1 \wedge \mathbf{u}_2, \mathbf{v})(\mathbf{u}_3, \mathbf{y}) - \frac{1}{3}(\mathbf{u}, \mathbf{v})(\mathbf{x}, \mathbf{y}). \end{aligned}$
0902.0431_E01081	183	0.733	$\begin{aligned} &S_1 \tau S_2 = \chi P_1 \tau \chi P_2 = \chi(X_1, Y_1, \xi_1, \eta_1) \tau \chi(X_2, Y_2, \xi_2, \eta_2) \\ &= \left(k \left(g X_1 - \frac{\xi_1}{2} E \right) + i k \left(g(\gamma Y_1) - \frac{\eta_1}{2} E \right) \right) J \tau \left(\left(k \left(g X_2 - \frac{\xi_2}{2} E \right) + i k \left(g(\gamma Y_2) - \frac{\eta_2}{2} E \right) \right) J \right) \end{aligned}$	$\begin{aligned} &S_1 \tau S_2 = \chi P_1 \tau \chi P_2 = \chi(X_1, Y_1, \xi_1, \eta_1) \tau \chi(X_2, Y_2, \xi_2, \eta_2) \\ &= \left(k \left(g X_1 - \frac{\xi_1}{2} E \right) + i k \left(g(\gamma Y_1) - \frac{\eta_1}{2} E \right) \right) J \tau \left(\left(k \left(g X_2 - \frac{\xi_2}{2} E \right) + i k \left(g(\gamma Y_2) - \frac{\eta_2}{2} E \right) \right) J \right) \end{aligned}$	$\begin{aligned} S_1 \tau S_2 &= \chi P_1 \tau \chi P_2 = \chi(X_1, Y_1, \xi_1, \eta_1) \tau \chi(X_2, Y_2, \xi_2, \eta_2) \\ &= \left(k \left(g X_1 - \frac{\xi_1}{2} E \right) + i k \left(g(\gamma Y_1) - \frac{\eta_1}{2} E \right) \right) J \tau \left(\left(k \left(g X_2 - \frac{\xi_2}{2} E \right) + i k \left(g(\gamma Y_2) - \frac{\eta_2}{2} E \right) \right) J \right) \end{aligned}$

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0902.0431_E01096	186	■ 0.736	$\begin{aligned} & \left\{ \left(\zeta \left(S_1 \right) I, \zeta \left(l \left(S_2 \right) I \right) \right) \right\} \setminus \left\{ \left(\Theta \left(0, 2 \lambda \gamma \chi^{-1} S_1, 2 \chi^{-1} S_1, 0, 0, 0 \right), \Theta \left(0, 2 \lambda \gamma \chi^{-1} S_2, 2 \chi^{-1} S_2, 0, 0, 0 \right) \right) \right\} \\ & = \Theta \left(2 \lambda \gamma \chi^{-1} S_1 \times 2 \chi^{-1} S_2 - 2 \lambda \gamma \chi^{-1} S_2 \times 2 \chi^{-1} S_1, 0, 0 \right) \\ & = \Theta \left(2 \lambda \gamma \chi^{-1} S_1, 2 \lambda \gamma \chi^{-1} S_2, 2 \chi^{-1} S_1, 2 \chi^{-1} S_2, 0, 0 \right), \end{aligned}$	$\begin{aligned} & [\zeta l(S_1)I, \zeta(l(S_2)I)] \\ & = [\Theta(0, 2\lambda\gamma\chi^{-1}S_1, 2\chi^{-1}S_1, 0, 0, 0), \Theta(0, 2\lambda\gamma\chi^{-1}S_2, 2\chi^{-1}S_2, 0, 0, 0)] \\ & = \Theta(2\lambda\gamma\chi^{-1}S_1 \times 2\chi^{-1}S_2 - 2\lambda\gamma\chi^{-1}S_2 \times 2\chi^{-1}S_1, 0, 0) \\ & = \Theta(2\lambda\gamma\chi^{-1}S_1, 2\lambda\gamma\chi^{-1}S_2, 2\chi^{-1}S_1, 2\chi^{-1}S_2, 0, 0), \end{aligned}$	$\begin{aligned} & [\zeta l(S_1)I, \zeta(l(S_2)I)] \\ & = [\Theta(0, 2\lambda\gamma\chi^{-1}S_1, 2\chi^{-1}S_1, 0, 0, 0), \Theta(0, 2\lambda\gamma\chi^{-1}S_2, 2\chi^{-1}S_2, 0, 0, 0)] \\ & = \Theta(2\lambda\gamma\chi^{-1}S_1 \times 2\chi^{-1}S_2 - 2\lambda\gamma\chi^{-1}S_2 \times 2\chi^{-1}S_1, 0, 0, \\ & \quad \frac{1}{8}(-\{2\lambda\gamma\chi^{-1}S_1, 2\chi^{-1}S_2\} + \{2\lambda\gamma\chi^{-1}S_2, 2\chi^{-1}S_1\}), \\ & \quad \frac{1}{4}\{2\lambda\gamma\chi^{-1}S_1, 2\lambda\gamma\chi^{-1}S_2\}, -\frac{1}{4}\{2\chi^{-1}S_1, 2\chi^{-1}S_2\}), \end{aligned}$
0902.0431_E00158	29	■ 0.746	$\left(\alpha_1 x \right) \left(\alpha_2 y \right) = \alpha_3 (xy), \quad x, y \in \mathfrak{C}.$	$(\alpha_1 x)(\alpha_2 y) = \alpha_3(xy), \quad x, y \in \mathfrak{C}.$	$(\alpha_1 x)(\alpha_2 y) = \alpha_3(xy), \quad x, y \in \mathfrak{C}.$
0902.0431_E00621	103	■ 0.747	$\begin{aligned} & \langle \varphi(A)X, \varphi(A)Y \rangle = \langle g\varphi(A)X, g\varphi(A)Y \rangle \\ & = \langle A(gX)A^*, A(gY)A^* \rangle = \langle gX, gY \rangle = \langle X, Y \rangle. \end{aligned}$	$\begin{aligned} & \langle \varphi(A)X, \varphi(A)Y \rangle = \langle g\varphi(A)X, g\varphi(A)Y \rangle \\ & = \langle A(gX)A^*, A(gY)A^* \rangle = \langle gX, gY \rangle = \langle X, Y \rangle. \end{aligned}$	$\begin{aligned} & \langle \varphi(A)X, \varphi(A)Y \rangle = \langle g\varphi(A)X, g\varphi(A)Y \rangle \\ & = \langle A(gX)A^*, A(gY)A^* \rangle = \langle gX, gY \rangle = \langle X, Y \rangle. \end{aligned}$
0902.0431_E01102	187	■ 0.751	$\begin{aligned} & \left(\left(\mathfrak{e}_{8(8)} \right)_{\tilde{\lambda}\gamma} \right)_{\tau\tilde{\lambda}} = \left(\left(\mathfrak{e}_{8(8)} \right)_{\tau\gamma} \right)_{\tilde{\lambda}\gamma} = \left(\mathfrak{e}_{8(8)} \right)_{\tilde{\lambda}\gamma} = \left(\mathfrak{e}_{8(8)} \right)^{\tilde{\lambda}\gamma}, \\ & \left(\left(\mathfrak{e}_{8(8)} \right)_{-\tilde{\lambda}\gamma} \right)_{\tau\tilde{\lambda}} = \left(\left(\mathfrak{e}_{8(8)} \right)_{\tau\gamma} \right)_{-\tilde{\lambda}\gamma} = \left(\mathfrak{e}_{8(8)} \right)_{-\tilde{\lambda}\gamma}, \end{aligned}$	$\begin{aligned} & \left(\left(\mathfrak{e}_{8(8)} \right)_{\tilde{\lambda}\gamma} \right)_{\tau\tilde{\lambda}} = \left(\left(\mathfrak{e}_{8(8)} \right)_{\tau\gamma} \right)_{\tilde{\lambda}\gamma} = \left(\mathfrak{e}_{8(8)} \right)_{\tilde{\lambda}\gamma} = \left(\mathfrak{e}_{8(8)} \right)^{\tilde{\lambda}\gamma}, \\ & \left(\left(\mathfrak{e}_{8(8)} \right)_{-\tilde{\lambda}\gamma} \right)_{\tau\tilde{\lambda}} = \left(\left(\mathfrak{e}_{8(8)} \right)_{\tau\gamma} \right)_{-\tilde{\lambda}\gamma} = \left(\mathfrak{e}_{8(8)} \right)_{-\tilde{\lambda}\gamma}, \end{aligned}$	$\begin{aligned} & \left(\left(\mathfrak{e}_{8(8)} \right)_{\tilde{\lambda}\gamma} \right)_{\tau\tilde{\lambda}} = \left(\left(\mathfrak{e}_{8(8)} \right)_{\tau\gamma} \right)_{\tilde{\lambda}\gamma} = \left(\mathfrak{e}_{8(8)} \right)_{\tilde{\lambda}\gamma} = \left(\mathfrak{e}_{8(8)} \right)^{\tilde{\lambda}\gamma}, \\ & \left(\left(\mathfrak{e}_{8(8)} \right)_{-\tilde{\lambda}\gamma} \right)_{\tau\tilde{\lambda}} = \left(\left(\mathfrak{e}_{8(8)} \right)_{\tau\gamma} \right)_{-\tilde{\lambda}\gamma} = \left(\mathfrak{e}_{8(8)} \right)_{-\tilde{\lambda}\gamma}, \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00250	41	0.754	$\begin{aligned} & \& (4) \quad \operatorname{tr}(\alpha X) = (\alpha X, E, E) = (\alpha X, \alpha E, \alpha E) = (X, E, E) \text{ (Lemma 2.2.2.(2))} \\ & \operatorname{tr}(X). \text{ Hence } \frac{1}{2}(\operatorname{tr}(X) \operatorname{tr}(Y) - (X, Y)) = (X, Y, E) \text{ (Lemma 2.1.1)} = (\alpha X, \alpha Y, \alpha E) \\ & = (\alpha X, \alpha Y, E) = \frac{1}{2}(\operatorname{tr}(\alpha X) \operatorname{tr}(\alpha Y) - (\alpha X, \alpha Y)) = \frac{1}{2}(\operatorname{tr}(X) \operatorname{tr}(Y) - (\alpha X, \alpha Y)). \text{ There-} \end{aligned}$	$(4) \Rightarrow (2) \operatorname{tr}(\alpha X) = (\alpha X, E, E) = (\alpha X, \alpha E, \alpha E) = (X, E, E) \text{ (Lemma 2.2.2.(2))}$ $= \operatorname{tr}(X). \text{ Hence } \frac{1}{2}(\operatorname{tr}(X) \operatorname{tr}(Y) - (X, Y)) = (X, Y, E) \text{ (Lemma 2.1.1)} = (\alpha X, \alpha Y, \alpha E)$ $= (\alpha X, \alpha Y, E) = \frac{1}{2}(\operatorname{tr}(\alpha X) \operatorname{tr}(\alpha Y) - (\alpha X, \alpha Y)) = \frac{1}{2}(\operatorname{tr}(X) \operatorname{tr}(Y) - (\alpha X, \alpha Y)). \text{ There-}$	$(4) \Rightarrow (2) \operatorname{tr}(\alpha X) = (\alpha X, E, E) = (\alpha X, \alpha E, \alpha E) = (X, E, E) \text{ (Lemma 2.2.2.(2))}$ $= \operatorname{tr}(X). \text{ Hence } \frac{1}{2}(\operatorname{tr}(X) \operatorname{tr}(Y) - (X, Y)) = (X, Y, E) \text{ (Lemma 2.1.1)} = (\alpha X, \alpha Y, \alpha E)$ $= (\alpha X, \alpha Y, E) = \frac{1}{2}(\operatorname{tr}(\alpha X) \operatorname{tr}(\alpha Y) - (\alpha X, \alpha Y)) = \frac{1}{2}(\operatorname{tr}(X) \operatorname{tr}(Y) - (\alpha X, \alpha Y)). \text{ There-}$
0902.0431_E00911	151	0.754	$E_{7,C} = \left\{ \alpha \in \operatorname{Iso}_C \left((\mathfrak{P}_C)^C \right) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q, \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle \right\}.$	$E_{7,C} = \left\{ \alpha \in \operatorname{Iso}_C \left((\mathfrak{P}_C)^C \right) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q, \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle \right\}.$	$E_{7,C} = \left\{ \alpha \in \operatorname{Iso}_C \left((\mathfrak{P}_C)^C \right) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q, \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle \right\}.$
0902.0431_E01107	187	0.754	$k^2 B_8(R, R') = B_8(\alpha R, \alpha R') = B_8(R, R'), \quad R, R' \in (\mathfrak{e}_8^C)_{-\widetilde{\lambda}_\gamma},$	$k^2 B_8(R, R') = B_8(\alpha R, \alpha R') = B_8(R, R'), \quad R, R' \in (\mathfrak{e}_8^C)_{-\widetilde{\lambda}_\gamma},$	$k^2 B_8(R, R') = B_8(\alpha R, \alpha R') = B_8(R, R'), \quad R, R' \in (\mathfrak{e}_8^C)_{-\widetilde{\lambda}_\gamma},$
0902.0431_E00448	73	0.755	$\phi = \delta + \widetilde{T}, \quad \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C.$	$\phi = \delta + \widetilde{T}, \quad \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C.$	$\phi = \delta + \widetilde{T}, \quad \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C.$
0902.0431_E00318	51	0.755	$\begin{aligned} & ([\delta, [\widetilde{A}, \widetilde{B}]], [\widetilde{C}, \widetilde{D}])_4 + ([\widetilde{A}, \widetilde{B}], [\delta, [\widetilde{C}, \widetilde{D}]])_4 \\ & = ([\widetilde{\delta A}, \widetilde{B}] + [\widetilde{A}, \widetilde{\delta B}], [\widetilde{C}, \widetilde{D}])_4 + ([\widetilde{A}, \widetilde{B}], [\widetilde{\delta C}, \widetilde{D}] + [\widetilde{C}, \widetilde{\delta D}])_4 \text{ (Lemma 2.4.5.(1))} \\ & = ([\widetilde{\delta A}, \widetilde{B}]C, D) + ([\widetilde{A}, \widetilde{\delta B}]C, D) + ([\widetilde{A}, \widetilde{B}]\delta C, D) + ([\widetilde{A}, \widetilde{B}]C, \delta D) \\ & = (\delta A \circ (B \circ C), D) - (B \circ (\delta A \circ C), D) + (A \circ (\delta B \circ C), D) - (\delta B \circ (A \circ C), D) \\ & \quad + (A \circ (B \circ \delta C), D) - (B \circ (A \circ \delta C), D) + (A \circ (B \circ C), \delta D) - (B \circ (A \circ C), \delta D), \end{aligned}$	$\begin{aligned} & ([\delta, [\widetilde{A}, \widetilde{B}]], [\widetilde{C}, \widetilde{D}])_4 + ([\widetilde{A}, \widetilde{B}], [\delta, [\widetilde{C}, \widetilde{D}]])_4 \\ & = ([\widetilde{\delta A}, \widetilde{B}] + [\widetilde{A}, \widetilde{\delta B}], [\widetilde{C}, \widetilde{D}])_4 + ([\widetilde{A}, \widetilde{B}], [\widetilde{\delta C}, \widetilde{D}] + [\widetilde{C}, \widetilde{\delta D}])_4 \text{ (Lemma 2.4.5.(1))} \\ & = ([\widetilde{\delta A}, \widetilde{B}]C, D) + ([\widetilde{A}, \widetilde{\delta B}]C, D) + ([\widetilde{A}, \widetilde{B}]\delta C, D) + ([\widetilde{A}, \widetilde{B}]C, \delta D) \\ & = (\delta A \circ (B \circ C), D) - (B \circ (\delta A \circ C), D) + (A \circ (\delta B \circ C), D) - (\delta B \circ (A \circ C), D) \\ & \quad + (A \circ (B \circ \delta C), D) - (B \circ (A \circ \delta C), D) + (A \circ (B \circ C), \delta D) - (B \circ (A \circ C), \delta D) \end{aligned}$	$\begin{aligned} & ([\delta, [\widetilde{A}, \widetilde{B}]], [\widetilde{C}, \widetilde{D}])_4 + ([\widetilde{A}, \widetilde{B}], [\delta, [\widetilde{C}, \widetilde{D}]])_4 \\ & = ([\widetilde{\delta A}, \widetilde{B}] + [\widetilde{A}, \widetilde{\delta B}], [\widetilde{C}, \widetilde{D}])_4 + ([\widetilde{A}, \widetilde{B}], [\widetilde{\delta C}, \widetilde{D}] + [\widetilde{C}, \widetilde{\delta D}])_4 \text{ (Lemma 2.4.5.(1))} \\ & = ([\widetilde{\delta A}, \widetilde{B}]C, D) + ([\widetilde{A}, \widetilde{\delta B}]C, D) + ([\widetilde{A}, \widetilde{B}]\delta C, D) + ([\widetilde{A}, \widetilde{B}]C, \delta D) \\ & = (\delta A \circ (B \circ C), D) - (B \circ (\delta A \circ C), D) + (A \circ (\delta B \circ C), D) - (\delta B \circ (A \circ C), D) \\ & \quad + (A \circ (B \circ \delta C), D) - (B \circ (A \circ \delta C), D) + (A \circ (B \circ C), \delta D) - (B \circ (A \circ C), \delta D), \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00117	22	■ 0.762	$\begin{aligned} & \left(\alpha e_4\right)\left(\alpha e_7\right)=a_4 a_7=a_4\left(a_3 a_4\right)=-a_4\left(a_4 a_3\right)=a_3=\alpha e_3=\alpha\left(e_4 e_7\right), \\ & \left(\alpha e_1\right)\left(\alpha e_6\right)=a_1 a_6=a_1\left(a_4 a_2\right)=-a_4\left(a_1 a_2\right)=-a_4 a_3=a_3 a_4=a_7 \\ & \quad =\alpha e_7=\alpha\left(e_1 e_6\right), \\ & \left(\alpha e_3\right)\left(\alpha e_6\right)=a_3 a_6=\left(a_1 a_2\right)\left(a_4 a_2\right)=-\left(a_2 a_1\right)\left(a_4 a_2\right)=-a_2\left(a_1 a_4\right) a_2 \\ & \quad =-a_2 a_5 a_2=a_2 a_2 a_5=-a_5=-\alpha e_5=\alpha\left(e_3 e_6\right), \text { etc. } \end{aligned}$	$\begin{aligned} & \left(\alpha e_4\right)\left(\alpha e_7\right)=a_4 a_7=a_4\left(a_3 a_4\right)=-a_4\left(a_4 a_3\right)=a_3=\alpha e_3=\alpha\left(e_4 e_7\right), \\ & \left(\alpha e_1\right)\left(\alpha e_6\right)=a_1 a_6=a_1\left(a_4 a_2\right)=-a_4\left(a_1 a_2\right)=-a_4 a_3=a_3 a_4=a_7 \\ & \quad =\alpha e_7=\alpha\left(e_1 e_6\right), \\ & \left(\alpha e_3\right)\left(\alpha e_6\right)=a_3 a_6=\left(a_1 a_2\right)\left(a_4 a_2\right)=-\left(a_2 a_1\right)\left(a_4 a_2\right)=-a_2\left(a_1 a_4\right) a_2 \\ & \quad =-a_2 a_5 a_2=a_2 a_2 a_5=-a_5=-\alpha e_5=\alpha\left(e_3 e_6\right), \text { etc. } \end{aligned}$	$\begin{aligned} & \left(\alpha e_4\right)\left(\alpha e_7\right)=a_4 a_7=a_4\left(a_3 a_4\right)=-a_4\left(a_4 a_3\right)=a_3=\alpha e_3=\alpha\left(e_4 e_7\right), \\ & \left(\alpha e_1\right)\left(\alpha e_6\right)=a_1 a_6=a_1\left(a_4 a_2\right)=-a_4\left(a_1 a_2\right)=-a_4 a_3=a_3 a_4=a_7 \\ & \quad =\alpha e_7=\alpha\left(e_1 e_6\right), \\ & \left(\alpha e_3\right)\left(\alpha e_6\right)=a_3 a_6=\left(a_1 a_2\right)\left(a_4 a_2\right)=-\left(a_2 a_1\right)\left(a_4 a_2\right)=-a_2\left(a_1 a_4\right) a_2 \\ & \quad =-a_2 a_5 a_2=a_2 a_2 a_5=-a_5=-\alpha e_5=\alpha\left(e_3 e_6\right), \text { etc. } \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00368	60	■ 0.766	$\begin{aligned} & \left\{ \begin{array}{l} \eta_1 = \xi_1 \\ \eta_2 = \frac{\xi_2 + \xi_3}{2} + \frac{\xi_2 - \xi_3}{2} \cos 2 a + \frac{(a, x_1)}{ a } \sin 2 a \\ \eta_3 = \frac{\xi_2 + \xi_3}{2} - \frac{\xi_2 - \xi_3}{2} \cos 2 a - \frac{(a, x_1)}{ a } \sin 2 a , \end{array} \right. \\ & \left\{ \begin{array}{l} y_1 = x_1 - \frac{(\xi_2 - \xi_3)a}{2 a } \sin 2 a - \frac{2(a, x_1)a}{ a ^2} \sin^2 a \\ y_2 = x_2 \cos a - \frac{x_3 \overline{a}}{ a } \sin a \\ y_3 = x_3 \cos a + \frac{ax_2}{ a } \sin a \end{array} \right. \end{aligned}$	$\left\{ \begin{array}{l} \eta_1 = \xi_1 \\ \eta_2 = \frac{\xi_2 + \xi_3}{2} + \frac{\xi_2 - \xi_3}{2} \cos 2 a + \frac{(a, x_1)}{ a } \sin 2 a \\ \eta_3 = \frac{\xi_2 + \xi_3}{2} - \frac{\xi_2 - \xi_3}{2} \cos 2 a - \frac{(a, x_1)}{ a } \sin 2 a , \end{array} \right.$ $\left\{ \begin{array}{l} y_1 = x_1 - \frac{(\xi_2 - \xi_3)a}{2 a } \sin 2 a - \frac{2(a, x_1)a}{ a ^2} \sin^2 a \\ y_2 = x_2 \cos a - \frac{x_3 \overline{a}}{ a } \sin a \\ y_3 = x_3 \cos a + \frac{ax_2}{ a } \sin a \end{array} \right.$	$\left\{ \begin{array}{l} \eta_1 = \xi_1 \\ \eta_2 = \frac{\xi_2 + \xi_3}{2} + \frac{\xi_2 - \xi_3}{2} \cos 2 a + \frac{(a, x_1)}{ a } \sin 2 a \\ \eta_3 = \frac{\xi_2 + \xi_3}{2} - \frac{\xi_2 - \xi_3}{2} \cos 2 a - \frac{(a, x_1)}{ a } \sin 2 a , \end{array} \right.$ $\left\{ \begin{array}{l} y_1 = x_1 - \frac{(\xi_2 - \xi_3)a}{2 a } \sin 2 a - \frac{2(a, x_1)a}{ a ^2} \sin^2 a \\ y_2 = x_2 \cos a - \frac{x_3 \overline{a}}{ a } \sin a \\ y_3 = x_3 \cos a + \frac{ax_2}{ a } \sin a \end{array} \right.$
0902.0431_E01100	187	■ 0.766	$\mathfrak{e}_{8(8)} = \left(\mathfrak{e}_{8(8)}^{\tau\gamma} = \{R \in \mathfrak{e}_{8(8)} \mid \tau\gamma R = R\} \right)^{\tau\gamma}$	$\mathfrak{e}_{8(8)} = (\mathfrak{e}_8^C)^{\tau\gamma} = \{R \in \mathfrak{e}_8^C \mid \tau\gamma R = R\}$	$\mathfrak{e}_{8(8)} = (\mathfrak{e}_8^C)^{\tau\gamma} = \{R \in \mathfrak{e}_8^C \mid \tau\gamma R = R\}$
0902.0431_E00816	136	■ 0.770	$\alpha = \left(\begin{array}{cccc} \beta & 0 & M & 0 \\ 0 & \delta & 0 & N \\ a & 0 & \mu & 0 \\ 0 & b & 0 & \nu \end{array} \right), \quad \begin{array}{l} \beta, \delta \in \text{Hom}_C(\mathfrak{J}^C), \\ a, b \in \text{Hom}_C(\mathfrak{J}^C, C), \\ M, N \in \mathfrak{J}^C, \\ \mu, \nu \in C. \end{array}$	$\alpha = \left(\begin{array}{cccc} \beta & 0 & M & 0 \\ 0 & \delta & 0 & N \\ a & 0 & \mu & 0 \\ 0 & b & 0 & \nu \end{array} \right), \quad \begin{array}{l} \beta, \delta \in \text{Hom}_C(\mathfrak{J}^C), \\ a, b \in \text{Hom}_C(\mathfrak{J}^C, C), \\ M, N \in \mathfrak{J}^C, \\ \mu, \nu \in C. \end{array}$	$\alpha = \left(\begin{array}{cccc} \beta & 0 & M & 0 \\ 0 & \delta & 0 & N \\ a & 0 & \mu & 0 \\ 0 & b & 0 & \nu \end{array} \right), \quad \begin{array}{l} \beta, \delta \in \text{Hom}_C(\mathfrak{J}^C), \\ a, b \in \text{Hom}_C(\mathfrak{J}^C, C), \\ M, N \in \mathfrak{J}^C, \\ \mu, \nu \in C. \end{array}$

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0902.0431_E00746	123	0.770	$\begin{aligned} & \pm (\lambda_k - \lambda_l), \quad \pm (\lambda_k + \lambda_l), \quad 0 \leq k < l \leq 3, \\ & \pm \lambda_k \pm \frac{1}{2} (\mu_2 - \mu_3), \quad 0 \leq k \leq 3, \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \left(\mu_j + \frac{2}{3} \nu \right), \quad 0 \leq j \leq 3, \\ & \pm \lambda_k \pm \left(\frac{1}{2} \mu_1 - \frac{2}{3} \nu \right), \quad 0 \leq k \leq 3, \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \left(\frac{1}{2} \mu_2 - \frac{2}{3} \nu \right), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \left(\frac{1}{2} \mu_2 - \frac{2}{3} \nu \right), \end{aligned}$	$\begin{aligned} & \pm (\lambda_k - \lambda_l), \quad \pm (\lambda_k + \lambda_l), \quad 0 \leq k < l \leq 3, \\ & \pm \lambda_k \pm \frac{1}{2} (\mu_2 - \mu_3), \quad 0 \leq k \leq 3, \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \left(\mu_j + \frac{2}{3} \nu \right), \quad 0 \leq j \leq 3, \\ & \pm \lambda_k \pm \left(\frac{1}{2} \mu_1 - \frac{2}{3} \nu \right), \quad 0 \leq k \leq 3, \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \left(\frac{1}{2} \mu_2 - \frac{2}{3} \nu \right), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \left(\frac{1}{2} \mu_2 - \frac{2}{3} \nu \right), \end{aligned}$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_EQ1119	190	■ 0.778	$\mathfrak{e}_{\{8\}^C}=\mathfrak{mathfrak{sl}}(3,C)\oplus\mathfrak{mathfrak{e}}_6^C\oplus(\mathfrak{J}^C)^3\oplus(\mathfrak{J}^C)^3,$	$\mathfrak{e}_8^C=\mathfrak{sl}(3,C)\oplus\mathfrak{e}_6^C\oplus(\mathfrak{J}^C)^3\oplus(\mathfrak{J}^C)^3,$	$\mathfrak{e}_8^C=\mathfrak{sl}(3,C)\oplus\mathfrak{e}_6^C\oplus(\mathfrak{J}^C)^3\oplus(\mathfrak{J}^C)^3,$
0902.0431_EQ0856	142	■ 0.779	$\begin{aligned} & 2\sin^2a\left(E_2,E_3\times X\right)+\tau\xi\sin a\cos a-\left(E_3,Y\right)\sin a\cos a-\eta\cos^2a \\ &= \eta\sin^2a+(\tau\xi-\xi)\sin a\cos a-\eta\cos^2a \\ &= \frac{1}{2}(\tau\xi-\xi)\sin 2a-\eta\cos 2a=0. \end{aligned}$	$\begin{aligned} & 2\sin^2a\left(E_2,E_3\times X\right)+\tau\xi\sin a\cos a-\left(E_3,Y\right)\sin a\cos a-\eta\cos^2a \\ &= \eta\sin^2a+(\tau\xi-\xi)\sin a\cos a-\eta\cos^2a \\ &= \frac{1}{2}(\tau\xi-\xi)\sin 2a-\eta\cos 2a=0. \end{aligned}$	$\begin{aligned} & 2\sin^2a\left(E_2,E_3\times X\right)+\tau\xi\sin a\cos a-\left(E_3,Y\right)\sin a\cos a-\eta\cos^2a \\ &= \eta\sin^2a+(\tau\xi-\xi)\sin a\cos a-\eta\cos^2a \\ &= \frac{1}{2}(\tau\xi-\xi)\sin 2a-\eta\cos 2a=0. \end{aligned}$
0902.0431_EQ1077	183	■ 0.782	$\begin{aligned} & g\left(2\left(X_1\vee\gamma X_2-X_2\vee\gamma X_1\right)X\right) \\ &= -4gX_2\circ\left(gX_1\circ gX\right)+4gX_1\circ\left(gX_2\circ gX\right) \\ &= -gX_2gX_1gX-gX_2gXgX_1-gX_1gXgX_2-gXgX_1gX_2 \\ &\quad +gX_1gX_2gX+gX_1gXgX_2+gX_2gXgX_1+gXgX_2gX_1 \\ &= \left[gX_1,gX_2\right]gX-gX\left[gX_1,gX_2\right]=g\left(\left(\varphi_*\left[gX_1,gX_2\right]\right)X\right). \end{aligned}$	$\begin{aligned} & g\left(2\left(X_1\vee\gamma X_2-X_2\vee\gamma X_1\right)X\right) \\ &= -4gX_2\circ\left(gX_1\circ gX\right)+4gX_1\circ\left(gX_2\circ gX\right) \\ &= -gX_2gX_1gX-gX_2gXgX_1-gX_1gXgX_2-gXgX_1gX_2 \\ &\quad +gX_1gX_2gX+gX_1gXgX_2+gX_2gXgX_1+gXgX_2gX_1 \\ &= \left[gX_1,gX_2\right]gX-gX\left[gX_1,gX_2\right]=g\left(\left(\varphi_*\left[gX_1,gX_2\right]\right)X\right). \end{aligned}$	$\begin{aligned} & g\left(2\left(X_1\vee\gamma X_2-X_2\vee\gamma X_1\right)X\right) \\ &= -4gX_2\circ\left(gX_1\circ gX\right)+4gX_1\circ\left(gX_2\circ gX\right) \\ &= -gX_2gX_1gX-gX_2gXgX_1-gX_1gXgX_2-gXgX_1gX_2 \\ &\quad +gX_1gX_2gX+gX_1gXgX_2+gX_2gXgX_1+gXgX_2gX_1 \\ &= \left[gX_1,gX_2\right]gX-gX\left[gX_1,gX_2\right]=g\left(\left(\varphi_*\left[gX_1,gX_2\right]\right)X\right). \end{aligned}$
0902.0431_EQ1137	192	■ 0.785	$\begin{aligned} & \left(\mathfrak{e}_{\{8\}}\right)^w\mid\mathfrak{R}=R,\tau\tilde{\lambda}R=R \\ &= \left\{\left(D,\phi,0,0\right)\in\mathfrak{e}_8^C\mid D\in\mathfrak{su}(3),\phi\in\mathfrak{e}_6\right\}\cong\mathfrak{su}(3)\oplus\mathfrak{e}_6, \end{aligned}$	$\begin{aligned} \left(\mathfrak{e}_8\right)^w &= \left\{R\in\mathfrak{e}_8^C\mid wR=R,\tau\tilde{\lambda}R=R\right\} \\ &= \left\{\left(D,\phi,0,0\right)\in\mathfrak{e}_8^C\mid D\in\mathfrak{su}(3),\phi\in\mathfrak{e}_6\right\}\cong\mathfrak{su}(3)\oplus\mathfrak{e}_6, \end{aligned}$	$\begin{aligned} \left(\mathfrak{e}_8\right)^w &= \left\{R\in\mathfrak{e}_8^C\mid wR=R,\tau\tilde{\lambda}R=R\right\} \\ &= \left\{\left(D,\phi,0,0\right)\in\mathfrak{e}_8^C\mid D\in\mathfrak{su}(3),\phi\in\mathfrak{e}_6\right\}\cong\mathfrak{su}(3)\oplus\mathfrak{e}_6, \end{aligned}$
0902.0431_EQ0299	48	■ 0.788	$\mathfrak{f}_{\{4\}^C}=\mathfrak{d}_{\{4\}^C}\oplus\widetilde{\mathfrak{A}}_1^C\oplus\widetilde{\mathfrak{A}}_2^C\oplus\widetilde{\mathfrak{A}}_3^C=\mathfrak{d}_4^C\oplus\widetilde{\mathfrak{A}}^C.$	$\mathfrak{f}_4^C=\mathfrak{d}_4^C\oplus\widetilde{\mathfrak{A}}_1^C\oplus\widetilde{\mathfrak{A}}_2^C\oplus\widetilde{\mathfrak{A}}_3^C=\mathfrak{d}_4^C\oplus\widetilde{\mathfrak{A}}^C.$	$\mathfrak{f}_4^C=\mathfrak{d}_4^C\oplus\widetilde{\mathfrak{A}}_1^C\oplus\widetilde{\mathfrak{A}}_2^C\oplus\widetilde{\mathfrak{A}}_3^C=\mathfrak{d}_4^C\oplus\widetilde{\mathfrak{A}}^C.$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E01034	176	0.789	$\begin{array}{r} \frac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)-\frac{1}{2}\mu_3+\frac{1}{3}\nu+r=1 \\ \frac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)-\frac{1}{2}\mu_3-\frac{1}{3}\nu-r=1 \\ \frac{1}{2}(\lambda_0-\lambda_1-\lambda_2+\lambda_3)+\frac{1}{2}\mu_3+\frac{1}{3}\nu+r=1 \\ \frac{1}{2}(\lambda_0-\lambda_1-\lambda_2+\lambda_3)+\frac{1}{2}\mu_3+\frac{1}{3}\nu-r=1 \\ \frac{1}{2}(\lambda_0-\lambda_1-\lambda_2+\lambda_3)-\frac{1}{2}\mu_3-\frac{1}{3}\nu+r=1 \\ \frac{1}{2}(\lambda_0-\lambda_1-\lambda_2+\lambda_3)-\frac{1}{2}\mu_3-\frac{1}{3}\nu-r=1 \end{array}$	$\begin{array}{r} \frac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)-\frac{1}{2}\mu_3-\frac{1}{3}\nu+r=1 \\ \frac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)-\frac{1}{2}\mu_3-\frac{1}{3}\nu-r=1 \\ \frac{1}{2}(\lambda_0-\lambda_1-\lambda_2+\lambda_3)+\frac{1}{2}\mu_3+\frac{1}{3}\nu+r=1 \end{array}$	$\begin{array}{r} \frac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)-\frac{1}{2}\mu_3-\frac{1}{3}\nu+r=1 \\ \frac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)-\frac{1}{2}\mu_3-\frac{1}{3}\nu-r=1 \\ \frac{1}{2}(\lambda_0-\lambda_1-\lambda_2+\lambda_3)+\frac{1}{2}\mu_3+\frac{1}{3}\nu+r=1 \\ \frac{1}{2}(\lambda_0-\lambda_1-\lambda_2+\lambda_3)+\frac{1}{2}\mu_3+\frac{1}{3}\nu-r=1 \\ \frac{1}{2}(\lambda_0-\lambda_1-\lambda_2+\lambda_3)-\frac{1}{2}\mu_3-\frac{1}{3}\nu+r=1 \\ \frac{1}{2}(\lambda_0-\lambda_1-\lambda_2+\lambda_3)-\frac{1}{2}\mu_3-\frac{1}{3}\nu-r=1 \end{array}$
0902.0431_E00899	149	0.790	$\begin{aligned} & \chi(\varphi_*(D)X) = \chi\left(\begin{pmatrix} \varphi_*(D)X \\ \gamma\varphi_*(D)\gamma Y \\ 0 \\ 0 \end{pmatrix}\right) = \chi\left(\begin{pmatrix} \varphi_*(D)X & 0 & 0 & 0 \\ 0 & \tau\varphi_*(D)\tau & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} X \\ Y \\ \xi \\ \eta \end{pmatrix}\right) \\ & = \chi(\Phi(\varphi_*(D), 0, 0, 0)P). \end{aligned}$	$\begin{aligned} & = \chi\left(\begin{pmatrix} \varphi_*(D)X \\ \gamma\varphi_*(D)\gamma Y \\ 0 \\ 0 \end{pmatrix}\right) = \chi\left(\begin{pmatrix} \varphi_*(D)X & 0 & 0 & 0 \\ 0 & \tau\varphi_*(D)\tau & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} X \\ Y \\ \xi \\ \eta \end{pmatrix}\right) \\ & = \chi(\Phi(\varphi_*(D), 0, 0, 0)P). \end{aligned}$	$\begin{aligned} & = \chi\left(\begin{pmatrix} \varphi_*(D)X \\ \gamma\varphi_*(D)\gamma Y \\ 0 \\ 0 \end{pmatrix}\right) = \chi\left(\begin{pmatrix} \varphi_*(D)X & 0 & 0 & 0 \\ 0 & \tau\varphi_*(D)\tau & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} X \\ Y \\ \xi \\ \eta \end{pmatrix}\right) \\ & = \chi(\Phi(\varphi_*(D), 0, 0, 0)P). \end{aligned}$
0902.0431_E00631	104	0.794	$\begin{aligned} & \varphi(B)(M+\mathbf{a}) = g^{-1}(Bg(M+\mathbf{a})B^*) \\ & = g^{-1}\left(\begin{pmatrix} p & 0 \\ 0 & D \end{pmatrix}\left(\begin{pmatrix} \frac{1}{2}\mathrm{tr}(M) & i\mathbf{a} \\ i\mathbf{a}^* & M-\frac{1}{2}\mathrm{tr}(M)E \end{pmatrix}\begin{pmatrix} \bar{p} & 0 \\ 0 & D^* \end{pmatrix}\right)\right) \\ & = g^{-1}\left(\begin{pmatrix} \frac{1}{2}\mathrm{tr}(M) & ip\mathbf{a}D^* \\ iD\mathbf{a}^*\bar{p} & DMD^*-\frac{1}{2}\mathrm{tr}(M)E \end{pmatrix}\right) \\ & = DMD^*+p\mathbf{a}D^* = \beta(M+\mathbf{a}). \end{aligned}$	$\begin{aligned} & \varphi(B)(M+\mathbf{a}) = g^{-1}(Bg(M+\mathbf{a})B^*) \\ & = g^{-1}\left(\begin{pmatrix} p & 0 \\ 0 & D \end{pmatrix}\left(\begin{pmatrix} \frac{1}{2}\mathrm{tr}(M) & i\mathbf{a} \\ i\mathbf{a}^* & M-\frac{1}{2}\mathrm{tr}(M)E \end{pmatrix}\begin{pmatrix} \bar{p} & 0 \\ 0 & D^* \end{pmatrix}\right)\right) \\ & = g^{-1}\left(\begin{pmatrix} \frac{1}{2}\mathrm{tr}(M) & ip\mathbf{a}D^* \\ iD\mathbf{a}^*\bar{p} & DMD^*-\frac{1}{2}\mathrm{tr}(M)E \end{pmatrix}\right) \\ & = DMD^*+p\mathbf{a}D^* = \beta(M+\mathbf{a}). \end{aligned}$	$\begin{aligned} & \varphi(B)(M+\mathbf{a}) = g^{-1}(Bg(M+\mathbf{a})B^*) \\ & = g^{-1}\left(\begin{pmatrix} p & 0 \\ 0 & D \end{pmatrix}\left(\begin{pmatrix} \frac{1}{2}\mathrm{tr}(M) & i\mathbf{a} \\ i\mathbf{a}^* & M-\frac{1}{2}\mathrm{tr}(M)E \end{pmatrix}\begin{pmatrix} \bar{p} & 0 \\ 0 & D^* \end{pmatrix}\right)\right) \\ & = g^{-1}\left(\begin{pmatrix} \frac{1}{2}\mathrm{tr}(M) & ip\mathbf{a}D^* \\ iD\mathbf{a}^*\bar{p} & DMD^*-\frac{1}{2}\mathrm{tr}(M)E \end{pmatrix}\right) \\ & = DMD^*+p\mathbf{a}D^* = \beta(M+\mathbf{a}). \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00108	21	■ 0.795	$\begin{aligned} (\alpha x)(\alpha y) &= (a+A\boldsymbol{m})(b+A\boldsymbol{n}) \\ &= (ab - \langle A\boldsymbol{m}, A\boldsymbol{n} \rangle) + (aA\boldsymbol{n} + \bar{b}A\boldsymbol{m} - \overline{A\boldsymbol{m} \times A\boldsymbol{n}}) \\ &= (ab - \langle \boldsymbol{m}, A^*A\boldsymbol{n} \rangle) + (aA\boldsymbol{n} + \bar{b}A\boldsymbol{m} - \tilde{t}\widetilde{A}(\boldsymbol{m} \times \boldsymbol{n})) \\ &= (ab - \langle \boldsymbol{m}, \boldsymbol{n} \rangle) + A(a\boldsymbol{n} + \bar{b}\boldsymbol{m} - \overline{\boldsymbol{m} \times \boldsymbol{n}}) \\ &= \varphi(A)((a + \boldsymbol{m})(b + \boldsymbol{n})) = \alpha(xy) \end{aligned}$	$\begin{aligned} (\alpha x)(\alpha y) &= (a + A\boldsymbol{m})(b + A\boldsymbol{n}) \\ &= (ab - \langle A\boldsymbol{m}, A\boldsymbol{n} \rangle) + (aA\boldsymbol{n} + \bar{b}A\boldsymbol{m} - \overline{A\boldsymbol{m} \times A\boldsymbol{n}}) \\ &= (ab - \langle \boldsymbol{m}, A^*A\boldsymbol{n} \rangle) + (aA\boldsymbol{n} + \bar{b}A\boldsymbol{m} - \tilde{t}\widetilde{A}(\boldsymbol{m} \times \boldsymbol{n})) \\ &= (ab - \langle \boldsymbol{m}, \boldsymbol{n} \rangle) + A(a\boldsymbol{n} + \bar{b}\boldsymbol{m} - \overline{\boldsymbol{m} \times \boldsymbol{n}}) \\ &= \varphi(A)((a + \boldsymbol{m})(b + \boldsymbol{n})) = \alpha(xy) \end{aligned}$	$\begin{aligned} (\alpha x)(\alpha y) &= (a + A\boldsymbol{m})(b + A\boldsymbol{n}) \\ &= (ab - \langle A\boldsymbol{m}, A\boldsymbol{n} \rangle) + (aA\boldsymbol{n} + \bar{b}A\boldsymbol{m} - \overline{A\boldsymbol{m} \times A\boldsymbol{n}}) \\ &= (ab - \langle \boldsymbol{m}, A^*A\boldsymbol{n} \rangle) + (aA\boldsymbol{n} + \bar{b}A\boldsymbol{m} - \tilde{t}\widetilde{A}(\boldsymbol{m} \times \boldsymbol{n})) \\ &= (ab - \langle \boldsymbol{m}, \boldsymbol{n} \rangle) + A(a\boldsymbol{n} + \bar{b}\boldsymbol{m} - \overline{\boldsymbol{m} \times \boldsymbol{n}}) \\ &= \varphi(A)((a + \boldsymbol{m})(b + \boldsymbol{n})) = \alpha(xy). \end{aligned}$
0902.0431_E00882	146	■ 0.798	$\alpha P = (X, Y, \xi, \eta), \quad X, Y \quad \text{are diagonal}, \quad \xi > 0.$	$\alpha P = (X, Y, \xi, \eta), \quad X, Y \quad \text{are diagonal}, \quad \xi > 0.$	$\alpha P = (X, Y, \xi, \eta), \quad X, Y \quad \text{are diagonal}, \quad \xi > 0.$
0902.0431_E00695	115	■ 0.803	$\Phi = \Phi_{-3} + \Phi_{-2} + \Phi_{-1} + \Phi_0 + \Phi_1 + \Phi_2 + \Phi_3, \quad \Phi_i \in \mathfrak{e}_7^C$	$\Phi = \Phi_{-3} + \Phi_{-2} + \Phi_{-1} + \Phi_0 + \Phi_1 + \Phi_2 + \Phi_3, \quad \Phi_i \in \mathfrak{e}_7^C$	$\Phi = \Phi_{-3} + \Phi_{-2} + \Phi_{-1} + \Phi_0 + \Phi_1 + \Phi_2 + \Phi_3, \quad \Phi_i \in \mathfrak{e}_7^C,$
0902.0431_E00931	154	■ 0.803	$\epsilon A = \overline{(\text{Ad } J_3) A}, \quad J_3 = \begin{pmatrix} 0 & E \\ -E & 0 \end{pmatrix},$	$\epsilon A = \overline{(\text{Ad } J_3) A}, \quad J_3 = \begin{pmatrix} 0 & E \\ -E & 0 \end{pmatrix},$	$\epsilon A = \overline{(\text{Ad } J_3) A}, \quad J_3 = \begin{pmatrix} 0 & E \\ -E & 0 \end{pmatrix},$
0902.0431_E00666	110	■ 0.804	$\begin{aligned} E_{6(6)} &\simeq \text{Sp}(4)/\boldsymbol{Z}_2 \times \boldsymbol{R}^{42}, \\ E_{6(2)} &\simeq (\text{Sp}(1) \times \text{SU}(6))/\boldsymbol{Z}_2 \times \boldsymbol{R}^{40}, \\ E_{6(-14)} &\simeq (U(1) \times \text{Spin}(10))/\boldsymbol{Z}_4 \times \boldsymbol{R}^{32}, \\ E_{6(-26)} &\simeq F_4 \times \boldsymbol{R}^{26}. \end{aligned}$	$\begin{aligned} E_{6(6)} &\simeq \text{Sp}(4)/\boldsymbol{Z}_2 \times \boldsymbol{R}^{42}, \\ E_{6(2)} &\simeq (\text{Sp}(1) \times \text{SU}(6))/\boldsymbol{Z}_2 \times \boldsymbol{R}^{40}, \\ E_{6(-14)} &\simeq (U(1) \times \text{Spin}(10))/\boldsymbol{Z}_4 \times \boldsymbol{R}^{32}, \\ E_{6(-26)} &\simeq F_4 \times \boldsymbol{R}^{26}. \end{aligned}$	$\begin{aligned} E_{6(6)} &\simeq \text{Sp}(4)/\boldsymbol{Z}_2 \times \boldsymbol{R}^{42}, \\ E_{6(2)} &\simeq (\text{Sp}(1) \times \text{SU}(6))/\boldsymbol{Z}_2 \times \boldsymbol{R}^{40}, \\ E_{6(-14)} &\simeq (U(1) \times \text{Spin}(10))/\boldsymbol{Z}_4 \times \boldsymbol{R}^{32}, \\ E_{6(-26)} &\simeq F_4 \times \boldsymbol{R}^{26}. \end{aligned}$
0902.0431_E01192	200	■ 0.804	$\begin{aligned} (a \times b)c &= -(a_1 \wedge c_1, b)a_1 \wedge c_2 + (a_1 \wedge c_2, b)a_2 \wedge c_1 \\ &\quad + (a_2 \wedge c_1, b)a_1 \wedge c_2 - (a_2 \wedge c_2, b)a_1 \wedge c_1 + \frac{4}{5}(a, b)c. \end{aligned}$	$\begin{aligned} (a \times b)c &= -(a_1 \wedge c_1, b)a_1 \wedge c_2 + (a_1 \wedge c_2, b)a_2 \wedge c_1 \\ &\quad + (a_2 \wedge c_1, b)a_1 \wedge c_2 - (a_2 \wedge c_2, b)a_1 \wedge c_1 + \frac{4}{5}(a, b)c. \end{aligned}$	$\begin{aligned} (a \times b)c &= -(a_1 \wedge c_1, b)a_1 \wedge c_2 + (a_1 \wedge c_2, b)a_2 \wedge c_1 \\ &\quad + (a_2 \wedge c_1, b)a_1 \wedge c_2 - (a_2 \wedge c_2, b)a_1 \wedge c_1 + \frac{4}{5}(a, b)c. \end{aligned}$

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0902.0431_E08059	14	■ 0.807	$\begin{array}{c} \backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{c}ll\} \backslash\mathrm{lambda} G_{23}+\backslash\mathrm{mu} G_{45}+\backslash\mathrm{nu} G_{67}, \\ \& -\backslash\mathrm{lambda} G_{13}-\backslash\mathrm{mu} G_{46}+\backslash\mathrm{nu} G_{57}, \& \backslash\backslash \backslash\mathrm{lambda} \\ G_{12}+\backslash\mathrm{mu} G_{47}+\backslash\mathrm{nu} G_{56}, \& -\backslash\mathrm{lambda} G_{15}+\backslash\mathrm{mu} \\ G_{26}-\backslash\mathrm{nu} G_{37}, \& \backslash\mathrm{lambda}, \backslash\mathrm{mu}, \backslash\mathrm{nu} \in \backslash\mathrm{boldsymbol}\{R\} \\ \backslash\backslash \backslash\mathrm{lambda} G_{14}-\backslash\mathrm{mu} G_{27}-\backslash\mathrm{nu} G_{36}, \& -\backslash\mathrm{lambda} \\ G_{17}-\backslash\mathrm{mu} G_{24}+\backslash\mathrm{nu} G_{35}, \& \backslash\mathrm{lambda}+\backslash\mathrm{mu}+\backslash\mathrm{nu}=0 \backslash\backslash \& \\ \backslash\mathrm{lambda} G_{16}+\backslash\mathrm{mu} G_{25}+\backslash\mathrm{nu} G_{34} \end{array}$	$\begin{array}{c} \lambda G_{23} + \mu G_{45} + \nu G_{67}, \quad -\lambda G_{13} - \mu G_{46} + \nu G_{57}, \\ \lambda G_{12} + \mu G_{47} + \nu G_{56}, \quad -\lambda G_{15} + \mu G_{26} - \nu G_{37}, \quad \lambda, \mu, \nu \in \boldsymbol{R} \\ \lambda G_{14} - \mu G_{27} - \nu G_{36}, \quad -\lambda G_{17} - \mu G_{24} + \nu G_{35}, \quad \lambda + \mu + \nu = 0 \\ \lambda G_{16} + \mu G_{25} + \nu G_{34} \end{array}$	$\begin{array}{c} \lambda G_{23} + \mu G_{45} + \nu G_{67}, \quad -\lambda G_{13} - \mu G_{46} + \nu G_{57}, \\ \lambda G_{12} + \mu G_{47} + \nu G_{56}, \quad -\lambda G_{15} + \mu G_{26} - \nu G_{37}, \quad \lambda, \mu, \nu \in \boldsymbol{R} \\ \lambda G_{14} - \mu G_{27} - \nu G_{36}, \quad -\lambda G_{17} - \mu G_{24} + \nu G_{35}, \quad \lambda + \mu + \nu = 0 \\ \lambda G_{16} + \mu G_{25} + \nu G_{34}. \end{array}$
0902.0431_E01023	171	■ 0.807	$\begin{array}{c} \backslash\mathrm{begin}\{\mathrm{aligned}\} \& \{[(\backslash\mathrm{Phi}(h), \ 0,0, \ r, \\ 0), (0,0,0,0,1,0)]= (0,0,0,0,2 \ r, \ 0) \} \backslash\backslash \& \{[(\backslash\mathrm{Phi}(h), \ 0,0, \\ r, \ 0), (0,0,0,0,0,1)]= (0,0,0,0,0, -2 \ r), \} \end{array}$	$\begin{array}{c} [(\Phi(h), 0, 0, r, 0), (0, 0, 0, 0, 1, 0)] = (0, 0, 0, 0, 2r, 0) \\ [(\Phi(h), 0, 0, r, 0), (0, 0, 0, 0, 0, 1)] = (0, 0, 0, 0, 0, -2r), \end{array}$	$\begin{array}{c} [(\Phi(h), 0, 0, r, 0), (0, 0, 0, 0, 1, 0)] = (0, 0, 0, 0, 2r, 0), \\ [(\Phi(h), 0, 0, r, 0), (0, 0, 0, 0, 0, 1)] = (0, 0, 0, 0, 0, -2r), \end{array}$
0902.0431_E08628	104	■ 0.808	$\begin{array}{c} \backslash\mathrm{begin}\{\mathrm{aligned}\} \backslash\mathrm{varphi}(A) \ E \ =\mathrm{g}^{\wedge\{-1\}}\backslash\mathrm{left}(A(\mathrm{g} \ E) \ A^{\wedge\{*\}} \\ \backslash\mathrm{right})=\mathrm{g}^{\wedge\{-1\}}\backslash\mathrm{left}(A\backslash\mathrm{left}(2 \ E_{1}-\backslash\mathrm{frac}\{1\}\{2\} \ E\backslash\mathrm{right}) \\ A^{\wedge\{*\}}\backslash\mathrm{right})=\mathrm{g}^{\wedge\{-1\}}\backslash\mathrm{left}(2 \ A \ E_{1}\ A^{\wedge\{*\}}-\backslash\mathrm{frac}\{1\}\{2\} \ E \\ \backslash\mathrm{right}) \backslash\backslash \& =\mathrm{g}^{\wedge\{-1\}}(\mathrm{g}(\backslash\mathrm{alpha} \ E))=\backslash\mathrm{alpha} \ E \end{array}$	$\begin{array}{c} \varphi(A)E = g^{-1}\left(A(gE)A^*\right) = g^{-1}\left(A\left(2E_1 - \frac{1}{2}E\right)A^*\right) = g^{-1}\left(2AE_1A^* - \frac{1}{2}E\right) \\ = g^{-1}(g(\alpha E)) = \alpha E \end{array}$	$\begin{array}{c} \varphi(A)E \ = \ g^{-1}(A(gE)A^*) = g^{-1}\left(A\left(2E_1 - \frac{1}{2}E\right)A^*\right) = g^{-1}\left(2AE_1A^* - \frac{1}{2}E\right) \\ = \ g^{-1}(g(\alpha E)) = \alpha E. \end{array}$
0902.0431_E08333	53	■ 0.813	$\begin{array}{c} \backslash\mathrm{mathfrak}\{f\}_{4}\{ \ }^{\wedge\{C\}}=\backslash\mathrm{mathfrak}\{d\}_{4}\{ \ }^{\wedge\{C\}} \\ \backslash\mathrm{oplus} \ \backslash\mathrm{widetilde}\{\backslash\mathrm{mathfrak}\{A\}\}_{1}\{ \ }^{\wedge\{C\}} \ \backslash\mathrm{oplus} \\ \backslash\mathrm{widetilde}\{\backslash\mathrm{mathfrak}\{A\}\}_{2}\{ \ }^{\wedge\{C\}} \ \backslash\mathrm{oplus} \ \backslash\mathrm{widetilde}\{ \\ \backslash\mathrm{mathfrak}\{A\}\}_{3}\{ \ }^{\wedge\{C\}} \ . \end{array}$	$\mathfrak{f}_4^C = \mathfrak{d}_4^C \oplus \widetilde{\mathfrak{A}}_1^C \oplus \widetilde{\mathfrak{A}}_2^C \oplus \widetilde{\mathfrak{A}}_3^C.$	$\mathfrak{f}_4^C = \mathfrak{d}_4^C \oplus \widetilde{\mathfrak{A}}_1^C \oplus \widetilde{\mathfrak{A}}_2^C \oplus \widetilde{\mathfrak{A}}_3^C.$
0902.0431_E08301	48	■ 0.817	$\begin{array}{c} \backslash\mathrm{mathfrak}\{a\} \ \backslash\mathrm{supset}\backslash\mathrm{left}[\backslash\mathrm{mathfrak}\{a\}, \ \backslash\mathrm{mathfrak}\{f\}_{4}\{ \\ \ }^{\wedge\{C\}}\backslash\mathrm{right}] \ \backslash\mathrm{supset}\backslash\mathrm{left}[\backslash\mathrm{mathfrak}\{d\}_{4}\{ \ }^{\wedge\{C\}}, \\ \backslash\mathrm{widetilde}\{\backslash\mathrm{mathfrak}\{A\}\}_{i}\{ \ }^{\wedge\{C\}}\backslash\mathrm{right}]=\backslash\mathrm{widetilde}\{ \\ \backslash\mathrm{mathfrak}\{A\}\}_{i}\{ \ }^{\wedge\{C\}}, \ \backslash\mathrm{quad} \ i=1,2,3 \ . \end{array}$	$\mathfrak{a} \supset [\mathfrak{a}, \mathfrak{f}_4^C] \supset [\mathfrak{d}_4^C, \widetilde{\mathfrak{A}}_i^C] = \widetilde{\mathfrak{A}}_i^C, \quad i = 1, 2, 3.$	$\mathfrak{a} \supset [\mathfrak{a}, \mathfrak{f}_4^C] \supset [\mathfrak{d}_4^C, \widetilde{\mathfrak{A}}_i^C] = \widetilde{\mathfrak{A}}_i^C, \quad i = 1, 2, 3.$
0902.0431_E08315	50	■ 0.827	$\begin{array}{c} (\backslash\mathrm{delta},[\backslash\mathrm{widetilde}\{A\}, \ \backslash\mathrm{widetilde}\{B\}])_{4}=(\backslash\mathrm{delta} \ A, \\ B), \ \backslash\mathrm{quad} \ \backslash\mathrm{delta} \ \in \ \backslash\mathrm{mathfrak}\{f\}_{4}\{ \ }^{\wedge\{C\}}, \ \backslash\mathrm{quad} \ A, \ B \\ \in \ \backslash\mathrm{mathfrak}\{J\}^{\wedge\{C\}} \ . \end{array}$	$(\delta, [\widetilde{A}, \widetilde{B}])_4 = (\delta A, B), \quad \delta \in \mathfrak{f}_4^C, \quad A, B \in \mathfrak{J}^C.$	$(\delta, [\widetilde{A}, \widetilde{B}])_4 = (\delta A, B), \quad \delta \in \mathfrak{f}_4^C, \quad A, B \in \mathfrak{J}^C.$
0902.0431_E08749	124	■ 0.827	$\begin{array}{c} \backslash\mathrm{mathfrak}\{h\}=\backslash\mathrm{left}\backslash\backslash\mathrm{Phi}\backslash\mathrm{left}(\backslash\mathrm{sum}_{k=0}^{\wedge\{3\}} \ \backslash\mathrm{lambda}_{k} \\ H_{k}+\backslash\mathrm{left}(\backslash\mathrm{sum}_{j=1}^{\wedge\{3\}} \ \backslash\mathrm{mu}_{j} \ E_{j})\backslash\mathrm{right})^{\wedge\{\sim\}}, \\ 0,0, \ \backslash\mathrm{nu}\backslash\mathrm{right}) \ \in \ \backslash\mathrm{mathfrak}\{e\}_{7}\{ \ }^{\wedge\{C\}} \ \backslash\mathrm{left}\backslash\mathrm{vert}\backslash, \\ \backslash\mathrm{begin}\{\mathrm{array}\}\{l\} \ \backslash\mathrm{lambda}_{k}, \ \backslash\mathrm{nu} \ \in \ C \ \backslash\backslash \ \backslash\mathrm{mu}_{j} \ \in \ C, \\ \backslash\mathrm{mu}_{1}+\backslash\mathrm{mu}_{2}+\backslash\mathrm{mu}_{3}=0 \ \end{array}$	$\mathfrak{h} = \left\{ \Phi \left(\sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim}, 0, 0, \nu \right) \in \mathfrak{e}_7^C \mid \begin{array}{l} \lambda_k, \nu \in C \\ \mu_j \in C, \mu_1 + \mu_2 + \mu_3 = 0 \end{array} \right\}$	$\mathfrak{h} = \left\{ \Phi \left(\sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim}, 0, 0, \nu \right) \in \mathfrak{e}_7^C \mid \begin{array}{l} \lambda_k, \nu \in C \\ \mu_j \in C, \mu_1 + \mu_2 + \mu_3 = 0 \end{array} \right\}$
0902.0431_E01076	183	■ 0.830	$\begin{array}{c} \backslash\mathrm{begin}\{\mathrm{aligned}\} \& \mathrm{g}\backslash\mathrm{left}(2\backslash\mathrm{left}(X_{1} \ \vee \ \backslash\mathrm{gamma} \ X_{2} \\ \backslash\mathrm{right}) \ X\backslash\mathrm{right}) \ \backslash\mathrm{quad} \ X \ \in \ \backslash\mathrm{mathfrak}\{J\}^{\wedge\{C\}} \ \backslash\backslash \& =\mathrm{g} \\ \backslash\mathrm{left}(\backslash\mathrm{left}(\backslash\mathrm{gamma} \ X_{2}, \ X\backslash\mathrm{right}) \ X_{1})+\backslash\mathrm{frac}\{1\}\{3\} \\ \backslash\mathrm{left}(X_{1}, \ \backslash\mathrm{gamma} \ X_{2})\backslash\mathrm{right}) \ X-4 \ \backslash\mathrm{gamma} \ X_{2} \ \backslash\mathrm{times} \\ \backslash\mathrm{left}(X_{1} \ \backslash\mathrm{times} \ X\backslash\mathrm{right})\backslash\mathrm{right})(\backslash\mathrm{text} \ \{ \ \mathrm{Lemma} \ 3.4.1 \}) \\ \backslash\backslash \& =\backslash\mathrm{left}(\backslash\mathrm{gamma} \ X_{2}, \ X\backslash\mathrm{right}) \ \mathrm{g} \ X_{1}+\backslash\mathrm{frac}\{1\}\{3\} \\ \backslash\mathrm{left}(X_{1}, \ \backslash\mathrm{gamma} \ X_{2})\backslash\mathrm{right}) \ \mathrm{g} \ X-4 \ \mathrm{g} \ X_{2} \ \backslash\mathrm{circ} \\ \backslash\mathrm{left}(\mathrm{g} \ X_{1} \ \backslash\mathrm{circ} \ \mathrm{g} \ X\backslash\mathrm{right})+\backslash\mathrm{left}(\backslash\mathrm{gamma} \ X_{1}, \ X \\ \backslash\mathrm{right}) \ \mathrm{g} \ X_{2} \ \backslash\backslash \& \ \backslash\mathrm{quad}+\backslash\mathrm{left}(\backslash\mathrm{gamma} \ X_{2}, \ \backslash\mathrm{gamma} \\ X_{1}, \ \backslash\mathrm{gamma} \ X\backslash\mathrm{right}) \ E(\backslash\mathrm{text} \ \{ \ \mathrm{Lemma} \ 3.12.1 \}), \\ \end{array}$	$\begin{array}{c} g \left(2 \left(X_1 \vee \gamma X_2 \right) X \right) \quad X \in \mathfrak{J}^C \\ = g \left(\left(\gamma X_2, X \right) X_1 + \frac{1}{3} \left(X_1, \gamma X_2 \right) X - 4 \gamma X_2 \times \left(X_1 \times X \right) \right) (\text{ Lemma 3.4.1 }) \\ = \left(\gamma X_2, X \right) g X_1 + \frac{1}{3} \left(X_1, \gamma X_2 \right) g X - 4 g X_2 \circ \left(g X_1 \circ g X \right) + \left(\gamma X_1, X \right) g X_2 \\ + \left(\gamma X_2, \gamma X_1, \gamma X \right) E (\text{ Lemma 3.12.1 }), \end{array}$	$\begin{array}{c} g(2(X_1 \vee \gamma X_2)X) \quad X \in \mathfrak{J}^C \\ = g((\gamma X_2, X)X_1 + \frac{1}{3}(X_1, \gamma X_2)X - 4\gamma X_2 \times (X_1 \times X)) \text{ (Lemma 3.4.1)} \\ = (\gamma X_2, X)gX_1 + \frac{1}{3}(X_1, \gamma X_2)gX - 4gX_2 \circ (gX_1 \circ gX) + (\gamma X_1, X)gX_2 \\ + (\gamma X_2, \gamma X_1, \gamma X)E \text{ (Lemma 3.12.1),} \end{array}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00548	92	■ 0.835	$\begin{aligned} 0 &= \left(\xi E_1 + \xi_2 E_2 + \xi_3 E_3 + F_1(x_1) \right) \times \left(\xi E_1 + \xi_2 E_2 + \xi_3 E_3 + F_1(x_1) \right) \\ &= (\xi_2 \xi_3 - (x_1, x_1)) E_1 + \xi \xi_3 E_2 + \xi \xi_2 E_3 - \xi F_1(x_1) \end{aligned}$	$0 = (\xi E_1 + \xi_2 E_2 + \xi_3 E_3 + F_1(x_1)) \times (\xi E_1 + \xi_2 E_2 + \xi_3 E_3 + F_1(x_1))$ $= (\xi_2 \xi_3 - (x_1, x_1)) E_1 + \xi \xi_3 E_2 + \xi \xi_2 E_3 - \xi F_1(x_1)$	$\begin{aligned} 0 &= (\xi E_1 + \xi_2 E_2 + \xi_3 E_3 + F_1(x_1)) \times (\xi E_1 + \xi_2 E_2 + \xi_3 E_3 + F_1(x_1)) \\ &= (\xi_2 \xi_3 - (x_1, x_1)) E_1 + \xi \xi_3 E_2 + \xi \xi_2 E_3 - \xi F_1(x_1). \end{aligned}$
0902.0431_E00379	63	■ 0.836	$\begin{aligned} \mathfrak{J}_{\sigma} &= \{X \in \mathfrak{J} \mid \sigma X = X\} = \left\{ \begin{pmatrix} \xi_1 & 0 & 0 \\ 0 & \xi_2 & x_1 \\ 0 & \bar{x}_1 & \xi_3 \end{pmatrix} \mid \xi_i \in \mathbf{R}, x_1 \in \mathfrak{C} \right\} \\ &= \{X \in \mathfrak{J} \mid E_1 \circ X = 0\} \oplus \mathfrak{E}_1 \text{ (where } \mathfrak{E}_1 = \{\xi E_1 \mid \xi \in \mathbf{R}\}) \\ &= \mathfrak{J}(2, \mathfrak{C}) \oplus \mathfrak{E}_1, \end{aligned}$ $\mathfrak{J}_{-\sigma} = \{X \in \mathfrak{J} \mid \sigma X = -X\} = \left\{ \begin{pmatrix} 0 & x_3 & \bar{x}_2 \\ \bar{x}_3 & 0 & 0 \\ x_2 & 0 & 0 \end{pmatrix} \mid x_i \in \mathfrak{C} \right\}$ $= \{X \in \mathfrak{J} \mid 2E_1 \circ X = X\}.$	$\mathfrak{J}_{\sigma} = \{X \in \mathfrak{J} \mid \sigma X = X\} = \left\{ \begin{pmatrix} \xi_1 & 0 & 0 \\ 0 & \xi_2 & x_1 \\ 0 & \bar{x}_1 & \xi_3 \end{pmatrix} \mid \xi_i \in \mathbf{R}, x_1 \in \mathfrak{C} \right\}$ $= \{X \in \mathfrak{J} \mid E_1 \circ X = 0\} \oplus \mathfrak{E}_1 \text{ (where } \mathfrak{E}_1 = \{\xi E_1 \mid \xi \in \mathbf{R}\})$ $= \mathfrak{J}(2, \mathfrak{C}) \oplus \mathfrak{E}_1,$ $\mathfrak{J}_{-\sigma} = \{X \in \mathfrak{J} \mid \sigma X = -X\} = \left\{ \begin{pmatrix} 0 & x_3 & \bar{x}_2 \\ \bar{x}_3 & 0 & 0 \\ x_2 & 0 & 0 \end{pmatrix} \mid x_i \in \mathfrak{C} \right\}$ $= \{X \in \mathfrak{J} \mid 2E_1 \circ X = X\}.$	
0902.0431_E00189	32	■ 0.837	$\begin{aligned} \left(\alpha e_1 \right) \left(\left(\alpha e_3 \right) b_0 \right) &= a_1 (a_3 b_0) = a_1 (a_2 (a_1 b_0)) = -a_1 (a_1 (a_2 b_0)) = a_2 b_0 \\ &= \alpha (e_2) b_0 = \alpha (e_1 e_3) b_0, \\ \left(\alpha e_2 \right) \left(\left(\alpha e_5 \right) b_0 \right) &= a_2 (a_5 b_0) = a_2 (a_1 (a_4 b_0)) = -a_1 (a_2 (a_4 b_0)) = -a_1 (a_6 b_0) \\ &= a_6 (a_1 b_0) = a_7 b_0 = \alpha (e_7) b_0 = \alpha (e_2 e_5) b_0. \end{aligned}$	$\begin{aligned} (\alpha e_1)((\alpha e_3) b_0) &= a_1(a_3 b_0) = a_1(a_2(a_1 b_0)) = -a_1(a_1(a_2 b_0)) = a_2 b_0 \\ &= \alpha(e_2) b_0 = \alpha(e_1 e_3) b_0, \\ (\alpha e_2)((\alpha e_5) b_0) &= a_2(a_5 b_0) = a_2(a_1(a_4 b_0)) = -a_1(a_2(a_4 b_0)) = -a_1(a_6 b_0) \\ &= a_6(a_1 b_0) = a_7 b_0 = \alpha(e_7) b_0 = \alpha(e_2 e_5) b_0. \end{aligned}$	$\begin{aligned} (\alpha e_1)((\alpha e_3) b_0) &= a_1(a_3 b_0) = a_1(a_2(a_1 b_0)) = -a_1(a_1(a_2 b_0)) = a_2 b_0 \\ &= \alpha(e_2) b_0 = \alpha(e_1 e_3) b_0, \\ (\alpha e_2)((\alpha e_5) b_0) &= a_2(a_5 b_0) = a_2(a_1(a_4 b_0)) = -a_1(a_2(a_4 b_0)) = -a_1(a_6 b_0) \\ &= a_6(a_1 b_0) = a_7 b_0 = \alpha(e_7) b_0 = \alpha(e_2 e_5) b_0. \end{aligned}$
0902.0431_E00643	106	■ 0.840	$\tau h(A, B) = h(B, A), \quad \overline{h(A, B)} = h(\bar{B}, \bar{A}), \quad h(A, B)^* = h(B^*, A^*).$	$\tau h(A, B) = h(B, A), \quad \overline{h(A, B)} = h(\bar{B}, \bar{A}), \quad h(A, B)^* = h(B^*, A^*).$	$\tau h(A, B) = h(B, A), \quad \overline{h(A, B)} = h(\bar{B}, \bar{A}), \quad h(A, B)^* = h(B^*, A^*).$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
9902.0431_E00603	101	0.841	$\begin{aligned} & \left(M + iF(\boldsymbol{ae}_4) \mid M \in \mathfrak{J}(3, \boldsymbol{H}), \boldsymbol{a} \in \boldsymbol{H}^3 \right) \\ &= \mathfrak{J}_{\boldsymbol{H}} \oplus i\boldsymbol{H}^3, \\ & (\mathfrak{J}^C)_{-\tau\gamma} = \{X \in \mathfrak{J}^C \mid \tau\gamma X = -X\} \\ &= \left\{ i \begin{pmatrix} \mu_1 & m_3 & \overline{m}_2 \\ \overline{m}_3 & \mu_2 & m_1 \\ m_2 & \overline{m}_1 & \mu_3 \end{pmatrix} + \begin{pmatrix} 0 & a_3 e_4 & -a_2 e_4 \\ -a_3 e_4 & 0 & a_1 e_4 \\ a_2 e_4 & -a_1 e_4 & 0 \end{pmatrix} \mid \begin{matrix} \mu_i \in \boldsymbol{R} \\ m_i, a_i \in \boldsymbol{H} \end{matrix} \right\} \\ &= \{iM + F(\boldsymbol{ae}_4) \mid M \in \mathfrak{J}(3, \boldsymbol{H}), \boldsymbol{a} \in \boldsymbol{H}^3\} \\ &= i\mathfrak{J}_{\boldsymbol{H}} \oplus \boldsymbol{H}^3, \end{aligned}$	$\begin{aligned} &= \{M + iF(\boldsymbol{ae}_4) \mid M \in \mathfrak{J}(3, \boldsymbol{H}), \boldsymbol{a} \in \boldsymbol{H}^3\} \\ &= \mathfrak{J}_{\boldsymbol{H}} \oplus i\boldsymbol{H}^3, \\ & (\mathfrak{J}^C)_{-\tau\gamma} = \{X \in \mathfrak{J}^C \mid \tau\gamma X = -X\} \\ &= \left\{ i \begin{pmatrix} \mu_1 & m_3 & \overline{m}_2 \\ \overline{m}_3 & \mu_2 & m_1 \\ m_2 & \overline{m}_1 & \mu_3 \end{pmatrix} + \begin{pmatrix} 0 & a_3 e_4 & -a_2 e_4 \\ -a_3 e_4 & 0 & a_1 e_4 \\ a_2 e_4 & -a_1 e_4 & 0 \end{pmatrix} \mid \begin{matrix} \mu_i \in \boldsymbol{R} \\ m_i, a_i \in \boldsymbol{H} \end{matrix} \right\} \\ &= \{iM + F(\boldsymbol{ae}_4) \mid M \in \mathfrak{J}(3, \boldsymbol{H}), \boldsymbol{a} \in \boldsymbol{H}^3\} \\ &= i\mathfrak{J}_{\boldsymbol{H}} \oplus \boldsymbol{H}^3, \end{aligned}$	$\begin{aligned} &= \{M + iF(\boldsymbol{ae}_4) \mid M \in \mathfrak{J}(3, \boldsymbol{H}), \boldsymbol{a} \in \boldsymbol{H}^3\} \\ &= \mathfrak{J}_{\boldsymbol{H}} \oplus i\boldsymbol{H}^3, \\ & (\mathfrak{J}^C)_{-\tau\gamma} = \{X \in \mathfrak{J}^C \mid \tau\gamma X = -X\} \\ &= \left\{ i \begin{pmatrix} \mu_1 & m_3 & \overline{m}_2 \\ \overline{m}_3 & \mu_2 & m_1 \\ m_2 & \overline{m}_1 & \mu_3 \end{pmatrix} + \begin{pmatrix} 0 & a_3 e_4 & -a_2 e_4 \\ -a_3 e_4 & 0 & a_1 e_4 \\ a_2 e_4 & -a_1 e_4 & 0 \end{pmatrix} \mid \begin{matrix} \mu_i \in \boldsymbol{R} \\ m_i, a_i \in \boldsymbol{H} \end{matrix} \right\} \\ &= \{iM + F(\boldsymbol{ae}_4) \mid M \in \mathfrak{J}(3, \boldsymbol{H}), \boldsymbol{a} \in \boldsymbol{H}^3\} \\ &= i\mathfrak{J}_{\boldsymbol{H}} \oplus \boldsymbol{H}^3, \end{aligned}$
9902.0431_E00348	56	0.842	$\begin{aligned} & \left(\alpha_1, \alpha_1 \right) = \left(\alpha_2, \alpha_2 \right) = \frac{1}{9}, \quad \left(\alpha_3, \alpha_3 \right) = \left(\alpha_4, \alpha_4 \right) = \frac{1}{18}, \\ & \left(\alpha_1, \alpha_2 \right) = -\frac{1}{18}, \quad \left(\alpha_2, \alpha_3 \right) = -\frac{1}{18}, \quad \left(\alpha_3, \alpha_4 \right) = -\frac{1}{36}, \\ & \left(\alpha_1, \alpha_3 \right) = \left(\alpha_1, \alpha_4 \right) = \left(\alpha_2, \alpha_4 \right) = 0, \\ & \left(-\mu, -\mu \right) = \frac{1}{9}, \quad \left(-\mu, \alpha_1 \right) = -\frac{1}{18}, \quad \left(-\mu, \alpha_i \right) = 0, \quad i = 2, 3, 4, \end{aligned}$	$\begin{aligned} & \left(\alpha_1, \alpha_1 \right) = \left(\alpha_2, \alpha_2 \right) = \frac{1}{9}, \quad \left(\alpha_3, \alpha_3 \right) = \left(\alpha_4, \alpha_4 \right) = \frac{1}{18} \\ & \left(\alpha_1, \alpha_2 \right) = -\frac{1}{18}, \quad \left(\alpha_2, \alpha_3 \right) = -\frac{1}{18}, \quad \left(\alpha_3, \alpha_4 \right) = -\frac{1}{36} \\ & \left(\alpha_1, \alpha_3 \right) = \left(\alpha_1, \alpha_4 \right) = \left(\alpha_2, \alpha_4 \right) = 0, \\ & \left(-\mu, -\mu \right) = \frac{1}{9}, \quad \left(-\mu, \alpha_1 \right) = -\frac{1}{18}, \quad \left(-\mu, \alpha_i \right) = 0, \quad i = 2, 3, 4 \end{aligned}$	$\begin{aligned} & \left(\alpha_1, \alpha_1 \right) = \left(\alpha_2, \alpha_2 \right) = \frac{1}{9}, \quad \left(\alpha_3, \alpha_3 \right) = \left(\alpha_4, \alpha_4 \right) = \frac{1}{18}, \\ & \left(\alpha_1, \alpha_2 \right) = -\frac{1}{18}, \quad \left(\alpha_2, \alpha_3 \right) = -\frac{1}{18}, \quad \left(\alpha_3, \alpha_4 \right) = -\frac{1}{36}, \\ & \left(\alpha_1, \alpha_3 \right) = \left(\alpha_1, \alpha_4 \right) = \left(\alpha_2, \alpha_4 \right) = 0, \\ & \left(-\mu, -\mu \right) = \frac{1}{9}, \quad \left(-\mu, \alpha_1 \right) = -\frac{1}{18}, \quad \left(-\mu, \alpha_i \right) = 0, \quad i = 2, 3, 4, \end{aligned}$
9902.0431_E00637	105	0.844	$\begin{aligned} & \left(\phi X, X, X \right) = 0, \left(\phi X, Y \right) + \left(X, \phi Y \right) = 0 \\ &= \left\{ \delta + i\tilde{T} \mid \delta \in \mathfrak{f}_{4,C}, T \in (\mathfrak{J}_C)_0 \right\} \end{aligned}$	$\begin{aligned} & \left(\phi X, X, X \right) = 0, \left(\phi X, Y \right) + \left(X, \phi Y \right) = 0 \\ &= \left\{ \delta + i\tilde{T} \mid \delta \in \mathfrak{f}_{4,C}, T \in (\mathfrak{J}_C)_0 \right\} \end{aligned}$	$\begin{aligned} & \left(\phi X, X, X \right) = 0, \left(\phi X, Y \right) + \left(X, \phi Y \right) = 0 \\ &= \left\{ \delta + i\tilde{T} \mid \delta \in \mathfrak{f}_{4,C}, T \in (\mathfrak{J}_C)_0 \right\} \end{aligned}$

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0902.0431_E00686	114	■ 0.844	$\begin{aligned} \left[\phi_1, \phi_2 \right] + 2A_1 \vee B_2 - 2A_2 \vee B_1 \\ A = \left(\phi_1 + \frac{2}{3}\nu_1 \right) A_2 - \left(\phi_2 + \frac{2}{3}\nu_2 \right) A_1 \\ B = - \left({}^t\phi_1 + \frac{2}{3}\nu_1 \right) B_2 + \left({}^t\phi_2 + \frac{2}{3}\nu_2 \right) B_1 \\ \nu = (A_1, B_2) - (B_1, A_2). \end{aligned}$	$\begin{cases} \phi = [\phi_1, \phi_2] + 2A_1 \vee B_2 - 2A_2 \vee B_1 \\ A = \left(\phi_1 + \frac{2}{3}\nu_1 \right) A_2 - \left(\phi_2 + \frac{2}{3}\nu_2 \right) A_1 \\ B = - \left({}^t\phi_1 + \frac{2}{3}\nu_1 \right) B_2 + \left({}^t\phi_2 + \frac{2}{3}\nu_2 \right) B_1 \\ \nu = (A_1, B_2) - (B_1, A_2). \end{cases}$	$\begin{cases} \phi = [\phi_1, \phi_2] + 2A_1 \vee B_2 - 2A_2 \vee B_1 \\ A = \left(\phi_1 + \frac{2}{3}\nu_1 \right) A_2 - \left(\phi_2 + \frac{2}{3}\nu_2 \right) A_1 \\ B = - \left({}^t\phi_1 + \frac{2}{3}\nu_1 \right) B_2 + \left({}^t\phi_2 + \frac{2}{3}\nu_2 \right) B_1 \\ \nu = (A_1, B_2) - (B_1, A_2). \end{cases}$
0902.0431_E00509	83	■ 0.847	$\begin{aligned} \left[h, \widetilde{F}_{23}(a) \right] &= \left[h_\delta, \widetilde{F}_{23}(a) \right] + \left[\widetilde{H}, \widetilde{F}_{23}(a) \right] \\ &= (h_\delta F_{23}(a))^\sim + \frac{1}{2} [H, F_{23}(a)]^\sim \quad (\text{Lemma 3.6.3}) \\ &= \widetilde{F}_{23}(h_\delta a) + \frac{1}{2} [H, F_{23}(a)]^\sim \\ &= \left(\lambda_k + \frac{1}{2} (\mu_2 - \mu_3) \right) \widetilde{F}_{23}(a), \end{aligned}$	$\begin{aligned} [h, \widetilde{F}_{23}(a)] &= [h_\delta, \widetilde{F}_{23}(a)] + [\widetilde{H}, \widetilde{F}_{23}(a)] \\ &= (h_\delta F_{23}(a))^\sim + \frac{1}{2} [H, F_{23}(a)]^\sim \quad (\text{Lemma 3.6.3}) \\ &= \widetilde{F}_{23}(h_\delta a) + \frac{1}{2} [H, F_{23}(a)]^\sim \\ &= \left(\lambda_k + \frac{1}{2} (\mu_2 - \mu_3) \right) \widetilde{F}_{23}(a), \end{aligned}$	$\begin{aligned} [h, \widetilde{F}_{23}(a)] &= [h_\delta, \widetilde{F}_{23}(a)] + [\widetilde{H}, \widetilde{F}_{23}(a)] \\ &= (h_\delta F_{23}(a))^\sim + \frac{1}{2} [H, F_{23}(a)]^\sim \quad (\text{Lemma 3.6.3}) \\ &= \widetilde{F}_{23}(h_\delta a) + \frac{1}{2} [H, F_{23}(a)]^\sim \\ &= \left(\lambda_k + \frac{1}{2} (\mu_2 - \mu_3) \right) \widetilde{F}_{23}(a), \end{aligned}$
0902.0431_E00091	18	■ 0.848	$\begin{aligned} \left[H_1, [H_1, S_1] \right] &= [H_1, S_2] = -S_1, & [H_1, [H_1, S_2]] &= [H_1, -S_1] = -S_2, \\ \left[H_1, [H_1, S_3] \right] &= [H_1, -S_4] = -S_3, & [H_1, [H_1, S_4]] &= [H_1, S_3] = -S_4, \\ \left[H_1, [H_1, S_5] \right] &= 0, & [H_1, [H_1, S_6]] &= 0 \end{aligned}$	$\begin{aligned} [H_1, [H_1, S_1]] &= [H_1, S_2] = -S_1, & [H_1, [H_1, S_2]] &= [H_1, -S_1] = -S_2, \\ [H_1, [H_1, S_3]] &= [H_1, -S_4] = -S_3, & [H_1, [H_1, S_4]] &= [H_1, S_3] = -S_4, \\ [H_1, [H_1, S_5]] &= 0, & [H_1, [H_1, S_6]] &= 0 \end{aligned}$	$\begin{aligned} [H_1, [H_1, S_1]] &= [H_1, S_2] = -S_1, & [H_1, [H_1, S_2]] &= [H_1, -S_1] = -S_2, \\ [H_1, [H_1, S_3]] &= [H_1, -S_4] = -S_3, & [H_1, [H_1, S_4]] &= [H_1, S_3] = -S_4, \\ [H_1, [H_1, S_5]] &= 0, & [H_1, [H_1, S_6]] &= 0 \end{aligned}$
0902.0431_E00683 (2)	113	■ 0.848	$\begin{aligned} \alpha P \wedge \alpha Q &= \frac{8}{3} ((\alpha P \times \alpha P) \alpha Q - (\alpha P \times \alpha Q) \alpha P) \quad (\text{Lemma 4.1.1.(2)}) \\ &= \frac{8}{3} (\alpha(P \times P) Q - \alpha(P \times Q) P) = \{P, Q\} \alpha P. \end{aligned}$	$\begin{aligned} \{\alpha P, \alpha Q\} \alpha P &= \frac{8}{3} ((\alpha P \times \alpha P) \alpha Q - (\alpha P \times \alpha Q) \alpha P) \quad (\text{Lemma 4.1.1.(2)}) \\ &= \frac{8}{3} (\alpha(P \times P) Q - \alpha(P \times Q) P) = \{P, Q\} \alpha P. \end{aligned}$	$\begin{aligned} \{\alpha P, \alpha Q\} \alpha P &= \frac{8}{3} ((\alpha P \times \alpha P) \alpha Q - (\alpha P \times \alpha Q) \alpha P) \quad (\text{Lemma 4.1.1.(2)}) \\ &= \frac{8}{3} (\alpha(P \times P) Q - \alpha(P \times Q) P) = \{P, Q\} \alpha P. \end{aligned}$
0902.0431_E00423	69	■ 0.848	$\beta X = B X B^* \quad \text{or} \quad \beta X = B \bar{X} B^*, \quad X \in \mathfrak{J}_C.$	$\beta X = B X B^* \quad \text{or} \quad \beta X = B \bar{X} B^*, \quad X \in \mathfrak{J}_C.$	$\beta X = B X B^* \quad \text{or} \quad \beta X = B \bar{X} B^*, \quad X \in \mathfrak{J}_C.$
0902.0431_E00995	165	■ 0.852	$E_8^C = \{ \alpha \in \text{Iso}_C(\mathfrak{e}_8^C) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \}.$	$E_8^C = \{ \alpha \in \text{Iso}_C(\mathfrak{e}_8^C) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \}.$	$E_8^C = \{ \alpha \in \text{Iso}_C(\mathfrak{e}_8^C) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \}.$

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0902.0431_E00772	130	■ 0.859	$\begin{aligned} & \widetilde{\alpha} P \times \widetilde{\alpha} Q = (\alpha X, \tau \alpha \tau Y, \xi, \eta) \times (\alpha Z, \tau \alpha \tau W, \zeta, \omega) \\ & = \cdots \left(\text{using } \alpha(X \vee Y) \alpha^{-1} = \alpha X \vee \tau \alpha \tau Y, (\alpha \phi \alpha^{-1})' = (\tau \alpha \tau) \phi' (\tau \alpha^{-1} \tau) \text{ etc.} \right) \cdots \\ & = \widetilde{\alpha} (P \times Q) \widetilde{\alpha}^{-1} \end{aligned}$	$\widetilde{\alpha} P \times \widetilde{\alpha} Q = (\alpha X, \tau \alpha \tau Y, \xi, \eta) \times (\alpha Z, \tau \alpha \tau W, \zeta, \omega)$ $= \cdots \left(\text{using } \alpha(X \vee Y) \alpha^{-1} = \alpha X \vee \tau \alpha \tau Y, (\alpha \phi \alpha^{-1})' = (\tau \alpha \tau) \phi' (\tau \alpha^{-1} \tau) \text{ etc.} \right) \cdots$ $= \widetilde{\alpha} (P \times Q) \widetilde{\alpha}^{-1}$	$\widetilde{\alpha} P \times \widetilde{\alpha} Q = (\alpha X, \tau \alpha \tau Y, \xi, \eta) \times (\alpha Z, \tau \alpha \tau W, \zeta, \omega)$ $= \cdots \left(\text{using } \alpha(X \vee Y) \alpha^{-1} = \alpha X \vee \tau \alpha \tau Y, (\alpha \phi \alpha^{-1})' = (\tau \alpha \tau) \phi' (\tau \alpha^{-1} \tau) \text{ etc.} \right) \cdots$ $= \widetilde{\alpha} (P \times Q) \widetilde{\alpha}^{-1}$
0902.0431_E00908	151	■ 0.860	$\alpha_1 \left(a_1 \right)^{-1} \alpha_2 \left(a_2 \right)^{-1} \alpha_3 \left(a_3 \right)^{-1} \beta \alpha \dot{1} = \dot{1},$	$\alpha_1 \left(a_1 \right)^{-1} \alpha_2 \left(a_2 \right)^{-1} \alpha_3 \left(a_3 \right)^{-1} \beta \alpha \dot{1} = \dot{1},$	$\alpha_1 \left(a_1 \right)^{-1} \alpha_2 \left(a_2 \right)^{-1} \alpha_3 \left(a_3 \right)^{-1} \beta \alpha \dot{1} = \dot{1},$
0902.0431_E01078	183	■ 0.862	$\begin{aligned} & \left[\begin{aligned} & \varphi_* \left((S_1 \tau S_2 - S_2 \tau S_1) - \frac{1}{8} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1) E \right) \\ & = 4 \left(\lambda \gamma \chi^{-1} S_1 \times \chi^{-1} S_2 - \lambda \gamma \chi^{-1} S_2 \times \chi^{-1} S_1 \right) \end{aligned} \right] \\ & \end{aligned}$	$\varphi_* \left((S_1 \tau S_2 - S_2 \tau S_1) - \frac{1}{8} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1) E \right)$ $= 4 \left(\lambda \gamma \chi^{-1} S_1 \times \chi^{-1} S_2 - \lambda \gamma \chi^{-1} S_2 \times \chi^{-1} S_1 \right)$	$\varphi_* \left((S_1 \tau S_2 - S_2 \tau S_1) - \frac{1}{8} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1) E \right)$ $= 4 \left(\lambda \gamma \chi^{-1} S_1 \times \chi^{-1} S_2 - \lambda \gamma \chi^{-1} S_2 \times \chi^{-1} S_1 \right).$
0902.0431_E01060	180	■ 0.863	$\mathfrak{W}_1 = \left\{ R \in \mathfrak{e}_8^C \mid R \times R = 0, \langle R, R \rangle = 4 \right\}$	$\mathfrak{W}_1 = \left\{ R \in \mathfrak{e}_8^C \mid R \times R = 0, \langle R, R \rangle = 4 \right\}$	$\mathfrak{W}_1 = \left\{ R \in \mathfrak{e}_8^C \mid R \times R = 0, \langle R, R \rangle = 4 \right\}$
0902.0431_E00535	90	■ 0.868	$\begin{aligned} & \det(\alpha X) = \det X = 1, \quad (\text{hence } \xi_i > 0, i = 1, 2, 3), \\ & \xi_1^2 + \xi_2^2 + \xi_3^2 = \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle = 3. \end{aligned}$	$\xi_1 \xi_2 \xi_3 = \det(\alpha X) = \det X = 1, \quad (\text{hence } \xi_i > 0, i = 1, 2, 3),$ $\xi_1^2 + \xi_2^2 + \xi_3^2 = \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle = 3.$	$\xi_1 \xi_2 \xi_3 = \det(\alpha X) = \det X = 1, \quad (\text{hence } \xi_i > 0, i = 1, 2, 3),$ $\xi_1^2 + \xi_2^2 + \xi_3^2 = \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle = 3.$
0902.0431_E00955	159	■ 0.868	$\begin{aligned} & f(P_C + (M + N)) = f(P_C) + \mu^{-1}(M + N), \\ & P_C + (M + N) \in (\mathfrak{P}_C)^C \oplus (M(3, \mathbf{C})^C \oplus M(3, \mathbf{C})^C) = \mathfrak{P}^C. \end{aligned}$	$f(P_C + (M + N)) = f(P_C) + \mu^{-1}(M + N),$ $P_C + (M + N) \in (\mathfrak{P}_C)^C \oplus (M(3, \mathbf{C})^C \oplus M(3, \mathbf{C})^C) = \mathfrak{P}^C.$	$f(P_C + (M + N)) = f(P_C) + \mu^{-1}(M + N),$ $P_C + (M + N) \in (\mathfrak{P}_C)^C \oplus (M(3, \mathbf{C})^C \oplus M(3, \mathbf{C})^C) = \mathfrak{P}^C.$

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0902.0431_E00521	86	■ 0.873	$\begin{aligned} B_6(h, h') &= \frac{4}{3} B_4 \left(\sum_{k=0}^3 \lambda_k H_k, \sum_{k=0}^3 \lambda_k' H_k \right) + 12 \left(\sum_{j=1}^3 \mu_j E_j, \sum_{j=1}^3 \mu_j' E_j \right) \\ &= \frac{4}{3} 18 \sum_{k=0}^3 \lambda_k \lambda_k' + 12 \sum_{j=1}^3 \mu_j \mu_j' \quad (\text{Theorem 2.6.2}) \\ &= 12 \left(2 \sum_{k=0}^3 \lambda_k \lambda_k' + \sum_{j=1}^3 \mu_j \mu_j' \right). \end{aligned}$	$B_6(h, h') = \frac{4}{3} B_4 \left(\sum_{k=0}^3 \lambda_k H_k, \sum_{k=0}^3 \lambda_k' H_k \right) + 12 \left(\sum_{j=1}^3 \mu_j E_j, \sum_{j=1}^3 \mu_j' E_j \right)$ $= \frac{4}{3} 18 \sum_{k=0}^3 \lambda_k \lambda_k' + 12 \sum_{j=1}^3 \mu_j \mu_j' \quad (\text{Theorem 2.6.2})$ $= 12 \left(2 \sum_{k=0}^3 \lambda_k \lambda_k' + \sum_{j=1}^3 \mu_j \mu_j' \right).$	
0902.0431_E00505	83	■ 0.876	$\mathfrak{e}_6^C = \mathfrak{f}_4^C \oplus \tilde{\mathfrak{J}}_0^C.$	$\mathfrak{e}_6^C = \mathfrak{f}_4^C \oplus \tilde{\mathfrak{J}}_0^C.$	$\mathfrak{e}_6^C = \mathfrak{f}_4^C \oplus \tilde{\mathfrak{J}}_0^C.$
0902.0431_E01105	187	■ 0.877	$z \left((E_8)^{\tilde{\lambda}\gamma} \right) = \{1, \tilde{\lambda}\gamma\}.$	$z \left((E_8)^{\tilde{\lambda}\gamma} \right) = \{1, \tilde{\lambda}\gamma\}.$	$z((E_8)^{\tilde{\lambda}\gamma}) = \{1, \tilde{\lambda}\gamma\}.$
0902.0431_E00279	45	■ 0.879	$\begin{aligned} &\left(D_1 x_1, \overline{x_2 x_3} \right) = - \left(x_1, D_1 (\overline{x_2 x_3}) \right) = - \left(x_1, \overline{(D_2 x_2) x_3 + x_2 (D_3 x_3)} \right) \\ &= - \left(x_1, \overline{x_3 (D_2 x_2) + (D_3 x_3) \overline{x_2}} \right) = - \left(x_3 x_1, \overline{D_2 x_2} \right) - \left(x_1 x_2, \overline{D_3 x_3} \right). \end{aligned}$	$\begin{aligned} (D_1 x_1, \overline{x_2 x_3}) &= - (x_1, D_1 (\overline{x_2 x_3})) = - \left(x_1, \overline{(D_2 x_2) x_3 + x_2 (D_3 x_3)} \right) \\ &= - \left(x_1, \overline{x_3 (D_2 x_2) + (D_3 x_3) \overline{x_2}} \right) = - (x_3 x_1, \overline{D_2 x_2}) - (x_1 x_2, \overline{D_3 x_3}). \end{aligned}$	$\begin{aligned} (D_1 x_1, \overline{x_2 x_3}) &= - (x_1, D_1 (\overline{x_2 x_3})) = - (x_1, \overline{(D_2 x_2) x_3 + x_2 (D_3 x_3)}) \\ &= - (x_1, \overline{x_3 (D_2 x_2) + (D_3 x_3) \overline{x_2}}) = - (x_3 x_1, \overline{D_2 x_2}) - (x_1 x_2, \overline{D_3 x_3}). \end{aligned}$
0902.0431_E00073	15	■ 0.881	$\begin{array}{l} e_1(E_{11} - E_{22}), \quad e_1(E_{22} - E_{33}), \quad E_{12} - E_{21}, \quad e_1(E_{12} + E_{21}), \\ e_{13} - E_{31}, \quad e_1(E_{13} + E_{31}), \quad E_{23} - E_{32}, \quad e_1(E_{23} + E_{32}) \end{array}$	$\begin{array}{l} e_1(E_{11} - E_{22}), \quad e_1(E_{22} - E_{33}), \quad E_{12} - E_{21}, \quad e_1(E_{12} + E_{21}), \\ E_{13} - E_{31}, \quad e_1(E_{13} + E_{31}), \quad E_{23} - E_{32}, \quad e_1(E_{23} + E_{32}) \end{array}$	$\begin{array}{l} e_1(E_{11} - E_{22}), \quad e_1(E_{22} - E_{33}), \quad E_{12} - E_{21}, \quad e_1(E_{12} + E_{21}), \\ E_{13} - E_{31}, \quad e_1(E_{13} + E_{31}), \quad E_{23} - E_{32}, \quad e_1(E_{23} + E_{32}) \end{array}$
0902.0431_E01126	191	■ 0.881	$\langle R_1, R_2 \rangle = 60 \text{tr}(D_1(\tau^t D_2)) + \frac{5}{2} B_6(\phi_1, \tau^t \phi_2 \tau) + 15 \langle \mathbf{X}_1, \mathbf{X}_2 \rangle + 15 \langle \mathbf{Y}_1, \mathbf{Y}_2 \rangle.$	$\langle R_1, R_2 \rangle = 60 \text{tr}(D_1(\tau^t D_2)) + \frac{5}{2} B_6(\phi_1, \tau^t \phi_2 \tau) + 15 \langle \mathbf{X}_1, \mathbf{X}_2 \rangle + 15 \langle \mathbf{Y}_1, \mathbf{Y}_2 \rangle.$	$\langle R_1, R_2 \rangle = 60 \text{tr}(D_1(\tau^t D_2)) + \frac{5}{2} B_6(\phi_1, \tau^t \phi_2 \tau) + 15 \langle \mathbf{X}_1, \mathbf{X}_2 \rangle + 15 \langle \mathbf{Y}_1, \mathbf{Y}_2 \rangle.$

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0902.0431_E00708	117	0.882	$\begin{aligned} & \{[\] \ \& \ \Phi, P \times P\} \ \& \ = \& \ \left[\right. \\ & \Phi(\phi, A, B, \nu), \Phi\left(-X \vee Y, -\frac{1}{2}(Y \times Y - \xi X), \frac{1}{2}(X \times X - \eta Y), \frac{1}{4}((X, Y) - 3\xi\eta)\right) \\ & \left. \frac{1}{2}(X \times Y - \xi X), \frac{1}{2}(X \times X - \eta Y), \frac{1}{4}((X, Y) - 3\xi\eta)\right\} \ \& \ = \& \ \\ & \cdots \left(\text{using } \phi(X \times Y) = \phi'X \times Y + X \times \phi'Y \text{ (Lemma 3.4.3.(1)), } [\phi, A \vee B] \right. \\ & \left. = \phi A \vee B + A \vee \phi' B \text{ (Lemma 3.4.4.(1)), the formula about } A \vee B \text{ (Lemma 3.4.1) etc.)} \cdots \right. \\ & \left. = 2\Phi P \times P. \right. \end{aligned}$	$[\Phi, P \times P]$ $= \left[\Phi(\phi, A, B, \nu), \Phi\left(-X \vee Y, -\frac{1}{2}(Y \times Y - \xi X), \frac{1}{2}(X \times X - \eta Y), \frac{1}{4}((X, Y) - 3\xi\eta)\right) \right]$ $= \cdots \left(\text{using } \phi(X \times Y) = \phi'X \times Y + X \times \phi'Y \text{ (Lemma 3.4.3.(1)), } [\phi, A \vee B] \right.$ $= \phi A \vee B + A \vee \phi' B \text{ (Lemma 3.4.4.(1)), the formula about } A \vee B \text{ (Lemma 3.4.1) etc.)} \cdots$ $= 2\Phi P \times P.$	
0902.0431_EQ1121	190	0.884	$\left\{ \begin{array}{l} D = \left[D_1, D_2 \right] + \frac{1}{4} \boldsymbol{X}_1 \cdot \boldsymbol{Y}_2 - \frac{1}{4} \boldsymbol{X}_2 \cdot \boldsymbol{Y}_1 \\ \phi = [\phi_1, \phi_2] + \frac{1}{2} \boldsymbol{X}_1 \vee \boldsymbol{Y}_2 - \frac{1}{2} \boldsymbol{X}_2 \vee \boldsymbol{Y}_1 \\ \boldsymbol{X} = \phi_1 \boldsymbol{X}_2 - \phi_2 \boldsymbol{X}_1 + D_1 \boldsymbol{X}_2 - D_2 \boldsymbol{X}_1 - \boldsymbol{Y}_1 \times \boldsymbol{Y}_2 \\ \boldsymbol{Y} = -{}^t \phi_1 \boldsymbol{Y}_2 + {}^t \phi_2 \boldsymbol{Y}_1 - {}^t D_1 \boldsymbol{Y}_2 + {}^t D_2 \boldsymbol{Y}_1 + \boldsymbol{X}_1 \times \boldsymbol{X}_2, \end{array} \right.$	$\left\{ \begin{array}{l} D = [D_1, D_2] + \frac{1}{4} \boldsymbol{X}_1 \cdot \boldsymbol{Y}_2 - \frac{1}{4} \boldsymbol{X}_2 \cdot \boldsymbol{Y}_1 \\ \phi = [\phi_1, \phi_2] + \frac{1}{2} \boldsymbol{X}_1 \vee \boldsymbol{Y}_2 - \frac{1}{2} \boldsymbol{X}_2 \vee \boldsymbol{Y}_1 \\ \boldsymbol{X} = \phi_1 \boldsymbol{X}_2 - \phi_2 \boldsymbol{X}_1 + D_1 \boldsymbol{X}_2 - D_2 \boldsymbol{X}_1 - \boldsymbol{Y}_1 \times \boldsymbol{Y}_2 \\ \boldsymbol{Y} = -{}^t \phi_1 \boldsymbol{Y}_2 + {}^t \phi_2 \boldsymbol{Y}_1 - {}^t D_1 \boldsymbol{Y}_2 + {}^t D_2 \boldsymbol{Y}_1 + \boldsymbol{X}_1 \times \boldsymbol{X}_2 \end{array} \right.$	$\left\{ \begin{array}{l} D = [D_1, D_2] + \frac{1}{4} \boldsymbol{X}_1 \cdot \boldsymbol{Y}_2 - \frac{1}{4} \boldsymbol{X}_2 \cdot \boldsymbol{Y}_1 \\ \phi = [\phi_1, \phi_2] + \frac{1}{2} \boldsymbol{X}_1 \vee \boldsymbol{Y}_2 - \frac{1}{2} \boldsymbol{X}_2 \vee \boldsymbol{Y}_1 \\ \boldsymbol{X} = \phi_1 \boldsymbol{X}_2 - \phi_2 \boldsymbol{X}_1 + D_1 \boldsymbol{X}_2 - D_2 \boldsymbol{X}_1 - \boldsymbol{Y}_1 \times \boldsymbol{Y}_2 \\ \boldsymbol{Y} = -{}^t \phi_1 \boldsymbol{Y}_2 + {}^t \phi_2 \boldsymbol{Y}_1 - {}^t D_1 \boldsymbol{Y}_2 + {}^t D_2 \boldsymbol{Y}_1 + \boldsymbol{X}_1 \times \boldsymbol{X}_2, \end{array} \right.$
0902.0431_EQ0821 (iii)	137	0.887	$\left\{ \begin{array}{l} 2\delta(X \times X) \times N = a(X)\beta X \\ \delta(X \times X) \times \delta(X \times X) = a(X)(\det X)M \\ (\beta X, \delta(X \times X)) + \det X = 3a(X)b(X \times X) \text{ (use (i))} \end{array} \right.$	$\left\{ \begin{array}{l} 2\delta(X \times X) \times N = a(X)\beta X \\ \delta(X \times X) \times \delta(X \times X) = a(X)(\det X)M \\ (\beta X, \delta(X \times X)) + \det X = 3a(X)b(X \times X) \text{ (use (i))} \end{array} \right.$	$\left\{ \begin{array}{l} 2\delta(X \times X) \times N = a(X)\beta X \\ \delta(X \times X) \times \delta(X \times X) = a(X)(\det X)M \\ (\beta X, \delta(X \times X)) + \det X = 3a(X)b(X \times X) \text{ (use (i))}. \end{array} \right.$

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0902.0431_E00172 (ii)	30	■ 0.890	$\overline{\bar{x}}(\bar{y}p) = (\overline{\bar{x}y})p, \quad x, y \in \mathfrak{C}.$	$\bar{x}(\bar{y}p) = (\overline{\bar{x}y})p, \quad x, y \in \mathfrak{C}.$	$\bar{x}(\bar{y}p) = (\overline{\bar{x}y})p, \quad x, y \in \mathfrak{C}.$
0902.0431_E00235	39	■ 0.891	$\begin{aligned} & \{ \}^{\{t\}} \alpha^{-1}(Y \times Y) \times \{ \}^{\{t\}} \alpha^{-1}(Y \times Y) = (\alpha Y \times \alpha Y) \times (\alpha Y \times \alpha Y) \quad (\text{Lemma 2.2.2.(4)}) \\ & = (\det \alpha Y) \alpha Y \quad (\text{Lemma 2.1.1.(3)}) = (\det Y) \alpha Y = \alpha((\det Y) Y) \\ & = \alpha((Y \times Y) \times (Y \times Y)), \quad Y \in \mathfrak{J}. \end{aligned}$	$\begin{aligned} & {}^t\alpha^{-1}(Y \times Y) \times {}^t\alpha^{-1}(Y \times Y) = (\alpha Y \times \alpha Y) \times (\alpha Y \times \alpha Y) \quad (\text{Lemma 2.2.2.(4)}) \\ & = (\det \alpha Y) \alpha Y \quad (\text{Lemma 2.1.1.(3)}) = (\det Y) \alpha Y = \alpha((\det Y) Y) \\ & = \alpha((Y \times Y) \times (Y \times Y)), \quad Y \in \mathfrak{J}. \end{aligned}$	$\begin{aligned} & {}^t\alpha^{-1}(Y \times Y) \times {}^t\alpha^{-1}(Y \times Y) = (\alpha Y \times \alpha Y) \times (\alpha Y \times \alpha Y) \quad (\text{Lemma 2.2.2.(4)}) \\ & = (\det \alpha Y) \alpha Y \quad (\text{Lemma 2.1.1.(3)}) = (\det Y) \alpha Y = \alpha((\det Y) Y) \\ & = \alpha((Y \times Y) \times (Y \times Y)), \quad Y \in \mathfrak{J}. \end{aligned}$
0902.0431_E00087	18	■ 0.892	$\begin{array}{l} \{\left[H_{11}, L_{12}\right]=2 L_{21},\} \\ \& \{\left[H_{11}, L_{21}\right]=-2 L_{12},\} \& \{\left[H_{11}, L_{13}\right]=L_{31}, \\ \& \{\left[H_{11}, L_{31}\right]=-L_{13},\} \& \{\left[H_{11}, L_{23}\right]=-L_{32}, \\ \& \{\left[H_{11}, L_{32}\right]=L_{23}\} . \end{array}$	$\begin{aligned} & [H_{11}, L_{12}] = 2L_{21}, \quad [H_{11}, L_{21}] = -2L_{12}, \quad [H_{11}, L_{13}] = L_{31}, \\ & [H_{11}, L_{31}] = -L_{13}, \quad [H_{11}, L_{23}] = -L_{32}, \quad [H_{11}, L_{32}] = L_{23}. \end{aligned}$	$\begin{aligned} & [H_{11}, L_{12}] = 2L_{21}, \quad [H_{11}, L_{21}] = -2L_{12}, \quad [H_{11}, L_{13}] = L_{31}, \\ & [H_{11}, L_{31}] = -L_{13}, \quad [H_{11}, L_{23}] = -L_{32}, \quad [H_{11}, L_{32}] = L_{23}. \end{aligned}$
0902.0431_E00417	68	■ 0.894	$\begin{aligned} & (X+M) \times (Y+N) \& = \left(X \times Y - \frac{1}{2}(M^*N + N^*M)\right) - \frac{1}{2}(MY + NX + \overline{M \times N}), \\ & (X+M, Y+N) = (X, Y) + 2(M, N), \\ & w(X+M) = X + \omega_1 M, \quad \left(\omega_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2}e_1\right), \end{aligned}$	$\begin{aligned} & (X+M) \times (Y+N) = \left(X \times Y - \frac{1}{2}(M^*N + N^*M)\right) - \frac{1}{2}(MY + NX + \overline{M \times N}), \\ & (X+M, Y+N) = (X, Y) + 2(M, N), \\ & w(X+M) = X + \omega_1 M, \quad \left(\omega_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2}e_1\right), \end{aligned}$	$\begin{aligned} & (X+M) \times (Y+N) = \left(X \times Y - \frac{1}{2}(M^*N + N^*M)\right) - \frac{1}{2}(MY + NX + \overline{M \times N}), \\ & (X+M, Y+N) = (X, Y) + 2(M, N), \\ & w(X+M) = X + \omega_1 M, \quad \left(\omega_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2}e_1\right), \end{aligned}$
0902.0431_E00690 (i)	114	■ 0.894	$\begin{cases} \Phi P \times P = 0, & P \in \mathfrak{M}^C \\ \{\Phi P, Q\} + \{P, \Phi Q\} = 0, & P, Q \in \mathfrak{P}^C. \end{cases}$	$\begin{cases} \Phi P \times P = 0, & P \in \mathfrak{M}^C \\ \{\Phi P, Q\} + \{P, \Phi Q\} = 0, & P, Q \in \mathfrak{P}^C. \end{cases}$	$\begin{cases} \Phi P \times P = 0, & P \in \mathfrak{M}^C \\ \{\Phi P, Q\} + \{P, \Phi Q\} = 0, & P, Q \in \mathfrak{P}^C. \end{cases}$
0902.0431_E00084	17	■ 0.895	$\mathfrak{g}_2 = \mathfrak{su}(3) \oplus \mathfrak{S} \quad (\text{Theorem 1.5.1}).$	$\mathfrak{g}_2 = \mathfrak{su}(3) \oplus \mathfrak{S} \quad (\text{Theorem 1.5.1}).$	$\mathfrak{g}_2 = \mathfrak{su}(3) \oplus \mathfrak{S} \quad (\text{Theorem 1.5.1}).$

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0902.0431_E00738	122	0.896	$\begin{aligned} & \begin{aligned} & \left(\Phi, P \times Q \right)_7 = \left(\Phi \begin{pmatrix} \phi \\ A \\ B \\ \nu \end{pmatrix}, \Phi \begin{pmatrix} -\frac{1}{2}(X \vee W + Z \vee Y) \\ -\frac{1}{4}(2Y \times W - \xi Z - \zeta X) \\ \frac{1}{4}(2X \times Z - \eta W - \omega Y) \\ \frac{1}{8}((X, W) + (Z, Y) - 3(\xi\omega + \zeta\eta)) \end{pmatrix} \right)_7 \\ & = (\phi, X \vee W + Z \vee Y)_6 - (A, 2X \times Z - \eta W - \omega Y) + (2Y \times W - \xi Z - \zeta X, B) \\ & \quad - \frac{1}{3}\nu((X, W) + (Z, Y) - 3(\xi\omega + \zeta\eta)) \\ & = (\phi X, W) + (\phi Z, Y) - 2(A, X, Z) + \eta(A, W) + \omega(A, Y) + 2(Y, W, B) \\ & \quad - \xi(Z, B) - \zeta(X, B) - \frac{1}{3}\nu(X, W) - \frac{1}{3}\nu(Z, Y) + \nu(\xi\omega + \zeta\eta) \\ & = \left\{ \begin{pmatrix} \phi X - \frac{1}{3}\nu X + 2B \times Y + \eta A \\ 2A \times X - {}^t\phi Y + \frac{1}{3}\nu Y + \xi B \\ (A, Y) + \nu\xi \\ (B, X) - \nu\eta \end{pmatrix}, \begin{pmatrix} Z \\ W \\ \zeta \\ \omega \end{pmatrix} \right\} = \{\Phi P, Q\}. \end{aligned} \end{aligned}$	$\begin{aligned} & (\Phi, P \times Q)_7 = \left(\Phi \begin{pmatrix} \phi \\ A \\ B \\ \nu \end{pmatrix}, \Phi \begin{pmatrix} -\frac{1}{2}(X \vee W + Z \vee Y) \\ -\frac{1}{4}(2Y \times W - \xi Z - \zeta X) \\ \frac{1}{4}(2X \times Z - \eta W - \omega Y) \\ \frac{1}{8}((X, W) + (Z, Y) - 3(\xi\omega + \zeta\eta)) \end{pmatrix} \right)_7 \\ & = (\phi, X \vee W + Z \vee Y)_6 - (A, 2X \times Z - \eta W - \omega Y) + (2Y \times W - \xi Z - \zeta X, B) \\ & \quad - \frac{1}{3}\nu((X, W) + (Z, Y) - 3(\xi\omega + \zeta\eta)) \\ & = (\phi X, W) + (\phi Z, Y) - 2(A, X, Z) + \eta(A, W) + \omega(A, Y) + 2(Y, W, B) \\ & \quad - \xi(Z, B) - \zeta(X, B) - \frac{1}{3}\nu(X, W) - \frac{1}{3}\nu(Z, Y) + \nu(\xi\omega + \zeta\eta) \\ & = \left\{ \begin{pmatrix} \phi X - \frac{1}{3}\nu X + 2B \times Y + \eta A \\ 2A \times X - {}^t\phi Y + \frac{1}{3}\nu Y + \xi B \\ (A, Y) + \nu\xi \\ (B, X) - \nu\eta \end{pmatrix}, \begin{pmatrix} Z \\ W \\ \zeta \\ \omega \end{pmatrix} \right\} = \{\Phi P, Q\}. \end{aligned}$	
0902.0431_E00122	23	0.896	$\alpha e_1 = e_1 \quad \text{or} \quad \alpha e_1 = -e_1.$	$\alpha e_1 = e_1 \quad \text{or} \quad \alpha e_1 = -e_1.$	$\alpha e_1 = e_1 \quad \text{or} \quad \alpha e_1 = -e_1.$
0902.0431_E00587	98	0.896	$\begin{aligned} & \begin{aligned} & \left(\phi \in \text{Hom}_C((\mathfrak{J}_H)^C) \mid (\phi M, M, M) = 0, \langle \phi M, N \rangle + \langle M, \phi N \rangle = 0 \right) \\ & = \left\{ \delta + i\tilde{T} \mid \delta \in \mathfrak{f}_{4, H}, T \in (\mathfrak{J}_H)_0 \right\} \end{aligned} \end{aligned}$	$\begin{aligned} \mathfrak{e}_{6, H} &= \left\{ \phi \in \text{Hom}_C((\mathfrak{J}_H)^C) \mid (\phi M, M, M) = 0, \langle \phi M, N \rangle + \langle M, \phi N \rangle = 0 \right\} \\ &= \left\{ \delta + i\tilde{T} \mid \delta \in \mathfrak{f}_{4, H}, T \in (\mathfrak{J}_H)_0 \right\} \end{aligned}$	$\begin{aligned} \mathfrak{e}_{6, H} &= \left\{ \phi \in \text{Hom}_C((\mathfrak{J}_H)^C) \mid (\phi M, M, M) = 0, \langle \phi M, N \rangle + \langle M, \phi N \rangle = 0 \right\} \\ &= \left\{ \delta + i\tilde{T} \mid \delta \in \mathfrak{f}_{4, H}, T \in (\mathfrak{J}_H)_0 \right\}, \end{aligned}$
0902.0431_E00615	102	0.896	$\langle gX, gY \rangle = \langle X, Y \rangle, \quad X, Y \in \mathfrak{J}^C.$	$\langle gX, gY \rangle = \langle X, Y \rangle, \quad X, Y \in \mathfrak{J}^C.$	$\langle gX, gY \rangle = \langle X, Y \rangle, \quad X, Y \in \mathfrak{J}^C.$

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0902.0431_E00307	49	■ 0.899	$\begin{aligned} & \&=\delta(A\circ(B\circ X)-B\circ(A\circ X))-(A\circ(B\circ\delta X)-B\circ(A\circ\delta X)) \\ & \&=\delta A\circ(B\circ X)+A\circ(\delta B\circ X)+A\circ(B\circ\delta X)-\delta B\circ(A\circ X) \\ & \&-B\circ(\delta A\circ X)-B\circ(A\circ\delta X)-A\circ(B\circ\delta X)+B\circ(A\circ\delta X) \\ & \&=\delta A\circ(B\circ X)-B\circ(\delta A\circ X)+A\circ(\delta B\circ X)-\delta B\circ(A\circ X) \\ & \&=[\widetilde{\delta A},\widetilde{B}]X+[\widetilde{A},\widetilde{\delta B}]X, \quad X\in\mathfrak{J}^C. \end{aligned}$	$\begin{aligned} & =\delta(A\circ(B\circ X)-B\circ(A\circ X))-(A\circ(B\circ\delta X)-B\circ(A\circ\delta X)) \\ & =\delta A\circ(B\circ X)+A\circ(\delta B\circ X)+A\circ(B\circ\delta X)-\delta B\circ(A\circ X) \\ & -B\circ(\delta A\circ X)-B\circ(A\circ\delta X)-A\circ(B\circ\delta X)+B\circ(A\circ\delta X) \\ & =\delta A\circ(B\circ X)-B\circ(\delta A\circ X)+A\circ(\delta B\circ X)-\delta B\circ(A\circ X) \\ & =[\widetilde{\delta A},\widetilde{B}]X+[\widetilde{A},\widetilde{\delta B}]X, \quad X\in\mathfrak{J}^C. \end{aligned}$	$\begin{aligned} & =\delta(A\circ(B\circ X)-B\circ(A\circ X))-(A\circ(B\circ\delta X)-B\circ(A\circ\delta X)) \\ & =\delta A\circ(B\circ X)+A\circ(\delta B\circ X)+A\circ(B\circ\delta X)-\delta B\circ(A\circ X) \\ & -B\circ(\delta A\circ X)-B\circ(A\circ\delta X)-A\circ(B\circ\delta X)+B\circ(A\circ\delta X) \\ & =\delta A\circ(B\circ X)-B\circ(\delta A\circ X)+A\circ(\delta B\circ X)-\delta B\circ(A\circ X) \\ & =[\widetilde{\delta A},\widetilde{B}]X+[\widetilde{A},\widetilde{\delta B}]X, \quad X\in\mathfrak{J}^C. \end{aligned}$
0902.0431_E01108	188	■ 0.902	$\left(\mathfrak{e}_8^C\right)^{\widetilde{\lambda}_\gamma}=\left\{\sum_{k,l}[R_k,R_l] \mid R_k,R_l\in\left(\mathfrak{e}_8^C\right)_{-\widetilde{\lambda}_\gamma}\right\}.$	$\left(\mathfrak{e}_8^C\right)^{\widetilde{\lambda}_\gamma}=\left\{\sum_{k,l}[R_k,R_l] \mid R_k,R_l\in\left(\mathfrak{e}_8^C\right)_{-\widetilde{\lambda}_\gamma}\right\}.$	$\left(\mathfrak{e}_8^C\right)^{\widetilde{\lambda}_\gamma}=\left\{\sum_{k,l}[R_k,R_l] \mid R_k,R_l\in\left(\mathfrak{e}_8^C\right)_{-\widetilde{\lambda}_\gamma}\right\}.$
0902.0431_E00372	62	■ 0.903	$\begin{aligned} & \&\xi_1+\xi_2+\xi_3=\mathrm{tr}(X) \\ & \&\xi_1^2+\xi_2^2+\xi_3^2=\mathrm{tr}(X,X) \\ & \&\xi_1^3+\xi_2^3+\xi_3^3=\mathrm{tr}(X,X,X) \end{aligned}$	$\begin{cases} \xi_1+\xi_2+\xi_3=\mathrm{tr}(X) \\ \xi_1^2+\xi_2^2+\xi_3^2=\mathrm{tr}(X,X) \\ \xi_1^3+\xi_2^3+\xi_3^3=\mathrm{tr}(X,X,X) \end{cases}$	$\begin{cases} \xi_1+\xi_2+\xi_3=\mathrm{tr}(X) \\ \xi_1^2+\xi_2^2+\xi_3^2=\mathrm{tr}(X,X) \\ \xi_1^3+\xi_2^3+\xi_3^3=\mathrm{tr}(X,X,X). \end{cases}$
0902.0431_E00023	9	■ 0.904	$\begin{aligned} & \&[F_{ij},F_{jk}]=F_{ik}, \quad i,j,k \text{ are distinct} \\ & \&[F_{ij},F_{kl}]=0, \quad i,j,k,l \text{ are distinct.} \end{aligned}$	$\begin{aligned} & [F_{ij},F_{jk}]=F_{ik}, \quad i,j,k \text{ are distinct} \\ & [F_{ij},F_{kl}]=0, \quad i,j,k,l \text{ are distinct.} \end{aligned}$	$\begin{aligned} & [F_{ij},F_{jk}]=F_{ik}, \quad i,j,k \text{ are distinct,} \\ & [F_{ij},F_{kl}]=0, \quad i,j,k,l \text{ are distinct.} \end{aligned}$
0902.0431_E01106	187	■ 0.904	$\alpha R=kR, \quad R\in\left(\mathfrak{e}_8^C\right)_{-\widetilde{\lambda}_\gamma}.$	$\alpha R=kR, \quad R\in\left(\mathfrak{e}_8^C\right)_{-\widetilde{\lambda}_\gamma}.$	$\alpha R=kR, \quad R\in\left(\mathfrak{e}_8^C\right)_{-\widetilde{\lambda}_\gamma}.$
0902.0431_E00947	157	■ 0.904	$\sum(A\boldsymbol{a}\wedge B\boldsymbol{a}\wedge C\boldsymbol{a})=\overline{a}\wedge\overline{b}\wedge\overline{c}, \quad a,b,c\in C^6$	$\sum(A\boldsymbol{a}\wedge B\boldsymbol{a}\wedge C\boldsymbol{a})=\overline{a}\wedge\overline{b}\wedge\overline{c}, \quad a,b,c\in C^6$	$\sum(A\boldsymbol{a}\wedge B\boldsymbol{a}\wedge C\boldsymbol{a})=\overline{a}\wedge\overline{b}\wedge\overline{c}, \quad a,b,c\in C^6$
0902.0431_E01009	168	■ 0.905	$U(248)=U\left(\mathfrak{e}_8^C\right)=\left\{\alpha\in\mathrm{Iso}_C\left(\mathfrak{e}_8^C\right)\mid\langle\alpha R_1,\alpha R_2\rangle=\langle R_1,R_2\rangle\right\}.$	$U(248)=U\left(\mathfrak{e}_8^C\right)=\left\{\alpha\in\mathrm{Iso}_C\left(\mathfrak{e}_8^C\right)\mid\langle\alpha R_1,\alpha R_2\rangle=\langle R_1,R_2\rangle\right\}.$	$U(248)=U\left(\mathfrak{e}_8^C\right)=\left\{\alpha\in\mathrm{Iso}_C\left(\mathfrak{e}_8^C\right)\mid\langle\alpha R_1,\alpha R_2\rangle=\langle R_1,R_2\rangle\right\}.$
0902.0431_E00981	163	■ 0.905	$\begin{aligned} & \&\mathfrak{a}\ni[1,[1_-, [1,R]]]=[1,[1_-, (0,P,-Q,0,2s,-2t)]] \\ & \&=[1,(0,0,P,-2s,0,0)]=-(0,0,P,0,0,0)=-P_- \end{aligned}$	$\begin{aligned} & \mathfrak{a}\ni[1,[1_-, [1,R]]]=[1,[1_-, (0,P,-Q,0,2s,-2t)]] \\ & = [1,(0,0,P,-2s,0,0)]=-(0,0,P,0,0,0)=-P_- \end{aligned}$	$\begin{aligned} & \mathfrak{a}\ni[1,[1_-, [1,R]]]=[1,[1_-, (0,P,-Q,0,2s,-2t)]] \\ & = [1,(0,0,P,-2s,0,0)]=-(0,0,P,0,0,0)=-P_- . \end{aligned}$
0902.0431_E00459	75	■ 0.908	$\mathfrak{a}\ni\left[\delta_1,\delta+\widetilde{T}\right]=[\delta_1,\delta]+\widetilde{\delta_1T} \text{ (Theorem 3.2.2),}$	$\mathfrak{a}\ni\left[\delta_1,\delta+\widetilde{T}\right]=[\delta_1,\delta]+\widetilde{\delta_1T} \text{ (Theorem 3.2.2),}$	$\mathfrak{a}\ni[\delta_1,\delta+\widetilde{T}]=[\delta_1,\delta]+\widetilde{\delta_1T} \text{ (Theorem 3.2.2),}$

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0902.0431_E01190	200	■ 0.909	$\begin{gathered} (*(*\boldsymbol{a} \wedge \boldsymbol{x}) \times \boldsymbol{y})\boldsymbol{v} = (\boldsymbol{y}, \boldsymbol{v}) * (*\boldsymbol{a} \wedge \boldsymbol{x}) - \frac{1}{5}(*\boldsymbol{a} \wedge \boldsymbol{x}), \boldsymbol{v})\boldsymbol{y}, \\ = (\boldsymbol{y}, \boldsymbol{v}) * (*\boldsymbol{a} \wedge \boldsymbol{x}) - \frac{1}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{v})\boldsymbol{y}, \\ -(*(*\boldsymbol{a} \wedge \boldsymbol{x}) \times \boldsymbol{y})\boldsymbol{v} = -(\boldsymbol{x}, \boldsymbol{v}) * (*\boldsymbol{a} \wedge \boldsymbol{y}) - \frac{1}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{v})\boldsymbol{y}. \end{gathered}$	$\begin{aligned} &(*(*\boldsymbol{a} \wedge \boldsymbol{x}) \times \boldsymbol{y})\boldsymbol{v} = (\boldsymbol{y}, \boldsymbol{v}) * (*\boldsymbol{a} \wedge \boldsymbol{x}) - \frac{1}{5}(*(*\boldsymbol{a} \wedge \boldsymbol{x}), \boldsymbol{v})\boldsymbol{y} \\ &= (\boldsymbol{y}, \boldsymbol{v}) * (*\boldsymbol{a} \wedge \boldsymbol{x}) - \frac{1}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{v})\boldsymbol{y} \\ &-(*(*\boldsymbol{a} \wedge \boldsymbol{x}) \times \boldsymbol{y})\boldsymbol{v} = -(\boldsymbol{x}, \boldsymbol{v}) * (*\boldsymbol{a} \wedge \boldsymbol{y}) - \frac{1}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{v})\boldsymbol{y} \end{aligned}$	$\begin{aligned} &(*(*\boldsymbol{a} \wedge \boldsymbol{x}) \times \boldsymbol{y})\boldsymbol{v} = (\boldsymbol{y}, \boldsymbol{v}) * (*\boldsymbol{a} \wedge \boldsymbol{x}) - \frac{1}{5}(*(*\boldsymbol{a} \wedge \boldsymbol{x}), \boldsymbol{v})\boldsymbol{y}, \\ &= (\boldsymbol{y}, \boldsymbol{v}) * (*\boldsymbol{a} \wedge \boldsymbol{x}) - \frac{1}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{v})\boldsymbol{y}, \\ &-(*(*\boldsymbol{a} \wedge \boldsymbol{x}) \times \boldsymbol{y})\boldsymbol{v} = -(\boldsymbol{x}, \boldsymbol{v}) * (*\boldsymbol{a} \wedge \boldsymbol{y}) - \frac{1}{5}(\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{v})\boldsymbol{y}. \end{aligned}$
0902.0431_E01172	197	■ 0.911	$\left(\mathfrak{e}_8\right)^{w_3} = \left\{R \in \mathfrak{e}_8^C \mid \tau \tilde{\lambda} R = R, w_3 R = R\right\} = \{(D, 0, 0) \in \mathfrak{e}_8^C \mid D \in \mathfrak{su}(9)\} \cong \mathfrak{su}(9),$	$(\mathfrak{e}_8)^{w_3} = \left\{R \in \mathfrak{e}_8^C \mid \tau \tilde{\lambda} R = R, w_3 R = R\right\} = \{(D, 0, 0) \in \mathfrak{e}_8^C \mid D \in \mathfrak{su}(9)\} \cong \mathfrak{su}(9),$	$(\mathfrak{e}_8)^{w_3} = \left\{R \in \mathfrak{e}_8^C \mid \tau \tilde{\lambda} R = R, w_3 R = R\right\} = \{(D, 0, 0) \in \mathfrak{e}_8^C \mid D \in \mathfrak{su}(9)\} \cong \mathfrak{su}(9),$
0902.0431_E00263	43	■ 0.912	$\begin{aligned} &a_{11} = -\{x_{12}, x_{23}, x_{31}\} - \{x_{13}, x_{32}, x_{21}\}, \\ &a_{22} = -\{x_{21}, x_{13}, x_{32}\} - \{x_{23}, x_{31}, x_{12}\}, \\ &a_{33} = -\{x_{31}, x_{12}, x_{23}\} - \{x_{32}, x_{21}, x_{13}\}, \end{aligned}$	$\begin{aligned} &a_{11} = -\{x_{12}, x_{23}, x_{31}\} - \{x_{13}, x_{32}, x_{21}\}, \\ &a_{22} = -\{x_{21}, x_{13}, x_{32}\} - \{x_{23}, x_{31}, x_{12}\}, \\ &a_{33} = -\{x_{31}, x_{12}, x_{23}\} - \{x_{32}, x_{21}, x_{13}\}, \end{aligned}$	$\begin{aligned} &a_{11} = -\{x_{12}, x_{23}, x_{31}\} - \{x_{13}, x_{32}, x_{21}\}, \\ &a_{22} = -\{x_{21}, x_{13}, x_{32}\} - \{x_{23}, x_{31}, x_{12}\}, \\ &a_{33} = -\{x_{31}, x_{12}, x_{23}\} - \{x_{32}, x_{21}, x_{13}\}, \end{aligned}$
0902.0431_E00452	74	■ 0.914	$\begin{aligned} &[\delta, \tilde{T}]X = \delta(T \circ X) - T \circ \delta X = \delta T \circ X + T \circ \delta X - T \circ \delta X \\ &= \delta T \circ X = \widetilde{\delta T} X, \quad X \in \mathfrak{J}^C. \end{aligned}$	$\begin{aligned} &[\delta, \tilde{T}]X = \delta(T \circ X) - T \circ \delta X = \delta T \circ X + T \circ \delta X - T \circ \delta X \\ &= \delta T \circ X = \widetilde{\delta T} X, \quad X \in \mathfrak{J}^C. \end{aligned}$	$\begin{aligned} &[\delta, \tilde{T}]X = \delta(T \circ X) - T \circ \delta X = \delta T \circ X + T \circ \delta X - T \circ \delta X \\ &= \delta T \circ X = \widetilde{\delta T} X, \quad X \in \mathfrak{J}^C. \end{aligned}$
0902.0431_E00689	114	■ 0.914	$\left\{ \begin{array}{l} \alpha P \times \alpha P = 0, \\ \{\alpha P, \alpha Q\} = \{P, Q\}, \end{array} \right. \quad \begin{array}{l} P \in \mathfrak{M}^C \\ P, Q \in \mathfrak{P}^C \end{array} \quad (\text{Proposition 4.2.2.(2)}),$	$\left\{ \begin{array}{l} \alpha P \times \alpha P = 0, \\ \{\alpha P, \alpha Q\} = \{P, Q\}, \end{array} \right. \quad \begin{array}{l} P \in \mathfrak{M}^C \\ P, Q \in \mathfrak{P}^C \end{array} \quad (\text{Proposition 4.2.2.(2)}),$	$\left\{ \begin{array}{l} \alpha P \times \alpha P = 0, \\ \{\alpha P, \alpha Q\} = \{P, Q\}, \end{array} \right. \quad \begin{array}{l} P \in \mathfrak{M}^C \\ P, Q \in \mathfrak{P}^C \end{array} \quad (\text{Proposition 4.2.2.(2)}),$
0902.0431_E00344	55	■ 0.915	$\begin{aligned} &HE_i = 0, \quad i = 1, 2, 3 \\ &HF_1(x) = F_1(Hx), HF_2(x) = F_2((\nu H)x), HF_3(x) = F_3((\kappa \pi H)x), \end{aligned}$	$\begin{aligned} &HE_i = 0, \quad i = 1, 2, 3 \\ &HF_1(x) = F_1(Hx), HF_2(x) = F_2((\nu H)x), HF_3(x) = F_3((\kappa \pi H)x), \end{aligned}$	$\begin{aligned} &HE_i = 0, \quad i = 1, 2, 3, \\ &HF_1(x) = F_1(Hx), HF_2(x) = F_2((\nu H)x), HF_3(x) = F_3((\kappa \pi H)x), \end{aligned}$

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0902.0431_E00508	83	■ 0.928	$\begin{aligned} \{[h, S]\} &= \left[h_{\delta} + \widetilde{H}, S\right] - [S, \widetilde{H}] \\ &= \alpha(h_{\delta})S - \widetilde{S\widetilde{H}} \text{ (Lemma 3.6.3.(1))} \\ &= \alpha(h_{\delta})S = (\pm\lambda_k \pm \lambda_l)S \end{aligned}$	$\begin{aligned} [h, S] &= \left[h_{\delta} + \widetilde{H}, S\right] = [h_{\delta}, S] - [S, \widetilde{H}] \\ &= \alpha(h_{\delta})S - \widetilde{S\widetilde{H}} \text{ (Lemma 3.6.3.(1))} \\ &= \alpha(h_{\delta})S = (\pm\lambda_k \pm \lambda_l)S \end{aligned}$	$\begin{aligned} [h, S] &= [h_{\delta} + \widetilde{H}, S] = [h_{\delta}, S] - [S, \widetilde{H}] \\ &= \alpha(h_{\delta})S - \widetilde{S\widetilde{H}} \text{ (Lemma 3.6.3.(1))} \\ &= \alpha(h_{\delta})S = (\pm\lambda_k \pm \lambda_l)S. \end{aligned}$
0902.0431_E00665	110	■ 0.935	$E_{\{6(6)\}} \cong \left(E_{\{6\}}^C\right)^{\tau\gamma}, E_{\{6(2)\}} \cong \left(E_{\{6\}}^C\right)^{\tau\lambda\gamma}, E_{\{6(-14)\}} \cong \left(E_{\{6\}}^C\right)^{\tau\lambda\sigma}, E_{\{6(-26)\}} \cong \left(E_{\{6\}}^C\right)^{\tau}.$	$E_{6(6)} \cong (E_6^C)^{\tau\gamma}, E_{6(2)} \cong (E_6^C)^{\tau\lambda\gamma}, E_{6(-14)} \cong (E_6^C)^{\tau\lambda\sigma}, E_{6(-26)} \cong (E_6^C)^{\tau}.$	$E_{6(6)} \cong (E_6^C)^{\tau\gamma}, E_{6(2)} \cong (E_6^C)^{\tau\lambda\gamma}, E_{6(-14)} \cong (E_6^C)^{\tau\lambda\sigma}, E_{6(-26)} \cong (E_6^C)^{\tau}.$
0902.0431_E01052	179	■ 0.936	$\left\{\beta_2 P, \beta_3 Q\right\} = \left\{\beta_2 P, \beta_2 Q\right\}, \quad P, Q \in \mathfrak{P}^C,$	$\{\beta_2 P, \beta_3 Q\} = \{\beta_2 P, \beta_2 Q\}, \quad P, Q \in \mathfrak{P}^C,$	$\{\beta_2 P, \beta_3 Q\} = \{\beta_2 P, \beta_2 Q\}, \quad P, Q \in \mathfrak{P}^C,$
0902.0431_E00512	84	■ 0.938	$\begin{aligned} \nu h_{\delta} &= \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3)H_0 + \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &+ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3, \\ \kappa\pi h_{\delta} &= \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3)H_0 + \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &+ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3 \end{aligned}$	$\begin{aligned} \nu h_{\delta} &= \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3)H_0 + \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &+ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3 \\ \kappa\pi h_{\delta} &= \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3)H_0 + \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &+ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3 \end{aligned}$	$\begin{aligned} \nu h_{\delta} &= \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3)H_0 + \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &+ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3, \\ \kappa\pi h_{\delta} &= \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3)H_0 + \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &+ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3 \end{aligned}$

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0902.0431_EQ1175	198	■ 0.940	$\begin{array}{ll} \{\left[\mathfrak{g}_{\{0\}}, \mathfrak{g}_{\{0\}}\right] \subset \mathfrak{g}_{\{0\}} \text{ \& } \left[\left(C_{\{1\}}, D_{\{1\}}\right), \left(C_{\{2\}}, D_{\{2\}}\right)\right] = \left[\left(C_{\{1\}}, C_{\{2\}}\right), \left(D_{\{1\}}, D_{\{2\}}\right)\right], \\ \left[\left(C, D\right), \boldsymbol{x} \otimes \boldsymbol{a}\right] = \left(C \boldsymbol{x}\right) \otimes \boldsymbol{a} + \boldsymbol{x} \otimes \left(D \boldsymbol{a}\right), \\ \left[\left(C, D\right), \boldsymbol{b} \otimes \boldsymbol{y}\right] = \left(C \boldsymbol{b}\right) \otimes \boldsymbol{y} + \boldsymbol{b} \otimes \left(-{}^t D \boldsymbol{a}\right), \\ \left[\left(C, D\right), \boldsymbol{c} \otimes \boldsymbol{z}\right] = \left(-{}^t C \boldsymbol{c}\right) \otimes \boldsymbol{z} + \boldsymbol{c} \otimes \left(D \boldsymbol{z}\right), \\ \left[\left(C, D\right), \boldsymbol{w} \otimes \boldsymbol{d}\right] = \left(-{}^t C \boldsymbol{w}\right) \otimes \boldsymbol{z} + \boldsymbol{c} \otimes \left(-{}^t D \boldsymbol{z}\right), \\ \left[\boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{w} \otimes \boldsymbol{d}\right] = \left(-\left(\boldsymbol{a}, \boldsymbol{d}\right) \boldsymbol{x} \times \boldsymbol{w},\left(\boldsymbol{x}, \boldsymbol{w}\right) \boldsymbol{a} \times \boldsymbol{d}\right), \\ \left[\boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}\right] = \left(\left(\boldsymbol{y}, \boldsymbol{z}\right) \boldsymbol{b} \times \boldsymbol{c},\left(\boldsymbol{b}, \boldsymbol{c}\right) \boldsymbol{z} \times \boldsymbol{y}\right), \\ \left[\boldsymbol{x}_1 \otimes \boldsymbol{a}_1, \boldsymbol{x}_2 \otimes \boldsymbol{a}_2\right] = \left(\boldsymbol{x}_1 \wedge \boldsymbol{x}_2\right) \otimes *\left(\boldsymbol{a}_1 \wedge \boldsymbol{a}_2\right), \\ \left[\boldsymbol{b}_1 \otimes \boldsymbol{y}_1, \boldsymbol{b}_2 \otimes \boldsymbol{y}_2\right] = *\left(\boldsymbol{b}_1 \wedge \boldsymbol{b}_2\right) \otimes \left(\boldsymbol{y}_1 \wedge \boldsymbol{y}_2\right), \\ \left[\boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}\right] = *\left(\boldsymbol{b} \wedge \boldsymbol{x}\right) \otimes *\left(* \boldsymbol{a} \wedge \boldsymbol{y}\right), \\ \left[\boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{w} \otimes \boldsymbol{d}\right] = *\left(* \boldsymbol{b} \wedge \boldsymbol{w}\right) \otimes *\left(\boldsymbol{d} \wedge \boldsymbol{y}\right). \end{array}$	$\begin{array}{ll} \left[\mathfrak{g}_0, \mathfrak{g}_0\right] \subset \mathfrak{g}_0 & \left[\left(C_1, D_1\right),\left(C_2, D_2\right)\right]=\left[\left(C_1, C_2\right),\left[D_1, D_2\right]\right), \\ \left[\mathfrak{g}_0, \mathfrak{g}_1\right] \subset \mathfrak{g}_1 & \left[\left(C, D\right), \boldsymbol{x} \otimes \boldsymbol{a}\right]=\left(C \boldsymbol{x}\right) \otimes \boldsymbol{a}+\boldsymbol{x} \otimes\left(D \boldsymbol{a}\right), \\ \left[\mathfrak{g}_0, \mathfrak{g}_2\right] \subset \mathfrak{g}_2 & \left[\left(C, D\right), \boldsymbol{b} \otimes \boldsymbol{y}\right]=\left(C \boldsymbol{b}\right) \otimes \boldsymbol{y}+\boldsymbol{b} \otimes\left(-{}^t D \boldsymbol{a}\right), \\ \left[\mathfrak{g}_0, \mathfrak{g}_{-2}\right] \subset \mathfrak{g}_{-2} & \left[\left(C, D\right), \boldsymbol{c} \otimes \boldsymbol{z}\right]=\left(-{}^t C \boldsymbol{c}\right) \otimes \boldsymbol{z}+\boldsymbol{c} \otimes\left(D \boldsymbol{z}\right), \\ \left[\mathfrak{g}_0, \mathfrak{g}_{-1}\right] \subset \mathfrak{g}_{-1} & \left[\left(C, D\right), \boldsymbol{w} \otimes \boldsymbol{d}\right]=\left(-{}^t C \boldsymbol{w}\right) \otimes \boldsymbol{z}+\boldsymbol{c} \otimes\left(-{}^t D \boldsymbol{z}\right), \\ \left[\mathfrak{g}_1, \mathfrak{g}_{-1}\right] \subset \mathfrak{g}_0 & \left[\boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{w} \otimes \boldsymbol{d}\right]=\left(-\left(\boldsymbol{a}, \boldsymbol{d}\right) \boldsymbol{x} \times \boldsymbol{w},\left(\boldsymbol{x}, \boldsymbol{w}\right) \boldsymbol{a} \times \boldsymbol{d}\right), \\ \left[\mathfrak{g}_2, \mathfrak{g}_{-2}\right] \subset \mathfrak{g}_0 & \left[\boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}\right]=\left(\left(\boldsymbol{y}, \boldsymbol{z}\right) \boldsymbol{b} \times \boldsymbol{c},\left(\boldsymbol{b}, \boldsymbol{c}\right) \boldsymbol{z} \times \boldsymbol{y}\right), \\ \left[\mathfrak{g}_1, \mathfrak{g}_1\right] \subset \mathfrak{g}_2 & \left[\boldsymbol{x}_1 \otimes \boldsymbol{a}_1, \boldsymbol{x}_2 \otimes \boldsymbol{a}_2\right]=\left(\boldsymbol{x}_1 \wedge \boldsymbol{x}_2\right) \otimes *\left(\boldsymbol{a}_1 \wedge \boldsymbol{a}_2\right), \\ \left[\mathfrak{g}_{-1}, \mathfrak{g}_{-1}\right] \subset \mathfrak{g}_{-2} & \left[\boldsymbol{b}_1 \otimes \boldsymbol{y}_1, \boldsymbol{b}_2 \otimes \boldsymbol{y}_2\right]=*\left(\boldsymbol{b}_1 \wedge \boldsymbol{b}_2\right) \otimes\left(\boldsymbol{y}_1 \wedge \boldsymbol{y}_2\right), \\ \left[\mathfrak{g}_2, \mathfrak{g}_2\right] \subset \mathfrak{g}_{-1} & \left[\boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}\right]=*\left(\boldsymbol{b} \wedge \boldsymbol{x}\right) \otimes *\left(* \boldsymbol{a} \wedge \boldsymbol{y}\right), \\ \left[\mathfrak{g}_{-2}, \mathfrak{g}_{-2}\right] \subset \mathfrak{g}_1 & \left[\boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{w} \otimes \boldsymbol{d}\right]=*\left(* \boldsymbol{b} \wedge \boldsymbol{w}\right) \otimes *\left(\boldsymbol{d} \wedge \boldsymbol{y}\right). \\ \left[\mathfrak{g}_1, \mathfrak{g}_2\right] \subset \mathfrak{g}_{-2} & \\ \left[\mathfrak{g}_{-1}, \mathfrak{g}_{-2}\right] \subset \mathfrak{g}_2 & \\ \left[\mathfrak{g}_2, \mathfrak{g}_{-1}\right] \subset \mathfrak{g}_1 & \\ \left[\mathfrak{g}_{-2}, \mathfrak{g}_1\right] \subset \mathfrak{g}_{-1} & \end{array}$	

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0902.0431_E00413	68	■ 0.941	$\left(\begin{array}{lll} \xi_1 & a_3 & \bar{a}_2 \\ \bar{a}_3 & \xi_2 & a_1 \\ a_2 & \bar{a}_1 & \xi_3 \end{array}\right) + (\boldsymbol{m}_1, \boldsymbol{m}_2, \boldsymbol{m}_3)$		
0902.0431_E00017	8	■ 0.942	$\left\{ \begin{array}{l} \{ 2 F_{- \{ 0 \, 1 \}} = G_{- \{ 0 \, 1 \}} + G_{- \{ 2 \, 3 \}} + G_{- \{ 4 \, 5 \}} + G_{- \{ 6 \, 7 \}} \\ \{ 2 F_{- \{ 2 \, 3 \}} = G_{- \{ 0 \, 1 \}} + G_{- \{ 2 \, 3 \}} - G_{- \{ 4 \, 5 \}} - G_{- \{ 6 \, 7 \}} \\ \{ 2 F_{- \{ 4 \, 5 \}} = G_{- \{ 0 \, 1 \}} - G_{- \{ 2 \, 3 \}} + G_{- \{ 4 \, 5 \}} - G_{- \{ 6 \, 7 \}} \\ \{ 2 F_{- \{ 6 \, 7 \}} = G_{- \{ 0 \, 1 \}} - G_{- \{ 2 \, 3 \}} - G_{- \{ 4 \, 5 \}} + G_{- \{ 6 \, 7 \}} \end{array} \right\} \end{array} \left\{ \begin{array}{l} 2 F_{02} = G_{02} - G_{13} - G_{46} + G_{57} \\ 2 F_{13} = -G_{02} + G_{13} - G_{46} + G_{57} \\ 2 F_{46} = -G_{02} - G_{13} + G_{46} + G_{57} \\ 2 F_{57} = G_{02} + G_{13} + G_{46} + G_{57} \end{array} \right.$		
0902.0431_E00927	153	■ 0.942	$D = \left(\begin{array}{cc} B & L \\ -L^* & C \end{array} \right) + \frac{\nu}{3} \left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array} \right), \quad B, C \in \mathfrak{su}(3), L \in M(3, C), \nu \in e_1 \mathbf{R}.$		
0902.0431_E00421	69	■ 0.942	$\varphi(A, 1)X = AXA^*, \quad \varphi(A, \epsilon)X = A\bar{X}A^*, \quad X \in \mathfrak{J}_C.$		
0902.0431_E00421	69	■ 0.942	$\varphi(A, 1)X = AXA^*, \quad \varphi(A, \epsilon)X = A\bar{X}A^*, \quad X \in \mathfrak{J}_C.$		
0902.0431_E00421	69	■ 0.942	$\varphi(A, 1)X = AXA^*, \quad \varphi(A, \epsilon)X = A\bar{X}A^*, \quad X \in \mathfrak{J}_C.$		
0902.0431_E00257	42	■ 0.943	$\begin{aligned} \mathfrak{f}_4 &= \left\{ \delta \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid \delta(X \circ Y) = \delta X \circ Y + X \circ \delta Y \right\} \\ &= \left\{ \delta \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid \begin{array}{l} \operatorname{tr}(\delta X, Y, Z) + \operatorname{tr}(X, \delta Y, Z) + \operatorname{tr}(X, Y, \delta Z) = 0 \\ (\delta X, Y) + (X, \delta Y) = 0 \end{array} \right\} \\ &= \left\{ \delta \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid (\delta X, X, X) = 0, (\delta X, Y) + (X, \delta Y) = 0 \right\} \\ &= \left\{ \delta \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid (\delta X, X, X) = 0, \delta E = 0 \right\} \\ &= \left\{ \delta \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid \delta(X \times Y) = \delta X \times Y + X \times \delta Y \right\}. \end{aligned}$		

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0902.0431_E00431	71	■ 0.943	$\begin{aligned} F_4^C &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, (\alpha X, \alpha Y) = (X, Y)\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, \alpha E = E\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha(X \times Y) = \alpha X \times \alpha Y\}. \end{aligned}$	$F_4^C = \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\}$ $= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, (\alpha X, \alpha Y) = (X, Y)\}$ $= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, \alpha E = E\}$ $= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha(X \times Y) = \alpha X \times \alpha Y\}.$	$F_4^C = \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\}$ $= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, (\alpha X, \alpha Y) = (X, Y)\}$ $= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, \alpha E = E\}$ $= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha(X \times Y) = \alpha X \times \alpha Y\}.$
0902.0431_E00645	106	■ 0.946	$\epsilon(A, B) = (\bar{B}, \bar{A}),$	$\epsilon(A, B) = (\bar{B}, \bar{A}),$	$\epsilon(A, B) = (\bar{B}, \bar{A}),$
0902.0431_E01049 (i)	179	■ 0.948	$\beta_1(P \times Q) = \beta_2 P \times \beta_3 Q, \quad \beta_1(P \times Q) = \beta_2 P \times \beta_3 Q, \quad \beta_1(P \times Q) = \beta_2 P \times \beta_3 Q,$	$\beta_1(P \times Q) = \beta_2 P \times \beta_3 Q, \quad \{\beta_2 P, \beta_3 Q\} = \{P, Q\},$	$\beta_1(P \times Q) = \beta_2 P \times \beta_3 Q, \quad \{\beta_2 P, \beta_3 Q\} = \{P, Q\},$
0902.0431_E00435	72	■ 0.948	$F_4^C \simeq (F_4^C \cap U(\mathfrak{J}^C)) \times \mathbf{R}^d = F_4 \times \mathbf{R}^d,$	$F_4^C \simeq (F_4^C \cap U(\mathfrak{J}^C)) \times \mathbf{R}^d = F_4 \times \mathbf{R}^d,$	$F_4^C \simeq (F_4^C \cap U(\mathfrak{J}^C)) \times \mathbf{R}^d = F_4 \times \mathbf{R}^d,$
0902.0431_E00105	20	■ 0.950	$\begin{aligned} &\left(\alpha_1, \alpha_1 \right) = B_2(H_{\alpha_1}, H_{\alpha_1}) = 8 \frac{1}{8} \frac{1}{8} (1 + 1) = \frac{1}{4}, \\ &\left(\alpha_2, \alpha_2 \right) = B_2(H_{\alpha_2}, H_{\alpha_2}) = 8 \frac{1}{24} \frac{1}{24} (1 + 4 + 1) = \frac{1}{12}, \\ &\left(\alpha_1, \alpha_2 \right) = B_2(H_{\alpha_1}, H_{\alpha_2}) = 8 \frac{1}{8} \frac{1}{24} (-1 - 2) = -\frac{1}{8}, \\ &(-\mu, -\mu) = \frac{1}{4}, \quad (-\mu, \alpha_1) = -\frac{1}{8}, \quad (-\mu, \alpha_2) = 0. \end{aligned}$	$\left(\alpha_1, \alpha_1 \right) = B_2(H_{\alpha_1}, H_{\alpha_1}) = 8 \frac{1}{8} \frac{1}{8} (1 + 1) = \frac{1}{4}$ $\left(\alpha_2, \alpha_2 \right) = B_2(H_{\alpha_2}, H_{\alpha_2}) = 8 \frac{1}{24} \frac{1}{24} (1 + 4 + 1) = \frac{1}{12}$ $\left(\alpha_1, \alpha_2 \right) = B_2(H_{\alpha_1}, H_{\alpha_2}) = 8 \frac{1}{8} \frac{1}{24} (-1 - 2) = -\frac{1}{8}$ $(-\mu, -\mu) = \frac{1}{4}, \quad (-\mu, \alpha_1) = -\frac{1}{8}, \quad (-\mu, \alpha_2) = 0$	$\left(\alpha_1, \alpha_1 \right) = B_2(H_{\alpha_1}, H_{\alpha_1}) = 8 \frac{1}{8} \frac{1}{8} (1 + 1) = \frac{1}{4},$ $\left(\alpha_2, \alpha_2 \right) = B_2(H_{\alpha_2}, H_{\alpha_2}) = 8 \frac{1}{24} \frac{1}{24} (1 + 4 + 1) = \frac{1}{12},$ $\left(\alpha_1, \alpha_2 \right) = B_2(H_{\alpha_1}, H_{\alpha_2}) = 8 \frac{1}{8} \frac{1}{24} (-1 - 2) = -\frac{1}{8},$ $(-\mu, -\mu) = \frac{1}{4}, \quad (-\mu, \alpha_1) = -\frac{1}{8}, \quad (-\mu, \alpha_2) = 0.$
0902.0431_E00272	44	■ 0.950	$\begin{aligned} &\left(\alpha_1, \alpha_1 \right) = B_2(H_{\alpha_1}, H_{\alpha_1}) = 8 \frac{1}{8} \frac{1}{8} (1 + 1) = \frac{1}{4}, \\ &\left(\alpha_2, \alpha_2 \right) = B_2(H_{\alpha_2}, H_{\alpha_2}) = 8 \frac{1}{24} \frac{1}{24} (1 + 4 + 1) = \frac{1}{12}, \\ &\left(\alpha_1, \alpha_2 \right) = B_2(H_{\alpha_1}, H_{\alpha_2}) = 8 \frac{1}{8} \frac{1}{24} (-1 - 2) = -\frac{1}{8}, \\ &(-\mu, -\mu) = \frac{1}{4}, \quad (-\mu, \alpha_1) = -\frac{1}{8}, \quad (-\mu, \alpha_2) = 0. \end{aligned}$	$\begin{cases} \tilde{A}_i(a)E_i = 0 \\ \tilde{A}_i(a)E_{i+1} = -\frac{1}{2}F_i(a) \\ \tilde{A}_i(a)E_{i+2} = \frac{1}{2}F_i(a), \end{cases} \quad \begin{cases} \tilde{A}_i(a)F_i(x) = (a, x)(E_{i+1} - E_{i+2}) \\ \tilde{A}_i(a)F_{i+1}(x) = \frac{1}{2}F_{i+2}(\overline{ax}) \\ \tilde{A}_i(a)F_{i+2}(x) = -\frac{1}{2}F_{i+1}(\overline{xa}). \end{cases}$	$\begin{cases} \tilde{A}_i(a)E_i = 0 \\ \tilde{A}_i(a)E_{i+1} = -\frac{1}{2}F_i(a) \\ \tilde{A}_i(a)E_{i+2} = \frac{1}{2}F_i(a), \end{cases} \quad \begin{cases} \tilde{A}_i(a)F_i(x) = (a, x)(E_{i+1} - E_{i+2}) \\ \tilde{A}_i(a)F_{i+1}(x) = \frac{1}{2}F_{i+2}(\overline{ax}) \\ \tilde{A}_i(a)F_{i+2}(x) = -\frac{1}{2}F_{i+1}(\overline{xa}). \end{cases}$
0902.0431_E00473	78	■ 0.951	$\left(\phi_1, \phi_2 \right)_6 = (\delta_1, \delta_2)_4 + (T_1, T_2),$	$(\phi_1, \phi_2)_6 = (\delta_1, \delta_2)_4 + (T_1, T_2),$	$(\phi_1, \phi_2)_6 = (\delta_1, \delta_2)_4 + (T_1, T_2),$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00726	120	■ 0.951	$\begin{aligned} & \& \mathfrak{a} \ni \left[\Phi(0, A, B, \nu), \Phi \left(0, 0, 0, -\frac{3}{2} \right) \right] = \Phi(0, A, -B, 0), \\ & \& \Phi(0, A, -B, 0), \Phi(0, 0, B_1, 0) = \Phi(2A \vee B_1, 0, 0, (A, B_1)), \\ & \& \Phi(2A \vee B_1, 0, 0, (A, B_1)), \Phi(\phi, 0, 0, 0) = \Phi(2[A \vee B_1 \cdot \phi], 0, 0, 0), \end{aligned}$	$\begin{aligned} & \& \Phi(0, A, B, \nu), \Phi \left(0, 0, 0, -\frac{3}{2} \right) = \Phi(0, A, -B, 0), \\ & \& \Phi(0, A, -B, 0), \Phi(0, 0, B_1, 0) = \Phi(2A \vee B_1, 0, 0, (A, B_1)), \\ & \& \Phi(2A \vee B_1, 0, 0, (A, B_1)), \Phi(\phi, 0, 0, 0) = \Phi(2[A \vee B_1 \cdot \phi], 0, 0, 0), \end{aligned}$	$\begin{aligned} & \& \Phi(0, A, B, \nu), \Phi \left(0, 0, 0, -\frac{3}{2} \right) = \Phi(0, A, -B, 0), \\ & \& \Phi(0, A, -B, 0), \Phi(0, 0, B_1, 0) = \Phi(2A \vee B_1, 0, 0, (A, B_1)), \\ & \& \Phi(2A \vee B_1, 0, 0, (A, B_1)), \Phi(\phi, 0, 0, 0) = \Phi(2[A \vee B_1 \cdot \phi], 0, 0, 0), \end{aligned}$
0902.0431_E00323	51	■ 0.953	$B_4(\delta, \delta) = \text{tr} \left((\text{ad} \tilde{A}_1(1))^2 \right) = (-1) \times 7 \times 2 + \left(-\frac{1}{4} \right) \times 8 \times 2 = -18.$	$B_4(\delta, \delta) = \text{tr} \left((\text{ad} \tilde{A}_1(1))^2 \right) = (-1) \times 7 \times 2 + \left(-\frac{1}{4} \right) \times 8 \times 2 = -18.$	$B_4(\delta, \delta) = \text{tr}((\text{ad} \tilde{A}_1(1))^2) = (-1) \times 7 \times 2 + \left(-\frac{1}{4} \right) \times 8 \times 2 = -18.$
0902.0431_E00161	29	■ 0.953	$\left(\alpha_1 x \right) \left(\alpha_2 y \right) = xy, \quad x, y \in \mathfrak{C}.$	$(\alpha_1 x)(\alpha_2 y) = xy, \quad x, y \in \mathfrak{C}.$	$(\alpha_1 x)(\alpha_2 y) = xy, \quad x, y \in \mathfrak{C}.$
0902.0431_E00489	80	■ 0.954	$\begin{aligned} & \& \left(\phi' A \times A, A \right) (\text{Lemma 3.4.3.(1)}) = 2(\phi' A, A \times A) = -2(A, \phi(A \times A)) \\ & \& = -2(\phi, (A \times A) \vee A)_6. \end{aligned}$	$\begin{aligned} & \& \left(\phi' A \times A, A \right) (\text{Lemma 3.4.3.(1)}) = 2(\phi' A, A \times A) = -2(A, \phi(A \times A)) \\ & \& = -2(\phi, (A \times A) \vee A)_6. \end{aligned}$	$\begin{aligned} & \& \left(\phi' A \times A, A \right) (\text{Lemma 3.4.3.(1)}) = 2(\phi' A, A \times A) = -2(A, \phi(A \times A)) \\ & \& = -2(\phi, (A \times A) \vee A)_6. \end{aligned}$
0902.0431_E00178	31	■ 0.954	$(ax\bar{a})(ay\bar{a}) = a(xy)\bar{a}, \quad x, y \in \mathfrak{C}.$	$(ax\bar{a})(ay\bar{a}) = a(xy)\bar{a}, \quad x, y \in \mathfrak{C}.$	$(ax\bar{a})(ay\bar{a}) = a(xy)\bar{a}, \quad x, y \in \mathfrak{C}.$
0902.0431_E00343	55	■ 0.954	$B_4(H, H') = 18 \sum_{k=0}^3 \lambda_k \lambda_k', \quad H = \sum_{k=0}^3 \lambda_k H_k, H' = \sum_{k=0}^3 \lambda_k' H_k \in \mathfrak{h}_R.$	$B_4(H, H') = 18 \sum_{k=0}^3 \lambda_k \lambda_k', \quad H = \sum_{k=0}^3 \lambda_k H_k, H' = \sum_{k=0}^3 \lambda_k' H_k \in \mathfrak{h}_R.$	$B_4(H, H') = 18 \sum_{k=0}^3 \lambda_k \lambda_k', \quad H = \sum_{k=0}^3 \lambda_k H_k, H' = \sum_{k=0}^3 \lambda_k' H_k \in \mathfrak{h}_R.$
0902.0431_E00495	81	■ 0.956	$\begin{aligned} & \& \left(\sum_{k=1}^3 ((\sigma_{ik} x_{ik}) y_{ki} + \bar{y}_{ki} (\overline{\sigma_{ik} x_{ik}})) + \sum_{k=1}^3 (x_{ik} (\sigma_{ki} y_{ki}) + (\overline{\sigma_{ki} y_{ki}}) \bar{x}_{ik}) \right) \\ & \& = 2R \left(\sum_k ((\sigma_{ik} x_{ik}) y_{ki} + x_{ik} (\sigma_{ki} y_{ki})) \right) \\ & \& = 2 \sum_k ((\sigma_{ik} x_{ik}, \bar{y}_{ki}) + (x_{ik}, \overline{\sigma_{ki} y_{ki}})) \\ & \& = 2 \sum_k ((x_{ik}, -\sigma_{ik} \bar{y}_{ki}) + (x_{ik}, \sigma_{ik} \bar{y}_{ki})) = 0. \end{aligned}$	$\begin{aligned} & \& \left(\sum_{k=1}^3 ((\sigma_{ik} x_{ik}) y_{ki} + \bar{y}_{ki} (\overline{\sigma_{ik} x_{ik}})) + \sum_{k=1}^3 (x_{ik} (\sigma_{ki} y_{ki}) + (\overline{\sigma_{ki} y_{ki}}) \bar{x}_{ik}) \right) \\ & \& = 2R \left(\sum_k ((\sigma_{ik} x_{ik}) y_{ki} + x_{ik} (\sigma_{ki} y_{ki})) \right) \\ & \& = 2 \sum_k ((\sigma_{ik} x_{ik}, \bar{y}_{ki}) + (x_{ik}, \overline{\sigma_{ki} y_{ki}})) \\ & \& = 2 \sum_k ((x_{ik}, -\sigma_{ik} \bar{y}_{ki}) + (x_{ik}, \sigma_{ik} \bar{y}_{ki})) = 0. \end{aligned}$	$\begin{aligned} & \& \left(\sum_{k=1}^3 ((\sigma_{ik} x_{ik}) y_{ki} + \bar{y}_{ki} (\overline{\sigma_{ik} x_{ik}})) + \sum_{k=1}^3 (x_{ik} (\sigma_{ki} y_{ki}) + (\overline{\sigma_{ki} y_{ki}}) \bar{x}_{ik}) \right) \\ & \& = 2R \left(\sum_k ((\sigma_{ik} x_{ik}) y_{ki} + x_{ik} (\sigma_{ki} y_{ki})) \right) \\ & \& = 2 \sum_k ((\sigma_{ik} x_{ik}, \bar{y}_{ki}) + (x_{ik}, \overline{\sigma_{ki} y_{ki}})) \\ & \& = 2 \sum_k ((x_{ik}, -\sigma_{ik} \bar{y}_{ki}) + (x_{ik}, \sigma_{ik} \bar{y}_{ki})) = 0. \end{aligned}$

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0902.0431_E00226	38	■ 0.957	$\left\{ \begin{array}{l} E_{-i} \times E_i = 0 \\ E_i \times F_i(x) = -\frac{1}{2}F_i(x) \\ F_i(x) \times F_i(y) = -(x,y)E_i, \end{array} \right. \quad \left\{ \begin{array}{l} E_i \times E_{i+1} = \frac{1}{2}E_{i+2} \\ E_i \times F_j(x) = 0, \quad i \neq j \\ F_i(x) \times F_{i+1}(y) = \frac{1}{2}F_{i+2}(\overline{xy}), \end{array} \right.$		
0902.0431_E00862	143	■ 0.957	$\begin{aligned} & \alpha(t) \left(\begin{pmatrix} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{pmatrix}, \begin{pmatrix} \eta_1 & y_3 & \bar{y}_2 \\ \bar{y}_3 & \eta_2 & y_1 \\ y_2 & \bar{y}_1 & \eta_3 \end{pmatrix}, \xi, \eta \right) \\ & = \left(\begin{pmatrix} e^{2it}\xi_1 & e^{it}x_3 & e^{it}\bar{x}_2 \\ e^{it}\bar{x}_3 & \xi_2 & x_1 \\ e^{it}x_2 & \bar{x}_1 & \xi_3 \end{pmatrix}, \begin{pmatrix} e^{-2it}\eta_1 & e^{-it}y_3 & e^{-it}\bar{y}_2 \\ e^{-it}\bar{y}_3 & \eta_2 & y_1 \\ e^{-it}y_2 & \bar{y}_1 & \eta_3 \end{pmatrix}, e^{-2it}\xi, e^{2it}\eta \right), \end{aligned}$		$\alpha(t) \left(\begin{pmatrix} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{pmatrix}, \begin{pmatrix} \eta_1 & y_3 & \bar{y}_2 \\ \bar{y}_3 & \eta_2 & y_1 \\ y_2 & \bar{y}_1 & \eta_3 \end{pmatrix}, \xi, \eta \right) \\ = \left(\begin{pmatrix} e^{2it}\xi_1 & e^{it}x_3 & e^{it}\bar{x}_2 \\ e^{it}\bar{x}_3 & \xi_2 & x_1 \\ e^{it}x_2 & \bar{x}_1 & \xi_3 \end{pmatrix}, \begin{pmatrix} e^{-2it}\eta_1 & e^{-it}y_3 & e^{-it}\bar{y}_2 \\ e^{-it}\bar{y}_3 & \eta_2 & y_1 \\ e^{-it}y_2 & \bar{y}_1 & \eta_3 \end{pmatrix}, e^{-2it}\xi, e^{2it}\eta \right),$
0902.0431_E00216	36	■ 0.960	$X = X(\xi, x) = \begin{pmatrix} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{pmatrix}, \quad \xi_i \in \mathbf{R}, x_i \in \mathfrak{C}.$		$X = X(\xi, x) = \begin{pmatrix} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{pmatrix}, \quad \xi_i \in \mathbf{R}, x_i \in \mathfrak{C}.$
0902.0431_E00164	29	■ 0.960	$\left(\alpha_1 x \right) \left(\alpha_2 y \right) = \overline{\alpha_3 (\overline{xy})}, \quad x, y \in \mathfrak{C}$		$(\alpha_1 x)(\alpha_2 y) = \overline{\alpha_3(\overline{xy})}, \quad x, y \in \mathfrak{C}$
0902.0431_E00656	108	■ 0.960	$\langle \alpha X, \alpha M \rangle = 0 = \langle X, M \rangle, \quad \langle \alpha M, \alpha N \rangle = (\tau \alpha M, \alpha N) = (P \tau M h(A, B)^*, P N \tau h(A, B)^*) = (\tau M, N) = \langle M, N \rangle,$		$\langle \alpha X, \alpha M \rangle = 0 = \langle X, M \rangle, \quad \langle \alpha M, \alpha N \rangle = (\tau \alpha M, \alpha N) = (P \tau M h(A, B)^*, P N \tau h(A, B)^*) = (\tau M, N) = \langle M, N \rangle,$
0902.0431_E01197	201	■ 0.960	$[(C, 0), (C, D, g_1, g_2, g_{-2}, g_{-1})] = (0, 0, [C, g_1], [C, g_2], [C, g_{-2}], [C, g_{-1}]) \in \mathfrak{q} \cap \mathfrak{a} = \{0\},$		$[(C, 0), (C, D, g_1, g_2, g_{-2}, g_{-1})] = (0, 0, [C, g_1], [C, g_2], [C, g_{-2}], [C, g_{-1}]) \in \mathfrak{q} \cap \mathfrak{a} = \{0\},$

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0902.0431_E00465	77	■ 0.962	$\begin{aligned} A \circ (B \circ X) + B \circ (X \circ A) + X \circ (A \circ B) - \operatorname{tr}(A)B \circ X - \operatorname{tr}(B)A \circ X \\ - \operatorname{tr}(X)A \circ B + \frac{1}{2}(\operatorname{tr}(A)\operatorname{tr}(B)X + \operatorname{tr}(B)\operatorname{tr}(X)A + \operatorname{tr}(X)\operatorname{tr}(A)B) \\ - \frac{1}{2}((A, B)X + (B, X)A + (X, A)B) = (A, B, X)E. \end{aligned}$	$A \circ (B \circ X) + B \circ (X \circ A) + X \circ (A \circ B) - \operatorname{tr}(A)B \circ X - \operatorname{tr}(B)A \circ X \\ - \operatorname{tr}(X)A \circ B + \frac{1}{2}(\operatorname{tr}(A)\operatorname{tr}(B)X + \operatorname{tr}(B)\operatorname{tr}(X)A + \operatorname{tr}(X)\operatorname{tr}(A)B) \\ - \frac{1}{2}((A, B)X + (B, X)A + (X, A)B) = (A, B, X)E.$	$A \circ (B \circ X) + B \circ (X \circ A) + X \circ (A \circ B) - \operatorname{tr}(A)B \circ X - \operatorname{tr}(B)A \circ X \\ - \operatorname{tr}(X)A \circ B + \frac{1}{2}(\operatorname{tr}(A)\operatorname{tr}(B)X + \operatorname{tr}(B)\operatorname{tr}(X)A + \operatorname{tr}(X)\operatorname{tr}(A)B) \\ - \frac{1}{2}((A, B)X + (B, X)A + (X, A)B) = (A, B, X)E.$
0902.0431_E00076	16	■ 0.963	$\mathfrak{g}_2 = \mathfrak{su}(3) \oplus \mathfrak{S}.$	$\mathfrak{g}_2 = \mathfrak{su}(3) \oplus \mathfrak{S}.$	$\mathfrak{g}_2 = \mathfrak{su}(3) \oplus \mathfrak{S}.$
0902.0431_E00520	86	■ 0.963	$B_6(h, h') = 12 \left(2 \sum_{k=0}^3 \lambda_k \lambda'_k + \sum_{j=1}^3 \mu_j \mu'_j \right)$	$B_6(h, h') = 12 \left(2 \sum_{k=0}^3 \lambda_k \lambda'_k + \sum_{j=1}^3 \mu_j \mu'_j \right)$	$B_6(h, h') = 12 \left(2 \sum_{k=0}^3 \lambda_k \lambda'_k + \sum_{j=1}^3 \mu_j \mu'_j \right)$
0902.0431_E00898	149	■ 0.964	$\begin{aligned} P = & (X, Y, \xi, \eta) \quad \text{and} \quad \chi \mapsto k \left(gX - \frac{\xi}{2} E \right) J + e_1 k \left(g(\gamma Y) - \frac{\eta}{2} E \right) J \\ & \longrightarrow k(D)k \left(gX - \frac{\xi}{2} E \right) J + e_1 k(D)k \left(g(\gamma Y) - \frac{\eta}{2} E \right) J \\ & + k \left(gX - \frac{\xi}{2} E \right) J^t k(D) + e_1 k \left(g(\gamma Y) - \frac{\eta}{2} E \right) J^t k(D) \\ = & k \left(D \left(gX - \frac{\xi}{2} E \right) \right) J + e_1 k \left(D \left(g(\gamma Y) - \frac{\eta}{2} E \right) \right) J \\ & + k \left(\left(gX - \frac{\xi}{2} E \right) D^* \right) J + e_1 k \left(\left(g(\gamma Y) - \frac{\eta}{2} E \right) D^* \right) J \\ = & k(D(gX) + (gX)D^*)J + e_1 k(D(g(\gamma Y) + (g(\gamma Y))D^*)J \\ = & k(g(\varphi_*(D)X))J + e_1 k(g(\varphi_*(D)(\gamma Y)))J \end{aligned}$	$\begin{aligned} P = & (X, Y, \xi, \eta) \\ & \xrightarrow{\chi} k \left(gX - \frac{\xi}{2} E \right) J + e_1 k \left(g(\gamma Y) - \frac{\eta}{2} E \right) J \\ & \longrightarrow k(D)k \left(gX - \frac{\xi}{2} E \right) J + e_1 k(D)k \left(g(\gamma Y) - \frac{\eta}{2} E \right) J \\ & + k \left(gX - \frac{\xi}{2} E \right) J^t k(D) + e_1 k \left(g(\gamma Y) - \frac{\eta}{2} E \right) J^t k(D) \\ = & k \left(D \left(gX - \frac{\xi}{2} E \right) \right) J + e_1 k \left(D \left(g(\gamma Y) - \frac{\eta}{2} E \right) \right) J \\ & + k \left(\left(gX - \frac{\xi}{2} E \right) D^* \right) J + e_1 k \left(\left(g(\gamma Y) - \frac{\eta}{2} E \right) D^* \right) J \\ = & k(D(gX) + (gX)D^*)J + e_1 k(D(g(\gamma Y) + (g(\gamma Y))D^*)J \\ = & k(g(\varphi_*(D)X))J + e_1 k(g(\varphi_*(D)(\gamma Y)))J \end{aligned}$	$\begin{aligned} P = & (X, Y, \xi, \eta) \\ & \xrightarrow{\chi} k \left(gX - \frac{\xi}{2} E \right) J + e_1 k \left(g(\gamma Y) - \frac{\eta}{2} E \right) J \\ & \longrightarrow k(D)k \left(gX - \frac{\xi}{2} E \right) J + e_1 k(D)k \left(g(\gamma Y) - \frac{\eta}{2} E \right) J \\ & + k \left(gX - \frac{\xi}{2} E \right) J^t k(D) + e_1 k \left(g(\gamma Y) - \frac{\eta}{2} E \right) J^t k(D) \\ = & k \left(D \left(gX - \frac{\xi}{2} E \right) \right) J + e_1 k \left(D \left(g(\gamma Y) - \frac{\eta}{2} E \right) \right) J \\ & + k \left(\left(gX - \frac{\xi}{2} E \right) D^* \right) J + e_1 k \left(\left(g(\gamma Y) - \frac{\eta}{2} E \right) D^* \right) J \\ = & k(D(gX) + (gX)D^*)J + e_1 k(D(g(\gamma Y) + (g(\gamma Y))D^*)J \\ = & k(g(\varphi_*(D)X))J + e_1 k(g(\varphi_*(D)(\gamma Y)))J \end{aligned}$
0902.0431_E00145	27	■ 0.964	$G_2 = \left\{ \alpha \in G_2^C \mid \tau \alpha = \alpha \tau \right\}.$	$G_2 = \left\{ \alpha \in G_2^C \mid \tau \alpha = \alpha \tau \right\}.$	$G_2 = \left\{ \alpha \in G_2^C \mid \tau \alpha = \alpha \tau \right\}.$
0902.0431_E01092 (2)	185	■ 0.964	$\begin{aligned} & \zeta[l(D), l(S)I] = \zeta(l(DS - S\tau D)I) = \zeta(l(DS + S^t D)I) \\ & = \Theta(0, 2\lambda\gamma\chi^{-1}(DS + S^t D), 2\chi^{-1}(DS + S^t D), 0, 0, 0) \end{aligned}$	$\begin{aligned} & \zeta[l(D), l(S)I] = \zeta(l(DS - S\tau D)I) = \zeta(l(DS + S^t D)I) \\ & = \Theta(0, 2\lambda\gamma\chi^{-1}(DS + S^t D), 2\chi^{-1}(DS + S^t D), 0, 0, 0). \end{aligned}$	$\begin{aligned} & \zeta[l(D), l(S)I] = \zeta(l(DS - S\tau D)I) = \zeta(l(DS + S^t D)I) \\ & = \Theta(0, 2\lambda\gamma\chi^{-1}(DS + S^t D), 2\chi^{-1}(DS + S^t D), 0, 0, 0). \end{aligned}$

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0902.0431_E00494	81	■ 0.964	$\begin{array}{l} \sigma_{23}=\delta_1, \text{ \& \sigma_{31}=\delta_2, \& \sigma_{12}=\delta_3} \\ \sigma_{32}=\kappa\delta_1, \text{ \& \sigma_{13}=\kappa\delta_2, \& \sigma_{21}=\kappa\delta_3.} \end{array}$	$\begin{array}{ccc} \sigma_{23} = \delta_1, & \sigma_{31} = \delta_2, & \sigma_{12} = \delta_3 \\ \sigma_{32} = \kappa\delta_1, & \sigma_{13} = \kappa\delta_2, & \sigma_{21} = \kappa\delta_3. \end{array}$	$\begin{array}{ccc} \sigma_{23} = \delta_1, & \sigma_{31} = \delta_2, & \sigma_{12} = \delta_3, \\ \sigma_{32} = \kappa\delta_1, & \sigma_{13} = \kappa\delta_2, & \sigma_{21} = \kappa\delta_3. \end{array}$
0902.0431_E00760	127	■ 0.964	$\begin{array}{l} \lambda_{\lambda_0-\frac{1}{2}\mu_1+\frac{2}{3}\nu=1} \& \lambda_{\lambda_1-\frac{1}{2}\mu_1+\frac{2}{3}\nu=0} \& \lambda_{\lambda_2-\frac{1}{2}\mu_1+\frac{2}{3}\nu=0} \& \lambda_{\lambda_3-\frac{1}{2}\mu_1+\frac{2}{3}\nu=0} \\ \lambda_{\lambda_0+\frac{1}{2}\mu_1-\frac{2}{3}\nu=1} \& \lambda_{\lambda_1+\frac{1}{2}\mu_1-\frac{2}{3}\nu=0} \& \lambda_{\lambda_2+\frac{1}{2}\mu_1-\frac{2}{3}\nu=0} \& \lambda_{\lambda_3+\frac{1}{2}\mu_1-\frac{2}{3}\nu=0} \end{array}$	$\begin{array}{l} \lambda_0 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_1 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_2 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_3 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_0 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_1 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_2 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_3 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1 \end{array}$	$\begin{array}{l} \lambda_0 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_1 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_2 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_3 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_0 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_1 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_2 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_3 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1 \end{array}$
0902.0431_E00433	71	■ 0.965	$F_4=\left\{\alpha\in F_4\mid \tau\alpha=\alpha\tau\right\}.$	$F_4 = \left\{ \alpha \in F_4^C \mid \tau\alpha = \alpha\tau \right\}.$	$F_4 = \left\{ \alpha \in F_4^C \mid \tau\alpha = \alpha\tau \right\}.$
0902.0431_E00713	118	■ 0.965	$\alpha\in E_7 \quad \text{if and only if} \quad \tau\lambda\alpha=\alpha\tau\lambda.$	$\alpha \in E_7 \quad \text{if and only if} \quad \tau\lambda\alpha = \alpha\tau\lambda.$	$\alpha \in E_7 \quad \text{if and only if} \quad \tau\lambda\alpha = \alpha\tau\lambda.$
0902.0431_E00511	84	■ 0.965	$\begin{array}{l} \left[h,\widetilde{F}_{31}(a)\right]=\left(h_{\delta}F_{31}(a)\right)^{\sim}+\frac{1}{2}\left[\widetilde{H},\widetilde{F}_{31}(a)\right]=\widetilde{F}_{31}\left((\nu h_{\delta})a\right)+\frac{1}{2}\left(\mu_3-\mu_1\right)\widetilde{F}_{31}(a), \\ \left[h,\widetilde{F}_{12}(a)\right]=\left(h_{\delta}F_{12}(a)\right)^{\sim}+\frac{1}{2}\left[\widetilde{H},\widetilde{F}_{12}(a)\right]=\widetilde{F}_{12}\left((\kappa\pi h_{\delta})a\right)+\frac{1}{2}\left(\mu_1-\mu_2\right)\widetilde{F}_{12}(a), \end{array}$	$\begin{array}{l} \left[h,\widetilde{F}_{31}(a)\right]=\left(h_{\delta}F_{31}(a)\right)^{\sim}+\frac{1}{2}\left[\widetilde{H},\widetilde{F}_{31}(a)\right]=\widetilde{F}_{31}\left((\nu h_{\delta})a\right)+\frac{1}{2}\left(\mu_3-\mu_1\right)\widetilde{F}_{31}(a), \\ \left[h,\widetilde{F}_{12}(a)\right]=\left(h_{\delta}F_{12}(a)\right)^{\sim}+\frac{1}{2}\left[\widetilde{H},\widetilde{F}_{12}(a)\right]=\widetilde{F}_{12}\left((\kappa\pi h_{\delta})a\right)+\frac{1}{2}\left(\mu_1-\mu_2\right)\widetilde{F}_{12}(a), \end{array}$	$\begin{array}{l} \left[h,\widetilde{F}_{31}(a)\right]=\left(h_{\delta}F_{31}(a)\right)^{\sim}+\frac{1}{2}\left[\widetilde{H},\widetilde{F}_{31}(a)\right]=\widetilde{F}_{31}\left((\nu h_{\delta})a\right)+\frac{1}{2}\left(\mu_3-\mu_1\right)\widetilde{F}_{31}(a), \\ \left[h,\widetilde{F}_{12}(a)\right]=\left(h_{\delta}F_{12}(a)\right)^{\sim}+\frac{1}{2}\left[\widetilde{H},\widetilde{F}_{12}(a)\right]=\widetilde{F}_{12}\left((\kappa\pi h_{\delta})a\right)+\frac{1}{2}\left(\mu_1-\mu_2\right)\widetilde{F}_{12}(a), \end{array}$

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0902.0431_E00906	151	■ 0.966	$\begin{gathered} \gamma \varphi(D) \gamma Y = g^{-1}(D(g(\gamma Y))D^*) \text{ is real diagonal,} \\ \varphi(D)X = \varphi(D) \left(\frac{1}{\xi} Y \times Y \right) = \frac{1}{\xi} (\gamma \varphi(D) \gamma Y \times \gamma \varphi(D) \gamma Y) \text{ is real diagonal.} \end{gathered}$	$\gamma \varphi(D) \gamma Y = g^{-1}(D(g(\gamma Y))D^*) \text{ is real diagonal,}$ $\varphi(D)X = \varphi(D) \left(\frac{1}{\xi} Y \times Y \right) = \frac{1}{\xi} (\gamma \varphi(D) \gamma Y \times \gamma \varphi(D) \gamma Y) \text{ is real diagonal.}$	$\gamma \varphi(D) \gamma Y = g^{-1}(D(g(\gamma Y))D^*) \text{ is real diagonal,}$ $\varphi(D)X = \varphi(D) \left(\frac{1}{\xi} Y \times Y \right) = \frac{1}{\xi} (\gamma \varphi(D) \gamma Y \times \gamma \varphi(D) \gamma Y) \text{ is real diagonal.}$
0902.0431_E00191	33	■ 0.966	$\widetilde{\alpha} e_i = b_i, \quad i = 0, 1, \dots, 7$	$\widetilde{\alpha} e_i = b_i, \quad i = 0, 1, \dots, 7$	$\widetilde{\alpha} e_i = b_i, \quad i = 0, 1, \dots, 7$
0902.0431_E00614	102	■ 0.967	$\begin{aligned} gX \circ gY &= g(\gamma(X \times Y)) + \frac{1}{4}(\gamma X, Y)E, \quad X, Y \in \mathfrak{J}^C \\ (gX, gY) &= (\gamma X, Y) \end{aligned}$	$gX \circ gY = g(\gamma(X \times Y)) + \frac{1}{4}(\gamma X, Y)E, \quad X, Y \in \mathfrak{J}^C$ $(gX, gY) = (\gamma X, Y)$	$gX \circ gY = g(\gamma(X \times Y)) + \frac{1}{4}(\gamma X, Y)E, \quad X, Y \in \mathfrak{J}^C.$ $(gX, gY) = (\gamma X, Y),$
0902.0431_E00730	120	■ 0.969	$W \ni \Phi(0, 0, X_1, 0)(X, 0, 0, 3\eta) = (0, 0, 0, (X_1, X)),$	$W \ni \Phi(0, 0, X_1, 0)(X, 0, 0, 3\eta) = (0, 0, 0, (X_1, X)),$	$W \ni \Phi(0, 0, X_1, 0)(X, 0, 0, 3\eta) = (0, 0, 0, (X_1, X)),$
0902.0431_E01004	167	■ 0.969	$\begin{aligned} &\langle R_1, R_2 \rangle \\ &= (\tau \lambda \Phi_1 \lambda^{-1} \tau, \Phi_2)_7 + \langle P_1, P_2 \rangle + \langle Q_1, Q_2 \rangle + 8(\tau r_1) r_2 + 4(\tau s_1) s_2 + 4(\tau t_1) t_2. \end{aligned}$	$\langle R_1, R_2 \rangle$ $= (\tau \lambda \Phi_1 \lambda^{-1} \tau, \Phi_2)_7 + \langle P_1, P_2 \rangle + \langle Q_1, Q_2 \rangle + 8(\tau r_1) r_2 + 4(\tau s_1) s_2 + 4(\tau t_1) t_2.$	$\langle R_1, R_2 \rangle$ $= (\tau \lambda \Phi_1 \lambda^{-1} \tau, \Phi_2)_7 + \langle P_1, P_2 \rangle + \langle Q_1, Q_2 \rangle + 8(\tau r_1) r_2 + 4(\tau s_1) s_2 + 4(\tau t_1) t_2.$

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0902.0431_E00595	99	■ 0.969	$\begin{aligned} & 2\langle \tau^t \varphi(p, A) \tau(M + \mathbf{a}), N + \mathbf{b} \rangle \quad M + \mathbf{a}, N + \mathbf{b} \in \mathfrak{J}^C \\ &= 2\langle M + \mathbf{a}, \varphi(p, A)(N + \mathbf{b}) \rangle \\ &= 2\langle M + \mathbf{a}, k_J^{-1}(A(k_J(N))^t A) + p \mathbf{b} k^{-1}(\tau^t A) \rangle \\ &= 2\langle M, k_J^{-1}(A(k_J(N))^t A) \rangle + 4\langle \mathbf{a}, p \mathbf{b} k^{-1}(\tau^t A) \rangle \\ &= \langle k_J M, A(k_J(N))^t A \rangle + 2\langle k(\bar{p} \mathbf{a}), (k \mathbf{b}) \tau^t A \rangle \\ &= \langle \tau^t A(k_J(M)) \tau A, k_J(N) \rangle + 2\langle k(\bar{p} \mathbf{a}) A, k \mathbf{b} \rangle \\ &= 2\langle k_J^{-1}(\tau^t A(k_J(M)) \tau A), N \rangle + 4\langle \bar{p} \mathbf{a} k^{-1}(A), \mathbf{b} \rangle \\ &= 2\langle k_J^{-1}(\tau^t A(k_J(M)) \tau A + \bar{p} \mathbf{a} k^{-1}(A)), N + \mathbf{b} \rangle \\ &= 2\langle \varphi(\bar{p}, \tau^t A)(M + \mathbf{a}), N + \mathbf{b} \rangle, \end{aligned}$	$\begin{aligned} & 2\langle \tau^t \varphi(p, A) \tau(M + \mathbf{a}), N + \mathbf{b} \rangle \quad M + \mathbf{a}, N + \mathbf{b} \in \mathfrak{J}^C \\ &= 2\langle M + \mathbf{a}, \varphi(p, A)(N + \mathbf{b}) \rangle \\ &= 2\langle M + \mathbf{a}, k_J^{-1}(A(k_J(N))^t A) + p \mathbf{b} k^{-1}(\tau^t A) \rangle \\ &= 2\langle M, k_J^{-1}(A(k_J(N))^t A) \rangle + 4\langle \mathbf{a}, p \mathbf{b} k^{-1}(\tau^t A) \rangle \\ &= \langle k_J M, A(k_J(N))^t A \rangle + 2\langle k(\bar{p} \mathbf{a}), (k \mathbf{b}) \tau^t A \rangle \\ &= \langle \tau^t A(k_J(M)) \tau A, k_J(N) \rangle + 2\langle k(\bar{p} \mathbf{a}) A, k \mathbf{b} \rangle \\ &= 2\langle k_J^{-1}(\tau^t A(k_J(M)) \tau A), N \rangle + 4\langle \bar{p} \mathbf{a} k^{-1}(A), \mathbf{b} \rangle \\ &= 2\langle k_J^{-1}(\tau^t A(k_J(M)) \tau A + \bar{p} \mathbf{a} k^{-1}(A)), N + \mathbf{b} \rangle \\ &= 2\langle \varphi(\bar{p}, \tau^t A)(M + \mathbf{a}), N + \mathbf{b} \rangle, \end{aligned}$	$\begin{aligned} & 2\langle \tau^t \varphi(p, A) \tau(M + \mathbf{a}), N + \mathbf{b} \rangle \quad M + \mathbf{a}, N + \mathbf{b} \in \mathfrak{J}^C \\ &= 2\langle M + \mathbf{a}, \varphi(p, A)(N + \mathbf{b}) \rangle \\ &= 2\langle M + \mathbf{a}, k_J^{-1}(A(k_J(N))^t A) + p \mathbf{b} k^{-1}(\tau^t A) \rangle \\ &= 2\langle M, k_J^{-1}(A(k_J(N))^t A) \rangle + 4\langle \mathbf{a}, p \mathbf{b} k^{-1}(\tau^t A) \rangle \\ &= \langle k_J M, A(k_J(N))^t A \rangle + 2\langle k(\bar{p} \mathbf{a}), (k \mathbf{b}) \tau^t A \rangle \\ &= \langle \tau^t A(k_J(M)) \tau A, k_J(N) \rangle + 2\langle k(\bar{p} \mathbf{a}) A, k \mathbf{b} \rangle \\ &= 2\langle k_J^{-1}(\tau^t A(k_J(M)) \tau A), N \rangle + 4\langle \bar{p} \mathbf{a} k^{-1}(A), \mathbf{b} \rangle \\ &= 2\langle k_J^{-1}(\tau^t A(k_J(M)) \tau A + \bar{p} \mathbf{a} k^{-1}(A)), N + \mathbf{b} \rangle \\ &= 2\langle \varphi(\bar{p}, \tau^t A)(M + \mathbf{a}), N + \mathbf{b} \rangle, \end{aligned}$
0902.0431_E00253	41	■ 0.969	$0(27)=0(\frac{\mathfrak{J}}{\mathfrak{J}})=\left\{\alpha\in\operatorname{Iso}_{\mathfrak{J}}(\frac{\mathfrak{J}}{\mathfrak{J}})\mid(\alpha X,\alpha Y)=(X,Y)\right\}\quad\text{(Lemma 2.2.4.(2)).}$	$O(27)=O(\mathfrak{J})=\{\alpha\in\operatorname{Iso}_{\mathbf{R}}(\mathfrak{J})\mid(\alpha X,\alpha Y)=(X,Y)\}\quad\text{(Lemma 2.2.4.(2)).}$	$O(27)=O(\mathfrak{J})=\{\alpha\in\operatorname{Iso}_{\mathbf{R}}(\mathfrak{J})\mid(\alpha X,\alpha Y)=(X,Y)\}\quad\text{(Lemma 2.2.4.(2)).}$
0902.0431_E00997	166	■ 0.970	$(R\times R)R_1=\Theta(R)^2R_1+\frac{1}{30}B_8(R,R_1)R,\quad R_1\in\mathfrak{e}_8^C,$	$(R\times R)R_1=\Theta(R)^2R_1+\frac{1}{30}B_8(R,R_1)R,\quad R_1\in\mathfrak{e}_8^C,$	$(R\times R)R_1=\Theta(R)^2R_1+\frac{1}{30}B_8(R,R_1)R,\quad R_1\in\mathfrak{e}_8^C,$

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0902.0431_EQ1016	170	■ 0.971	$\begin{gathered} \pm \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \left(\frac{1}{2}\mu_2 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_2 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{2}{3}\nu\right) \pm r, \\ \pm 2r, \\ \pm \nu \pm r \end{gathered}$	$\begin{gathered} \pm \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \left(\frac{1}{2}\mu_2 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_2 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{2}{3}\nu\right) \pm r, \\ \pm 2r, \\ \pm \nu \pm r \end{gathered}$	
0902.0431_EQ0230	38	■ 0.971	$\begin{aligned} & (\operatorname{tr}(\alpha X) - \operatorname{tr}(X))X^2 + \frac{1}{2}(\operatorname{tr}(X)^2 - \operatorname{tr}(\alpha X)^2 + \operatorname{tr}((\alpha X)^2) - \operatorname{tr}(X^2))X \\ & = (\det X - \det(\alpha X))E \end{aligned}$	$(\operatorname{tr}(\alpha X) - \operatorname{tr}(X))X^2 + \frac{1}{2}(\operatorname{tr}(X)^2 - \operatorname{tr}(\alpha X)^2 + \operatorname{tr}((\alpha X)^2) - \operatorname{tr}(X^2))X = (\det X - \det(\alpha X))E$	$(\operatorname{tr}(\alpha X) - \operatorname{tr}(X))X^2 + \frac{1}{2}(\operatorname{tr}(X)^2 - \operatorname{tr}(\alpha X)^2 + \operatorname{tr}((\alpha X)^2) - \operatorname{tr}(X^2))X = (\det X - \det(\alpha X))E.$
0902.0431_EQ0944	157	■ 0.971	$\overline{f^{-1}(A(fP))} = f^{-1}(\overline{(\operatorname{Ad} J_3) A f \bar{P}}), \quad A \in SU(6), P \in (\mathfrak{P}_C)^C.$	$\overline{f^{-1}(A(fP))} = f^{-1}(\overline{((\operatorname{Ad} J_3) A f \bar{P})}), \quad A \in SU(6), P \in (\mathfrak{P}_C)^C.$	$\overline{f^{-1}(A(fP))} = f^{-1}(\overline{((\operatorname{Ad} J_3) A f \bar{P})}), \quad A \in SU(6), P \in (\mathfrak{P}_C)^C.$
0902.0431_EQ0979	163	■ 0.974	$\mathfrak{a} \ni [\Phi_1, (\Phi, P, Q, r, s, t)] = ([\Phi_1, \Phi], \Phi_1 P, \Phi_1 Q, 0, 0, 0),$	$\mathfrak{a} \ni [\Phi_1, (\Phi, P, Q, r, s, t)] = ([\Phi_1, \Phi], \Phi_1 P, \Phi_1 Q, 0, 0, 0),$	$\mathfrak{a} \ni [\Phi_1, (\Phi, P, Q, r, s, t)] = ([\Phi_1, \Phi], \Phi_1 P, \Phi_1 Q, 0, 0, 0),$

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0902.0431_E00751	124	■ 0.975	$\begin{aligned} & \left[\Phi(h_\delta + \tilde{H}, 0, 0, \nu), \Phi(0, E_j, 0, 0) \right] \\ &= \Phi\left(0, \left(h_\delta + \tilde{H} + \frac{2}{3}\nu\right)E_j, 0, 0\right) = \left(\mu_j + \frac{2}{3}\nu\right)\Phi(0, E_j, 0, 0), \\ & \left[\Phi\left((h_\delta + \tilde{H}, 0, 0, \nu), \Phi(0, 0, E_j, 0)\right) = \Phi\left(0, 0, \left((h_\delta + \tilde{H})' - \frac{2}{3}\nu\right)E_j, 0\right) \right. \\ & \left. = \Phi\left(0, 0, \left(h_\delta - \tilde{H} - \frac{2}{3}\nu\right)E_j, 0\right) = \left(-\mu_j - \frac{2}{3}\nu\right)\Phi(0, 0, E_j, 0). \right. \end{aligned}$	$\begin{aligned} & [\Phi(h_\delta + \tilde{H}, 0, 0, \nu), \Phi(0, E_j, 0, 0)] \\ &= \Phi\left(0, \left(h_\delta + \tilde{H} + \frac{2}{3}\nu\right)E_j, 0, 0\right) = \left(\mu_j + \frac{2}{3}\nu\right)\Phi(0, E_j, 0, 0), \\ & [\Phi((h_\delta + \tilde{H}, 0, 0, \nu), \Phi(0, 0, E_j, 0))] = \Phi\left(0, 0, \left((h_\delta + \tilde{H})' - \frac{2}{3}\nu\right)E_j, 0\right) \\ &= \Phi\left(0, 0, \left(h_\delta - \tilde{H} - \frac{2}{3}\nu\right)E_j, 0\right) = \left(-\mu_j - \frac{2}{3}\nu\right)\Phi(0, 0, E_j, 0). \end{aligned}$	
0902.0431_E00894	148	■ 0.975	$\mathfrak{S}(8, \boldsymbol{C}) = \left\{ S \in M(8, \boldsymbol{C}) \mid {}^t S = -S \right\},$	$\mathfrak{S}(8, \boldsymbol{C}) = \left\{ S \in M(8, \boldsymbol{C}) \mid {}^t S = -S \right\},$	$\mathfrak{S}(8, \boldsymbol{C}) = \left\{ S \in M(8, \boldsymbol{C}) \mid {}^t S = -S \right\},$
0902.0431_E00608	101	■ 0.976	$\begin{aligned} & \boldsymbol{P}_3 \in \mathfrak{J}(4, \boldsymbol{H}) \mid P^2 = P, \operatorname{tr}(P) = 1 \\ &= \{AE_1A^* \mid A \in \operatorname{Sp}(4), E_1 = \operatorname{diag}(1, 0, 0, 0) \in M(4, \boldsymbol{H})\}. \end{aligned}$	$\begin{aligned} \boldsymbol{HP}_3 &= \{P \in \mathfrak{J}(4, \boldsymbol{H}) \mid P^2 = P, \operatorname{tr}(P) = 1\} \\ &= \{AE_1A^* \mid A \in \operatorname{Sp}(4), E_1 = \operatorname{diag}(1, 0, 0, 0) \in M(4, \boldsymbol{H})\}. \end{aligned}$	$\begin{aligned} \boldsymbol{HP}_3 &= \{P \in \mathfrak{J}(4, \boldsymbol{H}) \mid P^2 = P, \operatorname{tr}(P) = 1\} \\ &= \left\{ AE_1A^* \mid A \in Sp(4), E_1 = \operatorname{diag}(1, 0, 0, 0) \in M(4, \boldsymbol{H}) \right\}. \end{aligned}$
0902.0431_E00813	136	■ 0.976	$\widetilde{\iota}(\alpha) = \iota \alpha \iota^{-1}, \quad \alpha \in E_7.$	$\widetilde{\iota}(\alpha) = \iota \alpha \iota^{-1}, \quad \alpha \in E_7.$	$\widetilde{\iota}(\alpha) = \iota \alpha \iota^{-1}, \quad \alpha \in E_7.$
0902.0431_E00414	68	■ 0.976	$M \times N = \left(\begin{array}{ccc} \boldsymbol{m}_2 \times \boldsymbol{n}_3 & \boldsymbol{m}_3 \times \boldsymbol{n}_1 & \boldsymbol{m}_1 \times \boldsymbol{n}_2 \\ + & + & + \\ \boldsymbol{n}_2 \times \boldsymbol{m}_3 & \boldsymbol{n}_3 \times \boldsymbol{m}_1 & \boldsymbol{n}_1 \times \boldsymbol{m}_2 \end{array} \right) \in M(3, \boldsymbol{C}),$	$M \times N = \left(\begin{array}{ccc} \boldsymbol{m}_2 \times \boldsymbol{n}_3 & \boldsymbol{m}_3 \times \boldsymbol{n}_1 & \boldsymbol{m}_1 \times \boldsymbol{n}_2 \\ + & + & + \\ \boldsymbol{n}_2 \times \boldsymbol{m}_3 & \boldsymbol{n}_3 \times \boldsymbol{m}_1 & \boldsymbol{n}_1 \times \boldsymbol{m}_2 \end{array} \right) \in M(3, \boldsymbol{C}),$	$M \times N = \left(\begin{array}{ccc} \boldsymbol{m}_2 \times \boldsymbol{n}_3 & \boldsymbol{m}_3 \times \boldsymbol{n}_1 & \boldsymbol{m}_1 \times \boldsymbol{n}_2 \\ + & + & + \\ \boldsymbol{n}_2 \times \boldsymbol{m}_3 & \boldsymbol{n}_3 \times \boldsymbol{m}_1 & \boldsymbol{n}_1 \times \boldsymbol{m}_2 \end{array} \right) \in M(3, \boldsymbol{C}),$
0902.0431_E01008	167	■ 0.976	$\begin{aligned} & E_8 = \left\{ \alpha \in E_8^C \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle \right\} \\ &= \left\{ \alpha \in E_8^C \mid \tau \tilde{\lambda} \alpha = \alpha \tau \tilde{\lambda} \right\}. \end{aligned}$	$\begin{aligned} E_8 &= \left\{ \alpha \in E_8^C \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle \right\} \\ &= \left\{ \alpha \in E_8^C \mid \tau \tilde{\lambda} \alpha = \alpha \tau \tilde{\lambda} \right\}. \end{aligned}$	$\begin{aligned} E_8 &= \left\{ \alpha \in E_8^C \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle \right\} \\ &= \left\{ \alpha \in E_8^C \mid \tau \tilde{\lambda} \alpha = \alpha \tau \tilde{\lambda} \right\}. \end{aligned}$
0902.0431_E00088	18	■ 0.976	$B_2(D_1, D_2) = 4 \operatorname{tr}(D_1 D_2), \quad D_i \in \mathfrak{g}_2^C.$	$B_2(D_1, D_2) = 4 \operatorname{tr}(D_1 D_2), \quad D_i \in \mathfrak{g}_2^C.$	$B_2(D_1, D_2) = 4 \operatorname{tr}(D_1 D_2), \quad D_i \in \mathfrak{g}_2^C.$

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0902.0431_E08669	111	■ 0.977	$\begin{aligned} (P, Q) &= (X, Z) + (Y, W) + \xi \zeta + \eta \omega, \\ \langle P, Q \rangle &= \langle X, Z \rangle + \langle Y, W \rangle + (\tau \xi) \zeta + (\tau \eta) \omega, \\ \{P, Q\} &= (X, W) - (Z, Y) + \xi \omega - \zeta \eta, \end{aligned}$	$\begin{aligned} (P, Q) &= (X, Z) + (Y, W) + \xi \zeta + \eta \omega, \\ \langle P, Q \rangle &= \langle X, Z \rangle + \langle Y, W \rangle + (\tau \xi) \zeta + (\tau \eta) \omega, \\ \{P, Q\} &= (X, W) - (Z, Y) + \xi \omega - \zeta \eta, \end{aligned}$	$\begin{aligned} (P, Q) &= (X, Z) + (Y, W) + \xi \zeta + \eta \omega, \\ \langle P, Q \rangle &= \langle X, Z \rangle + \langle Y, W \rangle + (\tau \xi) \zeta + (\tau \eta) \omega, \\ \{P, Q\} &= (X, W) - (Z, Y) + \xi \omega - \zeta \eta, \end{aligned}$
0902.0431_E08684	113	■ 0.977	$\mathfrak{e}_7^C = \left\{ \Phi(\phi, A, B, \nu) \in \operatorname{Hom}_C(\mathfrak{P}^C) \mid \phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C \right\}.$	$\mathfrak{e}_7^C = \left\{ \Phi(\phi, A, B, \nu) \in \operatorname{Hom}_C(\mathfrak{P}^C) \mid \phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C \right\}.$	$\mathfrak{e}_7^C = \left\{ \Phi(\phi, A, B, \nu) \in \operatorname{Hom}_C(\mathfrak{P}^C) \mid \phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C \right\}.$
0902.0431_E08948	157	■ 0.979	$((X, Y, \xi, \eta), (M, N)) = (X + M, Y + N, \xi, \eta).$	$((X, Y, \xi, \eta), (M, N)) = (X + M, Y + N, \xi, \eta).$	$((X, Y, \xi, \eta), (M, N)) = (X + M, Y + N, \xi, \eta).$
0902.0431_E08994	19	■ 0.979	$\begin{array}{l} f_*(A) = \varepsilon A - \bar{\varepsilon}^t A, \quad \varepsilon = \frac{1}{2}(1 + ie_1) \\ \varphi_*(D)(a + \boldsymbol{m}) = D\boldsymbol{m}, \quad a + \boldsymbol{m} \in \boldsymbol{C}^C \oplus (\boldsymbol{C}^3)^C = \mathfrak{C}^C \end{array}$	$\begin{array}{l} f_*(A) = \varepsilon A - \bar{\varepsilon}^t A, \quad \varepsilon = \frac{1}{2}(1 + ie_1) \\ \varphi_*(D)(a + \boldsymbol{m}) = D\boldsymbol{m}, \quad a + \boldsymbol{m} \in \boldsymbol{C}^C \oplus (\boldsymbol{C}^3)^C = \mathfrak{C}^C \end{array}$	$\begin{array}{l} f_*(A) = \varepsilon A - \bar{\varepsilon}^t A, \quad \varepsilon = \frac{1}{2}(1 + ie_1), \\ \varphi_*(D)(a + \boldsymbol{m}) = D\boldsymbol{m}, \quad a + \boldsymbol{m} \in \boldsymbol{C}^C \oplus (\boldsymbol{C}^3)^C = \mathfrak{C}^C, \end{array}$
0902.0431_E08381	63	■ 0.979	$\begin{aligned} \alpha E_2 &= \alpha(-F_2(1) \times F_2(1)) = -\alpha F_2(1) \times \alpha F_2(1) \\ &= -(F_2(x_2) + F_3(x_3)) \times (F_2(x_2) + F_3(x_3)) \quad (\text{for some } x_2, x_3 \in \mathfrak{C}) \\ &= (x_2, x_2)E_2 + (x_3, x_3)E_3 - F_1(\overline{x_2 x_3}) \in \mathfrak{J}(2, \mathfrak{C}). \end{aligned}$	$\begin{aligned} \alpha E_2 &= \alpha(-F_2(1) \times F_2(1)) = -\alpha F_2(1) \times \alpha F_2(1) \\ &= -(F_2(x_2) + F_3(x_3)) \times (F_2(x_2) + F_3(x_3)) \quad (\text{for some } x_2, x_3 \in \mathfrak{C}) \\ &= (x_2, x_2)E_2 + (x_3, x_3)E_3 - F_1(\overline{x_2 x_3}) \in \mathfrak{J}(2, \mathfrak{C}) \end{aligned}$	$\begin{aligned} \alpha E_2 &= \alpha(-F_2(1) \times F_2(1)) = -\alpha F_2(1) \times \alpha F_2(1) \\ &= -(F_2(x_2) + F_3(x_3)) \times (F_2(x_2) + F_3(x_3)) \quad (\text{for some } x_2, x_3 \in \mathfrak{C}) \\ &= (x_2, x_2)E_2 + (x_3, x_3)E_3 - F_1(\overline{x_2 x_3}) \in \mathfrak{J}(2, \mathfrak{C}). \end{aligned}$
0902.0431_E01171	197	■ 0.979	$\begin{aligned} &\exp(\operatorname{ad}(X, 0, 0))(D, \boldsymbol{u}, \boldsymbol{v}) = (\exp(\operatorname{ad}(X)D, (\exp X)\boldsymbol{u}, (\exp(-^t X))\boldsymbol{v})) \\ &= (\operatorname{Ad}(\exp X)D, (\exp X)\boldsymbol{u}, {}^t(\exp X)^{-1}\boldsymbol{v}) \\ &= \varphi(\exp X)(D, \boldsymbol{u}, \boldsymbol{v}). \end{aligned}$	$\begin{aligned} &\exp(\operatorname{ad}(X, 0, 0))(D, \boldsymbol{u}, \boldsymbol{v}) = (\exp(\operatorname{ad}(X)D, (\exp X)\boldsymbol{u}, (\exp(-^t X))\boldsymbol{v})) \\ &= (\operatorname{Ad}(\exp X)D, (\exp X)\boldsymbol{u}, {}^t(\exp X)^{-1}\boldsymbol{v}) \\ &= \varphi(\exp X)(D, \boldsymbol{u}, \boldsymbol{v}). \end{aligned}$	$\begin{aligned} \exp(\operatorname{ad}(X, 0, 0))(D, \boldsymbol{u}, \boldsymbol{v}) &= (\exp(\operatorname{ad}(X)D, (\exp X)\boldsymbol{u}, (\exp(-^t X))\boldsymbol{v})) \\ &= (\operatorname{Ad}(\exp X)D, (\exp X)\boldsymbol{u}, {}^t(\exp X)^{-1}\boldsymbol{v}) \\ &= \varphi(\exp X)(D, \boldsymbol{u}, \boldsymbol{v}). \end{aligned}$
0902.0431_E08632	105	■ 0.979	$\alpha = \varphi(A)\beta = \varphi(A)\varphi(B) = \varphi(AB), \quad AB \in \operatorname{Sp}(4),$	$\alpha = \varphi(A)\beta = \varphi(A)\varphi(B) = \varphi(AB), \quad AB \in \operatorname{Sp}(4),$	$\alpha = \varphi(A)\beta = \varphi(A)\varphi(B) = \varphi(AB), \quad AB \in \operatorname{Sp}(4),$

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0902.0431_EQ0599	100	■ 0.980	$\begin{aligned} & \begin{aligned} & (\alpha \boldsymbol{a})^* (\alpha \boldsymbol{b}) = (\boldsymbol{p} \boldsymbol{a} k^{-1} (\tau^t A))^* (\boldsymbol{p} \boldsymbol{b} k^{-1} (\tau^t A)) \\ & = k^{-1} (A) \boldsymbol{a}^* \boldsymbol{b} k^{-1} (\tau^t A), \\ & \tau \alpha \tau (\boldsymbol{a}^* \boldsymbol{b}) = \tau \left(k_J^{-1} \left(A (k_J \tau (\boldsymbol{a}^* \boldsymbol{b}))^t A \right) \right) \\ & = -\tau k^{-1} \left(J \tau \left(A (k (\tau (\boldsymbol{a}^* \boldsymbol{b}))) J^t A \right) \right) \\ & = -\tau k^{-1} \left(J \tau \left(A J \tau k (\boldsymbol{a}^* \boldsymbol{b})^t A \right) \right) \\ & = -\tau k^{-1} \left(J \tau A J k (\boldsymbol{a}^* \boldsymbol{b}) \tau^t A \right) \\ & = -k^{-1} \left(A \tau k (\boldsymbol{a}^* \boldsymbol{b}) J^t A J \right) = k^{-1} (A) \boldsymbol{a}^* \boldsymbol{b} k^{-1} (\tau^t A), \\ & (\alpha \boldsymbol{a}) (\alpha N) = (\boldsymbol{p} \boldsymbol{a} k^{-1} (\tau^t A)) k_J^{-1} \left(A (k_J (N))^t A \right) \\ & = -\boldsymbol{p} \boldsymbol{a} k^{-1} (\tau^t A) k^{-1} \left(A (k(N)) J^t A J \right) \\ & = -\boldsymbol{p} \boldsymbol{a} N k^{-1} (J^t A J) = \boldsymbol{p} \boldsymbol{a} N \tau k^{-1} (\tau^t A), \\ & \tau \alpha \tau (\boldsymbol{a} N) = \tau (p \tau (\boldsymbol{a} N) k^{-1} (\tau^t A)) = \boldsymbol{p} \boldsymbol{a} N \tau k^{-1} (\tau^t A). \end{aligned} \end{aligned}$	$\begin{aligned} & (\alpha \boldsymbol{a})^* (\alpha \boldsymbol{b}) = (\boldsymbol{p} \boldsymbol{a} k^{-1} (\tau^t A))^* (\boldsymbol{p} \boldsymbol{b} k^{-1} (\tau^t A)) \\ & = k^{-1} (A) \boldsymbol{a}^* \boldsymbol{b} k^{-1} (\tau^t A), \\ & \tau \alpha \tau (\boldsymbol{a}^* \boldsymbol{b}) = \tau \left(k_J^{-1} \left(A (k_J \tau (\boldsymbol{a}^* \boldsymbol{b}))^t A \right) \right) \\ & = -\tau k^{-1} \left(J \tau \left(A (k (\tau (\boldsymbol{a}^* \boldsymbol{b}))) J^t A \right) \right) \\ & = -\tau k^{-1} \left(J \tau \left(A J \tau k (\boldsymbol{a}^* \boldsymbol{b})^t A \right) \right) \\ & = -\tau k^{-1} \left(J \tau A J k (\boldsymbol{a}^* \boldsymbol{b}) \tau^t A \right) \\ & = -k^{-1} \left(A \tau k (\boldsymbol{a}^* \boldsymbol{b}) J^t A J \right) = k^{-1} (A) \boldsymbol{a}^* \boldsymbol{b} k^{-1} (\tau^t A), \\ & (\alpha \boldsymbol{a}) (\alpha N) = (\boldsymbol{p} \boldsymbol{a} k^{-1} (\tau^t A)) k_J^{-1} \left(A (k_J (N))^t A \right) \\ & = -\boldsymbol{p} \boldsymbol{a} k^{-1} (\tau^t A) k^{-1} \left(A (k(N)) J^t A J \right) \\ & = -\boldsymbol{p} \boldsymbol{a} N k^{-1} (J^t A J) = \boldsymbol{p} \boldsymbol{a} N \tau k^{-1} (\tau^t A), \\ & \tau \alpha \tau (\boldsymbol{a} N) = \tau (p \tau (\boldsymbol{a} N) k^{-1} (\tau^t A)) = \boldsymbol{p} \boldsymbol{a} N \tau k^{-1} (\tau^t A). \end{aligned}$	
0902.0431_EQ1089	185	■ 0.980	$\begin{aligned} & \begin{aligned} & \left(\mathfrak{e}_8 \right)^{\tilde{\lambda} \gamma} = \left\{ \Theta \in \Theta (\mathfrak{e}_8^C) \mid \tilde{\lambda} \gamma \Theta = \Theta \tilde{\lambda} \gamma \right\} \\ & = \left\{ \Theta (\Phi, P, -\lambda \gamma P, 0, s, -s) \mid \Phi \in (\mathfrak{e}_7)^{\lambda \gamma}, P \in (\mathfrak{P}^C)_{\lambda \gamma}, s \in \boldsymbol{R} \right\} \end{aligned} \end{aligned}$	$\begin{aligned} & (\mathfrak{e}_8)^{\tilde{\lambda} \gamma} = \left\{ \Theta \in \Theta (\mathfrak{e}_8^C) \mid \tilde{\lambda} \gamma \Theta = \Theta \tilde{\lambda} \gamma \right\} \\ & = \left\{ \Theta (\Phi, P, -\lambda \gamma P, 0, s, -s) \mid \Phi \in (\mathfrak{e}_7)^{\lambda \gamma}, P \in (\mathfrak{P}^C)_{\lambda \gamma}, s \in \boldsymbol{R} \right\} \end{aligned}$	

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0902.0431_E00506	83	■ 0.980	$\mathfrak{h}=\left\{\begin{array}{l} h=h_{\delta}+\widetilde{H} \in \mathfrak{e}_6^C \\ \text{\textit{H}} \in \mathfrak{e}_6^C \end{array}\right\} \quad \left\{ \begin{array}{l} h_{\delta}=\sum_{k=0}^3 \lambda_k H_k=-\sum_{k=0}^3 \lambda_k i G_{k4+k}, \lambda_k \in C \\ H=\sum_{j=1}^3 \mu_j E_j, \mu_j \in C, \mu_1+\mu_2+\mu_3=0 \end{array} \right\},$		
0902.0431_E00831	139	■ 0.980	$\begin{aligned} \left(E_7\right)^{\kappa, \mu} &= \left\{\alpha \in E_7 \mid \kappa \alpha=\alpha \kappa, \mu \alpha=\alpha \mu\right\}, \\ \left(\left(E_7\right)^{\kappa, \mu}\right)_{(0, E_1, 0, 1)} &= \left\{\alpha \in\left(E_7\right)^{\kappa, \mu} \mid \alpha(0, E_1, 0, 1)=(0, E_1, 0, 1)\right\} . \end{aligned}$		$\begin{aligned} \left(E_7\right)^{\kappa, \mu} &= \left\{\alpha \in E_7 \mid \kappa \alpha=\alpha \kappa, \mu \alpha=\alpha \mu\right\}, \\ \left(\left(E_7\right)^{\kappa, \mu}\right)_{(0, E_1, 0, 1)} &= \left\{\alpha \in\left(E_7\right)^{\kappa, \mu} \mid \alpha(0, E_1, 0, 1)=(0, E_1, 0, 1)\right\} . \end{aligned}$
0902.0431_E01132	191	■ 0.981	$\begin{aligned} \operatorname{tr}\left(A D_{\{1\}} \tau^t A\right) &= \operatorname{tr}\left(A D_{\{1\}}\left(\tau^t A\right)\right) \\ \operatorname{tr}\left(A D_{\{1\}}\left(\tau^t A\right)\right) &= \operatorname{tr}\left(A D_{\{1\}}\left(\tau^t A\right)\right) \\ \operatorname{tr}\left(A D_{\{1\}}\left(\tau^t A\right)\right) &= \operatorname{tr}\left(A D_{\{1\}}\left(\tau^t A\right)\right) \end{aligned}$	$\operatorname{tr}\left(A D_1 \tau^t A\left(\tau^t\left(A D_2 \tau^t A\right)\right)\right)=\operatorname{tr}\left(A D_1\left(\tau^t D_2\right) A^{-1}\right)=\operatorname{tr}\left(D_1\left(\tau^t D_2\right)\right),$ $\langle A \boldsymbol{X}, A \boldsymbol{Y}\rangle=\langle \boldsymbol{X}, \boldsymbol{Y}\rangle,$	$\operatorname{tr}\left(A D_1 \tau^t A\left(\tau^t\left(A D_2 \tau^t A\right)\right)\right)=\operatorname{tr}\left(A D_1\left(\tau^t D_2\right) A^{-1}\right)=\operatorname{tr}\left(D_1\left(\tau^t D_2\right)\right),$ $\langle A \boldsymbol{X}, A \boldsymbol{Y}\rangle=\langle \boldsymbol{X}, \boldsymbol{Y}\rangle,$
0902.0431_E00641	64	■ 0.981	$\begin{aligned} \alpha \sigma X &= \alpha \sigma\left(X_1+X_2\right) \quad X_1 \in \mathfrak{J}_{\sigma}, \quad X_2 \in \mathfrak{J}_{-\sigma} \\ &= \alpha\left(X_1-X_2\right)=\alpha X_1-\alpha X_2=\sigma\left(\alpha X_1\right)+\sigma\left(\alpha X_2\right) \\ &= \sigma \alpha\left(X_1+X_2\right)=\sigma \alpha X, \quad X \in \mathfrak{J} . \end{aligned}$	$\begin{aligned} \alpha \sigma X &= \alpha \sigma\left(X_1+X_2\right) \quad X_1 \in \mathfrak{J}_{\sigma}, \quad X_2 \in \mathfrak{J}_{-\sigma} \\ &= \alpha\left(X_1-X_2\right)=\alpha X_1-\alpha X_2=\sigma\left(\alpha X_1\right)+\sigma\left(\alpha X_2\right) \\ &= \sigma \alpha\left(X_1+X_2\right)=\sigma \alpha X, \quad X \in \mathfrak{J} \end{aligned}$	$\begin{aligned} \alpha \sigma X &= \alpha \sigma\left(X_1+X_2\right) \quad X_1 \in \mathfrak{J}_{\sigma}, \quad X_2 \in \mathfrak{J}_{-\sigma} \\ &= \alpha\left(X_1-X_2\right)=\alpha X_1-\alpha X_2=\sigma\left(\alpha X_1\right)+\sigma\left(\alpha X_2\right) \\ &= \sigma \alpha\left(X_1+X_2\right)=\sigma \alpha X, \quad X \in \mathfrak{J} . \end{aligned}$
0902.0431_E00641	106	■ 0.981	$\begin{aligned} h(a, b) &= \frac{a+b}{2}+i \frac{a-b}{2} e_1=\iota a+\bar{\iota} b, \\ h(A, B) &= \frac{A+B}{2}+i \frac{A-B}{2} e_1=\iota A+\bar{\iota} B, \end{aligned} \quad \iota=\frac{1+i e_1}{2} .$	$\begin{aligned} h(a, b) &= \frac{a+b}{2}+i \frac{a-b}{2} e_1=\iota a+\bar{\iota} b, \\ h(A, B) &= \frac{A+B}{2}+i \frac{A-B}{2} e_1=\iota A+\bar{\iota} B, \end{aligned} \quad \iota=\frac{1+i e_1}{2} .$	$\begin{aligned} h(a, b) &= \frac{a+b}{2}+i \frac{a-b}{2} e_1=\iota a+\bar{\iota} b, \\ h(A, B) &= \frac{A+B}{2}+i \frac{A-B}{2} e_1=\iota A+\bar{\iota} B, \end{aligned} \quad \iota=\frac{1+i e_1}{2} .$
0902.0431_E00474	78	■ 0.981	$\left(\left[\phi, \phi_1\right], \phi_2\right)_6+\left(\phi_1,\left[\phi, \phi_2\right]\right)_6=0, \quad \phi, \phi_i \in \mathfrak{e}_6^C .$	$\left(\left[\phi, \phi_1\right], \phi_2\right)_6+\left(\phi_1,\left[\phi, \phi_2\right]\right)_6=0, \quad \phi, \phi_i \in \mathfrak{e}_6^C .$	$\left(\left[\phi, \phi_1\right], \phi_2\right)_6+\left(\phi_1,\left[\phi, \phi_2\right]\right)_6=0, \quad \phi, \phi_i \in \mathfrak{e}_6^C .$
0902.0431_E01173	197	■ 0.982	$\mathfrak{e}_8^C=\mathfrak{g}_0 \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1},$	$\mathfrak{e}_8^C=\mathfrak{g}_0 \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1},$	$\mathfrak{e}_8^C=\mathfrak{g}_0 \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1},$

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0902.0431_E00123	24	■ 0.985	$\overline{\omega_{\{1\}} \overline{\boldsymbol{m}}} = \overline{\gamma_{\{1\}} \boldsymbol{w}} = \overline{\gamma_{\{1\}} \boldsymbol{w} \boldsymbol{m}} = \overline{\gamma_{\{1\}} \boldsymbol{m}} = \overline{\gamma_{\{1\}} \boldsymbol{w} \boldsymbol{m}}$ for all $\boldsymbol{m} \in \boldsymbol{C}^3$,	$\omega_1 \overline{\boldsymbol{m}} = w(\gamma_1 \boldsymbol{m}) = \gamma_1(w \boldsymbol{m}) = \overline{\omega_1 \boldsymbol{m}} = \overline{\omega_1 \boldsymbol{m}}$ for all $\boldsymbol{m} \in \boldsymbol{C}^3$,	$\omega_1 \overline{\boldsymbol{m}} = w(\gamma_1 \boldsymbol{m}) = \gamma_1(w \boldsymbol{m}) = \overline{\omega_1 \boldsymbol{m}} = \overline{\omega_1 \boldsymbol{m}}$ for all $\boldsymbol{m} \in \boldsymbol{C}^3$,
0902.0431_E00740	122	■ 0.985	$B_{\{7\}}(\Phi_1, \Phi_2) = k(\Phi_1, \Phi_2)_7 = k' \text{tr}(\Phi_1 \Phi_2)$.	$B_7(\Phi_1, \Phi_2) = k(\Phi_1, \Phi_2)_7 = k' \text{tr}(\Phi_1 \Phi_2)$.	$B_7(\Phi_1, \Phi_2) = k(\Phi_1, \Phi_2)_7 = k' \text{tr}(\Phi_1 \Phi_2)$.
0902.0431_E01007	167	■ 0.985	$\begin{gathered} \sqrt{2} \left[\widetilde{E}_1, \widetilde{F}_2(e_i) \right], \quad \sqrt{2} \left[\widetilde{E}_1, \widetilde{F}_3(e_i) \right], \quad \sqrt{2} \left[\widetilde{E}_3, \widetilde{F}_1(e_i) \right], \quad 0 \leq i \leq 7, \\ \frac{1}{\sqrt{2}} \left[\widetilde{F}_1(e_i), \widetilde{F}_1(e_j) \right], \quad 0 \leq i < j \leq 7 \end{gathered}$	$\sqrt{2} \left[\widetilde{E}_1, \widetilde{F}_2(e_i) \right], \quad \sqrt{2} \left[\widetilde{E}_1, \widetilde{F}_3(e_i) \right], \quad \sqrt{2} \left[\widetilde{E}_3, \widetilde{F}_1(e_i) \right], \quad 0 \leq i \leq 7, \\ \frac{1}{\sqrt{2}} \left[\widetilde{F}_1(e_i), \widetilde{F}_1(e_j) \right], \quad 0 \leq i < j \leq 7$	$\sqrt{2} [\widetilde{E}_1, \widetilde{F}_2(e_i)], \quad \sqrt{2} [\widetilde{E}_1, \widetilde{F}_3(e_i)], \quad \sqrt{2} [\widetilde{E}_3, \widetilde{F}_1(e_i)], \quad 0 \leq i \leq 7, \\ \frac{1}{\sqrt{2}} [\widetilde{F}_1(e_i), \widetilde{F}_1(e_j)], \quad 0 \leq i < j \leq 7$
0902.0431_E01214	204	■ 0.985	$\text{Ker} \varphi = \{(E, E), (\zeta E, \zeta^2 E), (\zeta^2 E, \zeta^4 E), (\zeta^3 E, \zeta E), (\zeta^4 E, \zeta^3 E)\} = \boldsymbol{Z}_5$.	$\text{Ker} \varphi = \{(E, E), (\zeta E, \zeta^2 E), (\zeta^2 E, \zeta^4 E), (\zeta^3 E, \zeta E), (\zeta^4 E, \zeta^3 E)\} = \boldsymbol{Z}_5$.	$\text{Ker} \varphi = \{(E, E), (\zeta E, \zeta^2 E), (\zeta^2 E, \zeta^4 E), (\zeta^3 E, \zeta E), (\zeta^4 E, \zeta^3 E)\} = \boldsymbol{Z}_5$.
0902.0431_E00780	131	■ 0.985	$\beta_{\{1\}}(X \times X) = \beta X \times \beta X = \tau \beta \tau(X \times X)$,	$\beta_1(X \times X) = \beta X \times \beta X = \tau \beta \tau(X \times X)$,	$\beta_1(X \times X) = \beta X \times \beta X = \tau \beta \tau(X \times X)$,
0902.0431_E00152	28	■ 0.985	$\begin{aligned} G_{\{2(2)\}} \times \text{Simeq} \left(G_{\{2(2)\}} \cap O(8) \right) \times \boldsymbol{R}^d \\ = \left((G_{\{2\}}^C)^{\tau \gamma} \right)^\gamma \times \boldsymbol{R}^d = \left((G_{\{2\}}^C)^\tau \right)^\gamma \times \boldsymbol{R}^d = (G_2)^\gamma \times \boldsymbol{R}^d \\ = (\text{Sp}(1) \times \text{Sp}(1)) / \boldsymbol{Z}_2 \times \boldsymbol{R}^d \text{ (Theorem 1.10.1)}, \quad d = 8. \end{aligned}$	$\begin{aligned} G_{2(2)} \simeq (G_{2(2)} \cap O(8)) \times \boldsymbol{R}^d \\ = \left((G_2^C)^{\tau \gamma} \right)^\gamma \times \boldsymbol{R}^d = \left((G_2^C)^\tau \right)^\gamma \times \boldsymbol{R}^d = (G_2)^\gamma \times \boldsymbol{R}^d \\ = (\text{Sp}(1) \times \text{Sp}(1)) / \boldsymbol{Z}_2 \times \boldsymbol{R}^d \text{ (Theorem 1.10.1)}, \quad d = 8. \end{aligned}$	$\begin{aligned} G_{2(2)} \simeq (G_{2(2)} \cap O(8)) \times \boldsymbol{R}^d \\ = ((G_2^C)^{\tau \gamma})^\gamma \times \boldsymbol{R}^d = ((G_2^C)^\tau)^\gamma \times \boldsymbol{R}^d = (G_2)^\gamma \times \boldsymbol{R}^d \\ = (\text{Sp}(1) \times \text{Sp}(1)) / \boldsymbol{Z}_2 \times \boldsymbol{R}^d \text{ (Theorem 1.10.1)}, \quad d = 8. \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E08529	88	■ 0.985	$\begin{aligned} & \left\{ \begin{array}{l} \eta_1 = \xi_1 \\ \eta_2 = \frac{\xi_2 - \xi_3}{2} + \frac{\xi_2 + \xi_3}{2} \cos a + i \frac{(a, x_1)}{ a } \sin a \\ \eta_3 = -\frac{\xi_2 - \xi_3}{2} + \frac{\xi_2 + \xi_3}{2} \cos a + i \frac{(a, x_1)}{ a } \sin a , \end{array} \right. \\ & \left\{ \begin{array}{l} y_1 = x_1 + i \frac{(\xi_2 + \xi_3)a}{2 a } \sin a - \frac{2(a, x_1)a}{ a ^2} \sin^2 \frac{ a }{2} \\ y_2 = x_2 \cos \frac{ a }{2} + i \frac{\overline{x_3 a}}{ a } \sin \frac{ a }{2} \\ y_3 = x_3 \cos \frac{ a }{2} + i \frac{\overline{a x_2}}{ a } \sin \frac{ a }{2} \end{array} \right. \end{aligned}$	$\left\{ \begin{array}{l} \eta_1 = \xi_1 \\ \eta_2 = \frac{\xi_2 - \xi_3}{2} + \frac{\xi_2 + \xi_3}{2} \cos a + i \frac{(a, x_1)}{ a } \sin a \\ \eta_3 = -\frac{\xi_2 - \xi_3}{2} + \frac{\xi_2 + \xi_3}{2} \cos a + i \frac{(a, x_1)}{ a } \sin a \end{array} \right.$ $\left\{ \begin{array}{l} y_1 = x_1 + i \frac{(\xi_2 + \xi_3)a}{2 a } \sin a - \frac{2(a, x_1)a}{ a ^2} \sin^2 \frac{ a }{2} \\ y_2 = x_2 \cos \frac{ a }{2} + i \frac{\overline{x_3 a}}{ a } \sin \frac{ a }{2} \\ y_3 = x_3 \cos \frac{ a }{2} + i \frac{\overline{a x_2}}{ a } \sin \frac{ a }{2} \end{array} \right.$	$\left\{ \begin{array}{l} \eta_1 = \xi_1 \\ \eta_2 = \frac{\xi_2 - \xi_3}{2} + \frac{\xi_2 + \xi_3}{2} \cos a + i \frac{(a, x_1)}{ a } \sin a \\ \eta_3 = -\frac{\xi_2 - \xi_3}{2} + \frac{\xi_2 + \xi_3}{2} \cos a + i \frac{(a, x_1)}{ a } \sin a , \end{array} \right.$ $\left\{ \begin{array}{l} y_1 = x_1 + i \frac{(\xi_2 + \xi_3)a}{2 a } \sin a - \frac{2(a, x_1)a}{ a ^2} \sin^2 \frac{ a }{2} \\ y_2 = x_2 \cos \frac{ a }{2} + i \frac{\overline{x_3 a}}{ a } \sin \frac{ a }{2} \\ y_3 = x_3 \cos \frac{ a }{2} + i \frac{\overline{a x_2}}{ a } \sin \frac{ a }{2} \end{array} \right.$
0902.0431_E08314	50	■ 0.986	$\begin{aligned} & \begin{aligned} & \left([\widetilde{A}, \widetilde{B}]C, D \right) = (A \circ (B \circ C), D) - (B \circ (A \circ C), D) \\ & = (B \circ C, A \circ D) - (A \circ C, B \circ D) \\ & = (C \circ (D \circ A), B) - (D \circ (C \circ A), B) = ([\widetilde{C}, \widetilde{D}]A, B). \end{aligned} \end{aligned}$	$([\widetilde{A}, \widetilde{B}]C, D) = (A \circ (B \circ C), D) - (B \circ (A \circ C), D)$ $= (B \circ C, A \circ D) - (A \circ C, B \circ D)$ $= (C \circ (D \circ A), B) - (D \circ (C \circ A), B) = ([\widetilde{C}, \widetilde{D}]A, B).$	$([\widetilde{A}, \widetilde{B}]C, D) = (A \circ (B \circ C), D) - (B \circ (A \circ C), D)$ $= (B \circ C, A \circ D) - (A \circ C, B \circ D)$ $= (C \circ (D \circ A), B) - (D \circ (C \circ A), B) = ([\widetilde{C}, \widetilde{D}]A, B).$
0902.0431_E08449	73	■ 0.986	$\dim_C(\mathfrak{e}_6^C) = 52 + 26 = 78.$	$\dim_C(\mathfrak{e}_6^C) = 52 + 26 = 78.$	$\dim_C(\mathfrak{e}_6^C) = 52 + 26 = 78.$
0902.0431_E01090	185	■ 0.986	$\zeta(l(D) + l(S)I + l(icE)) = \Theta(\varphi_* D, 2\lambda\gamma\chi^{-1}S, 2\chi^{-1}S, 0, 2c, -2c),$	$\zeta(l(D) + l(S)I + l(icE)) = \Theta(\varphi_* D, 2\lambda\gamma\chi^{-1}S, 2\chi^{-1}S, 0, 2c, -2c),$	$\zeta(l(D) + l(S)I + l(icE)) = \Theta(\varphi_* D, 2\lambda\gamma\chi^{-1}S, 2\chi^{-1}S, 0, 2c, -2c),$
0902.0431_E08718	118	■ 0.987	$\left\{ \begin{array}{l} \phi = [\phi_1, \phi_2] - 2A_1 \vee \tau A_2 + 2A_2 \vee \tau A_1 \\ A = \left(\phi_1 + \frac{2}{3}\nu_1 \right) A_2 - \left(\phi_2 + \frac{2}{3}\nu_2 \right) A_1 \\ \nu = \langle A_1, A_2 \rangle - \langle A_2, A_1 \rangle. \end{array} \right.$	$\left\{ \begin{array}{l} \phi = [\phi_1, \phi_2] - 2A_1 \vee \tau A_2 + 2A_2 \vee \tau A_1 \\ A = \left(\phi_1 + \frac{2}{3}\nu_1 \right) A_2 - \left(\phi_2 + \frac{2}{3}\nu_2 \right) A_1 \\ \nu = \langle A_1, A_2 \rangle - \langle A_2, A_1 \rangle. \end{array} \right.$	$\left\{ \begin{array}{l} \phi = [\phi_1, \phi_2] - 2A_1 \vee \tau A_2 + 2A_2 \vee \tau A_1 \\ A = \left(\phi_1 + \frac{2}{3}\nu_1 \right) A_2 - \left(\phi_2 + \frac{2}{3}\nu_2 \right) A_1 \\ \nu = \langle A_1, A_2 \rangle - \langle A_2, A_1 \rangle. \end{array} \right.$
0902.0431_E08055	13	■ 0.988	$\mathfrak{g}_2 = \{D \in \mathfrak{D}_4 \mid \nu D = D, \pi D = D\}$ $= \{D \in \mathfrak{D}_4 \mid \lambda D = D, \lambda \in \mathfrak{S}_3\}$ $= \{D \in \mathfrak{b}_3 \mid \pi D = D\}.$	$\mathfrak{g}_2 = \{D \in \mathfrak{D}_4 \mid \nu D = D, \pi D = D\}$ $= \{D \in \mathfrak{D}_4 \mid \lambda D = D, \lambda \in \mathfrak{S}_3\}$ $= \{D \in \mathfrak{b}_3 \mid \pi D = D\}.$	$\mathfrak{g}_2 = \{D \in \mathfrak{D}_4 \mid \nu D = D, \pi D = D\}$ $= \{D \in \mathfrak{D}_4 \mid \lambda D = D, \lambda \in \mathfrak{S}_3\}$ $= \{D \in \mathfrak{b}_3 \mid \pi D = D\}.$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E01152	194	■0.988	$\text{ (ii) } \begin{aligned} &= \sum_{j,k} x_k \tilde{u}_{jk} \mathbf{u}_j - (\det U) \mathbf{x} = \sum_{i,j,k} x_k \tilde{u}_{jk} u_{ji} \mathbf{e}_i - (\det U) \mathbf{x} \\ &= \sum_{j,k} x_k (\det U) \delta_{ki} \mathbf{e}_i - (\det U) \mathbf{x} = 0. \end{aligned}$	$\begin{aligned} \text{(ii)} &= \sum_{j,k} x_k \tilde{u}_{jk} \mathbf{u}_j - (\det U) \mathbf{x} = \sum_{i,j,k} x_k \tilde{u}_{jk} u_{ji} \mathbf{e}_i - (\det U) \mathbf{x} \\ &= \sum_{j,k} x_k (\det U) \delta_{ki} \mathbf{e}_i - (\det U) \mathbf{x} = 0. \end{aligned}$	$\begin{aligned} \text{(ii)} &= \sum_{j,k} x_k \tilde{u}_{jk} \mathbf{u}_j - (\det U) \mathbf{x} = \sum_{i,j,k} x_k \tilde{u}_{jk} u_{ji} \mathbf{e}_i - (\det U) \mathbf{x} \\ &= \sum_{j,k} x_k (\det U) \delta_{ki} \mathbf{e}_i - (\det U) \mathbf{x} = 0. \end{aligned}$
0902.0431_E01022	170	■0.988	$[(\Phi(h), 0, 0, r, 0, 0), (0, 0, Q, 0, 0, 0)] = (0, 0, (\Phi(h) - r)Q, 0, 0, 0),$	$[(\Phi(h), 0, 0, r, 0, 0), (0, 0, Q, 0, 0, 0)] = (0, 0, (\Phi(h) - r)Q, 0, 0, 0),$	$[(\Phi(h), 0, 0, r, 0, 0), (0, 0, Q, 0, 0, 0)] = (0, 0, (\Phi(h) - r)Q, 0, 0, 0),$
0902.0431_E01066	181	■0.988	$\tilde{\lambda} \gamma(\Phi, P, Q, , s, t) = (\lambda \gamma \Phi \gamma \lambda^{-1}, \lambda \gamma Q, -\lambda \gamma P, -r, -t, -s).$	$\tilde{\lambda} \gamma(\Phi, P, Q, , s, t) = (\lambda \gamma \Phi \gamma \lambda^{-1}, \lambda \gamma Q, -\lambda \gamma P, -r, -t, -s).$	$\tilde{\lambda} \gamma(\Phi, P, Q, , s, t) = (\lambda \gamma \Phi \gamma \lambda^{-1}, \lambda \gamma Q, -\lambda \gamma P, -r, -t, -s).$
0902.0431_E00033	10	■0.988	$\pi L_{e_i} = \pi(2F_{i0}) = -2\pi F_{0i} = -2G_{0i} \text{ (Lemma 1.3.1)} = 2G_{i0} = T_{e_i},$	$\pi L_{e_i} = \pi(2F_{i0}) = -2\pi F_{0i} = -2G_{0i} \text{ (Lemma 1.3.1)} = 2G_{i0} = T_{e_i},$	$\pi L_{e_i} = \pi(2F_{i0}) = -2\pi F_{0i} = -2G_{0i} \text{ (Lemma 1.3.1)} = 2G_{i0} = T_{e_i},$
0902.0431_E00762	128	■0.989	$\mathfrak{h}_{\mathbf{R}} = \left\{ \Phi \left(\sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim}, 0, 0, \nu \right) \in \mathfrak{e}_7^C \mid \begin{array}{l} \lambda_k, \nu \in \mathbf{R} \\ \mu_j \in \mathbf{R}, \mu_1 + \mu_2 + \mu_3 = 0 \end{array} \right\}$	$\mathfrak{h}_{\mathbf{R}} = \left\{ \Phi \left(\sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim}, 0, 0, \nu \right) \in \mathfrak{e}_7^C \mid \begin{array}{l} \lambda_k, \nu \in \mathbf{R} \\ \mu_j \in \mathbf{R}, \mu_1 + \mu_2 + \mu_3 = 0 \end{array} \right\}$	$\mathfrak{h}_{\mathbf{R}} = \left\{ \Phi \left(\sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim}, 0, 0, \nu \right) \in \mathfrak{e}_7^C \mid \begin{array}{l} \lambda_k, \nu \in \mathbf{R} \\ \mu_j \in \mathbf{R}, \mu_1 + \mu_2 + \mu_3 = 0 \end{array} \right\}$
0902.0431_E00269	44	■0.989	$\begin{aligned} ([A, X], X X) &= (A X, X X) - (X A, X X) \\ &= (A, X(X X)) - (A, (X X) X) \quad (\text{Lemma 2.3.5}) \\ &= (A, [X, X X]) = (A, a E) = \frac{1}{2} \text{tr}(A a + \bar{a} A^*) \\ &= \frac{1}{2} \text{tr}(A a - a A) = \frac{1}{2} (\text{tr}(A) a - a \text{tr}(A)) = 0. \end{aligned}$	$\begin{aligned} ([A, X], X X) &= (A X, X X) - (X A, X X) \\ &= (A, X(X X)) - (A, (X X) X) \quad (\text{Lemma 2.3.5}) \\ &= (A, [X, X X]) = (A, a E) = \frac{1}{2} \text{tr}(A a + \bar{a} A^*) \\ &= \frac{1}{2} \text{tr}(A a - a A) = \frac{1}{2} (\text{tr}(A) a - a \text{tr}(A)) = 0. \end{aligned}$	$\begin{aligned} ([A, X], X X) &= (A X, X X) - (X A, X X) \\ &= (A, X(X X)) - (A, (X X) X) \quad (\text{Lemma 2.3.5}) \\ &= (A, [X, X X]) = (A, a E) = \frac{1}{2} \text{tr}(A a + \bar{a} A^*) \\ &= \frac{1}{2} \text{tr}(A a - a A) = \frac{1}{2} (\text{tr}(A) a - a \text{tr}(A)) = 0. \end{aligned}$
0902.0431_E00893	148	■0.989	$\left\{ \begin{array}{l} \phi = [\phi_1, \phi_2] - 2A_1 \vee \gamma A_2 + 2A_2 \vee \gamma A_1, \\ A = \phi_1 A_2 - \phi_2 A_1. \end{array} \right.$	$\left\{ \begin{array}{l} \phi = [\phi_1, \phi_2] - 2A_1 \vee \gamma A_2 + 2A_2 \vee \gamma A_1, \\ A = \phi_1 A_2 - \phi_2 A_1. \end{array} \right.$	$\left\{ \begin{array}{l} \phi = [\phi_1, \phi_2] - 2A_1 \vee \gamma A_2 + 2A_2 \vee \gamma A_1, \\ A = \phi_1 A_2 - \phi_2 A_1. \end{array} \right.$
0902.0431_E00267	43	■0.989	$\left\{ \begin{array}{l} ([A, X], Y) + (X, [A, Y]) = 0, \quad X, Y \in \mathfrak{J} \\ \text{tr}([A, X], Y, Z) + \text{tr}(X, [A, Y], Z) + \text{tr}(X, Y, [A, Z]) = 0, \quad X, Y, Z \in \mathfrak{J} \end{array} \right.$	$\left\{ \begin{array}{l} ([A, X], Y) + (X, [A, Y]) = 0, \quad X, Y \in \mathfrak{J} \\ \text{tr}([A, X], Y, Z) + \text{tr}(X, [A, Y], Z) + \text{tr}(X, Y, [A, Z]) = 0, \quad X, Y, Z \in \mathfrak{J} \end{array} \right.$	$\left\{ \begin{array}{l} ([A, X], Y) + (X, [A, Y]) = 0, \quad X, Y \in \mathfrak{J}, \\ \text{tr}([A, X], Y, Z) + \text{tr}(X, [A, Y], Z) + \text{tr}(X, Y, [A, Z]) = 0, \quad X, Y, Z \in \mathfrak{J}. \end{array} \right.$
0902.0431_E00195	33	■0.990	$\alpha(xy) = (\tilde{\alpha}x) \overline{(\tilde{\alpha}y)}, \quad x, y \in \mathfrak{C}.$	$\alpha(xy) = (\tilde{\alpha}x) \overline{(\tilde{\alpha}y)}, \quad x, y \in \mathfrak{C}.$	$\alpha(xy) = (\tilde{\alpha}x) \overline{(\tilde{\alpha}y)}, \quad x, y \in \mathfrak{C}.$

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0902.0431_E00409	67	■ 0.990	$\begin{aligned} & G_{ij}, \quad 0 \leq i < j \leq 3, 4 \leq i < j \leq 7, \\ & \widetilde{A}_k(e_j), \quad 0 \leq j \leq 3, k = 1, 2, 3 \end{aligned}$	$G_{ij}, \quad 0 \leq i < j \leq 3, 4 \leq i < j \leq 7, \\ \widetilde{A}_k(e_j), \quad 0 \leq j \leq 3, k = 1, 2, 3$	$G_{ij}, \quad 0 \leq i < j \leq 3, 4 \leq i < j \leq 7, \\ \widetilde{A}_k(e_j), \quad 0 \leq j \leq 3, k = 1, 2, 3$
0902.0431_E00508	98	■ 0.990	$\mathfrak{e}_{\{6\}} = \mathfrak{f}_{\{4\}} \oplus i \left(\widetilde{\mathfrak{J}}_{\{H\}} \right)_{\{0\}}.$	$\mathfrak{e}_{6,H} = \mathfrak{f}_{4,H} \oplus i \left(\widetilde{\mathfrak{J}}_H \right)_0.$	$\mathfrak{e}_{6,H} = \mathfrak{f}_{4,H} \oplus i \left(\widetilde{\mathfrak{J}}_H \right)_0.$
0902.0431_E00748	124	■ 0.990	$\mathfrak{e}_{\{7\}}^{\{C\}} = \mathfrak{e}_{\{6\}}^{\{C\}} \oplus \mathfrak{J}^C \oplus \mathfrak{J}^C \oplus C.$	$\mathfrak{e}_7^C = \mathfrak{e}_6^C \oplus \mathfrak{J}^C \oplus \mathfrak{J}^C \oplus C.$	$\mathfrak{e}_7^C = \mathfrak{e}_6^C \oplus \mathfrak{J}^C \oplus \mathfrak{J}^C \oplus C.$
0902.0431_E00750	124	■ 0.990	$\begin{aligned} & [h, S] = \left[\Phi \left(h_\delta + \widetilde{H}, 0, 0, \nu \right), \Phi(S, 0, 0, 0) \right] \\ & = \Phi \left([h_\delta + \widetilde{H}, S], 0, 0, 0 \right) = \Phi \left(\alpha \left(h_\delta + \widetilde{H} \right) S, 0, 0, 0 \right) = \alpha(h)S. \end{aligned}$	$[h, S] = \left[\Phi \left(h_\delta + \widetilde{H}, 0, 0, \nu \right), \Phi(S, 0, 0, 0) \right] \\ = \Phi \left([h_\delta + \widetilde{H}, S], 0, 0, 0 \right) = \Phi \left(\alpha \left(h_\delta + \widetilde{H} \right) S, 0, 0, 0 \right) = \alpha(h)S.$	$[h, S] = [\Phi(h_\delta + \widetilde{H}, 0, 0, \nu), \Phi(S, 0, 0, 0)] \\ = \Phi([h_\delta + \widetilde{H}, S], 0, 0, 0) = \Phi(\alpha(h_\delta + \widetilde{H})S, 0, 0, 0) = \alpha(h)S.$
0902.0431_E00969	161	■ 0.990	$E_{\{7(7)\}} \cong \left(E_{\{7\}}^{\{C\}} \right)^{\tau \gamma}, \quad \text{quad} \quad E_{\{7(-5)\}} \cong \left(E_{\{7\}}^{\{C\}} \right)^{\tau \lambda \sigma}, \quad \text{quad} \quad E_{\{7(-25)\}} \cong \left(E_{\{7\}}^{\{C\}} \right)^{\tau}.$	$E_{7(7)} \cong (E_7^C)^{\tau \gamma}, \quad E_{7(-5)} \cong (E_7^C)^{\tau \lambda \sigma}, \quad E_{7(-25)} \cong (E_7^C)^{\tau}.$	$E_{7(7)} \cong (E_7^C)^{\tau \gamma}, \quad E_{7(-5)} \cong (E_7^C)^{\tau \lambda \sigma}, \quad E_{7(-25)} \cong (E_7^C)^{\tau}.$
0902.0431_E00293	47	■ 0.990	$\widetilde{A}X = A \circ X, \quad X \in \mathfrak{J}^C.$	$\widetilde{A}X = A \circ X, \quad X \in \mathfrak{J}^C.$	$\widetilde{A}X = A \circ X, \quad X \in \mathfrak{J}^C.$

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0902.0431_E00018	9	■ 0.990	$\begin{gathered} \left\{ \begin{array}{l} 2F_{3} = G_{3} + G_{12} + G_{47} + G_{56} \\ 2F_{12} = G_{3} + G_{12} - G_{47} - G_{56} \\ 2F_{47} = G_{3} - G_{12} + G_{47} - G_{56} \\ 2F_{56} = G_{3} - G_{12} - G_{47} + G_{56}, \end{array} \right. \left\{ \begin{array}{l} 2F_{04} = G_{04} - G_{15} + G_{26} - G_{37} \\ 2F_{15} = -G_{04} + G_{15} + G_{26} - G_{37} \\ 2F_{26} = G_{04} + G_{15} + G_{26} + G_{37} \\ 2F_{37} = -G_{04} - G_{15} + G_{26} + G_{37}, \end{array} \right. \\ \left\{ \begin{array}{l} 2F_{05} = G_{05} + G_{14} - G_{27} - G_{36} \\ 2F_{14} = G_{05} + G_{14} + G_{27} + G_{36} \\ 2F_{27} = -G_{05} + G_{14} + G_{27} - G_{36} \\ 2F_{36} = -G_{05} + G_{14} - G_{27} + G_{36}, \end{array} \right. \left\{ \begin{array}{l} 2F_{06} = G_{06} - G_{17} - G_{24} + G_{35} \\ 2F_{17} = -G_{06} + G_{17} - G_{24} + G_{35} \\ 2F_{24} = -G_{06} - G_{17} + G_{24} + G_{35} \\ 2F_{35} = G_{06} + G_{17} + G_{24} + G_{35}, \end{array} \right. \\ \left\{ \begin{array}{l} 2F_{07} = G_{07} + G_{16} + G_{25} + G_{34} \\ 2F_{16} = G_{07} + G_{16} - G_{25} - G_{34} \\ 2F_{25} = G_{07} - G_{16} + G_{25} - G_{34} \\ 2F_{34} = G_{07} - G_{16} - G_{25} + G_{34}. \end{array} \right.$	$\begin{cases} 2F_{03} = G_{03} + G_{12} + G_{47} + G_{56} \\ 2F_{12} = G_{03} + G_{12} - G_{47} - G_{56} \\ 2F_{47} = G_{03} - G_{12} + G_{47} - G_{56} \\ 2F_{56} = G_{03} - G_{12} - G_{47} + G_{56}, \end{cases} \begin{cases} 2F_{04} = G_{04} - G_{15} + G_{26} - G_{37} \\ 2F_{15} = -G_{04} + G_{15} + G_{26} - G_{37} \\ 2F_{26} = G_{04} + G_{15} + G_{26} + G_{37} \\ 2F_{37} = -G_{04} - G_{15} + G_{26} + G_{37}, \end{cases} \\ \begin{cases} 2F_{05} = G_{05} + G_{14} - G_{27} - G_{36} \\ 2F_{14} = G_{05} + G_{14} + G_{27} + G_{36} \\ 2F_{27} = -G_{05} + G_{14} + G_{27} - G_{36} \\ 2F_{36} = -G_{05} + G_{14} - G_{27} + G_{36}, \end{cases} \begin{cases} 2F_{06} = G_{06} - G_{17} - G_{24} + G_{35} \\ 2F_{17} = -G_{06} + G_{17} - G_{24} + G_{35} \\ 2F_{24} = -G_{06} - G_{17} + G_{24} + G_{35} \\ 2F_{35} = G_{06} + G_{17} + G_{24} + G_{35}, \end{cases} \\ \begin{cases} 2F_{07} = G_{07} + G_{16} + G_{25} + G_{34} \\ 2F_{16} = G_{07} + G_{16} - G_{25} - G_{34} \\ 2F_{25} = G_{07} - G_{16} + G_{25} - G_{34} \\ 2F_{34} = G_{07} - G_{16} - G_{25} + G_{34}. \end{cases}$	$\begin{cases} 2F_{03} = G_{03} + G_{12} + G_{47} + G_{56} \\ 2F_{12} = G_{03} + G_{12} - G_{47} - G_{56} \\ 2F_{47} = G_{03} - G_{12} + G_{47} - G_{56} \\ 2F_{56} = G_{03} - G_{12} - G_{47} + G_{56}, \end{cases} \begin{cases} 2F_{04} = G_{04} - G_{15} + G_{26} - G_{37} \\ 2F_{15} = -G_{04} + G_{15} + G_{26} - G_{37} \\ 2F_{26} = G_{04} + G_{15} + G_{26} + G_{37} \\ 2F_{37} = -G_{04} - G_{15} + G_{26} + G_{37}, \end{cases} \\ \begin{cases} 2F_{05} = G_{05} + G_{14} - G_{27} - G_{36} \\ 2F_{14} = G_{05} + G_{14} + G_{27} + G_{36} \\ 2F_{27} = -G_{05} + G_{14} + G_{27} - G_{36} \\ 2F_{36} = -G_{05} + G_{14} - G_{27} + G_{36}, \end{cases} \begin{cases} 2F_{06} = G_{06} - G_{17} - G_{24} + G_{35} \\ 2F_{17} = -G_{06} + G_{17} - G_{24} + G_{35} \\ 2F_{24} = -G_{06} - G_{17} + G_{24} + G_{35} \\ 2F_{35} = G_{06} + G_{17} + G_{24} + G_{35}, \end{cases} \\ \begin{cases} 2F_{07} = G_{07} + G_{16} + G_{25} + G_{34} \\ 2F_{16} = G_{07} + G_{16} - G_{25} - G_{34} \\ 2F_{25} = G_{07} - G_{16} + G_{25} - G_{34} \\ 2F_{34} = G_{07} - G_{16} - G_{25} + G_{34}. \end{cases}$
0902.0431_E00722	119	■ 0.991	$\mathfrak{e}_{7\{ \}^{\{C\}}} = \mathfrak{e}_{6\{ \}^{\{C\}}} \oplus \mathfrak{N}^{\{C\}},$	$\mathfrak{e}_7^C = \mathfrak{e}_6^C \oplus \mathfrak{N}^C,$	$\mathfrak{e}_7^C = \mathfrak{e}_6^C \oplus \mathfrak{N}^C,$
0902.0431_E01001	166	■ 0.991	$\widetilde{\lambda} = \lambda \lambda' = \lambda' \lambda.$	$\widetilde{\lambda} = \lambda \lambda' = \lambda' \lambda.$	$\widetilde{\lambda} = \lambda \lambda' = \lambda' \lambda.$
0902.0431_E00162	29	■ 0.991	$(x\,p)(\overline{p}\,y) = x\,y, \quad x, y \in \mathfrak{C}.$	$(xp)(\overline{p}y) = xy, \quad x, y \in \mathfrak{C}.$	$(xp)(\overline{p}y) = xy, \quad x, y \in \mathfrak{C}.$

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0902.0431_E00296	47	■ 0.991	$\begin{array}{ll} \{\left[\widetilde{E}_{_i}, \right. \\ \left. \widetilde{E}_{_j}\right]=0, \} \& \{\left[\widetilde{E}_{_i}, \right. \\ \left. \widetilde{F}_{_i}(a)\right]=0, \} \quad \& \{\left[\right. \\ \left. \widetilde{E}_{_i}, \widetilde{F}_{_{i+1}}(a)\right]= \\ \frac{1}{2} \widetilde{A}_{_{i+1}}(a), \} \& \{\left[\right. \\ \left. \widetilde{E}_{_i}, \widetilde{F}_{_{i+2}}(a)\right]= \\ \frac{1}{2} \widetilde{A}_{_{i+2}}(a), \} \quad \& \{\left[\right. \\ \left. \widetilde{F}_{_i}(a), \widetilde{F}_{_i}(b)\right] \in \mathfrak{d}_4^C, \} \& \{\left[\right. \\ \left. \widetilde{F}_{_i}(a), \widetilde{F}_{_{i+1}}(b)\right] = \\ -\frac{1}{2} \widetilde{A}_{_{i+2}}(\overline{ab}), \} \quad \& \{\left[\right. \\ \left. D, \widetilde{A}_i(a)\right] = \widetilde{A}_i(D_i a), \} \quad \& \{\left[\right. \\ \left. \widetilde{A}_i(a), \widetilde{A}_i(b)\right] \in \mathfrak{d}_4^C, \} \quad \& \{\left[\right. \\ \left. \widetilde{F}_1(e_i), \widetilde{F}_1(e_j)\right] = G_{ij}. \} \end{array}$	$\begin{array}{ll} \left[\widetilde{E}_i, \widetilde{E}_j\right] = 0, & \left[\widetilde{E}_i, \widetilde{F}_i(a)\right] = 0, \\ \left[\widetilde{E}_i, \widetilde{F}_{i+1}(a)\right] = -\frac{1}{2} \widetilde{A}_{i+1}(a), & \left[\widetilde{E}_i, \widetilde{F}_{i+2}(a)\right] = \frac{1}{2} \widetilde{A}_{i+2}(a), \\ \left[\widetilde{F}_i(a), \widetilde{F}_i(b)\right] \in \mathfrak{d}_4^C, & \left[\widetilde{F}_i(a), \widetilde{F}_{i+1}(b)\right] = -\frac{1}{2} \widetilde{A}_{i+2}(\overline{ab}), \\ \left[D, \widetilde{A}_i(a)\right] = \widetilde{A}_i(D_i a), & \text{where } D = (D_1, D_2, D_3) \in \mathfrak{d}_4^C, \\ \left[\widetilde{A}_i(a), \widetilde{A}_i(b)\right] \in \mathfrak{d}_4^C, & \left[\widetilde{A}_i(a), \widetilde{A}_{i+1}(b)\right] = -\frac{1}{2} \widetilde{A}_{i+2}(\overline{ab}), \\ \left[\widetilde{F}_1(e_i), \widetilde{F}_1(e_j)\right] = G_{ij}. & \left[\widetilde{A}_i(a), \widetilde{A}_{i+2}(b)\right] = \frac{1}{2} \widetilde{A}_{i+1}(\overline{ba}). \end{array}$	$\begin{array}{ll} [\widetilde{E}_i, \widetilde{E}_j] = 0, & [\widetilde{E}_i, \widetilde{F}_i(a)] = 0, \\ [\widetilde{E}_i, \widetilde{F}_{i+1}(a)] = -\frac{1}{2} \widetilde{A}_{i+1}(a), & [\widetilde{E}_i, \widetilde{F}_{i+2}(a)] = \frac{1}{2} \widetilde{A}_{i+2}(a), \\ [\widetilde{F}_i(a), \widetilde{F}_i(b)] \in \mathfrak{d}_4^C, & [\widetilde{F}_i(a), \widetilde{F}_{i+1}(b)] = -\frac{1}{2} \widetilde{A}_{i+2}(\overline{ab}), \\ [D, \widetilde{A}_i(a)] = \widetilde{A}_i(D_i a), & \text{where } D = (D_1, D_2, D_3) \in \mathfrak{d}_4^C, \\ [\widetilde{A}_i(a), \widetilde{A}_i(b)] \in \mathfrak{d}_4^C, & [\widetilde{A}_i(a), \widetilde{A}_{i+1}(b)] = -\frac{1}{2} \widetilde{A}_{i+2}(\overline{ab}), \\ [\widetilde{F}_1(e_i), \widetilde{F}_1(e_j)] = G_{ij}. & [\widetilde{A}_i(a), \widetilde{A}_{i+2}(b)] = \frac{1}{2} \widetilde{A}_{i+1}(\overline{ba}). \end{array}$
0902.0431_E01151	194	■ 0.991	$\begin{array}{l} \& \left(\boldsymbol{u}_{_1} \wedge \cdots \wedge \boldsymbol{u}_{_j-1} \wedge \boldsymbol{x} \wedge \boldsymbol{u}_{_j+1} \wedge \cdots \wedge \boldsymbol{u}_9, e_1 \wedge \cdots \wedge e_9\right) = \sum_{k=1}^9 \widetilde{u}_{jk} x_k, \\ \& \left(\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_9, e_1 \wedge \cdots \wedge e_9\right) = \det U, \end{array}$	$\begin{array}{l} \& \left(\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_{_j-1} \wedge \boldsymbol{x} \wedge \boldsymbol{u}_{_j+1} \wedge \cdots \wedge \boldsymbol{u}_9, e_1 \wedge \cdots \wedge e_9\right) = \sum_{k=1}^9 \widetilde{u}_{jk} x_k, \\ \& \left(\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_9, e_1 \wedge \cdots \wedge e_9\right) = \det U, \end{array}$	$\begin{array}{l} (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_{j-1} \wedge \boldsymbol{x} \wedge \boldsymbol{u}_{j+1} \wedge \cdots \wedge \boldsymbol{u}_9, e_1 \wedge \cdots \wedge e_9) = \sum_{k=1}^9 \widetilde{u}_{jk} x_k, \\ (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_9, e_1 \wedge \cdots \wedge e_9) = \det U, \end{array}$

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0902.0431_E08484	80	■ 0.991	$\begin{aligned} & \& \left[\phi, \left[\phi, \widetilde{A}_2(e_i)\right]\right] = \left[\phi, -\frac{1}{2}\widetilde{F}_2(e_i)\right] = \frac{1}{4}\widetilde{A}_2(e_i), \\ & \& \left[\phi, \left[\phi, \widetilde{A}_3(e_i)\right]\right] = \left[\phi, \widetilde{F}_3(e_i)\right] = \widetilde{A}_3(e_i), \\ & \& \left[\phi, \left[\phi, \widetilde{F}_1(e_i)\right]\right] = \left[\phi, -\frac{1}{2}\widetilde{A}_1(e_i)\right] = \frac{1}{4}\widetilde{F}_1(e_i), \\ & \& \left[\phi, \left[\phi, \widetilde{F}_2(e_i)\right]\right] = \left[\phi, -\frac{1}{2}\widetilde{A}_2(e_i)\right] = \frac{1}{4}\widetilde{F}_2(e_i), \\ & \& \left[\phi, \left[\phi, \widetilde{F}_3(e_i)\right]\right] = \left[\phi, \widetilde{A}_3(e_i)\right] = \widetilde{F}_3(e_i), \end{aligned}$	$\begin{aligned} \left[\phi, \left[\phi, \widetilde{A}_2(e_i)\right]\right] &= \left[\phi, -\frac{1}{2}\widetilde{F}_2(e_i)\right] = \frac{1}{4}\widetilde{A}_2(e_i), \\ \left[\phi, \left[\phi, \widetilde{A}_3(e_i)\right]\right] &= \left[\phi, \widetilde{F}_3(e_i)\right] = \widetilde{A}_3(e_i), \\ \left[\phi, \left[\phi, \widetilde{F}_1(e_i)\right]\right] &= \left[\phi, -\frac{1}{2}\widetilde{A}_1(e_i)\right] = \frac{1}{4}\widetilde{F}_1(e_i), \\ \left[\phi, \left[\phi, \widetilde{F}_2(e_i)\right]\right] &= \left[\phi, -\frac{1}{2}\widetilde{A}_2(e_i)\right] = \frac{1}{4}\widetilde{F}_2(e_i), \\ \left[\phi, \left[\phi, \widetilde{F}_3(e_i)\right]\right] &= \left[\phi, \widetilde{A}_3(e_i)\right] = \widetilde{F}_3(e_i), \end{aligned}$	$\begin{aligned} [\phi, [\phi, \widetilde{A}_2(e_i)]] &= \left[\phi, -\frac{1}{2}\widetilde{F}_2(e_i)\right] = \frac{1}{4}\widetilde{A}_2(e_i), \\ [\phi, [\phi, \widetilde{A}_3(e_i)]] &= [\phi, \widetilde{F}_3(e_i)] = \widetilde{A}_3(e_i), \\ [\phi, [\phi, \widetilde{F}_1(e_i)]] &= \left[\phi, -\frac{1}{2}\widetilde{A}_1(e_i)\right] = \frac{1}{4}\widetilde{F}_1(e_i), \\ [\phi, [\phi, \widetilde{F}_2(e_i)]] &= \left[\phi, -\frac{1}{2}\widetilde{A}_2(e_i)\right] = \frac{1}{4}\widetilde{F}_2(e_i), \\ [\phi, [\phi, \widetilde{F}_3(e_i)]] &= [\phi, \widetilde{A}_3(e_i)] = \widetilde{F}_3(e_i), \end{aligned}$
0902.0431_E08497	81	■ 0.991	$\left(\sigma_{ik}x_{ik}\right)y_{kj}+x_{ik}\left(\sigma_{kj}y_{kj}\right)=\sigma_{ij}\left(x_{ik}y_{kj}\right).$	$\left(\sigma_{ik}x_{ik}\right)y_{kj}+x_{ik}\left(\sigma_{kj}y_{kj}\right)=\sigma_{ij}\left(x_{ik}y_{kj}\right).$	$\left(\sigma_{ik}x_{ik}\right)y_{kj}+x_{ik}\left(\sigma_{kj}y_{kj}\right)=\sigma_{ij}\left(x_{ik}y_{kj}\right).$
0902.0431_E08544	92	■ 0.991	$\begin{aligned} & \alpha E_2 = \alpha(-F_2(1) \times F_2(1)) = -\tau\alpha\tau F_2(1) \times \tau\alpha\tau F_2(1) \\ & = -(F_2(x_2) + F_3(x_3)) \times (F_2(x_2) + F_3(x_3)) \quad (\text{for some } x_2, x_3 \in \mathfrak{C}^C) \\ & = (x_2, x_2)E_2 + (x_3, x_3)E_3 - F_1(\overline{x_2x_3}) \in \mathfrak{J}(2, \mathfrak{C}^C). \end{aligned}$	$\begin{aligned} \alpha E_2 &= \alpha(-F_2(1) \times F_2(1)) = -\tau\alpha\tau F_2(1) \times \tau\alpha\tau F_2(1) \\ &= -(F_2(x_2) + F_3(x_3)) \times (F_2(x_2) + F_3(x_3)) \quad (\text{for some } x_2, x_3 \in \mathfrak{C}^C) \\ &= (x_2, x_2)E_2 + (x_3, x_3)E_3 - F_1(\overline{x_2x_3}) \in \mathfrak{J}(2, \mathfrak{C}^C) \end{aligned}$	$\begin{aligned} \alpha E_2 &= \alpha(-F_2(1) \times F_2(1)) = -\tau\alpha\tau F_2(1) \times \tau\alpha\tau F_2(1) \\ &= -(F_2(x_2) + F_3(x_3)) \times (F_2(x_2) + F_3(x_3)) \quad (\text{for some } x_2, x_3 \in \mathfrak{C}^C) \\ &= (x_2, x_2)E_2 + (x_3, x_3)E_3 - F_1(\overline{x_2x_3}) \in \mathfrak{J}(2, \mathfrak{C}^C). \end{aligned}$
0902.0431_E08787	132	■ 0.991	$\alpha P = (X, Y, \xi, \eta), \quad X, Y \text{ are diagonal, } \xi > 0.$	$\alpha P = (X, Y, \xi, \eta), \quad X, Y \text{ are diagonal, } \xi > 0.$	$\alpha P = (X, Y, \xi, \eta), \quad X, Y \text{ are diagonal, } \xi > 0.$
0902.0431_E01014	168	■ 0.991	$E_8^C \simeq (E_8^C \cap U(\mathfrak{e}_8^C)) \times \mathbf{R}^d = E_8 \times \mathbf{R}^d, \quad d = 248.$	$E_8^C \simeq (E_8^C \cap U(\mathfrak{e}_8^C)) \times \mathbf{R}^d = E_8 \times \mathbf{R}^d, \quad d = 248.$	$E_8^C \simeq (E_8^C \cap U(\mathfrak{e}_8^C)) \times \mathbf{R}^d = E_8 \times \mathbf{R}^d, \quad d = 248.$
0902.0431_E08234	39	■ 0.992	$\det({}^t\alpha^{-1}X) = \det({}^t\alpha X) = \det X, \quad \text{for all } X \in \mathfrak{J}.$	$\det({}^t\alpha^{-1}X) = \det({}^t\alpha X) = \det X, \quad \text{for all } X \in \mathfrak{J}.$	$\det({}^t\alpha^{-1}X) = \det({}^t\alpha X) = \det X, \quad \text{for all } X \in \mathfrak{J}.$
0902.0431_E08586	98	■ 0.992	$F_{4,\mathbf{H}} = \{\alpha \in \mathbf{Iso}_{\mathbf{R}}(\mathfrak{J}_{\mathbf{H}}) \mid \alpha(M \circ N) = \alpha M \circ \alpha N\}$	$F_{4,\mathbf{H}} = \{\alpha \in \mathbf{Iso}_{\mathbf{R}}(\mathfrak{J}_{\mathbf{H}}) \mid \alpha(M \circ N) = \alpha M \circ \alpha N\}$	$F_{4,\mathbf{H}} = \{\alpha \in \mathbf{Iso}_{\mathbf{R}}(\mathfrak{J}_{\mathbf{H}}) \mid \alpha(M \circ N) = \alpha M \circ \alpha N\}$

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0902.0431_E00928	154	■0.992	$\backslash[\ \mathfrak{e}_{\{7,\ C\}}\ \mathop{\cong}\ \mathfrak{su}(6)\ .\ \backslash]$ $\backslash\mathrm{end}\{\mathrm{itemize}\}$	$\mathfrak{e}_{7,C} \cong \mathfrak{su}(6).$	$\mathfrak{e}_{7,C} \cong \mathfrak{su}(6).$
0902.0431_E00936	155	■0.992	$=\backslash\mathrm{left}\{\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{rrrrrr}\}\ \mathrm{b}_{\{11\}}\ \&\ \mathrm{b}_{\{12\}}\ \&\ \mathrm{b}_{\{13\}}\ \&\ \mathrm{l}_{\{11\}}\ \&\ \mathrm{l}_{\{12\}}\ \&\ \mathrm{l}_{\{13\}}\ \backslash\backslash\ -\backslash\mathrm{bar}\{\mathrm{b}\}_{\{12\}}\ \&\ \mathrm{b}_{\{22\}}\ \&\ \mathrm{b}_{\{23\}}\ \&\ \mathrm{l}_{\{21\}}\ \&\ \mathrm{l}_{\{22\}}\ \&\ \mathrm{l}_{\{23\}}\ \backslash\backslash\ -\backslash\mathrm{bar}\{\mathrm{b}\}_{\{13\}}\ \&\ \mathrm{b}_{\{23\}}\ \&\ \mathrm{b}_{\{33\}}\ \&\ \mathrm{l}_{\{31\}}\ \&\ \mathrm{l}_{\{32\}}\ \&\ \mathrm{l}_{\{33\}}\ \backslash\backslash\ -\backslash\mathrm{bar}\{\mathrm{l}\}_{\{11\}}\ \&\ -\backslash\mathrm{bar}\{\mathrm{l}\}_{\{21\}}\ \&\ -\backslash\mathrm{bar}\{\mathrm{l}\}_{\{31\}}\ \&\ \mathrm{c}_{\{11\}}\ \&\ \mathrm{c}_{\{12\}}\ \&\ \mathrm{c}_{\{13\}}\ \backslash\backslash\ -\backslash\mathrm{bar}\{\mathrm{l}\}_{\{12\}}\ \&\ -\backslash\mathrm{bar}\{\mathrm{l}\}_{\{22\}}\ \&\ -\backslash\mathrm{bar}\{\mathrm{l}\}_{\{32\}}\ \&\ -\backslash\mathrm{bar}\{\mathrm{c}\}_{\{12\}}\ \&\ \mathrm{c}_{\{22\}}\ \&\ \mathrm{c}_{\{23\}}\ \backslash\backslash\ -\backslash\mathrm{bar}\{\mathrm{l}\}_{\{13\}}\ \&\ -\backslash\mathrm{bar}\{\mathrm{l}\}_{\{23\}}\ \&\ -\backslash\mathrm{bar}\{\mathrm{l}\}_{\{33\}}\ \&\ -\backslash\mathrm{bar}\{\mathrm{c}\}_{\{13\}}\ \&\ -\backslash\mathrm{bar}\{\mathrm{c}\}_{\{23\}}\ \&\ \mathrm{c}_{\{33\}}\ \backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right})+\backslash\mathrm{frac}\{\nu\}{3}\backslash\mathrm{left}\{\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{cc}\}\ \mathrm{E}\ \&\ \emptyset\ \backslash\backslash\ \emptyset\ \&\ -\mathrm{E}\ \backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}\}\ \backslash\mathrm{in}\ \mathfrak{su}(6),$	$=\left(\begin{array}{cccccc} b_{11} & b_{12} & b_{13} & l_{11} & l_{12} & l_{13} \\ -\bar{b}_{12} & b_{22} & b_{23} & l_{21} & l_{22} & l_{23} \\ -\bar{b}_{13} & b_{23} & b_{33} & l_{31} & l_{32} & l_{33} \\ -\bar{l}_{11} & -\bar{l}_{21} & -\bar{l}_{31} & c_{11} & c_{12} & c_{13} \\ -\bar{l}_{12} & -\bar{l}_{22} & -\bar{l}_{32} & -\bar{c}_{12} & c_{22} & c_{23} \\ -\bar{l}_{13} & -\bar{l}_{23} & -\bar{l}_{33} & -\bar{c}_{13} & -\bar{c}_{23} & c_{33} \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)\in\mathfrak{su}(6),$	$=\left(\begin{array}{cccccc} b_{11} & b_{12} & b_{13} & l_{11} & l_{12} & l_{13} \\ -\bar{b}_{12} & b_{22} & b_{23} & l_{21} & l_{22} & l_{23} \\ -\bar{b}_{13} & b_{23} & b_{33} & l_{31} & l_{32} & l_{33} \\ -\bar{l}_{11} & -\bar{l}_{21} & -\bar{l}_{31} & c_{11} & c_{12} & c_{13} \\ -\bar{l}_{12} & -\bar{l}_{22} & -\bar{l}_{32} & -\bar{c}_{12} & c_{22} & c_{23} \\ -\bar{l}_{13} & -\bar{l}_{23} & -\bar{l}_{33} & -\bar{c}_{13} & -\bar{c}_{23} & c_{33} \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)\in\mathfrak{su}(6),$
0902.0431_E00994	165	■0.992	$B_{\{8\}}(1,1)=56\ \times 2+4\ \times 2=120\ .$	$B_8(1,1)=56\times2+4\times2=120.$	$B_8(1,1)=56\times2+4\times2=120.$
0902.0431_E00998	166	■0.992	$\mathfrak{W}^C=\backslash\mathrm{left}\{\mathrm{R}\ \backslash\mathrm{in}\ \mathfrak{e}_{\{8\}}\{\ \}^C\ \backslash\mathrm{mid}\ \mathrm{R}\ \times\ \mathrm{R}=0,\ \mathrm{R}\ \mathrm{neq}\ 0\backslash\mathrm{right}\}\ .$	$\mathfrak{W}^C=\left\{R\in\mathfrak{e}_8^C\mid R\times R=0,R\neq0\right\}.$	$\mathfrak{W}^C=\left\{R\in\mathfrak{e}_8^C\mid R\times R=0,R\neq0\right\}.$
0902.0431_E00034	11	■0.992	$\backslash\mathrm{begin}\{\mathrm{aligned}\}\ \&\ \backslash\pi\ \mathrm{T}_{\{\mathrm{e}_{\{i\}}\}}=\backslash\pi\backslash\mathrm{left}(2\ \mathrm{G}_{\{i\}\ 0}\ \backslash\mathrm{right})=2\ \mathrm{F}_{\{i\}\ 0}=\mathrm{L}_{\{\mathrm{e}_{\{i\}}\}},\ \backslash\backslash\ \&\ \backslash\pi\ \mathrm{R}_{\{a\}}=\backslash\pi\ \backslash\mathrm{left}(\mathrm{T}_{\{a\}}-\mathrm{L}_{\{a\}}\backslash\mathrm{right})=\mathrm{L}_{\{a\}}-\mathrm{T}_{\{a\}}=-\mathrm{R}_{\{a\}}\ .\ \backslash\mathrm{end}\{\mathrm{aligned}\}$	$\pi T_{e_i}=\pi(2G_{i0})=2F_{i0}=L_{e_i},$ $\pi R_a=\pi(T_a-L_a)=L_a-T_a=-R_a.$	$\pi T_{e_i}=\pi(2G_{i0})=2F_{i0}=L_{e_i},$ $\pi R_a=\pi(T_a-L_a)=L_a-T_a=-R_a.$
0902.0431_E00065	14	■0.992	$\backslash\mathrm{begin}\{\mathrm{aligned}\}\ \mathrm{x}\ \&\ =\mathrm{x}_{\{0\}}+\mathrm{x}_{\{1\}}\ \mathrm{e}_{\{1\}}+\mathrm{x}_{\{2\}}\ \mathrm{e}_{\{2\}}+\mathrm{x}_{\{3\}}\ \mathrm{e}_{\{3\}}+\mathrm{x}_{\{4\}}\ \mathrm{e}_{\{4\}}+\mathrm{x}_{\{5\}}\ \mathrm{e}_{\{5\}}+\mathrm{x}_{\{6\}}\ \mathrm{e}_{\{6\}}+\mathrm{x}_{\{7\}}\ \mathrm{e}_{\{7\}}\ \backslash\mathrm{quad}\backslash\mathrm{left}(\mathrm{x}_{\{i\}}\ \backslash\mathrm{in}\ \boldsymbol{\mathrm{symbol}}\{\mathrm{R}\}\backslash\mathrm{right})\ \backslash\backslash\ \&\ =\backslash\mathrm{left}(\mathrm{x}_{\{0\}}+\mathrm{x}_{\{1\}}\ \mathrm{e}_{\{1\}}\backslash\mathrm{right})+\backslash\mathrm{left}(\mathrm{x}_{\{2\}}+\mathrm{x}_{\{3\}}\ \mathrm{e}_{\{1\}}\backslash\mathrm{right})\ \mathrm{e}_{\{2\}}+\backslash\mathrm{left}(\mathrm{x}_{\{4\}}+\mathrm{x}_{\{5\}}\ \mathrm{e}_{\{1\}}\backslash\mathrm{right})\ \mathrm{e}_{\{4\}}+\backslash\mathrm{left}(\mathrm{x}_{\{6\}}+\mathrm{x}_{\{7\}}\ \mathrm{e}_{\{1\}}\backslash\mathrm{right})\ \mathrm{e}_{\{6\}},\ \backslash\mathrm{end}\{\mathrm{aligned}\}$	$x=x_0+x_1e_1+x_2e_2+x_3e_3+x_4e_4+x_5e_5+x_6e_6+x_7e_7\quad(x_i\in\boldsymbol{R})$ $=\left(x_0+x_1e_1\right)+\left(x_2+x_3e_1\right)e_2+\left(x_4+x_5e_1\right)e_4+\left(x_6+x_7e_1\right)e_6,$	$x\ =\ x_0+x_1e_1+x_2e_2+x_3e_3+x_4e_4+x_5e_5+x_6e_6+x_7e_7\quad(x_i\in\boldsymbol{R})$ $\ =\ \left(x_0+x_1e_1\right)+\left(x_2+x_3e_1\right)e_2+\left(x_4+x_5e_1\right)e_4+\left(x_6+x_7e_1\right)e_6,$
0902.0431_E00812	136	■0.992	$\backslash\mathrm{iota}(\mathrm{X},\ \mathrm{Y},\ \backslash\mathrm{xi},\ \backslash\mathrm{eta})=(-\mathrm{i}\ \mathrm{X},\ \mathrm{i}\ \mathrm{Y},-\mathrm{i}\ \backslash\mathrm{xi},\ \mathrm{i}\ \backslash\mathrm{eta})\ .$	$\iota(X,Y,\xi,\eta)=(-iX,iY,-i\xi,i\eta).$	$\iota(X,Y,\xi,\eta)=(-iX,iY,-i\xi,i\eta).$
0902.0431_E01046	178	■0.992	$\emptyset=\backslash\mathrm{left}[\backslash\mathrm{left}(\backslash\mathrm{beta}_{\{1\}}\ \backslash\mathrm{Phi},\ 0,0,\ \mathrm{a}_{\{1\}}\ \backslash\mathrm{Phi},\ 0,0\ \backslash\mathrm{right}),(\emptyset,0,0,0,1,0)\backslash\mathrm{right}]=\backslash\mathrm{left}(\emptyset,0,0,0,2\ \mathrm{a}_{\{1\}}\ \backslash\mathrm{Phi},\ 0\backslash\mathrm{right}),$	$0=\left[(\beta_1\Phi,0,0,a_1\Phi,0,0),\left(0,0,0,0,1,0\right)\right]=\left(0,0,0,0,2a_1\Phi,0\right),$	$0=\left[(\beta_1\Phi,0,0,a_1\Phi,0,0),\left(0,0,0,0,1,0\right)\right]=\left(0,0,0,0,2a_1\Phi,0\right),$
0902.0431_E00835	139	■0.993	$(\mathrm{A},\ \mathrm{Y})=-\backslash\mathrm{left}(\mathrm{A},\ \backslash\mathrm{kappa}_{\{1\}}\ \mathrm{Y}\backslash\mathrm{right}),\ \backslash\mathrm{quad}\ \mathrm{Y}\ \backslash\mathrm{in}\ \mathfrak{J}^C\ .$	$(A,Y)=-\left(A,\kappa_1Y\right),\quad Y\in\mathfrak{J}^C.$	$(A,Y)=-\left(A,\kappa_1Y\right),\quad Y\in\mathfrak{J}^C.$
0902.0431_E01019	170	■0.993	$[(\backslash\mathrm{Phi}(\mathrm{h}),\ 0,0,\ r,\ 0,0),(\emptyset,\ \mathrm{P},\ 0,0,0,0)]=(\emptyset,(\backslash\mathrm{Phi}(\mathrm{h})+r)\ \mathrm{P},\ 0,0,0,0),$	$[(\Phi(h),0,0,r,0,0),\left(0,P,0,0,0,0\right)]=\left(0,\left(\Phi(h)+r\right)P,0,0,0,0\right),$	$[(\Phi(h),0,0,r,0,0),\left(0,P,0,0,0,0\right)]=\left(0,\left(\Phi(h)+r\right)P,0,0,0,0\right),$

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0902.0431_E00443	73	■ 0.993	$\begin{aligned} E_6^C &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X \\ &\quad \mid (\alpha X, \alpha Y, \alpha Z) = (X, Y, Z)\}, \\ E_6 &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid (\alpha X, \alpha Y, \alpha Z) = (X, Y, Z), \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha X \times \alpha Y = {}^t\alpha^{-1}(X \times Y), \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha X \times \alpha Y = \tau\alpha\tau(X \times Y), \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\}. \end{aligned}$	$\begin{aligned} E_6^C &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X \\ &\quad \mid (\alpha X, \alpha Y, \alpha Z) = (X, Y, Z)\}, \\ E_6 &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid (\alpha X, \alpha Y, \alpha Z) = (X, Y, Z), \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha X \times \alpha Y = {}^t\alpha^{-1}(X \times Y), \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha X \times \alpha Y = \tau\alpha\tau(X \times Y), \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\}. \end{aligned}$	$\begin{aligned} E_6^C &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X \\ &\quad \mid (\alpha X, \alpha Y, \alpha Z) = (X, Y, Z)\}, \\ E_6 &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid (\alpha X, \alpha Y, \alpha Z) = (X, Y, Z), \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha X \times \alpha Y = {}^t\alpha^{-1}(X \times Y), \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha X \times \alpha Y = \tau\alpha\tau(X \times Y), \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\}. \end{aligned}$
0902.0431_E00709	117	■ 0.993	$\begin{aligned} &(\exp t\Phi)(P \times Q)(\exp t\Phi)^{-1} \\ &= (\exp t(\operatorname{ad}\Phi))(P \times Q) \quad ((\operatorname{ad}\Phi)\Phi_1 = [\Phi, \Phi_1], \Phi_1 \in \mathfrak{e}_7^C) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} (t(\operatorname{ad}\Phi))^n (P \times Q) \\ &= \sum_{n=0}^{\infty} \frac{t^n}{n!} \left(\sum_{k+l=n} \frac{n!}{k!l!} \Phi^k P \times \Phi^l Q \right) \quad (\text{Proposition 4.3.2}) \\ &= \sum_{n=0}^{\infty} \left(\sum_{k+l=n} \frac{t^k t^l}{k!l!} \Phi^k P \times \Phi^l Q \right) \\ &= \left(\sum_{k=0}^{\infty} \frac{1}{k!} (t\Phi)^k P \right) \times \left(\sum_{l=0}^{\infty} \frac{1}{l!} (t\Phi)^l Q \right) \\ &= (\exp t\Phi)P \times (\exp t\Phi)Q. \end{aligned}$	$\begin{aligned} &(\exp t\Phi)(P \times Q)(\exp t\Phi)^{-1} \\ &= (\exp t(\operatorname{ad}\Phi))(P \times Q) \quad ((\operatorname{ad}\Phi)\Phi_1 = [\Phi, \Phi_1], \Phi_1 \in \mathfrak{e}_7^C) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} (t(\operatorname{ad}\Phi))^n (P \times Q) \\ &= \sum_{n=0}^{\infty} \frac{t^n}{n!} \left(\sum_{k+l=n} \frac{n!}{k!l!} \Phi^k P \times \Phi^l Q \right) \quad (\text{Proposition 4.3.2}) \\ &= \sum_{n=0}^{\infty} \left(\sum_{k+l=n} \frac{t^k t^l}{k!l!} \Phi^k P \times \Phi^l Q \right) \\ &= \left(\sum_{k=0}^{\infty} \frac{1}{k!} (t\Phi)^k P \right) \times \left(\sum_{l=0}^{\infty} \frac{1}{l!} (t\Phi)^l Q \right) \\ &= (\exp t\Phi)P \times (\exp t\Phi)Q. \end{aligned}$	$\begin{aligned} &(\exp t\Phi)(P \times Q)(\exp t\Phi)^{-1} \\ &= (\exp t(\operatorname{ad}\Phi))(P \times Q) \quad ((\operatorname{ad}\Phi)\Phi_1 = [\Phi, \Phi_1], \Phi_1 \in \mathfrak{e}_7^C) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} (t(\operatorname{ad}\Phi))^n (P \times Q) \\ &= \sum_{n=0}^{\infty} \frac{t^n}{n!} \left(\sum_{k+l=n} \frac{n!}{k!l!} \Phi^k P \times \Phi^l Q \right) \quad (\text{Proposition 4.3.2}) \\ &= \sum_{n=0}^{\infty} \left(\sum_{k+l=n} \frac{t^k t^l}{k!l!} \Phi^k P \times \Phi^l Q \right) \\ &= \left(\sum_{k=0}^{\infty} \frac{1}{k!} (t\Phi)^k P \right) \times \left(\sum_{l=0}^{\infty} \frac{1}{l!} (t\Phi)^l Q \right) \\ &= (\exp t\Phi)P \times (\exp t\Phi)Q. \end{aligned}$
0902.0431_E00555	93	■ 0.993	$\beta_{\alpha_{23}}(t_0)X = E_2 - E_3 \in S^8.$	$\beta_{\alpha_{23}}(t_0)X = E_2 - E_3 \in S^8.$	$\beta_{\alpha_{23}}(t_0)X = E_2 - E_3 \in S^8.$

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0902.0431_E00777	130	■0.993	$\backslash\mathrm{mathfrak{M}}^{\mathrm{C}}\backslash\mathrm{ni}\backslash\alpha(X,X\times X,\backslash\mathrm{operatorname{det}}X,1)=\backslash\mathrm{left}(\backslash\beta X,\backslash\beta_{\mathrm{1}}(X\times X),\backslash\mathrm{operatorname{det}}X,1\mathrm{right})$,	$\mathfrak{M}^C\ni\alpha(X,X\times X,\det X,1)=(\beta X,\beta_1(X\times X),\det X,1)$,	$\mathfrak{M}^C\ni\alpha(X,X\times X,\det X,1)=(\beta X,\beta_1(X\times X),\det X,1)$,
0902.0431_E00041 (ii)	12	■0.994	$\backslash\mathrm{left}(\mathrm{L}_{\mathrm{b}}\mathrm{x}\mathrm{right})\mathrm{y}+\mathrm{x}\backslash\mathrm{left}(\mathrm{R}_{\mathrm{b}}\mathrm{y}\mathrm{right})=\mathrm{T}_{\mathrm{b}}(\mathrm{x}\mathrm{y})$, $\quad\mathrm{x},\mathrm{y}\in\backslash\mathrm{mathfrak{C}}$.	$(L_bx)y+x(R_by)=T_b(xy),\quad x,y\in\mathfrak{C}.$	$(L_bx)y+x(R_by)=T_b(xy),\quad x,y\in\mathfrak{C}.$
0902.0431_E00077	16	■0.994	$\mathrm{B}=\mathrm{D}+\sum_{\mathrm{i}=1}^6\mathrm{x}_{\mathrm{i}}\mathrm{S}_{\mathrm{i}}$, $\quad\mathrm{D}\in\backslash\mathrm{mathfrak{s}}\mathrm{u}(3)$, $\quad\mathrm{x}_{\mathrm{i}}\in\backslash\mathrm{boldsymbol{R}}$	$B=D+\sum_{i=1}^6x_iS_i,\quad D\in\mathfrak{su}(3),x_i\in\boldsymbol{R}$	$B=D+\sum_{i=1}^6x_iS_i,\quad D\in\mathfrak{su}(3),x_i\in\boldsymbol{R}.$
0902.0431_E00996	166	■0.994	$\backslash\mathrm{operatorname{Aut}}\backslash\mathrm{left}(\backslash\mathrm{mathfrak{e}}_{\mathrm{8}}^{\mathrm{C}}\backslash\mathrm{right})=\backslash\mathrm{operatorname{Inn}}\backslash\mathrm{left}(\backslash\mathrm{mathfrak{e}}_{\mathrm{8}}^{\mathrm{C}}\backslash\mathrm{right})$.	$\mathrm{Aut}\left(\mathfrak{e}_8^C\right)=\mathrm{Inn}\left(\mathfrak{e}_8^C\right).$	$\mathrm{Aut}(\mathfrak{e}_8^C)=\mathrm{Inn}(\mathfrak{e}_8^C).$
0902.0431_E00999	166	■0.994	$\mathrm{E}_{\mathrm{8}}^{\mathrm{C}}\backslash\mathrm{left}(\mathrm{E}_{\mathrm{8}}^{\mathrm{C}}\mathrm{right})_{\mathrm{1}_{-}}\backslash\mathrm{simeq}\backslash\mathrm{mathfrak{W}}^{\mathrm{C}}$,	$E_8^C/(E_8^C)_{1_-}\simeq\mathfrak{W}^C,$	$E_8^C/(E_8^C)_{1_-}\simeq\mathfrak{W}^C,$
0902.0431_E00562	94	■0.994	$\backslash\mathrm{begin}{gathered}\backslash\mathrm{mathfrak{E}}_{\mathrm{1}}^{\mathrm{C}}=\backslash\mathrm{left}\backslash\mathrm{xi}\mathrm{E}_{\mathrm{1}}\backslash\mathrm{mid}\backslash\mathrm{xi}\backslash\mathrm{in}\mathrm{C}\mathrm{right}\backslash$, $\quad\backslash\mathrm{mathfrak{J}}(2,\mathfrak{C})^C=\backslash\mathrm{left}\backslash\mathrm{X}\backslash\mathrm{in}\backslash\mathrm{mathfrak{J}}^{\mathrm{C}}\backslash\mathrm{mid}4\mathrm{E}_{\mathrm{1}}\backslash\mathrm{times}\backslash\mathrm{left}(\mathrm{E}_{\mathrm{1}}\backslash\mathrm{times}\mathrm{X}\mathrm{right})=\mathrm{X}\mathrm{right}\backslash$, $\backslash\backslash\backslash\mathrm{left}(\backslash\mathrm{mathfrak{J}}^{\mathrm{C}}\mathrm{right})_{-\backslash\mathrm{sigma}}=\backslash\mathrm{left}\backslash\mathrm{X}\backslash\mathrm{in}\backslash\mathrm{mathfrak{J}}^{\mathrm{C}}\backslash\mathrm{mid}\mathrm{E}_{\mathrm{1}}\backslash\mathrm{times}\mathrm{X}=0,\backslash\mathrm{left}\backslash\mathrm{angle}\mathrm{E}_{\mathrm{1}},\mathrm{X}\mathrm{right}\backslash\mathrm{rangle}=0\mathrm{right}\backslash$. $\backslash\mathrm{end}{gathered}$	$\mathfrak{E}_1^C=\{\xi E_1\mid\xi\in C\},\quad\mathfrak{J}(2,\mathfrak{C})^C=\{X\in\mathfrak{J}^C\mid 4E_1\times(E_1\times X)=X\},$ $(\mathfrak{J}^C)_{-\sigma}=\{X\in\mathfrak{J}^C\mid E_1\times X=0,\langle E_1,X\rangle=0\}.$	$\mathfrak{E}_1^C=\{\xi E_1\mid\xi\in C\},\quad\mathfrak{J}(2,\mathfrak{C})^C=\{X\in\mathfrak{J}^C\mid 4E_1\times(E_1\times X)=X\},$ $(\mathfrak{J}^C)_{-\sigma}=\{X\in\mathfrak{J}^C\mid E_1\times X=0,\langle E_1,X\rangle=0\}.$
0902.0431_E00810	135	■0.994	$\backslash\alpha=\backslash\mathrm{left}(\backslash\mathrm{begin}{array}{cccc}\backslash\omega^{\backslash\mathrm{prime}}1&\&\&\&\backslash\omega^{\backslash\mathrm{prime}-1}1&\&\&\backslash\omega&\&\&\backslash\omega&\&\&1&\&\&\backslash\omega&\&\&1&\mathrm{end}{array}\mathrm{right})$, $\quad\backslash\omega^{\backslash\mathrm{prime}}=1$, $\backslash\omega\backslash\mathrm{text}\{\mathrm{or}\}\backslash\omega^2$.	$\alpha=\left(\begin{array}{cccc}\omega'1&0&0&0\\0&\omega'^{-1}1&0&0\\0&0&1&0\\0&0&0&1\end{array}\right),\quad\omega'=1,\omega\text{ or }\omega^2.$	$\alpha=\left(\begin{array}{cccc}\omega'1&0&0&0\\0&\omega'^{-1}1&0&0\\0&0&1&0\\0&0&0&1\end{array}\right),\quad\omega'=1,\omega\text{ or }\omega^2.$
0902.0431_E00012	8	■0.994	$\backslash\mathrm{begin}{aligned}(\backslash\alpha\mathrm{x},\backslash\alpha\mathrm{y})&=\mathrm{frac}{1}{2}((\backslash\alpha\mathrm{x})(\overline{\backslash\alpha\mathrm{y}})+(\backslash\alpha\mathrm{y})(\overline{\backslash\alpha\mathrm{x}}))=\mathrm{frac}{1}{2}((\backslash\alpha\mathrm{x})(\backslash\alpha\mathrm{\bar{y}})+(\backslash\alpha\mathrm{y})(\backslash\alpha\mathrm{\bar{x}}))\\&\backslash\&=\alpha\backslash\mathrm{left}(\mathrm{frac}{1}{2}(\mathrm{x}\mathrm{\bar{y}}+\mathrm{y}\mathrm{\bar{x}})\mathrm{right})=\alpha((\mathrm{x},\mathrm{y}))=(\mathrm{x},\mathrm{y})$.	$(\alpha x,\alpha y)=\frac{1}{2}((\alpha x)(\overline{\alpha y})+(\alpha y)(\overline{\alpha x}))=\frac{1}{2}((\alpha x)(\alpha \bar{y})+(\alpha y)(\alpha \bar{x}))$ $=\alpha\left(\frac{1}{2}(x\bar{y}+y\bar{x})\right)=\alpha((x,y))=(x,y).$	$(\alpha x,\alpha y)=\frac{1}{2}((\alpha x)(\overline{\alpha y})+(\alpha y)(\overline{\alpha x}))=\frac{1}{2}((\alpha x)(\alpha \bar{y})+(\alpha y)(\alpha \bar{x}))$ $=\alpha\left(\frac{1}{2}(x\bar{y}+y\bar{x})\right)=\alpha((x,y))=(x,y).$
0902.0431_E00412	68	■0.994	$\backslash\mathrm{left}(\backslash\mathrm{begin}{array}{ccc}\backslash\mathrm{x}_{\mathrm{1}}\&\mathrm{x}_{\mathrm{3}}\&\backslash\mathrm{bar{x}}_{\mathrm{2}}\backslash\backslash\backslash\mathrm{bar{x}}_{\mathrm{3}}\&\backslash\mathrm{x}_{\mathrm{2}}\&\backslash\mathrm{bar{x}}_{\mathrm{1}}\&\backslash\mathrm{x}_{\mathrm{3}}\mathrm{end}{array}\mathrm{right})$, $\quad\backslash\mathrm{x}_{\mathrm{i}}\in\backslash\mathrm{boldsymbol{R}}$, $\quad\mathrm{x}_{\mathrm{i}}=\mathrm{a}_{\mathrm{i}}+\backslash\mathrm{boldsymbol{m}}_{\mathrm{i}}\in\backslash\mathrm{boldsymbol{C}}\oplus\backslash\mathrm{boldsymbol{C}}^3=\backslash\mathrm{mathfrak{C}}$	$\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\\bar{x}_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{array}\right),\quad\xi_i\in\boldsymbol{R},x_i=a_i+\boldsymbol{m}_i\in\boldsymbol{C}\oplus\boldsymbol{C}^3=\mathfrak{C}$	$\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\\bar{x}_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{array}\right),\quad\xi_i\in\boldsymbol{R},x_i=a_i+\boldsymbol{m}_i\in\boldsymbol{C}\oplus\boldsymbol{C}^3=\mathfrak{C}$
0902.0431_E00673	112	■0.994	$\backslash\mathrm{begin}{aligned}&=\backslash\Phi\backslash\mathrm{left}(-\mathrm{X}\vee\mathrm{Y},-\mathrm{frac}{1}{2}(\mathrm{Y}\times\mathrm{Y}-\backslash\mathrm{xi}\mathrm{X}),\mathrm{frac}{1}{2}(\mathrm{X}\times\mathrm{X}-\backslash\mathrm{eta}\mathrm{Y}),\mathrm{frac}{1}{4}((\mathrm{X},\mathrm{Y})-3\backslash\mathrm{xi}\backslash\mathrm{eta})\mathrm{right})(\mathrm{Z},\mathrm{W},\backslash\mathrm{zeta},\backslash\mathrm{omega}),\\&\backslash\&-\mathrm{frac}{3}{8}((\mathrm{X},\mathrm{W})-(\mathrm{Z},\mathrm{Y})+\xi\omega-\zeta\eta)(\mathrm{X},\mathrm{Y},\xi,\eta)\\&=(\mathrm{P}\times\mathrm{P})\mathrm{Q}-\mathrm{frac}{3}{8}\{\mathrm{P},\mathrm{Q}\}\mathrm{P}.$	$=\Phi\left(-X\vee Y,-\frac{1}{2}(Y\times Y-\xi X),\frac{1}{2}(X\times X-\eta Y),\frac{1}{4}((X,Y)-3\xi\eta)\right)(Z,W,\zeta,\omega),$ $-\frac{3}{8}((X,W)-(Z,Y)+\xi\omega-\zeta\eta)(X,Y,\xi,\eta)$ $=(P\times P)Q-\frac{3}{8}\{P,Q\}P.$	$=\Phi\left(-X\vee Y,-\frac{1}{2}(Y\times Y-\xi X),\frac{1}{2}(X\times X-\eta Y),\frac{1}{4}((X,Y)-3\xi\eta)\right)(Z,W,\zeta,\omega),$ $-\frac{3}{8}((X,W)-(Z,Y)+\xi\omega-\zeta\eta)(X,Y,\xi,\eta)$ $=(P\times P)Q-\frac{3}{8}\{P,Q\}P.$

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0902.0431_E08136	25	■ 0.995	$\begin{array}{cc} 2 G_{12}-G_{47}-G_{56}, & \\ -G_{47}+G_{56}, & \backslash \backslash 2 G_{13}-G_{46}-G_{57}, & \\ G_{46}+G_{57}, & \backslash \backslash 2 G_{23}-G_{45}-G_{67}, & \\ -G_{45}+G_{67} & \end{array}$	$\begin{array}{cc} 2 G_{12}-G_{47}-G_{56}, & -G_{47}+G_{56}, \\ 2 G_{13}-G_{46}-G_{57}, & G_{46}+G_{57}, \\ 2 G_{23}-G_{45}-G_{67}, & -G_{45}+G_{67} \end{array}$	$\begin{array}{cc} 2 G_{12}-G_{47}-G_{56}, & -G_{47}+G_{56}, \\ 2 G_{13}-G_{46}-G_{57}, & G_{46}+G_{57}, \\ 2 G_{23}-G_{45}-G_{67}, & -G_{45}+G_{67} \end{array}$
0902.0431_E08282	45	■ 0.995	$\left(D_{\{i\}} x, y\right)+\left(x, D_{\{i\}} y\right)=0, \quad \text {quad} \\ x, y \in \mathfrak{C} \} .$	$(D_i x, y)+(x, D_i y)=0, \quad x, y \in \mathfrak{C} .$	$(D_i x, y)+(x, D_i y)=0, \quad x, y \in \mathfrak{C} .$
0902.0431_E01149 (i)	194	■ 0.995	$\begin{aligned} & (\boldsymbol{u} \times \boldsymbol{v}) \boldsymbol{x}=\left(\boldsymbol{x} \wedge \boldsymbol{u}_2 \wedge \boldsymbol{u}_3, \boldsymbol{v}\right) \boldsymbol{u}_1+\left(\boldsymbol{u}_1 \wedge \boldsymbol{x} \wedge \boldsymbol{u}_3, \boldsymbol{v}\right) \boldsymbol{u}_2 \\ & +\left(\boldsymbol{u}_1 \wedge \boldsymbol{u}_2 \wedge \boldsymbol{x}, \boldsymbol{v}\right) \boldsymbol{u}_3-\frac{1}{3}(\boldsymbol{u}, \boldsymbol{v}) \boldsymbol{x} . \end{aligned}$	$\begin{aligned} & (\boldsymbol{u} \times \boldsymbol{v}) \boldsymbol{x}=\left(\boldsymbol{x} \wedge \boldsymbol{u}_2 \wedge \boldsymbol{u}_3, \boldsymbol{v}\right) \boldsymbol{u}_1+\left(\boldsymbol{u}_1 \wedge \boldsymbol{x} \wedge \boldsymbol{u}_3, \boldsymbol{v}\right) \boldsymbol{u}_2 \\ & +\left(\boldsymbol{u}_1 \wedge \boldsymbol{u}_2 \wedge \boldsymbol{x}, \boldsymbol{v}\right) \boldsymbol{u}_3-\frac{1}{3}(\boldsymbol{u}, \boldsymbol{v}) \boldsymbol{x} . \end{aligned}$	$\begin{aligned} & (\boldsymbol{u} \times \boldsymbol{v}) \boldsymbol{x}=\left(\boldsymbol{x} \wedge \boldsymbol{u}_2 \wedge \boldsymbol{u}_3, \boldsymbol{v}\right) \boldsymbol{u}_1+\left(\boldsymbol{u}_1 \wedge \boldsymbol{x} \wedge \boldsymbol{u}_3, \boldsymbol{v}\right) \boldsymbol{u}_2 \\ & +\left(\boldsymbol{u}_1 \wedge \boldsymbol{u}_2 \wedge \boldsymbol{x}, \boldsymbol{v}\right) \boldsymbol{u}_3-\frac{1}{3}(\boldsymbol{u}, \boldsymbol{v}) \boldsymbol{x} . \end{aligned}$
0902.0431_E08292	46	■ 0.995	$\begin{aligned} & \mathfrak{e}_6^C=\left\{\phi \in \operatorname{Hom}_{\mathfrak{C}}\left(\mathfrak{J}^C\right) \mid\left(\phi X, X, X\right)=0\right\} \\ & =\left\{\phi \in \operatorname{Hom}_{\mathfrak{C}}\left(\mathfrak{J}^C\right) \mid-\phi(X \times Y)=\phi X \times Y+X \times \phi Y\right\}, \\ & \mathfrak{f}_4^C=\left\{\delta \in \operatorname{Hom}_{\mathfrak{C}}\left(\mathfrak{J}^C\right) \mid \delta(X \circ Y)=\delta X \circ Y+X \circ \delta Y\right\} \\ & =\left\{\delta \in \operatorname{Hom}_{\mathfrak{C}}\left(\mathfrak{J}^C\right) \mid \delta(X \times Y)=\delta X \times Y+X \times \delta Y\right\} . \end{aligned}$	$\begin{aligned} & \mathfrak{e}_6^C=\left\{\phi \in \operatorname{Hom}_{\mathfrak{C}}\left(\mathfrak{J}^C\right) \mid\left(\phi X, X, X\right)=0\right\} \\ & =\left\{\phi \in \operatorname{Hom}_{\mathfrak{C}}\left(\mathfrak{J}^C\right) \mid-\phi(X \times Y)=\phi X \times Y+X \times \phi Y\right\}, \\ & \mathfrak{f}_4^C=\left\{\delta \in \operatorname{Hom}_{\mathfrak{C}}\left(\mathfrak{J}^C\right) \mid \delta(X \circ Y)=\delta X \circ Y+X \circ \delta Y\right\} \\ & =\left\{\delta \in \operatorname{Hom}_{\mathfrak{C}}\left(\mathfrak{J}^C\right) \mid \delta(X \times Y)=\delta X \times Y+X \times \delta Y\right\} . \end{aligned}$	$\begin{aligned} \mathfrak{e}_6^C & =\left\{\phi \in \operatorname{Hom}_{\mathfrak{C}}\left(\mathfrak{J}^C\right) \mid\left(\phi X, X, X\right)=0\right\} \\ & =\left\{\phi \in \operatorname{Hom}_{\mathfrak{C}}\left(\mathfrak{J}^C\right) \mid-\phi(X \times Y)=\phi X \times Y+X \times \phi Y\right\}, \\ \mathfrak{f}_4^C & =\left\{\delta \in \operatorname{Hom}_{\mathfrak{C}}\left(\mathfrak{J}^C\right) \mid \delta(X \circ Y)=\delta X \circ Y+X \circ \delta Y\right\} \\ & =\left\{\delta \in \operatorname{Hom}_{\mathfrak{C}}\left(\mathfrak{J}^C\right) \mid \delta(X \times Y)=\delta X \times Y+X \times \delta Y\right\} . \end{aligned}$
0902.0431_E0822 (iv)	137	■ 0.995	$\begin{aligned} & a(X) \operatorname{det} X=a(X)(\operatorname{det} X)(M, N)=\left(\delta(X \times X) \times \delta(X \times X), N\right) \\ & =\left(\delta(X \times X), \delta(X \times X) \times N\right)=\frac{1}{2} a(X)(\delta(X \times X), \beta X) \\ & =\frac{1}{2} a(X)(3 a(X) b(X \times X)-\operatorname{det} X) . \end{aligned}$	$\begin{aligned} a(X) \operatorname{det} X & =a(X)(\operatorname{det} X)(M, N)=\left(\delta(X \times X) \times \delta(X \times X), N\right) \\ & =\left(\delta(X \times X), \delta(X \times X) \times N\right)=\frac{1}{2} a(X)(\delta(X \times X), \beta X) \\ & =\frac{1}{2} a(X)(3 a(X) b(X \times X)-\operatorname{det} X) . \end{aligned}$	$\begin{aligned} a(X) \operatorname{det} X & =a(X)(\operatorname{det} X)(M, N)=\left(\delta(X \times X) \times \delta(X \times X), N\right) \\ & =\left(\delta(X \times X), \delta(X \times X) \times N\right)=\frac{1}{2} a(X)(\delta(X \times X), \beta X) \\ & =\frac{1}{2} a(X)(3 a(X) b(X \times X)-\operatorname{det} X) . \end{aligned}$
0902.0431_E01842	178	■ 0.995	$\left(E_8^C\right)_{1,1^-,1_-}=\left\{\alpha \in E_8^C \mid \alpha 1=1, \alpha 1^-=1^-, \alpha 1_-=1_-\right\} .$	$\left(E_8^C\right)_{1,1^-,1_-}=\left\{\alpha \in E_8^C \mid \alpha 1=1, \alpha 1^-=1^-, \alpha 1_-=1_-\right\} .$	$\left(E_8^C\right)_{1,1^-,1_-}=\left\{\alpha \in E_8^C \mid \alpha 1=1, \alpha 1^-=1^-, \alpha 1_-=1_-\right\} .$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00387	64	■ 0.995	$\gamma\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1 & \gamma x_3 & \overline{\gamma x_2} \\ \overline{\gamma x_3} & \xi_2 & \gamma x_1 \\ \gamma x_2 & \overline{\gamma x_1} & \xi_3\end{array}\right).$	$\gamma\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1 & \gamma x_3 & \overline{\gamma x_2} \\ \overline{\gamma x_3} & \xi_2 & \gamma x_1 \\ \gamma x_2 & \overline{\gamma x_1} & \xi_3\end{array}\right).$	$\gamma\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1 & \gamma x_3 & \overline{\gamma x_2} \\ \overline{\gamma x_3} & \xi_2 & \gamma x_1 \\ \gamma x_2 & \overline{\gamma x_1} & \xi_3\end{array}\right).$
0902.0431_E00744	123	■ 0.995	$\Phi_0\Phi_0(X,Y,\xi,\eta)=\Phi_0\left(-\frac{X}{3},\frac{Y}{3},\xi,-\eta\right)=\left(\frac{X}{9},\frac{Y}{9},\xi,\eta\right),$	$\Phi_0\Phi_0(X,Y,\xi,\eta)=\Phi_0\left(-\frac{X}{3},\frac{Y}{3},\xi,-\eta\right)=\left(\frac{X}{9},\frac{Y}{9},\xi,\eta\right),$	$\Phi_0\Phi_0(X,Y,\xi,\eta)=\Phi_0\left(-\frac{X}{3},\frac{Y}{3},\xi,-\eta\right)=\left(\frac{X}{9},\frac{Y}{9},\xi,\eta\right),$
0902.0431_E01109	188	■ 0.995	$S\,0(16),\quad S\,s(16)\,.$	$SO(16),\quad Ss(16).$	$SO(16),\quad Ss(16).$
0902.0431_E01124	190	■ 0.995	$\tau\widetilde{\lambda}(D,\phi,\boldsymbol{X},\boldsymbol{Y})=(-\tau^tD,-\tau^t\phi\tau,-\tau\boldsymbol{Y},-\tau\boldsymbol{X}).$	$\tau\widetilde{\lambda}(D,\phi,\boldsymbol{X},\boldsymbol{Y})=(-\tau^tD,-\tau^t\phi\tau,-\tau\boldsymbol{Y},-\tau\boldsymbol{X}).$	$\tau\widetilde{\lambda}(D,\phi,\boldsymbol{X},\boldsymbol{Y})=(-\tau^tD,-\tau^t\phi\tau,-\tau\boldsymbol{Y},-\tau\boldsymbol{X}).$
0902.0431_E01044	178	■ 0.995	$\alpha=\left(\begin{array}{cccccc}\beta_1 & \beta_{12} & \beta_{13} & 0 & 0 & 0 \\ \beta_{21} & \beta_2 & \beta_{23} & 0 & 0 & 0 \\ \beta_{31} & \beta_{32} & \beta_3 & 0 & 0 & 0 \\ a_1 & b_1 & c_1 & 1 & 0 & 0 \\ a_2 & b_2 & c_2 & 0 & 1 & 0 \\ a_3 & b_3 & c_3 & 0 & 0 & 1\end{array}\right),$	$\alpha=\left(\begin{array}{cccccc}\beta_1 & \beta_{12} & \beta_{13} & 0 & 0 & 0 \\ \beta_{21} & \beta_2 & \beta_{23} & 0 & 0 & 0 \\ \beta_{31} & \beta_{32} & \beta_3 & 0 & 0 & 0 \\ a_1 & b_1 & c_1 & 1 & 0 & 0 \\ a_2 & b_2 & c_2 & 0 & 1 & 0 \\ a_3 & b_3 & c_3 & 0 & 0 & 1\end{array}\right),$	$\alpha=\left(\begin{array}{cccccc}\beta_1 & \beta_{12} & \beta_{13} & 0 & 0 & 0 \\ \beta_{21} & \beta_2 & \beta_{23} & 0 & 0 & 0 \\ \beta_{31} & \beta_{32} & \beta_3 & 0 & 0 & 0 \\ a_1 & b_1 & c_1 & 1 & 0 & 0 \\ a_2 & b_2 & c_2 & 0 & 1 & 0 \\ a_3 & b_3 & c_3 & 0 & 0 & 1\end{array}\right),$
0902.0431_E00410	68	■ 0.995	$w\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1 & \omega x_3 & \overline{\omega x_2} \\ \overline{\omega x_3} & \xi_2 & \omega x_1 \\ \omega x_2 & \overline{\omega x_1} & \xi_3\end{array}\right).$	$w\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1 & \omega x_3 & \overline{\omega x_2} \\ \overline{\omega x_3} & \xi_2 & \omega x_1 \\ \omega x_2 & \overline{\omega x_1} & \xi_3\end{array}\right).$	$w\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1 & \omega x_3 & \overline{\omega x_2} \\ \overline{\omega x_3} & \xi_2 & \omega x_1 \\ \omega x_2 & \overline{\omega x_1} & \xi_3\end{array}\right).$
0902.0431_E01134	191	■ 0.995	$(\mathrm{ad}\,\phi')(D,\phi,\boldsymbol{X},\boldsymbol{Y})=(0,(\mathrm{ad}\,\phi')\phi,\phi'\boldsymbol{X},-{}^t\phi'\boldsymbol{Y}),$	$(\mathrm{ad}\,\phi')(D,\phi,\boldsymbol{X},\boldsymbol{Y})=(0,(\mathrm{ad}\,\phi')\phi,\phi'\boldsymbol{X},-{}^t\phi'\boldsymbol{Y}),$	$(\mathrm{ad}\,\phi')(D,\phi,\boldsymbol{X},\boldsymbol{Y})=(0,(\mathrm{ad}\,\phi')\phi,\phi'\boldsymbol{X},-{}^t\phi'\boldsymbol{Y}),$
0902.0431_E00493	81	■ 0.996	$\delta(X\circ Y)=\delta X\circ Y+X\circ\delta Y,\quad X,Y\in(\mathfrak{M}^r)^C.$	$\delta(X\circ Y)=\delta X\circ Y+X\circ\delta Y,\quad X,Y\in(\mathfrak{M}^r)^C.$	$\delta(X\circ Y)=\delta X\circ Y+X\circ\delta Y,\quad X,Y\in(\mathfrak{M}^r)^C.$

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0902.0431_E00571	95	■ 0.996	J=\left(\begin{array}{ccc} J & \theta & \theta \\ \theta & J & \theta \\ \theta & \theta & J \end{array}\right) \in M(6, C), \quad J=\left(\begin{array}{cc} \theta & 1 \\ -1 & \theta \end{array}\right)	$J = \begin{pmatrix} J & 0 & 0 \\ 0 & J & 0 \\ 0 & 0 & J \end{pmatrix} \in M(6, C), \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$	$J = \begin{pmatrix} J & 0 & 0 \\ 0 & J & 0 \\ 0 & 0 & J \end{pmatrix} \in M(6, C), \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$
0902.0431_E00638	105	■ 0.996	\operatorname{dim}\left(\mathfrak{e}_{6,C}, \boldsymbol{\mathrm{C}}\right)=8+8=16 .	$\dim(\mathfrak{e}_{6,C}) = 8 + 8 = 16.$	$\dim(\mathfrak{e}_{6,C}) = 8 + 8 = 16.$
0902.0431_E00647	107	■ 0.996	\begin{aligned} \operatorname{det}(\alpha X) &= (\operatorname{det}(h(A, B))) (\operatorname{det} X) \\ \left(\operatorname{det}\left(h(A, B)^*\right)\right) &= \operatorname{det} X, \\ \left\langle\alpha X, \alpha Y\right\rangle &= \left\langle h(A, B) X h(A, B)^*, h(A, B) Y h(A, B)^*\right\rangle \\ &= \left\langle\tau h(A, B) \tau X \tau h(A, B)^*, h(A, B) Y h(A, B)^*\right\rangle \\ &= \left\langle\tau X, Y\right\rangle = \langle X, Y\rangle . \end{aligned}	$\det(\alpha X) = (\det(h(A, B))) (\det X) (\det(h(A, B)^*)) = \det X,$ $\langle\alpha X, \alpha Y\rangle = \langle h(A, B) X h(A, B)^*, h(A, B) Y h(A, B)^*\rangle$ $= \langle \tau h(A, B) \tau X \tau h(A, B)^*, h(A, B) Y h(A, B)^*\rangle$ $= \langle \tau X, Y \rangle = \langle X, Y \rangle.$	$\det(\alpha X) = (\det(h(A, B))) (\det X) (\det(h(A, B)^*)) = \det X,$ $\langle\alpha X, \alpha Y\rangle = \langle h(A, B) X h(A, B)^*, h(A, B) Y h(A, B)^*\rangle$ $= \langle \tau h(A, B) \tau X \tau h(A, B)^*, h(A, B) Y h(A, B)^*\rangle$ $= \langle \tau X, Y \rangle = \langle X, Y \rangle.$
0902.0431_E00687	114	■ 0.996	\operatorname{dim}_{\mathrm{C}}\left(\mathfrak{e}_{7}^{\mathrm{C}}\right)=78+27 \times 2+1=133 .	$\dim_C(\mathfrak{e}_7^C) = 78 + 27 \times 2 + 1 = 133.$	$\dim_C(\mathfrak{e}_7^C) = 78 + 27 \times 2 + 1 = 133.$
0902.0431_E00807	135	■ 0.996	\begin{aligned} &\left(\theta, 0, 0, \theta^{-3} \eta\right)=\varphi_1(\theta)(0, 0, 0, \eta)=\varphi_1(\theta) \alpha(0, 0, 1, 0) \\ &=\alpha \varphi_1(\theta)(0, 0, 1, 0)=\alpha\left(0, 0, \theta^3, 0\right)=\left(0, 0, 0, \theta^3 \eta\right), \end{aligned}	$(0, 0, 0, \theta^{-3} \eta) = \varphi_1(\theta)(0, 0, 0, \eta) = \varphi_1(\theta) \alpha(0, 0, 1, 0)$ $= \alpha \varphi_1(\theta)(0, 0, 1, 0) = \alpha(0, 0, \theta^3, 0) = (0, 0, 0, \theta^3 \eta),$	$(0, 0, 0, \theta^{-3} \eta) = \varphi_1(\theta)(0, 0, 0, \eta) = \varphi_1(\theta) \alpha(0, 0, 1, 0)$ $= \alpha \varphi_1(\theta)(0, 0, 1, 0) = \alpha(0, 0, \theta^3, 0) = (0, 0, 0, \theta^3 \eta),$
0902.0431_E00933	154	■ 0.996	\psi(A, 1) P=f^{-1}(A(f P)), \quad \psi(A, \epsilon) P=f^{-1}(A(f \bar{P})), \quad P \in\left(\mathfrak{P}_C\right)^C .	$\psi(A, 1) P = f^{-1}(A(f P)), \quad \psi(A, \epsilon) P = f^{-1}(A(f \bar{P})), \quad P \in (\mathfrak{P}_C)^C.$	$\psi(A, 1) P = f^{-1}(A(f P)), \quad \psi(A, \epsilon) P = f^{-1}(A(f \bar{P})), \quad P \in (\mathfrak{P}_C)^C.$
0902.0431_E00977	162	■ 0.996	\begin{aligned} \Phi &= (\Phi, \theta, 0, 0, \theta, \theta), \quad P^{-}=(\theta, P, \theta, 0, 0, \theta), \\ Q_{-} &=(0, 0, Q, 0, 0, \theta), \quad r=(0, 0, 0, r, 0, \theta), \\ s^{-} &=(0, 0, 0, 0, s, \theta), \quad t_{-}=(0, 0, 0, 0, 0, t) . \end{aligned}	$\Phi = (\Phi, \theta, 0, 0, \theta, \theta), \quad P^{-} = (\theta, P, \theta, 0, 0, \theta),$ $Q_{-} = (0, 0, Q, 0, 0, \theta), \quad r = (0, 0, 0, r, 0, \theta),$ $s^{-} = (0, 0, 0, 0, s, \theta), \quad t_{-} = (0, 0, 0, 0, 0, t).$	$\Phi = (\Phi, \theta, 0, 0, \theta, \theta), \quad P^{-} = (\theta, P, \theta, 0, 0, \theta),$ $Q_{-} = (0, 0, Q, 0, 0, \theta), \quad r = (0, 0, 0, r, 0, \theta),$ $s^{-} = (0, 0, 0, 0, s, \theta), \quad t_{-} = (0, 0, 0, 0, 0, t).$
0902.0431_E01098	186	■ 0.996	\begin{aligned} &\zeta\left[l\left(i c_1 E\right), l\left(i c_2 E\right)\right]=\zeta l\left[i c_1 E, i c_2 E\right]=\zeta 0=0 \\ &=\Theta\left(0, 0, 0, 0, 2 c_1,-2 c_1\right), \Theta\left(0, 0, 0, 0, 2 c_2,-2 c_2\right)) \\ &=\left[\zeta l\left(i c_1 E\right), \zeta l\left(i c_2 E\right)\right] . \end{aligned}	$\zeta[l(ic_1 E), l(ic_2 E)] = \zeta l[ic_1 E, ic_2 E] = \zeta 0 = 0$ $= \Theta(0, 0, 0, 0, 2c_1, -2c_1), \Theta(0, 0, 0, 0, 2c_2, -2c_2))]$ $= [\zeta l(ic_1 E), \zeta l(ic_2 E)].$	$\zeta[l(ic_1 E), l(ic_2 E)] = \zeta l[ic_1 E, ic_2 E] = \zeta 0 = 0$ $= \Theta(0, 0, 0, 0, 2c_1, -2c_1), \Theta(0, 0, 0, 0, 2c_2, -2c_2))]$ $= [\zeta l(ic_1 E), \zeta l(ic_2 E)].$
0902.0431_E01186	199	■ 0.996	\begin{aligned} &((\boldsymbol{a} \times *(\boldsymbol{b} \wedge \boldsymbol{x})) \boldsymbol{v}, \boldsymbol{w}) \\ &= (\boldsymbol{v} \wedge \boldsymbol{a}, \boldsymbol{w} \wedge *(\boldsymbol{b} \wedge \boldsymbol{x})) - \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) \end{aligned}	$((\boldsymbol{a} \times *(\boldsymbol{b} \wedge \boldsymbol{x})) \boldsymbol{v}, \boldsymbol{w})$ $= (\boldsymbol{v} \wedge \boldsymbol{a}, \boldsymbol{w} \wedge *(\boldsymbol{b} \wedge \boldsymbol{x})) - \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w})$	$((\boldsymbol{a} \times *(\boldsymbol{b} \wedge \boldsymbol{x})) \boldsymbol{v}, \boldsymbol{w})$ $= (\boldsymbol{v} \wedge \boldsymbol{a}, \boldsymbol{w} \wedge *(\boldsymbol{b} \wedge \boldsymbol{x})) - \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w})$

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0902.0431_E00724	119	■ 0.996	$\begin{aligned} \mathfrak{a} \supset & \left\{ \mathfrak{a}, \mathfrak{e}_7^C \right\} \supset \left\{ \Phi(\mathfrak{e}_6^C, 0, 0, 0), \Phi(0, \mathfrak{J}^C, 0, 0) \right\} \\ & = \Phi(0, \mathfrak{e}_6^C \mathfrak{J}^C, 0, 0) = \Phi(0, \mathfrak{J}^C, 0, 0) \text{ (Proposition 3.3.2.(3)).} \end{aligned}$	$\mathfrak{a} \supset [\mathfrak{a}, \mathfrak{e}_7^C] \supset [\Phi(\mathfrak{e}_6^C, 0, 0, 0), \Phi(0, \mathfrak{J}^C, 0, 0)]$ $= \Phi(0, \mathfrak{e}_6^C \mathfrak{J}^C, 0, 0) = \Phi(0, \mathfrak{J}^C, 0, 0) \text{ (Proposition 3.3.2.(3)).}$	$\mathfrak{a} \supset [\mathfrak{a}, \mathfrak{e}_7^C] \supset [\Phi(\mathfrak{e}_6^C, 0, 0, 0), \Phi(0, \mathfrak{J}^C, 0, 0)]$ $= \Phi(0, \mathfrak{e}_6^C \mathfrak{J}^C, 0, 0) = \Phi(0, \mathfrak{J}^C, 0, 0) \text{ (Proposition 3.3.2.(3)).}$
0902.0431_E00063	14	■ 0.997	$\begin{aligned} \lambda_{23} + \lambda_{45} + \lambda_{67} = 0, \quad & -\lambda_{13} - \lambda_{46} + \lambda_{57} = 0, \\ \lambda_{12} + \lambda_{47} + \lambda_{56} = 0, \quad & -\lambda_{15} + \lambda_{26} - \lambda_{37} = 0, \\ \lambda_{14} - \lambda_{27} - \lambda_{36} = 0, \quad & -\lambda_{17} - \lambda_{24} + \lambda_{35} = 0, \\ & \lambda_{16} + \lambda_{25} + \lambda_{34} = 0, \end{aligned}$	$\begin{aligned} \lambda_{23} + \lambda_{45} + \lambda_{67} = 0, \quad & -\lambda_{13} - \lambda_{46} + \lambda_{57} = 0, \\ \lambda_{12} + \lambda_{47} + \lambda_{56} = 0, \quad & -\lambda_{15} + \lambda_{26} - \lambda_{37} = 0, \\ \lambda_{14} - \lambda_{27} - \lambda_{36} = 0, \quad & -\lambda_{17} - \lambda_{24} + \lambda_{35} = 0, \\ & \lambda_{16} + \lambda_{25} + \lambda_{34} = 0, \end{aligned}$	$\begin{aligned} \lambda_{23} + \lambda_{45} + \lambda_{67} = 0, \quad & -\lambda_{13} - \lambda_{46} + \lambda_{57} = 0, \\ \lambda_{12} + \lambda_{47} + \lambda_{56} = 0, \quad & -\lambda_{15} + \lambda_{26} - \lambda_{37} = 0, \\ \lambda_{14} - \lambda_{27} - \lambda_{36} = 0, \quad & -\lambda_{17} - \lambda_{24} + \lambda_{35} = 0, \\ & \lambda_{16} + \lambda_{25} + \lambda_{34} = 0, \end{aligned}$
0902.0431_E00264	43	■ 0.997	$(X, Y) = \frac{1}{2} \operatorname{tr} \left(X Y + Y^* X^* \right) .$	$(X, Y) = \frac{1}{2} \operatorname{tr} (XY + Y^* X^*) .$	$(X, Y) = \frac{1}{2} \operatorname{tr} (XY + Y^* X^*) .$
0902.0431_E00335	53	■ 0.997	$\begin{aligned} H \left(e_k + i e_{4+k} \right) &= - \sum_{j=0}^3 \lambda_k i G_{j4+j} \left(e_k + i e_{4+k} \right) \\ &= -i \lambda_k \left(-e_{4+k} + i e_k \right) = \lambda_k \left(e_k + i e_{4+k} \right), \end{aligned}$	$\begin{aligned} H \left(e_k + i e_{4+k} \right) &= - \sum_{j=0}^3 \lambda_k i G_{j4+j} \left(e_k + i e_{4+k} \right) \\ &= -i \lambda_k \left(-e_{4+k} + i e_k \right) = \lambda_k \left(e_k + i e_{4+k} \right), \end{aligned}$	$\begin{aligned} H \left(e_k + i e_{4+k} \right) &= - \sum_{j=0}^3 \lambda_k i G_{j4+j} \left(e_k + i e_{4+k} \right) \\ &= -i \lambda_k \left(-e_{4+k} + i e_k \right) = \lambda_k \left(e_k + i e_{4+k} \right), \end{aligned}$
0902.0431_E00881	146	■ 0.997	$\alpha X = \begin{pmatrix} \xi_1 & 0 & 0 \\ 0 & \xi_2 & 0 \\ 0 & 0 & \xi_3 \end{pmatrix}, \quad \xi_i \in C.$	$\alpha X = \begin{pmatrix} \xi_1 & 0 & 0 \\ 0 & \xi_2 & 0 \\ 0 & 0 & \xi_3 \end{pmatrix}, \quad \xi_i \in C.$	$\alpha X = \begin{pmatrix} \xi_1 & 0 & 0 \\ 0 & \xi_2 & 0 \\ 0 & 0 & \xi_3 \end{pmatrix}, \quad \xi_i \in C.$
0902.0431_E01057	180	■ 0.997	$-21_- = \left[(\Phi, P, Q, r, s, t), 1_- \right] = (0, 0, -P, s, 0, -2r),$	$-21_- = [(\Phi, P, Q, r, s, t), 1_-] = (0, 0, -P, s, 0, -2r),$	$-21_- = [(\Phi, P, Q, r, s, t), 1_-] = (0, 0, -P, s, 0, -2r),$
0902.0431_E00081	17	■ 0.997	$\begin{aligned} H_1 &= -G_{23} + G_{45}, \quad L_{12} = G_{24} + G_{35}, \quad L_{21} = -G_{25} + G_{34}, \quad L_{13} = G_{26} + G_{37}, \\ L_{21} &= -G_{27} + G_{36}, \quad L_{23} = G_{46} + G_{57}, \quad L_{32} = -G_{47} + G_{56}. \end{aligned}$	$\begin{aligned} H_1 &= -G_{23} + G_{45}, \quad H_2 = -G_{45} + G_{67} \\ L_{12} &= G_{24} + G_{35}, \quad L_{21} = -G_{25} + G_{34}, \quad L_{13} = G_{26} + G_{37}, \\ L_{21} &= -G_{27} + G_{36}, \quad L_{23} = G_{46} + G_{57}, \quad L_{32} = -G_{47} + G_{56}. \end{aligned}$	$\begin{aligned} H_1 &= -G_{23} + G_{45}, \quad H_2 = -G_{45} + G_{67} \\ L_{12} &= G_{24} + G_{35}, \quad L_{21} = -G_{25} + G_{34}, \quad L_{13} = G_{26} + G_{37}, \\ L_{21} &= -G_{27} + G_{36}, \quad L_{23} = G_{46} + G_{57}, \quad L_{32} = -G_{47} + G_{56}. \end{aligned}$

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0902.0431_E00624	104	■ 0.997	$\begin{aligned} & (g(\alpha E))^2 = g(\alpha E) \circ g(\alpha E) = g(\gamma(\alpha E \times \alpha E)) + \frac{1}{4}(\gamma \alpha E, \alpha E)E \text{ (Lemma 3.12.1)} \\ & = g(\gamma \tau \alpha \tau E) + \frac{1}{4}\langle \tau \gamma \alpha E, \alpha E \rangle E = g(\alpha \tau \gamma E) + \frac{1}{4}\langle \alpha \tau \gamma E, \alpha E \rangle E \\ & = g(\alpha E) + \frac{1}{4}\langle \alpha E, \alpha E \rangle E = g(\alpha E) + \frac{1}{4}\langle E, E \rangle E = g(\alpha E) + \frac{3}{4}E. \end{aligned}$	$(g(\alpha E))^2 = g(\alpha E) \circ g(\alpha E) = g(\gamma(\alpha E \times \alpha E)) + \frac{1}{4}(\gamma \alpha E, \alpha E)E \text{ (Lemma 3.12.1)}$ $= g(\gamma \tau \alpha \tau E) + \frac{1}{4}\langle \tau \gamma \alpha E, \alpha E \rangle E = g(\alpha \tau \gamma E) + \frac{1}{4}\langle \alpha \tau \gamma E, \alpha E \rangle E$ $= g(\alpha E) + \frac{1}{4}\langle \alpha E, \alpha E \rangle E = g(\alpha E) + \frac{1}{4}\langle E, E \rangle E = g(\alpha E) + \frac{3}{4}E.$	$(g(\alpha E))^2 = g(\alpha E) \circ g(\alpha E) = g(\gamma(\alpha E \times \alpha E)) + \frac{1}{4}(\gamma \alpha E, \alpha E)E \text{ (Lemma 3.12.1)}$ $= g(\gamma \tau \alpha \tau E) + \frac{1}{4}\langle \tau \gamma \alpha E, \alpha E \rangle E = g(\alpha \tau \gamma E) + \frac{1}{4}\langle \alpha \tau \gamma E, \alpha E \rangle E$ $= g(\alpha E) + \frac{1}{4}\langle \alpha E, \alpha E \rangle E = g(\alpha E) + \frac{1}{4}\langle E, E \rangle E = g(\alpha E) + \frac{3}{4}E.$
0902.0431_E00374	62	■ 0.997	$F_4 / \operatorname{Spin}(9) \simeq \mathfrak{P}_2.$	$F_4 / \operatorname{Spin}(9) \simeq \mathfrak{CP}_2.$	$F_4 / \operatorname{Spin}(9) \simeq \mathfrak{CP}_2.$
0902.0431_E00434	71	■ 0.997	$F_4^C \simeq F_4 \times \mathbf{R}^{52}.$	$F_4^C \simeq F_4 \times \mathbf{R}^{52}.$	$F_4^C \simeq F_4 \times \mathbf{R}^{52}.$
0902.0431_E00717	118	■ 0.997	$[\Phi(\phi_1, A_1, -\tau A_1, \nu_1), \Phi(\phi_2, A_2, -\tau A_2, \nu_2)] = \Phi(\phi, A, -\tau A, \nu),$	$[\Phi(\phi_1, A_1, -\tau A_1, \nu_1), \Phi(\phi_2, A_2, -\tau A_2, \nu_2)] = \Phi(\phi, A, -\tau A, \nu),$	$[\Phi(\phi_1, A_1, -\tau A_1, \nu_1), \Phi(\phi_2, A_2, -\tau A_2, \nu_2)] = \Phi(\phi, A, -\tau A, \nu),$
0902.0431_E00914	152	■ 0.997	$\dim \mathfrak{e}_7 = 16 + 18 + 1 = 35.$	$\dim \mathfrak{e}_{7,C} = 16 + 18 + 1 = 35.$	$\dim \mathfrak{e}_{7,C} = 16 + 18 + 1 = 35.$
0902.0431_E00208	35	■ 0.997	$\begin{aligned} & \operatorname{Spin}(7) = \{\alpha \in \operatorname{Spin}(8) \mid \kappa \alpha = \alpha\}, \\ & G_2 = \{\alpha \in \operatorname{Spin}(8) \mid \lambda \alpha = \alpha, \lambda \in \mathfrak{S}_3\} \\ & = \{\alpha \in \operatorname{Spin}(8) \mid \pi \alpha = \alpha, \nu \alpha = \alpha\} \\ & = \{\alpha \in \operatorname{Spin}(8) \mid \nu \alpha = \alpha\} \\ & = \{\alpha \in \operatorname{Spin}(7) \mid \pi \alpha = \alpha\}. \end{aligned}$	$\begin{aligned} & \operatorname{Spin}(7) = \{\alpha \in \operatorname{Spin}(8) \mid \kappa \alpha = \alpha\}, \\ & G_2 = \{\alpha \in \operatorname{Spin}(8) \mid \lambda \alpha = \alpha, \lambda \in \mathfrak{S}_3\} \\ & = \{\alpha \in \operatorname{Spin}(8) \mid \pi \alpha = \alpha, \nu \alpha = \alpha\} \\ & = \{\alpha \in \operatorname{Spin}(8) \mid \nu \alpha = \alpha\} \\ & = \{\alpha \in \operatorname{Spin}(7) \mid \pi \alpha = \alpha\}. \end{aligned}$	$\begin{aligned} \operatorname{Spin}(7) &= \{\alpha \in \operatorname{Spin}(8) \mid \kappa \alpha = \alpha\}, \\ G_2 &= \{\alpha \in \operatorname{Spin}(8) \mid \lambda \alpha = \alpha, \lambda \in \mathfrak{S}_3\} \\ &= \{\alpha \in \operatorname{Spin}(8) \mid \pi \alpha = \alpha, \nu \alpha = \alpha\} \\ &= \{\alpha \in \operatorname{Spin}(8) \mid \nu \alpha = \alpha\} \\ &= \{\alpha \in \operatorname{Spin}(7) \mid \pi \alpha = \alpha\}. \end{aligned}$

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0902.0431_E00183	31	■ 0.998	$\widetilde{B}_3=\{\widetilde{\alpha}\in SO(8)\mid (\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),x,y\in \mathfrak{C}\text{ for some }\alpha\in SO(7)\}.$	$\widetilde{B}_3=\{\widetilde{\alpha}\in SO(8)\mid (\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),x,y\in \mathfrak{C}\text{ for some }\alpha\in SO(7)\}.$	$\widetilde{B}_3=\{\widetilde{\alpha}\in SO(8)\mid (\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),x,y\in \mathfrak{C}\text{ for some }\alpha\in SO(7)\}.$
0902.0431_E00255	41	■ 0.998	$\begin{aligned} E_{6(-26)}&=\left\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\det(\alpha X)=\det X\right.\\&=\left.\left\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid{}^t\alpha^{-1}(X\times Y)=\alpha X\times\alpha Y\right\}\right\}.$	$E_{6(-26)}=\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\det(\alpha X)=\det X\\=\left\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid{}^t\alpha^{-1}(X\times Y)=\alpha X\times\alpha Y\right\}\}.$	$E_{6(-26)}=\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\det(\alpha X)=\det X\\=\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid{}^t\alpha^{-1}(X\times Y)=\alpha X\times\alpha Y\}.$
0902.0431_E00617 (i)	103	■ 0.998	$(g\,X,\,g\,Y)=(\gamma X,\,Y),\quad X,\,Y\in\mathfrak{J}^C.$	$(gX,gY)=(\gamma X,Y),\quad X,Y\in\mathfrak{J}^C.$	$(gX,gY)=(\gamma X,Y),\quad X,Y\in\mathfrak{J}^C.$
0902.0431_E00635	105	■ 0.998	$E_{6,C}=\left\{\alpha\in\operatorname{Iso}_C\left((\mathfrak{J}_C)^C\right)\mid\det(\alpha X)=\det X,\langle\alpha X,\alpha Y\rangle=\langle X,Y\rangle\right\}.$	$E_{6,C}=\left\{\alpha\in\operatorname{Iso}_C\left((\mathfrak{J}_C)^C\right)\mid\det(\alpha X)=\det X,\langle\alpha X,\alpha Y\rangle=\langle X,Y\rangle\right\}.$	$E_{6,C}=\{\alpha\in\operatorname{Iso}_C((\mathfrak{J}_C)^C)\mid\det(\alpha X)=\det X,\langle\alpha X,\alpha Y\rangle=\langle X,Y\rangle\}.$
0902.0431_E00965	160	■ 0.998	$E_7^C\simeq(E_7^C\cap U(\mathfrak{P}^C))\times\boldsymbol{R}^d=E_7\times\boldsymbol{R}^d,\quad d=133.$	$E_7^C\simeq(E_7^C\cap U(\mathfrak{P}^C))\times\boldsymbol{R}^d=E_7\times\boldsymbol{R}^d,\quad d=133.$	$E_7^C\simeq(E_7^C\cap U(\mathfrak{P}^C))\times\boldsymbol{R}^d=E_7\times\boldsymbol{R}^d,\quad d=133.$
0902.0431_E01113	188	■ 0.998	$\left(E_8\right)^{\sigma}\cong Ss(16).$	$(E_8)^{\sigma}\cong Ss(16).$	$(E_8)^{\sigma}\cong Ss(16).$
0902.0431_E01116	189	■ 0.998	$\left(\mathfrak{J}^C\right)^3=\left\{\boldsymbol{X}=\begin{pmatrix}X_1\\X_2\\X_3\end{pmatrix}\middle X_i\in\mathfrak{J}^C\right\}.$	$(\mathfrak{J}^C)^3=\left\{\boldsymbol{X}=\begin{pmatrix}X_1\\X_2\\X_3\end{pmatrix}\middle X_i\in\mathfrak{J}^C\right\}.$	$(\mathfrak{J}^C)^3=\left\{\boldsymbol{X}=\begin{pmatrix}X_1\\X_2\\X_3\end{pmatrix}\middle X_i\in\mathfrak{J}^C\right\}.$
0902.0431_E01138	192	■ 0.998	$\begin{gathered}\left(\boldsymbol{x}_1\wedge\cdots\wedge\boldsymbol{x}_k,\boldsymbol{y}_1\wedge\cdots\wedge\boldsymbol{y}_k\right)=\det\left(\left(\boldsymbol{x}_i,\boldsymbol{y}_j\right)\right),\quad k\geq 1,\\(a,b)=ab,\quad a,b\in\Lambda^0(C^n)=C.\end{gathered}$	$\left(\boldsymbol{x}_1\wedge\cdots\wedge\boldsymbol{x}_k,\boldsymbol{y}_1\wedge\cdots\wedge\boldsymbol{y}_k\right)=\det\left(\left(\boldsymbol{x}_i,\boldsymbol{y}_j\right)\right),\quad k\geq 1,\\(a,b)=ab,\quad a,b\in\Lambda^0(C^n)=C.$	$\left(\boldsymbol{x}_1\wedge\cdots\wedge\boldsymbol{x}_k,\boldsymbol{y}_1\wedge\cdots\wedge\boldsymbol{y}_k\right)=\det\left(\left(\boldsymbol{x}_i,\boldsymbol{y}_j\right)\right),\quad k\geq 1,\\(a,b)=ab,\quad a,b\in\Lambda^0(C^n)=C.$
0902.0431_E00148	27	■ 0.998	$(m+ae_4')(n+be_4')=(mn+\bar{b}a)+(a\bar{n}+bm)e_4'.$	$(m+ae_4')(n+be_4')=(mn+\bar{b}a)+(a\bar{n}+bm)e_4'.$	$(m+ae_4')(n+be_4')=(mn+\bar{b}a)+(a\bar{n}+bm)e_4'.$
0902.0431_E00691 (ii)	114	■ 0.998	$\Phi=\begin{pmatrix}g&l&C&A\\k&h&B&D\\c&a&\nu&\lambda\\b&d&\kappa&\mu\end{pmatrix},\quad\begin{array}{l}g,h,k,l\in\operatorname{Hom}_C(\mathfrak{J}^C),\\a,b,c,d\in\operatorname{Hom}_C(\mathfrak{J}^C,C),\\A,B,C,D\in\mathfrak{J}^C,\\ \nu,\mu,\kappa,\lambda\in C.\end{array}$	$\Phi=\begin{pmatrix}g&l&C&A\\k&h&B&D\\c&a&\nu&\lambda\\b&d&\kappa&\mu\end{pmatrix},\quad\begin{array}{l}g,h,k,l\in\operatorname{Hom}_C(\mathfrak{J}^C),\\a,b,c,d\in\operatorname{Hom}_C(\mathfrak{J}^C,C),\\A,B,C,D\in\mathfrak{J}^C,\\ \nu,\mu,\kappa,\lambda\in C.\end{array}$	$\Phi=\begin{pmatrix}g&l&C&A\\k&h&B&D\\c&a&\nu&\lambda\\b&d&\kappa&\mu\end{pmatrix},\quad\begin{array}{l}g,h,k,l\in\operatorname{Hom}_C(\mathfrak{J}^C),\\a,b,c,d\in\operatorname{Hom}_C(\mathfrak{J}^C,C),\\A,B,C,D\in\mathfrak{J}^C,\\ \nu,\mu,\kappa,\lambda\in C.\end{array}$

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0902.0431_E00745	123	■ 0.998	$\operatorname{tr}\left(\Phi_0^2\right)=\frac{1}{9}\times 27\times 2+1+1=8$.	$\operatorname{tr}\left(\Phi_0^2\right)=\frac{1}{9}\times 27\times 2+1+1=8$.	$\operatorname{tr}\left(\Phi_0^2\right)=\frac{1}{9}\times 27\times 2+1+1=8$.
0902.0431_E00025	10	■ 0.998	$\begin{aligned} & \left[F_{i0}, F_{kl}\right] x = \left(F_{i0} F_{kl} - F_{kl} F_{i0}\right) x = \frac{1}{4} e_i\left(e_l\left(e_k x\right)\right) - \frac{1}{4} e_l\left(e_k\left(e_i x\right)\right) \\ & = -\frac{1}{4} e_l\left(e_i\left(e_k x\right)\right) + \frac{1}{4} e_l\left(e_i\left(e_k x\right)\right) = 0, \\ & \left[F_{ij}, F_{jk}\right] x = \left(F_{ij} F_{jk} - F_{jk} F_{ij}\right) x = \frac{1}{4} e_j\left(e_i\left(e_k\left(e_j x\right)\right)\right) - \frac{1}{4} e_k\left(e_j\left(e_j\left(e_i x\right)\right)\right) \\ & = \frac{1}{4} e_i\left(e_k\left(e_j\left(e_j x\right)\right)\right) + \frac{1}{4} e_k\left(e_i x\right) = -\frac{1}{4} e_i\left(e_k x\right) + \frac{1}{4} e_k\left(e_i x\right) \\ & = \frac{1}{2} e_k\left(e_i x\right) = F_{ik} x, \\ & \left[F_{ij}, F_{kl}\right] x = \left(F_{ij} F_{kl} - F_{kl} F_{ij}\right) x = \frac{1}{4} e_j\left(e_i\left(e_l\left(e_k x\right)\right)\right) - \frac{1}{4} e_l\left(e_k\left(e_j\left(e_i x\right)\right)\right) \\ & = \frac{1}{4} e_j\left(e_k\left(e_i\left(e_l x\right)\right)\right) - \frac{1}{4} e_j\left(e_l\left(e_k\left(e_i x\right)\right)\right) = \cdots = 0. \end{aligned}$	$\begin{aligned} & \left[F_{i0}, F_{kl}\right] x = \left(F_{i0} F_{kl} - F_{kl} F_{i0}\right) x = \frac{1}{4} e_i\left(e_l\left(e_k x\right)\right) - \frac{1}{4} e_l\left(e_k\left(e_i x\right)\right) \\ & = -\frac{1}{4} e_l\left(e_i\left(e_k x\right)\right) + \frac{1}{4} e_l\left(e_i\left(e_k x\right)\right) = 0, \\ & \left[F_{ij}, F_{jk}\right] x = \left(F_{ij} F_{jk} - F_{jk} F_{ij}\right) x = \frac{1}{4} e_j\left(e_i\left(e_k\left(e_j x\right)\right)\right) - \frac{1}{4} e_k\left(e_j\left(e_j\left(e_i x\right)\right)\right) \\ & = \frac{1}{4} e_i\left(e_k\left(e_j\left(e_j x\right)\right)\right) + \frac{1}{4} e_k\left(e_i x\right) = -\frac{1}{4} e_i\left(e_k x\right) + \frac{1}{4} e_k\left(e_i x\right) \\ & = \frac{1}{2} e_k\left(e_i x\right) = F_{ik} x, \\ & \left[F_{ij}, F_{kl}\right] x = \left(F_{ij} F_{kl} - F_{kl} F_{ij}\right) x = \frac{1}{4} e_j\left(e_i\left(e_l\left(e_k x\right)\right)\right) - \frac{1}{4} e_l\left(e_k\left(e_j\left(e_i x\right)\right)\right) \\ & = \frac{1}{4} e_j\left(e_k\left(e_i\left(e_l x\right)\right)\right) - \frac{1}{4} e_j\left(e_l\left(e_k\left(e_i x\right)\right)\right) = \cdots = 0. \end{aligned}$	$\begin{aligned} \left[F_{i0}, F_{kl}\right] x &= \left(F_{i0} F_{kl} - F_{kl} F_{i0}\right) x = \frac{1}{4} e_i\left(e_l\left(e_k x\right)\right) - \frac{1}{4} e_l\left(e_k\left(e_i x\right)\right) \\ &= -\frac{1}{4} e_l\left(e_i\left(e_k x\right)\right) + \frac{1}{4} e_l\left(e_i\left(e_k x\right)\right) = 0, \\ \left[F_{ij}, F_{jk}\right] x &= \left(F_{ij} F_{jk} - F_{jk} F_{ij}\right) x = \frac{1}{4} e_j\left(e_i\left(e_k\left(e_j x\right)\right)\right) - \frac{1}{4} e_k\left(e_j\left(e_j\left(e_i x\right)\right)\right) \\ &= \frac{1}{4} e_i\left(e_k\left(e_j\left(e_j x\right)\right)\right) + \frac{1}{4} e_k\left(e_i x\right) = -\frac{1}{4} e_i\left(e_k x\right) + \frac{1}{4} e_k\left(e_i x\right) \\ &= \frac{1}{2} e_k\left(e_i x\right) = F_{ik} x, \\ \left[F_{ij}, F_{kl}\right] x &= \left(F_{ij} F_{kl} - F_{kl} F_{ij}\right) x = \frac{1}{4} e_j\left(e_i\left(e_l\left(e_k x\right)\right)\right) - \frac{1}{4} e_l\left(e_k\left(e_j\left(e_i x\right)\right)\right) \\ &= \frac{1}{4} e_j\left(e_k\left(e_i\left(e_l x\right)\right)\right) - \frac{1}{4} e_j\left(e_l\left(e_k\left(e_i x\right)\right)\right) = \cdots = 0. \end{aligned}$
0902.0431_E00047 (iv)	12	■ 0.998	$x(Dy)=D(x\,y),\quad x,y\in\mathfrak{C}.$	$x(Dy)=D(xy),\quad x,y\in\mathfrak{C}.$	$x(Dy)=D(xy),\quad x,y\in\mathfrak{C}.$
0902.0431_E00138	26	■ 0.998	$\alpha=1\quad\text{or}\quad\alpha=\gamma.$	$\alpha=1\quad\text{or}\quad\alpha=\gamma.$	$\alpha=1\quad\text{or}\quad\alpha=\gamma.$
0902.0431_E00330	53	■ 0.998	$\mathfrak{D}_{4\{ \} ^C}\ni D\rightarrow(D,\nu D,\kappa\pi D)\in\mathfrak{d}_{4\{ \} ^C}\subset\mathfrak{f}_4^C$	$\mathfrak{D}_4^C\supset D\rightarrow(D,\nu D,\kappa\pi D)\in\mathfrak{d}_4^C\subset\mathfrak{f}_4^C$	$\mathfrak{D}_4^C\supset D\rightarrow(D,\nu D,\kappa\pi D)\in\mathfrak{d}_4^C\subset\mathfrak{f}_4^C$
0902.0431_E00457	75	■ 0.998	$\phi=\frac{\phi-\phi^*}{2}+i\frac{\phi+\phi^*}{2i},\quad\frac{\phi-\phi^*}{2},\frac{\phi+\phi^*}{2i}\in\mathfrak{e}_6,$	$\phi=\frac{\phi-\phi^*}{2}+i\frac{\phi+\phi^*}{2i},\quad\frac{\phi-\phi^*}{2},\frac{\phi+\phi^*}{2i}\in\mathfrak{e}_6.$	$\phi=\frac{\phi-\phi^*}{2}+i\frac{\phi+\phi^*}{2i},\quad\frac{\phi-\phi^*}{2},\frac{\phi+\phi^*}{2i}\in\mathfrak{e}_6.$
0902.0431_E00499	81	■ 0.998	$\widetilde{T}X=\frac{1}{2}\left(TX+XT^*\right),\quad\text{where}\quad T^*={}^t\overline{T}.$	$\widetilde{T}X=\frac{1}{2}\left(TX+XT^*\right),\quad\text{where}\quad T^*={}^t\overline{T}.$	$\widetilde{T}X=\frac{1}{2}\left(TX+XT^*\right),\quad\text{where}\quad T^*={}^t\overline{T}.$

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0902.0431_E00513	84	■ 0.998	$\begin{aligned} & \backslash\mathrm{begin}\{\mathrm{aligned}\} \ \& \ \alpha_{\mathrm{1}}=\lambda_{\mathrm{0}}-\lambda_{\mathrm{1}}, \\ & \quad \backslash\mathrm{quad} \ \alpha_{\mathrm{2}}=\lambda_{\mathrm{1}}-\lambda_{\mathrm{2}}, \quad \backslash\mathrm{quad} \\ & \quad \alpha_{\mathrm{3}}=\lambda_{\mathrm{2}}-\lambda_{\mathrm{3}}, \quad \backslash\backslash \ \& \ \alpha_{\mathrm{4}}= \\ & \quad \lambda_{\mathrm{3}}+\frac{\mathrm{1}}{\mathrm{2}}\left(\mu_{\mathrm{2}}-\mu_{\mathrm{3}}\right), \quad \backslash\backslash \\ & \quad \& \ \alpha_{\mathrm{5}}=\frac{\mathrm{1}}{\mathrm{2}}\left(-\lambda_{\mathrm{0}}-\lambda_{\mathrm{1}}-\lambda_{\mathrm{2}}+\lambda_{\mathrm{3}}\right)+\frac{\mathrm{1}}{\mathrm{2}}\left(\mu_{\mathrm{3}}-\mu_{\mathrm{1}}\right)- \\ & \quad \lambda_{\mathrm{2}}+\lambda_{\mathrm{3}}\right)+\frac{\mathrm{1}}{\mathrm{2}}\left(\mu_{\mathrm{3}}-\mu_{\mathrm{1}}\right)- \\ & \quad \mu_{\mathrm{1}}\right), \quad \backslash\backslash \ \& \ \alpha_{\mathrm{6}}=\frac{\mathrm{1}}{\mathrm{2}}\left(\lambda_{\mathrm{0}}+\lambda_{\mathrm{1}}+\lambda_{\mathrm{2}}+\lambda_{\mathrm{3}}\right)+\frac{\mathrm{1}}{\mathrm{2}}\left(\mu_{\mathrm{1}}-\mu_{\mathrm{2}}\right) \\ & \quad \lambda_{\mathrm{0}}+\lambda_{\mathrm{1}}+\lambda_{\mathrm{2}}+\lambda_{\mathrm{3}}\right)+\frac{\mathrm{1}}{\mathrm{2}}\left(\mu_{\mathrm{1}}-\mu_{\mathrm{2}}\right) \\ & \quad \backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\}\left(\mu_{\mathrm{1}}-\mu_{\mathrm{2}}\right) \end{aligned}$	$\begin{aligned} \alpha_1 &= \lambda_0 - \lambda_1, & \alpha_2 &= \lambda_1 - \lambda_2, & \alpha_3 &= \lambda_2 - \lambda_3, \\ \alpha_4 &= \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3), \\ \alpha_5 &= \frac{1}{2}(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1), \\ \alpha_6 &= \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) \end{aligned}$	$\begin{aligned} \alpha_1 &= \lambda_0 - \lambda_1, & \alpha_2 &= \lambda_1 - \lambda_2, & \alpha_3 &= \lambda_2 - \lambda_3, \\ \alpha_4 &= \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3), \\ \alpha_5 &= \frac{1}{2}(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1), \\ \alpha_6 &= \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) \end{aligned}$
0902.0431_E00679	113	■ 0.998	$\begin{aligned} & \backslash\mathrm{begin}\{\mathrm{aligned}\} \ E_{\mathrm{7}}^{\mathrm{C}} \ \& \ =\left\{\backslash\alpha \ \mathrm{in} \right. \\ & \quad \backslash\mathrm{operatorname}\{\mathrm{Iso}\}_{\mathrm{C}}\left.\backslash\mathrm{mathfrak}\{\mathrm{P}\}^{\mathrm{C}}\right\} \\ & \quad \backslash\mathrm{mid} \ \alpha(P \ \backslash\mathrm{times} \ Q) \ \alpha^{\mathrm{-1}}=\alpha P \ \backslash\mathrm{times} \\ & \quad \alpha Q\right\}, \quad \backslash\backslash \ E_{\mathrm{7}} \ \& \ =\left\{\backslash\alpha \ \mathrm{in} \right. \\ & \quad \backslash\mathrm{operatorname}\{\mathrm{Iso}\}_{\mathrm{C}}\left.\backslash\mathrm{mathfrak}\{\mathrm{P}\}^{\mathrm{C}}\right\} \\ & \quad \backslash\mathrm{mid} \ \alpha(P \ \backslash\mathrm{times} \ Q) \ \alpha^{\mathrm{-1}}=\alpha P \ \backslash\mathrm{times} \\ & \quad \alpha Q, \backslash\mathrm{angle}\alpha P, \ \backslash\alpha Q\mathrm{rangle}=\mathrm{angle} P, Q \\ & \quad \mathrm{rangle}\right\} . \end{aligned}$	$\begin{aligned} E_7^C &= \{ \alpha \in \mathrm{Iso}_C(\mathfrak{P}^C) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q \}, \\ E_7 &= \{ \alpha \in \mathrm{Iso}_C(\mathfrak{P}^C) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q, \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle \}. \end{aligned}$	$\begin{aligned} E_7^C &= \{ \alpha \in \mathrm{Iso}_C(\mathfrak{P}^C) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q \}, \\ E_7 &= \{ \alpha \in \mathrm{Iso}_C(\mathfrak{P}^C) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q, \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle \}. \end{aligned}$
0902.0431_E00959	159	■ 0.998	$\alpha P = f^{-1}(A(fP)) \quad \mathrm{or} \quad \alpha P = f^{-1}(A(f\bar{P})), \quad P \in (\mathfrak{P}_C)^C$	$\alpha P = f^{-1}(A(fP)) \quad \mathrm{or} \quad \alpha P = f^{-1}(A(f\bar{P})), \quad P \in (\mathfrak{P}_C)^C$	$\alpha P = f^{-1}(A(fP)) \quad \mathrm{or} \quad \alpha P = f^{-1}(A(f\bar{P})), \quad P \in (\mathfrak{P}_C)^C$
0902.0431_E00985	164	■ 0.998	$\mathrm{mathfrak}\{e\}_{\mathrm{8}}\{ \ }^{\mathrm{C}} \ \mathrm{cong} \ \Theta\left(\mathrm{mathfrak}\{e\}_{\mathrm{8}}\{ \ }^{\mathrm{C}}\right)=\left\{\Theta(R) \ \mathrm{mid} \ R \ \mathrm{in} \ \mathrm{mathfrak}\{e\}_{\mathrm{8}}\{ \ }^{\mathrm{C}}\right\}$	$\mathfrak{e}_8^C \cong \Theta(\mathfrak{e}_8^C) = \{ \Theta(R) \mid R \in \mathfrak{e}_8^C \}$	$\mathfrak{e}_8^C \cong \Theta(\mathfrak{e}_8^C) = \{ \Theta(R) \mid R \in \mathfrak{e}_8^C \}$
0902.0431_E01128	191	■ 0.998	$w(D, \phi, \boldsymbol{X}, \boldsymbol{Y}) = \left(D, \phi, \omega \boldsymbol{X}, \omega^2 \boldsymbol{Y} \right),$	$w(D, \phi, \boldsymbol{X}, \boldsymbol{Y}) = \left(D, \phi, \omega \boldsymbol{X}, \omega^2 \boldsymbol{Y} \right),$	$w(D, \phi, \boldsymbol{X}, \boldsymbol{Y}) = \left(D, \phi, \omega \boldsymbol{X}, \omega^2 \boldsymbol{Y} \right),$
0902.0431_E00007	7	■ 0.998	$\begin{aligned} & \backslash\mathrm{begin}\{\mathrm{aligned}\} \ \& \ (\mathrm{x} \ \mathrm{x}) \ \mathrm{y}+\mathrm{x}(\mathrm{y} \ \mathrm{a})=\mathrm{a}(\mathrm{x} \ \mathrm{y})+(\mathrm{x} \ \mathrm{y}) \ \mathrm{a}, \quad \backslash\backslash \ \& \\ & \quad (\mathrm{x} \ \mathrm{a}) \ \mathrm{y}+(\mathrm{x} \ \mathrm{y}) \ \mathrm{a}=\mathrm{x}(\mathrm{a} \ \mathrm{y})+\mathrm{x}(\mathrm{y} \ \mathrm{a}), \quad \backslash\backslash \ \& \ (\mathrm{x} \ \mathrm{x}) \ \mathrm{y}+(\mathrm{x} \ \mathrm{a}) \ \mathrm{y}=\mathrm{a}(\mathrm{x} \\ & \quad \mathrm{y})+\mathrm{x}(\mathrm{a} \ \mathrm{y}) . \end{aligned}$	$\begin{aligned} (ax)y + x(ya) &= a(xy) + (xy)a, \\ (xa)y + (xy)a &= x(ay) + x(ya), \\ (ax)y + (xa)y &= a(xy) + x(ay). \end{aligned}$	$\begin{aligned} (ax)y + x(ya) &= a(xy) + (xy)a, \\ (xa)y + (xy)a &= x(ay) + x(ya), \\ (ax)y + (xa)y &= a(xy) + x(ay). \end{aligned}$
0902.0431_E00147	27	■ 0.998	$G_{\mathrm{2}}\{ \ }^{\mathrm{C}} \ \mathrm{simeq} \left(G_{\mathrm{2}}\{ \ }^{\mathrm{C}}\right) \cap U \left(\mathfrak{C}^C\right) \times \boldsymbol{R}^d = G_2 \times \boldsymbol{R}^d,$	$G_2^C \simeq (G_2^C \cap U(\mathfrak{C}^C)) \times \boldsymbol{R}^d = G_2 \times \boldsymbol{R}^d,$	$G_2^C \simeq (G_2^C \cap U(\mathfrak{C}^C)) \times \boldsymbol{R}^d = G_2 \times \boldsymbol{R}^d,$

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0902.0431_E01029	172	0.998	$\begin{aligned} & \begin{aligned} & \lambda_2 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 1 \quad 2 \quad 2 \quad 1 \quad 1 \quad 0 \quad 1 \\ & \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 1 \quad 2 \quad 2 \quad 1 \quad 0 \quad 0 \quad 1 \\ & \lambda_0 - \frac{1}{2}(\mu_2 - \mu_3) = 2 \quad 3 \quad 4 \quad 3 \quad 3 \quad 2 \quad 1 \quad 2 \\ & \lambda_1 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \\ & \lambda_2 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \\ & \lambda_3 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \end{aligned} \\ & \begin{aligned} & \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) = 1 \quad 1 \quad 2 \quad 2 \quad 2 \quad 2 \quad 1 \quad 1 \\ & \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) = 1 \quad 1 \quad 2 \quad 2 \quad 2 \quad 1 \quad 1 \quad 1 \\ & \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) = 1 \quad 1 \quad 2 \quad 2 \quad 2 \quad 1 \quad 0 \quad 1 \\ & \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) = 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \\ & \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) = 1 \quad 3 \quad 4 \quad 3 \quad 2 \quad 2 \quad 1 \quad 2 \\ & \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) = 1 \quad 3 \quad 4 \quad 3 \quad 2 \quad 1 \quad 1 \quad 2 \\ & \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) = 1 \quad 3 \quad 4 \quad 3 \quad 2 \quad 1 \quad 0 \quad 2 \\ & \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) = 1 \quad 2 \quad 2 \quad 1 \quad 0 \quad 0 \quad 0 \quad 1 \\ & \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) = 1 \quad 2 \quad 2 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ & \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) = 1 \quad 2 \quad 2 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1 \\ & \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) = 1 \quad 2 \quad 2 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \\ & \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) = 1 \quad 3 \quad 4 \quad 3 \quad 3 \quad 2 \quad 1 \quad 2 \\ & \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 1 \quad 1 \quad 2 \quad 2 \quad 1 \quad 1 \quad 0 \quad 1 \\ & \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 1 \quad 1 \quad 2 \quad 2 \quad 1 \quad 0 \quad 0 \quad 1 \\ & \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 1 \quad 1 \quad 2 \quad 2 \quad 1 \quad 1 \quad 1 \quad 1 \\ & \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 1 \quad 2 \quad 4 \quad 4 \quad 3 \quad 2 \quad 1 \quad 2 \end{aligned} \\ & \begin{aligned} & \mu_1 + \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \\ & \mu_2 + \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 1 \\ & -\mu_3 - \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \end{aligned} \end{aligned}$		

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0902.0431_E00050	12	■ 0.999	$\begin{array}{ll} \left(D_{\{2\}}x\right)y+x\left(D_{\{3\}}y\right)=\left(\kappa D_{\{1\}}\right)(xy), & x,y\in\mathfrak{C}, \\ \left(D_{\{3\}}x\right)y+x\left(D_{\{1\}}y\right)=\left(\kappa D_{\{2\}}\right)(xy), & x,y\in\mathfrak{C}. \end{array}$	$\begin{array}{l} (D_2x)y+x(D_3y)=(\kappa D_1)(xy), \quad x,y\in\mathfrak{C}, \\ (D_3x)y+x(D_1y)=(\kappa D_2)(xy), \quad x,y\in\mathfrak{C}. \end{array}$	$\begin{array}{l} (D_2x)y+x(D_3y)=(\kappa D_1)(xy), \quad x,y\in\mathfrak{C}, \\ (D_3x)y+x(D_1y)=(\kappa D_2)(xy), \quad x,y\in\mathfrak{C}. \end{array}$
0902.0431_E00180	31	■ 0.999	$\alpha_{\bar{\omega}_1}(a+\boldsymbol{m})=\bar{\omega}_1(a+\boldsymbol{m})\omega_1=\bar{\omega}_1a\omega_1+\bar{\omega}_1^2\boldsymbol{m}=a+\omega_1\boldsymbol{m}=w(a+\boldsymbol{m}),$	$\alpha_{\bar{\omega}_1}(a+\boldsymbol{m})=\bar{\omega}_1(a+\boldsymbol{m})\omega_1=\bar{\omega}_1a\omega_1+\bar{\omega}_1^2\boldsymbol{m}=a+\omega_1\boldsymbol{m}=w(a+\boldsymbol{m}),$	$\alpha_{\bar{\omega}_1}(a+\boldsymbol{m})=\bar{\omega}_1(a+\boldsymbol{m})\omega_1=\bar{\omega}_1a\omega_1+\bar{\omega}_1^2\boldsymbol{m}=a+\omega_1\boldsymbol{m}=w(a+\boldsymbol{m}),$
0902.0431_E00345	56	■ 0.999	$\begin{aligned} & B_{\{4\}}\left(H,H^{\prime}\right)=\sum_{k,l=0}^3\lambda_k\lambda_l^{\prime}B_{\{4\}}\left(H_k,H_l\right) \\ & =3\sum_{k,l}\left(\lambda_k\lambda_l^{\prime}\left(\mathrm{tr}\left(H_kH_l\right)+\mathrm{tr}\left(\left(\nu H_k\right)\left(\nu H_l\right)\right)+\mathrm{tr}\left(\left(\kappa\pi H_k\right)\left(\kappa\pi H_l\right)\right)\right)\right) \\ & =3\sum_{k,l}\lambda_k\lambda_l^{\prime}\left(2+2+2\right)\delta_{kl}=18\sum_{k=0}^3\lambda_k\lambda_k^{\prime}. \end{aligned}$	$\begin{aligned} B_4\left(H,H^{\prime}\right) &= \sum_{k,l=0}^3\lambda_k\lambda_l^{\prime}B_4\left(H_k,H_l\right) \\ &= 3\sum_{k,l}\left(\lambda_k\lambda_l^{\prime}\left(\mathrm{tr}\left(H_kH_l\right)+\mathrm{tr}\left(\left(\nu H_k\right)\left(\nu H_l\right)\right)+\mathrm{tr}\left(\left(\kappa\pi H_k\right)\left(\kappa\pi H_l\right)\right)\right)\right) \\ &= 3\sum_{k,l}\lambda_k\lambda_l^{\prime}\left(2+2+2\right)\delta_{kl}=18\sum_{k=0}^3\lambda_k\lambda_k^{\prime} \end{aligned}$	$\begin{aligned} B_4(H,H') &= \sum_{k,l=0}^3\lambda_k\lambda_l^{\prime}B_4(H_k,H_l) \\ &= 3\sum_{k,l}\left(\lambda_k\lambda_l^{\prime}\left(\mathrm{tr}(H_kH_l)+\mathrm{tr}((\nu H_k)(\nu H_l))+\mathrm{tr}((\kappa\pi H_k)(\kappa\pi H_l))\right)\right) \\ &= 3\sum_{k,l}\lambda_k\lambda_l^{\prime}(2+2+2)\delta_{kl}=18\sum_{k=0}^3\lambda_k\lambda_k^{\prime}. \end{aligned}$
0902.0431_E00405	67	■ 0.999	$\varphi(p,A)((M+\boldsymbol{a})\times(N+\boldsymbol{b}))=\varphi(p,A)(M+\boldsymbol{a})\times\varphi(p,A)(N+\boldsymbol{b}),$	$\varphi(p,A)((M+\boldsymbol{a})\times(N+\boldsymbol{b}))=\varphi(p,A)(M+\boldsymbol{a})\times\varphi(p,A)(N+\boldsymbol{b}),$	$\varphi(p,A)((M+\boldsymbol{a})\times(N+\boldsymbol{b}))=\varphi(p,A)(M+\boldsymbol{a})\times\varphi(p,A)(N+\boldsymbol{b}),$
0902.0431_E00441	72	■ 0.999	$\begin{aligned} & F_{\{4(4)\}}\,\&\,\simeq\,(\operatorname{Sp}(1)\,\times\,\operatorname{Sp}(3))\,/\,\boldsymbol{Z}_2\,\times\,\boldsymbol{R}^{28}, \\ & F_{\{4(-20)\}}\,\&\,\simeq\,\operatorname{Spin}(9)\,\times\,\boldsymbol{R}^{16}. \end{aligned}$	$\begin{array}{l} F_{4(4)}\simeq(\operatorname{Sp}(1)\times\operatorname{Sp}(3))/\boldsymbol{Z}_2\times\boldsymbol{R}^{28}, \\ F_{4(-20)}\simeq\operatorname{Spin}(9)\times\boldsymbol{R}^{16}. \end{array}$	$\begin{array}{l} F_{4(4)}\simeq(\operatorname{Sp}(1)\times\operatorname{Sp}(3))/\boldsymbol{Z}_2\times\boldsymbol{R}^{28}, \\ F_{4(-20)}\simeq\operatorname{Spin}(9)\times\boldsymbol{R}^{16}. \end{array}$
0902.0431_E00485	80	■ 0.999	$B_{\{6\}}(\phi,\phi)=\operatorname{tr}\left(\left(\operatorname{ad}\phi\right)^2\right)=\left(\frac{1}{4}\times4+1\times2\right)\times8=24.$	$B_6(\phi,\phi)=\operatorname{tr}\left((\operatorname{ad}\phi)^2\right)=\left(\frac{1}{4}\times4+1\times2\right)\times8=24.$	$B_6(\phi,\phi)=\operatorname{tr}\left((\operatorname{ad}\phi)^2\right)=\left(\frac{1}{4}\times4+1\times2\right)\times8=24.$
0902.0431_E00496	81	■ 0.999	$\begin{aligned} & \sum_{k=1}^3\left((\sigma_{ik}x_{ik})y_{kj}+\overline{y_{ki}(\overline{\sigma_{jk}x_{jk}})}\right)+\sum_{k=1}^3\left(x_{ik}(\sigma_{kj}y_{kj})+\overline{(\sigma_{ki}y_{ki})\overline{x_{jk}}}\right) \\ & =\sum_k\left((\sigma_{ik}x_{ik})y_{kj}+y_{ik}(\sigma_{kj}x_{kj})\right)+\sum_k\left(\overline{(\sigma_{jk}x_{jk})y_{ki}+x_{jk}(\sigma_{ki}y_{ki})}\right) \end{aligned}$	$\begin{aligned} & \sum_{k=1}^3\left((\sigma_{ik}x_{ik})y_{kj}+\overline{y_{ki}(\overline{\sigma_{jk}x_{jk}})}\right)+\sum_{k=1}^3\left(x_{ik}(\sigma_{kj}y_{kj})+\overline{(\sigma_{ki}y_{ki})\overline{x_{jk}}}\right) \\ & =\sum_k\left((\sigma_{ik}x_{ik})y_{kj}+y_{ik}(\sigma_{kj}x_{kj})\right)+\sum_k\left(\overline{(\sigma_{jk}x_{jk})y_{ki}+x_{jk}(\sigma_{ki}y_{ki})}\right) \end{aligned}$	$\begin{aligned} & \sum_{k=1}^3\left((\sigma_{ik}x_{ik})y_{kj}+\overline{y_{ki}(\overline{\sigma_{jk}x_{jk}})}\right)+\sum_{k=1}^3\left(x_{ik}(\sigma_{kj}y_{kj})+\overline{(\sigma_{ki}y_{ki})\overline{x_{jk}}}\right) \\ & =\sum_k\left((\sigma_{ik}x_{ik})y_{kj}+y_{ik}(\sigma_{kj}x_{kj})\right)+\sum_k\left(\overline{(\sigma_{jk}x_{jk})y_{ki}+x_{jk}(\sigma_{ki}y_{ki})}\right) \end{aligned}$

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0902.0431_E00560	94	■ 0.999	$\phi(\theta)\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\theta^4\xi_1 & \theta x_3 & \theta\bar{x}_2 \\ \theta\bar{x}_3 & \theta^{-2}\xi_2 & \theta^{-2}x_1 \\ \theta x_2 & \theta^{-2}\bar{x}_1 & \theta^{-2}\xi_3\end{array}\right).$	$\phi(\theta)\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\theta^4\xi_1 & \theta x_3 & \theta\bar{x}_2 \\ \theta\bar{x}_3 & \theta^{-2}\xi_2 & \theta^{-2}x_1 \\ \theta x_2 & \theta^{-2}\bar{x}_1 & \theta^{-2}\xi_3\end{array}\right).$	$\phi(\theta)\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\theta^4\xi_1 & \theta x_3 & \theta\bar{x}_2 \\ \theta\bar{x}_3 & \theta^{-2}\xi_2 & \theta^{-2}x_1 \\ \theta x_2 & \theta^{-2}\bar{x}_1 & \theta^{-2}\xi_3\end{array}\right).$
0902.0431_E00578	96	■ 0.999	$k((a+be_2)+i(c+de_2))=0, \quad a,b,c,d\in\boldsymbol{C},$	$k((a+be_2)+i(c+de_2))=0, \quad a,b,c,d\in\boldsymbol{C},$	$k((a+be_2)+i(c+de_2))=0, \quad a,b,c,d\in\boldsymbol{C},$
0902.0431_E00607	101	■ 0.999	$A(P\circ Q)A^*=APA^*\circ AQA^*, \quad A\in\mathrm{Sp}(4), P,Q\in\mathfrak{J}(4,\boldsymbol{H}).$	$A(P\circ Q)A^*=APA^*\circ AQA^*, \quad A\in\mathrm{Sp}(4), P,Q\in\mathfrak{J}(4,\boldsymbol{H}).$	$A(P\circ Q)A^*=APA^*\circ AQA^*, \quad A\in\mathrm{Sp}(4), P,Q\in\mathfrak{J}(4,\boldsymbol{H}).$
0902.0431_E00693	114	■ 0.999	$rf_r(P\times Q)f_r^{-1}=f_rP\times f_rQ, \quad P,Q\in\mathfrak{P}^C.$	$rf_r(P\times Q)f_r^{-1}=f_rP\times f_rQ, \quad P,Q\in\mathfrak{P}^C.$	$rf_r(P\times Q)f_r^{-1}=f_rP\times f_rQ, \quad P,Q\in\mathfrak{P}^C.$
0902.0431_E00720	118	■ 0.999	$\Phi\in\mathfrak{e}_7 \quad \text{if and only if} \quad \tau\lambda\Phi\lambda^{-1}\tau=\Phi.$	$\Phi\in\mathfrak{e}_7 \quad \text{if and only if} \quad \tau\lambda\Phi\lambda^{-1}\tau=\Phi.$	$\Phi\in\mathfrak{e}_7 \quad \text{if and only if} \quad \tau\lambda\Phi\lambda^{-1}\tau=\Phi.$
0902.0431_E00836 (i)	139	■ 0.999	$(E_1,\phi X)-\frac{1}{3}\nu(E_1,X)=-\nu(E_1,X).$	$(E_1,\phi X)-\frac{1}{3}\nu(E_1,X)=-\nu(E_1,X).$	$(E_1,\phi X)-\frac{1}{3}\nu(E_1,X)=-\nu(E_1,X).$
0902.0431_E00883	147	■ 0.999	$\tau\gamma(X,Y,\xi,\eta)=(\tau\gamma X,\tau\gamma Y,\tau\xi,\tau\eta),$	$\tau\gamma(X,Y,\xi,\eta)=(\tau\gamma X,\tau\gamma Y,\tau\xi,\tau\eta),$	$\tau\gamma(X,Y,\xi,\eta)=(\tau\gamma X,\tau\gamma Y,\tau\xi,\tau\eta),$
0902.0431_E00962	160	■ 0.999	$\gamma_1:\mathfrak{P}^C\rightarrow\mathfrak{P}^C,\gamma_1(P_C+(M+N))=\overline{P_C}+(\bar{M}+\bar{N}), \quad P_C+(M+N)\in\mathfrak{P}^C.$	$\gamma_1:\mathfrak{P}^C\rightarrow\mathfrak{P}^C,\gamma_1(P_C+(M+N))=\overline{P_C}+(\bar{M}+\bar{N}), \quad P_C+(M+N)\in\mathfrak{P}^C.$	$\gamma_1:\mathfrak{P}^C\rightarrow\mathfrak{P}^C,\gamma_1(P_C+(M+N))=\overline{P_C}+(\bar{M}+\bar{N}), \quad P_C+(M+N)\in\mathfrak{P}^C.$
0902.0431_E00974	162	■ 0.999	$\left\{\begin{array}{l}\Phi=[\Phi_1,\Phi_2]+P_1\times Q_2-P_2\times Q_1\\P=\Phi_1P_2-\Phi_2P_1+r_1P_2-r_2P_1+s_1Q_2-s_2Q_1\\Q=\Phi_1Q_2-\Phi_2Q_1-r_1Q_2+r_2Q_1+t_1P_2-t_2P_1\\r=-\frac{1}{8}\{P_1,Q_2\}+\frac{1}{8}\{P_2,Q_1\}+s_1t_2-s_2t_1\\s=\frac{1}{4}\{P_1,P_2\}+2r_1s_2-2r_2s_1\\t=-\frac{1}{4}\{Q_1,Q_2\}-2r_1t_2+2r_2t_1\end{array}\right.$	$\left\{\begin{array}{l}\Phi=[\Phi_1,\Phi_2]+P_1\times Q_2-P_2\times Q_1\\P=\Phi_1P_2-\Phi_2P_1+r_1P_2-r_2P_1+s_1Q_2-s_2Q_1\\Q=\Phi_1Q_2-\Phi_2Q_1-r_1Q_2+r_2Q_1+t_1P_2-t_2P_1\\r=-\frac{1}{8}\{P_1,Q_2\}+\frac{1}{8}\{P_2,Q_1\}+s_1t_2-s_2t_1\\s=\frac{1}{4}\{P_1,P_2\}+2r_1s_2-2r_2s_1\\t=-\frac{1}{4}\{Q_1,Q_2\}-2r_1t_2+2r_2t_1\end{array}\right.$	$\left\{\begin{array}{l}\Phi=[\Phi_1,\Phi_2]+P_1\times Q_2-P_2\times Q_1\\P=\Phi_1P_2-\Phi_2P_1+r_1P_2-r_2P_1+s_1Q_2-s_2Q_1\\Q=\Phi_1Q_2-\Phi_2Q_1-r_1Q_2+r_2Q_1+t_1P_2-t_2P_1\\r=-\frac{1}{8}\{P_1,Q_2\}+\frac{1}{8}\{P_2,Q_1\}+s_1t_2-s_2t_1\\s=\frac{1}{4}\{P_1,P_2\}+2r_1s_2-2r_2s_1\\t=-\frac{1}{4}\{Q_1,Q_2\}-2r_1t_2+2r_2t_1\end{array}\right.$
0902.0431_E00987	164	■ 0.999	$(R_1,R_2)_8=(\Phi_1,\Phi_2)_7-\{Q_1,P_2\}+\{P_1,Q_2\}-8r_1r_2-4t_1s_2-s_1t_2,$	$(R_1,R_2)_8=(\Phi_1,\Phi_2)_7-\{Q_1,P_2\}+\{P_1,Q_2\}-8r_1r_2-4t_1s_2-s_1t_2,$	$(R_1,R_2)_8=(\Phi_1,\Phi_2)_7-\{Q_1,P_2\}+\{P_1,Q_2\}-8r_1r_2-4t_1s_2-s_1t_2,$
0902.0431_E01054	179	■ 0.999	$\beta(P\times Q)\beta^{-1}=\beta P\times\beta Q,$	$\beta(P\times Q)\beta^{-1}=\beta P\times\beta Q,$	$\beta(P\times Q)\beta^{-1}=\beta P\times\beta Q,$
0902.0431_E01099	186	■ 0.999	$\mathrm{Spin}(16), \quad S_0(16), \quad S_s(16), \quad S_0(16)/\boldsymbol{Z}_2.$	$\mathrm{Spin}(16), \quad SO(16), \quad Ss(16), \quad SO(16)/\boldsymbol{Z}_2.$	$Spin(16), \quad SO(16), \quad Ss(16), \quad SO(16)/\boldsymbol{Z}_2.$
0902.0431_E01136	192	■ 0.999	$\varphi(A,\alpha)=\varphi_1(A)\varphi_2(\alpha).$	$\varphi(A,\alpha)=\varphi_1(A)\varphi_2(\alpha).$	$\varphi(A,\alpha)=\varphi_1(A)\varphi_2(\alpha).$

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0902.0431_E01216	204	■ 0.999	$\begin{aligned} E_{8(8)} &= \left\{ \alpha \in \operatorname{Iso}_R(\mathfrak{e}_{8(8)}) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \right\}, \\ E_{8(-24)} &= \left\{ \alpha \in \operatorname{Iso}_R(\mathfrak{e}_{8(-24)}) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \right\}. \end{aligned}$	$E_{8(8)} = \left\{ \alpha \in \operatorname{Iso}_R(\mathfrak{e}_{8(8)}) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \right\},$ $E_{8(-24)} = \left\{ \alpha \in \operatorname{Iso}_R(\mathfrak{e}_{8(-24)}) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \right\}.$	$E_{8(8)} = \{ \alpha \in \operatorname{Iso}_R(\mathfrak{e}_{8(8)}) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \},$ $E_{8(-24)} = \{ \alpha \in \operatorname{Iso}_R(\mathfrak{e}_{8(-24)}) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \}.$
0902.0431_E01218	204	■ 0.999	$\begin{aligned} E_{8(8)} &\simeq Ss(16) \times \boldsymbol{R}^{128}, \\ E_{8(-24)} &\simeq (SU(2) \times E_7) / \boldsymbol{Z}_2 \times \boldsymbol{R}^{112}. \end{aligned}$	$E_{8(8)} \simeq Ss(16) \times \boldsymbol{R}^{128},$ $E_{8(-24)} \simeq (SU(2) \times E_7) / \boldsymbol{Z}_2 \times \boldsymbol{R}^{112}.$	$E_{8(8)} \simeq Ss(16) \times \boldsymbol{R}^{128},$ $E_{8(-24)} \simeq (SU(2) \times E_7) / \boldsymbol{Z}_2 \times \boldsymbol{R}^{112}.$
0902.0431_E00092	19	■ 0.999	$B_2(H_1, H_1) = \operatorname{tr}(\operatorname{ad} H_1)^2 = (-4) \times 2 + (-1) \times 8 = -16.$	$B_2(H_1, H_1) = \operatorname{tr}(\operatorname{ad} H_1)^2 = (-4) \times 2 + (-1) \times 8 = -16.$	$B_2(H_1, H_1) = \operatorname{tr}(\operatorname{ad} H_1)^2 = (-4) \times 2 + (-1) \times 8 = -16.$
0902.0431_E00165	29	■ 0.999	$\begin{array}{ll} \left(\alpha_2 x \right) \left(\alpha_3 y \right) = \overline{\alpha_1(\overline{xy})}, & x, y \in \mathfrak{C}, \\ \left(\alpha_3 x \right) \left(\alpha_1 y \right) = \overline{\alpha_2(\overline{xy})}, & x, y \in \mathfrak{C}. \end{array}$	$\begin{array}{ll} (\alpha_2 x) (\alpha_3 y) = \overline{\alpha_1(\overline{xy})}, & x, y \in \mathfrak{C}, \\ (\alpha_3 x) (\alpha_1 y) = \overline{\alpha_2(\overline{xy})}, & x, y \in \mathfrak{C}. \end{array}$	$\begin{array}{ll} (\alpha_2 x) (\alpha_3 y) = \overline{\alpha_1(\overline{xy})}, & x, y \in \mathfrak{C}, \\ (\alpha_3 x) (\alpha_1 y) = \overline{\alpha_2(\overline{xy})}, & x, y \in \mathfrak{C}. \end{array}$
0902.0431_E00450	74	■ 0.999	$\delta + \widetilde{T} = 0, \quad \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C \quad \text{implies} \quad \delta = 0, T = 0.$	$\delta + \widetilde{T} = 0, \quad \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C \quad \text{implies} \quad \delta = 0, T = 0.$	$\delta + \widetilde{T} = 0, \quad \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C \quad \text{implies} \quad \delta = 0, T = 0.$
0902.0431_E00541	91	■ 0.999	$\begin{aligned} &\left(\mathfrak{J}^C \right)_\sigma = \{ X \in \mathfrak{J}^C \mid \sigma X = X \} \\ &= \{ X \in \mathfrak{J}^C \mid 4E_1 \times (E_1 \times X) = X \} \oplus \mathfrak{E}_1^C, \\ &\left(\mathfrak{J}^C \right)_{-\sigma} = \{ X \in \mathfrak{J}^C \mid \sigma X = -X \} \\ &= \{ X \in \mathfrak{J}^C \mid E_1 \times X = 0, \langle E_1, X \rangle = 0 \}, \end{aligned}$	$\begin{aligned} (\mathfrak{J}^C)_\sigma &= \{ X \in \mathfrak{J}^C \mid \sigma X = X \} \\ &= \{ X \in \mathfrak{J}^C \mid 4E_1 \times (E_1 \times X) = X \} \oplus \mathfrak{E}_1^C, \\ (\mathfrak{J}^C)_{-\sigma} &= \{ X \in \mathfrak{J}^C \mid \sigma X = -X \} \\ &= \{ X \in \mathfrak{J}^C \mid E_1 \times X = 0, \langle E_1, X \rangle = 0 \}, \end{aligned}$	$\begin{aligned} (\mathfrak{J}^C)_\sigma &= \{ X \in \mathfrak{J}^C \mid \sigma X = X \} \\ &= \{ X \in \mathfrak{J}^C \mid 4E_1 \times (E_1 \times X) = X \} \oplus \mathfrak{E}_1^C, \\ (\mathfrak{J}^C)_{-\sigma} &= \{ X \in \mathfrak{J}^C \mid \sigma X = -X \} \\ &= \{ X \in \mathfrak{J}^C \mid E_1 \times X = 0, \langle E_1, X \rangle = 0 \}, \end{aligned}$
0902.0431_E00201	34	■ 0.999	$\left(\alpha_1 x \right) \left(\alpha_2 y \right) = \overline{\alpha_3(\overline{xy})}. \quad x, y \in \mathfrak{C}.$	$(\alpha_1 x) (\alpha_2 y) = \overline{\alpha_3(\overline{xy})}. \quad x, y \in \mathfrak{C}.$	$(\alpha_1 x) (\alpha_2 y) = \overline{\alpha_3(\overline{xy})}. \quad x, y \in \mathfrak{C}.$
0902.0431_E00002	5	■ 0.999	$\gamma, \quad \sigma, \quad \iota, \quad \kappa, \quad \lambda, \quad \mu, \quad \tau, \quad v, \quad \chi, \quad g, \quad w.$	$\gamma, \quad \sigma, \quad \iota, \quad \kappa, \quad \lambda, \quad \mu, \quad \tau, \quad v, \quad \chi, \quad g, \quad w.$	$\gamma, \quad \sigma, \quad \iota, \quad \kappa, \quad \lambda, \quad \mu, \quad \tau, \quad v, \quad \chi, \quad g, \quad w.$
0902.0431_E00004	6	■ 0.999	$e_i^2 = -1, \quad i \neq 0, \quad e_i e_j = -e_j e_i, \quad i \neq j, i \neq 0, j \neq 0,$	$e_i^2 = -1, \quad i \neq 0, \quad e_i e_j = -e_j e_i, \quad i \neq j, i \neq 0, j \neq 0,$	$e_i^2 = -1, \quad i \neq 0, \quad e_i e_j = -e_j e_i, \quad i \neq j, i \neq 0, j \neq 0,$

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0902.0431_E00068	15	■0.999	$\begin{aligned} (a+\boldsymbol{m})(b+\boldsymbol{n}) &= (a\,b-\langle\boldsymbol{m},\boldsymbol{n}\rangle) + (a\boldsymbol{n}+\bar{\boldsymbol{b}}\boldsymbol{m}-\overline{\boldsymbol{m}\times\boldsymbol{n}}), \\ (a+\boldsymbol{m},b+\boldsymbol{n}) &= (a,b) + (\boldsymbol{m},\boldsymbol{n}), \\ \overline{a+\boldsymbol{m}} &= \bar{a}-\boldsymbol{m}, \end{aligned}$	$(a+\boldsymbol{m})(b+\boldsymbol{n}) = (ab-\langle\boldsymbol{m},\boldsymbol{n}\rangle) + (a\boldsymbol{n}+\bar{\boldsymbol{b}}\boldsymbol{m}-\overline{\boldsymbol{m}\times\boldsymbol{n}}),$ $(a+\boldsymbol{m},b+\boldsymbol{n}) = (a,b) + (\boldsymbol{m},\boldsymbol{n}),$ $\overline{a+\boldsymbol{m}} = \bar{a}-\boldsymbol{m},$	$(a+\boldsymbol{m})(b+\boldsymbol{n}) = (ab-\langle\boldsymbol{m},\boldsymbol{n}\rangle) + (a\boldsymbol{n}+\bar{\boldsymbol{b}}\boldsymbol{m}-\overline{\boldsymbol{m}\times\boldsymbol{n}}),$ $(a+\boldsymbol{m},b+\boldsymbol{n}) = (a,b) + (\boldsymbol{m},\boldsymbol{n}),$ $\overline{a+\boldsymbol{m}} = \bar{a}-\boldsymbol{m},$
0902.0431_E00139	26	■0.999	$G_2^C = \left\{ \alpha \in \mathrm{Iso}_C(\mathfrak{C}^C) \mid \alpha(xy) = (\alpha x)(\alpha y) \right\}.$	$G_2^C = \left\{ \alpha \in \mathrm{Iso}_C(\mathfrak{C}^C) \mid \alpha(xy) = (\alpha x)(\alpha y) \right\}.$	$G_2^C = \left\{ \alpha \in \mathrm{Iso}_C(\mathfrak{C}^C) \mid \alpha(xy) = (\alpha x)(\alpha y) \right\}.$
0902.0431_E00182 (ii)	31	■0.999	$(\alpha x)(\widetilde{\alpha y}) = \widetilde{\alpha(xy)}, \quad x, y \in \mathfrak{C}.$	$(\alpha x)(\widetilde{\alpha y}) = \widetilde{\alpha(xy)}, \quad x, y \in \mathfrak{C}.$	$(\alpha x)(\widetilde{\alpha y}) = \widetilde{\alpha(xy)}, \quad x, y \in \mathfrak{C}.$
0902.0431_E00187 (i)	32	■0.999	$(\alpha x)((\alpha y)b_0) = (\alpha(xy))b_0, \quad x, y \in \mathfrak{C}.$	$(\alpha x)((\alpha y)b_0) = (\alpha(xy))b_0, \quad x, y \in \mathfrak{C}.$	$(\alpha x)((\alpha y)b_0) = (\alpha(xy))b_0, \quad x, y \in \mathfrak{C}.$
0902.0431_E00209	35	■0.999	$\mathrm{Spin}(8)/\{(1,1,1),(-1,-1,1)\} \cong \mathrm{Spin}(8)/\{(1,1,1),(1,-1,-1)\},$	$\mathrm{Spin}(8)/\{(1,1,1),(-1,-1,1)\} \cong \mathrm{Spin}(8)/\{(1,1,1),(1,-1,-1)\},$	$\mathrm{Spin}(8)/\{(1,1,1),(-1,-1,1)\} \cong \mathrm{Spin}(8)/\{(1,1,1),(1,-1,-1)\},$
0902.0431_E00211	35	■0.999	$G_2 = \{ \alpha \in \mathrm{Spin}(8) \mid \nu \alpha = \alpha \}.$	$G_2 = \{ \alpha \in \mathrm{Spin}(8) \mid \nu \alpha = \alpha \}.$	$G_2 = \{ \alpha \in \mathrm{Spin}(8) \mid \nu \alpha = \alpha \}.$
0902.0431_E00225	37	■0.999	$\begin{aligned} &\left\{ \begin{array}{l} E_- \circ E_+ = E_0 \\ E_+ \circ E_-(x) = 0 \\ F_-(x) \circ F_+(y) = (x,y)(E_{i+1} + E_{i+2}), \end{array} \right\} \quad \left\{ \begin{array}{l} E_i \circ E_j = 0, \quad i \neq j \\ E_i \circ F_j(x) = \frac{1}{2}F_j(x), \quad i \neq j \\ F_i(x) \circ F_{i+1}(y) = \frac{1}{2}F_{i+2}(\overline{xy}), \end{array} \right. \\ &\left\{ \begin{array}{l} E_- \circ E_+ = E_0 \\ E_+ \circ E_-(x) = 0 \\ F_-(x) \circ F_+(y) = (x,y)(E_{i+1} + E_{i+2}), \end{array} \right\} \quad \left\{ \begin{array}{l} E_i \circ E_j = 0, \quad i \neq j \\ E_i \circ F_j(x) = \frac{1}{2}F_j(x), \quad i \neq j \\ F_i(x) \circ F_{i+1}(y) = \frac{1}{2}F_{i+2}(\overline{xy}), \end{array} \right. \end{aligned}$	$\left\{ \begin{array}{l} E_i \circ E_i = E_i \\ E_i \circ F_i(x) = 0 \\ F_i(x) \circ F_i(y) = (x,y)(E_{i+1} + E_{i+2}), \end{array} \right\} \quad \left\{ \begin{array}{l} E_i \circ E_j = 0, \quad i \neq j \\ E_i \circ F_j(x) = \frac{1}{2}F_j(x), \quad i \neq j \\ F_i(x) \circ F_{i+1}(y) = \frac{1}{2}F_{i+2}(\overline{xy}), \end{array} \right.$	$\left\{ \begin{array}{l} E_i \circ E_i = E_i \\ E_i \circ F_i(x) = 0 \\ F_i(x) \circ F_i(y) = (x,y)(E_{i+1} + E_{i+2}), \end{array} \right\} \quad \left\{ \begin{array}{l} E_i \circ E_j = 0, \quad i \neq j \\ E_i \circ F_j(x) = \frac{1}{2}F_j(x), \quad i \neq j \\ F_i(x) \circ F_{i+1}(y) = \frac{1}{2}F_{i+2}(\overline{xy}), \end{array} \right.$
0902.0431_E00240 (3)	40	■0.999	$\det(\alpha X) = \det X, \quad (\alpha X, \alpha Y) = (X, Y).$	$\det(\alpha X) = \det X, \quad (\alpha X, \alpha Y) = (X, Y).$	$\det(\alpha X) = \det X, \quad (\alpha X, \alpha Y) = (X, Y).$
0902.0431_E00259	42	■0.999	$[X, Y] = XY - YX.$	$[X, Y] = XY - YX.$	$[X, Y] = XY - YX.$
0902.0431_E00280	45	■0.999	$\mathfrak{J}_i = \{ F_i(x) \mid x \in \mathfrak{C} \} = \{ X \in \mathfrak{J} \mid 2E_{i+1} \circ X = 2E_{i+2} \circ X = X \}.$	$\mathfrak{J}_i = \{ F_i(x) \mid x \in \mathfrak{C} \} = \{ X \in \mathfrak{J} \mid 2E_{i+1} \circ X = 2E_{i+2} \circ X = X \}.$	$\mathfrak{J}_i = \{ F_i(x) \mid x \in \mathfrak{C} \} = \{ X \in \mathfrak{J} \mid 2E_{i+1} \circ X = 2E_{i+2} \circ X = X \}.$
0902.0431_E00286	45	■0.999	$2\delta E_i \circ E_i = \delta E_i, \quad \delta E_i \circ E_j + E_i \circ \delta E_j = 0.$	$2\delta E_i \circ E_i = \delta E_i, \quad \delta E_i \circ E_j + E_i \circ \delta E_j = 0.$	$2\delta E_i \circ E_i = \delta E_i, \quad \delta E_i \circ E_j + E_i \circ \delta E_j = 0.$
0902.0431_E00328	52	■0.999	$\pm(\lambda_k - \lambda_l), \quad \pm(\lambda_k + \lambda_l), \quad 0 \leq k < l \leq 3.$	$\pm(\lambda_k - \lambda_l), \quad \pm(\lambda_k + \lambda_l), \quad 0 \leq k < l \leq 3.$	$\pm(\lambda_k - \lambda_l), \quad \pm(\lambda_k + \lambda_l), \quad 0 \leq k < l \leq 3.$
0902.0431_E00346	56	■0.999	$\begin{aligned} &H_{\alpha_1} = \frac{1}{18}(H_0 - H_1), \quad H_{\alpha_2} = \frac{1}{18}(H_1 - H_2), \\ &H_{\alpha_3} = \frac{1}{18}H_2, \quad H_{\alpha_4} = \frac{1}{36}(-H_0 - H_1 - H_2 + H_3). \end{aligned}$	$H_{\alpha_1} = \frac{1}{18}(H_0 - H_1), \quad H_{\alpha_2} = \frac{1}{18}(H_1 - H_2),$ $H_{\alpha_3} = \frac{1}{18}H_2, \quad H_{\alpha_4} = \frac{1}{36}(-H_0 - H_1 - H_2 + H_3).$	$H_{\alpha_1} = \frac{1}{18}(H_0 - H_1), \quad H_{\alpha_2} = \frac{1}{18}(H_1 - H_2),$ $H_{\alpha_3} = \frac{1}{18}H_2, \quad H_{\alpha_4} = \frac{1}{36}(-H_0 - H_1 - H_2 + H_3).$
0902.0431_E00351	57	■0.999	$\mathfrak{J}_i = \{ F_i(x) \mid x \in \mathfrak{C} \} = \{ X \in \mathfrak{J} \mid 2E_{i+1} \circ X = 2E_{i+2} \circ X = X \}, \quad i = 1, 2, 3.$	$\mathfrak{J}_i = \{ F_i(x) \mid x \in \mathfrak{C} \} = \{ X \in \mathfrak{J} \mid 2E_{i+1} \circ X = 2E_{i+2} \circ X = X \}, \quad i = 1, 2, 3.$	$\mathfrak{J}_i = \{ F_i(x) \mid x \in \mathfrak{C} \} = \{ X \in \mathfrak{J} \mid 2E_{i+1} \circ X = 2E_{i+2} \circ X = X \}, \quad i = 1, 2, 3.$

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0902.0431_E00362	58	■ 0.999	$\begin{array}{l} \backslash begin{array}{lll} \backslash alpha E=E, \& \backslash alpha E_{\{1\}}=E_{\{1\}}, \\ \& \backslash alpha \backslash left(E_{\{2\}}-E_{\{3\}}\backslash right)=A, \backslash \backslash \backslash alpha F_{\{1\}} \\ \backslash left(e_{\{i\}}\backslash right)=X_{\{i\}}, \& \backslash alpha F_{\{2\}}\backslash left(e_{\{i\}} \\ \backslash right)=Y_{\{i\}}, \& \backslash alpha F_{\{3\}}\backslash left(e_{\{i\}}\backslash right)=Z_{\{i\}}, \\ \backslash quad i=0,1, \backslash cdots, 7 \backslash end{array} \end{array}$	$\begin{array}{l} \alpha E = E, \qquad \alpha E_1 = E_1, \qquad \alpha (E_2 - E_3) = A, \\ \alpha F_1(e_i) = X_i, \quad \alpha F_2(e_i) = Y_i, \quad \alpha F_3(e_i) = Z_i, \quad i = 0, 1, \cdots, 7 \end{array}$	$\begin{array}{l} \alpha E = E, \qquad \alpha E_1 = E_1, \qquad \alpha (E_2 - E_3) = A, \\ \alpha F_1(e_i) = X_i, \quad \alpha F_2(e_i) = Y_i, \quad \alpha F_3(e_i) = Z_i, \quad i = 0, 1, \cdots, 7 \end{array}$
0902.0431_E00426	70	■ 0.999	$\backslash varphi(P, A)((X+M) \backslash times(Y+N))=\backslash varphi(P, A)(X+M) \backslash times \backslash varphi(P, A)(Y+N),$	$\varphi(P,A)((X+M)\times(Y+N))=\varphi(P,A)(X+M)\times\varphi(P,A)(Y+N),$	$\varphi(P,A)((X+M)\times(Y+N))=\varphi(P,A)(X+M)\times\varphi(P,A)(Y+N),$
0902.0431_E00438	72	■ 0.999	$\begin{array}{l} \backslash begin{aligned} \backslash mathfrak{J}\backslash left(3, \backslash mathfrak{C}\wedge\{ \\ \backslash prime\}\backslash right) \& \backslash cong\backslash left\{X \in \backslash mathfrak{J}\}\backslash left(3, \\ \backslash mathfrak{C}\wedge\{C\}\backslash right) \backslash mid \backslash tau \backslash gamma X=X\backslash right \\ \backslash \}=\backslash left(\backslash mathfrak{J}\backslash left(3, \backslash mathfrak{C}\wedge\{C\}\backslash right) \\ \backslash right)_{\backslash tau \backslash gamma}, \backslash \backslash \backslash mathfrak{J}(1,2, \backslash mathfrak{C}) \\ \& \backslash cong\backslash left\{X \in \backslash mathfrak{J}\}\backslash left(3, \backslash mathfrak{C}\wedge\{C \\ \backslash right) \backslash mid \backslash tau \backslash sigma X=X\backslash right\backslash \}=\backslash left(\backslash mathfrak{J} \\ \backslash left(3, \backslash mathfrak{C}\wedge\{C\}\backslash right)\backslash right)_{\backslash tau \backslash sigma} \\ \backslash end{aligned} \end{array}$	$\begin{array}{l} \mathfrak{J}(3, \mathfrak{C}') \cong \{X \in \mathfrak{J}(3, \mathfrak{C}^C) \mid \tau\gamma X = X\} = (\mathfrak{J}(3, \mathfrak{C}^C))_{\tau\gamma}, \\ \mathfrak{J}(1, 2, \mathfrak{C}) \cong \{X \in \mathfrak{J}(3, \mathfrak{C}^C) \mid \tau\sigma X = X\} = (\mathfrak{J}(3, \mathfrak{C}^C))_{\tau\sigma} \end{array}$	$\begin{array}{l} \mathfrak{J}(3, \mathfrak{C}') \cong \{X \in \mathfrak{J}(3, \mathfrak{C}^C) \mid \tau\gamma X = X\} = (\mathfrak{J}(3, \mathfrak{C}^C))_{\tau\gamma}, \\ \mathfrak{J}(1, 2, \mathfrak{C}) \cong \{X \in \mathfrak{J}(3, \mathfrak{C}^C) \mid \tau\sigma X = X\} = (\mathfrak{J}(3, \mathfrak{C}^C))_{\tau\sigma} \end{array}$
0902.0431_E00483	79	■ 0.999	$\backslash left[\backslash phi, \backslash left[\backslash phi, \backslash widetilde{A}_{\{1\}}\backslash left(e_{\{i\}} \\ \backslash right)\backslash right]\backslash right]=\backslash left[\backslash phi, -\backslash frac{1}{\{2\}} \\ \backslash widetilde{F}_{\{1\}}\backslash left(e_{\{i\}}\backslash right)\backslash right]=\backslash frac{1}{\{4\}} \\ \backslash widetilde{A}_{\{1\}}\backslash left(e_{\{i\}}\backslash right),$	$\left[\phi, \left[\phi, \widetilde{A}_1(e_i)\right]\right] = \left[\phi, -\frac{1}{2}\widetilde{F}_1(e_i)\right] = \frac{1}{4}\widetilde{A}_1(e_i),$	$[\phi, [\phi, \widetilde{A}_1(e_i)]] = \left[\phi, -\frac{1}{2}\widetilde{F}_1(e_i)\right] = \frac{1}{4}\widetilde{A}_1(e_i),$
0902.0431_E00487	80	■ 0.999	$\backslash operatorname{tr}(\backslash phi \backslash phi)=1 \backslash times 2+\backslash frac{1}{\{4\}} \backslash times 8 \backslash times 2=6 \; .$	$\mathrm{tr}(\phi\phi)=1\times2+\frac{1}{4}\times8\times2=6.$	$\mathrm{tr}(\phi\phi)=1\times2+\frac{1}{4}\times8\times2=6.$
0902.0431_E00597	99	■ 0.999	$(M+\backslash boldsymbol{a}) \backslash times(N+\backslash boldsymbol{b})=\backslash left(M \backslash times N-\backslash frac{1}{\{2\}}\backslash left(\backslash boldsymbol{a}\wedge\{*\} \backslash boldsymbol{b}+\backslash boldsymbol{b}\wedge\{*\} \backslash boldsymbol{a}\backslash right) \backslash right)-\backslash frac{1}{\{2\}}(\backslash boldsymbol{a} \backslash N+\backslash boldsymbol{b} \backslash M) \; .$	$(M+\boldsymbol{a})\times(N+\boldsymbol{b})=\left(M\times N-\frac{1}{2}(\boldsymbol{a}^*\boldsymbol{b}+\boldsymbol{b}^*\boldsymbol{a})\right)-\frac{1}{2}(\boldsymbol{a}N+\boldsymbol{b}M).$	$(M+\boldsymbol{a})\times(N+\boldsymbol{b})=\left(M\times N-\frac{1}{2}(\boldsymbol{a}^*\boldsymbol{b}+\boldsymbol{b}^*\boldsymbol{a})\right)-\frac{1}{2}(\boldsymbol{a}N+\boldsymbol{b}M).$
0902.0431_E00604	101	■ 0.999	$\backslash mathfrak{J}\wedge\{C\}=\backslash left(\backslash mathfrak{J}\wedge\{C\}\backslash right)_{\backslash \backslash tau \backslash gamma} \backslash oplus \backslash left(\backslash mathfrak{J}\wedge\{C\}\backslash right)_{\backslash \backslash \backslash tau \backslash gamma}=\backslash left(\backslash mathfrak{J}\wedge\{C\}\backslash right)_{\backslash \backslash \backslash tau \backslash gamma} \backslash oplus i \backslash left(\backslash mathfrak{J}\wedge\{C\}\backslash right)_{\backslash \backslash \backslash tau \backslash gamma} \; .$	$\mathfrak{J}^C=(\mathfrak{J}^C)_{\tau\gamma}\oplus(\mathfrak{J}^C)_{-\tau\gamma}=(\mathfrak{J}^C)_{\tau\gamma}\oplus i(\mathfrak{J}^C)_{\tau\gamma}.$	$\mathfrak{J}^C=(\mathfrak{J}^C)_{\tau\gamma}\oplus(\mathfrak{J}^C)_{-\tau\gamma}=(\mathfrak{J}^C)_{\tau\gamma}\oplus i(\mathfrak{J}^C)_{\tau\gamma}.$
0902.0431_E00668	111	■ 0.999	$\backslash mathfrak{P}\wedge\{C\}=\backslash mathfrak{J}\wedge\{C\} \backslash oplus \backslash mathfrak{J}\wedge\{C\} \backslash oplus C \backslash oplus C \; .$	$\mathfrak{P}^C=\mathfrak{J}^C\oplus\mathfrak{J}^C\oplus C\oplus C.$	$\mathfrak{P}^C=\mathfrak{J}^C\oplus\mathfrak{J}^C\oplus C\oplus C.$
0902.0431_E00672	112	■ 0.999	$\begin{array}{l} \backslash begin{aligned} \& (P \backslash times Q) P \backslash \backslash = \& \backslash Phi \backslash left(- \\ \backslash frac{1}{\{2\}}(X \vee W + Z \vee Y), -\backslash frac{1}{\{4\}}(2 Y \backslash times W - \\ \backslash xi Z - \backslash zeta X), \backslash frac{1}{\{4\}}(2 X \backslash times Z - \backslash eta W - \backslash omega \\ Y), \backslash right. \backslash \backslash \& \backslash left.\backslash frac{1}{\{8\}}((X, W) + (Z, Y) - 3(\xi \omega + \zeta \eta)) \backslash right)(X, Y, \backslash xi, \backslash eta) \backslash end{aligned} \end{array}$	$\begin{array}{l} (P\times Q)P \\ =\Phi\left(-\frac{1}{2}(X\vee W+Z\vee Y),-\frac{1}{4}(2Y\times W-\xi Z-\zeta X),\frac{1}{4}(2X\times Z-\eta W-\omega Y),\right. \\ \left.\frac{1}{8}((X,W)+(Z,Y)-3(\xi\omega+\zeta\eta))\right)(X,Y,\xi,\eta) \end{array}$	$\begin{array}{l} (P\times Q)P \\ =\Phi\left(-\frac{1}{2}(X\vee W+Z\vee Y),-\frac{1}{4}(2Y\times W-\xi Z-\zeta X),\frac{1}{4}(2X\times Z-\eta W-\omega Y),\right. \\ \left.\frac{1}{8}((X,W)+(Z,Y)-3(\xi\omega+\zeta\eta))\right)(X,Y,\xi,\eta) \end{array}$
0902.0431_E00715	118	■ 0.999	$\backslash mathfrak{e}_{\{7\}}=\backslash left\{\backslash Phi(\backslash phi, A, -\backslash tau A, \backslash nu) \backslash mid \backslash phi \backslash in \backslash mathfrak{e}_{\{6\}}, A \backslash in \backslash mathfrak{J}\wedge\{C\}, \backslash nu \backslash in i \backslash boldsymbol{R}\}\backslash right\backslash ,$	$\mathfrak{e}_7=\left\{\Phi(\phi,A,-\tau A,\nu)\mid\phi\in\mathfrak{e}_6,A\in\mathfrak{J}^C,\nu\in i\boldsymbol{R}\right\},$	$\mathfrak{e}_7=\left\{\Phi(\phi,A,-\tau A,\nu)\mid\phi\in\mathfrak{e}_6,A\in\mathfrak{J}^C,\nu\in i\boldsymbol{R}\right\},$

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0902.0431_E00243	40	■ 0.999	$\begin{aligned} & 3 \operatorname{det}(\alpha X) = \operatorname{tr}(\alpha X, \alpha X, \alpha X) - \frac{3}{2} \operatorname{tr}(\alpha X)(\alpha X, \alpha X) + \frac{1}{2} \operatorname{tr}(\alpha X)^3 \\ & = \operatorname{tr}(X, X, X) - \frac{3}{2} \operatorname{tr}(X)(X, X) + \frac{1}{2} \operatorname{tr}(X)^3 = 3 \operatorname{det} X. \end{aligned}$	$3 \operatorname{det}(\alpha X) = \operatorname{tr}(\alpha X, \alpha X, \alpha X) - \frac{3}{2} \operatorname{tr}(\alpha X)(\alpha X, \alpha X) + \frac{1}{2} \operatorname{tr}(\alpha X)^3$ $= \operatorname{tr}(X, X, X) - \frac{3}{2} \operatorname{tr}(X)(X, X) + \frac{1}{2} \operatorname{tr}(X)^3 = 3 \operatorname{det} X.$	$3 \operatorname{det}(\alpha X) = \operatorname{tr}(\alpha X, \alpha X, \alpha X) - \frac{3}{2} \operatorname{tr}(\alpha X)(\alpha X, \alpha X) + \frac{1}{2} \operatorname{tr}(\alpha X)^3$ $= \operatorname{tr}(X, X, X) - \frac{3}{2} \operatorname{tr}(X)(X, X) + \frac{1}{2} \operatorname{tr}(X)^3 = 3 \operatorname{det} X.$
0902.0431_E00706	116	■ 0.999	$\begin{aligned} \Phi &= \left(\begin{array}{cccc} \phi - \frac{1}{3}\nu & 2B & 0 & A \\ 2A & \phi' + \frac{1}{3}\nu & B & 0 \\ 0 & A & \nu & 0 \\ B & 0 & 0 & -\nu \end{array} \right) \\ &= \Phi(\phi, A, B, \nu), \quad \phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C. \end{aligned}$	$\Phi = \left(\begin{array}{cccc} \phi - \frac{1}{3}\nu & 2B & 0 & A \\ 2A & \phi' + \frac{1}{3}\nu & B & 0 \\ 0 & A & \nu & 0 \\ B & 0 & 0 & -\nu \end{array} \right)$ $= \Phi(\phi, A, B, \nu), \quad \phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C.$	$\Phi = \left(\begin{array}{cccc} \phi - \frac{1}{3}\nu & 2B & 0 & A \\ 2A & \phi' + \frac{1}{3}\nu & B & 0 \\ 0 & A & \nu & 0 \\ B & 0 & 0 & -\nu \end{array} \right)$ $= \Phi(\phi, A, B, \nu), \quad \phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C.$
0902.0431_E01071	182	■ 0.999	$\begin{aligned} B &= l(D') + l(S)I, & D' \in \mathfrak{u}(8), S \in \mathfrak{S}(8, C) \\ &= l(D) + l(S)I + l(icE), & D \in \mathfrak{su}(8), S \in \mathfrak{S}(8, C), c \in \mathbf{R}. \end{aligned}$	$B = l(D') + l(S)I, \quad D' \in \mathfrak{u}(8), S \in \mathfrak{S}(8, C)$ $= l(D) + l(S)I + l(icE), \quad D \in \mathfrak{su}(8), S \in \mathfrak{S}(8, C), c \in \mathbf{R}.$	$B = l(D') + l(S)I, \quad D' \in \mathfrak{u}(8), S \in \mathfrak{S}(8, C)$ $= l(D) + l(S)I + l(icE), \quad D \in \mathfrak{su}(8), S \in \mathfrak{S}(8, C), c \in \mathbf{R}.$
0902.0431_E01110	188	■ 0.999	$\begin{aligned} & \omega_1 \quad 2\omega_1 \quad \omega_2 \quad 2\omega_2 \quad \omega_3 \quad \omega_4 \quad \omega_5 \quad \omega_6 \quad \omega_7 \quad \omega_8 \quad \cdots \\ & 16 \quad 135 \quad 120 \quad 5304 \quad 560 \quad 1820 \quad 4368 \quad 8008 \quad 128 \quad 128 \quad \cdots \end{aligned}$	$\begin{aligned} & \omega_1 \quad 2\omega_1 \quad \omega_2 \quad 2\omega_2 \quad \omega_3 \quad \omega_4 \quad \omega_5 \quad \omega_6 \quad \omega_7 \quad \omega_8 \quad \cdots \\ & 16 \quad 135 \quad 120 \quad 5304 \quad 560 \quad 1820 \quad 4368 \quad 8008 \quad 128 \quad 128 \quad \cdots \end{aligned}$	$\begin{aligned} & \omega_1 \quad 2\omega_1 \quad \omega_2 \quad 2\omega_2 \quad \omega_3 \quad \omega_4 \quad \omega_5 \quad \omega_6 \quad \omega_7 \quad \omega_8 \quad \cdots \\ & 16 \quad 135 \quad 120 \quad 5304 \quad 560 \quad 1820 \quad 4368 \quad 8008 \quad 128 \quad 128 \quad \cdots \end{aligned}$
0902.0431_E00006	7	■ 1.000	$\{x, y, a\} = \{y, a, x\} = \{a, x, y\} = -\{y, x, a\} = -\{x, a, y\} = -\{a, y, x\}.$	$\{x, y, a\} = \{y, a, x\} = \{a, x, y\} = -\{y, x, a\} = -\{x, a, y\} = -\{a, y, x\}.$	$\{x, y, a\} = \{y, a, x\} = \{a, x, y\} = -\{y, x, a\} = -\{x, a, y\} = -\{a, y, x\}.$
0902.0431_E00011	8	■ 1.000	$\alpha 1 = 1, \quad \overline{\alpha e_i} = -\alpha e_i, \quad i = 1, 2, \dots, 7.$	$\alpha 1 = 1, \quad \overline{\alpha e_i} = -\alpha e_i, \quad i = 1, 2, \dots, 7.$	$\alpha 1 = 1, \quad \overline{\alpha e_i} = -\alpha e_i, \quad i = 1, 2, \dots, 7.$
0902.0431_E00019	9	■ 1.000	$\begin{aligned} (\kappa D)x &= \overline{Dx}, \quad x \in \mathfrak{C}, \\ \pi(G_{ij}) &= F_{ij}, \quad i, j = 0, 1, \dots, 7, i \neq j, \\ \nu &= \pi\kappa. \end{aligned}$	$\begin{aligned} (\kappa D)x &= \overline{Dx}, \quad x \in \mathfrak{C}, \\ \pi(G_{ij}) &= F_{ij}, \quad i, j = 0, 1, \dots, 7, i \neq j, \\ \nu &= \pi\kappa. \end{aligned}$	$\begin{aligned} (\kappa D)x &= \overline{Dx}, \quad x \in \mathfrak{C}, \\ \pi(G_{ij}) &= F_{ij}, \quad i, j = 0, 1, \dots, 7, i \neq j, \\ \nu &= \pi\kappa. \end{aligned}$
0902.0431_E00026	10	■ 1.000	$L_a x = ax, \quad R_a x = xa, \quad T_a x = ax + xa = (L_a + R_a)x, \quad x \in \mathfrak{C}.$	$L_a x = ax, \quad R_a x = xa, \quad T_a x = ax + xa = (L_a + R_a)x, \quad x \in \mathfrak{C}.$	$L_a x = ax, \quad R_a x = xa, \quad T_a x = ax + xa = (L_a + R_a)x, \quad x \in \mathfrak{C}.$
0902.0431_E00032	10	■ 1.000	$\begin{aligned} L_{e_i} x &= e_i x = 2F_{i0} x, \quad x \in \mathfrak{C}, \\ T_{e_i} x &= e_i x + x e_i = \begin{cases} 2e_i, & x = 1 \\ -2, & x = e_i \\ 0, & x = e_j, \quad j \neq 0, i. \end{cases} \end{aligned}$	$\begin{aligned} L_{e_i} x &= e_i x = 2F_{i0} x, \quad x \in \mathfrak{C}, \\ T_{e_i} x &= e_i x + x e_i = \begin{cases} 2e_i, & x = 1 \\ -2, & x = e_i \\ 0, & x = e_j, \quad j \neq 0, i. \end{cases} \end{aligned}$	$\begin{aligned} L_{e_i} x &= e_i x = 2F_{i0} x, \quad x \in \mathfrak{C}, \\ T_{e_i} x &= e_i x + x e_i = \begin{cases} 2e_i, & x = 1 \\ -2, & x = e_i \\ 0, & x = e_j, \quad j \neq 0, i. \end{cases} \end{aligned}$

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0902.0431_E00072	15	■ 1.000	$\varphi_{*}(D)(a+\boldsymbol{m})=D\ \boldsymbol{m},\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C},$ $\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C}.$	$\varphi_{*}(D)(a+\boldsymbol{m})=D\boldsymbol{m},\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C}.$	$\varphi_{*}(D)(a+\boldsymbol{m})=D\boldsymbol{m},\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C}.$
0902.0431_E00089	18	■ 1.000	$B_2\left(D_1,D_2\right)=k\operatorname{tr}\left(D_1D_2\right).$	$B_2(D_1,D_2)=k\operatorname{tr}(D_1D_2).$	$B_2(D_1,D_2)=k\operatorname{tr}(D_1D_2).$
0902.0431_E00090	18	■ 1.000	$\begin{array}{ll} \left[H_1,\left[H_1,L_{12}\right]\right]=\left[H_1,2L_{21}\right]=-4L_{12}, & \left[H_1,\left[H_1,L_{21}\right]\right]=\left[H_1,-2L_{12}\right]=-4L_{21}, \\ \left[H_1,\left[H_1,L_{13}\right]\right]=\left[H_1,L_{31}\right]=-L_{13}, & \left[H_1,\left[H_1,L_{31}\right]\right]=\left[H_1,-L_{13}\right]=-L_{31}, \\ \left[H_1,\left[H_1,L_{23}\right]\right]=\left[H_1,-L_{32}\right]=-L_{23}, & \left[H_1,\left[H_1,L_{32}\right]\right]=\left[H_1,L_{23}\right]=-L_{32}, \end{array}$	$\begin{array}{ll} \left[H_1,\left[H_1,L_{12}\right]\right]=\left[H_1,2L_{21}\right]=-4L_{12}, & \left[H_1,\left[H_1,L_{21}\right]\right]=\left[H_1,-2L_{12}\right]=-4L_{21}, \\ \left[H_1,\left[H_1,L_{13}\right]\right]=\left[H_1,L_{31}\right]=-L_{13}, & \left[H_1,\left[H_1,L_{31}\right]\right]=\left[H_1,-L_{13}\right]=-L_{31}, \\ \left[H_1,\left[H_1,L_{23}\right]\right]=\left[H_1,-L_{32}\right]=-L_{23}, & \left[H_1,\left[H_1,L_{32}\right]\right]=\left[H_1,L_{23}\right]=-L_{32}, \end{array}$	$\begin{array}{ll} \left[H_1,\left[H_1,L_{12}\right]\right]=\left[H_1,2L_{21}\right]=-4L_{12}, & \left[H_1,\left[H_1,L_{21}\right]\right]=\left[H_1,-2L_{12}\right]=-4L_{21}, \\ \left[H_1,\left[H_1,L_{13}\right]\right]=\left[H_1,L_{31}\right]=-L_{13}, & \left[H_1,\left[H_1,L_{31}\right]\right]=\left[H_1,-L_{13}\right]=-L_{31}, \\ \left[H_1,\left[H_1,L_{23}\right]\right]=\left[H_1,-L_{32}\right]=-L_{23}, & \left[H_1,\left[H_1,L_{32}\right]\right]=\left[H_1,L_{23}\right]=-L_{32}, \end{array}$
0902.0431_E00119	23	■ 1.000	$w(a+\boldsymbol{m})=a+\omega_1\boldsymbol{m},\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C}.$	$w(a+\boldsymbol{m})=a+\omega_1\boldsymbol{m},\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C}.$	$w(a+\boldsymbol{m})=a+\omega_1\boldsymbol{m},\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C}.$
0902.0431_E00146	27	■ 1.000	$G_2\{ \}^{\mathfrak{C}}\simeq G_2\ \times\ \boldsymbol{R}^{14}.$	$G_2^{\mathfrak{C}}\simeq G_2\times\boldsymbol{R}^{14}.$	$G_2^{\mathfrak{C}}\simeq G_2\times\boldsymbol{R}^{14}.$
0902.0431_E00176	31	■ 1.000	$(x\,\overline{a})(a\,y\,a)=(x\,y)\,a,\quad x,\,y\in\mathfrak{C}.$	$(x\overline{a})(aya)=(xy)a,\quad x,y\in\mathfrak{C}.$	$(x\overline{a})(aya)=(xy)a,\quad x,y\in\mathfrak{C}.$
0902.0431_E00181 (i)	31	■ 1.000	$(\alpha x)(\widetilde{\alpha}y)=\alpha^{\prime}(x\,y),\quad x,\,y\in\mathfrak{C}.$	$(\alpha x)(\widetilde{\alpha}y)=\alpha'(xy),\quad x,y\in\mathfrak{C}.$	$(\alpha x)(\widetilde{\alpha}y)=\alpha'(xy),\quad x,y\in\mathfrak{C}.$
0902.0431_E00185	32	■ 1.000	$\begin{array}{l} \left(a_2,a_6\right)=\left(a_2b_0,a_6b_0\right)=\left(a_2b_0,a_2\left(a_4b_0\right)\right)=\left(b_0,a_4b_0\right)=\left(1,a_4\right)=0, \\ \left(a_3,a_7\right)=\left(a_3b_0,a_7b_0\right)=\left(a_2\left(a_1b_0\right),a_6\left(a_1b_0\right)\right)=\left(a_1\left(a_2b_0\right),a_1\left(a_6b_0\right)\right) \\ \qquad\qquad\qquad=\left(a_2b_0,a_6b_0\right)=\left(a_2b_0,a_2\left(a_4b_0\right)\right)=\left(b_0,a_4b_0\right)=\left(1,a_4\right)=0. \end{array}$	$\begin{array}{l} \left(a_2,a_6\right)=\left(a_2b_0,a_6b_0\right)=\left(a_2b_0,a_2\left(a_4b_0\right)\right)=\left(b_0,a_4b_0\right)=\left(1,a_4\right)=0, \\ \left(a_3,a_7\right)=\left(a_3b_0,a_7b_0\right)=\left(a_2\left(a_1b_0\right),a_6\left(a_1b_0\right)\right)=\left(a_1\left(a_2b_0\right),a_1\left(a_6b_0\right)\right) \\ \qquad\qquad\qquad=\left(a_2b_0,a_6b_0\right)=\left(a_2b_0,a_2\left(a_4b_0\right)\right)=\left(b_0,a_4b_0\right)=\left(1,a_4\right)=0. \end{array}$	$\begin{array}{l} \left(a_2,a_6\right)=\left(a_2b_0,a_6b_0\right)=\left(a_2b_0,a_2\left(a_4b_0\right)\right)=\left(b_0,a_4b_0\right)=\left(1,a_4\right)=0, \\ \left(a_3,a_7\right)=\left(a_3b_0,a_7b_0\right)=\left(a_2\left(a_1b_0\right),a_6\left(a_1b_0\right)\right)=\left(a_1\left(a_2b_0\right),a_1\left(a_6b_0\right)\right) \\ \qquad\qquad\qquad=\left(a_2b_0,a_6b_0\right)=\left(a_2b_0,a_2\left(a_4b_0\right)\right)=\left(b_0,a_4b_0\right)=\left(1,a_4\right)=0. \end{array}$
0902.0431_E00192 (ii)	33	■ 1.000	$(\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(x\,y),\quad x,\,y\in\mathfrak{C}.$	$(\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),\quad x,y\in\mathfrak{C}.$	$(\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),\quad x,y\in\mathfrak{C}.$
0902.0431_E00194	33	■ 1.000	$(\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(x\,y),\quad x,\,y\in\mathfrak{C}.$	$(\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),\quad x,y\in\mathfrak{C}.$	$(\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),\quad x,y\in\mathfrak{C}.$
0902.0431_E00200	34	■ 1.000	$\left(\alpha_1,\alpha_2,\alpha_3\right)a=\alpha_1a,\quad a\in S^7.$	$(\alpha_1,\alpha_2,\alpha_3)a=\alpha_1a,\quad a\in S^7.$	$(\alpha_1,\alpha_2,\alpha_3)a=\alpha_1a,\quad a\in S^7.$

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0902.0431_E00222	37	■ 1.000	$\begin{aligned} (X, Y, Z) &= \sum_{i=1}^3 \left(\frac{1}{2} (\xi_i \eta_{i+1} \zeta_{i+2} + \xi_i \eta_{i+2} \zeta_{i+1}) + R(x_i y_{i+1} z_{i+2} + x_i z_{i+1} y_{i+2}) \right. \\ &\quad \left. - (\xi_i (y_i, z_i) + \eta_i (z_i, x_i) + \zeta_i (x_i, y_i)) \right) \\ \det X &= \xi_1 \xi_2 \xi_3 + 2R(x_1 x_2 x_3) - \xi_1 x_1 \bar{x}_1 - \xi_2 x_2 \bar{x}_2 - \xi_3 x_3 \bar{x}_3 \end{aligned}$	$(X, Y, Z) = \sum_{i=1}^3 \left(\frac{1}{2} (\xi_i \eta_{i+1} \zeta_{i+2} + \xi_i \eta_{i+2} \zeta_{i+1}) + R(x_i y_{i+1} z_{i+2} + x_i z_{i+1} y_{i+2}) \right. \\ \left. - (\xi_i (y_i, z_i) + \eta_i (z_i, x_i) + \zeta_i (x_i, y_i)) \right) \\ \det X = \xi_1 \xi_2 \xi_3 + 2R(x_1 x_2 x_3) - \xi_1 x_1 \bar{x}_1 - \xi_2 x_2 \bar{x}_2 - \xi_3 x_3 \bar{x}_3$	$(X, Y, Z) = \sum_{i=1}^3 \left(\frac{1}{2} (\xi_i \eta_{i+1} \zeta_{i+2} + \xi_i \eta_{i+2} \zeta_{i+1}) + R(x_i y_{i+1} z_{i+2} + x_i z_{i+1} y_{i+2}) \right. \\ \left. - (\xi_i (y_i, z_i) + \eta_i (z_i, x_i) + \zeta_i (x_i, y_i)) \right), \\ \det X = \xi_1 \xi_2 \xi_3 + 2R(x_1 x_2 x_3) - \xi_1 x_1 \bar{x}_1 - \xi_2 x_2 \bar{x}_2 - \xi_3 x_3 \bar{x}_3.$
0902.0431_E00237	39	■ 1.000	$\begin{aligned} &3 \operatorname{det} \left(\alpha^{-1} X \right) = \left(\alpha^{-1} X, \alpha^{-1} X \times \alpha^{-1} X \right) \\ &= \left(\alpha^{-1} X, \alpha(X \times X) \right) = (X, X \times X) = 3 \det X, \end{aligned}$	$3 \det \left({}^t \alpha^{-1} X \right) = \left({}^t \alpha^{-1} X, {}^t \alpha^{-1} X \times {}^t \alpha^{-1} X \right) \\ = \left({}^t \alpha^{-1} X, \alpha(X \times X) \right) = (X, X \times X) = 3 \det X,$	$3 \det \left({}^t \alpha^{-1} X \right) = \left({}^t \alpha^{-1} X, {}^t \alpha^{-1} X \times {}^t \alpha^{-1} X \right) \\ = \left({}^t \alpha^{-1} X, \alpha(X \times X) \right) = (X, X \times X) = 3 \det X,$
0902.0431_E00252	41	■ 1.000	$\operatorname{tr}(\alpha X, \alpha Y, \alpha Z) = \operatorname{tr}(X, Y, Z).$	$\operatorname{tr}(\alpha X, \alpha Y, \alpha Z) = \operatorname{tr}(X, Y, Z).$	$\operatorname{tr}(\alpha X, \alpha Y, \alpha Z) = \operatorname{tr}(X, Y, Z).$
0902.0431_E00256	42	■ 1.000	$\begin{aligned} \mathfrak{e}_{6(-26)} &= \left\{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid (\phi X, X, X) = 0 \right\} \\ &= \left\{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid (\phi X, Y, Z) + (X, \phi Y, Z) + (X, Y, \phi Z) = 0 \right\} \\ &= \left\{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid -{}^t \phi(X \times Y) = \phi X \times Y + X \times \phi Y \right\}. \end{aligned}$	$\mathfrak{e}_{6(-26)} = \{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid (\phi X, X, X) = 0 \} \\ = \{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid (\phi X, Y, Z) + (X, \phi Y, Z) + (X, Y, \phi Z) = 0 \} \\ = \{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid -{}^t \phi(X \times Y) = \phi X \times Y + X \times \phi Y \}.$	$\mathfrak{e}_{6(-26)} = \{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid (\phi X, X, X) = 0 \} \\ = \{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid (\phi X, Y, Z) + (X, \phi Y, Z) + (X, Y, \phi Z) = 0 \} \\ = \{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid -{}^t \phi(X \times Y) = \phi X \times Y + X \times \phi Y \}.$
0902.0431_E00260	42	■ 1.000	$\widetilde{A}X = \frac{1}{2}[A, X], \quad X \in \mathfrak{J}.$	$\widetilde{A}X = \frac{1}{2}[A, X], \quad X \in \mathfrak{J}.$	$\widetilde{A}X = \frac{1}{2}[A, X], \quad X \in \mathfrak{J}.$
0902.0431_E00265	43	■ 1.000	$(X Y, Z) = (Y Z, X) = (Z X, Y) = (X, Y Z) = (Y, Z X) = (Z, X Y).$	$(XY, Z) = (YZ, X) = (ZX, Y) = (X, YZ) = (Y, ZX) = (Z, XY).$	$(XY, Z) = (YZ, X) = (ZX, Y) = (X, YZ) = (Y, ZX) = (Z, XY).$
0902.0431_E00266	43	■ 1.000	$\begin{aligned} (XY, Z) &= R(XY, Z) = \frac{1}{2} R(\operatorname{tr}((XY)Z + Z^*(Y^* X^*))) \\ &= \frac{1}{2} R \left(\sum_{i,j,k} ((x_{ij} y_{jk}) z_{ki} + \bar{z}_{ji} (\bar{y}_{kj} \bar{x}_{ik})) \right) \\ &= \frac{1}{2} R \left(\sum_{i,j,k} ((y_{jk} z_{ki}) x_{ij} + \bar{x}_{ik} (\bar{z}_{ji} \bar{y}_{kj})) \right) \\ &= \frac{1}{2} R(\operatorname{tr}((YZ)X + X^*(Z^* Y^*))) = (YZ, X). \end{aligned}$	$(XY, Z) = R(XY, Z) = \frac{1}{2} R(\operatorname{tr}((XY)Z + Z^*(Y^* X^*))) \\ = \frac{1}{2} R \left(\sum_{i,j,k} ((x_{ij} y_{jk}) z_{ki} + \bar{z}_{ji} (\bar{y}_{kj} \bar{x}_{ik})) \right) \\ = \frac{1}{2} R \left(\sum_{i,j,k} ((y_{jk} z_{ki}) x_{ij} + \bar{x}_{ik} (\bar{z}_{ji} \bar{y}_{kj})) \right) \\ = \frac{1}{2} R(\operatorname{tr}((YZ)X + X^*(Z^* Y^*))) = (YZ, X).$	$(XY, Z) = R(XY, Z) = \frac{1}{2} R(\operatorname{tr}((XY)Z + Z^*(Y^* X^*))) \\ = \frac{1}{2} R \left(\sum_{i,j,k} ((x_{ij} y_{jk}) z_{ki} + \bar{z}_{ji} (\bar{y}_{kj} \bar{x}_{ik})) \right) \\ = \frac{1}{2} R \left(\sum_{i,j,k} ((y_{jk} z_{ki}) x_{ij} + \bar{x}_{ik} (\bar{z}_{ji} \bar{y}_{kj})) \right) \\ = \frac{1}{2} R(\operatorname{tr}((YZ)X + X^*(Z^* Y^*))) = (YZ, X).$

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0902.0431_E00283	45	■ 1.000	$\left(D_{\{1\}}x\right)y+x\left(D_{\{2\}}y\right)=\overline{\operatorname{D}_{\{3\}}(\overline{xy})}, \quad x,y\in\mathfrak{C}.$	$(D_1x)y+x(D_2y)=\overline{D_3(\overline{xy})}, \quad x,y\in\mathfrak{C}.$	$(D_1x)y+x(D_2y)=\overline{D_3(\overline{xy})}, \quad x,y\in\mathfrak{C}.$
0902.0431_E00295 (2)	47	■ 1.000	$\begin{aligned} & \& \{\widetilde{A},\widetilde{B}\}=\left[\left(A-\frac{1}{3}\operatorname{tr}(A)E\right)^{\sim},\left(B-\frac{1}{3}\operatorname{tr}(B)E\right)^{\sim}\right]\in e_6^C, \\ & \widetilde{[A,B]}E=\widetilde{A}(\widetilde{B}E)-\widetilde{B}(\widetilde{A}E)=A\circ B-B\circ A=0. \end{aligned}$	$[\widetilde{A},\widetilde{B}]=\left[\left(A-\frac{1}{3}\operatorname{tr}(A)E\right)^{\sim},\left(B-\frac{1}{3}\operatorname{tr}(B)E\right)^{\sim}\right]\in e_6^C,$ $[\widetilde{A},\widetilde{B}]E=\widetilde{A}(\widetilde{B}E)-\widetilde{B}(\widetilde{A}E)=A\circ B-B\circ A=0.$	$[\widetilde{A},\widetilde{B}]=\left[\left(A-\frac{1}{3}\operatorname{tr}(A)E\right)^{\sim},\left(B-\frac{1}{3}\operatorname{tr}(B)E\right)^{\sim}\right]\in e_6^C,$ $[\widetilde{A},\widetilde{B}]E=\widetilde{A}(\widetilde{B}E)-\widetilde{B}(\widetilde{A}E)=A\circ B-B\circ A=0.$
0902.0431_E00304	49	■ 1.000	$\emptyset\not\in\left[\widetilde{A}_{\{1\}}(1),\widetilde{A}_{\{1\}}\left(e_1\right)\right]\in\mathfrak{d}_4^C\cap\mathfrak{a}$	$0\neq\left[\widetilde{A}_1(1),\widetilde{A}_1\left(e_1\right)\right]\in\mathfrak{d}_4^C\cap\mathfrak{a}$	$0\neq\left[\widetilde{A}_1(1),\widetilde{A}_1\left(e_1\right)\right]\in\mathfrak{d}_4^C\cap\mathfrak{a}$
0902.0431_E00311	50	■ 1.000	$F_{\{3\}}(1)+F_{\{1\}}\left(z_{\{1\}}\right)\in W.$	$F_3(1)+F_1(z_1)\in W.$	$F_3(1)+F_1(z_1)\in W.$
0902.0431_E00320	51	■ 1.000	$B_{\{4\}}\left(\delta_{\{1\}},\delta_{\{2\}}\right)=k\left(\delta_{\{1\}},\delta_{\{2\}}\right)_{\{4\}}=k^{\prime}\operatorname{tr}\left(\delta_1\delta_2\right).$	$B_4\left(\delta_1,\delta_2\right)=k\left(\delta_1,\delta_2\right)_4=k'\operatorname{tr}\left(\delta_1\delta_2\right).$	$B_4(\delta_1,\delta_2)=k(\delta_1,\delta_2)_4=k'\operatorname{tr}(\delta_1\delta_2).$
0902.0431_E00321	51	■ 1.000	$\begin{aligned} & (\delta,\delta)_4=\left(\widetilde{A}_1(1),\widetilde{A}_1(1)\right)_4=-2\left(\widetilde{A}_1(1),\left[\widetilde{E}_3,\widetilde{F}_1(1)\right]\right)_4 \\ & \qquad\qquad=-2\left(\widetilde{A}_1(1)E_3,F_1(1)\right)=-(F_1(1),F_1(1))=-2. \end{aligned}$	$(\delta,\delta)_4=\left(\widetilde{A}_1(1),\widetilde{A}_1(1)\right)_4=-2\left(\widetilde{A}_1(1),\left[\widetilde{E}_3,\widetilde{F}_1(1)\right]\right)_4$ $=-2\left(\widetilde{A}_1(1)E_3,F_1(1)\right)=-(F_1(1),F_1(1))=-2.$	$(\delta,\delta)_4=\left(\widetilde{A}_1(1),\widetilde{A}_1(1)\right)_4=-2\left(\widetilde{A}_1(1),\left[\widetilde{E}_3,\widetilde{F}_1(1)\right]\right)_4$ $=-2\left(\widetilde{A}_1(1)E_3,F_1(1)\right)=-(F_1(1),F_1(1))=-2.$
0902.0431_E00325	52	■ 1.000	$\operatorname{tr}(\delta\delta)=\operatorname{tr}\left(\left(\widetilde{A}_1(1)\right)^2\right)=\left(-\frac{1}{2}\right)\times2-1+\left(-\frac{1}{4}\right)\times8\times2=-6.$	$\operatorname{tr}(\delta\delta)=\operatorname{tr}\left(\left(\widetilde{A}_1(1)\right)^2\right)=\left(-\frac{1}{2}\right)\times2-1+\left(-\frac{1}{4}\right)\times8\times2=-6.$	$\operatorname{tr}(\delta\delta)=\operatorname{tr}((\widetilde{A}_1(1))^2)=\left(-\frac{1}{2}\right)\times2-1+\left(-\frac{1}{4}\right)\times8\times2=-6.$
0902.0431_E00349	57	■ 1.000	$\varphi\left(\alpha_1,\alpha_2,\alpha_3\right)\left(\begin{array}{ccc}\xi_1&x_3&\overline{x_2}\\\overline{x_3}&\xi_2&x_1\\x_2&\overline{x_1}&\xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1&\alpha_3x_3&\overline{\alpha_2x_2}\\\overline{\alpha_3x_3}&\xi_2&\alpha_1x_1\\\alpha_2x_2&\overline{\alpha_1x_1}&\xi_3\end{array}\right).$	$\varphi\left(\alpha_1,\alpha_2,\alpha_3\right)\left(\begin{array}{ccc}\xi_1&x_3&\overline{x_2}\\\overline{x_3}&\xi_2&x_1\\x_2&\overline{x_1}&\xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1&\alpha_3x_3&\overline{\alpha_2x_2}\\\overline{\alpha_3x_3}&\xi_2&\alpha_1x_1\\\alpha_2x_2&\overline{\alpha_1x_1}&\xi_3\end{array}\right).$	$\varphi(\alpha_1,\alpha_2,\alpha_3)\left(\begin{array}{ccc}\xi_1&x_3&\overline{x_2}\\\overline{x_3}&\xi_2&x_1\\x_2&\overline{x_1}&\xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1&\alpha_3x_3&\overline{\alpha_2x_2}\\\overline{\alpha_3x_3}&\xi_2&\alpha_1x_1\\\alpha_2x_2&\overline{\alpha_1x_1}&\xi_3\end{array}\right).$
0902.0431_E00354	57	■ 1.000	$\left(\alpha_1x\right)\left(\alpha_2y\right)=\overline{\alpha_3(\overline{xy})}, \quad x,y\in\mathfrak{C}.$	$(\alpha_1x)(\alpha_2y)=\overline{\alpha_3(\overline{xy})}, \quad x,y\in\mathfrak{C}.$	$(\alpha_1x)(\alpha_2y)=\overline{\alpha_3(\overline{xy})}, \quad x,y\in\mathfrak{C}.$
0902.0431_E00361	58	■ 1.000	$\begin{aligned} & Y_i=-2Z_0\circ X_i, \quad i=4,5,6,7, \\ & Z_i=-2X_i\circ Y_0, \quad i=1,3,5,6,7. \end{aligned}$	$Y_i=-2Z_0\circ X_i, \quad i=4,5,6,7,$ $Z_i=-2X_i\circ Y_0, \quad i=1,3,5,6,7.$	$Y_i=-2Z_0\circ X_i, \quad i=4,5,6,7,$ $Z_i=-2X_i\circ Y_0, \quad i=1,3,5,6,7.$
0902.0431_E00364	59	■ 1.000	$p:\left(F_4\right)_{E_1}\rightarrow SO(9).$	$p:\left(F_4\right)_{E_1}\rightarrow SO(9).$	$p:\left(F_4\right)_{E_1}\rightarrow SO(9).$

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0902.0431_E00393	65	■ 1.000	$\begin{aligned} F_{4, \mathbf{H}} &= \{\alpha \in \mathbf{Iso}_{\mathbf{R}}(\mathfrak{J}_{\mathbf{H}}) \mid \alpha(M \circ N) = \alpha M \circ \alpha N\} \\ &= \{\alpha \in \mathbf{Iso}_{\mathbf{R}}(\mathfrak{J}_{\mathbf{H}}) \mid \alpha(M \times N) = \alpha M \times \alpha N\}. \end{aligned}$	$F_{4, \mathbf{H}} = \{\alpha \in \mathbf{Iso}_{\mathbf{R}}(\mathfrak{J}_{\mathbf{H}}) \mid \alpha(M \circ N) = \alpha M \circ \alpha N\} \\ = \{\alpha \in \mathbf{Iso}_{\mathbf{R}}(\mathfrak{J}_{\mathbf{H}}) \mid \alpha(M \times N) = \alpha M \times \alpha N\}.$	$F_{4, \mathbf{H}} = \{\alpha \in \mathbf{Iso}_{\mathbf{R}}(\mathfrak{J}_{\mathbf{H}}) \mid \alpha(M \circ N) = \alpha M \circ \alpha N\} \\ = \{\alpha \in \mathbf{Iso}_{\mathbf{R}}(\mathfrak{J}_{\mathbf{H}}) \mid \alpha(M \times N) = \alpha M \times \alpha N\}.$
0902.0431_E00402	66	■ 1.000	$\alpha = \varphi(A) \beta = \varphi(A) \varphi(B) = \varphi(AB), \quad AB \in \mathbf{Sp}(3).$	$\alpha = \varphi(A) \beta = \varphi(A) \varphi(B) = \varphi(AB), \quad AB \in \mathbf{Sp}(3).$	$\alpha = \varphi(A) \beta = \varphi(A) \varphi(B) = \varphi(AB), \quad AB \in Sp(3).$
0902.0431_E00447	73	■ 1.000	$\mathfrak{e}_{\{6\}}^C = \left\{ \phi \in \mathbf{Hom}_C(\mathfrak{J}^C) \mid (\phi X, X, X) = 0 \right\}.$	$\mathfrak{e}_6^C = \left\{ \phi \in \mathbf{Hom}_C(\mathfrak{J}^C) \mid (\phi X, X, X) = 0 \right\}.$	$\mathfrak{e}_6^C = \left\{ \phi \in \mathbf{Hom}_C(\mathfrak{J}^C) \mid (\phi X, X, X) = 0 \right\}.$
0902.0431_E00468	77	■ 1.000	$\left(\phi^{\prime} X, Y \right) = - (X, \phi Y), \quad X, Y \in \mathfrak{J}^C.$	$(\phi' X, Y) = -(X, \phi Y), \quad X, Y \in \mathfrak{J}^C.$	$(\phi' X, Y) = -(X, \phi Y), \quad X, Y \in \mathfrak{J}^C.$
0902.0431_E00469	77	■ 1.000	$\phi(X \times Y) = \phi' X \times Y + X \times \phi' Y, \quad X, Y \in \mathfrak{J}^C.$	$\phi(X \times Y) = \phi' X \times Y + X \times \phi' Y, \quad X, Y \in \mathfrak{J}^C.$	$\phi(X \times Y) = \phi' X \times Y + X \times \phi' Y, \quad X, Y \in \mathfrak{J}^C.$
0902.0431_E00478	79	■ 1.000	$\begin{aligned} (\phi, A \vee B)_6 &= \left(\delta + \widetilde{T}, [\widetilde{A}, \widetilde{B}] + \left(A \circ B - \frac{1}{3}(A, B)E \right)^{\sim} \right)_6 \\ &= (\delta, [\widetilde{A}, \widetilde{B}])_4 + \left(T, A \circ B - \frac{1}{3}(A, B)E \right) \\ &= (\delta A, B) + (\widetilde{T} A, B) = ((\delta + \widetilde{T}) A, B) = (\phi A, B). \end{aligned}$	$\begin{aligned} (\phi, A \vee B)_6 &= \left(\delta + \widetilde{T}, [\widetilde{A}, \widetilde{B}] + \left(A \circ B - \frac{1}{3}(A, B)E \right)^{\sim} \right)_6 \\ &= (\delta, [\widetilde{A}, \widetilde{B}])_4 + \left(T, A \circ B - \frac{1}{3}(A, B)E \right) \\ &= (\delta A, B) + (\widetilde{T} A, B) = ((\delta + \widetilde{T}) A, B) = (\phi A, B). \end{aligned}$	$\begin{aligned} (\phi, A \vee B)_6 &= \left(\delta + \widetilde{T}, [\widetilde{A}, \widetilde{B}] + \left(A \circ B - \frac{1}{3}(A, B)E \right)^{\sim} \right)_6 \\ &= (\delta, [\widetilde{A}, \widetilde{B}])_4 + \left(T, A \circ B - \frac{1}{3}(A, B)E \right) \\ &= (\delta A, B) + (\widetilde{T} A, B) = ((\delta + \widetilde{T}) A, B) = (\phi A, B). \end{aligned}$

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0902.0431_E00504	82	■1.000	$\begin{aligned} & \pm(\lambda_k - \lambda_l), \quad \pm(\lambda_k + \lambda_l), \quad 0 \leq k < l \leq 3, \\ & \pm \lambda_k \pm \frac{1}{2}(\mu_2 - \mu_3), \quad 0 \leq k \leq 3, \\ & \pm \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2}(\mu_3 - \mu_1), \\ & \pm \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2}(\mu_3 - \mu_1), \\ & \pm \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2}(\mu_3 - \mu_1), \\ & \pm \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2}(\mu_3 - \mu_1), \\ & \pm \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2}(\mu_1 - \mu_2), \\ & \pm \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2}(\mu_1 - \mu_2), \\ & \pm \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2}(\mu_1 - \mu_2), \\ & \pm \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2}(\mu_1 - \mu_2), \end{aligned}$	$\begin{aligned} & \pm(\lambda_k - \lambda_l), \quad \pm(\lambda_k + \lambda_l), \quad 0 \leq k < l \leq 3, \\ & \pm \lambda_k \pm \frac{1}{2}(\mu_2 - \mu_3), \quad 0 \leq k \leq 3, \\ & \pm \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2}(\mu_3 - \mu_1), \\ & \pm \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2}(\mu_3 - \mu_1), \\ & \pm \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2}(\mu_3 - \mu_1), \\ & \pm \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2}(\mu_3 - \mu_1), \\ & \pm \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2}(\mu_1 - \mu_2), \\ & \pm \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2}(\mu_1 - \mu_2), \\ & \pm \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2}(\mu_1 - \mu_2), \\ & \pm \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2}(\mu_1 - \mu_2), \end{aligned}$	
0902.0431_E00530	89	■1.000	$\alpha X=\left(\begin{array}{ccc}\xi_1&\theta&\theta\\\theta&\xi_2&\theta\\\theta&\theta&\xi_3\end{array}\right),\quad \xi_i\in C\;.$	$\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right),\quad \xi_i\in C.$	$\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right),\quad \xi_i\in C.$
0902.0431_E00531	89	■1.000	$\theta\neq x_{\scriptscriptstyle 1}=p+i\,q,\quad p,\,q\in\mathfrak{C}\;.$	$0\neq x_1=p+iq,\quad p,q\in\mathfrak{C}.$	$0\neq x_1=p+iq,\quad p,q\in\mathfrak{C}.$
0902.0431_E00569	95	■1.000	$k\left(a+b\,e_{\scriptscriptstyle 2}\right)=\left(\begin{array}{cc}a^{\prime}&b^{\prime}\\\tau a^{\prime}&\tau b^{\prime}\end{array}\right),\quad a,\,b\in\boldsymbol{C}\;.$	$k\left(a+b\,e_2\right)=\left(\begin{array}{cc}a'&b'\\-\tau b'&\tau a'\end{array}\right),\quad a,b\in\boldsymbol{C}.$	$k(a+b\,e_2)=\left(\begin{array}{cc}a'&b'\\-\tau b'&\tau a'\end{array}\right),\quad a,b\in\boldsymbol{C}.$

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0902.0431_E00743	122	■ 1.000	$B_7(\Phi_0, \Phi_0) = \operatorname{tr} \left((\operatorname{ad} \Phi_0)^2 \right) = \frac{4}{9} \times 27 \times 2 = 24.$		$B_7(\Phi_0, \Phi_0) = \operatorname{tr}((\operatorname{ad} \Phi_0)^2) = \frac{4}{9} \times 27 \times 2 = 24.$
0902.0431_E00747	124	■ 1.000	$\begin{aligned} & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2} \mu_2 - \frac{2}{3} \nu \right), \\ & \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2} \mu_2 - \frac{2}{3} \nu \right), \\ & \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \left(\frac{1}{2} \mu_3 - \frac{2}{3} \nu \right), \\ & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \left(\frac{1}{2} \mu_3 - \frac{2}{3} \nu \right), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2} \mu_3 - \frac{2}{3} \nu \right), \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2} \mu_3 - \frac{2}{3} \nu \right) \end{aligned}$		$\begin{aligned} & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2} \mu_2 - \frac{2}{3} \nu \right), \\ & \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2} \mu_2 - \frac{2}{3} \nu \right), \\ & \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \left(\frac{1}{2} \mu_3 - \frac{2}{3} \nu \right), \\ & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \left(\frac{1}{2} \mu_3 - \frac{2}{3} \nu \right), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2} \mu_3 - \frac{2}{3} \nu \right), \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2} \mu_3 - \frac{2}{3} \nu \right) \end{aligned}$
0902.0431_E00799	133	■ 1.000	$\xi = \cos r_1 \cos r_2 \cos r_3.$		$\xi = \cos r_1 \cos r_2 \cos r_3.$
0902.0431_E00829	138	■ 1.000	$\begin{aligned} & \kappa \left(\begin{array}{c} X \\ Y \\ \xi \\ \eta \end{array} \right) = \left(\begin{array}{c} -\kappa_1 X \\ \kappa_1 Y \\ -\xi \\ \eta \end{array} \right), \quad \kappa_1 X = (E_1, X) - 4E_1 \times (E_1 \times X), \\ & \mu \left(\begin{array}{c} X \\ Y \\ \xi \\ \eta \end{array} \right) = \left(\begin{array}{c} 2E_1 \times Y + \eta E_1 \\ 2E_1 \times X + \xi E_1 \\ (E_1, Y) \\ (E_1, X) \end{array} \right). \end{aligned}$		$\begin{aligned} & \kappa \left(\begin{array}{c} X \\ Y \\ \xi \\ \eta \end{array} \right) = \left(\begin{array}{c} -\kappa_1 X \\ \kappa_1 Y \\ -\xi \\ \eta \end{array} \right), \quad \kappa_1 X = (E_1, X) - 4E_1 \times (E_1 \times X), \\ & \mu \left(\begin{array}{c} X \\ Y \\ \xi \\ \eta \end{array} \right) = \left(\begin{array}{c} 2E_1 \times Y + \eta E_1 \\ 2E_1 \times X + \xi E_1 \\ (E_1, Y) \\ (E_1, X) \end{array} \right). \end{aligned}$
0902.0431_E00840	140	■ 1.000	$\phi(\nu) \left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array} \right) = \frac{\nu}{3} \left(\begin{array}{ccc} 4\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & -2\xi_2 & -2x_1 \\ x_2 & -2\bar{x}_1 & -2\xi_3 \end{array} \right)$		$\phi(\nu) \left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array} \right) = \frac{\nu}{3} \left(\begin{array}{ccc} 4\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & -2\xi_2 & -2x_1 \\ x_2 & -2\bar{x}_1 & -2\xi_3 \end{array} \right)$

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0902.0431_E08841	140	■ 1.000	$\left(\mathfrak{e}_7\right)^{\sigma} \cong \mathfrak{a}_1 \oplus \left(\mathfrak{e}_7\right)^{\kappa, \mu} .$	$\left(\mathfrak{e}_7\right)^{\sigma} \cong \mathfrak{a}_1 \oplus \left(\mathfrak{e}_7\right)^{\kappa, \mu} .$	$\left(\mathfrak{e}_7\right)^{\sigma} \cong \mathfrak{a}_1 \oplus \left(\mathfrak{e}_7\right)^{\kappa, \mu} .$
0902.0431_E08852	141	■ 1.000	$\begin{aligned} V^{\{11\}} &= \left\{ P \in \mathfrak{P}^C \mid \kappa P = P, \mu \tau \lambda P = P, P \times (0, E_1, 0, 1, 0) = 0 \right\} \\ &= \left\{ \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, \begin{pmatrix} \eta & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, 0, \tau \eta \right) \mid x \in \mathfrak{C}, \xi \in C, \eta \in i\mathbf{R} \right\} \end{aligned}$	$V^{11} = \{ P \in \mathfrak{P}^C \mid \kappa P = P, \mu \tau \lambda P = P, P \times (0, E_1, 0, 1, 0) = 0 \}$ $= \left\{ \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, \begin{pmatrix} \eta & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, 0, \tau \eta \right) \mid x \in \mathfrak{C}, \xi \in C, \eta \in i\mathbf{R} \right\}$	$V^{11} = \{ P \in \mathfrak{P}^C \mid \kappa P = P, \mu \tau \lambda P = P, P \times (0, E_1, 0, 1, 0) = 0 \}$ $= \left\{ \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, \begin{pmatrix} \eta & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, 0, \tau \eta \right) \mid x \in \mathfrak{C}, \xi \in C, \eta \in i\mathbf{R} \right\}$
0902.0431_E08872	145	■ 1.000	$z(\operatorname{Spin}(12))=\{1,-1,\,\,\sigma\!-\!\sigma\}\cong\{1,-1\}\times\{1,\sigma\}\cong\boldsymbol{Z}_2\times\boldsymbol{Z}_2$	$z(\mathbf{Spin}(12)) = \{1, -1, \sigma - \sigma\} \cong \{1, -1\} \times \{1, \sigma\} \cong \mathbf{Z}_2 \times \mathbf{Z}_2.$	$z(Spin(12)) = \{1, -1, \sigma - \sigma\} \cong \{1, -1\} \times \{1, \sigma\} \cong \mathbf{Z}_2 \times \mathbf{Z}_2.$
0902.0431_E08909	151	■ 1.000	$w(X,\,Y,\,\xi,\,\eta)=(w\,X,\,w\,Y,\,\xi,\,\eta)\,.$	$w(X,Y,\xi,\eta)=(wX,wY,\xi,\eta).$	$w(X,Y,\xi,\eta)=(wX,wY,\xi,\eta).$
0902.0431_E08923	153	■ 1.000	$\begin{aligned} &h(a,\,b)=\frac{a+b}{2}+i\,\frac{(b-a)}{2}e_1, \quad a,b\in\boldsymbol{C}, \\ &h\left(a+be_1\right)=a+bi, \quad a,b\in\boldsymbol{R}. \end{aligned}$	$h(a,b)=\frac{a+b}{2}+i\frac{(b-a)e_1}{2}, \quad a,b\in\boldsymbol{C},$ $h\left(a+be_1\right)=a+bi, \quad a,b\in\boldsymbol{R}.$	$h(a,b)=\frac{a+b}{2}+i\frac{(b-a)e_1}{2}, \quad a,b\in\boldsymbol{C},$ $h(a+be_1)=a+bi, \quad a,b\in\boldsymbol{R}.$
0902.0431_E08957	159	■ 1.000	$\psi(Q,\,A)\,P=f^{-1}((Q,\,A)(f\,P)), \quad P\in\mathfrak{P}^C.$	$\psi(Q,A)P=f^{-1}((Q,A)(fP)), \quad P\in\mathfrak{P}^C.$	$\psi(Q,A)P=f^{-1}((Q,A)(fP)), \quad P\in\mathfrak{P}^C.$
0902.0431_E08988	164	■ 1.000	$\left(\left[R,R_1\right],R_2\right)_8+\left(R_1,\left[R,R_2\right]\right)_8=0, \quad R,R_i\in\mathfrak{e}_8^C.$	$([R,R_1],R_2)_8+(R_1,[R,R_2])_8=0, \quad R,R_i\in\mathfrak{e}_8^C.$	$([R,R_1],R_2)_8+(R_1,[R,R_2])_8=0, \quad R,R_i\in\mathfrak{e}_8^C.$
0902.0431_E01003	167	■ 1.000	$\left\langle R_1,R_2\right\rangle =-\frac{1}{15}B_8\left(\tau\tilde{\lambda}R_1,R_2\right).$	$\langle R_1,R_2\rangle =-\frac{1}{15}B_8\left(\tau\tilde{\lambda}R_1,R_2\right).$	$\langle R_1,R_2\rangle =-\frac{1}{15}B_8(\tau\tilde{\lambda}R_1,R_2).$
0902.0431_E01013	168	■ 1.000	$E_8^{\{ \}^C} \simeq E_8 \times \boldsymbol{R}^{248}.$	$E_8^{\,C} \simeq E_8 \times \boldsymbol{R}^{248}.$	$E_8^{\,C} \simeq E_8 \times \boldsymbol{R}^{248}.$
0902.0431_E01050 (ii)	179	■ 1.000	$\left\{\beta_2P,\beta_2Q\right\}=\{P,Q\}.$	$\{\beta_2P,\beta_2Q\}=\{P,Q\}.$	$\{\beta_2P,\beta_2Q\}=\{P,Q\}.$
0902.0431_E01053	179	■ 1.000	$\beta_1\Phi=\beta\Phi\beta^{-1}.$	$\beta_1\Phi=\beta\Phi\beta^{-1}.$	$\beta_1\Phi=\beta\Phi\beta^{-1}.$
0902.0431_E01144	193	■ 1.000	$(\boldsymbol{u}\times\boldsymbol{v})\boldsymbol{x}=\ast(\boldsymbol{v}\wedge\ast(\boldsymbol{u}\wedge\boldsymbol{x}))+(-1)^{n-k}\frac{n-k}{n}(\boldsymbol{u},\boldsymbol{v})\boldsymbol{x}, \quad \boldsymbol{x}\in C^n.$	$(\boldsymbol{u}\times\boldsymbol{v})\boldsymbol{x}=\ast(\boldsymbol{v}\wedge\ast(\boldsymbol{u}\wedge\boldsymbol{x}))+(-1)^{n-k}\frac{n-k}{n}(\boldsymbol{u},\boldsymbol{v})\boldsymbol{x}, \quad \boldsymbol{x}\in C^n.$	$(\boldsymbol{u}\times\boldsymbol{v})\boldsymbol{x}=\ast(\boldsymbol{v}\wedge\ast(\boldsymbol{u}\wedge\boldsymbol{x}))+(-1)^{n-k}\frac{n-k}{n}(\boldsymbol{u},\boldsymbol{v})\boldsymbol{x}, \quad \boldsymbol{x}\in C^m.$
0902.0431_E01145	193	■ 1.000	$\mathfrak{e}_{\{8\}^{\{ \}^C}}=\mathfrak{sl}(9,C)\oplus\Lambda^3(C^9)\oplus\Lambda^3(C^9),$	$\mathfrak{e}_8^{\,C}=\mathfrak{sl}(9,C)\oplus\Lambda^3(C^9)\oplus\Lambda^3(C^9),$	$\mathfrak{e}_8^{\,C}=\mathfrak{sl}(9,C)\oplus\Lambda^3(C^9)\oplus\Lambda^3(C^9),$

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0902.0431_E01147	193	■ 1.000	$\left\{\begin{array}{l} D=[D_1,D_2]+\boldsymbol{u}_1\times\boldsymbol{v}_2-\boldsymbol{u}_2\times\boldsymbol{v}_1 \\ \boldsymbol{u}=D_1\boldsymbol{u}_2-D_2\boldsymbol{u}_1+*(\boldsymbol{v}_1\wedge\boldsymbol{v}_2) \\ \boldsymbol{v}=-{}^tD_1\boldsymbol{v}_2+{}^tD_2\boldsymbol{v}_1-*(\boldsymbol{u}_1\wedge\boldsymbol{u}_2), \end{array}\right.$	$\left\{\begin{array}{l} D=[D_1,D_2]+\boldsymbol{u}_1\times\boldsymbol{v}_2-\boldsymbol{u}_2\times\boldsymbol{v}_1 \\ \boldsymbol{u}=D_1\boldsymbol{u}_2-D_2\boldsymbol{u}_1+*(\boldsymbol{v}_1\wedge\boldsymbol{v}_2) \\ \boldsymbol{v}=-{}^tD_1\boldsymbol{v}_2+{}^tD_2\boldsymbol{v}_1-*(\boldsymbol{u}_1\wedge\boldsymbol{u}_2), \end{array}\right.$	$\begin{cases} D=[D_1,D_2]+\boldsymbol{u}_1\times\boldsymbol{v}_2-\boldsymbol{u}_2\times\boldsymbol{v}_1 \\ \boldsymbol{u}=D_1\boldsymbol{u}_2-D_2\boldsymbol{u}_1+*(\boldsymbol{v}_1\wedge\boldsymbol{v}_2) \\ \boldsymbol{v}=-{}^tD_1\boldsymbol{v}_2+{}^tD_2\boldsymbol{v}_1-*(\boldsymbol{u}_1\wedge\boldsymbol{u}_2), \end{cases}$
0902.0431_E01160	196	■ 1.000	$\begin{aligned} &0\neq\operatorname{ad}(T)\operatorname{ad}(S_{37})\operatorname{ad}(S_{27})\operatorname{ad}(S_{17})\operatorname{ad}(S_{36})\operatorname{ad}(S_{25})\operatorname{ad}(S_{14})R \\ &=(-E_{34},0,0)\in\mathfrak{sl}(9,C)\cap\mathfrak{a}. \end{aligned}$	$0\neq\operatorname{ad}(T)\operatorname{ad}(S_{37})\operatorname{ad}(S_{27})\operatorname{ad}(S_{17})\operatorname{ad}(S_{36})\operatorname{ad}(S_{25})\operatorname{ad}(S_{14})R \\ =(-E_{34},0,0)\in\mathfrak{sl}(9,C)\cap\mathfrak{a}.$	$0\neq\operatorname{ad}(T)\operatorname{ad}(S_{37})\operatorname{ad}(S_{27})\operatorname{ad}(S_{17})\operatorname{ad}(S_{36})\operatorname{ad}(S_{25})\operatorname{ad}(S_{14})R \\ =(-E_{34},0,0)\in\mathfrak{sl}(9,C)\cap\mathfrak{a}.$
0902.0431_E01165	196	■ 1.000	$\left\langle R_1,R_2\right\rangle =-B_8\left(R_1,\tau\widetilde{\lambda}R_2\right).$	$\langle R_1,R_2\rangle=-B_8\left(R_1,\tau\widetilde{\lambda}R_2\right).$	$\langle R_1,R_2\rangle=-B_8(R_1,\tau\widetilde{\lambda}R_2).$
0902.0431_E01168	197	■ 1.000	$w_3(D,\boldsymbol{u},\boldsymbol{v})=\left(D,\omega\boldsymbol{u},\omega^2\boldsymbol{v}\right),$	$w_3(D,\boldsymbol{u},\boldsymbol{v})=\left(D,\omega\boldsymbol{u},\omega^2\boldsymbol{v}\right),$	$w_3(D,\boldsymbol{u},\boldsymbol{v})=(D,\omega\boldsymbol{u},\omega^2\boldsymbol{v}),$
0902.0431_E01180	199	■ 1.000	$(*(\boldsymbol{x}\wedge\boldsymbol{y})\wedge\boldsymbol{z}),\boldsymbol{v})=(\boldsymbol{x}\wedge\boldsymbol{y},\boldsymbol{z}\wedge\boldsymbol{v})=(\boldsymbol{x},\boldsymbol{z})(\boldsymbol{y},\boldsymbol{v})-(\boldsymbol{y},\boldsymbol{z})(\boldsymbol{x},\boldsymbol{v}).$	$(*(\boldsymbol{x}\wedge\boldsymbol{y})\wedge\boldsymbol{z}),\boldsymbol{v})=(\boldsymbol{x}\wedge\boldsymbol{y},\boldsymbol{z}\wedge\boldsymbol{v})=(\boldsymbol{x},\boldsymbol{z})(\boldsymbol{y},\boldsymbol{v})-(\boldsymbol{y},\boldsymbol{z})(\boldsymbol{x},\boldsymbol{v}).$	$(*(\boldsymbol{x}\wedge\boldsymbol{y})\wedge\boldsymbol{z}),\boldsymbol{v})=(\boldsymbol{x}\wedge\boldsymbol{y},\boldsymbol{z}\wedge\boldsymbol{v})=(\boldsymbol{x},\boldsymbol{z})(\boldsymbol{y},\boldsymbol{v})-(\boldsymbol{y},\boldsymbol{z})(\boldsymbol{x},\boldsymbol{v}).$
0902.0431_E01201	202	■ 1.000	$\begin{aligned} &B_8(R_1,R_2)=60(\operatorname{tr}(C_1C_2)+\operatorname{tr}(D_1D_2)-(x_1,w_2)(a_1,d_2)-(x_2,w_1)(a_2,d_1) \\ &\quad -(y_1,z_2)(b_1,c_2)-(y_2,z_1)(b_2,c_1)), \end{aligned}$	$B_8(R_1,R_2)=60(\operatorname{tr}(C_1C_2)+\operatorname{tr}(D_1D_2)-(x_1,w_2)(a_1,d_2)-(x_2,w_1)(a_2,d_1) \\ - (y_1,z_2)(b_1,c_2)-(y_2,z_1)(b_2,c_1)),$	$B_8(R_1,R_2)=60(\operatorname{tr}(C_1C_2)+\operatorname{tr}(D_1D_2)-(x_1,w_2)(a_1,d_2)-(x_2,w_1)(a_2,d_1) \\ -(y_1,z_2)(b_1,c_2)-(y_2,z_1)(b_2,c_1)),$
0902.0431_E00001	5	■ 1.000	$\tau(u+iv)=u-iv.$	$\tau(u+iv)=u-iv.$	$\tau(u+iv)=u-iv.$
0902.0431_E00008	7	■ 1.000	$G_2=\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{C})\mid\alpha(xy)=(\alpha x)(\alpha y)\}.$	$G_2=\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{C})\mid\alpha(xy)=(\alpha x)(\alpha y)\}.$	$G_2=\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{C})\mid\alpha(xy)=(\alpha x)(\alpha y)\}.$
0902.0431_E00009	7	■ 1.000	$(\alpha x,\alpha y)=(x,y),\quad x,y\in\mathfrak{C}.$	$(\alpha x,\alpha y)=(x,y),\quad x,y\in\mathfrak{C}.$	$(\alpha x,\alpha y)=(x,y),\quad x,y\in\mathfrak{C}.$
0902.0431_E00010	7	■ 1.000	$\overline{\alpha x}=\alpha\overline{x}.$	$\overline{\alpha x}=\alpha\overline{x}.$	$\overline{\alpha x}=\alpha\overline{x}.$
0902.0431_E00013	8	■ 1.000	$O(8)=O(\mathfrak{C})=\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{C})\mid(\alpha x,\alpha y)=(x,y)\}.$	$O(8)=O(\mathfrak{C})=\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{C})\mid(\alpha x,\alpha y)=(x,y)\}.$	$O(8)=O(\mathfrak{C})=\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{C})\mid(\alpha x,\alpha y)=(x,y)\}.$

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0902.0431_E00014	8	■ 1.000	$\mathfrak{D}_4=\mathfrak{so}(8)=\mathfrak{so}(\mathfrak{C})=\left\{D\in\operatorname{Hom}_{\mathbf{R}}(\mathfrak{C})\mid(Dx,y)+(x,Dy)=0\right\}$	$\mathfrak{D}_4 = \mathfrak{so}(8) = \mathfrak{so}(\mathfrak{C}) = \{D \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{C}) \mid (Dx, y) + (x, Dy) = 0\}$	$\mathfrak{D}_4 = \mathfrak{so}(8) = \mathfrak{so}(\mathfrak{C}) = \{D \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{C}) \mid (Dx, y) + (x, Dy) = 0\}$
0902.0431_E00015	8	■ 1.000	$G_{ij}e_j=e_i, \quad G_{ij}e_i=-e_j, \quad G_{ij}e_k=0, \quad k\neq i, j.$	$G_{ij}e_j = e_i, \quad G_{ij}e_i = -e_j, \quad G_{ij}e_k = 0, k \neq i, j.$	$G_{ij}e_j = e_i, \quad G_{ij}e_i = -e_j, \quad G_{ij}e_k = 0, \quad k \neq i, j.$
0902.0431_E00016	8	■ 1.000	$F_{ij}x=\frac{1}{2}e_i\left(\bar{e}_jx\right), \quad x\in\mathfrak{C}.$	$F_{ij}x = \frac{1}{2}e_i(\bar{e}_jx), \quad x \in \mathfrak{C}.$	$F_{ij}x = \frac{1}{2}e_i(\bar{e}_jx), \quad x \in \mathfrak{C}.$
0902.0431_E00020	9	■ 1.000	$\kappa,\pi,\nu\in\operatorname{Aut}(\mathfrak{D}_4).$	$\kappa,\pi,\nu\in\operatorname{Aut}(\mathfrak{D}_4).$	$\kappa,\pi,\nu\in\operatorname{Aut}(\mathfrak{D}_4).$
0902.0431_E00022	9	■ 1.000	$\left[\pi G_{ij},\pi G_{kl}\right]=\pi\left[G_{ij},G_{kl}\right],$	$[\pi G_{ij}, \pi G_{kl}] = \pi [G_{ij}, G_{kl}],$	$[\pi G_{ij}, \pi G_{kl}] = \pi [G_{ij}, G_{kl}],$
0902.0431_E00027 (1)	10	■ 1.000	$L_a,R_a,T_a\in\mathfrak{D}_4.$	$L_a,R_a,T_a\in\mathfrak{D}_4.$	$L_a,R_a,T_a\in\mathfrak{D}_4.$
0902.0431_E00028 (2)	10	■ 1.000	$\kappa L_a=-R_a, \quad \kappa R_a=-L_a, \quad \kappa T_a=-T_a.$	$\kappa L_a = -R_a, \quad \kappa R_a = -L_a, \quad \kappa T_a = -T_a.$	$\kappa L_a = -R_a, \quad \kappa R_a = -L_a, \quad \kappa T_a = -T_a.$
0902.0431_E00029 (3)	10	■ 1.000	$\pi L_a=T_a, \quad \pi R_a=-R_a, \quad \pi T_a=L_a.$	$\pi L_a = T_a, \quad \pi R_a = -R_a, \quad \pi T_a = L_a.$	$\pi L_a = T_a, \quad \pi R_a = -R_a, \quad \pi T_a = L_a.$
0902.0431_E00030 (4)	10	■ 1.000	$\nu L_a=R_a, \quad \nu R_a=-T_a, \quad \nu T_a=-L_a.$	$\nu L_a = R_a, \quad \nu R_a = -T_a, \quad \nu T_a = -L_a.$	$\nu L_a = R_a, \quad \nu R_a = -T_a, \quad \nu T_a = -L_a.$
0902.0431_E00031	10	■ 1.000	$L_{e_i}=2F_{i0}, \quad T_{e_i}=2G_{i0}.$	$L_{e_i} = 2F_{i0}, \quad T_{e_i} = 2G_{i0}.$	$L_{e_i} = 2F_{i0}, \quad T_{e_i} = 2G_{i0}.$
0902.0431_E00035	11	■ 1.000	$D=L_a+\sum_i\left[L_{b_i},L_{c_i}\right], \quad a,b_i,c_i\in\mathfrak{C}_0.$	$D = L_a + \sum_i [L_{b_i}, L_{c_i}], \quad a, b_i, c_i \in \mathfrak{C}_0.$	$D = L_a + \sum_i [L_{b_i}, L_{c_i}], \quad a, b_i, c_i \in \mathfrak{C}_0.$
0902.0431_E00036	11	■ 1.000	$\kappa^2=1, \quad \pi^2=1, \quad \nu^3=1, \quad \nu=\pi\kappa.$	$\kappa^2 = 1, \quad \pi^2 = 1, \quad \nu^3 = 1, \quad \nu = \pi\kappa.$	$\kappa^2 = 1, \quad \pi^2 = 1, \quad \nu^3 = 1, \quad \nu = \pi\kappa.$
0902.0431_E00038	11	■ 1.000	$\left(D_1x\right)y+x\left(D_2y\right)=D_3(xy), \quad x,y\in\mathfrak{C}.$	$(D_1x)y + x(D_2y) = D_3(xy), \quad x, y \in \mathfrak{C}.$	$(D_1x)y + x(D_2y) = D_3(xy), \quad x, y \in \mathfrak{C}.$
0902.0431_E00039	11	■ 1.000	$D_2=\nu D_1, \quad D_3=\pi D_1.$	$D_2 = \nu D_1, \quad D_3 = \pi D_1.$	$D_2 = \nu D_1, \quad D_3 = \pi D_1.$
0902.0431_E00040 (i)	11	■ 1.000	$\left(L_ax\right)y+x\left(R_ay\right)=T_a(xy), \quad x,y\in\mathfrak{C}.$	$(L_ax)y + x(R_ay) = T_a(xy), \quad x, y \in \mathfrak{C}.$	$(L_ax)y + x(R_ay) = T_a(xy), \quad x, y \in \mathfrak{C}.$
0902.0431_E00042	12	■ 1.000	$\left(L_aL_bx\right)y+\left(L_bx\right)\left(R_ay\right)+\left(L_ax\right)\left(R_by\right)+x\left(R_aR_by\right)=T_aT_b(xy).$	$(L_aL_bx)y + (L_bx)(R_ay) + (L_ax)(R_by) + x(R_aR_by) = T_aT_b(xy).$	$(L_aL_bx)y + (L_bx)(R_ay) + (L_ax)(R_by) + x(R_aR_by) = T_aT_b(xy).$
0902.0431_E00043 (iii)	12	■ 1.000	$\left([L_a,L_b]x\right)y+x\left([R_a,R_b]y\right)=[T_a,T_b](xy).$	$([L_a, L_b]x)y + x([R_a, R_b]y) = [T_a, T_b](xy).$	$([L_a, L_b]x)y + x([R_a, R_b]y) = [T_a, T_b](xy).$
0902.0431_E00044	12	■ 1.000	$D_1=L_a+\sum\left[L_b,L_c\right], \quad a,b,c\in\mathfrak{C}_0$	$D_1 = L_a + \sum [L_b, L_c], \quad a, b, c \in \mathfrak{C}_0$	$D_1 = L_a + \sum [L_b, L_c], \quad a, b, c \in \mathfrak{C}_0$
0902.0431_E00045	12	■ 1.000	$\left(D_1x\right)y+x\left(D_2y\right)=D_3(xy), \quad x,y\in\mathfrak{C}.$	$(D_1x)y + x(D_2y) = D_3(xy), \quad x, y \in \mathfrak{C}.$	$(D_1x)y + x(D_2y) = D_3(xy), \quad x, y \in \mathfrak{C}.$
0902.0431_E00046	12	■ 1.000	$x\left(D_2y\right)=D_3(xy), \quad x,y\in\mathfrak{C}.$	$x(D_2y) = D_3(xy), \quad x, y \in \mathfrak{C}.$	$x(D_2y) = D_3(xy), \quad x, y \in \mathfrak{C}.$
0902.0431_E00048	12	■ 1.000	$x(y\,p)=(xy)\,p, \quad \text{for all } x,y\in\mathfrak{C}.$	$x(y p) = (xy)p, \quad \text{for all } x, y \in \mathfrak{C}.$	$x(y p) = (xy)p, \quad \text{for all } x, y \in \mathfrak{C}.$
0902.0431_E00049	12	■ 1.000	$\left(D_1x\right)y+x\left(D_2y\right)=\left(\kappa D_3\right)(xy), \quad x,y\in\mathfrak{C}$	$(D_1x)y + x(D_2y) = (\kappa D_3)(xy), \quad x, y \in \mathfrak{C}$	$(D_1x)y + x(D_2y) = (\kappa D_3)(xy), \quad x, y \in \mathfrak{C}$
0902.0431_E00052	13	■ 1.000	$\begin{aligned} &\&D_3^{\prime}=\nu D_2=\nu\nu D_1=\nu^{-1}D_1=\kappa\pi D_1=\kappa\kappa D_3=D_3\\ &\&D_1^{\prime}=\kappa\pi D_2=\nu^{-1}D_2=D_1 \end{aligned}$	$\begin{aligned} D_3' &= \nu D_2 = \nu\nu D_1 = \nu^{-1}D_1 = \kappa\pi D_1 = \kappa\kappa D_3 = D_3 \\ D_1' &= \kappa\pi D_2 = \nu^{-1}D_2 = D_1 \end{aligned}$	$\begin{aligned} D_3' &= \nu D_2 = \nu\nu D_1 = \nu^{-1}D_1 = \kappa\pi D_1 = \kappa\kappa D_3 = D_3, \\ D_1' &= \kappa\pi D_2 = \nu^{-1}D_2 = D_1. \end{aligned}$

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0902.0431_E00053	13	■ 1.000	$\mathfrak{g}_2=\left\{D\in\operatorname{Hom}_{\mathbf{R}}(\mathfrak{C})\mid D(xy)=(Dx)y+x(Dy)\right\}.$	$\mathfrak{g}_2=\{D\in\operatorname{Hom}_{\mathbf{R}}(\mathfrak{C})\mid D(xy)=(Dx)y+x(Dy)\}.$	$\mathfrak{g}_2=\{D\in\operatorname{Hom}_{\mathbf{R}}(\mathfrak{C})\mid D(xy)=(Dx)y+x(Dy)\}.$
0902.0431_E00054	13	■ 1.000	$\begin{aligned}\mathfrak{b}_3&=\left\{D\in\mathfrak{D}_4\mid D1=0\right\}\\&=\left\{D\in\mathfrak{D}_4\mid\kappa D=D\right\}=\left\{D\in\mathfrak{D}_4\mid\nu D=\pi D\right\}.\end{aligned}$	$\mathfrak{b}_3=\{D\in\mathfrak{D}_4\mid D1=0\}\\=\{D\in\mathfrak{D}_4\mid\kappa D=D\}=\{D\in\mathfrak{D}_4\mid\nu D=\pi D\}.$	$\mathfrak{b}_3=\{D\in\mathfrak{D}_4\mid D1=0\}\\=\{D\in\mathfrak{D}_4\mid\kappa D=D\}=\{D\in\mathfrak{D}_4\mid\nu D=\pi D\}.$
0902.0431_E00056	13	■ 1.000	$D\,x\in\mathfrak{C}_0,\quad x\in\mathfrak{C}.$	$Dx\in\mathfrak{C}_0,\quad x\in\mathfrak{C}.$	$Dx\in\mathfrak{C}_0,\quad x\in\mathfrak{C}.$
0902.0431_E00057	13	■ 1.000	$x+yx=-\left(x\,\overline{y}+y\,\overline{x}\right)=-2\left(x,\,y\right),\quad x,y\in\mathfrak{C}_0.$	$xy+yx=-\left(x\overline{y}+y\overline{x}\right)=-2\left(x,y\right),\quad x,y\in\mathfrak{C}_0.$	$xy+yx=-\left(x\overline{y}+y\overline{x}\right)=-2\left(x,y\right),\quad x,y\in\mathfrak{C}_0.$
0902.0431_E00058	13	■ 1.000	$(D\,x,\,y)+(x,\,D\,y)=0,\quad x,y\in\mathfrak{C}.$	$(Dx,y)+(x,Dy)=0,\quad x,y\in\mathfrak{C}.$	$(Dx,y)+(x,Dy)=0,\quad x,y\in\mathfrak{C}.$
0902.0431_E00060	14	■ 1.000	$\operatorname{dim}\mathfrak{g}_2=14.$	$\dim\mathfrak{g}_2=14.$	$\dim\mathfrak{g}_2=14.$
0902.0431_E00061	14	■ 1.000	$\sum_{0<i<j}\lambda_{ij}\,F_{ij}=\sum_{0<i<j}\lambda_{ij}\,G_{ij}.$	$\sum_{0<i<j}\lambda_{ij}F_{ij}=\sum_{0<i<j}\lambda_{ij}G_{ij}.$	$\sum_{0<i<j}\lambda_{ij}F_{ij}=\sum_{0<i<j}\lambda_{ij}G_{ij}.$
0902.0431_E00062	14	■ 1.000	$\sum_{0<i<j}\lambda_{ij}\,e_j\,e_i=0.$	$\sum_{0<i<j}\lambda_{ij}e_je_i=0.$	$\sum_{0<i<j}\lambda_{ij}e_je_i=0.$
0902.0431_E00064	14	■ 1.000	$\boldsymbol{C}=\left\{x_0+x_1e_1\mid x_i\in\mathbf{R}\right\}.$	$\boldsymbol{C}=\{x_0+x_1e_1\mid x_i\in\mathbf{R}\}.$	$\boldsymbol{C}=\{x_0+x_1e_1\mid x_i\in\mathbf{R}\}.$
0902.0431_E00066	15	■ 1.000	$x=a+m_1e_2+m_2e_4+m_3e_6,\quad a,m_i\in\boldsymbol{C}.$	$x=a+m_1e_2+m_2e_4+m_3e_6,\quad a,m_i\in\boldsymbol{C}.$	$x=a+m_1e_2+m_2e_4+m_3e_6,\quad a,m_i\in\boldsymbol{C}.$
0902.0431_E00070	15	■ 1.000	$\boldsymbol{C}\oplus\boldsymbol{C}^3=\mathfrak{C}.$	$\boldsymbol{C}\oplus\boldsymbol{C}^3=\mathfrak{C}.$	$\boldsymbol{C}\oplus\boldsymbol{C}^3=\mathfrak{C}.$
0902.0431_E00071	15	■ 1.000	$\left(\mathfrak{g}_2\right)_{e_1}=\left\{D\in\mathfrak{g}_2\mid De_1=0\right\}.$	$(\mathfrak{g}_2)_{e_1}=\{D\in\mathfrak{g}_2\mid De_1=0\}.$	$(\mathfrak{g}_2)_{e_1}=\{D\in\mathfrak{g}_2\mid De_1=0\}.$
0902.0431_E00074	16	■ 1.000	$\begin{array}{l}\varphi_*(\varphi_1(E_{11}-E_{22}))=-G_{23}+G_{45},\quad\varphi_*(\varphi_1(E_{22}-E_{33}))=-G_{45}+G_{67},\\ \varphi_*(E_{12}-E_{21})=G_{24}+G_{35},\quad\varphi_*(\varphi_1(E_{12}+E_{21}))=-G_{25}+G_{34},\\ \varphi_*(E_{13}-E_{31})=G_{26}+G_{37},\quad\varphi_*(\varphi_1(E_{13}+E_{31}))=-G_{27}+G_{36},\\ \varphi_*(E_{23}-E_{32})=G_{46}+G_{57},\quad\varphi_*(\varphi_1(E_{23}+E_{32}))=-G_{47}+G_{56}.\end{array}$	$\begin{array}{l}\varphi_*(\varphi_1(E_{11}-E_{22}))=-G_{23}+G_{45},\quad\varphi_*(\varphi_1(E_{22}-E_{33}))=-G_{45}+G_{67},\\ \varphi_*(E_{12}-E_{21})=G_{24}+G_{35},\quad\varphi_*(\varphi_1(E_{12}+E_{21}))=-G_{25}+G_{34},\\ \varphi_*(E_{13}-E_{31})=G_{26}+G_{37},\quad\varphi_*(\varphi_1(E_{13}+E_{31}))=-G_{27}+G_{36},\\ \varphi_*(E_{23}-E_{32})=G_{46}+G_{57},\quad\varphi_*(\varphi_1(E_{23}+E_{32}))=-G_{47}+G_{56}.\end{array}$	$\begin{array}{l}\varphi_*(\varphi_1(E_{11}-E_{22}))=-G_{23}+G_{45},\quad\varphi_*(\varphi_1(E_{22}-E_{33}))=-G_{45}+G_{67},\\ \varphi_*(E_{12}-E_{21})=G_{24}+G_{35},\quad\varphi_*(\varphi_1(E_{12}+E_{21}))=-G_{25}+G_{34},\\ \varphi_*(E_{13}-E_{31})=G_{26}+G_{37},\quad\varphi_*(\varphi_1(E_{13}+E_{31}))=-G_{27}+G_{36},\\ \varphi_*(E_{23}-E_{32})=G_{46}+G_{57},\quad\varphi_*(\varphi_1(E_{23}+E_{32}))=-G_{47}+G_{56}.\end{array}$
0902.0431_E00078	16	■ 1.000	$-2\,x_1e_2-2\,x_2e_3-2\,x_3e_4-2\,x_4e_5-2\,x_5e_6-2\,x_6e_7=0.$	$-2x_1e_2-2x_2e_3-2x_3e_4-2x_4e_5-2x_5e_6-2x_6e_7=0.$	$-2x_1e_2-2x_2e_3-2x_3e_4-2x_4e_5-2x_5e_6-2x_6e_7=0.$
0902.0431_E00079	16	■ 1.000	$\overline{x_1+ix_2}=\overline{x_1}+i\overline{x_2},\quad\tau(x_1+ix_2)=x_1-ix_2,\quad x_i\in\mathfrak{C}.$	$\overline{x_1+ix_2}=\overline{x_1}+i\overline{x_2},\quad\tau(x_1+ix_2)=x_1-ix_2,\quad x_i\in\mathfrak{C}.$	$\overline{x_1+ix_2}=\overline{x_1}+i\overline{x_2},\quad\tau(x_1+ix_2)=x_1-ix_2,\quad x_i\in\mathfrak{C}.$
0902.0431_E00080	16	■ 1.000	$\tau(x\,y)=(\tau x)(\tau y),\quad x,y\in\mathfrak{C}^C.$	$\tau(xy)=(\tau x)(\tau y),\quad x,y\in\mathfrak{C}^C.$	$\tau(xy)=(\tau x)(\tau y),\quad x,y\in\mathfrak{C}^C.$
0902.0431_E00083 (cdots ii)	17	■ 1.000	$-x_1S_6+x_2S_5\in W\quad\cdots\text{(i)}\quad\text{and}\quad-x_1S_5-x_2S_6\in W$	$-x_1S_6+x_2S_5\in W\quad\cdots\text{(i)}\quad\text{and}\quad-x_1S_5-x_2S_6\in W$	$-x_1S_6+x_2S_5\in W\quad\cdots\text{(i)}\quad\text{and}\quad-x_1S_5-x_2S_6\in W$
0902.0431_E00085	17	■ 1.000	$\mathfrak{a}\ni\left[D',D+S\right]=\left[D',D\right]+\left[D',S\right],\quad\left[D',S\right]\in\mathfrak{S}$	$\mathfrak{a}\ni[D',D+S]=[D',D]+[D',S],\quad [D',S]\in\mathfrak{S}$	$\mathfrak{a}\ni[D',D+S]=[D',D]+[D',S],\quad [D',S]\in\mathfrak{S}$

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0902.0431_E00096	19	■1.000	$\pm\left(\lambda_1-\lambda_2\right), \quad \pm\left(\lambda_1-\lambda_3\right), \quad \pm\left(\lambda_2-\lambda_3\right), \quad \pm\lambda_1, \quad \pm\lambda_2 \quad \pm\lambda_3$	$\pm\left(\lambda_1-\lambda_2\right), \quad \pm\left(\lambda_1-\lambda_3\right), \quad \pm\left(\lambda_2-\lambda_3\right), \quad \pm\lambda_1, \quad \pm\lambda_2 \quad \pm\lambda_3$	$\pm\left(\lambda_1-\lambda_2\right), \quad \pm\left(\lambda_1-\lambda_3\right), \quad \pm\left(\lambda_2-\lambda_3\right), \quad \pm\lambda_1, \quad \pm\lambda_2 \quad \pm\lambda_3$
0902.0431_E00099	20	■1.000	$\alpha_1=\lambda_1-\lambda_2, \quad \alpha_2=\lambda_2$	$\alpha_1=\lambda_1-\lambda_2, \quad \alpha_2=\lambda_2$	$\alpha_1=\lambda_1-\lambda_2, \quad \alpha_2=\lambda_2$
0902.0431_E00100	20	■1.000	$\mu=2\alpha_1+3\alpha_3$	$\mu=2\alpha_1+3\alpha_3$	$\mu=2\alpha_1+3\alpha_3$
0902.0431_E00102	20	■1.000	$\mathfrak{h}_{\boldsymbol{R}}=\left\{-i\lambda_1G_{23}-i\lambda_2G_{45}-i\lambda_3G_{67} \mid \lambda_i\in\boldsymbol{R}, \lambda_1+\lambda_2+\lambda_3=0\right\}$	$\mathfrak{h}_{\boldsymbol{R}}=\left\{-i\lambda_1G_{23}-i\lambda_2G_{45}-i\lambda_3G_{67} \mid \lambda_i\in\boldsymbol{R}, \lambda_1+\lambda_2+\lambda_3=0\right\}$	$\mathfrak{h}_{\boldsymbol{R}}=\left\{-i\lambda_1G_{23}-i\lambda_2G_{45}-i\lambda_3G_{67} \mid \lambda_i\in\boldsymbol{R}, \lambda_1+\lambda_2+\lambda_3=0\right\}$
0902.0431_E00103	20	■1.000	$B_2\left(H,H'\right)=8\sum_{k=1}^3\lambda_k\lambda_k',$	$B_2\left(H,H'\right)=8\sum_{k=1}^3\lambda_k\lambda_k',$	$B_2\left(H,H'\right)=8\sum_{k=1}^3\lambda_k\lambda_k',$
0902.0431_E00104	20	■1.000	$H_{\alpha_1}=-\frac{1}{8}iG_{23}+\frac{1}{8}iG_{45}, \quad H_{\alpha_2}=\frac{1}{24}iG_{23}-\frac{1}{12}iG_{45}+\frac{1}{24}iG_{67}.$	$H_{\alpha_1}=-\frac{1}{8}iG_{23}+\frac{1}{8}iG_{45}, \quad H_{\alpha_2}=\frac{1}{24}iG_{23}-\frac{1}{12}iG_{45}+\frac{1}{24}iG_{67}.$	$H_{\alpha_1}=-\frac{1}{8}iG_{23}+\frac{1}{8}iG_{45}, \quad H_{\alpha_2}=\frac{1}{24}iG_{23}-\frac{1}{12}iG_{45}+\frac{1}{24}iG_{67}.$
0902.0431_E00106	21	■1.000	$\left(G_2\right)_{e_1}=\left\{\alpha\in G_2 \mid \alpha e_1=e_1\right\}.$	$\left(G_2\right)_{e_1}=\left\{\alpha\in G_2 \mid \alpha e_1=e_1\right\}.$	$\left(G_2\right)_{e_1}=\left\{\alpha\in G_2 \mid \alpha e_1=e_1\right\}.$
0902.0431_E00107	21	■1.000	$\varphi(A)(a+\boldsymbol{m})=a+A\boldsymbol{m}, \quad a+\boldsymbol{m}\in\boldsymbol{C}\oplus\boldsymbol{C}^3=\mathfrak{C}.$	$\varphi(A)(a+\boldsymbol{m})=a+A\boldsymbol{m}, \quad a+\boldsymbol{m}\in\boldsymbol{C}\oplus\boldsymbol{C}^3=\mathfrak{C}.$	$\varphi(A)(a+\boldsymbol{m})=a+A\boldsymbol{m}, \quad a+\boldsymbol{m}\in\boldsymbol{C}\oplus\boldsymbol{C}^3=\mathfrak{C}.$
0902.0431_E00109	21	■1.000	$\alpha e_2=\boldsymbol{a}_1, \quad \alpha e_4=\boldsymbol{a}_2, \quad \alpha e_6=\boldsymbol{a}_3$	$\alpha e_2=\boldsymbol{a}_1, \quad \alpha e_4=\boldsymbol{a}_2, \quad \alpha e_6=\boldsymbol{a}_3$	$\alpha e_2=\boldsymbol{a}_1, \quad \alpha e_4=\boldsymbol{a}_2, \quad \alpha e_6=\boldsymbol{a}_3$
0902.0431_E00110	21	■1.000	$\left\langle\boldsymbol{a}_1,\boldsymbol{a}_2\right\rangle=0, \quad \boldsymbol{a}_3=\overline{\boldsymbol{a}_1\times\boldsymbol{a}_2}.$	$\left\langle\boldsymbol{a}_1,\boldsymbol{a}_2\right\rangle=0, \quad \boldsymbol{a}_3=\overline{\boldsymbol{a}_1\times\boldsymbol{a}_2}.$	$\left\langle\boldsymbol{a}_1,\boldsymbol{a}_2\right\rangle=0, \quad \boldsymbol{a}_3=\overline{\boldsymbol{a}_1\times\boldsymbol{a}_2}.$
0902.0431_E00111	22	■1.000	$a_3=a_1a_2.$	$a_3=a_1a_2.$	$a_3=a_1a_2.$
0902.0431_E00112	22	■1.000	$a_5=a_1a_4, \quad a_6=a_4a_2, \quad a_7=a_3a_4.$	$a_5=a_1a_4, \quad a_6=a_4a_2, \quad a_7=a_3a_4.$	$a_5=a_1a_4, \quad a_6=a_4a_2, \quad a_7=a_3a_4.$
0902.0431_E00114	22	■1.000	$\alpha e_i=a_i, \quad i=0,1,\cdots,7$	$\alpha e_i=a_i, \quad i=0,1,\cdots,7$	$\alpha e_i=a_i, \quad i=0,1,\cdots,7$
0902.0431_E00115	22	■1.000	$\alpha(xy)=(\alpha x)(\alpha y), \quad x,y\in\mathfrak{C}.$	$\alpha(xy)=(\alpha x)(\alpha y), \quad x,y\in\mathfrak{C}.$	$\alpha(xy)=(\alpha x)(\alpha y), \quad x,y\in\mathfrak{C}.$
0902.0431_E00116	22	■1.000	$\alpha\left(e_ie_j\right)=\left(\alpha e_i\right)\left(\alpha e_j\right), \quad i,j=0,1,\cdots,7$	$\alpha\left(e_ie_j\right)=\left(\alpha e_i\right)\left(\alpha e_j\right), \quad i,j=0,1,\cdots,7$	$\alpha\left(e_ie_j\right)=\left(\alpha e_i\right)\left(\alpha e_j\right), \quad i,j=0,1,\cdots,7$
0902.0431_E00118	23	■1.000	$w=\varphi(\operatorname{diag}(\omega_1,\omega_1,\omega_1))$	$w=\varphi(\operatorname{diag}(\omega_1,\omega_1,\omega_1))$	$w=\varphi(\operatorname{diag}(\omega_1,\omega_1,\omega_1))$
0902.0431_E00120	23	■1.000	$\left(G_2\right)^w=\left\{\alpha\in G_2 \mid w\alpha=\alpha w\right\}.$	$\left(G_2\right)^w=\left\{\alpha\in G_2 \mid w\alpha=\alpha w\right\}.$	$\left(G_2\right)^w=\left\{\alpha\in G_2 \mid w\alpha=\alpha w\right\}.$
0902.0431_E00121	23	■1.000	$\begin{aligned} w\varphi(A)(a+\boldsymbol{m}) &=w(a+A\boldsymbol{m})=a+\omega_1A\boldsymbol{m} \\ &=a+A\omega_1\boldsymbol{m}=\varphi(A)w(a+\boldsymbol{m}). \end{aligned}$	$\begin{aligned} w\varphi(A)(a+\boldsymbol{m}) &=w(a+A\boldsymbol{m})=a+\omega_1A\boldsymbol{m} \\ &=a+A\omega_1\boldsymbol{m}=\varphi(A)w(a+\boldsymbol{m}). \end{aligned}$	$\begin{aligned} w\varphi(A)(a+\boldsymbol{m}) &=w(a+A\boldsymbol{m})=a+\omega_1A\boldsymbol{m} \\ &=a+A\omega_1\boldsymbol{m}=\varphi(A)w(a+\boldsymbol{m}). \end{aligned}$
0902.0431_E00124	24	■1.000	$\boldsymbol{H}=\left\{x_0+x_1e_1+x_2e_2+x_3e_3 \mid x_i\in\boldsymbol{R}\right\}.$	$\boldsymbol{H}=\left\{x_0+x_1e_1+x_2e_2+x_3e_3 \mid x_i\in\boldsymbol{R}\right\}.$	$\boldsymbol{H}=\left\{x_0+x_1e_1+x_2e_2+x_3e_3 \mid x_i\in\boldsymbol{R}\right\}.$
0902.0431_E00126	24	■1.000	$x=m+ae_4, \quad m,a\in\boldsymbol{H}.$	$x=m+ae_4, \quad m,a\in\boldsymbol{H}.$	$x=m+ae_4, \quad m,a\in\boldsymbol{H}.$

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0902.0431_E00127	24	■ 1.000	$\begin{aligned} \left(m+a e_4\right) \left(n+b e_4\right) &= \left(m n-\bar{b} a\right)+\left(a \bar{n}+b m\right) e_4, \\ \left(m+a e_4, n+b e_4\right) &= (m, n)+(a, b), \\ \overline{m+a e_4} &= \bar{m}-a e_4, \\ \gamma\left(m+a e_4\right) &= m-a e_4. \end{aligned}$	$\begin{aligned} (m+a e_4)(n+b e_4) &= (mn-\bar{b}a) + (a\bar{n}+bm)e_4, \\ (m+a e_4, n+b e_4) &= (m, n) + (a, b), \\ \overline{m+a e_4} &= \bar{m} - a e_4, \\ \gamma(m+a e_4) &= m - a e_4. \end{aligned}$	$\begin{aligned} (m+a e_4)(n+b e_4) &= (mn-\bar{b}a) + (a\bar{n}+bm)e_4, \\ (m+a e_4, n+b e_4) &= (m, n) + (a, b), \\ \overline{m+a e_4} &= \bar{m} - a e_4, \\ \gamma(m+a e_4) &= m - a e_4. \end{aligned}$
0902.0431_E00128	24	■ 1.000	$\boldsymbol{H} \oplus \boldsymbol{H} e_4 = \mathfrak{C}.$	$\boldsymbol{H} \oplus \boldsymbol{H} e_4 = \mathfrak{C}.$	$\boldsymbol{H} \oplus \boldsymbol{H} e_4 = \mathfrak{C}.$
0902.0431_E00129	24	■ 1.000	$\left(G_2\right)^{\gamma}=\left\{\alpha \in G_2 \mid \gamma \alpha=\alpha \gamma\right\}.$	$\left(G_2\right)^{\gamma}=\left\{\alpha \in G_2 \mid \gamma \alpha=\alpha \gamma\right\}.$	$\left(G_2\right)^{\gamma}=\left\{\alpha \in G_2 \mid \gamma \alpha=\alpha \gamma\right\}.$
0902.0431_E00130	24	■ 1.000	$\varphi(p, q)\left(m+a e_4\right)=q m \bar{q}+(p a \bar{q}) e_4, \quad m+a e_4 \in \boldsymbol{H} \oplus \boldsymbol{H} e_4 = \mathfrak{C}.$	$\varphi(p, q)\left(m+a e_4\right)=q m \bar{q}+(p a \bar{q}) e_4, \quad m+a e_4 \in \boldsymbol{H} \oplus \boldsymbol{H} e_4 = \mathfrak{C}.$	$\varphi(p, q)\left(m+a e_4\right)=q m \bar{q}+(p a \bar{q}) e_4, \quad m+a e_4 \in \boldsymbol{H} \oplus \boldsymbol{H} e_4 = \mathfrak{C}.$
0902.0431_E00132	25	■ 1.000	$\alpha m=q m \bar{q}, \quad m \in \boldsymbol{H}$	$\alpha m=q m \bar{q}, \quad m \in \boldsymbol{H}$	$\alpha m=q m \bar{q}, \quad m \in \boldsymbol{H}$
0902.0431_E00135	25	■ 1.000	$f(p, q) x=p x \bar{q}, \quad x \in \boldsymbol{H}$	$f(p, q) x=p x \bar{q}, \quad x \in \boldsymbol{H}$	$f(p, q) x=p x \bar{q}, \quad x \in \boldsymbol{H}$
0902.0431_E00137	26	■ 1.000	$z\left(G_2\right)=\{1\}.$	$z\left(G_2\right)=\{1\}.$	$z\left(G_2\right)=\{1\}.$
0902.0431_E00140	26	■ 1.000	$(\alpha x, \alpha y)=(x, y), \quad x, y \in \mathfrak{C}^C.$	$(\alpha x, \alpha y)=(x, y), \quad x, y \in \mathfrak{C}^C.$	$(\alpha x, \alpha y)=(x, y), \quad x, y \in \mathfrak{C}^C.$
0902.0431_E00141	26	■ 1.000	$\overline{\alpha e_k}=-\alpha e_k, \quad k=1,2, \cdots, 7.$	$\overline{\alpha e_k}=-\alpha e_k, \quad k=1,2, \cdots, 7.$	$\overline{\alpha e_k}=-\alpha e_k, \quad k=1,2, \cdots, 7.$
0902.0431_E00144	26	■ 1.000	$\langle x, y\rangle=(\tau x, y),$	$\langle x, y\rangle=(\tau x, y),$	$\langle x, y\rangle=(\tau x, y),$
0902.0431_E00149	27	■ 1.000	$G_{2(2)}=\left\{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}\left(\mathfrak{C}^{\prime}\right) \mid \alpha(x y)=(\alpha x)(\alpha y)\right\}.$	$G_{2(2)}=\left\{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}\left(\mathfrak{C}^{\prime}\right) \mid \alpha(x y)=(\alpha x)(\alpha y)\right\}.$	$G_{2(2)}=\left\{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}\left(\mathfrak{C}^{\prime}\right) \mid \alpha(x y)=(\alpha x)(\alpha y)\right\}.$
0902.0431_E00150	28	■ 1.000	$G_{2(2)}=\left(G_2^C\right)^{\tau \gamma}=\left\{\alpha \in G_2^C \mid \tau \gamma \alpha=\alpha \gamma \tau\right\}.$	$G_{2(2)}=\left(G_2^C\right)^{\tau \gamma}=\left\{\alpha \in G_2^C \mid \tau \gamma \alpha=\alpha \gamma \tau\right\}.$	$G_{2(2)}=\left(G_2^C\right)^{\tau \gamma}=\left\{\alpha \in G_2^C \mid \tau \gamma \alpha=\alpha \gamma \tau\right\}.$
0902.0431_E00153	28	■ 1.000	$z\left(G_{2(2)}\right)=\{1\}.$	$z\left(G_{2(2)}\right)=\{1\}.$	$z\left(G_{2(2)}\right)=\{1\}.$
0902.0431_E00155	28	■ 1.000	$A=D_{a_m} \cdots D_{a_2} D_{a_1}, \quad a_i \in S^{n-1}$	$A=D_{a_m} \cdots D_{a_2} D_{a_1}, \quad a_i \in S^{n-1}$	$A=D_{a_m} \cdots D_{a_2} D_{a_1}, \quad a_i \in S^{n-1}.$
0902.0431_E00156	28	■ 1.000	$A=D_{a_{2 m}} \cdots D_{a_2} D_{a_1}, \quad a_i \in S^{n-1}$	$A=D_{a_{2 m}} \cdots D_{a_2} D_{a_1}, \quad a_i \in S^{n-1}$	$A=D_{a_{2 m}} \cdots D_{a_2} D_{a_1}, \quad a_i \in S^{n-1}.$
0902.0431_E00157	28	■ 1.000	$S O\left(\mathfrak{C}\right)=\left\{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}\left(\mathfrak{C}\right) \mid(\alpha x, \alpha y)=(x, y), \det \alpha=1\right\}.$	$S O\left(\mathfrak{C}\right)=\left\{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}\left(\mathfrak{C}\right) \mid(\alpha x, \alpha y)=(x, y), \det \alpha=1\right\}.$	$S O\left(\mathfrak{C}\right)=\left\{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}\left(\mathfrak{C}\right) \mid(\alpha x, \alpha y)=(x, y), \det \alpha=1\right\}.$
0902.0431_E00159	29	■ 1.000	$D_a x=x-2(x, a) a=x-(x \bar{a}+a \bar{x}) a=-a \bar{x} a, \quad x \in \mathfrak{C},$	$D_a x=x-2(x, a) a=x-(x \bar{a}+a \bar{x}) a=-a \bar{x} a, \quad x \in \mathfrak{C},$	$D_a x=x-2(x, a) a=x-(x \bar{a}+a \bar{x}) a=-a \bar{x} a, \quad x \in \mathfrak{C},$
0902.0431_E00160	29	■ 1.000	$\left(\alpha_1 x\right)\left(\alpha_2 y\right)=(b(\bar{a} x))((y \bar{a}) b)=b(\bar{a}(x y) \bar{a}) b=\alpha_3(x y), \quad x, y \in \mathfrak{C}.$	$\left(\alpha_1 x\right)\left(\alpha_2 y\right)=(b(\bar{a} x))((y \bar{a}) b)=b(\bar{a}(x y) \bar{a}) b=\alpha_3(x y), \quad x, y \in \mathfrak{C}.$	$\left(\alpha_1 x\right)\left(\alpha_2 y\right)=(b(\bar{a} x))((y \bar{a}) b)=b(\bar{a}(x y) \bar{a}) b=\alpha_3(x y), \quad x, y \in \mathfrak{C}.$
0902.0431_E00163	29	■ 1.000	$(x p) y=x(p y), \quad x, y \in \mathfrak{C}.$	$(x p) y=x(p y), \quad x, y \in \mathfrak{C}.$	$(x p) y=x(p y), \quad x, y \in \mathfrak{C}.$
0902.0431_E00167	30	■ 1.000	$\left(\alpha_2 y\right)\left(\alpha_3 z\right)=\overline{\alpha_1(\overline{y z})}.$	$\left(\alpha_2 y\right)\left(\alpha_3 z\right)=\overline{\alpha_1(\overline{y z})}.$	$\left(\alpha_2 y\right)\left(\alpha_3 z\right)=\overline{\alpha_1(\overline{y z})}.$
0902.0431_E00168	30	■ 1.000	$\left(\alpha_1 x\right)\left(\alpha_2 y\right)=\alpha_3(x y), \quad x, y \in \mathfrak{C},$	$\left(\alpha_1 x\right)\left(\alpha_2 y\right)=\alpha_3(x y), \quad x, y \in \mathfrak{C},$	$\left(\alpha_1 x\right)\left(\alpha_2 y\right)=\alpha_3(x y), \quad x, y \in \mathfrak{C},$

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0902.0431_E00245	40	■1.000	$\alpha X \circ \alpha E = \alpha(X \times E) = \frac{1}{2} \alpha(\mathrm{tr}(X)E - X),$ $\alpha(\operatorname{tr}(X) E - X),$	$\alpha X \circ \alpha E = \alpha(X \times E) = \frac{1}{2} \alpha(\mathrm{tr}(X)E - X),$	$\alpha X \circ \alpha E = \alpha(X \times E) = \frac{1}{2} \alpha(\mathrm{tr}(X)E - X),$
0902.0431_E00246	40	■1.000	$\alpha X \times P = \frac{1}{2} \operatorname{tr}(X) P - \frac{1}{2} \alpha X, \quad X \in \mathfrak{J}.$ $\alpha X \times P = \frac{1}{2} \operatorname{tr}(X) P - \frac{1}{2} \alpha X, \quad X \in \mathfrak{J}.$	$\alpha X \times P = \frac{1}{2} \operatorname{tr}(X) P - \frac{1}{2} \alpha X, \quad X \in \mathfrak{J}.$	$\alpha X \times P = \frac{1}{2} \operatorname{tr}(X) P - \frac{1}{2} \alpha X, \quad X \in \mathfrak{J}.$
0902.0431_E00247	40	■1.000	$\begin{aligned} & \frac{1}{2} \left(\rho_2 E_3 + \rho_3 E_2 - F_1(p_1) \right) \\ & = \frac{1}{2} \left(\lambda (\rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3 + F_1(p_1) + F_2(p_2) + F_3(p_3)) - E_1 \right) \end{aligned}$ $\frac{1}{2} (\rho_2 E_3 + \rho_3 E_2 - F_1(p_1))$ $= \frac{1}{2} (\lambda (\rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3 + F_1(p_1) + F_2(p_2) + F_3(p_3)) - E_1),$	$\frac{1}{2} (\rho_2 E_3 + \rho_3 E_2 - F_1(p_1))$ $= \frac{1}{2} (\lambda (\rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3 + F_1(p_1) + F_2(p_2) + F_3(p_3)) - E_1)$	$\frac{1}{2} (\rho_2 E_3 + \rho_3 E_2 - F_1(p_1))$ $= \frac{1}{2} (\lambda (\rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3 + F_1(p_1) + F_2(p_2) + F_3(p_3)) - E_1),$
0902.0431_E00248	41	■1.000	$\theta = \lambda \rho_1 - 1, \quad \rho_3 = \lambda \rho_2, \quad \rho_2 = \lambda \rho_3, \quad -p_1 = \lambda p_1, \quad 0 = \lambda p_2, \quad 0 = \lambda p_3.$ $\rho_3 = \lambda \rho_2, \quad \rho_2 = \lambda \rho_3, \quad -p_1 = \lambda p_1, \quad 0 = \lambda p_2, \quad 0 = \lambda p_3.$	$0 = \lambda \rho_1 - 1, \quad \rho_3 = \lambda \rho_2, \quad \rho_2 = \lambda \rho_3, \quad -p_1 = \lambda p_1, \quad 0 = \lambda p_2, \quad 0 = \lambda p_3.$	$0 = \lambda \rho_1 - 1, \quad \rho_3 = \lambda \rho_2, \quad \rho_2 = \lambda \rho_3, \quad -p_1 = \lambda p_1, \quad 0 = \lambda p_2, \quad 0 = \lambda p_3.$
0902.0431_E00249	41	■1.000	$-\frac{1}{2} \rho_1 F_1(1) = \frac{1}{2} (\mu (\rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3) - F_1(1)),$ $-\frac{1}{2} \rho_1 F_1(1) = \frac{1}{2} (\mu (\rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3) - F_1(1)),$	$-\frac{1}{2} \rho_1 F_1(1) = \frac{1}{2} (\mu (\rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3) - F_1(1)),$	$-\frac{1}{2} \rho_1 F_1(1) = \frac{1}{2} (\mu (\rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3) - F_1(1)),$
0902.0431_E00251	41	■1.000	$(\alpha X, \alpha Y) = (X, Y).$ $(\alpha X, \alpha Y) = (X, Y).$	$(\alpha X, \alpha Y) = (X, Y).$	$(\alpha X, \alpha Y) = (X, Y).$
0902.0431_E00254	41	■1.000	$\widetilde{\alpha} \left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array} \right) = \left(\begin{array}{ccc} \xi_1 & \alpha x_3 & \overline{\alpha x_2} \\ \overline{\alpha x_3} & \xi_2 & \alpha x_1 \\ \alpha x_2 & \overline{\alpha x_1} & \xi_3 \end{array} \right).$ $\widetilde{\alpha} \left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array} \right) = \left(\begin{array}{ccc} \xi_1 & \alpha x_3 & \overline{\alpha x_2} \\ \overline{\alpha x_3} & \xi_2 & \alpha x_1 \\ \alpha x_2 & \overline{\alpha x_1} & \xi_3 \end{array} \right).$	$\widetilde{\alpha} \left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array} \right) = \left(\begin{array}{ccc} \xi_1 & \alpha x_3 & \overline{\alpha x_2} \\ \overline{\alpha x_3} & \xi_2 & \alpha x_1 \\ \alpha x_2 & \overline{\alpha x_1} & \xi_3 \end{array} \right).$	$\widetilde{\alpha} \left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array} \right) = \left(\begin{array}{ccc} \xi_1 & \alpha x_3 & \overline{\alpha x_2} \\ \overline{\alpha x_3} & \xi_2 & \alpha x_1 \\ \alpha x_2 & \overline{\alpha x_1} & \xi_3 \end{array} \right).$
0902.0431_E00258	42	■1.000	$\mathfrak{M}^{\scriptscriptstyle -} = \{ A \in M(3, \mathfrak{C}) \mid A^* = -A \}.$ $\mathfrak{M}^{\scriptscriptstyle -} = \{ A \in M(3, \mathfrak{C}) \mid A^* = -A \}.$	$\mathfrak{M}^{\scriptscriptstyle -} = \{ A \in M(3, \mathfrak{C}) \mid A^* = -A \}.$	$\mathfrak{M}^{\scriptscriptstyle -} = \{ A \in M(3, \mathfrak{C}) \mid A^* = -A \}.$
0902.0431_E00261	42	■1.000	$[X, X X] = a E.$ $[X, X X] = a E.$	$[X, X X] = a E.$	$[X, X X] = a E.$
0902.0431_E00262	42	■1.000	$a_{ij} = \sum_{k,l} (x_{ik} (x_{kl} x_{lj}) - (x_{ik} x_{kl}) x_{lj}) = - \sum_{k,l} \{ x_{ik}, x_{kl}, x_{lj} \}, \quad i, j = 1, 2, 3.$ $a_{ij} = \sum_{k,l} (x_{ik} (x_{kl} x_{lj}) - (x_{ik} x_{kl}) x_{lj}) = - \sum_{k,l} \{ x_{ik}, x_{kl}, x_{lj} \}, \quad i, j = 1, 2, 3.$	$a_{ij} = \sum_{k,l} (x_{ik} (x_{kl} x_{lj}) - (x_{ik} x_{kl}) x_{lj}) = - \sum_{k,l} \{ x_{ik}, x_{kl}, x_{lj} \}, \quad i, j = 1, 2, 3.$	$a_{ij} = \sum_{k,l} (x_{ik} (x_{kl} x_{lj}) - (x_{ik} x_{kl}) x_{lj}) = - \sum_{k,l} \{ x_{ik}, x_{kl}, x_{lj} \}, \quad i, j = 1, 2, 3.$
0902.0431_E00268	43	■1.000	$([A, X], X X) = 0, \quad X \in \mathfrak{J}.$ $([A, X], X X) = 0, \quad X \in \mathfrak{J}.$	$([A, X], X X) = 0, \quad X \in \mathfrak{J}.$	$([A, X], X X) = 0, \quad X \in \mathfrak{J}.$
0902.0431_E00270	44	■1.000	$([A, X], Y Z + Z Y) + ([A, Y], X Z + Z X) + ([A, Z], X Y + Y X) = 0,$ $([A, X], Y Z + Z Y) + ([A, Y], X Z + Z X) + ([A, Z], X Y + Y X) = 0,$	$([A, X], Y Z + Z Y) + ([A, Y], X Z + Z X) + ([A, Z], X Y + Y X) = 0,$	$([A, X], Y Z + Z Y) + ([A, Y], X Z + Z X) + ([A, Z], X Y + Y X) = 0,$
0902.0431_E00271	44	■1.000	$A_1(a) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & a \\ 0 & -\bar{a} & 0 \end{pmatrix}, \quad A_2(a) = \begin{pmatrix} 0 & 0 & -\bar{a} \\ 0 & 0 & 0 \\ a & 0 & 0 \end{pmatrix}, \quad A_3(a) = \begin{pmatrix} 0 & a & 0 \\ -\bar{a} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$ $A_1(a) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & a \\ 0 & -\bar{a} & 0 \end{pmatrix}, \quad A_2(a) = \begin{pmatrix} 0 & 0 & -\bar{a} \\ 0 & 0 & 0 \\ a & 0 & 0 \end{pmatrix}, \quad A_3(a) = \begin{pmatrix} 0 & a & 0 \\ -\bar{a} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$	$A_1(a) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & a \\ 0 & -\bar{a} & 0 \end{pmatrix}, \quad A_2(a) = \begin{pmatrix} 0 & 0 & -\bar{a} \\ 0 & 0 & 0 \\ a & 0 & 0 \end{pmatrix}, \quad A_3(a) = \begin{pmatrix} 0 & a & 0 \\ -\bar{a} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$	$A_1(a) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & a \\ 0 & -\bar{a} & 0 \end{pmatrix}, \quad A_2(a) = \begin{pmatrix} 0 & 0 & -\bar{a} \\ 0 & 0 & 0 \\ a & 0 & 0 \end{pmatrix}, \quad A_3(a) = \begin{pmatrix} 0 & a & 0 \\ -\bar{a} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$
0902.0431_E00273	44	■1.000	$\mathfrak{d}_4 = \{ \delta \in \mathfrak{f}_4 \mid \delta E_i = 0, i = 1, 2, 3 \}$ $\mathfrak{d}_4 = \{ \delta \in \mathfrak{f}_4 \mid \delta E_i = 0, i = 1, 2, 3 \}$	$\mathfrak{d}_4 = \{ \delta \in \mathfrak{f}_4 \mid \delta E_i = 0, i = 1, 2, 3 \}$	$\mathfrak{d}_4 = \{ \delta \in \mathfrak{f}_4 \mid \delta E_i = 0, i = 1, 2, 3 \}$
0902.0431_E00274	44	■1.000	$\mathfrak{D}_4 = \{ D \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{C}) \mid (Dx, y) + (x, Dy) = 0 \}$ $\mathfrak{D}_4 = \{ D \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{C}) \mid (Dx, y) + (x, Dy) = 0 \}$	$\mathfrak{D}_4 = \{ D \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{C}) \mid (Dx, y) + (x, Dy) = 0 \}$	$\mathfrak{D}_4 = \{ D \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{C}) \mid (Dx, y) + (x, Dy) = 0 \}$

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0902.0431_E00275	44	■ 1.000	$\mathfrak{D}_{\{4\}} \ni D_{\{1\}} \longrightarrow \delta \in \mathfrak{D}_{\{4\}}$	$\mathfrak{D}_4 \ni D_1 \longrightarrow \delta \in \mathfrak{D}_4$	$\mathfrak{D}_4 \ni D_1 \longrightarrow \delta \in \mathfrak{D}_4$
0902.0431_E00276	44	■ 1.000	$\delta \left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array} \right) = \left(\begin{array}{ccc} 0 & D_3 x_3 & \overline{D_2 x_2} \\ \overline{D_3 x_3} & 0 & D_1 x_1 \\ D_2 x_2 & \overline{D_1 x_1} & 0 \end{array} \right),$	$\delta \left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array} \right) = \left(\begin{array}{ccc} 0 & D_3 x_3 & \overline{D_2 x_2} \\ \overline{D_3 x_3} & 0 & D_1 x_1 \\ D_2 x_2 & \overline{D_1 x_1} & 0 \end{array} \right),$	$\delta \left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array} \right) = \left(\begin{array}{ccc} 0 & D_3 x_3 & \overline{D_2 x_2} \\ \overline{D_3 x_3} & 0 & D_1 x_1 \\ D_2 x_2 & \overline{D_1 x_1} & 0 \end{array} \right),$
0902.0431_E00277	44	■ 1.000	$\left(D_{\{1\}} x \right) y + x \left(D_{\{2\}} y \right) = \overline{\overline{D_{\{3\}} (\overline{\overline{x y}})}} , \quad x, y \in \mathfrak{C}.$	$(D_1 x) y + x (D_2 y) = \overline{D_3 (\overline{xy})}, \quad x, y \in \mathfrak{C}.$	$(D_1 x) y + x (D_2 y) = \overline{D_3 (\overline{xy})}, \quad x, y \in \mathfrak{C}.$
0902.0431_E00281	45	■ 1.000	$\delta F_{\{i\}}(x) = F_{\{i\}} \left(D_{\{i\}} x \right), \quad x \in \mathfrak{C},$	$\delta F_i(x) = F_i(D_i x), \quad x \in \mathfrak{C},$	$\delta F_i(x) = F_i(D_i x), \quad x \in \mathfrak{C},$
0902.0431_E00284	45	■ 1.000	$\delta = D + \widetilde{A}, \quad D \in \mathfrak{D}_{\{4\}}, A \in \mathfrak{M}^-, \text{diag } A = 0,$	$\delta = D + \widetilde{A}, \quad D \in \mathfrak{D}_4, A \in \mathfrak{M}^-, \text{diag } A = 0,$	$\delta = D + \widetilde{A}, \quad D \in \mathfrak{D}_4, A \in \mathfrak{M}^-, \text{diag } A = 0,$
0902.0431_E00285	45	■ 1.000	$\text{operatorname{dim}} \mathfrak{f}_{\{4\}} = 28 + 24 = 52.$	$\dim \mathfrak{f}_4 = 28 + 24 = 52.$	$\dim \mathfrak{f}_4 = 28 + 24 = 52.$
0902.0431_E00287	46	■ 1.000	$\delta E_{\{1\}} = \left(\begin{array}{ccc} 0 & -a_3 & \bar{a}_2 \\ -\bar{a}_3 & 0 & 0 \\ a_2 & 0 & 0 \end{array} \right), \delta E_{\{2\}} = \left(\begin{array}{ccc} 0 & a_3 & 0 \\ \bar{a}_3 & 0 & -a_1 \\ 0 & -\bar{a}_1 & 0 \end{array} \right), \delta E_{\{3\}} = \left(\begin{array}{ccc} 0 & 0 & -\bar{a}_2 \\ 0 & 0 & a_1 \\ -a_2 & \bar{a}_1 & 0 \end{array} \right),$	$\delta E_1 = \left(\begin{array}{ccc} 0 & -a_3 & \bar{a}_2 \\ -\bar{a}_3 & 0 & 0 \\ a_2 & 0 & 0 \end{array} \right), \delta E_2 = \left(\begin{array}{ccc} 0 & a_3 & 0 \\ \bar{a}_3 & 0 & -a_1 \\ 0 & -\bar{a}_1 & 0 \end{array} \right), \delta E_3 = \left(\begin{array}{ccc} 0 & 0 & -\bar{a}_2 \\ 0 & 0 & a_1 \\ -a_2 & \bar{a}_1 & 0 \end{array} \right),$	$\delta E_1 = \left(\begin{array}{ccc} 0 & -a_3 & \bar{a}_2 \\ -\bar{a}_3 & 0 & 0 \\ a_2 & 0 & 0 \end{array} \right), \delta E_2 = \left(\begin{array}{ccc} 0 & a_3 & 0 \\ \bar{a}_3 & 0 & -a_1 \\ 0 & -\bar{a}_1 & 0 \end{array} \right), \delta E_3 = \left(\begin{array}{ccc} 0 & 0 & -\bar{a}_2 \\ 0 & 0 & a_1 \\ -a_2 & \bar{a}_1 & 0 \end{array} \right),$
0902.0431_E00288	46	■ 1.000	$\delta E_{\{i\}} = \widetilde{A} E_{\{i\}}, \quad i = 1, 2, 3.$	$\delta E_i = \widetilde{A} E_i, \quad i = 1, 2, 3.$	$\delta E_i = \widetilde{A} E_i, \quad i = 1, 2, 3.$
0902.0431_E00290	46	■ 1.000	$\overline{X_{\{1\}} + i X_{\{2\}}} = \bar{X}_{\{1\}} + i \bar{X}_{\{2\}}, \quad \tau \left(X_{\{1\}} + i X_{\{2\}} \right) = X_{\{1\}} - i X_{\{2\}}, \quad X_{\{i\}} \in \mathfrak{J}.$	$\overline{X_1 + i X_2} = \bar{X}_1 + i \bar{X}_2, \quad \tau(X_1 + i X_2) = X_1 - i X_2, \quad X_i \in \mathfrak{J}.$	$\overline{X_1 + i X_2} = \bar{X}_1 + i \bar{X}_2, \quad \tau(X_1 + i X_2) = X_1 - i X_2, \quad X_i \in \mathfrak{J}.$
0902.0431_E00291	46	■ 1.000	$\tau(X \circ Y) = \tau X \circ \tau Y, \quad \tau(X \times Y) = \tau X \times \tau Y, \quad X, Y \in \mathfrak{J}^C.$	$\tau(X \circ Y) = \tau X \circ \tau Y, \quad \tau(X \times Y) = \tau X \times \tau Y, \quad X, Y \in \mathfrak{J}^C.$	$\tau(X \circ Y) = \tau X \circ \tau Y, \quad \tau(X \times Y) = \tau X \times \tau Y, \quad X, Y \in \mathfrak{J}^C.$
0902.0431_E00294	47	■ 1.000	$= (A, (\text{operatorname{det}} X) E) = (\text{operatorname{det}} X) (A, E) = (\text{operatorname{det}} X) \text{operatorname{tr}}(A) = 0, \quad X \in \mathfrak{J}^C.$	$= (A, (\det X) E) = (\det X) (A, E) = (\det X) \text{tr}(A) = 0, \quad X \in \mathfrak{J}^C.$	$= (A, (\det X) E) = (\det X) (A, E) = (\det X) \text{tr}(A) = 0, \quad X \in \mathfrak{J}^C.$
0902.0431_E00303	48	■ 1.000	$\widetilde{A}_{\{1\}}(1) + \widetilde{A}_{\{2\}}(b) + \widetilde{A}_{\{3\}}(c) \in \mathfrak{a}.$	$\widetilde{A}_1(1) + \widetilde{A}_2(b) + \widetilde{A}_3(c) \in \mathfrak{a}.$	$\widetilde{A}_1(1) + \widetilde{A}_2(b) + \widetilde{A}_3(c) \in \mathfrak{a}.$
0902.0431_E00305	49	■ 1.000	$-\widetilde{A}_{\{3\}}(\bar{b}) + \widetilde{A}_{\{2\}}(\bar{c}) \in \mathfrak{a} \quad \text{and then} \quad \widetilde{A}_3(1) + \widetilde{A}_2(c') \in \mathfrak{a}$	$-\widetilde{A}_3(\bar{b}) + \widetilde{A}_2(\bar{c}) \in \mathfrak{a} \quad \text{and then} \quad \widetilde{A}_3(1) + \widetilde{A}_2(c') \in \mathfrak{a}$	$-\widetilde{A}_3(\bar{b}) + \widetilde{A}_2(\bar{c}) \in \mathfrak{a} \quad \text{and then} \quad \widetilde{A}_3(1) + \widetilde{A}_2(c') \in \mathfrak{a}$
0902.0431_E00306	49	■ 1.000	$[\delta, [\widetilde{A}, \widetilde{B}]] = [\widetilde{\delta A}, \widetilde{B}] + [\widetilde{A}, \widetilde{\delta B}].$	$[\delta, [\widetilde{A}, \widetilde{B}]] = [\widetilde{\delta A}, \widetilde{B}] + [\widetilde{A}, \widetilde{\delta B}].$	$[\delta, [\widetilde{A}, \widetilde{B}]] = [\widetilde{\delta A}, \widetilde{B}] + [\widetilde{A}, \widetilde{\delta B}].$
0902.0431_E00308	49	■ 1.000	$F_{\{i\}}(1) \xrightarrow{\widetilde{A}_{\{i\}}(1)} E_{i+1} - E_{i+2} \xrightarrow{\widetilde{A}_{\{i+2\}}(1)} F_{i+2}(1) \longrightarrow \cdots \longrightarrow F_{i+1}(1),$	$F_i(1) \xrightarrow{\widetilde{A}_i(1)} E_{i+1} - E_{i+2} \xrightarrow{\widetilde{A}_{i+2}(1)} F_{i+2}(1) \longrightarrow \cdots \longrightarrow F_{i+1}(1),$	$F_i(1) \xrightarrow{\widetilde{A}_i(1)} E_{i+1} - E_{i+2} \xrightarrow{\widetilde{A}_{i+2}(1)} F_{i+2}(1) \longrightarrow \cdots \longrightarrow F_{i+1}(1),$

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0902.0431_E00309	49	■ 1.000	$\theta \neq X=\backslash xi\left(E_{\{1\}}-E_{\{2\}}\right)+\eta \left(E_{\{2\}}-E_{\{3\}}\right)+F_{\{1\}}\left(x_{\{1\}}\right)+F_{\{2\}}\left(x_{\{2\}}\right)+F_{\{3\}}\left(x_{\{3\}}\right) \in W .$	$0 \neq X=\xi\left(E_1-E_2\right)+\eta\left(E_2-E_3\right)+F_1\left(x_1\right)+F_2\left(x_2\right)+F_3\left(x_3\right) \in W.$	$0 \neq X=\xi\left(E_1-E_2\right)+\eta\left(E_2-E_3\right)+F_1\left(x_1\right)+F_2\left(x_2\right)+F_3\left(x_3\right) \in W.$
0902.0431_E00310	50	■ 1.000	$F_{\{1\}}(1)+F_{\{2\}}\left(y_{\{2\}}\right)+F_{\{3\}}\left(y_{\{3\}}\right) \in W .$	$F_1(1)+F_2\left(y_2\right)+F_3\left(y_3\right) \in W.$	$F_1(1)+F_2\left(y_2\right)+F_3\left(y_3\right) \in W.$
0902.0431_E00312	50	■ 1.000	$X=\backslash xi\left(E_{\{1\}}-E_{\{2\}}\right)+\eta \left(E_{\{2\}}-E_{\{3\}}\right) \in W,$	$X=\xi\left(E_1-E_2\right)+\eta\left(E_2-E_3\right) \in W,$	$X=\xi\left(E_1-E_2\right)+\eta\left(E_2-E_3\right) \in W,$
0902.0431_E00313	50	■ 1.000	$([\widetilde{A}, \widetilde{B}] C, D)=([\widetilde{C}, \widetilde{D}] A, B) .$	$([\tilde{A}, \tilde{B}] C, D)=([\tilde{C}, \tilde{D}] A, B).$	$([\tilde{A}, \tilde{B}] C, D)=([\tilde{C}, \tilde{D}] A, B).$
0902.0431_E00316	50	■ 1.000	$\left(\delta_1, \delta_2\right)_4=\sum_{i, j}\left(\left[\widetilde{A}_i, \widetilde{B}_i\right] C_j, D_j\right)=\sum_{i, j}\left(\left[\widetilde{C}_j, \widetilde{D}_j\right] A_i, B_i\right),$	$\left(\delta_1, \delta_2\right)_4=\sum_{i, j}\left(\left[\tilde{A}_i, \tilde{B}_i\right] C_j, D_j\right)=\sum_{i, j}\left(\left[\tilde{C}_j, \tilde{D}_j\right] A_i, B_i\right),$	$\left(\delta_1, \delta_2\right)_4=\sum_{i, j}\left([\tilde{A}_i, \tilde{B}_i] C_j, D_j\right)=\sum_{i, j}\left([\tilde{C}_j, \tilde{D}_j] A_i, B_i\right),$
0902.0431_E00326	52	■ 1.000	$\mathfrak{D}_{\{4\}}^{\{C\}}=\left\{D \in \operatorname{Hom}_{\{C\}}\left(\mathfrak{C}^{\{C\}}\right) \mid(D x, y)+(x, D y)=0\right\} .$	$\mathfrak{D}_4^C=\left\{D \in \operatorname{Hom}_C\left(\mathfrak{C}^C\right) \mid(D x, y)+(x, D y)=0\right\} .$	$\mathfrak{D}_4^C=\left\{D \in \operatorname{Hom}_C\left(\mathfrak{C}^C\right) \mid(D x, y)+(x, D y)=0\right\} .$
0902.0431_E00327	52	■ 1.000	$\mathfrak{h}=\left\{H=\sum_{k=0}^3 \lambda_k H_k \mid \lambda_k \in C\right\}$	$\mathfrak{h}=\left\{H=\sum_{k=0}^3 \lambda_k H_k \mid \lambda_k \in C\right\}$	$\mathfrak{h}=\left\{H=\sum_{k=0}^3 \lambda_k H_k \mid \lambda_k \in C\right\}$
0902.0431_E00331	53	■ 1.000	$\nu=\pi \quad \kappa=\frac{1}{2} \left(\begin{array}{rrrr} -1 & -1 & 1 & -1 \\ 1 & 1 & 1 & -1 \\ -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \end{array}\right), \quad \pi=\frac{1}{2} \left(\begin{array}{rrrr} 1 & -1 & 1 & -1 \\ -1 & 1 & 1 & -1 \\ 1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 \end{array}\right)$	$\nu=\pi \kappa=\frac{1}{2} \left(\begin{array}{rrrr} -1 & -1 & 1 & -1 \\ 1 & 1 & 1 & -1 \\ -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \end{array}\right), \quad \pi=\frac{1}{2} \left(\begin{array}{rrrr} 1 & -1 & 1 & -1 \\ -1 & 1 & 1 & -1 \\ 1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 \end{array}\right)$	$\nu=\pi \kappa=\frac{1}{2} \left(\begin{array}{rrrr} -1 & -1 & 1 & -1 \\ 1 & 1 & 1 & -1 \\ -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \end{array}\right), \quad \pi=\frac{1}{2} \left(\begin{array}{rrrr} 1 & -1 & 1 & -1 \\ -1 & 1 & 1 & -1 \\ 1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 \end{array}\right)$
0902.0431_E00334	53	■ 1.000	$\pm\left(\lambda_k-\lambda_l\right), \quad \pm\left(\lambda_k+\lambda_l\right), \quad 0 \leq k<l \leq 3$	$\pm\left(\lambda_k-\lambda_l\right), \quad \pm\left(\lambda_k+\lambda_l\right), \quad 0 \leq k<l \leq 3$	$\pm\left(\lambda_k-\lambda_l\right), \quad \pm\left(\lambda_k+\lambda_l\right), \quad 0 \leq k<l \leq 3$
0902.0431_E00336	54	■ 1.000	$\begin{aligned} \nu H= & \quad \nu \left(\sum_{k=0}^3 \lambda_k H_k\right) \\ & =\frac{1}{2}\left(\lambda_0\left(-H_0+H_1-H_2+H_3\right)+\lambda_1\left(-H_0+H_1+H_2-H_3\right)\right. \\ & \quad \left.+\lambda_2\left(H_0+H_1+H_2+H_3\right)+\lambda_3\left(-H_0-H_1+H_2+H_3\right)\right) \\ & =\frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) H_0+\frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) H_1 \\ & \quad +\frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) H_2+\frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) H_3, \end{aligned}$	$\begin{aligned} \nu H= & \nu \left(\sum_{k=0}^3 \lambda_k H_k\right) \\ & =\frac{1}{2}\left(\lambda_0\left(-H_0+H_1-H_2+H_3\right)+\lambda_1\left(-H_0+H_1+H_2-H_3\right)\right. \\ & \quad \left.+\lambda_2\left(H_0+H_1+H_2+H_3\right)+\lambda_3\left(-H_0-H_1+H_2+H_3\right)\right) \\ & =\frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) H_0+\frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) H_1 \\ & \quad +\frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) H_2+\frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) H_3, \end{aligned}$	$\begin{aligned} \nu H= & \nu \left(\sum_{k=0}^3 \lambda_k H_k\right) \\ & =\frac{1}{2}\left(\lambda_0\left(-H_0+H_1-H_2+H_3\right)+\lambda_1\left(-H_0+H_1+H_2-H_3\right)\right. \\ & \quad \left.+\lambda_2\left(H_0+H_1+H_2+H_3\right)+\lambda_3\left(-H_0-H_1+H_2+H_3\right)\right) \\ & =\frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) H_0+\frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) H_1 \\ & \quad +\frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) H_2+\frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) H_3, \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00338	54	■ 1.000	$\backslash alpha_{\{1\}}=\backslash lambda_{\{0\}}-\backslash lambda_{\{1\}}, \quad \backslash quad \backslash alpha_{\{2\}}=\backslash lambda_{\{1\}}-\backslash lambda_{\{2\}}, \quad \backslash quad \backslash alpha_{\{3\}}=\backslash lambda_{\{2\}}, \quad \backslash quad \backslash alpha_{\{4\}}=\frac{1}{2}\{2\}\backslash left(-\backslash lambda_{\{0\}}-\backslash lambda_{\{1\}}-\backslash lambda_{\{2\}}+\backslash lambda_{\{3\}}\backslash right)$	$\alpha_1 = \lambda_0 - \lambda_1, \quad \alpha_2 = \lambda_1 - \lambda_2, \quad \alpha_3 = \lambda_2, \quad \alpha_4 = \frac{1}{2}(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3)$	$\alpha_1 = \lambda_0 - \lambda_1, \quad \alpha_2 = \lambda_1 - \lambda_2, \quad \alpha_3 = \lambda_2, \quad \alpha_4 = \frac{1}{2}(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3)$
0902.0431_E00339	54	■ 1.000	$\backslash mu=2 \backslash alpha_{\{1\}}+3 \backslash alpha_{\{2\}}+4 \backslash alpha_{\{3\}}+2 \backslash alpha_{\{4\}}$	$\mu = 2\alpha_1 + 3\alpha_2 + 4\alpha_3 + 2\alpha_4$	$\mu = 2\alpha_1 + 3\alpha_2 + 4\alpha_3 + 2\alpha_4$
0902.0431_E00342	55	■ 1.000	$\backslash mathfrak{h}_{\{\boldsymbol{R}\}}=\backslash left\{H=\sum_{k=0}^{\{3\}} \backslash lambda_{\{k\}} H_{\{k\}} \mid \backslash lambda_{\{k\}} \in \backslash boldsymbol{R}\} \backslash right\} \text{ .}$	$\mathfrak{h}_{\boldsymbol{R}} = \left\{ H = \sum_{k=0}^3 \lambda_k H_k \mid \lambda_k \in \boldsymbol{R} \right\}.$	$\mathfrak{h}_{\boldsymbol{R}} = \{ H = \sum_{k=0}^3 \lambda_k H_k \mid \lambda_k \in \boldsymbol{R} \}.$
0902.0431_E00347	56	■ 1.000	$\backslash left(\backslash alpha_{\{1\}}, \backslash alpha_{\{1\}}\backslash right)=B_{\{4\}}\backslash left(H_{\{ \backslash alpha_{\{1\}} \}}, H_{\{ \backslash alpha_{\{1\}} \}}\backslash right)=18 \frac{1}{18} \frac{1}{18} \backslash frac{1}{18} \{1+1\}=\backslash frac{1}{9}$	$(\alpha_1, \alpha_1) = B_4(H_{\alpha_1}, H_{\alpha_1}) = 18 \frac{1}{18} \frac{1}{18} (1 + 1) = \frac{1}{9}$	$(\alpha_1, \alpha_1) = B_4(H_{\alpha_1}, H_{\alpha_1}) = 18 \frac{1}{18} \frac{1}{18} (1 + 1) = \frac{1}{9}$
0902.0431_E00352	57	■ 1.000	$\backslash alpha \text{ F}_{\{i\}}(x)=F_{\{i\}}\backslash left(\backslash alpha_{\{i\}} \text{ x}\backslash right), \quad \backslash quad \text{ x} \backslash in \backslash mathfrak{C},$	$\alpha F_i(x) = F_i(\alpha_i x), \quad x \in \mathfrak{C},$	$\alpha F_i(x) = F_i(\alpha_i x), \quad x \in \mathfrak{C},$
0902.0431_E00353	57	■ 1.000	$\backslash left(\backslash alpha_{\{i\}} \text{ x}, \backslash alpha_{\{i\}} \text{ y}\backslash right)=(x, y), \quad \backslash quad \text{ x}, \text{ y} \backslash in \backslash mathfrak{C} \text{ .}$	$(\alpha_i x, \alpha_i y) = (x, y), \quad x, y \in \mathfrak{C}.$	$(\alpha_i x, \alpha_i y) = (x, y), \quad x, y \in \mathfrak{C}.$
0902.0431_E00355	57	■ 1.000	$\backslash left(F_{\{4\}}\backslash right)_{\{E_{\{1\}}\}}=\backslash left\{\backslash alpha \backslash in F_{\{4\}} \mid \backslash alpha \text{ E}_{\{1\}}=E_{\{1\}}\backslash right\} \text{ .}$	$(F_4)_{E_1} = \{ \alpha \in F_4 \mid \alpha E_1 = E_1 \}.$	$(F_4)_{E_1} = \{ \alpha \in F_4 \mid \alpha E_1 = E_1 \}.$
0902.0431_E00356	57	■ 1.000	$\backslash mathfrak{J}_{\{01\}}=\backslash left\{X \backslash in \backslash mathfrak{J} \mid \backslash mid E_{\{1\}} \backslash circ X=0, \backslash operatorname{tr}(X)=0\backslash right\}=\backslash left\{\backslash left. \backslash left(\backslash begin{array}{ccc} 0 \& 0 \& 0 \backslash \ 0 \& \backslash xi \& x \backslash \ 0 \& \backslash bar{x} \& -\backslash xi \backslash end{array}\backslash right) \backslash right\backslash rvert\, \backslash xi \backslash in \backslash boldsymbol{R}, \text{ x} \backslash in \backslash mathfrak{C}\backslash right\},$	$\mathfrak{J}_{01} = \{ X \in \mathfrak{J} \mid E_1 \circ X = 0, \mathrm{tr}(X) = 0 \} = \left\{ \left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\xi \end{array} \right) \middle \xi \in \boldsymbol{R}, x \in \mathfrak{C} \right\},$	$\mathfrak{J}_{01} = \{ X \in \mathfrak{J} \mid E_1 \circ X = 0, \mathrm{tr}(X) = 0 \} = \left\{ \left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\xi \end{array} \right) \middle \xi \in \boldsymbol{R}, x \in \mathfrak{C} \right\},$
0902.0431_E00357	58	■ 1.000	$\backslash mathfrak{J}_{\{23\}}=\backslash left\{Y \backslash in \backslash mathfrak{J} \mid \backslash mid 2 \text{ E}_{\{1\}} \backslash circ Y=Y\backslash right\}=\backslash left\{\backslash left.\backslash left(\backslash begin{array}{ccc} 0 \& y_{\{3\}} \& \backslash bar{y}_{\{2\}} \backslash \ \backslash bar{y}_{\{3\}} \& 0 \& 0 \backslash \ y_{\{2\}} \& 0 \& 0 \backslash end{array}\backslash right) \backslash right\backslash rvert\, y_{\{2\}}, y_{\{3\}} \backslash in \backslash mathfrak{C}\backslash right\} \text{ .}$	$\mathfrak{J}_{23} = \{ Y \in \mathfrak{J} \mid 2E_1 \circ Y = Y \} = \left\{ \left(\begin{array}{ccc} 0 & y_3 & \bar{y}_2 \\ \bar{y}_3 & 0 & 0 \\ y_2 & 0 & 0 \end{array} \right) \middle y_2, y_3 \in \mathfrak{C} \right\}.$	$\mathfrak{J}_{23} = \{ Y \in \mathfrak{J} \mid 2E_1 \circ Y = Y \} = \left\{ \left(\begin{array}{ccc} 0 & y_3 & \bar{y}_2 \\ \bar{y}_3 & 0 & 0 \\ y_2 & 0 & 0 \end{array} \right) \middle y_2, y_3 \in \mathfrak{C} \right\}.$
0902.0431_E00358	58	■ 1.000	$Z_{\{0\}}=2 \text{ X}_{\{0\}} \backslash circ Y_{\{0\}} \text{ .}$	$Z_0 = 2X_0 \circ Y_0.$	$Z_0 = 2X_0 \circ Y_0.$
0902.0431_E00359	58	■ 1.000	$Y_{\{1\}}=-2 \text{ Z}_{\{0\}} \backslash circ X_{\{1\}}, \quad \backslash quad \text{ Z}_{\{2\}}=-2 \text{ X}_{\{2\}} \backslash circ Y_{\{0\}}, \quad \backslash quad \text{ X}_{\{3\}}=-2 \text{ Y}_{\{1\}} \backslash circ \text{ Z}_{\{2\}} \text{ .}$	$Y_1 = -2Z_0 \circ X_1, \quad Z_2 = -2X_2 \circ Y_0, \quad X_3 = -2Y_1 \circ Z_2.$	$Y_1 = -2Z_0 \circ X_1, \quad Z_2 = -2X_2 \circ Y_0, \quad X_3 = -2Y_1 \circ Z_2.$
0902.0431_E00360	58	■ 1.000	$\backslash begin{array}{lll} \text{ Z}_{\{4\}}=-2 \text{ X}_{\{4\}} \backslash circ Y_{\{0\}}, \& \text{ Y}_{\{2\}}=-2 \text{ Z}_{\{0\}} \backslash circ X_{\{2\}}, \& \text{ Y}_{\{3\}}=-2 \text{ Z}_{\{0\}} \backslash circ X_{\{3\}}, \backslash \backslash \text{ X}_{\{5\}}=-2 \text{ Y}_{\{1\}} \backslash circ \text{ Z}_{\{4\}}, \& \text{ X}_{\{6\}}=2 \text{ Y}_{\{2\}} \backslash circ \text{ Z}_{\{4\}}, \& \text{ X}_{\{7\}}=-2 \text{ Y}_{\{3\}} \backslash circ \text{ Z}_{\{4\}} \backslash end{array}$	$Z_4 = -2X_4 \circ Y_0, \quad Y_2 = -2Z_0 \circ X_2, \quad Y_3 = -2Z_0 \circ X_3, \quad X_5 = -2Y_1 \circ Z_4, \quad X_6 = 2Y_2 \circ Z_4, \quad X_7 = -2Y_3 \circ Z_4$	$Z_4 = -2X_4 \circ Y_0, \quad Y_2 = -2Z_0 \circ X_2, \quad Y_3 = -2Z_0 \circ X_3, \quad X_5 = -2Y_1 \circ Z_4, \quad X_6 = 2Y_2 \circ Z_4, \quad X_7 = -2Y_3 \circ Z_4$
0902.0431_E00363	58	■ 1.000	$\backslash alpha(X \backslash circ Y)=\backslash alpha X \backslash circ \backslash alpha Y, \quad \backslash quad X, Y \backslash in \backslash mathfrak{J} \text{ .}$	$\alpha(X \circ Y) = \alpha X \circ \alpha Y, \quad X, Y \in \mathfrak{J}.$	$\alpha(X \circ Y) = \alpha X \circ \alpha Y, \quad X, Y \in \mathfrak{J}.$
0902.0431_E00366	59	■ 1.000	$\backslash left(F_{\{4\}}\backslash right)_{\{E_{\{1\}}\}} \wedge \{1, \backslash sigma\} \backslash cong S \text{ 0}(9) \text{ .}$	$(F_4)_{E_1} / \{1, \sigma\} \cong SO(9).$	$(F_4)_{E_1} / \{1, \sigma\} \cong SO(9).$
0902.0431_E00367	60	■ 1.000	$f\backslash left(\backslash alpha^{\backslash prime}\backslash right)=\backslash varepsilon \backslash alpha^{\backslash prime} \backslash varepsilon \backslash left\{-1\}, \quad \backslash quad(\backslash epsilon \text{ x}=\backslash bar{x}, \text{ x} \backslash in \backslash mathfrak{C})$	$f(\alpha') = \varepsilon \alpha' \varepsilon^{-1}, \quad (\epsilon x = \bar{x}, x \in \mathfrak{C})$	$f(\alpha') = \varepsilon \alpha' \varepsilon^{-1}, \quad (\epsilon x = \bar{x}, x \in \mathfrak{C})$
0902.0431_E00369	61	■ 1.000	$\backslash alpha \text{ X}=\backslash left(\backslash begin{array}{ccc} \backslash xi_{\{1\}} \& 0 \& 0 \backslash \ 0 \& \backslash xi_{\{2\}} \& 0 \backslash \ 0 \& 0 \& \backslash xi_{\{3\}} \backslash end{array}\backslash right), \quad \backslash quad \backslash xi_{\{i\}} \backslash in \backslash boldsymbol{R}$	$\alpha X = \left(\begin{array}{ccc} \xi_1 & 0 & 0 \\ 0 & \xi_2 & 0 \\ 0 & 0 & \xi_3 \end{array} \right), \quad \xi_i \in \boldsymbol{R}$	$\alpha X = \left(\begin{array}{ccc} \xi_1 & 0 & 0 \\ 0 & \xi_2 & 0 \\ 0 & 0 & \xi_3 \end{array} \right), \quad \xi_i \in \boldsymbol{R}.$

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0902.0431_E00373	62	■ 1.000	$\mathfrak{C}P_2=\left\{X\in\mathfrak{J}\mid X^2=X,\operatorname{tr}(X)=1\right\}.$	$\mathfrak{C}P_2=\{X\in\mathfrak{J}\mid X^2=X,\operatorname{tr}(X)=1\}.$	$\mathfrak{C}P_2=\{X\in\mathfrak{J}\mid X^2=X,\operatorname{tr}(X)=1\}.$
0902.0431_E00375	62	■ 1.000	$\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right),\quad \xi_i\in\boldsymbol{R}$	$\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right),\quad \xi_i\in\boldsymbol{R}$	$\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right),\quad \xi_i\in\boldsymbol{R}$
0902.0431_E00376	62	■ 1.000	$\left(\begin{array}{ccc}\xi_1^2&0&0\\0&\xi_2^2&0\\0&0&\xi_3^2\end{array}\right)=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right).$	$\left(\begin{array}{ccc}\xi_1^2&0&0\\0&\xi_2^2&0\\0&0&\xi_3^2\end{array}\right)=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right).$	$\left(\begin{array}{ccc}\xi_1^2&0&0\\0&\xi_2^2&0\\0&0&\xi_3^2\end{array}\right)=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right).$
0902.0431_E00377	63	■ 1.000	$\sigma\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\ \bar{x}_3&\xi_2&x_1\\ x_2&\bar{x}_1&\xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1&-x_3&-\bar{x}_2\\ -\bar{x}_3&\xi_2&x_1\\ -x_2&\bar{x}_1&\xi_3\end{array}\right).$	$\sigma\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\ \bar{x}_3&\xi_2&x_1\\ x_2&\bar{x}_1&\xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1&-x_3&-\bar{x}_2\\ -\bar{x}_3&\xi_2&x_1\\ -x_2&\bar{x}_1&\xi_3\end{array}\right).$	$\sigma\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\ \bar{x}_3&\xi_2&x_1\\ x_2&\bar{x}_1&\xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1&-x_3&-\bar{x}_2\\ -\bar{x}_3&\xi_2&x_1\\ -x_2&\bar{x}_1&\xi_3\end{array}\right).$
0902.0431_E00378	63	■ 1.000	$\left(F_4\right)^{\sigma}=\left\{\alpha\in F_4\mid\sigma\alpha=\alpha\sigma\right\}.$	$(F_4)^{\sigma}=\{\alpha\in F_4\mid\sigma\alpha=\alpha\sigma\}.$	$(F_4)^{\sigma}=\{\alpha\in F_4\mid\sigma\alpha=\alpha\sigma\}.$
0902.0431_E00380	63	■ 1.000	$\alpha E_2,\alpha E_3\in\mathfrak{J}(2,\mathfrak{C}).$	$\alpha E_2,\alpha E_3\in\mathfrak{J}(2,\mathfrak{C}).$	$\alpha E_2,\alpha E_3\in\mathfrak{J}(2,\mathfrak{C}).$
0902.0431_E00382	63	■ 1.000	$\alpha E_1=E_1+\xi_2E_2+\xi_3E_3+F_1(x_1).$	$\alpha E_1=E_1+\xi_2E_2+\xi_3E_3+F_1(x_1).$	$\alpha E_1=E_1+\xi_2E_2+\xi_3E_3+F_1(x_1).$
0902.0431_E00385	64	■ 1.000	$z\left(F_4\right)=\{1\}.$	$z\left(F_4\right)=\{1\}.$	$z(F_4)=\{1\}.$
0902.0431_E00386	64	■ 1.000	$\alpha=1\quad\text{or}\quad\alpha=\sigma.$	$\alpha=1\quad\text{or}\quad\alpha=\sigma.$	$\alpha=1\quad\text{or}\quad\alpha=\sigma.$
0902.0431_E00388	65	■ 1.000	$\left(F_4\right)^{\gamma}=\left\{\alpha\in F_4\mid\gamma\alpha=\alpha\gamma\right\}.$	$(F_4)^{\gamma}=\{\alpha\in F_4\mid\gamma\alpha=\alpha\gamma\}.$	$(F_4)^{\gamma}=\{\alpha\in F_4\mid\gamma\alpha=\alpha\gamma\}.$
0902.0431_E00389	65	■ 1.000	$X=\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\ \bar{x}_3&\xi_2&x_1\\ x_2&\bar{x}_1&\xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1&m_3&\bar{m}_2\\ \bar{m}_3&\xi_2&m_1\\ m_2&\bar{m}_1&\xi_3\end{array}\right)+\left(\begin{array}{ccc}0&a_3e_4&-a_2e_4\\ -a_3e_4&0&a_1e_4\\ a_2e_4&-a_1e_4&0\end{array}\right),$	$X=\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\ \bar{x}_3&\xi_2&x_1\\ x_2&\bar{x}_1&\xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1&m_3&\bar{m}_2\\ \bar{m}_3&\xi_2&m_1\\ m_2&\bar{m}_1&\xi_3\end{array}\right)+\left(\begin{array}{ccc}0&a_3e_4&-a_2e_4\\ -a_3e_4&0&a_1e_4\\ a_2e_4&-a_1e_4&0\end{array}\right),$	$X=\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\ \bar{x}_3&\xi_2&x_1\\ x_2&\bar{x}_1&\xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1&m_3&\bar{m}_2\\ \bar{m}_3&\xi_2&m_1\\ m_2&\bar{m}_1&\xi_3\end{array}\right)+\left(\begin{array}{ccc}0&a_3e_4&-a_2e_4\\ -a_3e_4&0&a_1e_4\\ a_2e_4&-a_1e_4&0\end{array}\right),$
0902.0431_E00390	65	■ 1.000	$\left(\begin{array}{ccc}\xi_1&m_3&\bar{m}_2\\ \bar{m}_3&\xi_2&m_1\\ m_2&\bar{m}_1&\xi_3\end{array}\right)+(a_1,a_2,a_3).$	$\left(\begin{array}{ccc}\xi_1&m_3&\bar{m}_2\\ \bar{m}_3&\xi_2&m_1\\ m_2&\bar{m}_1&\xi_3\end{array}\right)+(a_1,a_2,a_3).$	$\left(\begin{array}{ccc}\xi_1&m_3&\bar{m}_2\\ \bar{m}_3&\xi_2&m_1\\ m_2&\bar{m}_1&\xi_3\end{array}\right)+(a_1,a_2,a_3).$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00391	65	■ 1.000	$\begin{aligned} (M+\boldsymbol{a}) \times (N+\boldsymbol{b}) &= \left(M \times N - \frac{1}{2}(\boldsymbol{a}^* \boldsymbol{b} + \boldsymbol{b}^* \boldsymbol{a})\right) - \frac{1}{2}(\boldsymbol{a} N + \boldsymbol{b} M), \\ (M+\boldsymbol{a}, N+\boldsymbol{b}) &= (M, N) + 2(\boldsymbol{a}, \boldsymbol{b}), \\ \gamma(M+\boldsymbol{a}) &= M - \boldsymbol{a}. \end{aligned}$	$(M+\boldsymbol{a}) \times (N+\boldsymbol{b}) = \left(M \times N - \frac{1}{2}(\boldsymbol{a}^* \boldsymbol{b} + \boldsymbol{b}^* \boldsymbol{a})\right) - \frac{1}{2}(\boldsymbol{a} N + \boldsymbol{b} M),$ $(M+\boldsymbol{a}, N+\boldsymbol{b}) = (M, N) + 2(\boldsymbol{a}, \boldsymbol{b}),$ $\gamma(M+\boldsymbol{a}) = M - \boldsymbol{a}.$	$(M+\boldsymbol{a}) \times (N+\boldsymbol{b}) = \left(M \times N - \frac{1}{2}(\boldsymbol{a}^* \boldsymbol{b} + \boldsymbol{b}^* \boldsymbol{a})\right) - \frac{1}{2}(\boldsymbol{a} N + \boldsymbol{b} M),$ $(M+\boldsymbol{a}, N+\boldsymbol{b}) = (M, N) + 2(\boldsymbol{a}, \boldsymbol{b}),$ $\gamma(M+\boldsymbol{a}) = M - \boldsymbol{a}.$
0902.0431_E00392	65	■ 1.000	$\mathfrak{J}(3, \boldsymbol{H}) \oplus \boldsymbol{H}^3 = \mathfrak{J}.$	$\mathfrak{J}(3, \boldsymbol{H}) \oplus \boldsymbol{H}^3 = \mathfrak{J}.$	$\mathfrak{J}(3, \boldsymbol{H}) \oplus \boldsymbol{H}^3 = \mathfrak{J}.$
0902.0431_E00394	65	■ 1.000	$\varphi(A)M = AMA^*, \quad M \in \mathfrak{J}_H.$	$\varphi(A)M = AMA^*, \quad M \in \mathfrak{J}_H.$	$\varphi(A)M = AMA^*, \quad M \in \mathfrak{J}_H.$
0902.0431_E00395	66	■ 1.000	$\alpha E_i = AE_i A^*, \quad i = 1, 2, 3.$	$\alpha E_i = AE_i A^*, \quad i = 1, 2, 3.$	$\alpha E_i = AE_i A^*, \quad i = 1, 2, 3.$
0902.0431_E00396	66	■ 1.000	$\beta E_i = E_i, \quad i = 1, 2, 3.$	$\beta E_i = E_i, \quad i = 1, 2, 3.$	$\beta E_i = E_i, \quad i = 1, 2, 3.$
0902.0431_E00397 (i)	66	■ 1.000	$\left(\beta_1 m\right) \left(\beta_2 n\right) = \overline{\beta_3(\overline{mn})}, \quad m, n \in \boldsymbol{H}.$	$(\beta_1 m)(\beta_2 n) = \overline{\beta_3(\overline{mn})}, \quad m, n \in \boldsymbol{H}.$	$(\beta_1 m)(\beta_2 n) = \overline{\beta_3(\overline{mn})}, \quad m, n \in \boldsymbol{H}.$
0902.0431_E00398	66	■ 1.000	$\beta_2 m = \bar{p}(\beta_1 m) q, \quad \beta_3 m = \overline{(\beta_1 \bar{m})} q.$	$\beta_2 m = \bar{p}(\beta_1 m) q, \quad \beta_3 m = \overline{(\beta_1 \bar{m})} q.$	$\beta_2 m = \bar{p}(\beta_1 m) q, \quad \beta_3 m = \overline{(\beta_1 \bar{m})} q.$
0902.0431_E00399	66	■ 1.000	$(\zeta m)(\zeta n) = \zeta(mn), \quad m, n \in \boldsymbol{H},$	$(\zeta m)(\zeta n) = \zeta(mn), \quad m, n \in \boldsymbol{H},$	$(\zeta m)(\zeta n) = \zeta(mn), \quad m, n \in \boldsymbol{H},$
0902.0431_E00401	66	■ 1.000	$\beta M = BMB^*, \quad M \in \mathfrak{J}_H,$	$\beta M = BMB^*, \quad M \in \mathfrak{J}_H,$	$\beta M = BMB^*, \quad M \in \mathfrak{J}_H,$
0902.0431_E00403	67	■ 1.000	$\varphi(p, A)(M+\boldsymbol{a}) = AMA^* + p\boldsymbol{a}A^*, \quad M+\boldsymbol{a} \in \mathfrak{J}_H \oplus \boldsymbol{H}^3 = \mathfrak{J}.$	$\varphi(p, A)(M+\boldsymbol{a}) = AMA^* + p\boldsymbol{a}A^*, \quad M+\boldsymbol{a} \in \mathfrak{J}_H \oplus \boldsymbol{H}^3 = \mathfrak{J}.$	$\varphi(p, A)(M+\boldsymbol{a}) = AMA^* + p\boldsymbol{a}A^*, \quad M+\boldsymbol{a} \in \mathfrak{J}_H \oplus \boldsymbol{H}^3 = \mathfrak{J}.$
0902.0431_E00406	67	■ 1.000	$\alpha M = AMA^*, \quad M \in \mathfrak{J}_H$	$\alpha M = AMA^*, \quad M \in \mathfrak{J}_H$	$\alpha M = AMA^*, \quad M \in \mathfrak{J}_H$
0902.0431_E00407	67	■ 1.000	$\beta(m+ae_4) = m + (pa)e_4, \quad m+ae_4 \in \boldsymbol{H} \oplus \boldsymbol{H}e_4 = \mathfrak{C},$	$\beta(m+ae_4) = m + (pa)e_4, \quad m+ae_4 \in \boldsymbol{H} \oplus \boldsymbol{H}e_4 = \mathfrak{C},$	$\beta(m+ae_4) = m + (pa)e_4, \quad m+ae_4 \in \boldsymbol{H} \oplus \boldsymbol{H}e_4 = \mathfrak{C},$
0902.0431_E00408	67	■ 1.000	$\beta(M+\boldsymbol{a}) = M + p\boldsymbol{a} = \varphi(p, E)(M+\boldsymbol{a}),$	$\beta(M+\boldsymbol{a}) = M + p\boldsymbol{a} = \varphi(p, E)(M+\boldsymbol{a}),$	$\beta(M+\boldsymbol{a}) = M + p\boldsymbol{a} = \varphi(p, E)(M+\boldsymbol{a}),$
0902.0431_E00411	68	■ 1.000	$(F_4)^w = \{\alpha \in F_4 \mid w\alpha = \alpha w\}.$	$(F_4)^w = \{\alpha \in F_4 \mid w\alpha = \alpha w\}.$	$(F_4)^w = \{\alpha \in F_4 \mid w\alpha = \alpha w\}.$
0902.0431_E00415	68	■ 1.000	$PM \times PN = {}^t \widetilde{P}(M \times N), \quad MA \times NA = (M \times N) {}^t \widetilde{A},$	$PM \times PN = {}^t \widetilde{P}(M \times N), \quad MA \times NA = (M \times N) {}^t \widetilde{A},$	$PM \times PN = {}^t \widetilde{P}(M \times N), \quad MA \times NA = (M \times N) {}^t \widetilde{A},$
0902.0431_E00416	68	■ 1.000	$(M, N) = \frac{1}{2} \operatorname{tr}(M^* N + N^* M) = \sum_{i,j} (m_{ij}, n_{ij}),$	$(M, N) = \frac{1}{2} \operatorname{tr}(M^* N + N^* M) = \sum_{i,j} (m_{ij}, n_{ij}),$	$(M, N) = \frac{1}{2} \operatorname{tr}(M^* N + N^* M) = \sum_{i,j} (m_{ij}, n_{ij}),$
0902.0431_E00418	69	■ 1.000	$\mathfrak{J}(3, \boldsymbol{C}) \oplus M(3, \boldsymbol{C}) = \mathfrak{J}$	$\mathfrak{J}(3, \boldsymbol{C}) \oplus M(3, \boldsymbol{C}) = \mathfrak{J}$	$\mathfrak{J}(3, \boldsymbol{C}) \oplus M(3, \boldsymbol{C}) = \mathfrak{J}.$

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0902.0431_E00419	69	■ 1.000	$\begin{aligned} F_{4, \boldsymbol{C}} &= \{\alpha \in \mathbf{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{C}}) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\} \\ &= \{\alpha \in \mathbf{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{C}}) \mid \alpha(X \times Y) = \alpha X \times \alpha Y\}. \end{aligned}$	$F_{4, \boldsymbol{C}} = \{\alpha \in \mathbf{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{C}}) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\} \\ = \{\alpha \in \mathbf{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{C}}) \mid \alpha(X \times Y) = \alpha X \times \alpha Y\}.$	$F_{4, \boldsymbol{C}} = \{\alpha \in \mathbf{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{C}}) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\} \\ = \{\alpha \in \mathbf{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{C}}) \mid \alpha(X \times Y) = \alpha X \times \alpha Y\}.$
0902.0431_E00420	69	■ 1.000	$\epsilon A = \bar{A}, \quad A \in SU(3),$	$\epsilon A = \bar{A}, \quad A \in SU(3),$	$\epsilon A = \bar{A}, \quad A \in SU(3),$
0902.0431_E00422	69	■ 1.000	$\begin{aligned} \beta_1 x &= p x, \beta_2 x = x q, \beta_3 x = \bar{q} x \\ \bar{p} x &= \bar{p} x, \bar{\beta}_2 x = x \bar{q}, \bar{\beta}_3 x = \bar{q} x \end{aligned}$	$\beta_1 x = p x, \beta_2 x = x q, \beta_3 x = \bar{q} x \quad \text{or} \quad \beta_1 x = p \bar{x}, \beta_2 x = \bar{x} q, \beta_3 x = \bar{q} \bar{x}.$	$\beta_1 x = p x, \beta_2 x = x q, \beta_3 x = \bar{q} x \quad \text{or} \quad \beta_1 x = p \bar{x}, \beta_2 x = \bar{x} q, \beta_3 x = \bar{q} \bar{x}.$
0902.0431_E00424	69	■ 1.000	$\varphi(P, A)(X + M) = AXA^* + PMA^*, \quad X + M \in \mathfrak{J}(3, \boldsymbol{C}) \oplus M(3, \boldsymbol{C}) = \mathfrak{J}.$	$\varphi(P, A)(X + M) = AXA^* + PMA^*, \quad X + M \in \mathfrak{J}(3, \boldsymbol{C}) \oplus M(3, \boldsymbol{C}) = \mathfrak{J}.$	$\varphi(P, A)(X + M) = AXA^* + PMA^*, \quad X + M \in \mathfrak{J}(3, \boldsymbol{C}) \oplus M(3, \boldsymbol{C}) = \mathfrak{J}.$
0902.0431_E00427	70	■ 1.000	$\alpha X = A X A^* \quad \text{or} \quad \alpha X = A \bar{X} A^*, \quad X \in \mathfrak{J}_{\boldsymbol{C}}$	$\alpha X = A X A^* \quad \text{or} \quad \alpha X = A \bar{X} A^*, \quad X \in \mathfrak{J}_{\boldsymbol{C}}$	$\alpha X = A X A^* \quad \text{or} \quad \alpha X = A \bar{X} A^*, \quad X \in \mathfrak{J}_{\boldsymbol{C}}$
0902.0431_E00428	70	■ 1.000	$\beta(X + M) = X + PM = \varphi(P, E)(X + M), \quad X + M \in \mathfrak{J}_{\boldsymbol{C}} \oplus M(3, \boldsymbol{C}) = \mathfrak{J}.$	$\beta(X + M) = X + PM = \varphi(P, E)(X + M), \quad X + M \in \mathfrak{J}_{\boldsymbol{C}} \oplus M(3, \boldsymbol{C}) = \mathfrak{J}.$	$\beta(X + M) = X + PM = \varphi(P, E)(X + M), \quad X + M \in \mathfrak{J}_{\boldsymbol{C}} \oplus M(3, \boldsymbol{C}) = \mathfrak{J}.$
0902.0431_E00429	70	■ 1.000	$\alpha = \varphi(E, A)\beta = \varphi(E, A)\varphi(P, E) = \varphi(P, A).$	$\alpha = \varphi(E, A)\beta = \varphi(E, A)\varphi(P, E) = \varphi(P, A).$	$\alpha = \varphi(E, A)\beta = \varphi(E, A)\varphi(P, E) = \varphi(P, A).$
0902.0431_E00430	70	■ 1.000	$\begin{aligned} &G_{01}, \quad G_{23}, \quad G_{45}, \quad G_{67}, \quad G_{26} + G_{37}, \quad -G_{27} + G_{36}, \\ &G_{24} + G_{35}, \quad -G_{25} + G_{34}, \quad G_{46} + G_{57}, \quad -G_{47} + G_{56}, \\ &\tilde{A}_1(1), \quad \tilde{A}_2(1), \quad \tilde{A}_3(1), \quad \tilde{A}_1(e_1), \quad \tilde{A}_2(e_1), \quad \tilde{A}_3(e_1) \end{aligned}$	$\begin{aligned} &G_{01}, \quad G_{23}, \quad G_{45}, \quad G_{67}, \quad G_{26} + G_{37}, \quad -G_{27} + G_{36}, \\ &G_{24} + G_{35}, \quad -G_{25} + G_{34}, \quad G_{46} + G_{57}, \quad -G_{47} + G_{56}, \\ &\tilde{A}_1(1), \quad \tilde{A}_2(1), \quad \tilde{A}_3(1), \quad \tilde{A}_1(e_1), \quad \tilde{A}_2(e_1), \quad \tilde{A}_3(e_1) \end{aligned}$	$\begin{aligned} &G_{01}, \quad G_{23}, \quad G_{45}, \quad G_{67}, \quad G_{26} + G_{37}, \quad -G_{27} + G_{36}, \\ &G_{24} + G_{35}, \quad -G_{25} + G_{34}, \quad G_{46} + G_{57}, \quad -G_{47} + G_{56}, \\ &\tilde{A}_1(1), \quad \tilde{A}_2(1), \quad \tilde{A}_3(1), \quad \tilde{A}_1(e_1), \quad \tilde{A}_2(e_1), \quad \tilde{A}_3(e_1) \end{aligned}$
0902.0431_E00432	71	■ 1.000	$\langle X, Y \rangle = (\tau X, Y).$	$\langle X, Y \rangle = (\tau X, Y).$	$\langle X, Y \rangle = (\tau X, Y).$
0902.0431_E00437	72	■ 1.000	$X \circ Y = \frac{1}{2}(XY + YX)$	$X \circ Y = \frac{1}{2}(XY + YX)$	$X \circ Y = \frac{1}{2}(XY + YX).$
0902.0431_E00439	72	■ 1.000	$\begin{aligned} F_{4(4)} &= \{\alpha \in \mathbf{Iso}_{\boldsymbol{R}}(\mathfrak{J}(3, \mathfrak{C}')) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\}, \\ F_{4(-20)} &= \{\alpha \in \mathbf{Iso}_{\boldsymbol{R}}(\mathfrak{J}(1, 2, \mathfrak{C})) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\}. \end{aligned}$	$\begin{aligned} F_{4(4)} &= \{\alpha \in \mathbf{Iso}_{\boldsymbol{R}}(\mathfrak{J}(3, \mathfrak{C}')) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\}, \\ F_{4(-20)} &= \{\alpha \in \mathbf{Iso}_{\boldsymbol{R}}(\mathfrak{J}(1, 2, \mathfrak{C})) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\}. \end{aligned}$	$\begin{aligned} F_{4(4)} &= \{\alpha \in \mathbf{Iso}_{\boldsymbol{R}}(\mathfrak{J}(3, \mathfrak{C}')) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\}, \\ F_{4(-20)} &= \{\alpha \in \mathbf{Iso}_{\boldsymbol{R}}(\mathfrak{J}(1, 2, \mathfrak{C})) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\}. \end{aligned}$
0902.0431_E00442	72	■ 1.000	$z(F_{4(4)}) = \{1\}, \quad z(F_{4(-20)}) = \{1\}.$	$z(F_{4(4)}) = \{1\}, \quad z(F_{4(-20)}) = \{1\}.$	$z(F_{4(4)}) = \{1\}, \quad z(F_{4(-20)}) = \{1\}.$
0902.0431_E00444	73	■ 1.000	$\lambda(\alpha) = {}^t \alpha^{-1}, \quad \alpha \in E_6^C$	$\lambda(\alpha) = {}^t \alpha^{-1}, \quad \alpha \in E_6^C$	$\lambda(\alpha) = {}^t \alpha^{-1}, \quad \alpha \in E_6^C$
0902.0431_E00445	73	■ 1.000	$E_6 = \{\alpha \in E_6^C \mid \tau \lambda(\alpha) \tau = \alpha\} = (E_6^C)^{\tau \lambda}.$	$E_6 = \{\alpha \in E_6^C \mid \tau \lambda(\alpha) \tau = \alpha\} = (E_6^C)^{\tau \lambda}.$	$E_6 = \{\alpha \in E_6^C \mid \tau \lambda(\alpha) \tau = \alpha\} = (E_6^C)^{\tau \lambda}.$

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0902.0431_E00446	73	■ 1.000	$U(27)=U\left(\frac{J}{J^C}\right)=\left\{\alpha\in\operatorname{Iso}_C\left(\frac{J}{J^C}\right)\mid\langle\alpha X,\alpha Y\rangle=\langle X,Y\rangle\right\}.$	$U(27)=U\left(\mathfrak{J}^C\right)=\left\{\alpha\in\operatorname{Iso}_C\left(\mathfrak{J}^C\right)\mid\langle\alpha X,\alpha Y\rangle=\langle X,Y\rangle\right\}.$	$U(27)=U(\mathfrak{J}^C)=\left\{\alpha\in\operatorname{Iso}_C(\mathfrak{J}^C)\mid\langle\alpha X,\alpha Y\rangle=\langle X,Y\rangle\right\}.$
0902.0431_E00451	74	■ 1.000	$\left[\delta_1+\widetilde{T}_1,\delta_2+\widetilde{T}_2\right]=\left([\delta_1,\delta_2]+\left[\widetilde{T}_1,\widetilde{T}_2\right]\right)+\left(\widetilde{\delta_1T_2}-\widetilde{\delta_2T_1}\right),$	$\left[\delta_1+\widetilde{T}_1,\delta_2+\widetilde{T}_2\right]=\left([\delta_1,\delta_2]+\left[\widetilde{T}_1,\widetilde{T}_2\right]\right)+\left(\widetilde{\delta_1T_2}-\widetilde{\delta_2T_1}\right),$	$[\delta_1+\widetilde{T}_1,\delta_2+\widetilde{T}_2]=([\delta_1,\delta_2]+[\widetilde{T}_1,\widetilde{T}_2])+(\widetilde{\delta_1T_2}-\widetilde{\delta_2T_1}),$
0902.0431_E00453	74	■ 1.000	$\lambda(\phi)=-^t\phi=-^t(\delta+\widetilde{T})=\delta-\widetilde{T}.$	$\lambda(\phi)=-^t\phi=-^t(\delta+\widetilde{T})=\delta-\widetilde{T}.$	$\lambda(\phi)=-^t\phi=-^t(\delta+\widetilde{T})=\delta-\widetilde{T}.$
0902.0431_E00455	74	■ 1.000	$\mathfrak{e}_6=\left\{\phi\in\operatorname{Hom}_C\left(\mathfrak{J}^C\right)\mid\left(\phi X,X,X\right)=0,\left\langle\phi X,Y\right\rangle+\left\langle X,\phi Y\right\rangle=0\right\}.$	$\mathfrak{e}_6=\left\{\phi\in\operatorname{Hom}_C\left(\mathfrak{J}^C\right)\mid\left(\phi X,X,X\right)=0,\left\langle\phi X,Y\right\rangle+\left\langle X,\phi Y\right\rangle=0\right\}.$	$\mathfrak{e}_6=\left\{\phi\in\operatorname{Hom}_C(\mathfrak{J}^C)\mid(\phi X,X,X)=0,\langle\phi X,Y\rangle+\langle X,\phi Y\rangle=0\right\}.$
0902.0431_E00456	74	■ 1.000	$\phi=\delta+i\widetilde{T},\quad\delta\in\mathfrak{f}_4,T\in\mathfrak{J}_0.$	$\phi=\delta+i\widetilde{T},\quad\delta\in\mathfrak{f}_4,T\in\mathfrak{J}_0.$	$\phi=\delta+i\widetilde{T},\quad\delta\in\mathfrak{f}_4,T\in\mathfrak{J}_0.$
0902.0431_E00461	76	■ 1.000	$A\circ X=\left(A-\frac{1}{3}\operatorname{tr}(A)E\right)^{\sim}X+\frac{1}{3}\operatorname{tr}(A)X\in W.$	$A\circ X=\left(A-\frac{1}{3}\operatorname{tr}(A)E\right)^{\sim}X+\frac{1}{3}\operatorname{tr}(A)X\in W.$	$A\circ X=\left(A-\frac{1}{3}\operatorname{tr}(A)E\right)^{\sim}X+\frac{1}{3}\operatorname{tr}(A)X\in W.$
0902.0431_E00462	76	■ 1.000	$A\vee B=[\widetilde{A},\widetilde{B}]+\left(A\circ B-\frac{1}{3}(A,B)E\right)^{\sim}$	$A\vee B=[\widetilde{A},\widetilde{B}]+\left(A\circ B-\frac{1}{3}(A,B)E\right)^{\sim}$	$A\vee B=[\widetilde{A},\widetilde{B}]+\left(A\circ B-\frac{1}{3}(A,B)E\right)^{\sim}$
0902.0431_E00463	76	■ 1.000	$(A\vee B)X=\frac{1}{2}(B,X)A+\frac{1}{6}(A,B)X-2B\times(A\times X),\quad X\in\mathfrak{J}^C.$	$(A\vee B)X=\frac{1}{2}(B,X)A+\frac{1}{6}(A,B)X-2B\times(A\times X),\quad X\in\mathfrak{J}^C.$	$(A\vee B)X=\frac{1}{2}(B,X)A+\frac{1}{6}(A,B)X-2B\times(A\times X),\quad X\in\mathfrak{J}^C.$
0902.0431_E00464	76	■ 1.000	$X\circ(X\circ X)-\operatorname{tr}(X)X\circ X+\frac{1}{2}(\operatorname{tr}(X)^2-(X,X))X=\frac{1}{3}(X,X,X)E.$	$X\circ(X\circ X)-\operatorname{tr}(X)X\circ X+\frac{1}{2}(\operatorname{tr}(X)^2-(X,X))X=\frac{1}{3}(X,X,X)E.$	$X\circ(X\circ X)-\operatorname{tr}(X)X\circ X+\frac{1}{2}(\operatorname{tr}(X)^2-(X,X))X=\frac{1}{3}(X,X,X)E.$
0902.0431_E00467	77	■ 1.000	$\left[\left(E_1\vee E_1\right)\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\\bar{x}_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{array}\right)=\frac{1}{6}\left(\begin{array}{ccc}4\xi_1&x_3&\bar{x}_2\\\bar{x}_3&-2\xi_2&-2x_1\\x_2&-2\bar{x}_1&-2\xi_3\end{array}\right)\right].$	$(E_1\vee E_1)\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\\bar{x}_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{array}\right)=\frac{1}{6}\left(\begin{array}{ccc}4\xi_1&x_3&\bar{x}_2\\\bar{x}_3&-2\xi_2&-2x_1\\x_2&-2\bar{x}_1&-2\xi_3\end{array}\right).$	$(E_1\vee E_1)\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\\bar{x}_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{array}\right)=\frac{1}{6}\left(\begin{array}{ccc}4\xi_1&x_3&\bar{x}_2\\\bar{x}_3&-2\xi_2&-2x_1\\x_2&-2\bar{x}_1&-2\xi_3\end{array}\right).$
0902.0431_E00470	77	■ 1.000	$(A\vee B)^{\prime}=-B\vee A.$	$(A\vee B)^{\prime}=-B\vee A.$	$(A\vee B)^{\prime}=-B\vee A.$
0902.0431_E00471	78	■ 1.000	$[\phi,A\vee B]=\phi A\vee B+A\vee\phi^{\prime}B.$	$[\phi,A\vee B]=\phi A\vee B+A\vee\phi^{\prime}B.$	$[\phi,A\vee B]=\phi A\vee B+A\vee\phi^{\prime}B.$
0902.0431_E00475	78	■ 1.000	$(\phi,A\vee B)_6=(\phi A,B).$	$(\phi,A\vee B)_6=(\phi A,B).$	$(\phi,A\vee B)_6=(\phi A,B).$
0902.0431_E00476	78	■ 1.000	$([\phi,\phi_1],\phi_2)_6=([\delta+\widetilde{T},\delta_1+\widetilde{T}_1],\delta_2+\widetilde{T}_2)_6$	$([\phi,\phi_1],\phi_2)_6=([\delta+\widetilde{T},\delta_1+\widetilde{T}_1],\delta_2+\widetilde{T}_2)_6$	$([\phi,\phi_1],\phi_2)_6=([\delta+\widetilde{T},\delta_1+\widetilde{T}_1],\delta_2+\widetilde{T}_2)_6$
0902.0431_E00481	79	■ 1.000	$B_6(\phi_1,\phi_2)=k(\phi_1,\phi_2)_6=k^{\prime}\operatorname{tr}(\phi_1\phi_2).$	$B_6(\phi_1,\phi_2)=k(\phi_1,\phi_2)_6=k^{\prime}\operatorname{tr}(\phi_1\phi_2).$	$B_6(\phi_1,\phi_2)=k(\phi_1,\phi_2)_6=k^{\prime}\operatorname{tr}(\phi_1\phi_2).$

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0902.0431_E00482	79	■ 1.000	$(\phi_i, \phi_j)_{\{6\}} = \left(\left(\left(E_{\{1\}} - E_{\{2\}} \right)^{\sim} \right)^{\sim} \right)^{\sim} \left(\left(\left(E_{\{1\}} - E_{\{2\}} \right)^{\sim} \right)^{\sim} \right)^{\sim} \right)_{\{6\}} = \left(\left(E_{\{1\}} - E_{\{2\}} \right)^{\sim} \right)^{\sim} \left(\left(E_{\{1\}} - E_{\{2\}} \right)^{\sim} \right)^{\sim} = 2$.	$(\phi, \phi)_6 = ((E_1 - E_2)^{\sim}, (E_1 - E_2)^{\sim})_6 = (E_1 - E_2, E_1 - E_2) = 2.$	$(\phi, \phi)_6 = ((E_1 - E_2)^{\sim}, (E_1 - E_2)^{\sim})_6 = (E_1 - E_2, E_1 - E_2) = 2.$
0902.0431_E00486	80	■ 1.000	$\begin{array}{ll} \phi_i \phi_j E_{\{1\}} = \phi_i E_{\{1\}} = E_{\{1\}}, & \phi_i \phi_j \widetilde{F}_1(e_i) = -\frac{1}{2} \phi_i \widetilde{F}_1(e_i) = \frac{1}{4} \widetilde{F}_1(e_i), \\ \phi_i \phi_j E_{\{2\}} = -\phi_i E_{\{2\}} = E_{\{2\}}, & \phi_i \phi_j \widetilde{F}_2(e_i) = \frac{1}{2} \phi_i \widetilde{F}_2(e_i) = \frac{1}{4} \widetilde{F}_2(e_i), \\ \phi_i \phi_j E_{\{3\}} = \phi_i \phi_j \widetilde{F}_3(e_i) = \phi_i \phi_j \widetilde{F}_3(e_i) = 0, & \phi_i \phi_j \widetilde{F}_3(e_i) = \phi_i \phi_j \widetilde{F}_3(e_i) = 0. \end{array}$	$\phi_i \phi_j E_{\{1\}} = \phi_i E_{\{1\}} = E_{\{1\}}, \quad \phi_i \phi_j \widetilde{F}_1(e_i) = -\frac{1}{2} \phi_i \widetilde{F}_1(e_i) = \frac{1}{4} \widetilde{F}_1(e_i), \\ \phi_i \phi_j E_{\{2\}} = -\phi_i E_{\{2\}} = E_{\{2\}}, \quad \phi_i \phi_j \widetilde{F}_2(e_i) = \frac{1}{2} \phi_i \widetilde{F}_2(e_i) = \frac{1}{4} \widetilde{F}_2(e_i), \\ \phi_i \phi_j E_{\{3\}} = \phi_i \phi_j \widetilde{F}_3(e_i) = \phi_i \phi_j \widetilde{F}_3(e_i) = 0, \quad \phi_i \phi_j \widetilde{F}_3(e_i) = \phi_i \phi_j \widetilde{F}_3(e_i) = 0.$	$\phi_i \phi_j E_{\{1\}} = \phi_i E_{\{1\}} = E_{\{1\}}, \quad \phi_i \phi_j \widetilde{F}_1(e_i) = -\frac{1}{2} \phi_i \widetilde{F}_1(e_i) = \frac{1}{4} \widetilde{F}_1(e_i), \\ \phi_i \phi_j E_{\{2\}} = -\phi_i E_{\{2\}} = E_{\{2\}}, \quad \phi_i \phi_j \widetilde{F}_2(e_i) = \frac{1}{2} \phi_i \widetilde{F}_2(e_i) = \frac{1}{4} \widetilde{F}_2(e_i), \\ \phi_i \phi_j E_{\{3\}} = \phi_i \phi_j \widetilde{F}_3(e_i) = \phi_i \phi_j \widetilde{F}_3(e_i) = 0, \quad \phi_i \phi_j \widetilde{F}_3(e_i) = \phi_i \phi_j \widetilde{F}_3(e_i) = 0.$
0902.0431_E00488	80	■ 1.000	$A \vee (B \times C) + B \vee (C \times A) + C \vee (A \times B) = 0, \quad A, B, C \in \mathfrak{J}^C.$	$A \vee (B \times C) + B \vee (C \times A) + C \vee (A \times B) = 0, \quad A, B, C \in \mathfrak{J}^C.$	$A \vee (B \times C) + B \vee (C \times A) + C \vee (A \times B) = 0, \quad A, B, C \in \mathfrak{J}^C.$
0902.0431_E00490	80	■ 1.000	$\mathfrak{M}^r = \{X \in M(3, \mathfrak{C}) \mid \text{all diagonal elements of } X \text{ are real}\},$	$\mathfrak{M}^r = \{X \in M(3, \mathfrak{C}) \mid \text{all diagonal elements of } X \text{ are real}\},$	$\mathfrak{M}^r = \{X \in M(3, \mathfrak{C}) \mid \text{all diagonal elements of } X \text{ are real}\},$
0902.0431_E00491	81	■ 1.000	$X \circ Y = \frac{1}{2} (XY + Y^* X^*)$	$X \circ Y = \frac{1}{2} (XY + Y^* X^*)$	$X \circ Y = \frac{1}{2} (XY + Y^* X^*),$
0902.0431_E00498	81	■ 1.000	$\begin{aligned} & \sum_k \sigma_{ij} (x_{ik} y_{kj}) + \sum_k \overline{\sigma_{ji} (x_{jk} y_{ki})} \\ & = \sum_k \sigma_{ij} (x_{ik} y_{kj}) + \sum_k \sigma_{ij} (\overline{x_{jk} y_{ki}}) = (i, j)\text{-element of } \delta(X \circ Y). \end{aligned}$	$\begin{aligned} & = \sum_k \sigma_{ij} (x_{ik} y_{kj}) + \sum_k \overline{\sigma_{ji} (x_{jk} y_{ki})} \\ & = \sum_k \sigma_{ij} (x_{ik} y_{kj}) + \sum_k \sigma_{ij} (\overline{x_{jk} y_{ki}}) = (i, j)\text{-element of } \delta(X \circ Y). \end{aligned}$	$\begin{aligned} & = \sum_k \sigma_{ij} (x_{ik} y_{kj}) + \sum_k \overline{\sigma_{ji} (x_{jk} y_{ki})} \\ & = \sum_k \sigma_{ij} (x_{ik} y_{kj}) + \sum_k \sigma_{ij} (\overline{x_{jk} y_{ki}}) = (i, j)\text{-element of } \delta(X \circ Y). \end{aligned}$
0902.0431_E00500	82	■ 1.000	$[\delta, \widetilde{R}] = \widetilde{\delta R}.$	$[\delta, \widetilde{R}] = \widetilde{\delta R}.$	$[\delta, \widetilde{R}] = \widetilde{\delta R}.$
0902.0431_E00501	82	■ 1.000	$[\widetilde{H}, \widetilde{T}] = \frac{1}{2} [H, T]^{\sim}.$	$[\widetilde{H}, \widetilde{T}] = \frac{1}{2} [H, T]^{\sim}.$	$[\widetilde{H}, \widetilde{T}] = \frac{1}{2} [H, T]^{\sim}.$
0902.0431_E00503	82	■ 1.000	$= \frac{1}{4} ([H, T]X + X[H, T]^*) = \frac{1}{2} [H, T]^{\sim} X, \quad X \in \mathfrak{J}^C.$	$= \frac{1}{4} ([H, T]X + X[H, T]^*) = \frac{1}{2} [H, T]^{\sim} X, \quad X \in \mathfrak{J}^C.$	$= \frac{1}{4} ([H, T]X + X[H, T]^*) = \frac{1}{2} [H, T]^{\sim} X, \quad X \in \mathfrak{J}^C.$
0902.0431_E00510	84	■ 1.000	$[h, \widetilde{F}_{32}(a)] = \left(-\lambda_k + \frac{1}{2} (\mu_3 - \mu_2) \right) \widetilde{F}_{32}(a), \quad a = e_k + i e_{4+k},$	$[h, \widetilde{F}_{32}(a)] = \left(-\lambda_k + \frac{1}{2} (\mu_3 - \mu_2) \right) \widetilde{F}_{32}(a), \quad a = e_k + i e_{4+k},$	$[h, \widetilde{F}_{32}(a)] = \left(-\lambda_k + \frac{1}{2} (\mu_3 - \mu_2) \right) \widetilde{F}_{32}(a), \quad a = e_k + i e_{4+k},$
0902.0431_E00514	84	■ 1.000	$\mu = \alpha_1 + 2\alpha_2 + 3\alpha_3 + 2\alpha_4 + 2\alpha_5 + \alpha_6$	$\mu = \alpha_1 + 2\alpha_2 + 3\alpha_3 + 2\alpha_4 + 2\alpha_5 + \alpha_6$	$\mu = \alpha_1 + 2\alpha_2 + 3\alpha_3 + 2\alpha_4 + 2\alpha_5 + \alpha_6$
0902.0431_E00519	86	■ 1.000	$\mathfrak{h}_{\mathbf{R}} = \left\{ \sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim} \mid \lambda_k, \mu_j \in \mathbf{R}, \mu_1 + \mu_2 + \mu_3 = 0 \right\}.$	$\mathfrak{h}_{\mathbf{R}} = \left\{ \sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim} \mid \lambda_k, \mu_j \in \mathbf{R}, \mu_1 + \mu_2 + \mu_3 = 0 \right\}.$	$\mathfrak{h}_{\mathbf{R}} = \left\{ \sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim} \mid \lambda_k, \mu_j \in \mathbf{R}, \mu_1 + \mu_2 + \mu_3 = 0 \right\}.$
0902.0431_E00523	86	■ 1.000	$(\alpha_1, \alpha_1) = B_6(H_{\alpha_1}, H_{\alpha_1}) = 24 \frac{1}{24} \frac{1}{24} 2 = \frac{1}{12},$	$(\alpha_1, \alpha_1) = B_6(H_{\alpha_1}, H_{\alpha_1}) = 24 \frac{1}{24} \frac{1}{24} 2 = \frac{1}{12},$	$(\alpha_1, \alpha_1) = B_6(H_{\alpha_1}, H_{\alpha_1}) = 24 \frac{1}{24} \frac{1}{24} 2 = \frac{1}{12},$

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0902.0431_E08525	87	■ 1.000	$\begin{aligned} \left(E_6\right)^{\tau} &= \left\{\alpha \in E_6 \mid \tau \alpha = \alpha \tau\right\} \\ &= \left\{\alpha \in E_6 \mid \lambda(\alpha) = \alpha\right\} = \left(E_6\right)^{\lambda} . \end{aligned}$	$\begin{aligned} \left(E_6\right)^{\tau} &= \left\{\alpha \in E_6 \mid \tau \alpha = \alpha \tau\right\} \\ &= \left\{\alpha \in E_6 \mid \lambda(\alpha) = \alpha\right\} = \left(E_6\right)^{\lambda} . \end{aligned}$	$\begin{aligned} \left(E_6\right)^{\tau} &= \left\{\alpha \in E_6 \mid \tau \alpha = \alpha \tau\right\} \\ &= \left\{\alpha \in E_6 \mid \lambda(\alpha) = \alpha\right\} = \left(E_6\right)^{\lambda} . \end{aligned}$
0902.0431_E08526	87	■ 1.000	$\begin{aligned} \left(E_6\right)^{\tau} &= \left\{\alpha \in E_6 \mid (\alpha X, \alpha Y) = (X, Y), X, Y \in \mathfrak{J}^C\right\} \\ &= \left\{\alpha \in E_6 \mid \alpha E = E\right\} = \left(E_6\right)_E . \end{aligned}$	$\begin{aligned} \left(E_6\right)^{\tau} &= \left\{\alpha \in E_6 \mid (\alpha X, \alpha Y) = (X, Y), X, Y \in \mathfrak{J}^C\right\} \\ &= \left\{\alpha \in E_6 \mid \alpha E = E\right\} = \left(E_6\right)_E . \end{aligned}$	$\begin{aligned} \left(E_6\right)^{\tau} &= \left\{\alpha \in E_6 \mid (\alpha X, \alpha Y) = (X, Y), X, Y \in \mathfrak{J}^C\right\} \\ &= \left\{\alpha \in E_6 \mid \alpha E = E\right\} = \left(E_6\right)_E . \end{aligned}$
0902.0431_E08527	88	■ 1.000	$\begin{aligned} F_4 &= \left\{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}) \mid \det(\alpha X) = \det X, (\alpha X, \alpha Y) = (X, Y)\right\} \\ &= \left\{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}) \mid \alpha(X \times Y) = \alpha X \times \alpha Y\right\} . \end{aligned}$	$\begin{aligned} F_4 &= \left\{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}) \mid \det(\alpha X) = \det X, (\alpha X, \alpha Y) = (X, Y)\right\} \\ &= \left\{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}) \mid \alpha(X \times Y) = \alpha X \times \alpha Y\right\} . \end{aligned}$	$\begin{aligned} F_4 &= \left\{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}) \mid \det(\alpha X) = \det X, (\alpha X, \alpha Y) = (X, Y)\right\} \\ &= \left\{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}) \mid \alpha(X \times Y) = \alpha X \times \alpha Y\right\} . \end{aligned}$
0902.0431_E08528	88	■ 1.000	$\begin{aligned} \alpha_{12}(t) &\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right) = \left(\begin{array}{ccc} e^{it}\xi_1 & x_3 & e^{it/2}\bar{x}_2 \\ \bar{x}_3 & e^{-it}\xi_2 & e^{-it/2}x_1 \\ e^{it/2}x_2 & e^{-it/2}\bar{x}_1 & \xi_3 \end{array}\right) , \end{aligned}$	$\alpha_{12}(t) \left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right) = \left(\begin{array}{ccc} e^{it}\xi_1 & x_3 & e^{it/2}\bar{x}_2 \\ \bar{x}_3 & e^{-it}\xi_2 & e^{-it/2}x_1 \\ e^{it/2}x_2 & e^{-it/2}\bar{x}_1 & \xi_3 \end{array}\right) ,$	$\alpha_{12}(t) \left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right) = \left(\begin{array}{ccc} e^{it}\xi_1 & x_3 & e^{it/2}\bar{x}_2 \\ \bar{x}_3 & e^{-it}\xi_2 & e^{-it/2}x_1 \\ e^{it/2}x_2 & e^{-it/2}\bar{x}_1 & \xi_3 \end{array}\right) ,$
0902.0431_E08533	90	■ 1.000	$\begin{aligned} EIV &= \left\{X \in \mathfrak{J}^C \mid \det X = 1, \langle X, X \rangle = 3\right\} . \end{aligned}$	$EIV = \left\{X \in \mathfrak{J}^C \mid \det X = 1, \langle X, X \rangle = 3\right\} .$	$EIV = \left\{X \in \mathfrak{J}^C \mid \det X = 1, \langle X, X \rangle = 3\right\} .$
0902.0431_E08534	90	■ 1.000	$\begin{aligned} \alpha X &= \left(\begin{array}{ccc} \xi_1 & 0 & 0 \\ 0 & \xi_2 & 0 \\ 0 & 0 & \xi_3 \end{array}\right) , \quad \xi_1 \in C, \xi_2 \geq 0, \xi_3 \geq 0 \end{aligned}$	$\alpha X = \left(\begin{array}{ccc} \xi_1 & 0 & 0 \\ 0 & \xi_2 & 0 \\ 0 & 0 & \xi_3 \end{array}\right) , \quad \xi_1 \in C, \xi_2 \geq 0, \xi_3 \geq 0$	$\alpha X = \left(\begin{array}{ccc} \xi_1 & 0 & 0 \\ 0 & \xi_2 & 0 \\ 0 & 0 & \xi_3 \end{array}\right) , \quad \xi_1 \in C, \xi_2 \geq 0, \xi_3 \geq 0$
0902.0431_E08536	90	■ 1.000	$\begin{aligned} z(E_6) &= \{1, \omega 1, \omega^2 1\}, \quad \omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i \in C. \end{aligned}$	$z(E_6) = \{1, \omega 1, \omega^2 1\}, \quad \omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i \in C.$	$z(E_6) = \{1, \omega 1, \omega^2 1\}, \quad \omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i \in C.$
0902.0431_E08537	90	■ 1.000	$\beta Y = Y, \quad \text{for all } \beta \in F_4 .$	$\beta Y = Y, \quad \text{for all } \beta \in F_4 .$	$\beta Y = Y, \quad \text{for all } \beta \in F_4 .$
0902.0431_E08538	90	■ 1.000	$\beta X = T X T^{-1}, \quad X \in \mathfrak{J}^C,$	$\beta X = T X T^{-1}, \quad X \in \mathfrak{J}^C,$	$\beta X = T X T^{-1}, \quad X \in \mathfrak{J}^C,$
0902.0431_E08539	91	■ 1.000	$\alpha E = Y = \omega E, \quad \omega \in C,$	$\alpha E = Y = \omega E, \quad \omega \in C,$	$\alpha E = Y = \omega E, \quad \omega \in C,$
0902.0431_E08540	91	■ 1.000	$\left(E_6\right)^{\sigma} = \left\{\alpha \in E_6 \mid \sigma \alpha = \alpha \sigma\right\} .$	$\left(E_6\right)^{\sigma} = \left\{\alpha \in E_6 \mid \sigma \alpha = \alpha \sigma\right\} .$	$\left(E_6\right)^{\sigma} = \left\{\alpha \in E_6 \mid \sigma \alpha = \alpha \sigma\right\} .$
0902.0431_E08542	91	■ 1.000	$\alpha E_1 = \xi E_1, \quad (\tau \xi) \xi = 1 .$	$\alpha E_1 = \xi E_1, \quad (\tau \xi) \xi = 1 .$	$\alpha E_1 = \xi E_1, \quad (\tau \xi) \xi = 1 .$
0902.0431_E08543	91	■ 1.000	$\alpha E_2, \alpha E_3 \in \mathfrak{J}(2, \mathfrak{C}^C) .$	$\alpha E_2, \alpha E_3 \in \mathfrak{J}(2, \mathfrak{C}^C) .$	$\alpha E_2, \alpha E_3 \in \mathfrak{J}(2, \mathfrak{C}^C) .$
0902.0431_E08545	92	■ 1.000	$\alpha E_1 \notin \mathfrak{J}(2, \mathfrak{C}^C) .$	$\alpha E_1 \notin \mathfrak{J}(2, \mathfrak{C}^C) .$	$\alpha E_1 \notin \mathfrak{J}(2, \mathfrak{C}^C) .$
0902.0431_E08547	92	■ 1.000	$\alpha E_1 = \xi E_1 + \xi_2 E_2 + \xi_3 E_3 + F_1(x_1), \quad \xi \neq 0 .$	$\alpha E_1 = \xi E_1 + \xi_2 E_2 + \xi_3 E_3 + F_1(x_1), \quad \xi \neq 0 .$	$\alpha E_1 = \xi E_1 + \xi_2 E_2 + \xi_3 E_3 + F_1(x_1), \quad \xi \neq 0 .$

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0902.0431_E00549	92	■ 1.000	$\alpha E_1=\xi E_1, \quad \xi \neq 0$.	$\alpha E_1 = \xi E_1, \quad \xi \neq 0.$	$\alpha E_1 = \xi E_1, \quad \xi \neq 0.$
0902.0431_E00550	92	■ 1.000	$1=\left\langle \text{\texttt{l}}\text{\texttt{angle}} E_1, E_1\right\rangle \text{\texttt{right}}\text{\texttt{rangle}}=\left\langle \text{\texttt{l}}\text{\texttt{angle}} \alpha E_1, \alpha E_1\right\rangle \text{\texttt{right}}\text{\texttt{rangle}}=\left\langle \text{\texttt{l}}\text{\texttt{angle}} \xi E_1, \xi E_1\right\rangle \text{\texttt{right}}\text{\texttt{rangle}}=(\tau \xi)\xi \left\langle E_1, E_1\right\rangle =(\tau \xi)\xi ,$	$1 = \langle E_1, E_1 \rangle = \langle \alpha E_1, \alpha E_1 \rangle = \langle \xi E_1, \xi E_1 \rangle = (\tau \xi) \xi \langle E_1, E_1 \rangle = (\tau \xi) \xi,$	$1 = \langle E_1, E_1 \rangle = \langle \alpha E_1, \alpha E_1 \rangle = \langle \xi E_1, \xi E_1 \rangle = (\tau \xi) \xi \langle E_1, E_1 \rangle = (\tau \xi) \xi,$
0902.0431_E00551	92	■ 1.000	$\left(E_6\right)_{E_1}=\left\{\alpha \in E_6 \mid \alpha E_1=E_1\right\}.$	$(E_6)_{E_1} = \{ \alpha \in E_6 \mid \alpha E_1 = E_1 \}.$	$(E_6)_{E_1} = \{ \alpha \in E_6 \mid \alpha E_1 = E_1 \}.$
0902.0431_E00552	93	■ 1.000	$V^{\{10\}}=\left\{X \in \mathfrak{J}^C \mid 2 E_1 \times X=-\tau X\right\}=\left\{\left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{array}\right) \mid \xi \in C, x \in \mathfrak{C}\right\}.$	$V^{10} = \{ X \in \mathfrak{J}^C \mid 2E_1 \times X = -\tau X \} = \left\{ \left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{array} \right) \middle \xi \in C, x \in \mathfrak{C} \right\}.$	$V^{10} = \{ X \in \mathfrak{J}^C \mid 2E_1 \times X = -\tau X \} = \left\{ \left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{array} \right) \middle \xi \in C, x \in \mathfrak{C} \right\}.$
0902.0431_E00553	93	■ 1.000	$\begin{gathered} 2 E_1 \times \alpha X=2 \alpha E_1 \times \alpha X=2 \tau \alpha \tau\left(E_1 \times X\right)=\tau \alpha \tau(-\tau X)=-\tau(\alpha X), \\ \left\langle \alpha X, \alpha X\right\rangle=\langle X, X\rangle=2 . \end{gathered}$	$2E_1 \times \alpha X = 2\alpha E_1 \times \alpha X = 2\tau\alpha\tau(E_1 \times X) = \tau\alpha\tau(-\tau X) = -\tau(\alpha X),$ $\langle \alpha X, \alpha X \rangle = \langle X, X \rangle = 2.$	$2E_1 \times \alpha X = 2\alpha E_1 \times \alpha X = 2\tau\alpha\tau(E_1 \times X) = \tau\alpha\tau(-\tau X) = -\tau(\alpha X),$ $\langle \alpha X, \alpha X \rangle = \langle X, X \rangle = 2.$
0902.0431_E00554	93	■ 1.000	$\alpha_{23}\left(t_0\right) X \in S^8=\left\{X \in V^9 \mid\langle X, X\rangle=2\right\}$	$\alpha_{23}\left(t_0\right) X \in S^8=\left\{X \in V^9 \mid\langle X, X\rangle=2\right\}$	$\alpha_{23}\left(t_0\right) X \in S^8=\left\{X \in V^9 \mid\langle X, X\rangle=2\right\}$
0902.0431_E00556	93	■ 1.000	$\alpha_{23}(\pi / 2) \beta \alpha_{23}\left(t_0\right) X=i\left(E_2+E_3\right) .$	$\alpha_{23}(\pi / 2) \beta \alpha_{23}\left(t_0\right) X=i\left(E_2+E_3\right) .$	$\alpha_{23}(\pi / 2) \beta \alpha_{23}\left(t_0\right) X=i\left(E_2+E_3\right) .$
0902.0431_E00557	93	■ 1.000	$p:\left(E_6\right)_{E_1} \rightarrow S O(10)=S O\left(V^{10}\right)$	$p:\left(E_6\right)_{E_1} \rightarrow S O(10)=S O\left(V^{10}\right)$	$p:\left(E_6\right)_{E_1} \rightarrow S O(10)=S O\left(V^{10}\right)$
0902.0431_E00559	94	■ 1.000	$\left(E_6\right)_{E_1} /\{1, \sigma\} \cong S O(10) .$	$(E_6)_{E_1} /\{1, \sigma\} \cong SO(10).$	$(E_6)_{E_1} /\{1, \sigma\} \cong SO(10).$
0902.0431_E00561	94	■ 1.000	$U(1)=\left\{\alpha_{12}(t) \alpha_{13}(t)=\exp i t\left(2 E_1-E_2-E_3\right)^{\sim} \mid t \in \boldsymbol{R}\right\}=\left\{\phi(\theta) \mid \theta \in C,(\tau \theta) \theta=1\right\}$	$U(1) = \{ \alpha_{12}(t) \alpha_{13}(t) = \exp i t (2E_1 - E_2 - E_3)^{\sim} \mid t \in \boldsymbol{R} \}$ $= \{ \phi(\theta) \mid \theta \in C, (\tau \theta) \theta = 1 \}$	$U(1) = \{ \alpha_{12}(t) \alpha_{13}(t) = \exp i t (2E_1 - E_2 - E_3)^{\sim} \mid t \in \boldsymbol{R} \}$ $= \{ \phi(\theta) \mid \theta \in C, (\tau \theta) \theta = 1 \}$
0902.0431_E00563	94	■ 1.000	$\phi(\theta) \mid \mathfrak{E}_1^C=\theta^4 1, \quad \phi(\theta) \mid \mathfrak{J}(2, \mathfrak{C})^C=\theta^{-2} 1, \quad \phi(\theta) \mid\left(\mathfrak{J}^C\right)_{-\sigma}=\theta 1 .$	$\phi(\theta) \mid \mathfrak{E}_1^C = \theta^4 1, \quad \phi(\theta) \mid \mathfrak{J}(2, \mathfrak{C})^C = \theta^{-2} 1, \quad \phi(\theta) \mid (\mathfrak{J}^C)_{-\sigma} = \theta 1.$	$\phi(\theta) \mid \mathfrak{E}_1^C = \theta^4 1, \quad \phi(\theta) \mid \mathfrak{J}(2, \mathfrak{C})^C = \theta^{-2} 1, \quad \phi(\theta) \mid (\mathfrak{J}^C)_{-\sigma} = \theta 1.$
0902.0431_E00564	95	■ 1.000	$\varphi(\theta, \beta)=\phi(\theta) \beta .$	$\varphi(\theta, \beta) = \phi(\theta) \beta.$	$\varphi(\theta, \beta) = \phi(\theta) \beta.$
0902.0431_E00565	95	■ 1.000	$\alpha E_1=\theta^4 E_1=\phi(\theta) E_1$	$\alpha E_1 = \theta^4 E_1 = \phi(\theta) E_1$	$\alpha E_1 = \theta^4 E_1 = \phi(\theta) E_1$
0902.0431_E00566	95	■ 1.000	$\operatorname{Ker} \varphi=\left\{(\theta, \phi(\theta)^{-1}) \mid \theta \in C, \theta^4=1\right\}=\left\{(\theta, \phi\left(\theta^{-1}\right)) \mid \theta=\pm 1, \pm i\right\}=\boldsymbol{Z}_4 .$	$\operatorname{Ker} \varphi = \{ (\theta, \phi(\theta)^{-1}) \mid \theta \in C, \theta^4 = 1 \} = \{ (\theta, \phi(\theta^{-1})) \mid \theta = \pm 1, \pm i \} = \boldsymbol{Z}_4.$	$\operatorname{Ker} \varphi = \{ (\theta, \phi(\theta)^{-1}) \mid \theta \in C, \theta^4 = 1 \} = \{ (\theta, \phi(\theta^{-1})) \mid \theta = \pm 1, \pm i \} = \boldsymbol{Z}_4.$
0902.0431_E00567	95	■ 1.000	$\left(E_6\right)^{\gamma}=\left\{\alpha \in E_6 \mid \gamma \alpha=\alpha \gamma\right\} .$	$(E_6)^{\gamma} = \{ \alpha \in E_6 \mid \gamma \alpha = \alpha \gamma \}.$	$(E_6)^{\gamma} = \{ \alpha \in E_6 \mid \gamma \alpha = \alpha \gamma \}.$
0902.0431_E00568	95	■ 1.000	$\mathfrak{J}^C=\mathfrak{J}(3, \boldsymbol{H})^C \oplus\left(\boldsymbol{H}^3\right)^C,$	$\mathfrak{J}^C = \mathfrak{J}(3, \boldsymbol{H})^C \oplus (\boldsymbol{H}^3)^C,$	$\mathfrak{J}^C = \mathfrak{J}(3, \boldsymbol{H})^C \oplus (\boldsymbol{H}^3)^C,$
0902.0431_E00570	95	■ 1.000	$k: M(3, \boldsymbol{H}) \rightarrow M(6, C) \quad \text { and } \quad k: \boldsymbol{H}^3 \rightarrow M(2, 6, C) .$	$k: M(3, \boldsymbol{H}) \rightarrow M(6, C) \quad \text{and} \quad k: \boldsymbol{H}^3 \rightarrow M(2, 6, C).$	$k: M(3, \boldsymbol{H}) \rightarrow M(6, C) \quad \text{and} \quad k: \boldsymbol{H}^3 \rightarrow M(2, 6, C).$
0902.0431_E00574	96	■ 1.000	$k_J(M)=k(M) J .$	$k_J(M) = k(M) J.$	$k_J(M) = k(M) J.$

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0902.0431_E00576	96	■ 1.000	$\backslash\langle S, T\rangle=\operatorname{tr}\left(\left(\tau^t S\right) T\right), \quad \backslash\langle P, Q\rangle=\operatorname{tr}\left(\left(\tau^t P\right) Q\right) .$	$\langle S, T\rangle=\operatorname{tr}\left(\left(\tau^t S\right) T\right), \quad \langle P, Q\rangle=\operatorname{tr}\left(\left(\tau^t P\right) Q\right) .$	$\langle S, T\rangle=\operatorname{tr}\left(\left(\tau^t S\right) T\right), \quad \langle P, Q\rangle=\operatorname{tr}\left(\left(\tau^t P\right) Q\right) .$
0902.0431_E00577	96	■ 1.000	$\left[\begin{array}{ll} \backslash\langle k_J(M), k_J(N) \\ \backslash\langle M, N\rangle, & \& M, N \in \mathfrak{J}(3, \boldsymbol{H})^C, \\ \backslash\langle k(\boldsymbol{a}), k(\boldsymbol{b})\rangle=2\langle\boldsymbol{a}, \boldsymbol{b}\rangle, & \boldsymbol{a}, \boldsymbol{b} \in(\boldsymbol{H}^3)^C . \\ \det \left(k_J(M)\right)=\left(\det M\right)^2, & M \in \mathfrak{J}(3, \boldsymbol{H})^C . \end{array}\right]$	$\langle k_J(M), k_J(N)\rangle=2\langle M, N\rangle, \quad M, N \in \mathfrak{J}(3, \boldsymbol{H})^C, \\ \langle k(\boldsymbol{a}), k(\boldsymbol{b})\rangle=2\langle\boldsymbol{a}, \boldsymbol{b}\rangle, \quad \boldsymbol{a}, \boldsymbol{b} \in(\boldsymbol{H}^3)^C . \\ \det \left(k_J(M)\right)=\left(\det M\right)^2, \quad M \in \mathfrak{J}(3, \boldsymbol{H})^C .$	$\langle k_J(M), k_J(N)\rangle=2\langle M, N\rangle, \quad M, N \in \mathfrak{J}(3, \boldsymbol{H})^C, \\ \langle k(\boldsymbol{a}), k(\boldsymbol{b})\rangle=2\langle\boldsymbol{a}, \boldsymbol{b}\rangle, \quad \boldsymbol{a}, \boldsymbol{b} \in(\boldsymbol{H}^3)^C . \\ \det \left(k_J(M)\right)=\left(\det M\right)^2, \quad M \in \mathfrak{J}(3, \boldsymbol{H})^C .$
0902.0431_E00579	96	■ 1.000	$\left(\begin{array}{cc} a^{\prime} & b^{\prime} \\ -\tau b^{\prime} & \tau a^{\prime} \end{array}\right)+i\left(\begin{array}{cc} c^{\prime} & d^{\prime} \\ -\tau d^{\prime} & \tau c^{\prime} \end{array}\right)=0 .$	$\left(\begin{array}{cc} a^{\prime} & b^{\prime} \\ -\tau b^{\prime} & \tau a^{\prime} \end{array}\right)+i\left(\begin{array}{cc} c^{\prime} & d^{\prime} \\ -\tau d^{\prime} & \tau c^{\prime} \end{array}\right)=0 .$	$\left(\begin{array}{cc} a^{\prime} & b^{\prime} \\ -\tau b^{\prime} & \tau a^{\prime} \end{array}\right)+i\left(\begin{array}{cc} c^{\prime} & d^{\prime} \\ -\tau d^{\prime} & \tau c^{\prime} \end{array}\right)=0 .$
0902.0431_E00580	97	■ 1.000	$\backslash\langle k\left(m_1+i n_1\right), k\left(m_2+i n_2\right)\rangle=2\langle m_1+i n_1, m_2+i n_2\rangle .$	$\langle k\left(m_1+i n_1\right), k\left(m_2+i n_2\right)\rangle=2\langle m_1+i n_1, m_2+i n_2\rangle .$	$\langle k\left(m_1+i n_1\right), k\left(m_2+i n_2\right)\rangle=2\langle m_1+i n_1, m_2+i n_2\rangle .$
0902.0431_E00583	97	■ 1.000	$\begin{aligned} & \& \operatorname{det}\left(\begin{array}{ccc} \xi_1 & m_3+n_3 e_2 & \tau\left(m_2+n_2 e_2\right) \\ \tau\left(m_3+n_3 e_2\right) & \xi_2 & m_1+n_1 e_2 \\ m_2+n_2 e_2 & \tau\left(m_1+n_1 e_2\right) & \xi_3 \end{array}\right) \\ & =\xi_1 \xi_2 \xi_3+\left(m_1+n_1 e_2\right)\left(m_2+n_2 e_2\right)\left(m_3+n_3 e_2\right) \\ & \quad +\tau\left(\left(m_1+n_1 e_2\right)\left(m_2+n_2 e_2\right)\left(m_3+n_3 e_2\right)\right)-\sum_{i=1}^3 \xi_i\left(m_i+n_i e_2\right) \tau\left(m_i+n_i e_2\right) \\ & =\text { the contents of in the parenthesis above. } \end{aligned}$	$\begin{aligned} & \det \left(\begin{array}{ccc} \xi_1 & m_3+n_3 e_2 & \tau\left(m_2+n_2 e_2\right) \\ \tau\left(m_3+n_3 e_2\right) & \xi_2 & m_1+n_1 e_2 \\ m_2+n_2 e_2 & \tau\left(m_1+n_1 e_2\right) & \xi_3 \end{array}\right) \\ & =\xi_1 \xi_2 \xi_3+\left(m_1+n_1 e_2\right)\left(m_2+n_2 e_2\right)\left(m_3+n_3 e_2\right) \\ & \quad +\tau\left(\left(m_1+n_1 e_2\right)\left(m_2+n_2 e_2\right)\left(m_3+n_3 e_2\right)\right)-\sum_{i=1}^3 \xi_i\left(m_i+n_i e_2\right) \tau\left(m_i+n_i e_2\right) \\ & =\text { the contents of in the parenthesis above. } \end{aligned}$	$\det \left(\begin{array}{ccc} \xi_1 & m_3+n_3 e_2 & \tau\left(m_2+n_2 e_2\right) \\ \tau\left(m_3+n_3 e_2\right) & \xi_2 & m_1+n_1 e_2 \\ m_2+n_2 e_2 & \tau\left(m_1+n_1 e_2\right) & \xi_3 \end{array}\right) \\ =\xi_1 \xi_2 \xi_3+\left(m_1+n_1 e_2\right)\left(m_2+n_2 e_2\right)\left(m_3+n_3 e_2\right) \\ +\tau\left(\left(m_1+n_1 e_2\right)\left(m_2+n_2 e_2\right)\left(m_3+n_3 e_2\right)\right)-\sum_{i=1}^3 \xi_i\left(m_i+n_i e_2\right) \tau\left(m_i+n_i e_2\right) \\ =\text { the contents of in the parenthesis above. }$
0902.0431_E00584	98	■ 1.000	$E_{\{6, \boldsymbol{H}\}}=\left\{\alpha \in \operatorname{Iso}_C\left(\left(\mathfrak{J}_{\boldsymbol{H}}\right)^C\right) \mid \operatorname{det}(\alpha M)=\operatorname{det} M,\langle\alpha M, \alpha N\rangle=\langle M, N\rangle\right\} .$	$E_{6, \boldsymbol{H}}=\left\{\alpha \in \operatorname{Iso}_C\left(\left(\mathfrak{J}_{\boldsymbol{H}}\right)^C\right) \mid \operatorname{det}(\alpha M)=\operatorname{det} M,\langle\alpha M, \alpha N\rangle=\langle M, N\rangle\right\} .$	$E_{6, \boldsymbol{H}}=\left\{\alpha \in \operatorname{Iso}_C\left(\left(\mathfrak{J}_{\boldsymbol{H}}\right)^C\right) \mid \operatorname{det}(\alpha M)=\operatorname{det} M,\langle\alpha M, \alpha N\rangle=\langle M, N\rangle\right\} .$
0902.0431_E00585	98	■ 1.000	$F_{\{4, \boldsymbol{H}\}}=\left\{\alpha \in E_{6, \boldsymbol{H}} \mid \alpha E=E\right\} ,$	$F_{4, \boldsymbol{H}}=\left\{\alpha \in E_{6, \boldsymbol{H}} \mid \alpha E=E\right\} ,$	$F_{4, \boldsymbol{H}}=\left\{\alpha \in E_{6, \boldsymbol{H}} \mid \alpha E=E\right\} ,$
0902.0431_E00589	98	■ 1.000	$\varphi_*\left(C\right) M=C M+M C^*=\left[C, M\right], \quad M \in \mathfrak{J}_{\boldsymbol{H}}$	$\varphi_*\left(C\right) M=C M+M C^*=\left[C, M\right], \quad M \in \mathfrak{J}_{\boldsymbol{H}}$	$\varphi_*\left(C\right) M=C M+M C^*=\left[C, M\right], \quad M \in \mathfrak{J}_{\boldsymbol{H}}$
0902.0431_E00590	98	■ 1.000	$E_{\{6, \boldsymbol{H}\}} / F_{\{4, \boldsymbol{H}\}} \simeq E I V_{\boldsymbol{H}}=\left\{X \in\left(\mathfrak{J}_{\boldsymbol{H}}\right)^C \mid \operatorname{det} M=1,\langle M, M\rangle=3\right\}$	$E_{6, \boldsymbol{H}} / F_{4, \boldsymbol{H}} \simeq E I V_{\boldsymbol{H}}=\left\{X \in\left(\mathfrak{J}_{\boldsymbol{H}}\right)^C \mid \operatorname{det} M=1,\langle M, M\rangle=3\right\}$	$E_{6, \boldsymbol{H}} / F_{4, \boldsymbol{H}} \simeq E I V_{\boldsymbol{H}}=\left\{X \in\left(\mathfrak{J}_{\boldsymbol{H}}\right)^C \mid \operatorname{det} M=1,\langle M, M\rangle=3\right\}$

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0902.0431_E00592	98	■ 1.000	$\operatorname{det}(\varphi(A)M)=\operatorname{det}M$.	$\det(\varphi(A)M)=\det M$.	$\det(\varphi(A)M)=\det M$.
0902.0431_E00596	99	■ 1.000	$\alpha X\times\alpha Y=\alpha^{-1}(X\times Y),\quad X,Y\in\mathfrak{J}^C.$ $\quad X,Y\in\mathfrak{J}^C.$	$\alpha X\times\alpha Y=\alpha^{-1}(X\times Y),\quad X,Y\in\mathfrak{J}^C.$	$\alpha X\times\alpha Y=\alpha^{-1}(X\times Y),\quad X,Y\in\mathfrak{J}^C.$
0902.0431_E00598	99	■ 1.000	$k_J^{-1}(M)=-k^{-1}(MJ)=-k^{-1}(J(\tau M)),\quad \tau k^{-1}(M)=-k^{-1}(JMJ),$ $k^{-1}(M)=-k^{-1}(MJ)=-k^{-1}(J(\tau M)),\quad \tau k^{-1}(M)=-k^{-1}(JMJ),$	$k_J^{-1}(M)=-k^{-1}(MJ)=-k^{-1}(J(\tau M)),\quad \tau k^{-1}(M)=-k^{-1}(JMJ),$	$k_J^{-1}(M)=-k^{-1}(MJ)=-k^{-1}(J(\tau M)),\quad \tau k^{-1}(M)=-k^{-1}(JMJ),$
0902.0431_E00600	100	■ 1.000	$\alpha=\varphi(1,A)\beta=\varphi(1,A)\varphi(p,E)=\varphi(p,A).$ $\alpha=\varphi(1,A)\beta=\varphi(1,A)\varphi(p,E)=\varphi(p,A).$	$\alpha=\varphi(1,A)\beta=\varphi(1,A)\varphi(p,E)=\varphi(p,A).$	$\alpha=\varphi(1,A)\beta=\varphi(1,A)\varphi(p,E)=\varphi(p,A).$
0902.0431_E00601	100	■ 1.000	$\begin{aligned} \left(E_6\right)^{\tau\gamma}&=\left\{\alpha\in E_6\mid\tau\gamma\alpha=\alpha\tau\gamma\right\}\\&=\left\{\alpha\in E_6\mid\gamma\lambda(\alpha)\gamma=\alpha\right\}=\left(E_6\right)^{\lambda\gamma}.\begin{aligned} \left(E_6\right)^{\tau\gamma}&=\left\{\alpha\in E_6\mid\tau\gamma\alpha=\alpha\tau\gamma\right\}\\&=\left\{\alpha\in E_6\mid\gamma\lambda(\alpha)\gamma=\alpha\right\}=\left(E_6\right)^{\lambda\gamma}.$	$\begin{aligned} \left(E_6\right)^{\tau\gamma}&=\left\{\alpha\in E_6\mid\tau\gamma\alpha=\alpha\tau\gamma\right\}\\&=\left\{\alpha\in E_6\mid\gamma\lambda(\alpha)\gamma=\alpha\right\}=\left(E_6\right)^{\lambda\gamma}.$	$\begin{aligned} \left(E_6\right)^{\tau\gamma}&=\left\{\alpha\in E_6\mid\tau\gamma\alpha=\alpha\tau\gamma\right\}\\&=\left\{\alpha\in E_6\mid\gamma\lambda(\alpha)\gamma=\alpha\right\}=\left(E_6\right)^{\lambda\gamma}.$
0902.0431_E00605	101	■ 1.000	$\mathfrak{J}(4,\boldsymbol{H})=\left\{P\in M(4,\boldsymbol{H})\mid P^*=P\right\},$ $\mathfrak{J}(4,\boldsymbol{H})=\left\{P\in M(4,\boldsymbol{H})\mid P^*=P\right\},$	$\mathfrak{J}(4,\boldsymbol{H})=\left\{P\in M(4,\boldsymbol{H})\mid P^*=P\right\},$	$\mathfrak{J}(4,\boldsymbol{H})=\left\{P\in M(4,\boldsymbol{H})\mid P^*=P\right\},$
0902.0431_E00606	101	■ 1.000	$P\circ Q=\frac{1}{2}(PQ+QP),\quad (P,Q)=\operatorname{tr}(P\circ Q).$ $P\circ Q=\frac{1}{2}(PQ+QP),\quad (P,Q)=\operatorname{tr}(P\circ Q).$	$P\circ Q=\frac{1}{2}(PQ+QP),\quad (P,Q)=\operatorname{tr}(P\circ Q).$	$P\circ Q=\frac{1}{2}(PQ+QP),\quad (P,Q)=\operatorname{tr}(P\circ Q).$
0902.0431_E00609	101	■ 1.000	$\langle P,Q\rangle=(\tau P,Q).$ $\langle P,Q\rangle=(\tau P,Q).$	$\langle P,Q\rangle=(\tau P,Q).$	$\langle P,Q\rangle=(\tau P,Q).$
0902.0431_E00610	101	■ 1.000	$A\left(X_1+iX_2\right)A^*=AX_1A^*+iAX_2A^*,\quad A\in\operatorname{Sp}(4),X_1,X_2\in\mathfrak{J}(4,\boldsymbol{H}).$ $A\left(X_1+iX_2\right)A^*=AX_1A^*+iAX_2A^*,\quad A\in\operatorname{Sp}(4),X_1,X_2\in\mathfrak{J}(4,\boldsymbol{H}).$	$A\left(X_1+iX_2\right)A^*=AX_1A^*+iAX_2A^*,\quad A\in\operatorname{Sp}(4),X_1,X_2\in\mathfrak{J}(4,\boldsymbol{H}).$	$A\left(X_1+iX_2\right)A^*=AX_1A^*+iAX_2A^*,\quad A\in\operatorname{Sp}(4),X_1,X_2\in\mathfrak{J}(4,\boldsymbol{H}).$
0902.0431_E00611	101	■ 1.000	$\left\langle APA^*,AQA^*\right\rangle=\left\langle P,Q\right\rangle,\quad P,Q\in\mathfrak{J}(4,\boldsymbol{H})^C.$ $\left\langle APA^*,AQA^*\right\rangle=\left\langle P,Q\right\rangle,\quad P,Q\in\mathfrak{J}(4,\boldsymbol{H})^C.$	$\left\langle APA^*,AQA^*\right\rangle=\left\langle P,Q\right\rangle,\quad P,Q\in\mathfrak{J}(4,\boldsymbol{H})^C.$	$\left\langle APA^*,AQA^*\right\rangle=\left\langle P,Q\right\rangle,\quad P,Q\in\mathfrak{J}(4,\boldsymbol{H})^C.$
0902.0431_E00612	102	■ 1.000	$g(M+\boldsymbol{a})=\left(\begin{array}{cc}\frac{1}{2}\operatorname{tr}(M)&\boldsymbol{a}\\\boldsymbol{ia}^*&M-\frac{1}{2}\operatorname{tr}(M)E\end{array}\right),\quad M+\boldsymbol{a}\in(\mathfrak{J}_H)^C\oplus(\boldsymbol{H}^3)^C=\mathfrak{J}^C.$ $g(M+\boldsymbol{a})=\left(\begin{array}{cc}\frac{1}{2}\operatorname{tr}(M)&\boldsymbol{a}\\\boldsymbol{ia}^*&M-\frac{1}{2}\operatorname{tr}(M)E\end{array}\right),\quad M+\boldsymbol{a}\in(\mathfrak{J}_H)^C\oplus(\boldsymbol{H}^3)^C=\mathfrak{J}^C.$	$g(M+\boldsymbol{a})=\left(\begin{array}{cc}\frac{1}{2}\operatorname{tr}(M)&\boldsymbol{a}\\\boldsymbol{ia}^*&M-\frac{1}{2}\operatorname{tr}(M)E\end{array}\right),\quad M+\boldsymbol{a}\in(\mathfrak{J}_H)^C\oplus(\boldsymbol{H}^3)^C=\mathfrak{J}^C.$	$g(M+\boldsymbol{a})=\left(\begin{array}{cc}\frac{1}{2}\operatorname{tr}(M)&\boldsymbol{a}\\\boldsymbol{ia}^*&M-\frac{1}{2}\operatorname{tr}(M)E\end{array}\right),\quad M+\boldsymbol{a}\in(\mathfrak{J}_H)^C\oplus(\boldsymbol{H}^3)^C=\mathfrak{J}^C.$
0902.0431_E00613	102	■ 1.000	$g(M+i\boldsymbol{a})=\left(\begin{array}{cc}\operatorname{tr}(M)&-\boldsymbol{a}\\\boldsymbol{-a}^*&M\end{array}\right)-\frac{1}{2}\operatorname{tr}(M)E,\quad M+i\boldsymbol{a}\in\mathfrak{J}_H\oplus i\boldsymbol{H}^3=(\mathfrak{J}^C)_{\tau\gamma}.$ $g(M+i\boldsymbol{a})=\left(\begin{array}{cc}\operatorname{tr}(M)&-\boldsymbol{a}\\\boldsymbol{-a}^*&M\end{array}\right)-\frac{1}{2}\operatorname{tr}(M)E,\quad M+i\boldsymbol{a}\in\mathfrak{J}_H\oplus i\boldsymbol{H}^3=(\mathfrak{J}^C)_{\tau\gamma}.$	$g(M+i\boldsymbol{a})=\left(\begin{array}{cc}\operatorname{tr}(M)&-\boldsymbol{a}\\\boldsymbol{-a}^*&M\end{array}\right)-\frac{1}{2}\operatorname{tr}(M)E,\quad M+i\boldsymbol{a}\in\mathfrak{J}_H\oplus i\boldsymbol{H}^3=(\mathfrak{J}^C)_{\tau\gamma}.$	$g(M+i\boldsymbol{a})=\left(\begin{array}{cc}\operatorname{tr}(M)&-\boldsymbol{a}\\\boldsymbol{-a}^*&M\end{array}\right)-\frac{1}{2}\operatorname{tr}(M)E,\quad M+i\boldsymbol{a}\in\mathfrak{J}_H\oplus i\boldsymbol{H}^3=(\mathfrak{J}^C)_{\tau\gamma}.$
0902.0431_E00618	103	■ 1.000	$\left\langle g\left(X_1+iX_2\right),g\left(Y_1+iY_2\right)\right\rangle=\left\langle X_1+iX_2,Y_1+iY_2\right\rangle,\quad X_i,Y_i\in\left(\mathfrak{J}^C\right)_{\tau\gamma}$ $\left\langle g\left(X_1+iX_2\right),g\left(Y_1+iY_2\right)\right\rangle=\left\langle X_1+iX_2,Y_1+iY_2\right\rangle,\quad X_i,Y_i\in\left(\mathfrak{J}^C\right)_{\tau\gamma}$	$\left\langle g\left(X_1+iX_2\right),g\left(Y_1+iY_2\right)\right\rangle=\left\langle X_1+iX_2,Y_1+iY_2\right\rangle,\quad X_i,Y_i\in\left(\mathfrak{J}^C\right)_{\tau\gamma}$	$\left\langle g\left(X_1+iX_2\right),g\left(Y_1+iY_2\right)\right\rangle=\left\langle X_1+iX_2,Y_1+iY_2\right\rangle,\quad X_i,Y_i\in\left(\mathfrak{J}^C\right)_{\tau\gamma}$
0902.0431_E00619	103	■ 1.000	$\varphi(A)X=g^{-1}(A(gX)A^*),\quad X\in\mathfrak{J}^C.$ $\varphi(A)X=g^{-1}(A(gX)A^*),\quad X\in\mathfrak{J}^C.$	$\varphi(A)X=g^{-1}(A(gX)A^*),\quad X\in\mathfrak{J}^C.$	$\varphi(A)X=g^{-1}(A(gX)A^*),\quad X\in\mathfrak{J}^C.$
0902.0431_E00622	103	■ 1.000	$\tau\gamma\varphi(A)\tau\gamma X=\varphi(A)X,\quad X\in\left(\mathfrak{J}^C\right)_{\tau\gamma},$ $\tau\gamma\varphi(A)\tau\gamma X=\varphi(A)X,\quad X\in\left(\mathfrak{J}^C\right)_{\tau\gamma},$	$\tau\gamma\varphi(A)\tau\gamma X=\varphi(A)X,\quad X\in\left(\mathfrak{J}^C\right)_{\tau\gamma},$	$\tau\gamma\varphi(A)\tau\gamma X=\varphi(A)X,\quad X\in\left(\mathfrak{J}^C\right)_{\tau\gamma},$
0902.0431_E00623	103	■ 1.000	$(g(\alpha E))^2=g(\alpha E)+\frac{3}{4}E.$ $(g(\alpha E))^2=g(\alpha E)+\frac{3}{4}E.$	$(g(\alpha E))^2=g(\alpha E)+\frac{3}{4}E.$	$(g(\alpha E))^2=g(\alpha E)+\frac{3}{4}E.$

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0902.0431_E00625	104	■ 1.000	$P=\frac{1}{4}(2\,g(\alpha\,E)+E),$	$P=\frac{1}{4}(2g(\alpha E)+E),$	$P=\frac{1}{4}(2g(\alpha E)+E),$
0902.0431_E00626	104	■ 1.000	$\begin{aligned} P^2 &= \frac{1}{16} \left(4(g(\alpha E))^2 + 4g(\alpha E) + E \right) = \frac{1}{4}(2g(\alpha E) + E) = P, \\ \mathrm{tr}(P) &= \frac{1}{4} \mathrm{tr}(2g(\alpha E)) + \frac{1}{4} \mathrm{tr}(E) = 0 + 1 = 1. \end{aligned}$	$P^2 = \frac{1}{16} \left(4(g(\alpha E))^2 + 4g(\alpha E) + E \right) = \frac{1}{4}(2g(\alpha E) + E) = P,$ $\mathrm{tr}(P) = \frac{1}{4} \mathrm{tr}(2g(\alpha E)) + \frac{1}{4} \mathrm{tr}(E) = 0 + 1 = 1.$	$P^2 = \frac{1}{16} (4(g(\alpha E))^2 + 4g(\alpha E) + E) = \frac{1}{4}(2g(\alpha E) + E) = P,$ $\mathrm{tr}(P) = \frac{1}{4} \mathrm{tr}(2g(\alpha E)) + \frac{1}{4} \mathrm{tr}(E) = 0 + 1 = 1.$
0902.0431_E00627	104	■ 1.000	$P=A\,E_{\scriptscriptstyle 1}\,A^{\scriptscriptstyle *}\,.$	$P=AE_1A^*.$	$P=AE_1A^*.$
0902.0431_E00629	104	■ 1.000	$\beta(M+\boldsymbol{a})=DMD^*+paD^*,\quad M+\boldsymbol{a}\in\mathfrak{J}_H\oplus\boldsymbol{H}^3=\mathfrak{J},$ $\quad M+\boldsymbol{a}\in\mathfrak{J}_H\oplus\boldsymbol{H}^3=\mathfrak{J}.$	$\beta(M+\boldsymbol{a})=DMD^*+paD^*,\quad M+\boldsymbol{a}\in\mathfrak{J}_H\oplus\boldsymbol{H}^3=\mathfrak{J}.$	$\beta(M+\boldsymbol{a})=DMD^*+paD^*,\quad M+\boldsymbol{a}\in\mathfrak{J}_H\oplus\boldsymbol{H}^3=\mathfrak{J}.$
0902.0431_E00630	104	■ 1.000	$\beta=\varphi(B)\,.$	$\beta=\varphi(B).$	$\beta=\varphi(B).$
0902.0431_E00633	105	■ 1.000	$\left(E_6\right)^w=\left\{\alpha\in E_6\mid w\alpha=\alpha w\right\}.$	$(E_6)^w=\{\alpha\in E_6\mid w\alpha=\alpha w\}.$	$(E_6)^w=\{\alpha\in E_6\mid w\alpha=\alpha w\}.$
0902.0431_E00634	105	■ 1.000	$\mathfrak{J}(3,\boldsymbol{C})^C\oplus M(3,\boldsymbol{C})^C=\mathfrak{J}^C.$	$\mathfrak{J}(3,\boldsymbol{C})^C\oplus M(3,\boldsymbol{C})^C=\mathfrak{J}^C.$	$\mathfrak{J}(3,\boldsymbol{C})^C\oplus M(3,\boldsymbol{C})^C=\mathfrak{J}^C.$
0902.0431_E00636	105	■ 1.000	$F_{4,\boldsymbol{C}}=\left\{\alpha\in E_{6,\boldsymbol{C}}\mid\alpha E=E\right\},$	$F_{4,\boldsymbol{C}}=\{\alpha\in E_{6,\boldsymbol{C}}\mid\alpha E=E\},$	$F_{4,\boldsymbol{C}}=\{\alpha\in E_{6,\boldsymbol{C}}\mid\alpha E=E\},$
0902.0431_E00639	105	■ 1.000	$E\,I\,V_{\scriptscriptstyle\{\boldsymbol{C}\}}=\left\{X\in\left(\mathfrak{J}_{\scriptscriptstyle\{\boldsymbol{C}\}}\right)^C\mid\operatorname{det}X=1,\right.$ $\left.\angle X,X\angle=3\right\}$	$EIV_C=\left\{X\in(\mathfrak{J}_C)^C\mid\det X=1,\langle X,X\rangle=3\right\}$	$EIV_C=\{X\in(\mathfrak{J}_C)^C\mid\det X=1,\langle X,X\rangle=3\}$
0902.0431_E00644	106	■ 1.000	$\phi_{\boldsymbol{C}}(C,D)X=h(C,D)X+Xh(C,D)^*,\quad X\in(\mathfrak{J}_{\boldsymbol{C}})^C$	$\phi_C(C,D)X=h(C,D)X+Xh(C,D)^*,\quad X\in(\mathfrak{J}_C)^C$	$\phi_C(C,D)X=h(C,D)X+Xh(C,D)^*,\quad X\in(\mathfrak{J}_C)^C$
0902.0431_E00650	107	■ 1.000	$h(A,B)\,X\,h(A,B)^*=\overline{X},\quad X\in(\mathfrak{J}_{\boldsymbol{C}})^C$	$h(A,B)Xh(A,B)^*=\bar{X},\quad X\in(\mathfrak{J}_C)^C$	$h(A,B)Xh(A,B)^*=\overline{X},\quad X\in(\mathfrak{J}_C)^C$
0902.0431_E00653	108	■ 1.000	$\begin{aligned} \tau\,\alpha\,\tau((X+M)\times(Y+N)) &= \alpha(X+M)\times\alpha(Y+N), \\ \langle\alpha(X+M),\alpha(Y+N)\rangle &= \langle X+M,Y+M\rangle. \end{aligned}$	$\tau\alpha\tau((X+M)\times(Y+N))=\alpha(X+M)\times\alpha(Y+N),$ $\langle\alpha(X+M),\alpha(Y+N)\rangle=\langle X+M,Y+M\rangle.$	$\tau\alpha\tau((X+M)\times(Y+N))=\alpha(X+M)\times\alpha(Y+N),$ $\langle\alpha(X+M),\alpha(Y+N)\rangle=\langle X+M,Y+M\rangle.$
0902.0431_E00654	108	■ 1.000	$\tau\,\alpha\,\tau(X\times Y)=\alpha X\times\alpha Y,\quad\langle\alpha X,\alpha Y\rangle=\langle X,Y\rangle$	$\tau\alpha\tau(X\times Y)=\alpha X\times\alpha Y,\quad\langle\alpha X,\alpha Y\rangle=\langle X,Y\rangle$	$\tau\alpha\tau(X\times Y)=\alpha X\times\alpha Y,\quad\langle\alpha X,\alpha Y\rangle=\langle X,Y\rangle$
0902.0431_E00657	108	■ 1.000	$\alpha X=h(A,B)\,X\,h(A,B)^*\quad\text{or}\quad\alpha X=h(A,B)\bar{X}h(A,B)^*,\quad X\in(\mathfrak{J}_{\boldsymbol{C}})^C$	$\alpha X=h(A,B)Xh(A,B)^*\quad\text{or}\quad\alpha X=h(A,B)\bar{X}h(A,B)^*,\quad X\in(\mathfrak{J}_C)^C$	$\alpha X=h(A,B)Xh(A,B)^*\quad\text{or}\quad\alpha X=h(A,B)\bar{X}h(A,B)^*,\quad X\in(\mathfrak{J}_C)^C$
0902.0431_E00659	109	■ 1.000	$\alpha=\varphi(E,A,B)\beta=\varphi(E,A,B)\,\varphi(P,E,E)=\varphi(P,A,B).$	$\alpha=\varphi(E,A,B)\beta=\varphi(E,A,B)\varphi(P,E,E)=\varphi(P,A,B).$	$\alpha=\varphi(E,A,B)\beta=\varphi(E,A,B)\varphi(P,E,E)=\varphi(P,A,B).$

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0902.0431_E00680	113	■ 1.000	$U(56)=U\left(\frac{P}{\wedge^3 C}\right)=\left\{\alpha\in\operatorname{Iso}\left(\frac{P}{\wedge^3 C}\right)\mid\langle\alpha P,\alpha Q\rangle=\langle P,Q\rangle\right\}.$	$U(56)=U\left(\mathfrak{P}^C\right)=\left\{\alpha\in\operatorname{Iso}_C\left(\mathfrak{P}^C\right)\mid\langle\alpha P,\alpha Q\rangle=\langle P,Q\rangle\right\}.$	$U(56)=U(\mathfrak{P}^C)=\{\alpha\in\operatorname{Iso}_C(\mathfrak{P}^C)\mid\langle\alpha P,\alpha Q\rangle=\langle P,Q\rangle\}.$
0902.0431_E00681 (1)	113	■ 1.000	$\alpha\frac{M}{\wedge^3 C}=\frac{M}{\wedge^3 C}.$	$\alpha\mathfrak{M}^C=\mathfrak{M}^C.$	$\alpha\mathfrak{M}^C=\mathfrak{M}^C.$
0902.0431_E00697	115	■ 1.000	$l(Y)=2\,B\times Y,\quad Y\in\frac{J}{\wedge^3 C}$	$l(Y)=2B\times Y,\quad Y\in\mathfrak{J}^C$	$l(Y)=2B\times Y,\quad Y\in\mathfrak{J}^C.$
0902.0431_E00698	115	■ 1.000	$2(B\times(X\times X),X\times X)+(\operatorname{det}X)(X,B)=3(\operatorname{det}X)b(X),$	$2(B\times(X\times X),X\times X)+(\det X)(X,B)=3(\det X)b(X),$	$2(B\times(X\times X),X\times X)+(\det X)(X,B)=3(\det X)b(X),$
0902.0431_E00699	115	■ 1.000	$b(X)=(B,X),\quad X\in\frac{J}{\wedge^3 C}.$	$b(X)=(B,X),\quad X\in\mathfrak{J}^C.$	$b(X)=(B,X),\quad X\in\mathfrak{J}^C.$
0902.0431_E00700 (i)	116	■ 1.000	$k(X)=2\,A\times X,\quad a(Y)=(A,Y),\quad X,Y\in\frac{J}{\wedge^3 C}.$	$k(X)=2A\times X,\quad a(Y)=(A,Y),\quad X,Y\in\mathfrak{J}^C.$	$k(X)=2A\times X,\quad a(Y)=(A,Y),\quad X,Y\in\mathfrak{J}^C.$
0902.0431_E00701 (ii)	116	■ 1.000	$\begin{gathered} 2\,g(X)\times X=\mu\,X\times X+h(X\times X),\\ 2\,h(X\times X)\times(X\times X)=(\operatorname{det}X)(\nu\,X+g(X)),\\ (\operatorname{det}X)(\nu\,X+g(X)),\\ (g(X),X\times X)+(h(X\times X),X)=3(\nu+\mu)\operatorname{det}X. \end{gathered}$	$\begin{gathered} 2g(X)\times X=\mu X\times X+h(X\times X),\\ 2h(X\times X)\times(X\times X)=(\det X)(\nu X+g(X)),\\ (g(X),X\times X)+(h(X\times X),X)=3(\nu+\mu)\det X. \end{gathered}$	$\begin{gathered} 2g(X)\times X=\mu X\times X+h(X\times X),\\ 2h(X\times X)\times(X\times X)=(\det X)(\nu X+g(X)),\\ (g(X),X\times X)+(h(X\times X),X)=3(\nu+\mu)\det X. \end{gathered}$
0902.0431_E00703	116	■ 1.000	$\phi\in\frac{\mathfrak{e}_6}{\wedge^3 C}.$	$\phi\in\mathfrak{e}_6^C.$	$\phi\in\mathfrak{e}_6^C.$
0902.0431_E00705	116	■ 1.000	$\psi^{\prime}=\phi.$	$\psi'=\phi.$	$\psi'=\phi.$
0902.0431_E00707	117	■ 1.000	$[\Phi,P\times Q]=\Phi P\times Q+P\times\Phi Q.$	$[\Phi,P\times Q]=\Phi P\times Q+P\times\Phi Q.$	$[\Phi,P\times Q]=\Phi P\times Q+P\times\Phi Q.$
0902.0431_E00710	117	■ 1.000	$\lambda(X,Y,\xi,\eta)=(Y,-X,\eta,-\xi).$	$\lambda(X,Y,\xi,\eta)=(Y,-X,\eta,-\xi).$	$\lambda(X,Y,\xi,\eta)=(Y,-X,\eta,-\xi).$
0902.0431_E00711	117	■ 1.000	$(P,Q)=\{\lambda P,Q\}=-\{P,\lambda Q\},\quad \langle P,Q\rangle=\{\tau\lambda P,Q\}.$	$(P,Q)=\{\lambda P,Q\}=-\{P,\lambda Q\},\quad \langle P,Q\rangle=\{\tau\lambda P,Q\}.$	$(P,Q)=\{\lambda P,Q\}=-\{P,\lambda Q\},\quad \langle P,Q\rangle=\{\tau\lambda P,Q\}.$
0902.0431_E00716	118	■ 1.000	$\Phi(\phi,A,-\tau A,\nu)=\left(\begin{array}{cccc} \phi-\frac{1}{3}\nu & -2\tau A & 0 & A \\ 2A & \tau\phi\tau+\frac{1}{3}\nu & -\tau A & 0 \\ 0 & A & \nu & 0 \\ -\tau A & 0 & 0 & -\nu \end{array}\right).$	$\Phi(\phi,A,-\tau A,\nu)=\left(\begin{array}{cccc} \phi-\frac{1}{3}\nu & -2\tau A & 0 & A \\ 2A & \tau\phi\tau+\frac{1}{3}\nu & -\tau A & 0 \\ 0 & A & \nu & 0 \\ -\tau A & 0 & 0 & -\nu \end{array}\right).$	$\Phi(\phi,A,-\tau A,\nu)=\left(\begin{array}{cccc} \phi-\frac{1}{3}\nu & -2\tau A & 0 & A \\ 2A & \tau\phi\tau+\frac{1}{3}\nu & -\tau A & 0 \\ 0 & A & \nu & 0 \\ -\tau A & 0 & 0 & -\nu \end{array}\right).$
0902.0431_E00719	118	■ 1.000	$\operatorname{dim}\left(\frac{\mathfrak{e}_7}{\wedge^3 C}\right)=78+54+1=133.$	$\dim(\mathfrak{e}_7)=78+54+1=133.$	$\dim(\mathfrak{e}_7)=78+54+1=133.$
0902.0431_E00721	119	■ 1.000	$\Phi=\frac{\Phi-\Phi^*}{2}+i\frac{\Phi+\Phi^*}{2i},\quad \frac{\Phi-\Phi^*}{2},\frac{\Phi+\Phi^*}{2i}\in\mathfrak{e}_7.$	$\Phi=\frac{\Phi-\Phi^*}{2}+i\frac{\Phi+\Phi^*}{2i},\quad \frac{\Phi-\Phi^*}{2},\frac{\Phi+\Phi^*}{2i}\in\mathfrak{e}_7.$	$\Phi=\frac{\Phi-\Phi^*}{2}+i\frac{\Phi+\Phi^*}{2i},\quad \frac{\Phi-\Phi^*}{2},\frac{\Phi+\Phi^*}{2i}\in\mathfrak{e}_7.$
0902.0431_E00723	119	■ 1.000	$\frac{a}{\wedge^3 C}\ni\left[\Phi\left(\phi_1,0,0,0\right),\Phi\left(\phi,A,B,\nu\right)\right]=\Phi\left([\phi_1,\phi],\phi_1A,\phi_1'B,0\right)$	$\mathfrak{a}\ni[\Phi(\phi_1,0,0,0),\Phi(\phi,A,B,\nu)]=\Phi([\phi_1,\phi],\phi_1A,\phi_1'B,0)$	$\mathfrak{a}\ni[\Phi(\phi_1,0,0,0),\Phi(\phi,A,B,\nu)]=\Phi([\phi_1,\phi],\phi_1A,\phi_1'B,0)$
0902.0431_E00725	119	■ 1.000	$\frac{a}{\wedge^3 C}\ni\left[\Phi\left(0,E_1,0,0\right),\Phi\left(0,0,E_1,0\right)\right]=\Phi\left(2E_1\vee E_1,0,0,1\right),$	$\mathfrak{a}\ni[\Phi(0,E_1,0,0),\Phi(0,0,E_1,0)]=\Phi(2E_1\vee E_1,0,0,1),$	$\mathfrak{a}\ni[\Phi(0,E_1,0,0),\Phi(0,0,E_1,0)]=\Phi(2E_1\vee E_1,0,0,1),$
0902.0431_E00727	120	■ 1.000	$\left[\frac{a}{\wedge^3 C}\ni\left[\Phi\left(0,0,0,\nu\right),\Phi\left(0,A,0,0\right)\right]=\Phi\left(0,\frac{2}{3}\nu A,0,0\right).\right]$	$\mathfrak{a}\ni[\Phi(0,0,0,\nu),\Phi(0,A,0,0)]=\Phi\left(0,\frac{2}{3}\nu A,0,0\right).$	$\mathfrak{a}\ni[\Phi(0,0,0,\nu),\Phi(0,A,0,0)]=\Phi\left(0,\frac{2}{3}\nu A,0,0\right).$

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0902.0431_E00729 (c)	120	■ 1.000	$\begin{gathered} W \ni \Phi(0,0,0,3)(X, Y, \xi, \eta) = (-X, Y, 3\xi, -3\eta), \\ \eta) = (-X, Y, 3\xi, -3\eta), \quad W \ni \Phi(0,0,0,3)(-X, Y, 3\xi, -3\eta) = (X, Y, 9\xi, 9\eta). \end{gathered}$	$W \ni \Phi(0,0,0,3)(X, Y, \xi, \eta) = (-X, Y, 3\xi, -3\eta), \\ W \ni \Phi(0,0,0,3)(-X, Y, 3\xi, -3\eta) = (X, Y, 9\xi, 9\eta).$	$W \ni \Phi(0,0,0,3)(X, Y, \xi, \eta) = (-X, Y, 3\xi, -3\eta), \\ W \ni \Phi(0,0,0,3)(-X, Y, 3\xi, -3\eta) = (X, Y, 9\xi, 9\eta).$
0902.0431_E00731	121	■ 1.000	$\left[W \ni \Phi(0,0, B, 0)(0, Y, \xi, \eta) = (2B \times Y, \xi B, 0, 0), \right. \\ \left. \xi B, 0, 0) \right] \end{itemize}$	$W \ni \Phi(0,0, B, 0)(0, Y, \xi, \eta) = (2B \times Y, \xi B, 0, 0).$	$W \ni \Phi(0,0, B, 0)(0, Y, \xi, \eta) = (2B \times Y, \xi B, 0, 0).$
0902.0431_E00732	121	■ 1.000	$\left[W \ni \Phi(0,0, B, 0)(0,0, \xi, \eta) = (0, \xi B, 0, 0) \right. \\ \left. \right] \end{itemize}$	$W \ni \Phi(0,0, B, 0)(0,0, \xi, \eta) = (0, \xi B, 0, 0).$	$W \ni \Phi(0,0, B, 0)(0,0, \xi, \eta) = (0, \xi B, 0, 0).$
0902.0431_E00733	121	■ 1.000	$\left(\Phi_1, \Phi_2 \right)_7 = -2 \left(\phi_1, \phi_2 \right)_6 - 4 \left(A_1, B_2 \right) - 4 \left(A_2, B_1 \right) - \frac{8}{3} \nu_1 \nu_2$	$(\Phi_1, \Phi_2)_7 = -2(\phi_1, \phi_2)_6 - 4(A_1, B_2) - 4(A_2, B_1) - \frac{8}{3}\nu_1\nu_2$	$(\Phi_1, \Phi_2)_7 = -2(\phi_1, \phi_2)_6 - 4(A_1, B_2) - 4(A_2, B_1) - \frac{8}{3}\nu_1\nu_2,$
0902.0431_E00735	121	■ 1.000	$\left[(\Phi, P \times Q)_7 = \{ \Phi P, Q \} \right. \\ \left. \right] \end{itemize}$	$(\Phi, P \times Q)_7 = \{ \Phi P, Q \}.$	$(\Phi, P \times Q)_7 = \{ \Phi P, Q \}.$
0902.0431_E00736	121	■ 1.000	$= \left(\Phi \left(\begin{array}{c} [\phi, \phi_1] + 2A \vee B_1 - 2A_1 \vee B \\ \left(\phi + \frac{2}{3}\nu \right) A_1 - \left(\phi_1 + \frac{2}{3}\nu_1 \right) A \\ - \left({}^t\phi + \frac{2}{3}\nu \right) B_1 + \left({}^t\phi_1 + \frac{2}{3}\nu_1 \right) B \\ (A, B_1) - (B, A_1) \end{array} \right), \Phi \left(\begin{array}{c} \phi_2 \\ A_2 \\ B_2 \\ \nu_2 \end{array} \right) \right)_7$	$= \left(\Phi \left(\begin{array}{c} [\phi, \phi_1] + 2A \vee B_1 - 2A_1 \vee B \\ \left(\phi + \frac{2}{3}\nu \right) A_1 - \left(\phi_1 + \frac{2}{3}\nu_1 \right) A \\ - \left({}^t\phi + \frac{2}{3}\nu \right) B_1 + \left({}^t\phi_1 + \frac{2}{3}\nu_1 \right) B \\ (A, B_1) - (B, A_1) \end{array} \right), \Phi \left(\begin{array}{c} \phi_2 \\ A_2 \\ B_2 \\ \nu_2 \end{array} \right) \right)_7$	$= \left(\Phi \left(\begin{array}{c} [\phi, \phi_1] + 2A \vee B_1 - 2A_1 \vee B \\ \left(\phi + \frac{2}{3}\nu \right) A_1 - \left(\phi_1 + \frac{2}{3}\nu_1 \right) A \\ - \left({}^t\phi + \frac{2}{3}\nu \right) B_1 + \left({}^t\phi_1 + \frac{2}{3}\nu_1 \right) B \\ (A, B_1) - (B, A_1) \end{array} \right), \Phi \left(\begin{array}{c} \phi_2 \\ A_2 \\ B_2 \\ \nu_2 \end{array} \right) \right)_7$
0902.0431_E00739	122	■ 1.000	$\begin{aligned} B_7 \left(\Phi_1, \Phi_2 \right) &= -9 \left(\Phi_1, \Phi_2 \right)_7 \\ &= 18 \left(\phi_1, \phi_2 \right)_6 + 36 \left(A_1, B_2 \right) + 36 \left(A_2, B_1 \right) + 24 \nu_1 \nu_2 \\ &= \frac{3}{2} B_6 \left(\phi_1, \phi_2 \right) + 36 \left(A_1, B_2 \right) + 36 \left(A_2, B_1 \right) + 24 \nu_1 \nu_2 \\ &= 3 \operatorname{tr} \left(\Phi_1 \Phi_2 \right) \end{aligned}$	$B_7(\Phi_1, \Phi_2) = -9(\Phi_1, \Phi_2)_7 \\ = 18(\phi_1, \phi_2)_6 + 36(A_1, B_2) + 36(A_2, B_1) + 24\nu_1\nu_2 \\ = \frac{3}{2}B_6(\phi_1, \phi_2) + 36(A_1, B_2) + 36(A_2, B_1) + 24\nu_1\nu_2 \\ = 3\operatorname{tr}(\Phi_1\Phi_2)$	$B_7(\Phi_1, \Phi_2) = -9(\Phi_1, \Phi_2)_7 \\ = 18(\phi_1, \phi_2)_6 + 36(A_1, B_2) + 36(A_2, B_1) + 24\nu_1\nu_2 \\ = \frac{3}{2}B_6(\phi_1, \phi_2) + 36(A_1, B_2) + 36(A_2, B_1) + 24\nu_1\nu_2 \\ = 3\operatorname{tr}(\Phi_1\Phi_2),$
0902.0431_E00742	122	■ 1.000	$\left[\Phi_0, [\Phi_0, \Phi(\phi, A, B, \nu)] \right] = \left[\Phi_0, \Phi \left(0, \frac{2}{3}A, -\frac{2}{3}B, 0 \right) \right] = \Phi \left(0, \frac{4}{9}A, \frac{4}{9}B, 0 \right).$	$[\Phi_0, [\Phi_0, \Phi(\phi, A, B, \nu)]] = \left[\Phi_0, \Phi \left(0, \frac{2}{3}A, -\frac{2}{3}B, 0 \right) \right] = \Phi \left(0, \frac{4}{9}A, \frac{4}{9}B, 0 \right).$	$[\Phi_0, [\Phi_0, \Phi(\phi, A, B, \nu)]] = \left[\Phi_0, \Phi \left(0, \frac{2}{3}A, -\frac{2}{3}B, 0 \right) \right] = \Phi \left(0, \frac{4}{9}A, \frac{4}{9}B, 0 \right).$
0902.0431_E00752	124	■ 1.000	$\begin{aligned} &\left[\Phi \left(h_\delta + \tilde{H}, 0, 0, \nu \right), \Phi \left(0, F_1(a), 0, 0 \right) \right] \quad a = e_k \pm ie_{4+k} \\ &= \Phi \left(0, \left(h_\delta + \tilde{H} + \frac{2}{3}\nu \right) F_1(a), 0, 0 \right) \\ &= \Phi \left(0, F_1(h_\delta a) + \frac{1}{2}(\mu_2 + \mu_3)F_1(a) + \frac{2}{3}\nu F_1(a), 0, 0 \right) \end{aligned}$	$\left[\Phi \left(h_\delta + \tilde{H}, 0, 0, \nu \right), \Phi \left(0, F_1(a), 0, 0 \right) \right] \quad a = e_k \pm ie_{4+k} \\ = \Phi \left(0, \left(h_\delta + \tilde{H} + \frac{2}{3}\nu \right) F_1(a), 0, 0 \right) \\ = \Phi \left(0, F_1(h_\delta a) + \frac{1}{2}(\mu_2 + \mu_3)F_1(a) + \frac{2}{3}\nu F_1(a), 0, 0 \right)$	$\left[\Phi(h_\delta + \tilde{H}, 0, 0, \nu), \Phi(0, F_1(a), 0, 0) \right] \quad a = e_k \pm ie_{4+k} \\ = \Phi \left(0, \left(h_\delta + \tilde{H} + \frac{2}{3}\nu \right) F_1(a), 0, 0 \right) \\ = \Phi \left(0, F_1(h_\delta a) + \frac{1}{2}(\mu_2 + \mu_3)F_1(a) + \frac{2}{3}\nu F_1(a), 0, 0 \right)$

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0902.0431_E00753	125	■ 1.000	$=\left(\pm\lambda_k-\frac{1}{2}\mu_1+\frac{2}{3}\nu\right)\Phi(0,F_1(a),0,0).$ $\backslash nu\right)\backslash Phi\left(0,F_{\{1\}}(a),0,0\right).$	$=\left(\pm\lambda_k-\frac{1}{2}\mu_1+\frac{2}{3}\nu\right)\Phi(0,F_1(a),0,0).$	$=\left(\pm\lambda_k-\frac{1}{2}\mu_1+\frac{2}{3}\nu\right)\Phi(0,F_1(a),0,0).$
0902.0431_E00754	125	■ 1.000	$\begin{aligned} &\&\alpha_{\{1\}}=\lambda_{\{0\}}-\lambda_{\{1\}},\\ &\quad\alpha_{\{2\}}=\lambda_{\{1\}}-\lambda_{\{2\}},\quad\quad\alpha_{\{3\}}=\lambda_{\{2\}}-\lambda_{\{3\}},\quad\&\alpha_{\{4\}}=\\ &\frac{1}{2}\left(-\lambda_{\{0\}}-\lambda_{\{1\}}-\lambda_{\{2\}}+\lambda_{\{3\}}\right)+\frac{1}{2}\left(\mu_{\{3\}}-\mu_{\{1\}}\right),\\ &\quad\&\alpha_{\{5\}}=\frac{1}{2}\left(\lambda_{\{0\}}+\lambda_{\{1\}}+\lambda_{\{2\}}+\lambda_{\{3\}}\right)+\frac{1}{2}\left(\mu_{\{1\}}-\mu_{\{2\}}\right),\\ &\quad\&\alpha_{\{6\}}=\mu_{\{2\}}+\frac{3}{2}\nu,\quad\alpha_{\{7\}}=-\mu_{\{3\}}-\frac{3}{2}\nu\end{aligned}$	$\alpha_1=\lambda_0-\lambda_1,\quad\alpha_2=\lambda_1-\lambda_2,\quad\alpha_3=\lambda_2-\lambda_3,$ $\alpha_4=\frac{1}{2}(-\lambda_0-\lambda_1-\lambda_2+\lambda_3)+\frac{1}{2}(\mu_3-\mu_1),$ $\alpha_5=\frac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)+\frac{1}{2}(\mu_1-\mu_2),$ $\alpha_6=\mu_2+\frac{3}{2}\nu,\quad\alpha_7=-\mu_3-\frac{3}{2}\nu$	$\alpha_1=\lambda_0-\lambda_1,\quad\alpha_2=\lambda_1-\lambda_2,\quad\alpha_3=\lambda_2-\lambda_3,$ $\alpha_4=\frac{1}{2}(-\lambda_0-\lambda_1-\lambda_2+\lambda_3)+\frac{1}{2}(\mu_3-\mu_1),$ $\alpha_5=\frac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)+\frac{1}{2}(\mu_1-\mu_2),$ $\alpha_6=\mu_2+\frac{3}{2}\nu,\quad\alpha_7=-\mu_3-\frac{3}{2}\nu$
0902.0431_E00755	125	■ 1.000	$\mu=\alpha_{\{1\}}+2\alpha_{\{2\}}+3\alpha_{\{3\}}+4\alpha_{\{4\}}+3\alpha_{\{5\}}+2\alpha_{\{6\}}+2\alpha_{\{7\}}$	$\mu=\alpha_1+2\alpha_2+3\alpha_3+4\alpha_4+3\alpha_5+2\alpha_6+2\alpha_7$	$\mu=\alpha_1+2\alpha_2+3\alpha_3+4\alpha_4+3\alpha_5+2\alpha_6+2\alpha_7$
0902.0431_E00763	128	■ 1.000	$B_{\{7\}}\left(h,h'\right)=6\left(6\sum_{k=0}^3\lambda_k\lambda_k'+3\sum_{j=1}^3\mu_j\mu_j'+4\nu\nu'\right)$	$B_7(h,h')=6\left(6\sum_{k=0}^3\lambda_k\lambda_k'+3\sum_{j=1}^3\mu_j\mu_j'+4\nu\nu'\right)$	$B_7(h,h')=6\left(6\sum_{k=0}^3\lambda_k\lambda_k'+3\sum_{j=1}^3\mu_j\mu_j'+4\nu\nu'\right)$
0902.0431_E00766	128	■ 1.000	$\left(\alpha_{\{1\}},\alpha_{\{1\}}\right)=B_{\{7\}}\left(H_{\{\alpha_{\{1\}}\}},H_{\{\alpha_{\{1\}}\}}\right)=36\frac{1}{36}\frac{1}{36}2=\frac{1}{18}$	$(\alpha_1,\alpha_1)=B_7(H_{\alpha_1},H_{\alpha_1})=36\frac{1}{36}\frac{1}{36}2=\frac{1}{18}$	$(\alpha_1,\alpha_1)=B_7(H_{\alpha_1},H_{\alpha_1})=36\frac{1}{36}\frac{1}{36}2=\frac{1}{18}$
0902.0431_E00768	129	■ 1.000	$\left(E_{\{7\}}\right)_{\left(0,0,1,0\right)}=\left\{\alpha\in E_7\mid\alpha(0,0,1,0)=(0,0,1,0)\right\}.$	$(E_7)_{(0,0,1,0)}=\{\alpha\in E_7\mid\alpha(0,0,1,0)=(0,0,1,0)\}.$	$(E_7)_{(0,0,1,0)}=\{\alpha\in E_7\mid\alpha(0,0,1,0)=(0,0,1,0)\}.$
0902.0431_E00769	129	■ 1.000	$\alpha(0,0,0,1)=\alpha(-\tau\lambda(0,0,1,0))=-\tau\lambda\alpha(0,0,1,0)=-\tau\lambda(0,0,1,0)=(0,0,0,1).$	$\alpha(0,0,0,1)=\alpha(-\tau\lambda(0,0,1,0))=-\tau\lambda\alpha(0,0,1,0)=-\tau\lambda(0,0,1,0)=(0,0,0,1).$	$\alpha(0,0,0,1)=\alpha(-\tau\lambda(0,0,1,0))=-\tau\lambda\alpha(0,0,1,0)=-\tau\lambda(0,0,1,0)=(0,0,0,1).$
0902.0431_E00770	129	■ 1.000	$\left(E_{\{7\}}\right)_{\left(0,0,1,0\right)}\cong E_6.$	$(E_7)_{(0,0,1,0)}\cong E_6.$	$(E_7)_{(0,0,1,0)}\cong E_6.$
0902.0431_E00771	130	■ 1.000	$\widetilde{\alpha}=\left(\begin{array}{cccc}\alpha&\theta&\theta&\theta\\0&\tau\alpha\tau&0&0\\0&0&1&0\\0&0&0&1\end{array}\right)\in(E_7)_{(0,0,1,0)}\subset E_7.$	$\widetilde{\alpha}=\left(\begin{array}{cccc}\alpha&0&0&0\\0&\tau\alpha\tau&0&0\\0&0&1&0\\0&0&0&1\end{array}\right)\in(E_7)_{(0,0,1,0)}\subset E_7.$	$\widetilde{\alpha}=\left(\begin{array}{cccc}\alpha&0&0&0\\0&\tau\alpha\tau&0&0\\0&0&1&0\\0&0&0&1\end{array}\right)\in(E_7)_{(0,0,1,0)}\subset E_7.$
0902.0431_E00775	130	■ 1.000	$\mathfrak{M}^C\ni\alpha\left(\begin{array}{c}X\\ \frac{1}{\eta}X\times X\\ \frac{1}{\eta^2}\det X\\ \eta\end{array}\right)=\left(\begin{array}{c}\beta X+\frac{1}{\eta}\epsilon(X\times X)\\ \delta X+\frac{1}{\eta}\beta_1(X\times X)\\ \frac{1}{\eta^2}\det X\\ \eta\end{array}\right),$	$\mathfrak{M}^C\ni\alpha\left(\begin{array}{c}X\\ \frac{1}{\eta}X\times X\\ \frac{1}{\eta^2}\det X\\ \eta\end{array}\right)=\left(\begin{array}{c}\beta X+\frac{1}{\eta}\epsilon(X\times X)\\ \delta X+\frac{1}{\eta}\beta_1(X\times X)\\ \frac{1}{\eta^2}\det X\\ \eta\end{array}\right),$	$\mathfrak{M}^C\ni\alpha\left(\begin{array}{c}X\\ \frac{1}{\eta}X\times X\\ \frac{1}{\eta^2}\det X\\ \eta\end{array}\right)=\left(\begin{array}{c}\beta X+\frac{1}{\eta}\epsilon(X\times X)\\ \delta X+\frac{1}{\eta}\beta_1(X\times X)\\ \frac{1}{\eta^2}\det X\\ \eta\end{array}\right),$

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0902.0431_E00776	130	■ 1.000	$\left(\beta X + \frac{1}{\eta} \epsilon(X \times X)\right) \times \left(\beta X + \frac{1}{\eta} \epsilon(X \times X)\right) \epsilon(X \times X) = \eta \left(\delta X + \frac{1}{\eta} \beta_1(X \times X)\right)$	$\left(\beta X + \frac{1}{\eta} \epsilon(X \times X)\right) \times \left(\beta X + \frac{1}{\eta} \epsilon(X \times X)\right) = \eta \left(\delta X + \frac{1}{\eta} \beta_1(X \times X)\right)$	$\left(\beta X + \frac{1}{\eta} \epsilon(X \times X)\right) \times \left(\beta X + \frac{1}{\eta} \epsilon(X \times X)\right) = \eta \left(\delta X + \frac{1}{\eta} \beta_1(X \times X)\right)$
0902.0431_E00778	130	■ 1.000	$\beta X \times \beta X = \beta_1(X \times X), \quad \text{\texttt{\textbackslash quad}} \left(\beta X, \beta_1(X \times X)\right) = 3 \operatorname{det} X.$	$\beta X \times \beta X = \beta_1(X \times X), \quad (\beta X, \beta_1(X \times X)) = 3 \det X.$	$\beta X \times \beta X = \beta_1(X \times X), \quad (\beta X, \beta_1(X \times X)) = 3 \det X.$
0902.0431_E00779	131	■ 1.000	$\operatorname{det}(\beta X) = \frac{1}{3}(\beta X, \beta X \times \beta X) = \frac{1}{3}(\beta X, \beta_1(X \times X)) = \det X,$	$\det(\beta X) = \frac{1}{3}(\beta X, \beta X \times \beta X) = \frac{1}{3}(\beta X, \beta_1(X \times X)) = \det X,$	$\det(\beta X) = \frac{1}{3}(\beta X, \beta X \times \beta X) = \frac{1}{3}(\beta X, \beta_1(X \times X)) = \det X,$
0902.0431_E00781	131	■ 1.000	$(\operatorname{det} X) \beta_1 X = (\operatorname{det} X) \tau \beta \tau X.$	$(\det X) \beta_1 X = (\det X) \tau \beta \tau X.$	$(\det X) \beta_1 X = (\det X) \tau \beta \tau X.$
0902.0431_E00782	131	■ 1.000	$\varphi_1(\theta)(X, Y, \xi, \eta) = \left(\theta^{-1} X, \theta Y, \theta^3 \xi, \theta^{-3} \eta\right).$	$\varphi_1(\theta)(X, Y, \xi, \eta) = (\theta^{-1} X, \theta Y, \theta^3 \xi, \theta^{-3} \eta).$	$\varphi_1(\theta)(X, Y, \xi, \eta) = (\theta^{-1} X, \theta Y, \theta^3 \xi, \theta^{-3} \eta).$
0902.0431_E00783	131	■ 1.000	$U(1) = \left\{ \varphi_1(\theta) \mid \theta \in C, (\tau \theta) \theta = 1 \right\}$	$U(1) = \{ \varphi_1(\theta) \mid \theta \in C, (\tau \theta) \theta = 1 \}$	$U(1) = \{ \varphi_1(\theta) \mid \theta \in C, (\tau \theta) \theta = 1 \}$
0902.0431_E00784	131	■ 1.000	$\alpha_i(a) = \left(\begin{array}{cccc} 1 + (\cos a - 1) p_i & 2a \frac{\sin a }{ a } E_i & 0 & -\tau a \frac{\sin a }{ a } E_i \\ -2\tau a \frac{\sin a }{ a } E_i & 1 + (\cos a - 1) p_i & a \frac{\sin a }{ a } E_i & 0 \\ 0 & -\tau a \frac{\sin a }{ a } E_i & \cos a & 0 \\ a \frac{\sin a }{ a } E_i & 0 & 0 & \cos a \end{array} \right)$	$\alpha_i(a) = \left(\begin{array}{cccc} 1 + (\cos a - 1) p_i & 2a \frac{\sin a }{ a } E_i & 0 & -\tau a \frac{\sin a }{ a } E_i \\ -2\tau a \frac{\sin a }{ a } E_i & 1 + (\cos a - 1) p_i & a \frac{\sin a }{ a } E_i & 0 \\ 0 & -\tau a \frac{\sin a }{ a } E_i & \cos a & 0 \\ a \frac{\sin a }{ a } E_i & 0 & 0 & \cos a \end{array} \right)$	$\alpha_i(a) = \left(\begin{array}{cccc} 1 + (\cos a - 1) p_i & 2a \frac{\sin a }{ a } E_i & 0 & -\tau a \frac{\sin a }{ a } E_i \\ -2\tau a \frac{\sin a }{ a } E_i & 1 + (\cos a - 1) p_i & a \frac{\sin a }{ a } E_i & 0 \\ 0 & -\tau a \frac{\sin a }{ a } E_i & \cos a & 0 \\ a \frac{\sin a }{ a } E_i & 0 & 0 & \cos a \end{array} \right)$
0902.0431_E00785	132	■ 1.000	$p_i \left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array} \right) = \left(\begin{array}{ccc} \xi_1 & \delta_{i3} x_3 & \delta_{i2} \bar{x}_2 \\ \delta_{i3} \bar{x}_3 & \xi_2 & \delta_{i1} x_1 \\ \delta_{i2} x_2 & \delta_{i1} \bar{x}_1 & \xi_3 \end{array} \right),$	$p_i \left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array} \right) = \left(\begin{array}{ccc} \xi_1 & \delta_{i3} x_3 & \delta_{i2} \bar{x}_2 \\ \delta_{i3} \bar{x}_3 & \xi_2 & \delta_{i1} x_1 \\ \delta_{i2} x_2 & \delta_{i1} \bar{x}_1 & \xi_3 \end{array} \right),$	$p_i \left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array} \right) = \left(\begin{array}{ccc} \xi_1 & \delta_{i3} x_3 & \delta_{i2} \bar{x}_2 \\ \delta_{i3} \bar{x}_3 & \xi_2 & \delta_{i1} x_1 \\ \delta_{i2} x_2 & \delta_{i1} \bar{x}_1 & \xi_3 \end{array} \right),$
0902.0431_E00786	132	■ 1.000	$\Phi_i(a) = \Phi(0, -\tau a E_i, \tau a E_i, 0) = \left(\begin{array}{cccc} 0 & 2a E_i & 0 & -\tau a E_i \\ -2\tau a E_i & 0 & a E_i & 0 \\ 0 & -\tau a E_i & 0 & 0 \\ a E_i & 0 & 0 & 0 \end{array} \right) \in \mathfrak{e}_7$	$\Phi_i(a) = \Phi(0, -\tau a E_i, \tau a E_i, 0) = \left(\begin{array}{cccc} 0 & 2a E_i & 0 & -\tau a E_i \\ -2\tau a E_i & 0 & a E_i & 0 \\ 0 & -\tau a E_i & 0 & 0 \\ a E_i & 0 & 0 & 0 \end{array} \right) \in \mathfrak{e}_7$	$\Phi_i(a) = \Phi(0, -\tau a E_i, \tau a E_i, 0) = \left(\begin{array}{cccc} 0 & 2a E_i & 0 & -\tau a E_i \\ -2\tau a E_i & 0 & a E_i & 0 \\ 0 & -\tau a E_i & 0 & 0 \\ a E_i & 0 & 0 & 0 \end{array} \right) \in \mathfrak{e}_7$
0902.0431_E00788	132	■ 1.000	$\beta X = \frac{1}{\xi} \beta(Y \times Y) = \frac{1}{\xi} \tau \beta \tau Y \times \tau \beta \tau Y$	$\beta X = \frac{1}{\xi} \beta(Y \times Y) = \frac{1}{\xi} \tau \beta \tau Y \times \tau \beta \tau Y$	$\beta X = \frac{1}{\xi} \beta(Y \times Y) = \frac{1}{\xi} \tau \beta \tau Y \times \tau \beta \tau Y$

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0902.0431_E00789	132	■ 1.000	$\tau \beta \tau Y = \left(\begin{array}{ccc} \eta_1 & 0 & 0 \\ 0 & \eta_2 & 0 \\ 0 & 0 & \eta_3 \end{array} \right), \quad \eta_i \in C$		
0902.0431_E00790	132	■ 1.000	$\alpha_i(-\pi/2)\beta P = \left(\begin{array}{cccc} 1-p_i & -2E_i & 0 & E_i \\ 2E_i & 1-p_i & -E_i & 0 \\ 0 & E_i & 0 & 0 \\ -E_i & 0 & 0 & 0 \end{array} \right) \left(\begin{array}{c} \beta X \\ \tau \beta \tau Y \\ 0 \\ \eta \end{array} \right) = \left(\begin{array}{c} * \\ * \\ \eta_i \\ * \end{array} \right), \quad \eta_i \neq 0,$		
0902.0431_E00791	132	■ 1.000	$\alpha_{i+1}(-\pi/2)\beta P = (*, \xi_i E_{i+2} + \xi_{i+2} E_i, 0, *) , \quad \xi_i \neq 0,$		
0902.0431_E00792	133	■ 1.000	$\alpha_1(-\pi/2)P = (\eta E_1, 0, 0, 0), \quad \eta \neq 0,$		
0902.0431_E00793	133	■ 1.000	$\mathfrak{M}_1 = \{P \in \mathfrak{P}^C \mid P \times P = 0, \langle P, P \rangle = 1\}.$		
0902.0431_E00794	133	■ 1.000	$E_7/E_6 \simeq \mathfrak{M}_1.$		
0902.0431_E00795	133	■ 1.000	$\alpha P = \left(\frac{1}{\xi} \left(\begin{array}{ccc} \eta_2 \eta_3 & 0 & 0 \\ 0 & \eta_3 \eta_1 & 0 \\ 0 & 0 & \eta_1 \eta_2 \end{array} \right), \left(\begin{array}{ccc} \eta_1 & 0 & 0 \\ 0 & \eta_2 & 0 \\ 0 & 0 & \eta_3 \end{array} \right), \xi, \frac{1}{\xi^2} \eta_1 \eta_2 \eta_3 \right), \quad \xi > 0$		
0902.0431_E00796	133	■ 1.000	$\frac{1}{\xi^2} (\eta_2 \eta_3 ^2 + \eta_3 \eta_1 ^2 + \eta_1 \eta_2 ^2) + (\eta_1 ^2 + \eta_2 ^2 + \eta_3 ^2) + \xi^2 + \frac{1}{\xi^4} \eta_1 \eta_2 \eta_3 ^2 = 1,$		
0902.0431_E00797 (i)	133	■ 1.000	$\left(1 + \frac{ \eta_1 ^2}{\xi^2}\right) \left(1 + \frac{ \eta_2 ^2}{\xi^2}\right) \left(1 + \frac{ \eta_3 ^2}{\xi^2}\right) = \frac{1}{\xi^2}.$		
0902.0431_E00798	133	■ 1.000	$\tan r_i = \frac{ \eta_i }{\xi}, \quad i = 1, 2, 3,$		
0902.0431_E00800	134	■ 1.000	$a_i = \frac{\eta_i}{ \eta_i } r_i, \quad i = 1, 2, 3$		
0902.0431_E00801	134	■ 1.000	$r_i = a_i , \quad \eta_i = \frac{1}{ a_i } \frac{\eta_i}{ \eta_i } r_i \frac{ \eta_i }{\xi} \xi = \frac{a_i}{ a_i } \tan r_i \cos r_1 \cos r_2 \cos r_3.$		

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0902.0431_E00837	139	■ 1.000	$\backslash [-4 \backslash \tau A \backslash \times \backslash \left(E_{\{1\}} \backslash \times X \right) + \backslash \left(E_{\{1\}}, X \right) A = 4 E_{\{1\}} \backslash \times (A \backslash \times X) - \backslash \langle A, X \rangle E_1, \quad X \in \mathfrak{J}^C . \backslash] \backslash \end{itemize}$	$-4\tau A \times (E_1 \times X) + (E_1, X) A = 4E_1 \times (A \times X) - \langle A, X \rangle E_1, \quad X \in \mathfrak{J}^C.$	$-4\tau A \times (E_1 \times X) + (E_1, X) A = 4E_1 \times (A \times X) - \langle A, X \rangle E_1, \quad X \in \mathfrak{J}^C.$
0902.0431_E00838	140	■ 1.000	$\backslash \nu = - \backslash \left(A, E_{\{1\}} \right) = - \backslash \tau \backslash \left(A, E_{\{1\}} \right),$	$\nu = - (A, E_1) = -\tau (A, E_1),$	$\nu = - (A, E_1) = -\tau (A, E_1),$
0902.0431_E00839	140	■ 1.000	$\backslash \phi(\backslash \nu) = 2 \backslash \nu E_{\{1\}} \vee E_{\{1\}},$	$\phi(\nu) = 2\nu E_1 \vee E_1,$	$\phi(\nu) = 2\nu E_1 \vee E_1,$
0902.0431_E00842	140	■ 1.000	$\backslash \varphi_{\{*\}} \backslash \left(\backslash \Phi \backslash \left(\backslash \phi(\backslash \nu), a E_{\{1\}}, - \backslash \tau a E_{\{1\}}, \backslash \nu \right) \right) = \backslash \left(\backslash \begin{array}{cc} \backslash \nu & a \\ - \backslash \tau a & - \backslash \nu \end{array} \right)$	$\varphi_*(\Phi(\phi(\nu), aE_1, -\tau aE_1, \nu)) = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix}$	$\varphi_*(\Phi(\phi(\nu), aE_1, -\tau aE_1, \nu)) = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix}$
0902.0431_E00843	140	■ 1.000	$\backslash \left(\backslash \left(\backslash \begin{array}{cc} \backslash \nu & a \\ - \backslash \tau a & - \backslash \nu \end{array} \right), \backslash \left(\backslash \begin{array}{cc} \backslash \rho & b \\ - \backslash \tau b & - \backslash \rho \end{array} \right) \right) = \backslash \left(\backslash \begin{array}{cc} b(\tau a) - a(\tau b) & 2(b\nu - a\rho) \\ -2\tau(b\nu - a\rho) & a(\tau b) - b(\tau a) \end{array} \right),$	$\left[\begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix}, \begin{pmatrix} \rho & b \\ -\tau b & -\rho \end{pmatrix}\right] = \begin{pmatrix} b(\tau a) - a(\tau b) & 2(b\nu - a\rho) \\ -2\tau(b\nu - a\rho) & a(\tau b) - b(\tau a) \end{pmatrix},$	$\left[\begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix}, \begin{pmatrix} \rho & b \\ -\tau b & -\rho \end{pmatrix}\right] = \begin{pmatrix} b(\tau a) - a(\tau b) & 2(b\nu - a\rho) \\ -2\tau(b\nu - a\rho) & a(\tau b) - b(\tau a) \end{pmatrix},$
0902.0431_E00844	140	■ 1.000	$\backslash \begin{aligned} & \& \backslash \left(\backslash \Phi \backslash \left(\backslash \phi(\backslash \nu), a E_{\{1\}}, - \backslash \tau a E_{\{1\}}, \backslash \nu \right), \backslash \Phi \backslash \left(\backslash \phi(\backslash \rho), b E_{\{1\}}, - \backslash \tau b E_{\{1\}}, \backslash \rho \right) \right) \& \backslash \left(= \backslash \Phi \backslash \left(\backslash \phi(b\tau a) - a(\tau b), 2(b\nu - a\rho) E_1, -2\tau(b\nu - a\rho) E_1, (\tau a)b - a(\tau b)\right).$	$[\Phi(\phi(\nu), aE_1, -\tau aE_1, \nu), \Phi(\phi(\rho), bE_1, -\tau bE_1, \rho)] \\ = \Phi(\phi(b\tau a) - a(\tau b), 2(b\nu - a\rho)E_1, -2\tau(b\nu - a\rho)E_1, (\tau a)b - a(\tau b)).$	$\left[\Phi(\phi(\nu), aE_1, -\tau aE_1, \nu), \Phi(\phi(\rho), bE_1, -\tau bE_1, \rho)\right] \\ = \Phi(\phi(b\tau a) - a(\tau b), 2(b\nu - a\rho)E_1, -2\tau(b\nu - a\rho)E_1, (\tau a)b - a(\tau b)).$
0902.0431_E00845	140	■ 1.000	$\backslash \left(\backslash \mathfrak{e}_{\{7\}} \right)^{\backslash \sigma} \backslash \ni \backslash \Phi \backslash \left(\backslash \begin{array}{c} \backslash \phi \backslash \backslash A \backslash \backslash - \backslash \tau A \backslash \backslash \backslash \nu \end{array} \right) = \backslash \Phi \backslash \left(\backslash \begin{array}{c} \backslash \phi \backslash \left(\backslash \nu^{\backslash \prime} \right) \backslash \backslash a E_{\{1\}} \backslash \backslash - \backslash \tau a E_{\{1\}} \backslash \backslash \backslash \nu^{\backslash \prime} \end{array} \right) + \backslash \Phi \backslash \left(\backslash \begin{array}{c} \backslash \phi - \backslash \phi(\nu') \backslash \backslash A - a E_{\{1\}} \backslash \backslash - \backslash \tau A + \tau a E_{\{1\}} \backslash \backslash \backslash \nu - \backslash \nu' \end{array} \right) \in \mathfrak{a}_1 \oplus (\mathfrak{e}_7)^{\kappa, \mu},$	$(\mathfrak{e}_7)^\sigma \ni \Phi \begin{pmatrix} \phi \\ A \\ -\tau A \\ \nu \end{pmatrix} = \Phi \begin{pmatrix} \phi(\nu') \\ aE_1 \\ -\tau aE_1 \\ \nu' \end{pmatrix} + \Phi \begin{pmatrix} \phi - \phi(\nu') \\ A - aE_1 \\ -\tau A + \tau aE_1 \\ \nu - \nu' \end{pmatrix} \in \mathfrak{a}_1 \oplus (\mathfrak{e}_7)^{\kappa, \mu},$	$(\mathfrak{e}_7)^\sigma \ni \Phi \begin{pmatrix} \phi \\ A \\ -\tau A \\ \nu \end{pmatrix} = \Phi \begin{pmatrix} \phi(\nu') \\ aE_1 \\ -\tau aE_1 \\ \nu' \end{pmatrix} + \Phi \begin{pmatrix} \phi - \phi(\nu') \\ A - aE_1 \\ -\tau A + \tau aE_1 \\ \nu - \nu' \end{pmatrix} \in \mathfrak{a}_1 \oplus (\mathfrak{e}_7)^{\kappa, \mu},$
0902.0431_E00846	141	■ 1.000	$\backslash \alpha_{\{23\}}(a) = \backslash \alpha_{\{2\}}(a) \backslash \alpha_{\{3\}}(\backslash \tau a) \backslash \in \backslash \left(\backslash \left(E_{\{7\}} \right)^{\backslash \kappa, \mu} \right)_{\backslash \left(0, E_{\{1\}}, 0, 1\right)},$	$\alpha_{23}(a) = \alpha_2(a)\alpha_3(\tau a) \in ((E_7)^{\kappa, \mu})_{(0, E_1, 0, 1)},$	$\alpha_{23}(a) = \alpha_2(a)\alpha_3(\tau a) \in ((E_7)^{\kappa, \mu})_{(0, E_1, 0, 1)},$
0902.0431_E00847	141	■ 1.000	$\backslash \begin{aligned} & \backslash \alpha_{\{23\}}(a) \& = \backslash \alpha_{\{2\}}(a) \backslash \alpha_{\{3\}}(\backslash \tau a) \backslash \backslash \& = \exp \backslash \Phi \backslash \left(0, - \backslash \tau a E_2 - a E_3, a E_2 + \tau a E_3, 0\right) . \end{aligned}$	$\alpha_{23}(a) = \alpha_2(a)\alpha_3(\tau a) \\ = \exp \Phi(0, -\tau aE_2 - aE_3, aE_2 + \tau aE_3, 0).$	$\alpha_{23}(a) = \alpha_2(a)\alpha_3(\tau a) \\ = \exp \Phi(0, -\tau aE_2 - aE_3, aE_2 + \tau aE_3, 0).$
0902.0431_E00848	141	■ 1.000	$\backslash \begin{aligned} & \backslash \operatorname{Spin}(10) \& = \backslash \left(\backslash \alpha \backslash \mid \backslash \alpha E_{\{1\}} = E_{\{1\}} \right) \backslash \backslash \& = \backslash \left(\backslash \alpha \backslash \mid \backslash \sigma \alpha = \alpha \sigma, \alpha E_{\{1\}} = E_{\{1\}} \right) \backslash \subset E_7 \end{aligned}$	$\operatorname{Spin}(10) = \{\alpha \in E_6 \mid \alpha E_1 = E_1\} \\ = \{\alpha \in E_6 \mid \sigma \alpha = \alpha \sigma, \alpha E_1 = E_1\} \subset E_7$	$\operatorname{Spin}(10) = \{\alpha \in E_6 \mid \alpha E_1 = E_1\} \\ = \{\alpha \in E_6 \mid \sigma \alpha = \alpha \sigma, \alpha E_1 = E_1\} \subset E_7$
0902.0431_E00849	141	■ 1.000	$S^{\backslash \{9\}} = \backslash \left(\backslash \left(\backslash \left(\backslash \left(\backslash \begin{array}{ccc} \backslash \theta & \backslash \theta & \backslash \theta \backslash \backslash \backslash \theta \& \backslash \xi \& x \backslash \backslash \backslash \theta \& \backslash \bar{x} \& - \backslash \tau \xi \end{array} \right) \right) \right) \right) \backslash \right\backslash \operatorname{rvert}\backslash , \backslash \xi \backslash \in C, x \backslash \in \backslash \mathfrak{C}, \backslash \bar{x} + (\backslash \tau \xi) \xi = 1 \right) \backslash .$	$S^9 = \left\{ \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, 0, 0, 0 \right) \middle \xi \in C, x \in \mathfrak{C}, \bar{x}x + (\tau \xi)\xi = 1 \right\}.$	$S^9 = \left\{ \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, 0, 0, 0 \right) \middle \xi \in C, x \in \mathfrak{C}, \bar{x}x + (\tau \xi)\xi = 1 \right\}.$

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0902.0431_E00850	141	■ 1.000	$\alpha\left(\theta,-E_{\{1\}},\theta,1\right)=\left(\theta,-E_{\{1\}},\theta,1\right)\quad\text{if and only if}\quad\alpha(0,0,1,0)=(0,0,1,0).$ $\alpha(0,0,1,0)=(0,0,1,0)$.	$\alpha(0,-E_1,0,1)=(0,-E_1,0,1)\quad\text{if and only if}\quad\alpha(0,0,1,0)=(0,0,1,0).$	$\alpha(0,-E_1,0,1)=(0,-E_1,0,1)\quad\text{if and only if}\quad\alpha(0,0,1,0)=(0,0,1,0).$
0902.0431_E00851	141	■ 1.000	$\left\{\alpha\in((E_7)^{\kappa,\mu})_{(0,E_1,0,1)}\mid\alpha(0,-E_1,0,1)=(0,-E_1,0,1)\right\}\cong\text{Spin}(10).$	$\left\{\alpha\in((E_7)^{\kappa,\mu})_{(0,E_1,0,1)}\mid\alpha(0,-E_1,0,1)=(0,-E_1,0,1)\right\}\cong\text{Spin}(10).$	$\left\{\alpha\in((E_7)^{\kappa,\mu})_{(0,E_1,0,1)}\mid\alpha(0,-E_1,0,1)=(0,-E_1,0,1)\right\}\cong\text{Spin}(10).$
0902.0431_E00854	142	■ 1.000	$P=\left(\left(\begin{array}{ccc}\theta&\theta&\theta\\0&\xi&x\\0&\bar{x}&-\tau\xi\end{array}\right),\left(\begin{array}{ccc}\eta&0&0\\0&0&0\\0&0&0\end{array}\right),0,\tau\eta\right)\in S^{10},$	$P=\left(\left(\begin{array}{ccc}\theta&\theta&\theta\\0&\xi&x\\0&\bar{x}&-\tau\xi\end{array}\right),\left(\begin{array}{ccc}\eta&0&0\\0&0&0\\0&0&0\end{array}\right),0,\tau\eta\right)\in S^{10},$	$P=\left(\left(\begin{array}{ccc}\theta&\theta&\theta\\0&\xi&x\\0&\bar{x}&-\tau\xi\end{array}\right),\left(\begin{array}{ccc}\eta&0&0\\0&0&0\\0&0&0\end{array}\right),0,\tau\eta\right)\in S^{10},$
0902.0431_E00858	142	■ 1.000	$\beta\alpha_{23}(a)P=(i(E_2+E_3),0,0,0).$	$\beta\alpha_{23}(a)P=(i(E_2+E_3),0,0,0).$	$\beta\alpha_{23}(a)P=(i(E_2+E_3),0,0,0).$
0902.0431_E00859	142	■ 1.000	$\alpha_{23}(-\pi/4)\beta\alpha_{23}(a)P=(0,-iE_1,0,i).$	$\alpha_{23}(-\pi/4)\beta\alpha_{23}(a)P=(0,-iE_1,0,i).$	$\alpha_{23}(-\pi/4)\beta\alpha_{23}(a)P=(0,-iE_1,0,i).$
0902.0431_E00865	144	■ 1.000	$P=\left(\left(\begin{array}{ccc}\theta&\theta&\theta\\0&\xi&x\\0&\bar{x}&-\tau\xi\end{array}\right),\left(\begin{array}{ccc}\eta&0&0\\0&0&0\\0&0&0\end{array}\right),0,\tau\eta\right)\in S^{11},$	$P=\left(\left(\begin{array}{ccc}\theta&\theta&\theta\\0&\xi&x\\0&\bar{x}&-\tau\xi\end{array}\right),\left(\begin{array}{ccc}\eta&0&0\\0&0&0\\0&0&0\end{array}\right),0,\tau\eta\right)\in S^{11},$	$P=\left(\left(\begin{array}{ccc}\theta&\theta&\theta\\0&\xi&x\\0&\bar{x}&-\tau\xi\end{array}\right),\left(\begin{array}{ccc}\eta&0&0\\0&0&0\\0&0&0\end{array}\right),0,\tau\eta\right)\in S^{11},$
0902.0431_E00866	144	■ 1.000	$\alpha(t)P\in S^{10}.$	$\alpha(t)P\in S^{10}.$	$\alpha(t)P\in S^{10}.$
0902.0431_E00867	144	■ 1.000	$\beta\alpha(t)P=(0,-iE_1,0,i).$	$\beta\alpha(t)P=(0,-iE_1,0,i).$	$\beta\alpha(t)P=(0,-iE_1,0,i).$
0902.0431_E00868	144	■ 1.000	$\alpha(-\pi/4)\beta\alpha(t)P=(0,E_1,0,1).$	$\alpha(-\pi/4)\beta\alpha(t)P=(0,E_1,0,1).$	$\alpha(-\pi/4)\beta\alpha(t)P=(0,E_1,0,1).$
0902.0431_E00869	144	■ 1.000	$p:(E_7)^{\kappa,\mu}\rightarrow SO(12)=SO(V^{12})$	$p:(E_7)^{\kappa,\mu}\rightarrow SO(12)=SO(V^{12})$	$p:(E_7)^{\kappa,\mu}\rightarrow SO(12)=SO(V^{12})$
0902.0431_E00871	144	■ 1.000	$(E_7)^{\kappa,\mu}/\{1,\sigma\}\cong SO(12).$	$(E_7)^{\kappa,\mu}/\{1,\sigma\}\cong SO(12).$	$(E_7)^{\kappa,\mu}/\{1,\sigma\}\cong SO(12).$
0902.0431_E00874	145	■ 1.000	$\varphi_2(SU(2))=\{\varphi_2(A)\in E_7\mid A\in SU(2)\}$	$\varphi_2(SU(2))=\{\varphi_2(A)\in E_7\mid A\in SU(2)\}$	$\varphi_2(SU(2))=\{\varphi_2(A)\in E_7\mid A\in SU(2)\}$
0902.0431_E00877	145	■ 1.000	$\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu)(X,Y,\xi,\eta)=(X',Y',\xi',\eta')$	$\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu)(X,Y,\xi,\eta)=(X',Y',\xi',\eta')$	$\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu)(X,Y,\xi,\eta)=(X',Y',\xi',\eta')$
0902.0431_E00879	145	■ 1.000	$\exp(\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu))=\varphi_2(A)\in\varphi_2(SU(2))\subset(E_7)^\sigma.$	$\exp(\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu))=\varphi_2(A)\in\varphi_2(SU(2))\subset(E_7)^\sigma.$	$\exp(\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu))=\varphi_2(A)\in\varphi_2(SU(2))\subset(E_7)^\sigma.$
0902.0431_E00880	146	■ 1.000	$\varphi(A,\beta)=\varphi_2(A)\beta.$	$\varphi(A,\beta)=\varphi_2(A)\beta.$	$\varphi(A,\beta)=\varphi_2(A)\beta.$
0902.0431_E00884	147	■ 1.000	$(E_7)^{\tau\gamma}=\{\alpha\in E_7\mid\tau\gamma\alpha=\alpha\tau\gamma\}$ $=\{\alpha\in E_7\mid\lambda\gamma\alpha=\alpha\lambda\gamma\}=(E_7)^{\lambda\gamma}.$	$(E_7)^{\tau\gamma}=\{\alpha\in E_7\mid\tau\gamma\alpha=\alpha\tau\gamma\}$ $=\{\alpha\in E_7\mid\lambda\gamma\alpha=\alpha\lambda\gamma\}=(E_7)^{\lambda\gamma}.$	$(E_7)^{\tau\gamma}=\{\alpha\in E_7\mid\tau\gamma\alpha=\alpha\tau\gamma\}$ $=\{\alpha\in E_7\mid\lambda\gamma\alpha=\alpha\lambda\gamma\}=(E_7)^{\lambda\gamma}.$

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0902.0431_E0885	147	■1.000	$\begin{aligned} \left(\mathfrak{P}^C\right)_{\tau\gamma} &= \left\{P \in \mathfrak{P}^C \mid \tau\gamma P = P\right\} \\ &= \left\{(X, Y, \xi, \eta) \in \mathfrak{P}^C \mid X, Y \in (\mathfrak{J}^C)_{\tau\gamma}, \xi, \eta \in \boldsymbol{R}\right\}, \\ \left(\mathfrak{P}^C\right)_{-\tau\gamma} &= \left\{P \in \mathfrak{P}^C \mid \tau\gamma P = -P\right\} \\ &= \left\{(X, Y, \xi, \eta) \in \mathfrak{P}^C \mid X, Y \in (\mathfrak{J}^C)_{-\tau\gamma}, \xi, \eta \in i\boldsymbol{R}\right\}, \\ &= i\left(\mathfrak{P}^C\right)_{\tau\gamma}. \end{aligned}$	$\begin{aligned} (\mathfrak{P}^C)_{\tau\gamma} &= \{P \in \mathfrak{P}^C \mid \tau\gamma P = P\} \\ &= \left\{(X, Y, \xi, \eta) \in \mathfrak{P}^C \mid X, Y \in (\mathfrak{J}^C)_{\tau\gamma}, \xi, \eta \in \boldsymbol{R}\right\}, \\ (\mathfrak{P}^C)_{-\tau\gamma} &= \{P \in \mathfrak{P}^C \mid \tau\gamma P = -P\} \\ &= \left\{(X, Y, \xi, \eta) \in \mathfrak{P}^C \mid X, Y \in (\mathfrak{J}^C)_{-\tau\gamma}, \xi, \eta \in i\boldsymbol{R}\right\}, \\ &= i\left(\mathfrak{P}^C\right)_{\tau\gamma}. \end{aligned}$	$\begin{aligned} (\mathfrak{P}^C)_{\tau\gamma} &= \{P \in \mathfrak{P}^C \mid \tau\gamma P = P\} \\ &= \{(X, Y, \xi, \eta) \in \mathfrak{P}^C \mid X, Y \in (\mathfrak{J}^C)_{\tau\gamma}, \xi, \eta \in \boldsymbol{R}\}, \\ (\mathfrak{P}^C)_{-\tau\gamma} &= \{P \in \mathfrak{P}^C \mid \tau\gamma P = -P\} \\ &= \{(X, Y, \xi, \eta) \in \mathfrak{P}^C \mid X, Y \in (\mathfrak{J}^C)_{-\tau\gamma}, \xi, \eta \in i\boldsymbol{R}\}, \\ &= i(\mathfrak{P}^C)_{\tau\gamma}. \end{aligned}$
0902.0431_E0887	147	■1.000	$k\left(\left(a+b e_2\right)\right)=\left(\left(\begin{array}{cc} a & b \\ -\bar{b} & \bar{a} \end{array}\right)\right), \quad a, b \in \boldsymbol{C}.$	$k\left(\left(a+b e_2\right)\right)=\left(\left(\begin{array}{cc} a & b \\ -\bar{b} & \bar{a} \end{array}\right)\right), \quad a, b \in \boldsymbol{C}.$	$k\left(\left(a+b e_2\right)\right)=\left(\left(\begin{array}{cc} a & b \\ -\bar{b} & \bar{a} \end{array}\right)\right), \quad a, b \in \boldsymbol{C}.$
0902.0431_E0888	147	■1.000	$B=k(D)+e_1 k(T), \quad D \in \mathfrak{sp}(4), T \in \mathfrak{J}(4, \boldsymbol{H})_0.$	$B=k(D)+e_1 k(T), \quad D \in \mathfrak{sp}(4), T \in \mathfrak{J}(4, \boldsymbol{H})_0.$	$B=k(D)+e_1 k(T), \quad D \in \mathfrak{sp}(4), T \in \mathfrak{J}(4, \boldsymbol{H})_0.$
0902.0431_E0890	147	■1.000	$D_1+e_1 T_1=0, D_1 \in k(\mathfrak{sp}(4)), T_1 \in k(\mathfrak{J}(4, \boldsymbol{H})_0) \quad \text{implies} \quad C_1=T_1=0.$	$D_1+e_1 T_1=0, D_1 \in k(\mathfrak{sp}(4)), T_1 \in k(\mathfrak{J}(4, \boldsymbol{H})_0) \quad \text{implies} \quad C_1=T_1=0.$	$D_1+e_1 T_1=0, D_1 \in k(\mathfrak{sp}(4)), T_1 \in k(\mathfrak{J}(4, \boldsymbol{H})_0) \quad \text{implies} \quad C_1=T_1=0.$
0902.0431_E0891	148	■1.000	$\begin{aligned} \left(\mathfrak{e}_7\right)^{\tau\gamma} &= \left\{\Phi \in \mathfrak{e}_7 \mid \tau\gamma \Phi = \Phi\tau\gamma\right\} \\ &= \left\{\Phi(\phi, A, -\gamma A, 0) \in \mathfrak{e}_7 \mid \phi \in\left(\mathfrak{e}_6\right)^{\tau\gamma}, A \in\left(\mathfrak{J}^C\right)_{\tau\gamma}\right\} \\ &= \left\{\Phi\left(\varphi_*(D), g^{-1}(T),-\gamma g^{-1}(T), 0\right) \in \mathfrak{e}_7 \mid D \in \mathfrak{sp}(4), T \in \mathfrak{J}(4, \boldsymbol{H})_0\right\}. \end{aligned}$	$\begin{aligned} \left(\mathfrak{e}_7\right)^{\tau\gamma} &= \{\Phi \in \mathfrak{e}_7 \mid \tau\gamma \Phi = \Phi\tau\gamma\} \\ &= \left\{\Phi(\phi, A, -\gamma A, 0) \in \mathfrak{e}_7 \mid \phi \in (\mathfrak{e}_6)^{\tau\gamma}, A \in (\mathfrak{J}^C)_{\tau\gamma}\right\} \\ &= \left\{\Phi\left(\varphi_*(D), g^{-1}(T), -\gamma g^{-1}(T), 0\right) \in \mathfrak{e}_7 \mid D \in \mathfrak{sp}(4), T \in \mathfrak{J}(4, \boldsymbol{H})_0\right\}. \end{aligned}$	$\begin{aligned} \left(\mathfrak{e}_7\right)^{\tau\gamma} &= \{\Phi \in \mathfrak{e}_7 \mid \tau\gamma \Phi = \Phi\tau\gamma\} \\ &= \{\Phi(\phi, A, -\gamma A, 0) \in \mathfrak{e}_7 \mid \phi \in (\mathfrak{e}_6)^{\tau\gamma}, A \in (\mathfrak{J}^C)_{\tau\gamma}\} \\ &= \{\Phi(\varphi_*(D), g^{-1}(T), -\gamma g^{-1}(T), 0) \in \mathfrak{e}_7 \mid D \in \mathfrak{sp}(4), T \in \mathfrak{J}(4, \boldsymbol{H})_0\}. \end{aligned}$
0902.0431_E0895	148	■1.000	$k_J\left(M_1+i M_2\right)=k\left(M_1\right) J+i k\left(M_2\right) J, \quad M_1, M_2 \in \mathfrak{J}(4, \boldsymbol{H}),$	$k_J\left(M_1+i M_2\right)=k\left(M_1\right) J+i k\left(M_2\right) J, \quad M_1, M_2 \in \mathfrak{J}(4, \boldsymbol{H}),$	$k_J\left(M_1+i M_2\right)=k\left(M_1\right) J+i k\left(M_2\right) J, \quad M_1, M_2 \in \mathfrak{J}(4, \boldsymbol{H}),$
0902.0431_E0896	148	■1.000	$\chi(X, Y, \xi, \eta)=k_J\left(g X-\frac{\xi}{2} E\right)+e_1 k_J\left(g(\gamma Y)-\frac{\eta}{2} E\right).$	$\chi(X, Y, \xi, \eta)=k_J\left(g X-\frac{\xi}{2} E\right)+e_1 k_J\left(g(\gamma Y)-\frac{\eta}{2} E\right).$	$\chi(X, Y, \xi, \eta)=k_J\left(g X-\frac{\xi}{2} E\right)+e_1 k_J\left(g(\gamma Y)-\frac{\eta}{2} E\right).$
0902.0431_E0897	148	■1.000	$\varphi_*(B) P=\chi^{-1}\left(B(\chi P)+(\chi P)^t B\right), \quad P \in \mathfrak{P}^C.$	$\varphi_*(B) P=\chi^{-1}\left(B(\chi P)+(\chi P)^t B\right), \quad P \in \mathfrak{P}^C.$	$\varphi_*(B) P=\chi^{-1}\left(B(\chi P)+(\chi P)^t B\right), \quad P \in \mathfrak{P}^C.$
0902.0431_E0891	150	■1.000	$\varphi(A) P=\chi^{-1}\left(A(\chi P)^t A\right), \quad P \in \mathfrak{P}^C.$	$\varphi(A) P=\chi^{-1}\left(A(\chi P)^t A\right), \quad P \in \mathfrak{P}^C.$	$\varphi(A) P=\chi^{-1}\left(A(\chi P)^t A\right), \quad P \in \mathfrak{P}^C.$
0902.0431_E0892	150	■1.000	$\varphi_*(D) P=\chi^{-1}\left(D(\chi P)+(\chi P)^t D\right), \quad P \in \mathfrak{P}^C,$	$\varphi_*(D) P=\chi^{-1}\left(D(\chi P)+(\chi P)^t D\right), \quad P \in \mathfrak{P}^C,$	$\varphi_*(D) P=\chi^{-1}\left(D(\chi P)+(\chi P)^t D\right), \quad P \in \mathfrak{P}^C,$
0902.0431_E0894	150	■1.000	$\tau\gamma Y=Y, \quad X=\frac{1}{\xi}(Y \times Y), \quad \tau \xi=\xi, \quad \tau \eta=\eta.$	$\tau\gamma Y=Y, \quad X=\frac{1}{\xi}(Y \times Y), \quad \tau \xi=\xi, \quad \tau \eta=\eta.$	$\tau\gamma Y=Y, \quad X=\frac{1}{\xi}(Y \times Y), \quad \tau \xi=\xi, \quad \tau \eta=\eta.$

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0902.0431_E00910	151	■ 1.000	$\left(E_7\right)^w=\left\{\alpha\in E_7\mid w\alpha=\alpha w\right\}.$ $\alpha=\alpha\ w\right\}.$	$(E_7)^w = \{\alpha \in E_7 \mid w\alpha = \alpha w\}.$	$(E_7)^w = \{\alpha \in E_7 \mid w\alpha = \alpha w\}.$
0902.0431_E00912	151	■ 1.000	$E_{\{6,\,\boldsymbol{C}\}}=\left\{\alpha\in E_7,\right.$ $\left.\boldsymbol{C}\right\}\mid\alpha(0,0,1,0)=(0,0,1,0)\right\},$	$E_{6,C} = \{\alpha \in E_{7,C} \mid \alpha(0,0,1,0) = (0,0,1,0)\},$	$E_{6,C} = \{\alpha \in E_{7,C} \mid \alpha(0,0,1,0) = (0,0,1,0)\},$
0902.0431_E00913	152	■ 1.000	$\mathfrak{e}_{\{7,\,\boldsymbol{C}\}}=\left\{\Phi(\phi,\right.$ $A,-\tau A,\nu)\mid\phi\in\mathfrak{e}_{6,C},A\in(\mathfrak{J}_C)^C,\nu\in i\boldsymbol{R}\}$ $\left.\boldsymbol{C}\right\},A\in\left(\mathfrak{J}_{\boldsymbol{C}}\right)^C,\nu\in i\boldsymbol{R}\right\}$	$\mathfrak{e}_{7,C} = \left\{\Phi(\phi, A, -\tau A, \nu) \mid \phi \in \mathfrak{e}_{6,C}, A \in (\mathfrak{J}_C)^C, \nu \in i\boldsymbol{R}\right\}$	$\mathfrak{e}_{7,C} = \{\Phi(\phi, A, -\tau A, \nu) \mid \phi \in \mathfrak{e}_{6,C}, A \in (\mathfrak{J}_C)^C, \nu \in i\boldsymbol{R}\}$
0902.0431_E00915	152	■ 1.000	$\left(\mathfrak{M}_{\boldsymbol{C}}\right)_1=\left\{P\in\left(\mathfrak{M}_{\boldsymbol{C}}\right)^C\mid P\times P=0,\left\langle P,P\right\rangle=1\right\}$ $\left.\mathfrak{M}_{\boldsymbol{C}}\right)^C\mid P\times P=0,\left\langle P,P\right\rangle=1\right\}$	$(\mathfrak{M}_C)_1 = \left\{P \in (\mathfrak{M}_C)^C \mid P \times P = 0, \langle P, P \rangle = 1\right\}$	$(\mathfrak{M}_C)_1 = \{P \in (\mathfrak{M}_C)^C \mid P \times P = 0, \langle P, P \rangle = 1\}$
0902.0431_E00917	152	■ 1.000	$h^{\prime}(a+b\,i)=a+b\,e_{\{1\}},\quad a,\,b\in$ $\boldsymbol{R}.$	$h'(a+bi) = a + be_1, \quad a, b \in \boldsymbol{R}.$	$h'(a+bi) = a + be_1, \quad a, b \in \boldsymbol{R}.$
0902.0431_E00918	152	■ 1.000	$f(a\,v)=h^{\prime}(a)\,f(v),\quad a\in C,\,v\in V.$	$f(av) = h'(a)f(v), \quad a \in C, v \in V.$	$f(av) = h'(a)f(v), \quad a \in C, v \in V.$
0902.0431_E00921	153	■ 1.000	$\begin{array}{l} x_{156}=h^{\prime}\left(\xi_1\right), \& x_{164}=h^{\prime}\left(x_3\right), \& x_{145}=h^{\prime}\left(\bar{x}_2\right), \\ x_{256}=h^{\prime}\left(\bar{x}_3\right), \& x_{264}=h^{\prime}\left(\xi_2\right), \& x_{245}=h^{\prime}\left(x_1\right), \\ x_{356}=h^{\prime}\left(x_2\right), \& x_{364}=h^{\prime}\left(\bar{x}_1\right), \& x_{345}=h^{\prime}\left(\xi_3\right), \\ x_{423}=h^{\prime}\left(\eta_1\right), \& x_{431}=h^{\prime}\left(y_3\right), \& x_{412}=h^{\prime}\left(\bar{y}_2\right), \\ x_{523}=h^{\prime}\left(\bar{y}_3\right), \& x_{531}=h^{\prime}\left(\eta_2\right), \& x_{512}=h^{\prime}\left(y_1\right), \\ x_{623}=h^{\prime}\left(y_2\right), \& x_{631}=h^{\prime}\left(\bar{y}_1\right), \& x_{612}=h^{\prime}\left(\eta_3\right), \\ & x_{123}=h^{\prime}(\xi), \\ & x_{456}=h^{\prime}(\eta) . \end{array}$	$\begin{array}{lll} x_{156}=h'(\xi_1), & x_{164}=h'(x_3), & x_{145}=h'(\bar{x}_2), \\ x_{256}=h'(\bar{x}_3), & x_{264}=h'(\xi_2), & x_{245}=h'(x_1), \\ x_{356}=h'(x_2), & x_{364}=h'(\bar{x}_1), & x_{345}=h'(\xi_3), \\ x_{423}=h'(\eta_1), & x_{431}=h'(y_3), & x_{412}=h'(\bar{y}_2), \\ x_{523}=h'(\bar{y}_3), & x_{531}=h'(\eta_2), & x_{512}=h'(y_1), \\ x_{623}=h'(y_2), & x_{631}=h'(\bar{y}_1), & x_{612}=h'(\eta_3), \\ & x_{123}=h'(\xi), \\ & x_{456}=h'(\eta). \end{array}$	$\begin{array}{lll} x_{156}=h'(\xi_1), & x_{164}=h'(x_3), & x_{145}=h'(\bar{x}_2), \\ x_{256}=h'(\bar{x}_3), & x_{264}=h'(\xi_2), & x_{245}=h'(x_1), \\ x_{356}=h'(x_2), & x_{364}=h'(\bar{x}_1), & x_{345}=h'(\xi_3), \\ x_{423}=h'(\eta_1), & x_{431}=h'(y_3), & x_{412}=h'(\bar{y}_2), \\ x_{523}=h'(\bar{y}_3), & x_{531}=h'(\eta_2), & x_{512}=h'(y_1), \\ x_{623}=h'(y_2), & x_{631}=h'(\bar{y}_1), & x_{612}=h'(\eta_3), \\ & x_{123}=h'(\xi), \\ & x_{456}=h'(\eta). \end{array}$
0902.0431_E00925	153	■ 1.000	$A(\boldsymbol{a}\wedge\boldsymbol{b}\wedge\boldsymbol{c})=A\boldsymbol{a}\wedge A\boldsymbol{b}\wedge A\boldsymbol{c}.$	$A(\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c}) = A\boldsymbol{a} \wedge A\boldsymbol{b} \wedge A\boldsymbol{c}.$	$A(\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c}) = A\boldsymbol{a} \wedge A\boldsymbol{b} \wedge A\boldsymbol{c}.$
0902.0431_E00926	153	■ 1.000	$D(\boldsymbol{a}\wedge\boldsymbol{b}\wedge\boldsymbol{c})=D\boldsymbol{a}\wedge\boldsymbol{b}\wedge\boldsymbol{c}+\boldsymbol{a}\wedge D\boldsymbol{b}\wedge\boldsymbol{c}+\boldsymbol{a}\wedge\boldsymbol{b}\wedge D\boldsymbol{c}.$	$D(\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c}) = D\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c} + \boldsymbol{a} \wedge D\boldsymbol{b} \wedge \boldsymbol{c} + \boldsymbol{a} \wedge \boldsymbol{b} \wedge D\boldsymbol{c}.$	$D(\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c}) = D\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c} + \boldsymbol{a} \wedge D\boldsymbol{b} \wedge \boldsymbol{c} + \boldsymbol{a} \wedge \boldsymbol{b} \wedge D\boldsymbol{c}.$
0902.0431_E00929	154	■ 1.000	$\varphi_{\boldsymbol{C}}\left(\left(\begin{array}{cc} B & L \\ -L^* & C \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)\right)=\Phi\left(\phi_{\boldsymbol{C}}(B, C), h(L),-\tau h(L),-i\nu e_1\right)$	$\varphi_C\left(\left(\begin{array}{cc} B & L \\ -L^* & C \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)\right)=\Phi(\phi_C(B,C),h(L),-\tau h(L),-i\nu e_1)$	$\varphi_C\left(\left(\begin{array}{cc} B & L \\ -L^* & C \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)\right)=\Phi(\phi_C(B,C),h(L),-\tau h(L),-i\nu e_1)$

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0902.0431_E00930	154	■ 1.000	$\nu=\operatorname{tr}\left(B^{\prime}\right)=-\operatorname{tr}\left(C^{\prime}\right), \quad B=B^{\prime}-\frac{\nu}{3} E, \quad C=C^{\prime}+\frac{\nu}{3} E,$	$\nu=\operatorname{tr}\left(B^{\prime}\right)=-\operatorname{tr}\left(C^{\prime}\right), \quad B=B^{\prime}-\frac{\nu}{3} E, \quad C=C^{\prime}+\frac{\nu}{3} E,$	$\nu=\operatorname{tr}\left(B^{\prime}\right)=-\operatorname{tr}\left(C^{\prime}\right), \quad B=B^{\prime}-\frac{\nu}{3} E, \quad C=C^{\prime}+\frac{\nu}{3} E,$
0902.0431_E00934	154	■ 1.000	$\psi_{*}(D) P=f^{-1}(D(f P)), \quad P \in\left(\mathfrak{P}_C\right)^C$	$\psi_{*}(D) P=f^{-1}(D(f P)), \quad P \in\left(\mathfrak{P}_C\right)^C$	$\psi_{*}(D) P=f^{-1}(D(f P)), \quad P \in\left(\mathfrak{P}_C\right)^C$
0902.0431_E00935	154	■ 1.000	$D=\left(\begin{array}{cc} B & L \\ -L^{*} & C \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)$	$D=\left(\begin{array}{cc} B & L \\ -L^{*} & C \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)$	$D=\left(\begin{array}{cc} B & L \\ -L^{*} & C \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)$
0902.0431_E00942	157	■ 1.000	$=\left(\begin{array}{c} \phi_{\boldsymbol{C}}(B, C) F_1(1)+\frac{i \nu e_1}{3} F_1(1) \\ 2 h(L) \times F_1(1) \\ 0 \\ -\left(h(L), F_1(1)\right) \end{array}\right)=\Phi\left(\phi_{\boldsymbol{C}}(B, C), h(L),-\tau h(L),-i \nu e_1\right)\left(\begin{array}{c} F_1(1) \\ 0 \\ 0 \\ 0 \end{array}\right).$	$=\left(\begin{array}{c} \phi_{\boldsymbol{C}}(B, C) F_1(1)+\frac{i \nu e_1}{3} F_1(1) \\ 2 h(L) \times F_1(1) \\ 0 \\ -\left(h(L), F_1(1)\right) \end{array}\right)=\Phi\left(\phi_{\boldsymbol{C}}(B, C), h(L),-\tau h(L),-i \nu e_1\right)\left(\begin{array}{c} F_1(1) \\ 0 \\ 0 \\ 0 \end{array}\right).$	$=\left(\begin{array}{c} \phi_{\boldsymbol{C}}(B, C) F_1(1)+\frac{i \nu e_1}{3} F_1(1) \\ 2 h(L) \times F_1(1) \\ 0 \\ -\left(h(L), F_1(1)\right) \end{array}\right)=\Phi\left(\phi_{\boldsymbol{C}}(B, C), h(L),-\tau h(L),-i \nu e_1\right)\left(\begin{array}{c} F_1(1) \\ 0 \\ 0 \\ 0 \end{array}\right).$
0902.0431_E00943	157	■ 1.000	$f^{-1}(D(f P))=\Phi\left(\phi_{\boldsymbol{C}}(B, C), h(L),-\tau h(L),-i \nu e_1\right) P.$	$f^{-1}(D(f P))=\Phi\left(\phi_{\boldsymbol{C}}(B, C), h(L),-\tau h(L),-i \nu e_1\right) P.$	$f^{-1}(D(f P))=\Phi\left(\phi_{\boldsymbol{C}}(B, C), h(L),-\tau h(L),-i \nu e_1\right) P.$
0902.0431_E00950	158	■ 1.000	$\mu^{-1}\left(M_1+i M_2, N_1+i N_2\right)=\left(\begin{array}{cc} -N_2-N_1 e_1 & M_2+M_1 e_1 \\ M_2-M_1 e_1 & N_2-N_1 e_1 \end{array}\right), \quad M_i, N_i \in M(3, \boldsymbol{C}).$	$\mu^{-1}\left(M_1+i M_2, N_1+i N_2\right)=\left(\begin{array}{cc} -N_2-N_1 e_1 & M_2+M_1 e_1 \\ M_2-M_1 e_1 & N_2-N_1 e_1 \end{array}\right), \quad M_i, N_i \in M(3, \boldsymbol{C}).$	$\mu^{-1}\left(M_1+i M_2, N_1+i N_2\right)=\left(\begin{array}{cc} -N_2-N_1 e_1 & M_2+M_1 e_1 \\ M_2-M_1 e_1 & N_2-N_1 e_1 \end{array}\right), \quad M_i, N_i \in M(3, \boldsymbol{C}).$
0902.0431_E00951	158	■ 1.000	$\mu\left(\widetilde{M} D^{*}\right)=\psi_{*}(D)\left(\mu \widetilde{M}\right).$	$\mu\left(\widetilde{M} D^{*}\right)=\psi_{*}(D)\left(\mu \widetilde{M}\right).$	$\mu\left(\widetilde{M} D^{*}\right)=\psi_{*}(D)\left(\mu \widetilde{M}\right).$
0902.0431_E00954	159	■ 1.000	$=\left(\begin{array}{cc} -N_2-N_1 e_1 & M_2+M_1 e_1 \\ M_2+M_1 e_1 & N_2+N_1 e_1 \end{array}\right)\left(\left(\begin{array}{cc} -B & -L \\ L^{*} & -C \end{array}\right)-\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)\right)=\widetilde{M} D^{*}.$	$=\left(\begin{array}{cc} -N_2-N_1 e_1 & M_2+M_1 e_1 \\ M_2+M_1 e_1 & N_2+N_1 e_1 \end{array}\right)\left(\left(\begin{array}{cc} -B & -L \\ L^{*} & -C \end{array}\right)-\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)\right)=\widetilde{M} D^{*}.$	$=\left(\begin{array}{cc} -N_2-N_1 e_1 & M_2+M_1 e_1 \\ M_2+M_1 e_1 & N_2+N_1 e_1 \end{array}\right)\left(\left(\begin{array}{cc} -B & -L \\ L^{*} & -C \end{array}\right)-\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)\right)=\widetilde{M} D^{*}.$
0902.0431_E00956	159	■ 1.000	$(Q, A)\left(\sum\left(\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c}\right)+\widetilde{M}\right)=\sum\left(A \boldsymbol{a} \wedge A \boldsymbol{b} \wedge A \boldsymbol{c}\right)+Q \widetilde{M} A^{*},$	$(Q, A)\left(\sum\left(\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c}\right)+\widetilde{M}\right)=\sum\left(A \boldsymbol{a} \wedge A \boldsymbol{b} \wedge A \boldsymbol{c}\right)+Q \widetilde{M} A^{*},$	$(Q, A)\left(\sum\left(\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c}\right)+\widetilde{M}\right)=\sum\left(A \boldsymbol{a} \wedge A \boldsymbol{b} \wedge A \boldsymbol{c}\right)+Q \widetilde{M} A^{*},$
0902.0431_E00958	159	■ 1.000	$\Phi\left(\phi_{\boldsymbol{C}}(B, C), h(L),-\tau h(L),-i \nu e_1\right) \in \mathfrak{e}_7.$	$\Phi\left(\phi_{\boldsymbol{C}}(B, C), h(L),-\tau h(L),-i \nu e_1\right) \in \mathfrak{e}_7.$	$\Phi\left(\phi_{\boldsymbol{C}}(B, C), h(L),-\tau h(L),-i \nu e_1\right) \in \mathfrak{e}_7.$
0902.0431_E00961	160	■ 1.000	$\alpha=\psi(E, A) \beta=\psi(E, A) \psi(Q, E)=\psi(Q, A).$	$\alpha=\psi(E, A) \beta=\psi(E, A) \psi(Q, E)=\psi(Q, A).$	$\alpha=\psi(E, A) \beta=\psi(E, A) \psi(Q, E)=\psi(Q, A).$
0902.0431_E00966	161	■ 1.000	$\mathfrak{P}=\mathfrak{J}(3, \mathfrak{C}) \oplus \mathfrak{J}(3, \mathfrak{C}) \oplus \boldsymbol{R} \oplus \boldsymbol{R},$ $\mathfrak{P}'=\mathfrak{J}(3, \mathfrak{C}') \oplus \mathfrak{J}(3, \mathfrak{C}') \oplus \boldsymbol{R} \oplus \boldsymbol{R}.$	$\mathfrak{P}=\mathfrak{J}(3, \mathfrak{C}) \oplus \mathfrak{J}(3, \mathfrak{C}) \oplus \boldsymbol{R} \oplus \boldsymbol{R},$ $\mathfrak{P}'=\mathfrak{J}(3, \mathfrak{C}') \oplus \mathfrak{J}(3, \mathfrak{C}') \oplus \boldsymbol{R} \oplus \boldsymbol{R}.$	$\mathfrak{P}=\mathfrak{J}(3, \mathfrak{C}) \oplus \mathfrak{J}(3, \mathfrak{C}) \oplus \boldsymbol{R} \oplus \boldsymbol{R},$ $\mathfrak{P}'=\mathfrak{J}(3, \mathfrak{C}') \oplus \mathfrak{J}(3, \mathfrak{C}') \oplus \boldsymbol{R} \oplus \boldsymbol{R}.$
0902.0431_E00967	161	■ 1.000	$\langle P, Q\rangle_{\sigma}=\langle\sigma P, Q\rangle.$	$\langle P, Q\rangle_{\sigma}=\langle\sigma P, Q\rangle.$	$\langle P, Q\rangle_{\sigma}=\langle\sigma P, Q\rangle.$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00970	161	■ 1.000	$\begin{aligned} E_{7(7)} &\simeq SU(8)/\mathbf{Z}_2 \times \mathbf{R}^{70}, \\ E_{7(-5)} &\simeq (SU(2) \times \operatorname{Spin}(12))/\mathbf{Z}_2 \times \mathbf{R}^{64}, \\ E_{7(-25)} &\simeq (U(1) \times E_6)/\mathbf{Z}_3 \times \mathbf{R}^{54}. \end{aligned}$	$\begin{aligned} E_{7(7)} &\simeq SU(8)/\mathbf{Z}_2 \times \mathbf{R}^{70}, \\ E_{7(-5)} &\simeq (SU(2) \times \operatorname{Spin}(12))/\mathbf{Z}_2 \times \mathbf{R}^{64}, \\ E_{7(-25)} &\simeq (U(1) \times E_6)/\mathbf{Z}_3 \times \mathbf{R}^{54}. \end{aligned}$	$\begin{aligned} E_{7(7)} &\simeq SU(8)/\mathbf{Z}_2 \times \mathbf{R}^{70}, \\ E_{7(-5)} &\simeq (SU(2) \times \operatorname{Spin}(12))/\mathbf{Z}_2 \times \mathbf{R}^{64}, \\ E_{7(-25)} &\simeq (U(1) \times E_6)/\mathbf{Z}_3 \times \mathbf{R}^{54}. \end{aligned}$
0902.0431_E00971	161	■ 1.000	$\begin{aligned} z(E_{7(7)}) &= \mathbf{Z}_2, \quad z(E_{7(-5)}) = \mathbf{Z}_2, \quad z(E_{7(-25)}) = \mathbf{Z}_2. \end{aligned}$	$z(E_{7(7)}) = \mathbf{Z}_2, \quad z(E_{7(-5)}) = \mathbf{Z}_2, \quad z(E_{7(-25)}) = \mathbf{Z}_2.$	$z(E_{7(7)}) = \mathbf{Z}_2, \quad z(E_{7(-5)}) = \mathbf{Z}_2, \quad z(E_{7(-25)}) = \mathbf{Z}_2.$
0902.0431_E00972	162	■ 1.000	$\mathfrak{e}_8^C = \mathfrak{e}_7^C \oplus \mathfrak{P}^C \oplus \mathfrak{P}^C \oplus C \oplus C \oplus C,$	$\mathfrak{e}_8^C = \mathfrak{e}_7^C \oplus \mathfrak{P}^C \oplus \mathfrak{P}^C \oplus C \oplus C \oplus C,$	$\mathfrak{e}_8^C = \mathfrak{e}_7^C \oplus \mathfrak{P}^C \oplus \mathfrak{P}^C \oplus C \oplus C \oplus C,$
0902.0431_E00975	162	■ 1.000	$\begin{aligned} &\left[R_1, R_2 + R_3 \right] = \left[R_1, R_2 \right] + \left[R_1, R_3 \right], \\ &\left[k R_1, R_2 \right] = k \left[R_1, R_2 \right], \quad k \in C, \\ &\left[R_1, R_2 \right] = - \left[R_2, R_1 \right] \end{aligned}$	$\begin{aligned} &\left[R_1, R_2 + R_3 \right] = \left[R_1, R_2 \right] + \left[R_1, R_3 \right], \\ &\left[k R_1, R_2 \right] = k \left[R_1, R_2 \right], \quad k \in C, \\ &\left[R_1, R_2 \right] = - \left[R_2, R_1 \right] \end{aligned}$	$\begin{aligned} &\left[R_1, R_2 + R_3 \right] = \left[R_1, R_2 \right] + \left[R_1, R_3 \right], \\ &\left[k R_1, R_2 \right] = k \left[R_1, R_2 \right], \quad k \in C, \\ &\left[R_1, R_2 \right] = - \left[R_2, R_1 \right] \end{aligned}$
0902.0431_E00978	163	■ 1.000	$\mathfrak{e}_8^C = \mathfrak{e}_7^C \oplus \mathfrak{K}^C,$	$\mathfrak{e}_8^C = \mathfrak{e}_7^C \oplus \mathfrak{K}^C,$	$\mathfrak{e}_8^C = \mathfrak{e}_7^C \oplus \mathfrak{K}^C,$
0902.0431_E00980	163	■ 1.000	$\begin{aligned} &\ni \left[\Phi(0,0,0,1), (0,0,1,0)^- \right] = (0,0,1,0)^-, \\ &\ni \left[\Phi(0,0,0,1), (0,0,0,-1)_- \right] = (0,0,0,1)_-, \\ &\ni \left[(0,0,1,0)^-, (0,0,0,4)^- \right] = 1^-, \\ &\ni \left[(0,0,0,1)_-, (0,0,4,0)_- \right] = 1_-, \\ &\ni \left[1^-, 1_- \right] = 1, \\ &\ni \left[1^- + 1_-, Q^- + P_- \right] = P^- + Q_-, \end{aligned}$	$\begin{aligned} &\mathfrak{a} \ni \left[\Phi(0,0,0,1), (0,0,1,0)^- \right] = (0,0,1,0)^-, \\ &\mathfrak{a} \ni \left[\Phi(0,0,0,1), (0,0,0,-1)_- \right] = (0,0,0,1)_-, \\ &\mathfrak{a} \ni \left[(0,0,1,0)^-, (0,0,0,4)^- \right] = 1^-, \\ &\mathfrak{a} \ni \left[(0,0,0,1)_-, (0,0,4,0)_- \right] = 1_-, \\ &\mathfrak{a} \ni \left[1^-, 1_- \right] = 1, \\ &\mathfrak{a} \ni \left[1^- + 1_-, Q^- + P_- \right] = P^- + Q_-, \end{aligned}$	$\begin{aligned} &\mathfrak{a} \ni \left[\Phi(0,0,0,1), (0,0,1,0)^- \right] = (0,0,1,0)^-, \\ &\mathfrak{a} \ni \left[\Phi(0,0,0,1), (0,0,0,-1)_- \right] = (0,0,0,1)_-, \\ &\mathfrak{a} \ni \left[(0,0,1,0)^-, (0,0,0,4)^- \right] = 1^-, \\ &\mathfrak{a} \ni \left[(0,0,0,1)_-, (0,0,4,0)_- \right] = 1_-, \\ &\mathfrak{a} \ni \left[1^-, 1_- \right] = 1, \\ &\mathfrak{a} \ni \left[1^- + 1_-, Q^- + P_- \right] = P^- + Q_-, \end{aligned}$
0902.0431_E00984	164	■ 1.000	$\left[\ni \left[1_-^-, R \right] = (0,0,0,-s, 0,0) \right].$	$\mathfrak{a} \ni \left[1_-, R \right] = (0,0,0,-s,0,0).$	$\mathfrak{a} \ni \left[1_-, R \right] = (0,0,0,-s,0,0).$
0902.0431_E00990	165	■ 1.000	$\begin{aligned} &B_8(R_1, R_2) \\ &= -15(R_1, R_2)_8 \\ &= -15(\Phi_1, \Phi_2)_7 + 15\{Q_1, P_2\} - 15\{P_1, Q_2\} + 120r_1r_2 + 60t_1s_2 + 60s_1t_2 \\ &= \frac{5}{3}B_7(\Phi_1, \Phi_2) + 15\{Q_1, P_2\} - 15\{P_1, Q_2\} + 120r_1r_2 + 60t_1s_2 + 60s_1t_2 \end{aligned}$	$\begin{aligned} &B_8(R_1, R_2) \\ &= -15(R_1, R_2)_8 \\ &= -15(\Phi_1, \Phi_2)_7 + 15\{Q_1, P_2\} - 15\{P_1, Q_2\} + 120r_1r_2 + 60t_1s_2 + 60s_1t_2 \\ &= \frac{5}{3}B_7(\Phi_1, \Phi_2) + 15\{Q_1, P_2\} - 15\{P_1, Q_2\} + 120r_1r_2 + 60t_1s_2 + 60s_1t_2 \end{aligned}$	$\begin{aligned} &B_8(R_1, R_2) \\ &= -15(R_1, R_2)_8 \\ &= -15(\Phi_1, \Phi_2)_7 + 15\{Q_1, P_2\} - 15\{P_1, Q_2\} + 120r_1r_2 + 60t_1s_2 + 60s_1t_2 \\ &= \frac{5}{3}B_7(\Phi_1, \Phi_2) + 15\{Q_1, P_2\} - 15\{P_1, Q_2\} + 120r_1r_2 + 60t_1s_2 + 60s_1t_2, \end{aligned}$

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0902.0431_E00991	165	■ 1.000	$B_{\{8\}}\left(R_{\{1\}}, R_{\{2\}}\right)=k\left(R_{\{1\}}, R_{\{2\}}\right)_{\{8\}}, \quad \text{\texttt{\textbackslash quad}} R_{\{i\}} \in \mathfrak{e}_{\{8\}}^C.$	$B_8(R_1, R_2) = k(R_1, R_2)_8, \quad R_i \in \mathfrak{e}_8^C.$	$B_8(R_1, R_2) = k(R_1, R_2)_8, \quad R_i \in \mathfrak{e}_8^C.$
0902.0431_E00992	165	■ 1.000	$(1,1)_{\{8\}}=-8.$	$(1,1)_8 = -8.$	$(1,1)_8 = -8.$
0902.0431_E00993	165	■ 1.000	$[1,[1,(\Phi, P, Q, r, s, t)]]=[1,(\theta, P,-Q, \theta,2 s,-2 t)]=(\theta, P, Q, \theta,4 s,4 t),$	$[1,[1,(\Phi, P, Q, r, s, t)]] = [1, (0, P, -Q, 0, 2s, -2t)] = (0, P, Q, 0, 4s, 4t),$	$[1,[1,(\Phi, P, Q, r, s, t)]] = [1, (0, P, -Q, 0, 2s, -2t)] = (0, P, Q, 0, 4s, 4t),$
0902.0431_E01002	167	■ 1.000	$\tau(\Phi, P, Q, r, s, t)=(\tau \Phi \tau, \tau P, \tau Q, \tau r, \tau s, \tau t),$	$\tau(\Phi, P, Q, r, s, t) = (\tau\Phi\tau, \tau P, \tau Q, \tau r, \tau s, \tau t),$	$\tau(\Phi, P, Q, r, s, t) = (\tau\Phi\tau, \tau P, \tau Q, \tau r, \tau s, \tau t),$
0902.0431_E01005	167	■ 1.000	$\left(\tau \lambda \Phi_1 \lambda^{-1} \tau, \Phi_2\right)_7=2\left(\tau^t \phi_1 \tau, \phi_2\right)_6+4\left\langle A_1, A_2\right\rangle+4\left\langle B_1, B_2\right\rangle+\frac{8}{3}\left(\tau \nu_1\right) \nu_2.$	$(\tau\lambda\Phi_1\lambda^{-1}\tau, \Phi_2)_7 = 2(\tau^t\phi_1\tau, \phi_2)_6 + 4\langle A_1, A_2\rangle + 4\langle B_1, B_2\rangle + \frac{8}{3}(\tau\nu_1)\nu_2.$	$(\tau\lambda\Phi_1\lambda^{-1}\tau, \Phi_2)_7 = 2(\tau^t\phi_1\tau, \phi_2)_6 + 4\langle A_1, A_2\rangle + 4\langle B_1, B_2\rangle + \frac{8}{3}(\tau\nu_1)\nu_2.$
0902.0431_E01006	167	■ 1.000	$\left(\tau^t \phi_1 \tau, \phi_2\right)_6=-\left(\tau \delta_1 \tau, \delta_2\right)_4+\left\langle T_1, T_2\right\rangle.$	$(\tau^t\phi_1\tau, \phi_2)_6 = -(\tau\delta_1\tau, \delta_2)_4 + \langle T_1, T_2\rangle.$	$(\tau^t\phi_1\tau, \phi_2)_6 = -(\tau\delta_1\tau, \delta_2)_4 + \langle T_1, T_2\rangle.$
0902.0431_E01011	168	■ 1.000	$\tau \widetilde{\lambda} R=\left(\tau \lambda \Phi \lambda^{-1} \tau, \tau \lambda Q,-\tau \lambda P,-\tau r,-\tau t,-\tau s\right) .$	$\tau\widetilde{\lambda}R = (\tau\lambda\Phi\lambda^{-1}\tau, \tau\lambda Q, -\tau\lambda P, -\tau r, -\tau t, -\tau s).$	$\tau\widetilde{\lambda}R = (\tau\lambda\Phi\lambda^{-1}\tau, \tau\lambda Q, -\tau\lambda P, -\tau r, -\tau t, -\tau s).$
0902.0431_E01012	168	■ 1.000	$R=\frac{R-R^*}{2}+i \frac{R+R^*}{2 i}, \quad \frac{R-R^*}{2}, \frac{R+R^*}{2 i} \in \mathfrak{e}_8.$	$R = \frac{R-R^*}{2} + i\frac{R+R^*}{2i}, \quad \frac{R-R^*}{2}, \frac{R+R^*}{2i} \in \mathfrak{e}_8.$	$R = \frac{R-R^*}{2} + i\frac{R+R^*}{2i}, \quad \frac{R-R^*}{2}, \frac{R+R^*}{2i} \in \mathfrak{e}_8.$
0902.0431_E01017	170	■ 1.000	$\mathfrak{h}=\left\{\left(\Phi(h), 0,0, r, 0,0\right) \mid \Phi(h) \in \mathfrak{h}_7, r \in C\right\}$	$\mathfrak{h} = \{(\Phi(h), 0, 0, r, 0, 0) \mid \Phi(h) \in \mathfrak{h}_7, r \in C\}$	$\mathfrak{h} = \{(\Phi(h), 0, 0, r, 0, 0) \mid \Phi(h) \in \mathfrak{h}_7, r \in C\}$
0902.0431_E01018	170	■ 1.000	$\left[\left(\Phi(h), 0,0, r, 0,0\right),\left(\Phi, 0,0,0,0,0\right)\right]=\left([\Phi(h), \Phi], 0,0,0,0,0\right) .$	$[(\Phi(h), 0, 0, r, 0, 0), (\Phi, 0, 0, 0, 0, 0)] = ([\Phi(h), \Phi], 0, 0, 0, 0, 0).$	$[(\Phi(h), 0, 0, r, 0, 0), (\Phi, 0, 0, 0, 0, 0)] = ([\Phi(h), \Phi], 0, 0, 0, 0, 0).$
0902.0431_E01020	170	■ 1.000	$\begin{aligned} & \left(\Phi\left(h_{\delta}+\tilde{H}, 0,0, \nu\right)+r 1\right)(X, Y, \xi, \eta) \\ & =\left(\left(h_{\delta}+\tilde{H}-\frac{1}{3} \nu+r\right) X,\left(h_{\delta}-\tilde{H}+\frac{1}{3} \nu+r\right) Y,(\nu+r) \xi,(-\nu+r) \eta\right) . \end{aligned}$	$\begin{aligned} & \left(\Phi\left(h_{\delta}+\tilde{H}, 0,0, \nu\right)+r 1\right)(X, Y, \xi, \eta) \\ & =\left(\left(h_{\delta}+\tilde{H}-\frac{1}{3} \nu+r\right) X,\left(h_{\delta}-\tilde{H}+\frac{1}{3} \nu+r\right) Y,(\nu+r) \xi,(-\nu+r) \eta\right) . \end{aligned}$	$\begin{aligned} & (\Phi(h_{\delta}+\tilde{H}, 0, 0, \nu)+r1)(X, Y, \xi, \eta) \\ & =\left(\left(h_{\delta}+\tilde{H}-\frac{1}{3}\nu+r\right) X,\left(h_{\delta}-\tilde{H}+\frac{1}{3}\nu+r\right) Y,(\nu+r) \xi,(-\nu+r) \eta\right) . \end{aligned}$
0902.0431_E01024	171	■ 1.000	$\begin{aligned} & \alpha_1=\frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2-\lambda_3\right)+\frac{1}{2}\left(\mu_3-\mu_1\right), \\ & \alpha_2=\mu_1-\frac{1}{3} \nu-r, \quad \alpha_3=2 r, \\ & \alpha_4=\mu_2-\frac{1}{3} \nu-r, \quad \alpha_5=\lambda_3-\frac{1}{2}\left(\mu_2-\mu_3\right), \\ & \alpha_6=\lambda_2-\lambda_3, \quad \alpha_7=\lambda_1-\lambda_2, \quad \alpha_8=\nu-r \end{aligned}$	$\begin{aligned} \alpha_1 &= \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1), \\ \alpha_2 &= \mu_1 - \frac{1}{3}\nu - r, \quad \alpha_3 = 2r, \\ \alpha_4 &= \mu_2 - \frac{1}{3}\nu - r, \quad \alpha_5 = \lambda_3 - \frac{1}{2}(\mu_2 - \mu_3), \\ \alpha_6 &= \lambda_2 - \lambda_3, \quad \alpha_7 = \lambda_1 - \lambda_2, \quad \alpha_8 = \nu - r \end{aligned}$	$\begin{aligned} \alpha_1 &= \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1), \\ \alpha_2 &= \mu_1 - \frac{1}{3}\nu - r, \quad \alpha_3 = 2r, \\ \alpha_4 &= \mu_2 - \frac{1}{3}\nu - r, \quad \alpha_5 = \lambda_3 - \frac{1}{2}(\mu_2 - \mu_3), \\ \alpha_6 &= \lambda_2 - \lambda_3, \quad \alpha_7 = \lambda_1 - \lambda_2, \quad \alpha_8 = \nu - r \end{aligned}$
0902.0431_E01025	171	■ 1.000	$\mu=2 \alpha_1+4 \alpha_2+6 \alpha_3+5 \alpha_4+4 \alpha_5+3 \alpha_6+2 \alpha_7+3 \alpha_8$	$\mu = 2\alpha_1 + 4\alpha_2 + 6\alpha_3 + 5\alpha_4 + 4\alpha_5 + 3\alpha_6 + 2\alpha_7 + 3\alpha_8$	$\mu = 2\alpha_1 + 4\alpha_2 + 6\alpha_3 + 5\alpha_4 + 4\alpha_5 + 3\alpha_6 + 2\alpha_7 + 3\alpha_8$
0902.0431_E01035	176	■ 1.000	$\mathfrak{h}_{\{\boldsymbol{R}\}}=\left\{\left(\Phi(h), 0,0, r, 0\right) \mid \Phi(h) \in \mathfrak{h}_7\right\}_{\{\boldsymbol{R}\}}, \quad r \in \boldsymbol{R},$	$\mathfrak{h}_{\boldsymbol{R}} = \{(\Phi(h), 0, 0, r, 0) \mid \Phi(h) \in (\mathfrak{h}_7)_{\boldsymbol{R}}, r \in \boldsymbol{R}\},$	$\mathfrak{h}_{\boldsymbol{R}} = \{(\Phi(h), 0, 0, r, 0) \mid \Phi(h) \in (\mathfrak{h}_7)_{\boldsymbol{R}}, r \in \boldsymbol{R}\},$

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0902.0431_EQ1040	177	■ 1.000	$\left(\alpha_1, \alpha_1\right)=B_7\left(H_{\alpha_1}, H_{\alpha_1}\right)=60 \frac{1}{120} \frac{1}{120} 4+30 \frac{1}{120} \frac{1}{120} 8=\frac{1}{30},$	$(\alpha_1, \alpha_1)=B_7\left(H_{\alpha_1}, H_{\alpha_1}\right)=60 \frac{1}{120} \frac{1}{120} 4+30 \frac{1}{120} \frac{1}{120} 8=\frac{1}{30},$	$(\alpha_1, \alpha_1)=B_7\left(H_{\alpha_1}, H_{\alpha_1}\right)=60 \frac{1}{120} \frac{1}{120} 4+30 \frac{1}{120} \frac{1}{120} 8=\frac{1}{30},$
0902.0431_EQ1043	178	■ 1.000	$\widetilde{\beta}=\left(\begin{array}{cccccc} \operatorname{Ad} \beta & 0 & 0 & 0 & 0 & 0 \\ 0 & \beta & 0 & 0 & 0 & 0 \\ 0 & 0 & \beta & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right),$	$\widetilde{\beta}=\left(\begin{array}{cccccc} \operatorname{Ad} \beta & 0 & 0 & 0 & 0 & 0 \\ 0 & \beta & 0 & 0 & 0 & 0 \\ 0 & 0 & \beta & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right),$	$\widetilde{\beta}=\left(\begin{array}{cccccc} \operatorname{Ad} \beta & 0 & 0 & 0 & 0 & 0 \\ 0 & \beta & 0 & 0 & 0 & 0 \\ 0 & 0 & \beta & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right),$
0902.0431_EQ1045	178	■ 1.000	$\begin{gathered} 0=\left[\left(\beta_1 \Phi, \beta_{21} \Phi, \beta_{31} \Phi, a_1 \Phi, a_2 \Phi, a_3 \Phi\right),\left(0,0,0,1,0,0\right)\right] \\ =\left(0,-\beta_{21} \Phi, \beta_{31} \Phi, 0,-2 a_2 \Phi, 2 a_3 \Phi\right) \end{gathered}$	$0=\left[\left(\beta_1 \Phi, \beta_{21} \Phi, \beta_{31} \Phi, a_1 \Phi, a_2 \Phi, a_3 \Phi\right),\left(0,0,0,1,0,0\right)\right]$ $=\left(0,-\beta_{21} \Phi, \beta_{31} \Phi, 0,-2 a_2 \Phi, 2 a_3 \Phi\right)$	$0=\left[\left(\beta_1 \Phi, \beta_{21} \Phi, \beta_{31} \Phi, a_1 \Phi, a_2 \Phi, a_3 \Phi\right),\left(0,0,0,1,0,0\right)\right]$ $=\left(0,-\beta_{21} \Phi, \beta_{31} \Phi, 0,-2 a_2 \Phi, 2 a_3 \Phi\right),$
0902.0431_EQ1048	179	■ 1.000	$\alpha=\left(\begin{array}{cccc} \beta_1 & 0 & 0 & 0 \\ 0 & \beta_2 & 0 & 0 \\ 0 & 0 & \beta_3 & 0 \\ 0 & 0 & 0 & E \end{array}\right).$	$\alpha=\left(\begin{array}{cccc} \beta_1 & 0 & 0 & 0 \\ 0 & \beta_2 & 0 & 0 \\ 0 & 0 & \beta_3 & 0 \\ 0 & 0 & 0 & E \end{array}\right).$	$\alpha=\left(\begin{array}{cccc} \beta_1 & 0 & 0 & 0 \\ 0 & \beta_2 & 0 & 0 \\ 0 & 0 & \beta_3 & 0 \\ 0 & 0 & 0 & E \end{array}\right).$
0902.0431_EQ1051 (iii)	179	■ 1.000	$\left(\beta_1 \Phi\right)\left(\beta_2 P\right)=\beta_2(\Phi P).$	$\left(\beta_1 \Phi\right)\left(\beta_2 P\right)=\beta_2(\Phi P).$	$\left(\beta_1 \Phi\right)\left(\beta_2 P\right)=\beta_2(\Phi P).$
0902.0431_EQ1055	179	■ 1.000	$v(\Phi, P, Q, r, s, t)=(\Phi,-P,-Q, r, s, t).$	$v(\Phi, P, Q, r, s, t)=(\Phi,-P,-Q, r, s, t).$	$v(\Phi, P, Q, r, s, t)=(\Phi,-P,-Q, r, s, t).$
0902.0431_EQ1056	179	■ 1.000	$\left(E_8\right)^v=\left\{\alpha \in E_8 \mid v \alpha=\alpha v\right\}.$	$\left(E_8\right)^v=\left\{\alpha \in E_8 \mid v \alpha=\alpha v\right\}.$	$\left(E_8\right)^v=\left\{\alpha \in E_8 \mid v \alpha=\alpha v\right\}.$
0902.0431_EQ1058	180	■ 1.000	$\left(E_8\right)_{1-}=\left\{\alpha \in E_8 \mid \alpha 1_{-}=1_{-}\right\}.$	$\left(E_8\right)_{1-}=\left\{\alpha \in E_8 \mid \alpha 1_{-}=1_{-}\right\}.$	$\left(E_8\right)_{1-}=\left\{\alpha \in E_8 \mid \alpha 1_{-}=1_{-}\right\}.$
0902.0431_EQ1061	180	■ 1.000	$E_8 / E_7 \simeq \mathfrak{W}_1.$	$E_8 / E_7 \simeq \mathfrak{W}_1.$	$E_8 / E_7 \simeq \mathfrak{W}_1.$
0902.0431_EQ1062	180	■ 1.000	$\varphi_3(S U(2))=\left\{\varphi_3(A) \in E_8 \mid A \in S U(2)\right\}$	$\varphi_3(S U(2))=\left\{\varphi_3(A) \in E_8 \mid A \in S U(2)\right\}$	$\varphi_3(S U(2))=\left\{\varphi_3(A) \in E_8 \mid A \in S U(2)\right\}$
0902.0431_EQ1063	180	■ 1.000	$\varphi_3(A)=\left(\begin{array}{cccccc} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & a 1 & -\tau b 1 & 0 & 0 & 0 \\ 0 & b 1 & \tau a 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & (\tau a) a-(\tau b) b & -(\tau a) b & a(\tau b) \\ 0 & 0 & 0 & 2 a(\tau b) & a^2 & -(\tau b)^2 \\ 0 & 0 & 0 & 2(\tau a) b & -b^2 & (\tau a)^2 \end{array}\right).$	$\varphi_3(A)=\left(\begin{array}{cccccc} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & a 1 & -\tau b 1 & 0 & 0 & 0 \\ 0 & b 1 & \tau a 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & (\tau a) a-(\tau b) b & -(\tau a) b & a(\tau b) \\ 0 & 0 & 0 & 2 a(\tau b) & a^2 & -(\tau b)^2 \\ 0 & 0 & 0 & 2(\tau a) b & -b^2 & (\tau a)^2 \end{array}\right).$	$\varphi_3(A)=\left(\begin{array}{cccccc} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & a 1 & -\tau b 1 & 0 & 0 & 0 \\ 0 & b 1 & \tau a 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & (\tau a) a-(\tau b) b & -(\tau a) b & a(\tau b) \\ 0 & 0 & 0 & 2 a(\tau b) & a^2 & -(\tau b)^2 \\ 0 & 0 & 0 & 2(\tau a) b & -b^2 & (\tau a)^2 \end{array}\right).$
0902.0431_EQ1064	181	■ 1.000	$\varphi(A, \beta)=\varphi_3(A) \beta.$	$\varphi(A, \beta)=\varphi_3(A) \beta.$	$\varphi(A, \beta)=\varphi_3(A) \beta.$

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0902.0431_EQ1065	181	■1.000	$\begin{aligned} \left(\mathfrak{e}_8\right)^v &= \left(\Theta(R) \in \Theta(\mathfrak{e}_8) \mid v\Theta(R) = \Theta(R)v\right) \cong \{R \in \mathfrak{e}_8 \mid vR = R\} \\ &= \{(\Phi, 0, 0, r, s, -\tau s) \mid \Phi \in \mathfrak{e}_7, r \in i\boldsymbol{R}, s \in C\} . \end{aligned}$	$\begin{aligned} (\mathfrak{e}_8)^v &= \{\Theta(R) \in \Theta(\mathfrak{e}_8) \mid v\Theta(R) = \Theta(R)v\} \cong \{R \in \mathfrak{e}_8 \mid vR = R\} \\ &= \{(\Phi, 0, 0, r, s, -\tau s) \mid \Phi \in \mathfrak{e}_7, r \in i\boldsymbol{R}, s \in C\} . \end{aligned}$	$\begin{aligned} (\mathfrak{e}_8)^v &= \{\Theta(R) \in \Theta(\mathfrak{e}_8) \mid v\Theta(R) = \Theta(R)v\} \cong \{R \in \mathfrak{e}_8 \mid vR = R\} \\ &= \{(\Phi, 0, 0, r, s, -\tau s) \mid \Phi \in \mathfrak{e}_7, r \in i\boldsymbol{R}, s \in C\} . \end{aligned}$
0902.0431_EQ1068	181	■1.000	$l\left(\left(\left(x_{kl}+iy_{kl}\right)\right)\right)=\left(\left(\begin{array}{cc} x_{kl} & y_{kl} \\ -y_{kl} & x_{kl} \end{array}\right)\right), \quad x_{kl}, y_{kl} \in \boldsymbol{R}.$	$l\left(\left(x_{kl}+iy_{kl}\right)\right)=\left(\left(\begin{array}{cc} x_{kl} & y_{kl} \\ -y_{kl} & x_{kl} \end{array}\right)\right), \quad x_{kl}, y_{kl} \in \boldsymbol{R}.$	$l\left(\left(x_{kl}+iy_{kl}\right)\right)=\left(\left(\begin{array}{cc} x_{kl} & y_{kl} \\ -y_{kl} & x_{kl} \end{array}\right)\right), \quad x_{kl}, y_{kl} \in \boldsymbol{R}.$
0902.0431_EQ1069	182	■1.000	$I=\operatorname{diag}(I,\cdots,I), \quad \text{\rm quad } I=\left(\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array}\right), \\ \text{\rm quad } J=\operatorname{diag}(J,\cdots,J), \quad \text{\rm quad } J=\left(\begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array}\right),$	$I=\operatorname{diag}(I,\cdots,I), \quad I=\left(\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array}\right), \quad J=\operatorname{diag}(J,\cdots,J), \quad J=\left(\begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array}\right),$	$I=\operatorname{diag}(I,\cdots,I), \quad I=\left(\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array}\right), \quad J=\operatorname{diag}(J,\cdots,J), \quad J=\left(\begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array}\right),$
0902.0431_EQ1070	182	■1.000	$l(\mathfrak{S}(8,C))\,I=\{B\in\mathfrak{so}(16)\mid JB=-BJ\}.$	$l(\mathfrak{S}(8,C))I=\{B\in\mathfrak{so}(16)\mid JB=-BJ\}.$	$l(\mathfrak{S}(8,C))I=\{B\in\mathfrak{so}(16)\mid JB=-BJ\}.$
0902.0431_EQ1072	182	■1.000	$l(-D)=-B=\{ \}^{\{t\}} B=\{ \}^{\{t\}} l(D)=l(\tau^t D)$	$l(-D)=-B=\mathop{^t} B=\mathop{^t} l(D)=l(\tau^t D)$	$l(-D)=-B=\mathop{^t} B=\mathop{^t} l(D)=l(\tau^t D)$
0902.0431_EQ1074	182	■1.000	$l(-S)\,I=-B=\{ \}^{\{t\}} B=\{ \}^{\{t\}} I^{\{t\}}(l(S))=I\,l(\tau^t S)=l(\mathop{^t} S)\,I$	$l(-S)I=-B=\mathop{^t} B=\mathop{^t} I^{\mathop{^t}}(l(S))=I l(\tau^{\mathop{^t}} S)=l(\mathop{^t} S)I$	$l(-S)I=-B=\mathop{^t} B=\mathop{^t} I^{\mathop{^t}}(l(S))=I l(\tau^{\mathop{^t}} S)=l(\mathop{^t} S)I$
0902.0431_EQ1075	182	■1.000	$\varphi_*([gX_1,gX_2])=2(X_1\vee\gamma X_2-X_2\vee\gamma X_1), \quad X_1,X_2\in\mathfrak{J}^C.$	$\varphi_*([gX_1,gX_2])=2(X_1\vee\gamma X_2-X_2\vee\gamma X_1), \quad X_1,X_2\in\mathfrak{J}^C.$	$\varphi_*([gX_1,gX_2])=2(X_1\vee\gamma X_2-X_2\vee\gamma X_1), \quad X_1,X_2\in\mathfrak{J}^C.$
0902.0431_EQ1084	184	■1.000	$\begin{aligned} &\text{\rm quad } A=2\,\gamma X_1\times Y_2-2\gamma X_2\times Y_1+\eta_1X_2-\eta_2X_1-\xi_1\gamma X_2+\xi_2\gamma Y_1\in\mathfrak{J}^C) \\ &=kD+ik(gA)+\frac{i}{2}\{P_1,P_2\}E. \end{aligned}$	$\begin{aligned} A &= 2\gamma X_1 \times Y_2 - 2\gamma X_2 \times Y_1 + \eta_1 X_2 - \eta_2 X_1 - \xi_1 \gamma X_2 + \xi_2 \gamma Y_1 \in \mathfrak{J}^C) \\ &= kD + ik(gA) + \frac{i}{2} \{P_1, P_2\} E. \end{aligned}$	$\begin{aligned} A &= 2\gamma X_1 \times Y_2 - 2\gamma X_2 \times Y_1 + \eta_1 X_2 - \eta_2 X_1 - \xi_1 \gamma X_2 + \xi_2 \gamma Y_1 \in \mathfrak{J}^C) \\ &= kD + ik(gA) + \frac{i}{2} \{P_1, P_2\} E. \end{aligned}$
0902.0431_EQ1085	184	■1.000	$\operatorname{tr}\left(S_1\right)\tau S_2-S_2\tau S_1=4i\left\{P_1,P_2\right\}=4i\left\{\chi^{-1}S_1,\chi^{-1}S_2\right\}.$	$\operatorname{tr}(S_1\tau S_2-S_2\tau S_1)=4i\{P_1,P_2\}=4i\{\chi^{-1}S_1,\chi^{-1}S_2\}.$	$\operatorname{tr}(S_1\tau S_2-S_2\tau S_1)=4i\{P_1,P_2\}=4i\{\chi^{-1}S_1,\chi^{-1}S_2\}.$
0902.0431_EQ1086	184	■1.000	$S_1\tau S_2-S_2\tau S_1-\frac{1}{8}\operatorname{tr}(S_1\tau S_2-S_2\tau S_1)=kD+ik(gA)$	$S_1\tau S_2-S_2\tau S_1-\frac{1}{8}\operatorname{tr}(S_1\tau S_2-S_2\tau S_1)=kD+ik(gA)$	$S_1\tau S_2-S_2\tau S_1-\frac{1}{8}\operatorname{tr}(S_1\tau S_2-S_2\tau S_1)=kD+ik(gA),$

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0902.0431_EQ1087	184	■ 1.000	$\begin{aligned} & \lambda \gamma P_1 \times P_2 = \left(\begin{array}{c} \gamma Y_1 \\ -\gamma X_1 \\ \eta_1 \\ -\xi_1 \end{array} \right) \times \left(\begin{array}{c} X_2 \\ Y_2 \\ \xi_2 \\ \eta_2 \end{array} \right) \\ & = \Phi \left(\begin{array}{c} -\frac{1}{2}(\gamma Y_1 \vee Y_2 - X_2 \vee \gamma X_1) \\ \frac{1}{4}(2\gamma X_1 \times Y_2 + \eta_1 X_2 + \xi_2 \gamma Y_1) \\ \frac{1}{4}(2\gamma Y_1 \times X_2 + \xi_1 Y_2 + \eta_2 \gamma X_1) \\ \frac{1}{8}((\gamma Y_1, Y_2) - (X_2, \gamma X_1) - 3(\eta_1 \eta_2 - \xi_2 \xi_1)) \end{array} \right). \end{aligned}$	$\lambda \gamma P_1 \times P_2 = \left(\begin{array}{c} \gamma Y_1 \\ -\gamma X_1 \\ \eta_1 \\ -\xi_1 \end{array} \right) \times \left(\begin{array}{c} X_2 \\ Y_2 \\ \xi_2 \\ \eta_2 \end{array} \right)$	$\lambda \gamma P_1 \times P_2 = \left(\begin{array}{c} \gamma Y_1 \\ -\gamma X_1 \\ \eta_1 \\ -\xi_1 \end{array} \right) \times \left(\begin{array}{c} X_2 \\ Y_2 \\ \xi_2 \\ \eta_2 \end{array} \right)$
0902.0431_EQ1094	186	■ 1.000	$= \left[\Theta(\varphi_* D, 0, 0, 0, 0, 0), \Theta(0, 0, 0, 0, 2c, -2c) = [\zeta l(D), \zeta l(icE)] \right].$	$= [\Theta(\varphi_* D, 0, 0, 0, 0, 0), \Theta(0, 0, 0, 0, 2c, -2c) = [\zeta l(D), \zeta l(icE)]].$	$= [\Theta(\varphi_* D, 0, 0, 0, 0, 0), \Theta(0, 0, 0, 0, 2c, -2c) = [\zeta l(D), \zeta l(icE)]].$
0902.0431_EQ1111	188	■ 1.000	$\sigma(\Phi, P, Q, r, s, t) = (\sigma \Phi \sigma, \sigma P, \sigma Q, r, s, t).$	$\sigma(\Phi, P, Q, r, s, t) = (\sigma \Phi \sigma, \sigma P, \sigma Q, r, s, t).$	$\sigma(\Phi, P, Q, r, s, t) = (\sigma \Phi \sigma, \sigma P, \sigma Q, r, s, t).$
0902.0431_EQ1112	188	■ 1.000	$(E_8)^\sigma = \{\alpha \in E_8 \mid \sigma \alpha = \alpha \sigma\}.$	$(E_8)^\sigma = \{\alpha \in E_8 \mid \sigma \alpha = \alpha \sigma\}.$	$(E_8)^\sigma = \{\alpha \in E_8 \mid \sigma \alpha = \alpha \sigma\}.$
0902.0431_EQ1114	189	■ 1.000	$z(E_8) = \{1\}.$	$z(E_8) = \{1\}.$	$z(E_8) = \{1\}.$
0902.0431_EQ1115	189	■ 1.000	$\alpha = \varphi(E, 1) = 1 \quad \text{or} \quad \alpha = \varphi(E, -1) = v.$	$\alpha = \varphi(E, 1) = 1 \quad \text{or} \quad \alpha = \varphi(E, -1) = v.$	$\alpha = \varphi(E, 1) = 1 \quad \text{or} \quad \alpha = \varphi(E, -1) = v.$
0902.0431_EQ1120	190	■ 1.000	$[(D_1, \phi_1, \mathbf{X}_1, \mathbf{Y}_1), (D_2, \phi_2, \mathbf{X}_2, \mathbf{Y}_2)] = (D, \phi, \mathbf{X}, \mathbf{Y}),$	$[(D_1, \phi_1, \mathbf{X}_1, \mathbf{Y}_1), (D_2, \phi_2, \mathbf{X}_2, \mathbf{Y}_2)] = (D, \phi, \mathbf{X}, \mathbf{Y}),$	$[(D_1, \phi_1, \mathbf{X}_1, \mathbf{Y}_1), (D_2, \phi_2, \mathbf{X}_2, \mathbf{Y}_2)] = (D, \phi, \mathbf{X}, \mathbf{Y}),$
0902.0431_EQ1123	190	■ 1.000	$B_8(R_1, R_2) = 60 \operatorname{tr}(D_1 D_2) + \frac{5}{2} B_6(\phi_1, \phi_2) + 15(\mathbf{X}_1, \mathbf{Y}_2) + 15(\mathbf{X}_2, \mathbf{Y}_1)$	$B_8(R_1, R_2) = 60 \operatorname{tr}(D_1 D_2) + \frac{5}{2} B_6(\phi_1, \phi_2) + 15(\mathbf{X}_1, \mathbf{Y}_2) + 15(\mathbf{X}_2, \mathbf{Y}_1)$	$B_8(R_1, R_2) = 60 \operatorname{tr}(D_1 D_2) + \frac{5}{2} B_6(\phi_1, \phi_2) + 15(\mathbf{X}_1, \mathbf{Y}_2) + 15(\mathbf{X}_2, \mathbf{Y}_1)$
0902.0431_EQ1125	190	■ 1.000	$\langle R_1, R_2 \rangle = -B_8(R_1, \tau \tilde{\lambda} R_2).$	$\langle R_1, R_2 \rangle = -B_8(R_1, \tau \tilde{\lambda} R_2).$	$\langle R_1, R_2 \rangle = -B_8(R_1, \tau \tilde{\lambda} R_2).$
0902.0431_EQ1127	191	■ 1.000	$E_8 = \{\alpha \in \operatorname{Aut}(\mathfrak{e}_8^C) \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle\}$	$E_8 = \{\alpha \in \operatorname{Aut}(\mathfrak{e}_8^C) \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle\}$	$E_8 = \{\alpha \in \operatorname{Aut}(\mathfrak{e}_8^C) \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle\}$
0902.0431_EQ1129	191	■ 1.000	$(E_8)^w = \{\alpha \in E_8 \mid w \alpha = \alpha w\}.$	$(E_8)^w = \{\alpha \in E_8 \mid w \alpha = \alpha w\}.$	$(E_8)^w = \{\alpha \in E_8 \mid w \alpha = \alpha w\}.$
0902.0431_EQ1130	191	■ 1.000	$\varphi_1(A)(D, \phi, \mathbf{X}, \mathbf{Y}) = (ADA^{-1}, \phi, A\mathbf{X}, {}^t A^{-1} \mathbf{Y}).$	$\varphi_1(A)(D, \phi, \mathbf{X}, \mathbf{Y}) = (ADA^{-1}, \phi, A\mathbf{X}, {}^t A^{-1} \mathbf{Y}).$	$\varphi_1(A)(D, \phi, \mathbf{X}, \mathbf{Y}) = (ADA^{-1}, \phi, A\mathbf{X}, {}^t A^{-1} \mathbf{Y}).$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E01133	191	■ 1.000	$\varphi_2(\alpha)(D, \phi, \boldsymbol{X}, \boldsymbol{Y}) = \left(D, \alpha \phi \alpha^{-1}, \alpha \boldsymbol{X}, {}^t \alpha^{-1} \boldsymbol{Y} \right).$	$\varphi_2(\alpha)(D, \phi, \boldsymbol{X}, \boldsymbol{Y}) = \left(D, \alpha \phi \alpha^{-1}, \alpha \boldsymbol{X}, {}^t \alpha^{-1} \boldsymbol{Y} \right).$	$\varphi_2(\alpha)(D, \phi, \boldsymbol{X}, \boldsymbol{Y}) = \left(D, \alpha \phi \alpha^{-1}, \alpha \boldsymbol{X}, {}^t \alpha^{-1} \boldsymbol{Y} \right).$
0902.0431_E01135	192	■ 1.000	$\begin{aligned} B_6 \left(\alpha \phi_1 \alpha^{-1}, \tau^t \left(\alpha \phi_2 \alpha^{-1} \right) \tau \right) &= B_6 \left(\alpha \phi_1 \alpha^{-1}, \alpha \tau^t \phi_2 \tau \alpha^{-1} \right) = B_6 \left(\phi_1, \tau^t \phi_2 \tau \right), \\ \langle \alpha \boldsymbol{X}, {}^t \alpha^{-1} \boldsymbol{Y} \rangle &= \langle \boldsymbol{X}, \boldsymbol{Y} \rangle, \end{aligned}$	$B_6 \left(\alpha \phi_1 \alpha^{-1}, \tau^t \left(\alpha \phi_2 \alpha^{-1} \right) \tau \right) = B_6 \left(\alpha \phi_1 \alpha^{-1}, \alpha \tau^t \phi_2 \tau \alpha^{-1} \right) = B_6 \left(\phi_1, \tau^t \phi_2 \tau \right),$ $\langle \alpha \boldsymbol{X}, {}^t \alpha^{-1} \boldsymbol{Y} \rangle = \langle \boldsymbol{X}, \boldsymbol{Y} \rangle,$	$B_6(\alpha \phi_1 \alpha^{-1}, \tau^t(\alpha \phi_2 \alpha^{-1}) \tau) = B_6(\alpha \phi_1 \alpha^{-1}, \alpha \tau^t \phi_2 \tau \alpha^{-1}) = B_6(\phi_1, \tau^t \phi_2 \tau),$ $\langle \alpha \boldsymbol{X}, {}^t \alpha^{-1} \boldsymbol{Y} \rangle = \langle \boldsymbol{X}, \boldsymbol{Y} \rangle,$
0902.0431_E01139	192	■ 1.000	$(\boldsymbol{u} \wedge \boldsymbol{v}) = (\boldsymbol{u} \wedge \boldsymbol{v}, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_n), \quad \boldsymbol{v} \in \Lambda^{n-k}(C^n).$	$(\boldsymbol{u}, \boldsymbol{v}) = (\boldsymbol{u} \wedge \boldsymbol{v}, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_n), \quad \boldsymbol{v} \in \Lambda^{n-k}(C^n).$	$(\boldsymbol{u}, \boldsymbol{v}) = (\boldsymbol{u} \wedge \boldsymbol{v}, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_n), \quad \boldsymbol{v} \in \Lambda^{n-k}(C^n).$
0902.0431_E01140	192	■ 1.000	$\ast: \Lambda^k(C^n) \rightarrow \Lambda^{n-k}(C^n)$	$\ast: \Lambda^k(C^n) \rightarrow \Lambda^{n-k}(C^n)$	$\ast: \Lambda^k(C^n) \rightarrow \Lambda^{n-k}(C^n)$
0902.0431_E01141	192	■ 1.000	$\ast^2 \boldsymbol{u} = (-1)^{k(n-k)} \boldsymbol{u}, \quad \boldsymbol{u} \in \Lambda^k(C^n).$	$\ast^2 \boldsymbol{u} = (-1)^{k(n-k)} \boldsymbol{u}, \quad \boldsymbol{u} \in \Lambda^k(C^n).$	$\ast^2 \boldsymbol{u} = (-1)^{k(n-k)} \boldsymbol{u}, \quad \boldsymbol{u} \in \Lambda^k(C^n).$
0902.0431_E01142	192	■ 1.000	$A(\boldsymbol{x}_1 \wedge \cdots \wedge \boldsymbol{x}_k) = A \boldsymbol{x}_1 \wedge \cdots \wedge A \boldsymbol{x}_k, \quad A1 = 1.$	$A(\boldsymbol{x}_1 \wedge \cdots \wedge \boldsymbol{x}_k) = A \boldsymbol{x}_1 \wedge \cdots \wedge A \boldsymbol{x}_k, \quad A1 = 1.$	$A(\boldsymbol{x}_1 \wedge \cdots \wedge \boldsymbol{x}_k) = A \boldsymbol{x}_1 \wedge \cdots \wedge A \boldsymbol{x}_k, \quad A1 = 1.$
0902.0431_E01143	193	■ 1.000	$D(\boldsymbol{x}_1 \wedge \cdots \wedge \boldsymbol{x}_k) = \sum_{j=1}^k \boldsymbol{x}_1 \wedge \cdots \wedge D \boldsymbol{x}_j \wedge \cdots \wedge \boldsymbol{x}_k, \quad D1 = 0.$	$D(\boldsymbol{x}_1 \wedge \cdots \wedge \boldsymbol{x}_k) = \sum_{j=1}^k \boldsymbol{x}_1 \wedge \cdots \wedge D \boldsymbol{x}_j \wedge \cdots \wedge \boldsymbol{x}_k, \quad D1 = 0.$	$D(\boldsymbol{x}_1 \wedge \cdots \wedge \boldsymbol{x}_k) = \sum_{j=1}^k \boldsymbol{x}_1 \wedge \cdots \wedge D \boldsymbol{x}_j \wedge \cdots \wedge \boldsymbol{x}_k, \quad D1 = 0.$
0902.0431_E01146	193	■ 1.000	$[(D_1, \boldsymbol{u}_1, \boldsymbol{v}_1), (D_2, \boldsymbol{u}_2, \boldsymbol{v}_2)] = (D, \boldsymbol{u}, \boldsymbol{v}),$	$[(D_1, \boldsymbol{u}_1, \boldsymbol{v}_1), (D_2, \boldsymbol{u}_2, \boldsymbol{v}_2)] = (D, \boldsymbol{u}, \boldsymbol{v}),$	$[(D_1, \boldsymbol{u}_1, \boldsymbol{v}_1), (D_2, \boldsymbol{u}_2, \boldsymbol{v}_2)] = (D, \boldsymbol{u}, \boldsymbol{v}),$
0902.0431_E01156	195	■ 1.000	$\boldsymbol{e}_I = \boldsymbol{e}_i \wedge \boldsymbol{e}_j \wedge \boldsymbol{e}_k \in \Lambda^3(C^9).$	$\boldsymbol{e}_I = \boldsymbol{e}_i \wedge \boldsymbol{e}_j \wedge \boldsymbol{e}_k \in \Lambda^3(C^9).$	$\boldsymbol{e}_I = \boldsymbol{e}_i \wedge \boldsymbol{e}_j \wedge \boldsymbol{e}_k \in \Lambda^3(C^9).$
0902.0431_E01158	195	■ 1.000	$D = \frac{1}{3}(E_{ii} + E_{jj} + E_{kk}) - E_{ll}, \quad l \neq i, j, k$	$D = \frac{1}{3}(E_{ii} + E_{jj} + E_{kk}) - E_{ll}, \quad l \neq i, j, k$	$D = \frac{1}{3}(E_{ii} + E_{jj} + E_{kk}) - E_{ll}, \quad l \neq i, j, k.$
0902.0431_E01159	195	■ 1.000	$\begin{aligned} & \left((D, 0, 0), (0, \boldsymbol{e}_i \wedge \boldsymbol{e}_j \wedge \boldsymbol{e}_k, 0) \right) \in \mathfrak{a}, \\ & (0, 0, \boldsymbol{e}_i \wedge \boldsymbol{e}_j \wedge \boldsymbol{e}_k) = [(D, 0, 0), (0, 0, -\boldsymbol{e}_i \wedge \boldsymbol{e}_j \wedge \boldsymbol{e}_k)] \in \mathfrak{a}. \end{aligned}$	$\begin{aligned} (0, \boldsymbol{e}_i \wedge \boldsymbol{e}_j \wedge \boldsymbol{e}_k, 0) &= [(D, 0, 0), (0, \boldsymbol{e}_i \wedge \boldsymbol{e}_j \wedge \boldsymbol{e}_k, 0)] \in \mathfrak{a}, \\ (0, 0, \boldsymbol{e}_i \wedge \boldsymbol{e}_j \wedge \boldsymbol{e}_k) &= [(D, 0, 0), (0, 0, -\boldsymbol{e}_i \wedge \boldsymbol{e}_j \wedge \boldsymbol{e}_k)] \in \mathfrak{a}. \end{aligned}$	$\begin{aligned} (0, \boldsymbol{e}_i \wedge \boldsymbol{e}_j \wedge \boldsymbol{e}_k, 0) &= [(D, 0, 0), (0, \boldsymbol{e}_i \wedge \boldsymbol{e}_j \wedge \boldsymbol{e}_k, 0)] \in \mathfrak{a}, \\ (0, 0, \boldsymbol{e}_i \wedge \boldsymbol{e}_j \wedge \boldsymbol{e}_k) &= [(D, 0, 0), (0, 0, -\boldsymbol{e}_i \wedge \boldsymbol{e}_j \wedge \boldsymbol{e}_k)] \in \mathfrak{a}. \end{aligned}$

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0902.0431_E01161	196	■ 1.000	$B_{\{8\}}\left(\left(D_{\{1\}},\boldsymbol{u}_{\{1\}},\boldsymbol{v}_{\{1\}}\right),\left(D_{\{2\}},\boldsymbol{u}_{\{2\}},\boldsymbol{v}_{\{2\}}\right)\right)=60\left(\mathrm{tr}\left(D_{\{1\}}D_{\{2\}}\right)+\left(\boldsymbol{u}_1,\boldsymbol{v}_2\right)+\left(\boldsymbol{u}_2,\boldsymbol{v}_1\right)\right).$	$B_8\left(\left(D_1,\boldsymbol{u}_1,\boldsymbol{v}_1\right),\left(D_2,\boldsymbol{u}_2,\boldsymbol{v}_2\right)\right)=60\left(\mathrm{tr}\left(D_1D_2\right)+\left(\boldsymbol{u}_1,\boldsymbol{v}_2\right)+\left(\boldsymbol{u}_2,\boldsymbol{v}_1\right)\right).$	$B_8\left(\left(D_1,\boldsymbol{u}_1,\boldsymbol{v}_1\right),\left(D_2,\boldsymbol{u}_2,\boldsymbol{v}_2\right)\right)=60\left(\mathrm{tr}\left(D_1D_2\right)+\left(\boldsymbol{u}_1,\boldsymbol{v}_2\right)+\left(\boldsymbol{u}_2,\boldsymbol{v}_1\right)\right).$
0902.0431_E01162	196	■ 1.000	$B\left(\left(D_{\{1\}},\boldsymbol{u}_{\{1\}},\boldsymbol{v}_{\{1\}}\right),\left(D_{\{2\}},\boldsymbol{u}_{\{2\}},\boldsymbol{v}_{\{2\}}\right)\right)=\mathrm{tr}\left(D_{\{1\}}D_{\{2\}}\right)+\left(\boldsymbol{u}_1,\boldsymbol{v}_2\right)+\left(\boldsymbol{u}_2,\boldsymbol{v}_1\right).$	$B\left(\left(D_1,\boldsymbol{u}_1,\boldsymbol{v}_1\right),\left(D_2,\boldsymbol{u}_2,\boldsymbol{v}_2\right)\right)=\mathrm{tr}\left(D_1D_2\right)+\left(\boldsymbol{u}_1,\boldsymbol{v}_2\right)+\left(\boldsymbol{u}_2,\boldsymbol{v}_1\right).$	$B\left(\left(D_1,\boldsymbol{u}_1,\boldsymbol{v}_1\right),\left(D_2,\boldsymbol{u}_2,\boldsymbol{v}_2\right)\right)=\mathrm{tr}\left(D_1D_2\right)+\left(\boldsymbol{u}_1,\boldsymbol{v}_2\right)+\left(\boldsymbol{u}_2,\boldsymbol{v}_1\right).$
0902.0431_E01163	196	■ 1.000	$B_{\{8\}}(R,R)=120,\quad B(R,R)=2.$	$B_8(R,R)=120,\quad B(R,R)=2.$	$B_8(R,R)=120,\quad B(R,R)=2.$
0902.0431_E01164	196	■ 1.000	$\tau\widetilde{\lambda}(D,\boldsymbol{u},\boldsymbol{v})=\left(-\tau^tD,-\tau\boldsymbol{v},-\tau\boldsymbol{u}\right).$	$\tau\widetilde{\lambda}(D,\boldsymbol{u},\boldsymbol{v})=\left(-\tau^tD,-\tau\boldsymbol{v},-\tau\boldsymbol{u}\right).$	$\tau\widetilde{\lambda}(D,\boldsymbol{u},\boldsymbol{v})=\left(-\tau^tD,-\tau\boldsymbol{v},-\tau\boldsymbol{u}\right).$
0902.0431_E01166	196	■ 1.000	$\left\langle R_1,R_2\right\rangle =60\left(\mathrm{tr}\left(D_1\tau^tD_2\right)+\left(\boldsymbol{u}_1,\tau\boldsymbol{u}_2\right)+\left(\boldsymbol{v}_2,\tau\boldsymbol{v}_1\right)\right).$	$\langle R_1,R_2\rangle =60\left(\mathrm{tr}\left(D_1\tau^tD_2\right)+\left(\boldsymbol{u}_1,\tau\boldsymbol{u}_2\right)+\left(\boldsymbol{v}_2,\tau\boldsymbol{v}_1\right)\right).$	$\langle R_1,R_2\rangle =60\left(\mathrm{tr}\left(D_1\tau^tD_2\right)+\left(\boldsymbol{u}_1,\tau\boldsymbol{u}_2\right)+\left(\boldsymbol{v}_2,\tau\boldsymbol{v}_1\right)\right).$
0902.0431_E01167	196	■ 1.000	$E_{\{8\}}=\left\{\alpha\in\mathrm{Aut}\left(\mathrm{e}_{\{8\}}^C\right)\mid\left\langle\alpha R_1,\alpha R_2\right\rangle=\left\langle R_1,R_2\right\rangle\right\}$	$E_8=\left\{\alpha\in\mathrm{Aut}\left(\mathfrak{e}_8^C\right)\mid\left\langle\alpha R_1,\alpha R_2\right\rangle=\left\langle R_1,R_2\right\rangle\right\}$	$E_8=\left\{\alpha\in\mathrm{Aut}(\mathfrak{e}_8^C)\mid\left\langle\alpha R_1,\alpha R_2\right\rangle=\left\langle R_1,R_2\right\rangle\right\}$
0902.0431_E01169	197	■ 1.000	$\left(E_{\{8\}}\right)^{w_3}=\left\{\alpha\in E_{\{8\}}\mid w_3\alpha=\alpha w_3\right\}.$	$\left(E_8\right)^{w_3}=\left\{\alpha\in E_8\mid w_3\alpha=\alpha w_3\right\}.$	$\left(E_8\right)^{w_3}=\left\{\alpha\in E_8\mid w_3\alpha=\alpha w_3\right\}.$
0902.0431_E01170	197	■ 1.000	$\varphi(A)(D,\boldsymbol{u},\boldsymbol{v})=\left(ADA^{-1},A\boldsymbol{u},{}^tA^{-1}\boldsymbol{v}\right).$	$\varphi(A)(D,\boldsymbol{u},\boldsymbol{v})=\left(ADA^{-1},A\boldsymbol{u},{}^tA^{-1}\boldsymbol{v}\right).$	$\varphi(A)(D,\boldsymbol{u},\boldsymbol{v})=\left(ADA^{-1},A\boldsymbol{u},{}^tA^{-1}\boldsymbol{v}\right).$
0902.0431_E01177	198	■ 1.000	$*(*\boldsymbol{a}\wedge*(\boldsymbol{b}\wedge\boldsymbol{c})+*\boldsymbol{b}\wedge*(\boldsymbol{c}\wedge\boldsymbol{a})+*\boldsymbol{c}\wedge*(\boldsymbol{a}\wedge\boldsymbol{b}))$	$*(*\boldsymbol{a}\wedge*(\boldsymbol{b}\wedge\boldsymbol{c})+*\boldsymbol{b}\wedge*(\boldsymbol{c}\wedge\boldsymbol{a})+*\boldsymbol{c}\wedge*(\boldsymbol{a}\wedge\boldsymbol{b}))$	$*(*\boldsymbol{a}\wedge*(\boldsymbol{b}\wedge\boldsymbol{c})+*\boldsymbol{b}\wedge*(\boldsymbol{c}\wedge\boldsymbol{a})+*\boldsymbol{c}\wedge*(\boldsymbol{a}\wedge\boldsymbol{b}))$
0902.0431_E01184	199	■ 1.000	$\left((\boldsymbol{x}\times\boldsymbol{y})\boldsymbol{z},\boldsymbol{v}\right)=-\left(\boldsymbol{x}\wedge\boldsymbol{z},\boldsymbol{y}\wedge\boldsymbol{v}\right)+\frac{4}{5}\left(\boldsymbol{x},\boldsymbol{y}\right)\left(\boldsymbol{z},\boldsymbol{v}\right)=\left(\boldsymbol{y},\boldsymbol{z}\right)\left(\boldsymbol{x},\boldsymbol{v}\right)-\frac{1}{5}\left(\boldsymbol{x},\boldsymbol{y}\right)\left(\boldsymbol{z},\boldsymbol{v}\right),$	$\left((\boldsymbol{x}\times\boldsymbol{y})\boldsymbol{z},\boldsymbol{v}\right)=-\left(\boldsymbol{x}\wedge\boldsymbol{z},\boldsymbol{y}\wedge\boldsymbol{v}\right)+\frac{4}{5}\left(\boldsymbol{x},\boldsymbol{y}\right)\left(\boldsymbol{z},\boldsymbol{v}\right)=\left(\boldsymbol{y},\boldsymbol{z}\right)\left(\boldsymbol{x},\boldsymbol{v}\right)-\frac{1}{5}\left(\boldsymbol{x},\boldsymbol{y}\right)\left(\boldsymbol{z},\boldsymbol{v}\right),$	$\left((\boldsymbol{x}\times\boldsymbol{y})\boldsymbol{z},\boldsymbol{v}\right)=-\left(\boldsymbol{x}\wedge\boldsymbol{z},\boldsymbol{y}\wedge\boldsymbol{v}\right)+\frac{4}{5}\left(\boldsymbol{x},\boldsymbol{y}\right)\left(\boldsymbol{z},\boldsymbol{v}\right)=\left(\boldsymbol{y},\boldsymbol{z}\right)\left(\boldsymbol{x},\boldsymbol{v}\right)-\frac{1}{5}\left(\boldsymbol{x},\boldsymbol{y}\right)\left(\boldsymbol{z},\boldsymbol{v}\right),$
0902.0431_E01185 (i)	199	■ 1.000	$\left(\boldsymbol{x}\times\boldsymbol{y}\right)\boldsymbol{z}=\left(\boldsymbol{y},\boldsymbol{z}\right)\boldsymbol{x}-\frac{1}{5}\left(\boldsymbol{x},\boldsymbol{y}\right)\boldsymbol{z}.$	$\left(\boldsymbol{x}\times\boldsymbol{y}\right)\boldsymbol{z}=\left(\boldsymbol{y},\boldsymbol{z}\right)\boldsymbol{x}-\frac{1}{5}\left(\boldsymbol{x},\boldsymbol{y}\right)\boldsymbol{z}.$	$\left(\boldsymbol{x}\times\boldsymbol{y}\right)\boldsymbol{z}=\left(\boldsymbol{y},\boldsymbol{z}\right)\boldsymbol{x}-\frac{1}{5}\left(\boldsymbol{x},\boldsymbol{y}\right)\boldsymbol{z}.$

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0902.0431_E01191	200	■ 1.000	$\begin{aligned} & ((\boldsymbol{a} \times \boldsymbol{b})\boldsymbol{v}, \boldsymbol{w}) = (\boldsymbol{a} \wedge \boldsymbol{v}, \boldsymbol{b} \wedge \boldsymbol{w}) - \frac{3}{5}(\boldsymbol{a}, \boldsymbol{b})(\boldsymbol{v}, \boldsymbol{w}) \\ & = -(\boldsymbol{a}_1, \boldsymbol{w})(\boldsymbol{x} \wedge \boldsymbol{a}_2, \boldsymbol{b}) + (\boldsymbol{a}_2, \boldsymbol{w})(\boldsymbol{x} \wedge \boldsymbol{a}_1, \boldsymbol{b}) + \frac{2}{5}(\boldsymbol{a}, \boldsymbol{b})(\boldsymbol{v}, \boldsymbol{w}), \end{aligned}$	$((\boldsymbol{a} \times \boldsymbol{b})\boldsymbol{v}, \boldsymbol{w}) = (\boldsymbol{a} \wedge \boldsymbol{v}, \boldsymbol{b} \wedge \boldsymbol{w}) - \frac{3}{5}(\boldsymbol{a}, \boldsymbol{b})(\boldsymbol{v}, \boldsymbol{w})$ $= -(\boldsymbol{a}_1, \boldsymbol{w})(\boldsymbol{x} \wedge \boldsymbol{a}_2, \boldsymbol{b}) + (\boldsymbol{a}_2, \boldsymbol{w})(\boldsymbol{x} \wedge \boldsymbol{a}_1, \boldsymbol{b}) + \frac{2}{5}(\boldsymbol{a}, \boldsymbol{b})(\boldsymbol{v}, \boldsymbol{w})$	$((\boldsymbol{a} \times \boldsymbol{b})\boldsymbol{v}, \boldsymbol{w}) = (\boldsymbol{a} \wedge \boldsymbol{v}, \boldsymbol{b} \wedge \boldsymbol{w}) - \frac{3}{5}(\boldsymbol{a}, \boldsymbol{b})(\boldsymbol{v}, \boldsymbol{w})$ $= -(\boldsymbol{a}_1, \boldsymbol{w})(\boldsymbol{x} \wedge \boldsymbol{a}_2, \boldsymbol{b}) + (\boldsymbol{a}_2, \boldsymbol{w})(\boldsymbol{x} \wedge \boldsymbol{a}_1, \boldsymbol{b}) + \frac{2}{5}(\boldsymbol{a}, \boldsymbol{b})(\boldsymbol{v}, \boldsymbol{w}),$
0902.0431_E01194	201	■ 1.000	$(\boldsymbol{x} \times \boldsymbol{y})\boldsymbol{a} = (\boldsymbol{y}, \boldsymbol{a}_1)\boldsymbol{x} \wedge \boldsymbol{a}_2 - (\boldsymbol{y}, \boldsymbol{a}_2)\boldsymbol{x} \wedge \boldsymbol{a}_1 - \frac{2}{5}(\boldsymbol{x}, \boldsymbol{y})\boldsymbol{a}.$	$(\boldsymbol{x} \times \boldsymbol{y})\boldsymbol{a} = (\boldsymbol{y}, \boldsymbol{a}_1)\boldsymbol{x} \wedge \boldsymbol{a}_2 - (\boldsymbol{y}, \boldsymbol{a}_2)\boldsymbol{x} \wedge \boldsymbol{a}_1 - \frac{2}{5}(\boldsymbol{x}, \boldsymbol{y})\boldsymbol{a}.$	$(\boldsymbol{x} \times \boldsymbol{y})\boldsymbol{a} = (\boldsymbol{y}, \boldsymbol{a}_1)\boldsymbol{x} \wedge \boldsymbol{a}_2 - (\boldsymbol{y}, \boldsymbol{a}_2)\boldsymbol{x} \wedge \boldsymbol{a}_1 - \frac{2}{5}(\boldsymbol{x}, \boldsymbol{y})\boldsymbol{a}.$
0902.0431_E01198	201	■ 1.000	$\left[(C, D), \left(E_{45}, 0\right)\right] = \left(5E_{45}, 0\right) \in \mathfrak{g}_{01} \cap \mathfrak{a},$	$[(C, D), (E_{45}, 0)] = (5E_{45}, 0) \in \mathfrak{g}_{01} \cap \mathfrak{a},$	$[(C, D), (E_{45}, 0)] = (5E_{45}, 0) \in \mathfrak{g}_{01} \cap \mathfrak{a},$
0902.0431_E01199	202	■ 1.000	$\mathfrak{a} \supset \left[\mathfrak{g}_1, \mathfrak{g}_{-1}\right] \ni \left[\boldsymbol{e}_1 \otimes (\boldsymbol{e}_1 \wedge \boldsymbol{e}_2), \boldsymbol{e}_1 \otimes (\boldsymbol{e}_1 \wedge \boldsymbol{e}_3)\right] = (0, -E_{23}),$	$\mathfrak{a} \supset [\mathfrak{g}_1, \mathfrak{g}_{-1}] \ni [\boldsymbol{e}_1 \otimes (\boldsymbol{e}_1 \wedge \boldsymbol{e}_2), \boldsymbol{e}_1 \otimes (\boldsymbol{e}_1 \wedge \boldsymbol{e}_3)] = (0, -E_{23}),$	$\mathfrak{a} \supset [\mathfrak{g}_1, \mathfrak{g}_{-1}] \ni [\boldsymbol{e}_1 \otimes (\boldsymbol{e}_1 \wedge \boldsymbol{e}_2), \boldsymbol{e}_1 \otimes (\boldsymbol{e}_1 \wedge \boldsymbol{e}_3)] = (0, -E_{23}),$
0902.0431_E01200	202	■ 1.000	$\operatorname{ad}(T) \operatorname{ad}(S_{1523}) \operatorname{ad}(S_{1415}) \operatorname{ad}(S_{1314}) \operatorname{ad}(S_{1213}) R = (-E_{12}, 0) \in \mathfrak{g}_{01} \cap \mathfrak{a}.$	$\operatorname{ad}(T) \operatorname{ad}(S_{1523}) \operatorname{ad}(S_{1415}) \operatorname{ad}(S_{1314}) \operatorname{ad}(S_{1213}) R = (-E_{12}, 0) \in \mathfrak{g}_{01} \cap \mathfrak{a}.$	$\operatorname{ad}(T) \operatorname{ad}(S_{1523}) \operatorname{ad}(S_{1415}) \operatorname{ad}(S_{1314}) \operatorname{ad}(S_{1213}) R = (-E_{12}, 0) \in \mathfrak{g}_{01} \cap \mathfrak{a}.$
0902.0431_E01203	202	■ 1.000	$B_8(R, R) = 120, \quad B(R, R) = 2.$	$B_8(R, R) = 120, \quad B(R, R) = 2.$	$B_8(R, R) = 120, \quad B(R, R) = 2.$
0902.0431_E01204	202	■ 1.000	$\begin{aligned} & \tau \widetilde{\lambda}(C, D, \boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}, \boldsymbol{w} \otimes \boldsymbol{d}) \\ & = (-\tau^t C, -\tau^t D, \tau \boldsymbol{w} \otimes \tau \boldsymbol{d}, \tau \boldsymbol{c} \otimes \tau \boldsymbol{z}, \tau \boldsymbol{b} \otimes \tau \boldsymbol{y}, \tau \boldsymbol{x} \otimes \tau \boldsymbol{a}). \end{aligned}$	$\tau \widetilde{\lambda}(C, D, \boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}, \boldsymbol{w} \otimes \boldsymbol{d})$ $= (-\tau^t C, -\tau^t D, \tau \boldsymbol{w} \otimes \tau \boldsymbol{d}, \tau \boldsymbol{c} \otimes \tau \boldsymbol{z}, \tau \boldsymbol{b} \otimes \tau \boldsymbol{y}, \tau \boldsymbol{x} \otimes \tau \boldsymbol{a}).$	$\tau \widetilde{\lambda}(C, D, \boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}, \boldsymbol{w} \otimes \boldsymbol{d})$ $= (-\tau^t C, -\tau^t D, \tau \boldsymbol{w} \otimes \tau \boldsymbol{d}, \tau \boldsymbol{c} \otimes \tau \boldsymbol{z}, \tau \boldsymbol{b} \otimes \tau \boldsymbol{y}, \tau \boldsymbol{x} \otimes \tau \boldsymbol{a}).$
0902.0431_E01205	203	■ 1.000	$\langle R_1, R_2 \rangle = -B_8(R_1, \tau \widetilde{\lambda} R_2).$	$\langle R_1, R_2 \rangle = -B_8(R_1, \tau \widetilde{\lambda} R_2).$	$\langle R_1, R_2 \rangle = -B_8(R_1, \tau \widetilde{\lambda} R_2).$
0902.0431_E01207	203	■ 1.000	$E_8 = \{\alpha \in \operatorname{Aut}(\mathfrak{e}_8^C) \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle\}$	$E_8 = \{\alpha \in \operatorname{Aut}(\mathfrak{e}_8^C) \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle\}$	$E_8 = \{\alpha \in \operatorname{Aut}(\mathfrak{e}_8^C) \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle\}$

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0902.0431_E01208	203	■ 1.000	$z_{\{5\}}\left(C, D, g_{\{1\}}, g_{\{2\}}, g_{\{-2\}}, g_{\{-1\}}\right)=\left(C, D, \zeta\left(g_{\{1\}}\right), \zeta^2\left(g_{\{2\}}\right)\left(g_{\{2\}}\right), \zeta^3\left(g_{\{-2\}}\right), \zeta^4\left(g_{\{-1\}}\right)\left(g_{\{-1\}}\right)\right) .$	$z_5(C, D, g_1, g_2, g_{-2}, g_{-1}) = (C, D, \zeta(g_1), \zeta^2(g_2), \zeta^3(g_{-2}), \zeta^4(g_{-1})).$	$z_5(C, D, g_1, g_2, g_{-2}, g_{-1}) = (C, D, \zeta(g_1), \zeta^2(g_2), \zeta^3(g_{-2}), \zeta^4(g_{-1})).$
0902.0431_E01209	203	■ 1.000	$\left(E_{\{8\}}\right)^{z_{\{5\}}}=\left\{\alpha \in E_{\{8\}} \mid z_{\{5\}} \alpha=\alpha z_{\{5\}}\right\} .$	$\left(E_8\right)^{z_5}=\left\{\alpha \in E_8 \mid z_5 \alpha=\alpha z_5\right\} .$	$\left(E_8\right)^{z_5}=\left\{\alpha \in E_8 \mid z_5 \alpha=\alpha z_5\right\} .$
0902.0431_E01212	203	■ 1.000	$\left\langle\varphi(A) R_1, \varphi_1(A) R_2\right\rangle=\left\langle R_1, R_2\right\rangle .$	$\langle\varphi(A)R_1, \varphi_1(A)R_2\rangle = \langle R_1, R_2\rangle .$	$\langle\varphi(A)R_1, \varphi_1(A)R_2\rangle = \langle R_1, R_2\rangle .$
0902.0431_E01213	204	■ 1.000	$\varphi(A, B)=\varphi_{\{1\}}(A) \varphi_{\{2\}}(B) .$	$\varphi(A, B)=\varphi_1(A) \varphi_2(B).$	$\varphi(A, B)=\varphi_1(A) \varphi_2(B).$
0902.0431_E01215	204	■ 1.000	$\begin{aligned} \mathfrak{e}_{\{8(8)\}} &\&= \mathfrak{e}_{\{7(7)\}} \oplus \mathfrak{P}^{\prime} \oplus \mathfrak{P}^{\prime} \oplus \boldsymbol{R} \oplus \boldsymbol{R} \oplus \boldsymbol{R}, \quad\left(\text { where } \mathfrak{e}_{7(7)}=\left(\mathfrak{e}_7^C\right)^{\tau \gamma}\right), \\ \mathfrak{e}_{\{8(-24)\}} &= \mathfrak{e}_{7(-25)} \oplus \mathfrak{P} \oplus \mathfrak{P} \oplus \boldsymbol{R} \oplus \boldsymbol{R} \oplus \boldsymbol{R}, \quad\left(\text { where } \mathfrak{e}_{7(-25)}=\left(\mathfrak{e}_7^C\right)^{\tau}\right) . \end{aligned}$	$\begin{aligned} \mathfrak{e}_{8(8)} &= \mathfrak{e}_{7(7)} \oplus \mathfrak{P}' \oplus \mathfrak{P}' \oplus \boldsymbol{R} \oplus \boldsymbol{R} \oplus \boldsymbol{R}, \quad\left(\text { where } \mathfrak{e}_{7(7)}=\left(\mathfrak{e}_7^C\right)^{\tau \gamma}\right), \\ \mathfrak{e}_{8(-24)} &= \mathfrak{e}_{7(-25)} \oplus \mathfrak{P} \oplus \mathfrak{P} \oplus \boldsymbol{R} \oplus \boldsymbol{R} \oplus \boldsymbol{R}, \quad\left(\text { where } \mathfrak{e}_{7(-25)}=\left(\mathfrak{e}_7^C\right)^{\tau}\right) . \end{aligned}$	$\begin{aligned} \mathfrak{e}_{8(8)} &= \mathfrak{e}_{7(7)} \oplus \mathfrak{P}' \oplus \mathfrak{P}' \oplus \boldsymbol{R} \oplus \boldsymbol{R} \oplus \boldsymbol{R}, \quad\left(\text { where } \mathfrak{e}_{7(7)}=\left(\mathfrak{e}_7^C\right)^{\tau \gamma}\right), \\ \mathfrak{e}_{8(-24)} &= \mathfrak{e}_{7(-25)} \oplus \mathfrak{P} \oplus \mathfrak{P} \oplus \boldsymbol{R} \oplus \boldsymbol{R} \oplus \boldsymbol{R}, \quad\left(\text { where } \mathfrak{e}_{7(-25)}=\left(\mathfrak{e}_7^C\right)^{\tau}\right). \end{aligned}$
0902.0431_E01219	204	■ 1.000	$z\left(E_{\{8(8)\}}\right)=\{1\}, \quad \text { quad } z\left(E_{\{8(-24)\}}\right)=\{1\} .$	$z\left(E_{8(8)}\right)=\{1\}, \quad z\left(E_{8(-24)}\right)=\{1\} .$	$z\left(E_{8(8)}\right)=\{1\}, \quad z\left(E_{8(-24)}\right)=\{1\} .$
0902.0431_E00341	55	—	—	—	$\begin{aligned} \lambda_0-\lambda_1 &= \alpha_1 \\ \lambda_0-\lambda_2 &= \alpha_1 + \alpha_2 \\ -\lambda_0+\lambda_3 &= \alpha_2 + 2\alpha_3 + 2\alpha_4 \\ \lambda_1-\lambda_2 &= \alpha_2 \\ -\lambda_1+\lambda_3 &= \alpha_1 + \alpha_2 + 2\alpha_3 + 2\alpha_4 \\ -\lambda_2+\lambda_3 &= \alpha_1 + 2\alpha_2 + 2\alpha_3 + 2\alpha_4 \\ \lambda_0+\lambda_1 &= \alpha_1 + 2\alpha_2 + 2\alpha_3 \\ \lambda_0+\lambda_2 &= \alpha_1 + \alpha_2 + 2\alpha_3 \\ \lambda_0+\lambda_3 &= 2\alpha_1 + 3\alpha_2 + 4\alpha_3 + 2\alpha_4 \\ \lambda_1+\lambda_2 &= \alpha_2 + 2\alpha_3 \\ \lambda_1+\lambda_3 &= \alpha_1 + 3\alpha_2 + 4\alpha_3 + 2\alpha_4 \\ \lambda_2+\lambda_3 &= \alpha_1 + 2\alpha_2 + 4\alpha_3 + 4\alpha_4, \\ \frac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3) &= \alpha_1 + 2\alpha_2 + 3\alpha_3 + \alpha_4 \\ \frac{1}{2}(-\lambda_0-\lambda_1-\lambda_2+\lambda_3) &= \alpha_4 \\ \frac{1}{2}(\lambda_0+\lambda_1-\lambda_2+\lambda_3) &= \alpha_1 + 2\alpha_2 + 2\alpha_3 + \alpha_4 \\ \frac{1}{2}(\lambda_0-\lambda_1+\lambda_2+\lambda_3) &= \alpha_1 + \alpha_2 + 2\alpha_3 + \alpha_4 \\ \frac{1}{2}(-\lambda_0-\lambda_1+\lambda_2+\lambda_3) &= \alpha_3 + \alpha_4 \\ \frac{1}{2}(-\lambda_0+\lambda_1-\lambda_2+\lambda_3) &= \alpha_2 + \alpha_3 + \alpha_4 \\ \frac{1}{2}(\lambda_0-\lambda_1-\lambda_2+\lambda_3) &= \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 \\ \frac{1}{2}(-\lambda_0+\lambda_1+\lambda_2+\lambda_3) &= \alpha_2 + 2\alpha_3 + \alpha_4. \end{aligned}$

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0902.0431_E00758	126	—	—	—	$\begin{aligned} \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 2 & 3 & 3 & 2 & 1 & 1 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 0 & 1 & 2 & 3 & 2 & 1 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 1 & 2 & 3 & 2 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 2 & 2 & 3 & 2 & 1 & 1 \\ \frac{1}{2}(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 1 & 2 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 1 & 1 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) &= 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 2 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 0 & 1 & 1 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 0 & 1 & 2 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 3 & 4 & 3 & 2 & 2 \\ \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 0 & 0 & 1 & 2 & 1 & 1 & 1 \\ -\mu_1 - \frac{2}{3}\nu &= 1 & 2 & 3 & 4 & 2 & 1 & 2 \\ \mu_2 + \frac{2}{3}\nu &= 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ -\mu_3 - \frac{2}{3}\nu &= 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00940	156	—	—	—	$ \begin{aligned} &P = (F_1(1), 0, 0, 0) \\ &\xrightarrow{f} \mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 + \mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 \\ &\xrightarrow{D} (D\mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 + \mathbf{e}_2 \wedge D\mathbf{e}_4 \wedge \mathbf{e}_5 + \mathbf{e}_2 \wedge \mathbf{e}_4 \wedge D\mathbf{e}_5) \\ &\quad + (D\mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 + \mathbf{e}_3 \wedge D\mathbf{e}_6 \wedge \mathbf{e}_4 + \mathbf{e}_3 \wedge \mathbf{e}_6 \wedge D\mathbf{e}_4) \\ &= \left(b_{12}\mathbf{e}_1 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 + \left(b_{22} + \frac{\nu}{3} \right) \mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 - \bar{b}_{23}\mathbf{e}_3 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 - \bar{l}_{23}\mathbf{e}_6 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 \right. \\ &\quad + l_{11}\mathbf{e}_2 \wedge \mathbf{e}_1 \wedge \mathbf{e}_5 + l_{31}\mathbf{e}_2 \wedge \mathbf{e}_3 \wedge \mathbf{e}_5 + \left(c_{11} - \frac{\nu}{3} \right) \mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 - \bar{c}_{13}\mathbf{e}_2 \wedge \mathbf{e}_6 \wedge \mathbf{e}_5 \\ &\quad + l_{12}\mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_1 + l_{32}\mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_3 + \left(c_{22} - \frac{\nu}{3} \right) \mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 - \bar{c}_{23}\mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_6 \Big) \\ &\quad + \left(b_{13}\mathbf{e}_1 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 + b_{23}\mathbf{e}_2 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 + \left(b_{33} + \frac{\nu}{3} \right) \mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 - \bar{l}_{32}\mathbf{e}_5 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 \right. \\ &\quad + l_{13}\mathbf{e}_3 \wedge \mathbf{e}_1 \wedge \mathbf{e}_4 + l_{23}\mathbf{e}_3 \wedge \mathbf{e}_2 \wedge \mathbf{e}_4 + c_{23}\mathbf{e}_3 \wedge \mathbf{e}_5 \wedge \mathbf{e}_4 + \left(c_{33} - \frac{\nu}{3} \right) \mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 \\ &\quad \left. + l_{11}\mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_1 + l_{21}\mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_2 + \left(c_{11} - \frac{\nu}{3} \right) \mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 - \bar{c}_{12}\mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_5 \right) \\ &\xrightarrow{f^{-1}} \left(\begin{array}{c} \begin{pmatrix} 0 & h(b_{13}, c_{13}) & 0 \\ * & h(b_{23} + \bar{c}_{23}) & h(b_{22} - c_{33} - \frac{\nu}{3}, -b_{33} + c_{33} + \frac{\nu}{3}) \\ h(\bar{c}_{12}, \bar{b}_{12}) & * & -h(\bar{b}_{23} + c_{23}) \end{pmatrix} \\ \begin{pmatrix} -h(l_{23} + l_{32}) & h(l_{13}, \bar{l}_{31}) & * \\ * & 0 & -h(l_{11}) \\ h(l_{21}, \bar{l}_{12}) & * & 0 \end{pmatrix} \\ 0 \\ -h(\bar{l}_{23} + \bar{l}_{32}) \end{array} \right) \end{aligned} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E01028	172	—	—	—	$\begin{aligned} \lambda_2 + \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 1 & 2 & 2 & 1 & 1 & 0 & 1 \\ \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 1 & 2 & 2 & 1 & 0 & 0 & 1 \\ \lambda_0 - \frac{1}{2}(\mu_2 - \mu_3) &= 2 & 3 & 4 & 3 & 3 & 2 & 1 & 2 \\ \lambda_1 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ \lambda_2 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ \lambda_3 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 1 & 2 & 2 & 2 & 2 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 1 & 2 & 2 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 1 & 2 & 2 & 2 & 1 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 3 & 4 & 3 & 2 & 2 & 1 & 2 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 3 & 4 & 3 & 2 & 1 & 1 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 3 & 4 & 3 & 2 & 1 & 0 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 2 & 2 & 1 & 0 & 0 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 2 & 1 & 1 & 1 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 2 & 1 & 1 & 0 & 0 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 2 & 1 & 1 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 3 & 4 & 3 & 3 & 2 & 1 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 1 & 2 & 2 & 1 & 1 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 1 & 2 & 2 & 1 & 0 & 0 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 1 & 2 & 2 & 1 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 4 & 4 & 3 & 2 & 1 & 2 \\ \mu_1 + \frac{2}{3}\nu &= 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ \mu_2 + \frac{2}{3}\nu &= 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ -\mu_3 - \frac{2}{3}\nu &= 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \end{aligned}$